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Arthur Cayley

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## 630.

### ON AN EXPRESSION FOR $1 \pm \sin(2p + 1)u$ IN TERMS OF $\sin u$ .

[From the *Messenger of Mathematics*, vol. v. (1876), pp. 7, 8.]

Write  $\sin u = x$ , then we have

$$\begin{array}{ll} \sin u = x, & \cos u = \sqrt{(1 - x^2)}, \\ \sin 3u = 3x - 4x^3, & \cos 3u = (1 - 4x^2)\sqrt{(1 - x^2)}, \\ \sin 5u = 5x - 20x^3 + 16x^5, & \cos 5u = (1 - 12x^2 + 16x^4)\sqrt{(1 - x^2)}, \\ \&c. & \&c. \end{array}$$

It is hence clear, that in general

$$\begin{aligned} 1 - \sin(2p + 1)u &= (1 \pm x)\{(1, x)^p\}^2, \\ 1 + \sin(2p + 1)u &= (1 \mp x)\{(1, -x)^p\}^2, \end{aligned}$$

where  $(1, x)^p$  denotes a rational and integral function of  $x$  of the order  $p$ , and  $(1, -x)^p$  the same function of  $-x$ ; for it is only in this manner that we can have

$$\cos^2(2p + 1)u = (1 - x^2) \{(1, x)^p\}^2.$$

We, in fact, find

$$\begin{aligned} 1 + \sin u &= 1 + x, \\ 1 - \sin 3u &= (1 + x)(1 - 2x)^2, \\ 1 + \sin 5u &= (1 + x)(1 + 2x - 4x^2)^2, \\ 1 - \sin 7u &= (1 + x)(1 - 4x - 4x^2 + 8x^3)^2, \\ &\text{\&c.} \end{aligned}$$

and it thus appears that the form is

$$1 + (-)^p \sin(2p + 1)u = (1 + x) \{(1, x)^p\}^2.$$

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OF  $\sin u$ .* [630

To find herein the expression of the factor  $(1, x)^p$ , write  $u = \frac{1}{2}\pi - \theta$  and consequently  $x = \cos \theta$ ; we have therefore

$$1 + \cos(2p+1)\theta = (1 + \cos \theta) \{(1, x)^p\}^2,$$

where in the second factor on the right-hand side  $x$  is retained to stand for its value  $\cos \theta$ . This gives

$$2 \cos^2(p + \frac{1}{2})\theta = 2 \cos^2 \frac{1}{2}\theta \cdot \{(1, x)^p\}^2,$$

or, what is the same thing,

$$(1, x)^p = \frac{\cos(p + \frac{1}{2})\theta}{\cos \frac{1}{2}\theta},$$

viz. this is

$$= \cos p\theta - \sin p\theta \cdot \frac{\sin \frac{1}{2}\theta}{\cos \frac{1}{2}\theta},$$

which is

$$= \cos p\theta - \sin p\theta \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}.$$

We have

$$\cos p\theta + i \sin p\theta = \{x + i\sqrt{(1-x^2)}\}^p = X + i\sqrt{(1-x^2)} Y, \text{ suppose,}$$

where  $X, Y$  are rational and integral functions of  $x$  of the orders  $p$  and  $p-1$  respectively; that is,

$$\cos p\theta = X, \quad \sin p\theta = \sin \theta \cdot Y,$$

and we have therefore

$$(1, x)^p = X - Y(1-x),$$

which is the required expression for  $(1, x)^p$ . For instance  
 $p = 3$ ,

$$X + i\sqrt{(1-x^2)} Y = \{x + i\sqrt{(1-x^2)}\}^3;$$

that is,

$$X = -3x + 4x^3, \quad Y = -1 + 4x^2,$$

and  $\therefore -(1-x)Y = 1 - x - 4x^2 + 4x^3$ , so that

$$X - (1-x)Y = 1 - 4x - 4x^2 + 8x^3 = (1, x)^3,$$

and hence

$$1 - \sin 7u = (1+x)(1-4x-4x^2+8x^3)^2,$$

which agrees with a result already obtained.

The foregoing value of  $(1, x)^p$  may also be written

$$(1, x)^p = \frac{1}{x+1} \left\{ \left( \frac{x+1}{2} \right)^{p+1} - \left( \frac{x-1}{2} \right)^{p+1} \right\},$$

which however is not practically so convenient.

The formula corresponds to a like formula in elliptic functions, viz. writing  $\operatorname{sn} u = x$ , the numerator of  $1 + (-)^p \operatorname{sn}(2p+1)u$  is

$$= (1+x) \{(1, x)^p\}^2,$$

which is  $(1+x)$  multiplied by the square of a rational and integral function of  $x$ .

## 631.

### SYNOPSIS OF THE THEORY OF EQUATIONS.

[From the *Messenger of Mathematics*, vol. v. (1876),  
pp. 39—49.]

The following was proposed, as one of the subjects of a Dissertation for the Trinity Fellowships:

*Synopsis of the theory of equations*; i.e. a statement in a logical order, of the divisions of the subject and the leading questions and theorems, but without demonstrations.

In the subject "Theory of Equations," the term *equation* is used to denote an equation of the form  $x^n - p_1x^{n-1} + \dots + p_n = 0$ , where  $p_1, p_2, \dots, p_n$  are regarded as known, and  $x$  as a quantity to be determined; for shortness, the equation is written  $f(x) = 0$ .

The equation may be *numerical*; that is, the coefficients  $p_1, p_2, \dots, p_n$  are then numbers; understanding by number, a quantity of the form  $\alpha + \beta i$ , where  $\alpha$  and  $\beta$  have any positive or negative real values whatever; or say, each of these is regarded as susceptible of continuous variation from an indefinitely large negative to an indefinitely large positive value: and  $i$  denotes  $\sqrt{(-1)}$ .

Or the equation may be *algebraic*; viz. the coefficients are then not restricted to denote, or are not explicitly considered as denoting, numbers.

I. We consider first numerical equations.

A number  $a$  (real or imaginary), such that substituted for  $x$  it makes the function  $x^n - p_1x^{n-1} + \dots \pm p_n$  to be  $= 0$ , or say, such that it satisfies the equation, is said to be a *root* of the equation; viz.  $a$  being a root, we have

$$a^n - p_1a^{n-1} + \dots \pm p_n = 0, \quad \text{or say } f(a) = 0;$$

and it is then shown that  $x - a$  is a factor of the function  $f(x)$ , viz. that we have

$$f(x) = (x - a) f_1(x),$$

where  $f_1(x)$  is a function  $x^{n-1} - q_1x^{n-2} + \dots \pm q_{n-1}$ , of the order  $n - 1$ , with numerical coefficients  $q_1, q_2, \dots, q_{n-1}$ .

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In general,  $a$  is *not* a root of the equation  $f_1(x) = 0$ ; but it may be so, viz.  $f_1(x)$  may contain the factor  $x - a$ ; when this is so,  $f(x)$  will contain the factor  $(x - a)^2$ ; writing then  $f(x) = (x - a)^2 f_2(x)$ , and assuming that  $a$  is not a root of the equation  $f_2(x) = 0$ ,  $x = a$  is then said to be a *double root* of the equation. Similarly,  $f(x)$  may contain the factor  $(x - a)^3$  and no higher power, and then  $x = a$  is said to be a *triple root*; and so on.

Supposing, in general, that  $f(x) = (x - a)^\alpha F(x)$ , where  $\alpha$  is a positive integer which may be  $= 1$ , and  $F(x)$  is of the order  $n - \alpha$ , then if  $b$  is a root different from  $a$ , we shall have  $x - b$  a factor (in general a simple one, but it may be a multiple one) of  $F(x)$ , and  $f(x)$  will in this case become

$$f(x) = (x - a)^\alpha (x - b)^\beta \Phi(x),$$

where  $\beta$  is a positive integer which may be  $= 1$ , and  $\Phi(x)$  is of the order  $n - \alpha - \beta$ . The original equation  $f(x) = 0$  is in this case said to have  $\alpha$  roots each  $= a$ ,  $\beta$  roots each  $= b$ , and so on.

We have the theorem, a numerical equation of the order  $n$  has in every case  $n$  roots, viz. there exist  $n$  numbers  $a, b, \dots$  (in general, all of them distinct, but they may arrange themselves in groups of equal values) such that

$$f(x) = (x - a)(x - b)(x - c) \dots \text{ identically.}$$

If an equation has equal roots, these can in general be determined; the case is at any rate a special one, which may be here omitted from consideration. It is therefore, in general, assumed that the equation  $f(x) = 0$  under consideration has all its roots unequal.

If the coefficients  $p_1, p_2, \dots$  are all or any one or more of them imaginary, then the equation  $f(x) = 0$ , separating the real and imaginary parts, may be written  $F(x) + i\Psi(x) = 0$ , where  $F(x)$ ,  $\Psi(x)$  are each of them a function with real coefficients; and it thus appears that the equation  $f(x) = 0$  with imaginary coefficients has not in general any real root; supposing it to have a

real root  $a$ , this must be at once a root of each of the equations  $F(x) = 0$  and  $\Psi(x) = 0$ .

But an equation with real coefficients may have as well imaginary as real roots; and we have further the theorem that for such an equation the imaginary roots enter in pairs, viz.  $\alpha + \beta i$  being a root, then will also  $\alpha - \beta i$  be a root.

Considering an equation with real coefficients, the question arises as to the *number* and *situation* of its real roots; this is completely resolved by means of *Sturm's theorem*, viz. we form a series of functions  $f(x)$ ,  $f'(x)$ ,  $f_1(x)$ ,  $\dots$ ,  $f_n(x)$  (a constant) of the degrees  $n$ ,  $n - 1$ ,  $\dots$ ,  $2$ ,  $1$ , respectively; and substituting therein for  $x$  any two real values  $a$  and  $b$ , we find by means of the resulting signs of these functions how many real roots of  $f(x)$  lie between the limits  $a$ ,  $b$ .

The same thing can frequently be effected with greater facility by other means, but the only general method is the one just referred to.

In the general case of an equation with imaginary (it may be real) coefficients, the like question arises as to the *situation* of the (real or imaginary) roots, viz. if for facility of conception we regard the constituents  $\alpha$ ,  $\beta$  of a root  $\alpha + \beta i$  as the coordinates of a point *in plano*, and accordingly represent the root by such point; then drawing in the plane any closed curve or “contour,” the question is how many roots lie within such contour.



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This is solved theoretically by means of a theorem of Cauchy's, viz. writing in the original equation  $x + iy$  in place of  $x$ , the function  $f(x + iy)$  becomes  $= P + iQ$ , where  $P$  and  $Q$  are each of them a rational and integral function (with real coefficients) of  $(x, y)$ . Imagining the point  $(x, y)$  to travel along the contour, and considering the number of changes of sign from  $-$  to  $+$  and from  $+$  to  $-$  of the fraction  $\frac{Q}{P}$  corresponding to passages of the fraction through zero (that is, to values for which  $P$  becomes  $= 0$ , disregarding those for which  $Q$  becomes  $= 0$ ), the difference of these numbers determines the number of roots within the contour. The investigation leads to a proof of the before-mentioned theorem, that a numerical equation of the order  $n$  has precisely  $n$  roots.

But, for the actual determination, it is necessary to consider a rectangular contour, and to apply to each of its sides separately a method such as that of Sturm's theorem; and thus the actual determination ultimately depends on a method such as that of Sturm's theorem.

Recurring to the case of an equation with real coefficients, it is important to separate the real roots, viz. to determine limits, such that each real root lies alone by itself between two limits  $l$  and  $m$ . This can be done (with more or less difficulty according to the nearness of the real roots) by repeated applications of Sturm's theorem, or otherwise.

The same thing would be useful, and can theoretically be effected, in regard to the roots of an equation generally, viz. we may, by lines parallel to the axes of  $X$  and  $Y$  respectively, divide the plane into rectangles such that each (real or imaginary) root lies alone by itself in a given rectangle; but the ulterior theory, even as regards the imaginary roots of an equation with real coefficients, has not been developed, and the remarks which immediately follow have reference only to equations with real coefficients, and to the real roots of such equations.

Supposing the roots separated as above, so that a certain root is known to lie alone by itself between two given limits, then it

is possible by various processes (Horner's, or Lagrange's method of continued fractions) to obtain to any degree of approximation the numerical value of the real root in question, and thus to obtain (approximately as above) the values of the several real roots.

The real roots can also frequently be obtained, without the necessity of a previous separation of the roots, by other processes of approximation—Newton's, as completed by Fourier, or by a method given by Encke—and the problem of their determination to any degree of approximation may be regarded as completely solved. But this is far from being practically the case even as regards the imaginary roots of such equations, or as regards the roots of an equation with imaginary coefficients.

A class of numerical equations which need to be considered, are the *binomial* equations  $x^n - \alpha = 0$ , where  $\alpha = a + \beta i$ , is a complex number. The foregoing conclusions apply, viz. there are always  $n$  roots, which it may be shown are all unequal. Supposing

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one of these is  $\theta$ , so that  $\theta^n = \alpha$ , then, assuming  $\theta = \alpha^{1/n}y$ , we have  $y^n - 1 = 0$ , which equation (like the more general one  $x^n - \alpha = 0$ ) has precisely  $n$  roots; it is shown that these are  $1, \omega, \omega^2, \dots, \omega^{n-1}$ , where  $\omega$  is a complex number  $a + \beta i$  such that  $a^2 + \beta^2 = 1$ , or, what is the same thing, a complex number of the form  $\cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ ; and it then at once appears that  $\omega$  may be taken  $= e^{2\pi i/n}$ . We have thus the trigonometrical solution of the equation  $x^n - 1 = 0$ . We may also obtain a like trigonometrical solution of the first-mentioned equation  $x^n - \alpha = 0$ . We are thus led to the notion (a numerical) of the radical  $\alpha^{1/n}$ , regarded as an  $n$ -valued function, viz. any one of these being denoted by  $\sqrt[n]{\alpha}$ , then the series of values is

$$\sqrt[n]{\alpha}, \omega \sqrt[n]{\alpha}, \dots, \omega^{n-1} \sqrt[n]{\alpha}.$$

Or we may, if we please, use  $\sqrt[n]{\alpha}$ , instead of  $\alpha^{1/n}$ , as a symbol to denote the  $n$ -valued function.

It is not necessary, as regards the equation  $x^n - 1 = 0$ , to refer here to the distinctions between the cases  $n$  a prime, and a composite, number.

As the coefficients of an algebraical equation may be numerical, all which follows in regard to algebraical equations, is (with, it may be, some few modifications) applicable to numerical equations; and hence, concluding for the present this subject, it will be convenient to pass on to algebraical equations.

**II.** We consider, secondly, an algebraical equation

$$x^n - p_1 x^{n-1} + \dots = 0,$$

and we here assume the existence of roots, viz. we assume that there are  $n$  quantities  $a, b, c, \dots$  (in general, all of them different, but in particular cases they may become equal in sets in any manner), such that

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots$$

Or, looking at the question in a different point of view, and starting with the roots  $a, b, c, \dots$  as given, we express the product of the  $n$  factors  $x - a, x - b, \dots$  in the foregoing form, and thus arrive at an equation of the order  $n$  having the  $n$  roots  $a, b, c, \dots$ . In either case, we have

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \dots, \quad p_n = abc \dots,$$

viz. regarding the coefficients  $p_1, p_2, \dots, p_n$  as given, then we assume the existence of roots  $a, b, c, \dots$  such that  $p_1 = \Sigma a$ , &c., or regarding the roots as given, then we write  $p_1, p_2$ , &c., to denote the functions  $\Sigma a, \Sigma ab$ , &c.

It is to be noticed that, in virtue of

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots, \quad \&c.,$$

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or of the equivalent equations  $p_1 = \Sigma a$ , &c., then

$$\begin{aligned} a^n - p_1 a^{n-1} + \dots &= 0, \\ b^n - p_1 b^{n-1} + \dots &= 0, \end{aligned}$$

(viz. it is for this reason that  $a, b, \dots$  are said to be roots of  $x^n - p_1 x^{n-1} + \dots = 0$ ); and, moreover, that conversely from the last-mentioned equations, assuming that  $a, b, \dots$  are all different, we deduce

$$p_1 = \Sigma a, \quad p_2 = \Sigma ab, \quad \&c.,$$

and

$$x^n - p_1 x^{n-1} + \dots = (x - a)(x - b) \dots$$

Observe that, if for instance  $a = b$ , then the two equations  $a^n - p_1 a^{n-1} + \dots = 0$ ,  $b^n - p_1 b^{n-1} + \dots = 0$  would reduce themselves to a single equation, which would not of itself express that  $a$  was a double root, that is, that  $(x - a)^2$  was a factor of  $x^n - p_1 x^{n-1} + \dots$ ; but by considering  $b$  as the limit of  $a + h$ ,  $h$  indefinitely small, we obtain a second equation

$$na^{n-1} - (n - 1)p_1 a^{n-2} + \dots = 0,$$

which, with the first, expresses that  $a$  is a double root; and then the whole system of equations leads, as before, to the equations  $p_1 = \Sigma a$ , &c. But this in passing: the general case is when the roots are all unequal.

We have then the theorem that every rational symmetrical function of the roots is a rational function of the coefficients; this is an easy consequence from the less general theorem, every rational and integral symmetrical function of the roots is a rational and integral function of the coefficients.

In particular, the sums of powers  $\Sigma a^2$ ,  $\Sigma a^3$ , &c., are rational and integral functions of the coefficients.

An ordinary process, as regards the expression of other functions  $\Sigma a^p b^q$ , &c., in terms of the coefficients, is to make them

depend on the functions  $\Sigma a^n$ , &c., but this is very objectionable; the true theory consists in showing that we have systems of equations

$$\begin{aligned} p_1 &= \Sigma a, \\ \left\{ \begin{aligned} p_2 &= \Sigma ab, \\ p_1^2 &= \Sigma a^2 + 2\Sigma ab, \end{aligned} \right. \\ \left\{ \begin{aligned} p_3 &= \Sigma abc, \\ p_1 p_2 &= \Sigma a^2 b + 3\Sigma abc, \\ p_1^3 &= \Sigma a^3 + 3\Sigma a^2 b + 6\Sigma abc, \end{aligned} \right. \\ &\quad \&c., \quad \&c., \end{aligned}$$

where, in each system, there are precisely as many equations as there are root-functions on the right-hand side, e.g. 3 equations and 3 functions  $\Sigma abc$ ,  $\Sigma a^2 b$ ,  $\Sigma a^3$ . Hence, in each system, the root-functions can be determined linearly in terms of the powers and products of the coefficients.

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It follows that it is possible to determine an equation (of an assignable order) having for roots any given (unsymmetrical) functions of the roots of a given equation. For example, in the case of a quartic equation, roots  $(a, b, c, d)$ , it is possible to find an equation having the roots  $ab, ac, ad, bc, bd, cd$ , being therefore a sextic equation viz. in the product  $(y - ab)(y - ac)(y - ad)(y - bc)(y - bd)(y - cd)$ , the coefficients of the several powers of  $y$  will be symmetrical functions of  $a, b, c, d$ , and therefore rational and integral functions of the coefficients of the original quartic equation.

In connexion herewith, the question arises as to the number of values (obtained by permutations of the roots) of given *unsymmetrical* functions of the roots; for instance, with roots  $(a, b, c, d)$  as before, how many values are there of the function  $ab + cd$ ; or, better, how many functions are there of this form; the answer is 3, viz.  $ab + cd, ac + bd, ad + bc$ ; or, again, we may ask whether it is possible to obtain functions of a given number of values, 3-valued, 4-valued functions, &c.

We have, moreover, the very important theorem that, given the value of any unsymmetrical function, e.g.  $ab + cd$ , it is in general possible to determine rationally the value of any similar function, e.g.  $(a + b)^2 + (c + d)^2$ .

The *à priori* ground of this theorem may be illustrated by means of a numerical equation. Suppose, e.g. that the roots of a quartic equation are 1, 2, 3, 4; then if it is given that  $ab + cd = 14$ , this in effect determines  $a, b$  to be 1, 2 (viz.  $a = 1, b = 2$ , or else  $a = 2, b = 1$ ) and  $c, d$  to be 3, 4 (viz.  $c = 3, d = 4$ , or else  $c = 4, d = 3$ ); and it therefore in effect determines  $(a + b)^2 + (c + d)^2$  to be = 370, and not any other value. And we can in the same way account for cases of failure as regards particular equations; thus, the roots being 1, 2, 3, 4, as above,  $a - b = 2$  determines  $a$  to be = 1 and  $b$  to be = 2; but if the roots had been 1, 2, 4, 16, then  $a^2b = 16$  does not uniquely determine  $a$  and  $b$ , but only makes them to be 1 and 16, or else 2 and 4, respectively.

As to the *à posteriori* proof, assume, for instance,  $t_1 = ab + cd$ ,  $y_1 = (a + b)^2 + (c + d)^2$ , and so  $t_2 = ac + db$ ,  $y_2 = (a + c)^2 + (d + b)^2$ ,

&c.—in the present case there are only the functions  $t_1, t_2, t_3$  and  $y_1, y_2, y_3$ —then  $y_1 + y_2 + y_3, t_1y_1 + t_2y_2 + t_3y_3, t_1^2y_1 + t_2^2y_2 + t_3^2y_3$  will be respectively symmetrical functions of the roots of the quartic, and therefore rational and integral functions of its coefficients, that is, they will be known.

Imagine, in the first instance, that  $t_1, t_2, t_3$  are all known; then the equations being linear in  $y_1, y_2, y_3$ , these can be expressed rationally in terms of known functions of the coefficients and of  $t_1, t_2, t_3$ , that is,  $y_1, y_2, y_3$  will be known. But observe further, that  $y_1$  is obtained as a function of  $t_1, t_2, t_3$  symmetrical as regards  $t_2, t_3$ : it can consequently be expressed as a rational function of  $t_1$  and of  $t_2 + t_3, t_2t_3$ , or, what is the same thing, of  $t_1$  and  $t_1 + t_2 + t_3, t_1t_2 + t_1t_3 + t_2t_3, t_1t_2t_3$ ; but these last will be symmetrical functions of the roots, and as such expressible rationally in terms of the coefficients: that is,  $y_1$  will be expressed as a rational function of  $t_1$  and of the coefficients, or,  $t_1$  being known,  $y_1$  will be rationally determined.

We may consider now the question of the algebraical solution of equations, or, more accurately, that of the solution of equations by radicals.



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In the case of a quadric equation  $x^2 + px + q = 0$ , we can find for  $x$ , by the assistance of the sign  $\sqrt{(\ )}$  or  $(\ )^{\frac{1}{2}}$  an expression for  $x$  as a two-valued function of the coefficients  $p, q$ , such that, substituting this value in the equation, the equation is thereby identically satisfied, viz. we have

$$x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)},$$

giving

$$\begin{array}{rcl} x^2 & = & \frac{1}{4}p^2 - q \mp p\sqrt{(\frac{1}{4}p^2 - q)} \\ +px & = & -\frac{1}{2}p^2 \pm p\sqrt{(\frac{1}{4}p^2 - q)} \\ +q & = & +q \\ \hline x^2 + px + q & = & 0, \end{array}$$

and the equation is on this account said to be algebraically solvable, or, more accurately, to be solvable by radicals. Or we may, by writing  $x = -\frac{1}{2}p + z$ , reduce the equation to  $z^2 = \frac{1}{4}p^2 - q$ , viz. to an equation of the form  $z^2 = a$ , and, in virtue of its being thus reducible, we may say that the equation is solvable by radicals. And the question for an equation of any higher order is, say of the order  $n$ , can we by means of radicals, that is, by aid of the sign  $\sqrt[n]{(\ )}$  or  $(\ )^{1/n}$ , using as many as we please of such signs and with any values of  $m$ , find an  $n$ -valued function (or any function) of the coefficients, which substituted for  $x$  in the equation shall satisfy it identically.

It will be observed that the coefficients  $p, q, \dots$  are not explicitly considered as numbers, but that even if they do denote numbers, the question whether a numerical equation admits of solution by radicals is wholly unconnected with the before-mentioned theorem of the existence of the  $n$  roots of such an equation. It does not even follow that, in the case of a numerical equation solvable by radicals, the algebraical expression of  $x$  gives the numerical solution; but this requires explanation. Consider, first, a numerical quadric equation with imaginary coefficients; in the formula  $x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$ , substituting for  $p, q$  their given numerical values we obtain for  $x$  an expression of the form  $x = \alpha + \beta i \pm \sqrt{(\gamma + \delta i)}$ , where  $\alpha, \beta, \gamma, \delta$  are real numbers;

this value substituted in the numerical equation would satisfy it identically and it is thus an algebraical solution; but there is no obvious *à priori* reason why the expression  $\sqrt{(\gamma + \delta i)}$  should have a value  $= c + di$ , where  $c$  and  $d$  are real numbers calculable by the extraction of a root or roots of real numbers; it appears upon investigation that  $\sqrt{(\gamma + \delta i)}$  has such a value calculable by means of the radical expression  $\sqrt{\{\sqrt{(\gamma^2 + \delta^2)} \pm \gamma\}}$ ; and hence that the algebraical solution of a quadric equation does in every case give the numerical solution of a numerical quadric. The case of a numerical cubic will be considered presently.

A cubic equation can be solved by radicals, viz. taking for greater simplicity the cubic in the reduced form  $x^3 - qx - r = 0$ , and writing  $x = a + b$ , this will be a solution if only  $3ab = q$ , and  $a^3 + b^3 = r$ , or say  $\frac{1}{2}(a^3 + b^3) = \frac{1}{2}r$ ;

$$\frac{1}{2}(a^3 - b^3) = \pm \sqrt{(\frac{1}{4}r^2 - \frac{1}{27}q^3)},$$

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and therefore

$$a = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

a six-valued function of  $q, r$ . But then writing  $b = \frac{q}{3a}$ , we have, as may be shown,  $a + b$  a three-valued function of the coefficients; it would have been wrong to complete the solution by writing  $b = \sqrt[3]{\left\{\frac{1}{2}r \pm \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}}$ , since here  $(a + b)$  would be given as a 9-valued function, having only 3 of its values roots, and the other 6 values being irrelevant. An interesting variation of the solution is to write  $x = ab(a + b)$ , giving  $a^3b^3(a^3 + b^3) = r$  and  $3a^2b^2 = q$ , or say  $\frac{1}{2}(a^3 + b^3) = \frac{r}{2}$ ,  $a^3b^3 = \frac{q^3}{27}$ ; whence

$$\left\{\frac{1}{2}(a^3 - b^3)\right\}^2 = \frac{1}{4}r^2 - \frac{1}{27}q^3,$$

and therefore

$$a = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}}, \quad b = \sqrt[3]{\left\{\frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

and here although  $a, b$  are each of them a 6-valued function, yet, as may be shown,  $ab(a + b)$  is only a 3-valued function.

In the case of a numerical cubic, even when the coefficients are real, substituting their values in the expression

$$x = \sqrt[3]{\left\{\frac{1}{2}r + \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}} + \sqrt[3]{\left\{\frac{1}{2}r - \sqrt{\left(\frac{1}{4}r^2 - \frac{1}{27}q^3\right)}\right\}},$$

this may depend on an expression of the form  $\sqrt[3]{(\gamma + \delta i)}$ , where  $\gamma$  and  $\delta$  are real numbers (viz. it will do so if  $\frac{1}{4}r^2 - \frac{1}{27}q^3$  is a negative number), and here we cannot by the extraction of any root or roots of real numbers reduce  $\sqrt[3]{(\gamma + \delta i)}$  to the form  $c + di$ ,  $c$  and  $d$  real numbers; hence, here the algebraical solution does not give the numerical solution. It is to be added that the case in question, called the “irreducible case,” is that wherein the three roots of the cubic equation are all real; if the roots are one real and two imaginary, then, contrariwise, the quantity

under the cube root is real, and the algebraical solution gives the numerical one.

The irreducible case is solvable by a trigonometrical formula, but this is not a solution by radicals; it consists, in effect, in reducing the given numerical cubic (not to a cubic of the form  $z^3 = a$ , solvable by the extraction of a cube root, but) to a cubic of the form  $4x^3 - 3x = a$ , corresponding to the equation  $4\cos^3\theta - 3\cos\theta = \cos 3\theta$  which serves to determine  $\cos\theta$  when  $\cos 3\theta$  is known.

A quartic equation is solvable by radicals; and it may be remarked, that the existence of such a solution depends on the existence of 3-valued functions such as  $ab + cd$ , of the four roots  $(a, b, c, d)$ ; by what precedes,  $ab + cd$  is the root of a cubic equation, which equation is solvable by radicals; hence  $ab + cd$  can be found by radicals; and since  $abcd$  is a given value,  $ab$  and  $cd$  can each be found by radicals. But by what precedes, if  $ab$  be known, then any similar function, say  $a + b$ , is obtainable rationally; and, consequently, from the values of  $a + b$  and  $ab$  we may by radicals

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obtain the value of  $a$  or  $b$ , that is, an expression for a root of the given quartic expression; the expression finally obtained is 4-valued, corresponding to the different values of the several radicals which enter therein, and we have therefore the expression by radicals of each of the four roots of the quartic equation. But when the quartic is numerical, the same thing arises as in the cubic: the algebraical expression does not in every case give the numerical one.

It will be understood from the foregoing explanation as to the quartic, how in the next following case, that of a quintic equation, the question of the solvability by radicals depends on the existence or non-existence of  $r$ -valued functions of the five roots  $(a, b, c, d, e)$ ; a fundamental theorem on the subject is that a rational function of 5 letters, if it has less than 5, cannot have more than 2 values; viz. that there are no 3-valued, or 4-valued, functions of 5 letters; and by reasoning, depending in part upon this theorem, Abel showed that a general quintic equation is not solvable by radicals: and *à fortiori* the general equation of any order higher than 5 is not solvable by radicals.

The general theory of the solvability of an equation by radicals depends very much on Vandermonde's remark, that supposing an equation is solvable (by radicals) and that we have therefore an algebraical expression of  $x$  in terms of the coefficients, then substituting for the coefficients their values in terms of the roots, the resulting value of the expression must reduce itself to any one at pleasure of the roots  $a, b, c, \dots$ ; thus in the case of the quadric equation where the solution is  $x = -\frac{1}{2}p \pm \sqrt{(\frac{1}{4}p^2 - q)}$ , writing for  $p, q$  their values  $a + b, ab$ , this is  $x = \frac{1}{2}(a + b) \pm \sqrt{\{(a - b)^2\}}$ ,  $= a$  or  $b$  according to the value of the radical. But it is not considered necessary in the present sketch to go further into the theory of the solvability of an equation by radicals. It may be proper to remark that, for quintic equations, there are solutions analogous to the trigonometrical solution of a cubic equation, viz. the quintic equation is here in effect reduced to some special form of quintic equation; for instance, to Jerrard's form  $x^5 + ax + b = 0$  or to some form presenting itself in the

theory of elliptic functions; but the solutions in question are not solutions by radicals. And there are various other interesting parts of the theory which have been excluded from consideration.

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## 632.

### ON ARONHOLD'S INTEGRATION-FORMULA.

[From the *Messenger of Mathematics*, vol. v. (1876),  
pp. 88—90.]

The fundamental theorem in Aronhold's Memoir, "Ueber eine neue algebraische Behandlungsweise der Integrale  $\int \Pi(x, y) dx$ , &c.," *Crelle*, t. LXI. (1863), pp. 95—145, is a theorem of indefinite integration. The form is

$$\Lambda \int \frac{dx}{(\alpha x + \beta y + \gamma)(hx + by + f)} = \log \frac{(a\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi}{\alpha x + \beta y + \gamma}$$

where  $y$  is a certain irrational function of  $x$ , determined by a quadric equation, and the other symbols denote constants connected by certain relations; viz. writing, for shortness,

$$U = (a, b, c, f, g, h \parallel x, y, 1)^2 = (a, \dots \parallel x, y, 1)^2, \text{ for shortness,}$$

that is,

$$U = ax^2 + 2hxy + by^2 + 2fy + 2gx + c;$$

$$W = (a, b, c, f, g, h \parallel x, y, 1 \parallel \xi, \eta, 1) = (a, \dots \parallel x, y, 1 \parallel \xi, \eta, 1),$$

that is,

$$W = (ax + hy + g)\xi + (hx + by + f)\eta + gx + fy + c,$$

or

$$= (a\xi + h\eta + g)x + (h\xi + b\eta + f)y + g\xi + f\eta + c;$$

$$(P, Q, R) = (ax + hy + g, hx + by + f, gx + fy + c),$$

$$(P_0, Q_0, R_0) = (a\xi + h\eta + g, h\xi + b\eta + f, g\xi + f\eta + c),$$

$$\Omega = \alpha x + \beta y + \gamma,$$

$$\Omega_0 = \alpha\xi + \beta\eta + \gamma,$$

$$(A, B, C, F, G, H) = (bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch),$$

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then  $y$  is determined as a function of  $x$  by the equation  $U = 0$ , that is,

$$(a, b, c, f, g, h \parallel x, y, 1)^2 = 0;$$

or, what is the same thing,

$$by = -(hx + f) + \sqrt{-Cx^2 + 2Gx - A};$$

the constants  $\alpha, \beta, \gamma, \xi, \eta$  are such that

$$(a, b, c, f, g, h \parallel \xi, \eta, 1)^2 = 0, \quad \alpha\xi + \beta\eta + \gamma = 0,$$

that is,

$$U_0 = 0, \quad \Omega_0 = 0;$$

and the value of  $\Lambda$  is given by

$$\Lambda^2 = -(A, B, C, F, G, H \parallel \alpha, \beta, \gamma)^2.$$

The theorem may therefore be written

$$\Lambda \int \frac{dx}{\Omega \cdot Q} = \log \frac{W}{\Omega},$$

where the several symbols have the significations explained above.

The verification is as follows. We ought to have

$$\frac{\Lambda dx}{\Omega Q} = \frac{P_0 dx + Q_0 dy}{W} - \frac{\alpha dx + \beta dy}{\Omega},$$

when  $dx, dy$  satisfy the relation  $P dx + Q dy = 0$ , viz. substituting for  $dy$  the value  $-\frac{P}{Q} dx$ , the equation becomes

$$\frac{\Lambda}{\Omega \cdot Q} = \frac{P_0 Q - P Q_0}{W \cdot Q} - \frac{\alpha Q - \beta P}{\Omega \cdot Q},$$

that is, substituting for  $\Omega$  its value,

$$\Lambda W = (P_0 Q - P Q_0)(\alpha x + \beta y + \gamma) - (\alpha Q - \beta P) W.$$



On the right-hand side, substituting for  $W$  its value,

$$\text{coeff. } \alpha = x(P_0Q - PQ_0) - Q(P_0x + Q_0y + R_0), \quad = Q_0R - QR_0,$$

$$\text{coeff. } \beta = y(P_0Q - PQ_0) + P(P_0x + Q_0y + R_0), \quad = R_0P - RP_0,$$

(as at once appears by aid of the relation  $U = Px + Qy + R = 0$ ),

$$\text{coeff. } \gamma = P_0Q - PQ_0.$$

The equation to be verified thus is

$$\Lambda W = \begin{vmatrix} \alpha & \beta & \gamma \\ P_0 & Q_0 & R_0 \\ P & Q & R \end{vmatrix},$$

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which, substituting therein for  $P, Q, R, P_0, Q_0, R_0$ , their values, and writing

$$(\lambda, \mu, \nu) = (\eta - y, x - \xi, \xi y - \eta x),$$

is in fact

$$\Lambda W = (A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma).$$

We have identically

$$(a, \dots \backslash x, y, 1)^2 (a, \dots \backslash \xi, \eta, 1)^2 - W^2 = (A, \dots \backslash \lambda, \mu, \nu)^2,$$

which, in virtue of  $(a, \dots \backslash \xi, \eta, 1)^2 = 0$ , gives

$$W^2 = -(A, \dots \backslash \lambda, \mu, \nu)^2,$$

and since  $\Lambda^2 = -(A, \dots \backslash \alpha, \beta, \gamma)^2$ , the equation is thus

$$\sqrt{\{-(A, \dots \backslash \alpha, \beta, \gamma)^2\}} \sqrt{\{-(A, \dots \backslash \lambda, \mu, \nu)^2\}} = (A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma),$$

that is,

$$(A, \dots \backslash \alpha, \beta, \gamma)^2 (A, \dots \backslash \lambda, \mu, \nu)^2 - \{(A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma)\}^2 = 0.$$

The left-hand side is here identically

$$= K (\alpha\mu - \beta\lambda, \alpha\nu - \gamma\lambda, \beta\nu - \gamma\mu)^2,$$

substituting for  $\lambda, \mu, \nu$  their values, we find

$$(\alpha\mu - \beta\lambda, \alpha\nu - \gamma\lambda, \beta\nu - \gamma\mu) = (\xi\alpha + \eta\beta + \gamma)(\alpha, \beta, \gamma) - \Omega(\xi, \eta, 1),$$

viz. in virtue of  $\Omega_0 = 0$ , these are  $= -\xi\Omega, -\eta\Omega, -\Omega$ , and the quadric function is  $= \Omega^2(a, \dots \backslash \xi, \eta, 1)^2$ , vanishing in virtue of the relation  $(a, \dots \backslash \xi, \eta, 1)^2 = 0$ .

The equation in question

$$\sqrt{\{-(A, \dots \backslash \alpha, \beta, \gamma)^2\}} \sqrt{\{-(A, \dots \backslash \lambda, \mu, \nu)^2\}} = (A, \dots \backslash \lambda, \mu, \nu \backslash \alpha, \beta, \gamma)$$

is thus verified, and the theorem is proved.

**633.**

**NOTE ON MR. MARTIN'S PAPER, "ON THE  
INTEGRALS OF SOME DIFFERENTIALS."**

[From the *Messenger of Mathematics*, vol. v. (1876), p. 163.]

The Note refers to a detail in a process of integration.

## 634.

THEOREMS IN TRIGONOMETRY AND ON  
PARTITIONS.

[From the *Messenger of Mathematics*, vol. v. (1876), p. 164,  
and p. 188.]

If

$$A + B + C + F + G + H = 0,$$

then

$$\begin{vmatrix} \sin \overline{A + F} & \sin \overline{B + F} & \sin \overline{C + F} & \cos F & \sin F \\ \sin \overline{A + G} & \sin \overline{B + G} & \sin \overline{C + G} & \cos G & \sin G \\ \sin \overline{A + H} & \sin \overline{B + H} & \sin \overline{C + H} & \cos H & \sin H \end{vmatrix} = 0.$$

(The determinant is to be read with the convention that, for instance,  $\sin \overline{A + F}$  denotes  $\sin(A + F)$ .)

Let  $u_n$  = number of partitions of  $n$ , no part less than 2, the order attended to; e.g. if  $n = 7$ , the partitions are 7, 52, 25, 43, 34, 322, 232, 223,  $u_7 = 8$ ; the series is

$$u_2 = 1,$$

$$u_3 = 1,$$

$$u_4 = 2,$$

$$u_5 = 3,$$

$$u_6 = 5,$$

$$u_7 = 8,$$

$$u_8 = 13,$$

$$u_9 = 21,$$

where each term is the sum of the next preceding two terms.

## 635.

### NOTE ON THE DEMONSTRATION OF CLAIRAUT'S THEOREM.

[From the *Messenger of Mathematics*, vol. v. (1876), pp. 166,  
167.]

It seems worth while to indicate what the leading steps of the demonstration are.

The potential of the Earth's mass upon an external or superficial point is taken to be

$$V = \frac{V_0}{r} + \frac{V_1}{r^2} + \frac{V_2}{r^3} + \&c.,$$

where  $V_1, V_2, V_3, \dots$  are Laplace's functions of the angular coordinates.

The surface is assumed to be a nearly spherical surface  $r = a(1 + u)$ , where  $u = u_1 + u_2 + \&c.$ , and  $u_1, u_2, \dots$  are Laplace's functions of the angular coordinates. To be a surface of equilibrium, with an equation  $V + \frac{1}{2}\omega^2 r^2 \sin^2 \theta = C$ , the latter must be equivalent to the equation  $r = a(1 + u)$ , and it follows that we have

$$\begin{aligned} V_1 &= V_0 a u_1, \\ V_2 &= V_0 a^2 u_2 - \frac{1}{2}\omega^2 a^4 \left(\frac{1}{3} - \cos^2 \theta\right), \\ &\&c., \end{aligned}$$

which values are to be substituted in the expression for  $V$ .

The whole force of gravity (due to the attraction and the centrifugal force) is taken to be  $g = -\frac{d}{dr}(V + \frac{1}{2}\omega^2 r^2 \sin^2 \theta)$ , and it follows that

$$g = \frac{V_0}{a^2}(1 + u_1 + 2u_2 + \dots) - \frac{2}{3}\omega^2 a - \omega^2 a \left(\frac{1}{3} - \cos^2 \theta\right),$$

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which is of the form

$$g = G \left\{ 1 + u_1 - \frac{\omega^2 a}{G} \left( \frac{1}{3} - \cos^2 \theta \right) + 2u_2 + \dots \right\}.$$

Taking the Earth to be the spheroid of revolution

$$r = a \{ 1 + \epsilon \left( \frac{1}{3} - \cos^2 \theta \right) \},$$

then

$$u_1 = \epsilon \left( \frac{1}{3} - \cos^2 \theta \right), \quad u_2 = 0, \quad \&c.,$$

and the equation is

$$g = G \left\{ 1 - \left( \frac{\omega^2 a}{G} - \epsilon \right) \left( \frac{1}{3} - \cos^2 \theta \right) \right\},$$

or say

$$g = G \{ 1 - \left( \frac{5}{2} m - \epsilon \right) \left( \frac{1}{3} - \cos^2 \theta \right) \},$$

where  $m = \frac{\omega^2 a}{G}$ , is the ratio of the centrifugal force at the equator to the force of gravity, which is the theorem in question. The expression “it follows” has been twice used as meaning it follows as a mere analytical consequence, in the proper degree of approximation, the steps of the deduction being purposely omitted.

## 636.

ON THE THEORY OF THE SINGULAR  
SOLUTIONS OF DIFFERENTIAL EQUATIONS OF  
THE FIRST ORDER.

[From the *Messenger of Mathematics*, vol. vi. (1877),  
pp. 23—27.]

In continuation of the former paper with this title (*Messenger*, vol. ii., 1873, pp. 6—12, [545]), I propose to discuss various particular examples, chiefly of cases in which the differential equation is of the form  $(L, M, N \setminus\setminus p, 1)^2 = 0$ , where  $L, M, N$  are rational and integral functions of  $(x, y)$ , and whether it admits or does not admit of an integral equation  $(P, Q, R \setminus\setminus c, 1)^2 = 0$ , where  $P, Q, R$  are rational and integral functions of  $(x, y)$ .

The singular solution of the differential equation

$$(L, M, N \setminus\setminus p, 1)^2 = 0,$$

if there be a singular solution, is  $S = 0$ , where  $S$  is either  $= LN - M^2$ , or a factor of  $LN - M^2$ . But in general  $LN - M^2$  is an indecomposable function, such that  $LN - M^2 = 0$  is not a solution of the differential equation, and this being so, there is no singular solution; viz. a differential equation  $(L, M, N \setminus\setminus p, 1)^2 = 0$ , where  $L, M, N$  are rational and integral functions of  $(x, y)$ , has not in general any singular solution.

Consider now a system of algebraical curves  $U = 0$ , where  $U$  is as regards  $(x, y)$  a rational and integral function of the order  $m$ , and depends in any manner on an arbitrary parameter  $c^1$ . I say that there is always a proper envelope, which envelope is the singular solution of the differential equation obtained by the elimination of  $c$  from the equation  $U = 0$ , and the derived equation in regard to  $(x, y)$ . It follows that the differential equation  $(L, M, N \setminus\setminus p, 1)^2 = 0$ , which has no singular solution, does

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<sup>1</sup>The expressions in the text may be understood as extending to the case where  $U$  is a function of any number ( $o$ ) of constants  $c_1, c_2, \dots, c_o$ , connected by an  $(o - 1)$ -fold relation,  $U$  thus virtually depending on a single arbitrary parameter.

not admit of an integral of the form in question  $U = 0$ , viz. an integral representing a system of algebraic curves.

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**Ex. 1.**

$$p^2 - (1 - y^2) = 0, \text{ that is, } dy^2 - (1 - y^2) dx^2 = 0.$$

Here there is no algebraical integral, but there is a quasi-algebraical integral of the form  $(P, Q, R \setminus c, 1)^2 = 0$ ; viz. starting with the form  $y = \sin(x + c)$  and expressing  $\sin c$  and  $\cos c$  rationally in terms of a new parameter, this is

$$c^2(y + \cos x) - 2c \sin x + (y - \cos x) = 0,$$

where the coefficients are one-valued functions of  $(x, y)$ . The discriminant of the differential equation in regard to  $p$  and that of the integral equation in regard to  $c$  are each  $= y^2 - 1$ , and we have a true singular solution  $y^2 - 1 = 0$ .

**Ex. 2.**

$$(1 - x^2)p^2 - (1 - y^2) = 0,$$

that is,

$$(1 - x^2) dy^2 - (1 - y^2) dx^2 = 0.$$

We have here an algebraic integral of the proper form, which is at once derived from the circular form

$$C = \cos^{-1} x + \cos^{-1} y$$

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by changing the constant, viz. this is

$$c^2 - 2cxy - (1 - x^2 - y^2) = 0.$$

The two discriminants are here each  $= (x^2 - 1)(y^2 - 1)$ , and we have

$$(x^2 - 1)(y^2 - 1) = 0$$

as a true singular solution. The curves are in fact the system of conics (ellipses and hyperbolas) each touching the four lines  $x = 1, x = -1, y = 1, y = -1$ .

**Ex. 3.**

$$(1 - y^2)p^2 - 1 = 0, \quad \text{that is,} \quad (1 - y^2)dy^2 - dx^2 = 0.$$

This is an extremely interesting example: the curve is the orthogonal trajectory of the system of sinusoids  $y = \sin(x + c)$ , which is the integral of Example 1; and we thus at once see that the real portion of the curve is wholly included between the lines  $y = -1, y = +1$ , being an infinite continuous curve, having a series of equidistant cusps alternately at the one and the other line, and obtained by the continued repetition of the finite portion included between two consecutive cusps on the same line. The discriminant of the differential equation equated to zero gives  $y^2 - 1 = 0$ , the equation of the two lines in question; but this does not satisfy the differential equation, and it is consequently not a singular solution; by what precedes, it appears that it is, in fact, a cusp-locus.

We thus see that the curves which represent the integral equation have no real envelope; but it is to be further shown that there is no imaginary envelope, and that the curve obtained by the elimination of the parameter is, in fact, made up of a (imaginary) node-locus and of the foregoing cusp-locus.

The curve is properly represented by taking  $x, y$  each of them a one-valued function of the parameter  $\theta$ , viz. we may write

$$\begin{aligned} y &= \cos \theta, \\ x &= c + \frac{1}{2}\theta - \frac{1}{4}\sin 2\theta. \end{aligned}$$

In fact, these values give

$$\frac{dy}{d\theta} = -\sin \theta, \quad \frac{dx}{d\theta} = \frac{1}{2}(1 - \cos 2\theta) = \sin^2 \theta,$$

and therefore

$$p = \frac{-1}{\sin \theta} = \frac{-1}{\sqrt{1 - y^2}},$$

that is,  $(1 - y^2)p^2 - 1 = 0$ , the differential equation.

It is obvious that to a given value of the parameter there corresponds a single point of the curve; and it is to be shown that, conversely, to a given point of the curve corresponds in general a single value of the parameter.

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Suppose the coordinates of the given point are  $y = \cos \alpha$ ,  $x = c + \frac{1}{2}\alpha - \frac{1}{4}\sin 2\alpha$ , where  $\alpha$  is a determinate quantity; then, to find  $\theta$ , we have

$$\cos \theta = \cos \alpha, \quad 2\theta - \sin 2\theta = 2\alpha - \sin 2\alpha.$$

The first equation gives  $\theta = 2m\pi \pm \alpha$ , and the second equation then is

$$4m\pi \pm 2\alpha \mp \sin 2\alpha = 2\alpha - \sin 2\alpha,$$

viz. taking the upper signs, this is  $4m\pi = 0$ , giving  $m = 0$  and  $\theta = \alpha$ ; and, taking the lower signs, it is  $m\pi = \alpha - \sin \alpha$ , which,  $\alpha$  being given, is not in general satisfied; hence to the given point there corresponds only the value  $\alpha$  of the parameter  $\theta$ . If, however,  $\alpha$  is such that  $\alpha - \sin \alpha$  is equal to a multiple of  $\pi$ , say  $r\pi$ , then the last-mentioned equation is satisfied by the value  $m = r$ , so that to the given point of the curve correspond the two values  $\alpha$  and  $2r\pi - \alpha$  of the parameter; these values are in general unequal, and the point is then a node; but they may be equal, viz. this is so if  $\alpha = r\pi$  (the point on the curve being then  $y = \cos r\pi, = \pm 1, x = c + \frac{1}{2}r\pi$ ), and the point is then a cusp; showing what was known, that there are on each of the lines  $y = -1, y = +1$ , an infinite series of equidistant cusps.

More definitely, suppose  $\alpha = r\pi \pm \beta$ , where  $\beta$  is a root of the equation  $2\beta - \sin 2\beta = 0$ , then

$$\sin 2\alpha = \pm \sin 2\beta, \quad 2\alpha - \sin 2\alpha = 2r\pi + (2\beta - \sin 2\beta) = 2r\pi,$$

and to the given point on the curve correspond the two values  $\alpha$  and  $2r\pi - \alpha$  of the parameter. If  $\beta = 0$ , we have, as above, the cusps on the two lines  $y = +1, y = -1$  respectively; but if  $\beta$  be an imaginary root of the equation  $2\beta - \sin 2\beta = 0$ , then we have an infinite series of nodes on the imaginary line  $y = \cos r\pi \cos \beta$ ; and there are an infinite number of such lines corresponding to the different imaginary roots of the equation  $2\beta - \sin 2\beta = 0$ .

From the form in which the equation of the curve is given, we cannot directly form the equation of the envelope by equating

to zero the discriminant in regard to the constant  $c$ ; but we may determine the intersections of the curve by the consecutive curve (corresponding to a value  $c + \delta c$  of the constant), and thus determine the locus of these intersections.

Consider for a moment the curves belonging to the constants  $c$ ,  $c_1$ , and let  $\theta, \theta_1$  be the values of the parameter  $\theta$  belonging to the points of intersection; we have  $\cos \theta = \cos \theta_1$ ,  $4c + 2\theta - \sin 2\theta = 4c_1 + 2\theta_1 - \sin 2\theta_1$ , we have  $\theta_1 = 2r\pi + \theta$ , but we cannot thereby satisfy the second equation; or else  $\theta_1 = 2r\pi - \theta$ , giving

$$4c + 2\theta - \sin 2\theta = 4c_1 + 4r\pi - 2\theta + \sin 2\theta,$$

that is,  $2\theta - \sin 2\theta = 2c_1 - 2c + 2r\pi$ ; and we have thus corresponding to any given value of  $r$  a series of values of  $\theta$ , viz. these are  $\theta = r\pi + \beta$ , where  $\beta$  is any root of the equation

$$2\beta - \sin 2\beta = 2c_1 - 2c.$$

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In particular, taking  $c_1 = c$ , the intersections are given by  $\theta = r\pi + \beta$ , where  $\beta$  is any root of the equation  $2\beta - \sin 2\beta = 0$ ; viz. we have thus an infinite number of intersections lying on each of the lines  $y = \cos r\pi \cos \beta$ . If  $\beta = 0$ , the intersections lie on the two lines  $y = 1, y = -1$  respectively; if  $\beta$  be an imaginary root of the equation  $2\beta - \sin 2\beta = 0$ , then they lie on the imaginary lines  $y = \cos r\pi \cos \beta$ . But by what precedes, it is clear that in the former case the intersections are nothing else than the cusps on the lines  $y = 1, y = -1$ ; and in the latter case nothing else than the nodes on the lines  $y = \cos r\pi \cos \beta$ ; viz. there is no proper envelope, but instead thereof we have lines of cusps and of nodes.

**Ex. 4.**

$$(1 - y^2)p^2 - (1 - x^2) = 0,$$

that is,

$$(1 - y^2) dy^2 - (1 - x^2) dx^2 = 0.$$

I have not examined this; the curve is the series of orthogonal trajectories of the conics of Example 2, and the integral equation may be represented by  $y = \cos \theta, x = \cos \phi$ , where  $c = (2\theta - \sin 2\theta) - (2\phi - \sin 2\phi)$ .

Equating to zero the discriminant of the differential equation, we have  $(1 - y^2)(1 - x^2) = 0$ , viz. the four lines  $x = 1, x = -1, y = 1, y = -1$ ; this is not an envelope, but a locus of cusps.

## 637.

ON A DIFFERENTIAL EQUATION IN THE  
THEORY OF ELLIPTIC FUNCTIONS.

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 29.]

In the differential equation

$$Q^2 - Q\left(k + \frac{1}{k}\right) - 3 = 3(1 - k^2)\frac{dQ}{dk},$$

considered *Messenger*, t. iv., pp. 69 and 110, [594] and [597], writing  $Q = x$  and  $k + \frac{1}{k} = y$ , the equation becomes

$$dy = \frac{3(y^2 - 4)dx}{3 + xy - x^3},$$

and we have, as a particular solution,

$$y = \frac{1}{4}\left(x^2 - 6x - \frac{3}{x}\right).$$

To verify this, observe that from the value of  $y$

$$dy = \frac{3}{4x^2}(x^3 - 1)^2 dx, \quad 3 + xy - x^3 = \frac{1}{4}(x^3 - 1)(x^3 - 9),$$

and the equation becomes

$$\frac{3}{4x^2}(x^3 - 1)^2 = \frac{3}{16x^2} \frac{\{(x^3 - 6x^2 - 3)^2 - 64x^3\}}{\frac{1}{4}(x^3 - 1)(x^3 - 9)},$$

viz. this is

$$(x^3 - 1)^2(x^3 - 9) = (x^3 - 6x^2 - 3)^2 - 64x^3,$$

which is right.



## 638.

### ON A $q$ -FORMULA LEADING TO AN EXPRESSION FOR $E_1$ .

[From the *Messenger of Mathematics*, vol. vi. (1877),  
pp. 63—66.]

It is to be shown that we have identically

$$\begin{aligned} (1 + 2q + 2q^4 + \dots)^4 - 16 \left( \frac{q}{1 - q^2} + \frac{2q^2}{1 - q^4} + \frac{3q^3}{1 - q^6} + \dots \right) \\ = \frac{1 - 9q - 25q^2 + 49q^4 + 81q^6 - \dots}{1 - q - q^2 + q^5 + q^7 - \dots} \quad (A); \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} (1 - 2q + 2q^4 - 2q^9 + \dots)^4 - 16 \left( \frac{-q}{1 - q^2} + \frac{2q^2}{1 - q^4} - \frac{3q^3}{1 - q^6} + \dots \right) \\ = \frac{1 + 9q + 25q^2 + 49q^4 + 81q^6 + \dots}{1 + q + q^2 + q^4 + q^6 + \dots} \quad (B), \end{aligned}$$

where the form (A) is that intended to be made use of, but the form (B) is rather more convenient for the demonstration.

We have

$$(1 - 2q + 2q^4 - 2q^9 + \dots)^4 = 1 + 8 \left\{ \frac{-q}{1 + q} + \frac{2q^2}{1 + q^2} - \frac{3q^3}{1 + q^3} + \dots \right\},$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. i., p. 239), taking the formula as there written down, and changing  $q$  into  $-q$ .

Also, if for a moment

$$X = 1 + q + q^2 + q^4 + q^6 + \&c.,$$

and

$$X' = \frac{dX}{dq},$$

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Arthur Cayley

Cambridge, 1896

[638] ON A  $q$ -FORMULA LEADING TO AN EXPRESSION FOR  $E_1$  (CONCLUSION).

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 63–66; continued from preceding pages.]

(Page 26.) so that

$$qX'' = q + 3q^2 + 6q^3 + 10q^4 + \&c.,$$

then

$$X + 8qX = 1 + (9 + 25)q^2 + 49q^3 + 81q^4 + \&c.;$$

so that the right-hand side of (B) is

$$\frac{X + 8qX'}{X} = -1 + 8q \frac{X'}{X}.$$

But (*Fund. Nova*, p. 185, *Ges. Werke*, t. i, p. 237),

$$X = \frac{1 - q^3}{1 - q} \cdot \frac{1 - q^7}{1 - q^5} \cdot \frac{1 - q^{11}}{1 - q^9} \cdots,$$

so that

$$\frac{X'}{X} = \frac{-2q}{1 - q^3} - \frac{4q^5}{1 - q^7} - \frac{6q^9}{1 - q^{11}} - \cdots + \frac{1}{1 - q} + \frac{3q^5}{1 - q^9} + \frac{5q^9}{1 - q^{13}} + \cdots.$$

And the equation (B) intended to be proved thus becomes

$$\begin{aligned} & 1 + 8 \left\{ \frac{-q}{1 + q} + \frac{2q^2}{1 + q^2} - \frac{3q^3}{1 + q^3} + \cdots \right\} \\ & - 16 \left\{ \frac{-q}{1 - q^3} + \frac{2q^2}{1 - q^7} - \frac{3q^3}{1 - q^{11}} + \cdots \right\} \\ & = 1 + 8q \left\{ \frac{-2q}{1 - q^3} - \frac{4q^5}{1 - q^7} - \frac{6q^9}{1 - q^{11}} - \cdots \right. \\ & \quad \left. + \frac{1}{1 - q} + \frac{3q^5}{1 - q^9} + \frac{5q^9}{1 - q^{13}} + \cdots \right\}; \end{aligned}$$

viz. omitting the terms unity, dividing by  $8q$ , and transposing, this is

$$\begin{aligned} & -\frac{1}{1+q} + \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} + \dots \\ & + \frac{2}{1-q^3} - \frac{4q}{1-q^7} + \frac{6q^2}{1-q^{11}} - \dots \\ & + \frac{2q}{1-q^3} + \frac{4q^5}{1-q^7} + \frac{6q^9}{1-q^{11}} + \dots \\ & - \frac{1}{1-q} - \frac{3q^5}{1-q^9} - \frac{5q^9}{1-q^{13}} - \dots = 0. \end{aligned}$$

The second and third lines unite together, and the equation becomes

$$\begin{aligned} & -\frac{1}{1+q} + \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} + \frac{4q^3}{1+q^4} - \dots \\ & + \frac{2}{1-q^3} + \frac{4q}{1+q^7} + \frac{6q^2}{1-q^{11}} + \frac{8q^3}{1+q^{15}} + \dots \\ & - \frac{1}{1-q} - \frac{3q^5}{1-q^9} - \frac{5q^9}{1-q^{13}} - \frac{7q^{13}}{1-q^{17}} - \dots = 0; \end{aligned}$$

(Page 27.) or, collecting and arranging,

$$\begin{aligned} & -\frac{1}{1+q} + \frac{2q}{1+q^2} - \frac{3q^2}{1+q^3} - \frac{4q^3}{1+q^4} - \frac{5q^4}{1+q^5} - \dots \\ & + \frac{1}{1-q} + \frac{3q^2}{1-q^3} + \frac{5q^4}{1-q^5} + \dots = 0, \end{aligned}$$

an identity which it is easy to verify to any number of terms. But to prove it directly, we have only to add the pairs of terms in the alternate columns; calling the left-hand side  $Fq$ , we thus obtain

$$Fq = 2q \left\{ -\frac{1}{1+q^2} - \frac{2q^2}{1+q^4} - \frac{3q^4}{1+q^6} - \dots + \frac{1}{1-q^2} + \frac{3q^4}{1-q^6} + \dots \right\};$$

viz. this equation is  $Fq = 2qF(q^2)$ ; and thence

$$Fq = 2^n q^{1+2+4+\dots} F(q^{2^n}) = \&c.;$$

we thus have  $Fq = 0$ .

The equation (B), or, what is the same thing, the equation (A) is thus proved. Reverting to the equation (A), we have

$$(1 + 2q + 2q^4 + \dots)^4 = \frac{4K^2}{\pi^2}.$$

(Jacobi, *Fund. Nova*, p. 188, *Ges. Werke*, t. I, p. 239),

$$\left( \frac{q}{1-q^2} + \frac{2q^2}{1-q^4} + \dots \right) = \frac{K^2}{2\pi^2} \left( 1 - \frac{E_1}{K} \right),$$

(ib., p. 135; ib., p. 189),

if  $q = e^{-\frac{\pi K'}{K}}$ , and  $K$ ,  $E_1$  are the complete functions  $F_1 k$ ,  $E_1 k$ .

The left-hand side of the equation is thus

$$\frac{4K^2}{\pi^2} - \frac{8K^2}{\pi^2} \left( 1 - \frac{E_1}{K} \right) = \frac{4K^2}{\pi^2} \left( -1 + \frac{2E_1}{K} \right),$$

and we have

$$\left(-1 + \frac{2E_1}{K}\right) = \frac{\pi^2}{4K^2} \cdot \frac{1 - 9q^2 - 25q^6 + 40q^4 + 81q^{10} - \dots}{1 - q^2 - q^4 + q^6 + q^{14} - \dots},$$

which is a new expression for  $E_1$  as a  $q$ -function. The expression on the right-hand side presents itself, Clebsch, *Theorie der Elasticität* (Leipzig, 1862), p. 162, and must have been obtained by him as a value for  $\left(-1 + \frac{2E_1}{K}\right)$ ; but there is no statement that this is so, nor anything to show how this form of  $q$ -function was arrived at. Mr Todhunter called my attention to the passage in Clebsch.

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### 639.

#### AN ELEMENTARY CONSTRUCTION IN OPTICS.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 81, 82.]

(Page 28.) CONSIDER two lines meeting at a point  $P$ , and a point  $A$ ; through  $A$ , drawn at right angles to  $AP$ , a line meeting the two lines in the points  $U, V$  respectively; and through the same point  $A$  draw any other line meeting the two lines in the points  $U', V'$  respectively; also let the points  $u', v'$  be the feet of the perpendiculars let fall from  $U', V'$  respectively on the line  $UV$ ; then we have

$$\frac{1}{Au'} + \frac{1}{Av'} = \frac{1}{AU} + \frac{1}{AV}.$$

The theorem can be proved at once without any difficulty. It answers to the optical construction, according to which, if  $UPV$  represents the path of a ray through a convex lens  $AP$ , then the thin pencil, axis  $U'P$  and centre  $U'$ , converges after refraction to the point  $V'$ , where  $U'V'$  are in lined with  $A$  the centre of the lens; considering as usual the inclinations to the axis as small, we have approximately  $AV' = Av'$ ,  $AU' = Au'$ , and the theorem is

$$\frac{1}{AU} + \frac{1}{AV'} = \frac{1}{AU} + \frac{1}{AV},$$

if  $AF$  is the focal length of the lens.

In the original theorem, the line  $UV$  need not be at right angles to  $AP$ , but may be any line whatever; the projecting lines  $U'u'$  and  $V'v'$  must then be parallel to  $AP$ , and the theorem remains true.

### 640.

#### FURTHER NOTE ON MR MARTIN'S PAPER "ON THE INTEGRALS OF SOME DIFFERENTIALS."

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 82.]

(Page 29.) See paper, Number 639; this further note relates also to a detail.

## 641.

## ON THE FLEXURE OF A SPHERICAL SURFACE.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 88–90.]

(Page 30.) IT is known that an inextensible spherical surface, or to fix the ideas the spherical quadrilateral included between two arcs of meridian and two arcs of parallel, may be bent in such wise as to be part of a surface of revolution, the meridians and parallels of the spherical surface being meridians and parallels of the new surface, and, moreover, the radius of each parallel of the spherical surface being in the new surface altered in the constant ratio  $k$  to 1. We have, in fact, on the spherical surface, writing  $p$  for the latitude and  $q$  for the longitude, and the radius being unity,

$$x = \cos p \cos q,$$

$$y = \cos p \sin q,$$

$$z = \sin p,$$

values which give

$$dx^2 + dy^2 + dz^2 = dp^2 + \cos^2 p dq^2.$$

This last equation is satisfied by the values

$$x = \cos p \sin \frac{q}{k},$$

$$y = \cos p \sin \frac{q}{k},$$

$$z = E(k, p),$$

where  $E(k, p) = \int_0^p \sqrt{(1 - k^2 \sin^2 p)} dp$ , is the elliptic function of the second kind; or rather, this is so when  $k < 1$ , but the same notation may be used when  $k > 1$ . These values give the deformation in question.

The two cases to be considered are  $k < 1$ , and  $k > 1$ ; we take in each case a spherical quadrilateral  $ABCD$  (fig. 1), bounded by  $AB$  (an arc of the equator), the arc of parallel  $CD$ , and the two arcs of meridian  $AD$  and  $BC$ . In the first case, there is no limit to the latitude  $AD$ ,  $= BC$ , or taking these  $= 90^\circ$ , we may in place

(Page 31.) of the quadrilateral  $ABCD$  consider the directangular triangle  $ABE$ ; the new form of this is  $A'B'E'$ , where the radius  $OA' = k \cdot OA = k$ , is less than the original radius

FIG. 1.

[Figure: spherical quadrilateral with deformation. Original.]

unity, but  $OE' = E_1 k$ , is greater than the same radius unity. The surface has at  $E'$  a conical point, the semi-aperture of the cone being  $= \tan^{-1} \frac{1}{k'}$ , if as usual  $k' = \sqrt{(1 - k^2)}$ ; to verify this, writing for convenience  $q = 0$ , we have for the meridian section  $x = \cos p$ ,  $z = E(k, p)$ , and thence  $\frac{dx}{dz} = -\frac{\sqrt{(1 - k^2 \sin^2 p)}}{\sin p}$ , which for  $p = 90^\circ$  gives  $\frac{dx}{dz} = -k'$ . Observe also that for  $p = 0$ ,  $\frac{dx}{dz} = \infty$ , viz. the surface of revolution cuts the plane of the equator at right angles.

There is no limit to the arc  $AB$ , it may be  $= 360^\circ$ , viz. we must in this case cut the hemisphere along a meridian to allow of the deformation; or it may exceed  $360^\circ$ , the hemisphere spherical surface being in this case conceived of as wrapping indefinitely over itself, and we may instead of the half lune  $EA'B'$ , consider the lune included between two meridians extending from pole to pole, and therefore the whole spherical surface, conceived of as wrapping indefinitely over itself; the result is, that this may be deformed into a surface of revolution, which, in its general form,

resembles that obtained by the revolution of an arc less than a semi-circle round its chord; the half-chord being greater, and the versed-sine less than the radius of the original sphere.

If  $k > 1$ , there is obviously a limit to the latitude  $AD, = BC$ , of the spherical quadrilateral; viz. this is equal to  $\sin^{-1} \frac{1}{k}$ . Supposing that in the quadrilateral  $ABCD$  (fig. 1) the latitude has this limiting value, then (see fig. 2) the new form is  $A'B'C''D''$ , where along the bounding arc  $C''D''$  the tangent plane is horizontal; viz. as before  $\frac{dx}{dz} = -\frac{\sqrt{(1-k^2 \sin^2 p)}}{\sin p}$ ,  $= 0$  for  $p = \sin^{-1} \frac{1}{k}$ . It is to be observed, that the radii for the parallels  $A''B''$  and  $C''D''$  are  $k$  and  $k \cos p$  respectively; the difference of

(Page 32.) these in  $k(1 - \cos p)$ , which, however great  $k$  is, must be less than the arc of meridian  $A''D''$ ,  $= p$ ; substituting for  $k$  the value  $\frac{1}{\sin p}$ , the condition is  $\frac{1-\cos p}{\sin p} < p$ , viz. this

FIG. 2.

[Figure: bounded surface with deformation; bell-like shape.]

is  $\tan \frac{1}{2}p < p$ , which is true for every value up to  $p = 90^\circ$ . But, more than this, we should have

$$k^2(1 - \cos p)^2 + E^2(k, p) < p^2,$$

viz. writing as before  $k = \frac{1}{\sin p}$ , this is

$$E^2\left(\frac{1}{\sin p}, p\right) < p^2 - \tan^2 \frac{1}{2}p;$$

this must be true, although (relating as it does to a form of  $E$  for which  $k$  is greater than 1) there might be some difficulty in verifying it.

There is, as in the first case, no limit to the value of  $AB$ , viz. this may be  $= 360^\circ$ , the spherical zone being then cut along a meridian, or it may be greater than  $360^\circ$ ; and, moreover, the spherical quadrilateral may extend south of the equator, but of course so that the limiting south latitude does not extend beyond the foregoing value  $\sin^{-1} \frac{1}{k}$ ; viz. we may have a zone between the latitudes  $\pm \sin^{-1} \frac{1}{k}$ , which may be a complete zone from longitude  $0^\circ$  to  $360^\circ$  or to any greater value than  $360^\circ$ . The result is, that the zone is deformed into a surface of revolution, which in its general form resembles that obtained by the revolution of a half-circle or half-ellipse about a line parallel to and beyond its bounding diameter, the bounding half-diameter being less, and the greatest radius of rotation greater, than the radius of the original sphere.

## 642.

### ON A DIFFERENTIAL RELATION BETWEEN THE SIDES OF A QUADRANGLE.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 99–101.]

(Page 33.) LET the sides and diagonals  $YZ, ZX, XY, OX, OY, OZ$  of a quadrangle be  $f, g, h, a, b, c$ , and let the component triangles be denoted as follows:

$$\begin{aligned} A &= \triangle YZO, &= (b, c, f), \\ B &= \triangle Z XO, &= (c, a, g), \\ C &= \triangle X YO, &= (a, b, h), \\ \Omega &= \triangle X Y Z, &= (f, g, h), \end{aligned}$$

[Figure: quadrangle with vertices  $X, Y, Z$ , and interior point  $O$ ; subtriangles labelled  $A, B, C$  with  $\Omega$  the outer triangle.]

viz.  $A, B, C, \Omega$  are the triangles whose sides are  $(b, c, f), (c, a, g), (a, b, h), (f, g, h)$  respectively, so that  $\Omega = A + B + C$ . Then we have between  $(a, b, c, f, g, h)$  an equation giving rise to a differential relation, which may be written

$$\Omega (Aa da + Bb db + Cc dc) - (BCff^2 + CAgg^2 + ABh dh) = 0.$$

This may be proved geometrically and analytically. First, for the geometrical proof, it is enough to prove that, when  $a$  and  $b$  alone vary, the relation between the increments is  $Aa da + Bb db = 0$ ; for then  $a$  and  $g$  alone varying, the relation between

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(Page 34.) the increments will be  $\Omega ada - Cgdg = 0$  (as to the negative sign it is clear from the figure that  $a, g$  will increase or diminish together); and we thence at once infer the general relation.

$$Aada + Bbdb = 0;$$

or, what is the same thing,

$$ada : -bdb = XOZ : YOZ.$$

The points  $XYZ$  remain fixed; but  $O$  moves through the infinitesimal arc  $OO'$ , centre  $Z$ , which may be considered as situate in the right line  $OM$  drawn from  $O$  at right angles to  $ZO$ , and meeting  $XY$  produced in the point  $M$ . And then, writing for a moment  $\angle OXY = X, \angle OYX = Y, \angle OMY = M$ , we find at once

$$da = OO' \cos(X + M),$$

$$-db = OO' \cos(Y - M);$$

that is,

$$\frac{da}{db} = \frac{\cos(X + M)}{\cos(Y - M)}, \text{ or } \frac{ada}{bdb} = \frac{a \cos(X + M)}{b \cos(Y - M)}.$$

But drawing  $X\alpha, Y\beta$  each of them at right angles to  $ZO$ , we have  $a \cos(X + M) = X\alpha, b \cos(Y - M) = Y\beta$ , and evidently  $XOZ : YOZ = X\alpha : Y\beta$ ; whence the equation is  $\frac{ada}{bdb} = \frac{XOZ}{YOZ}$ , which is the required relation.

For the analytical proof, it is to be observed that the relation between  $a, b, c, f, g, h$  is a quadric relation in the quantities  $a^2, b^2, c^2, f^2, g^2, h^2$  respectively; this may be written

$$\left| \begin{array}{cccccc} 1 & a^2 & a^4 & & & \\ 1 & b^2g^2 + c^2h^2 + a^2f^2 - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2 & -(b^2 - c^2)(g^2 - h^2) & & & \\ \nu = & + (b^2 - h^2)(c^2 - g^2) & -b^2 - c^2 - g^2 - h^2 & +1 & & \\ f^2 & & & & & \\ f^4 & & & & & 1 \end{array} \right| = 0;$$

say for a moment this is  $A + Ba^2 + Ca^4 = 0$ , where

$$\begin{aligned} A &= b^2g^4 + b^4g^2 + c^4h^2 + c^2h^4 + f^4(g^2 - h^2)(c^2 - g^2) \\ &\quad - (b^2 + c^2)g^2h^2 - (g^2 + h^2)b^2c^2 + f^2(-b^2 - c^2 - g^2 - h^2) + f^4, \\ B &= -(b^2 - c^2)(g^2 - h^2) + f^2(-b^2 - c^2 - g^2 - h^2) + f^4, \\ C &= 1; \end{aligned}$$

then we have as usual

$$\frac{da}{d \cdot a^2} ada + \frac{da}{d \cdot a^2} bdb + \&c. = 0,$$

(Page 35.) where

$$4 \frac{da}{d \cdot a^2} = Ca^2 + \frac{1}{2}B.$$

But in virtue of  $u = 0$ , we have

$$(Ca^2 + \frac{1}{2}B)^2 = C(Ca^4 + Ba^2 + A) + \frac{1}{4}(B^2 - 4AC),$$

that is,  $\frac{da}{d\alpha^2} = \sqrt{(B^2 - 4AC)}$ ; and hence  $B^2 - 4AC$  is a quartic function of  $f^2$ , which is easily seen to reduce itself to the form

$$f^2 - (g + h)^2 f^2 - (g - h)^2 f^2 - (b + c)^2 f^2 - (b - c)^2;$$

The coefficients of  $bdb$ ,  $cdc$ , &c., are given as expressions of the like form; substituting their values, the differential relation is

$$\sqrt{(f^2 - (g + h)^2 f^2 - (g - h)^2 f^2 - (b + c)^2 f^2 - (b - c)^2)} ada + \&c. = 0,$$

which is, in fact, the foregoing result.

It is right to notice that there are in all 16 linear factors,

$$\begin{array}{ccccccc} f + g + h, & b + c + f, & c + a + g, & a + b + h & \text{say } d, f, g, h, \\ -f + g + h, & -b + c + f, & -c + a + g, & -a + b + h & d', f', g', h', \\ f - g + h, & b - c + f, & c - a + g, & a - b + h & d'', f'', g'', h'', \\ f + g - h, & b + c - f, & c + a - g, & a + b - h & d''', f''', g''', h'''; \end{array}$$

and this being so, the coefficients of  $ada$ ,  $bdb$ ,  $cdc$ ,  $fdf$ ,  $gdg$ ,  $hdh$ , are

$$\begin{array}{l} \sqrt{(dd'd''d''' \cdot f f' f'' f''')}, \sqrt{(gg'g''g''' \cdot hh'h''h''')}, \\ \sqrt{(dd'd''d''' \cdot gg'g''g''')}, \sqrt{(hh'h''h''' \cdot f f' f'' f''')}, \\ \sqrt{(dd'd''d''' \cdot hh'h''h''')}, \sqrt{(f f' f'' f''' \cdot gg'g''g''')}, \end{array}$$

respectively.

We may imagine the quadrilateral  $ZOXY$  composed of the four rods  $ZO$ ,  $OX$ ,  $XY$ ,  $YZ$  (lengths  $c$ ,  $a$ ,  $h$ ,  $f$  as before) jointed together at the angles, and kept in equilibrium by forces  $B$ ,  $G$  acting along the diagonals  $OY(=b)$ ,  $ZX(=a)$  respectively. We have  $c$ ,  $a$ ,  $h$ ,  $f$  given constants, and the relation  $\phi(a, b, c, f, g, h) = 0$ , which connects the six quantities  $b$  and  $g$ ; by what precedes, the differential relation  $\phi_y, dg + \phi_\delta \cdot db = 0$  is equivalent to the foregoing four equations equal to the differential  $dt$  of a new variable, the system of the equations being expressed by  $\delta\delta b + Gdg = 0$ ; by virtual velocities we have as the condition of equilibrium  $Bdb + Gdg = 0$ ; hence, eliminating  $db$ ,  $dg$  we have

$$\frac{B}{\Omega Bbb} = \frac{G}{CAgg},$$

or, say

$$\frac{B}{b} = \frac{G}{g} = \frac{1}{\sqrt{\Delta XYO \cdot \Delta ZYO}} : \frac{1}{\sqrt{\Delta XYZ \cdot \Delta ZXO}},$$

viz. the forces, divided by the diagonals along which they act, are proportional to the reciprocals of the products of the two pairs of triangles which stand on these diagonals respectively. The negative sign shows, what is obvious, that the forces must be, one of them a pull, the other a push.



## 643.

## ON A QUARTIC CURVE WITH TWO ODD BRANCHES.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 107, 108.]

(Page 36.) IT is a known theorem that the branches of a plane curve are even or odd; viz. two even branches, or an even and an odd branch (whether of the same curve or of different curves) intersect in an even number of real points; but two odd branches (of the same curve or of different curves) intersect in an odd number of real points\*.

In particular, a right line is an odd branch, and hence it meets any even branch of a curve in an even number of real points, and an odd branch in an odd number of real points; or (what is the same thing) an even branch is one which can be drawn throughout) of real points; an odd branch is one which is met by any right line whatever in an odd number of real points.

A curve without nodes has at most one odd branch; for if there were two, one of them being an asymptote (it is by no means certain what has been called a *complete* branch, viz. the partial branches which touch an asymptote at its opposite extremities are considered as parts of one and the same branch, and so in other cases. Thus a quadric curve, whether ellipse, parabola, or hyperbola, is one even branch; a cubic curve, in either one odd branch, or else it is an odd branch and an even branch; and generally a curve of an odd order has always an odd branch of odd branches, and a curve of an even order has always an even number of odd branches.

A curve without nodes has at most one odd branch; for if there were two, they would intersect in a real point, which would be a real node on the curve. In particular, a quartic curve having two odd branches must have a real node; this however may be, as in the instance about to be given, an isolated point of infinity.

A simple instance of a quartic curve with two odd branches is that represented by the equation

$$(x^2 - 1)(y^2 + 1) - 2nxy = 0,$$

\* The two branches must be distinct branches; a branch whether odd or even does not of necessity intersect itself (have upon it any real node), but it may intersect itself (have upon it any odd, or an even, number of real points.)

(Page 37.) or, what is the same thing,

$$y = \frac{1}{x^2 - 1} \left[ mx \pm \sqrt{\left\{ -(x^2 - 1) \left( x^2 - \frac{1}{n} \right) \right\}} \right],$$

where

$$n^2 + \frac{1}{n^2} = 2 + m^2,$$

or say

$$n^2 = \frac{1}{2} \left[ 2 + m^2 + m\sqrt{(4 + m^2)} \right],$$

$$\frac{1}{n^2} = \frac{1}{2} \left[ 2 + m^2 - m\sqrt{(4 + m^2)} \right],$$

so that  $m$  being positive  $n > 1$ , and the curve consists of two real branches included between the lines  $x = a$ ,  $x = -\frac{1}{a}$ , and the lines  $x = -a$ ,  $x = -\frac{1}{a}$  respectively; each of these lines touches the curve in a real point, viz.  $a$  having any one of the last-mentioned values, the value of  $y$  at the point of contact is  $y = \frac{ma}{a^2 - 1}$ ; and between each pair of lines we have the asymptote  $x = +1$  or  $x = -1$ . Hence the curve has the form shown in the figure, and it is thereby evident, that each branch of the curve is met by

[Figure: quartic curve showing two odd branches with asymptotes  $x = \pm 1$  and the nodes.]

any real right line whatever in one real point, or else in three real points. The numerical values in the figure are  $a = \frac{2}{3}$ ,  $m = \frac{1}{2}$ , whence also  $x = a$  or  $-\frac{1}{a}$ ,  $y = 1$ , and  $x = -a$  or  $\frac{1}{a}$ ,  $y = -1$ .

The curve has two nodes at infinity, viz. writing the equation in the form

$$(x^2 - z^2)(y^2 + z^2) - 2nxyz^2 = 0,$$

that is,

$$x^2y^2 + z^2(x^2 - y^2 - mxy) + z^4 = 0,$$

it appears that the points  $(x = 0, z = 0)$ ,  $(z = 0, y = 0)$  are each of them a node. The first of these  $(x = 0, z = 0)$  is the real intersection of the two odd branches; the other of them is a conjugate point.

## 644.

### NOTE ON MAGIC SQUARES.

[From the *Messenger of Mathematics*, vol. vi. (1877), p. 168.]

(Page 38.) IN a magic square of any odd order, formed according to the ordinary process, there is a tolerably simple analytical expression for the number which occupies any given compartment; thus taking the square of 21, let the dexter diagonals (N.W. to S.E.) commencing from the N.E. corner compartment, be numbered  $1, 2, 3, \dots, 20, 21, 20', \dots, 2', 1'$ , the diagonals of course containing these numbers of compartments respectively; and in any diagonal let the compartments reckoning from the top line be numbered  $1, 2, 3, \dots$ , respectively; then if  $D_{\theta, \phi}$  (or  $D'_{\theta, \phi}$  as the case may be) denotes the number in the compartment  $\phi$  of the diagonal  $\theta$  or  $\theta'$ , we have

$$\begin{aligned} D_{\text{odd}, \phi} &= 20\theta + 10 + \phi, \\ D_{\text{even}, \phi} &= 20\theta + 231 + \phi(-21), \\ D'_{\text{odd}, \phi} &= -22\theta + 430 + \phi, \\ D'_{\text{even}, \phi} &= -22\theta + 231 + \phi(-21), \end{aligned}$$

where in the second and fourth expressions the term  $-21$  is to be retained only if  $\phi > \theta$ ; if  $\phi \not> \theta$ , it is to be omitted. There would be a like formula for a square of any odd order, and it would be easy to write down the formulae for the general value  $2n + 1$ ; but I have preferred to give them for a specific case.

## 645.

### A SMITH'S PRIZE PAPER, 1877.

[From the *Messenger of Mathematics*, vol. vi. (1877), pp. 173–182.]

(Page 39.) THE paper was as follows:

1. Show (independently of the theory of roots) how, if  $x$  satisfies an equation of the order  $n$ , a given rational function of  $x$  can in general be expressed as a rational and integral function of the order  $n - 1$ . State the theorem in a more precise form, so as to make it true universally.
2. Investigate the form of the factors of  $1 \pm \sin(2n + 1)x$  considered as a function of  $\sin x$ ; and give the formulae in the two cases,  $2n + 1 = 3$  and  $5$  respectively.

3. Write down the substitutions which do not alter the function  $ab + cd$ ; and explain the constitution of the group.
4. Find in a form adapted for calculation an approximate value for the sum of the middle  $2x + 1$  terms of the expansion of  $(1 + 1)^n$ ,  $n$  being a large number, and  $x$  small in comparison therewith.  
Obtain thence a complete and precise statement of the theorem that in a large number of tosses the numbers of heads and tails will probably be nearly equal.
5. A point in space is represented on a given plane by its projections from two fixed points. Show how a problem relating to points, lines, and planes, is thereby reduced to a problem *in plano*; and apply the method to construct the line of intersection of two given planes each passing through three given points.
6. A weight is supported on a tripod of three unequal legs resting on a smooth horizontal plane, their feet connected in pairs by strings of given lengths. Show how to determine the tensions of the several strings.
7. Explain the ordinary configuration of a system of isoparametric lines on a spherical surface; for instance, what is the configuration when there are two points of minimum value, and one point of maximum value, of the parameter?

(Page 40.)

8. Find the attraction of an infinite circular cylinder, of uniform density, on a given exterior or interior point.
9. Determine the number of arbitrary constants contained in the equation of a surface of the order  $r$  which passes through the curve of intersection of two given surfaces of the orders  $m$  and  $n$  respectively.
10. Find, for the several values of  $p$ , the number of the conics passing through  $p$  given points and touching  $5 - p$  given lines; and, in each case, show how to obtain (in point-coordinates or line-coordinates, as may be most simple) the equations of the conics satisfying the conditions in question.
11. Investigate the theory of the linear transformation of a ternary quadric function into itself.
12. Explain the theory of the solution of a partial differential equation, given function of  $x, y, z, p, q, r = 0$ , viz. the equation involves only the dependent variable  $u$  in regard to the independent variables  $x, y, z$  respectively.

I propose, not (as in former years) to give complete solutions, but only to notice in more or less detail the leading points in the several questions.

1. The expression is of course required, not only for a given integral function of  $x$ , but for a given fractional function. The case where the given function is integral presents no difficulty; when the given function is fractional, the most simple case is where it is  $= \frac{1}{x-a}$ , supposing the equation to be  $f(x) = 0$ , here dividing  $f(x)$  by  $x - a$ , we have a quotient  $X(x)$  which is a rational and integral function of an order not exceeding  $n - 1$ , and a remainder  $f(a)$  which is a rational and integral function of  $a$ . Hence dividing

$$\frac{f(x)}{x - a} = R(x) + \frac{f(a)}{x - a},$$

or, in virtue of the given equation  $\frac{f(x)}{x-a} = -R(x)$ , viz. we have thus  $\frac{1}{x-a}$  in the required form. But if  $f(a) = 0$ , then we do not obtain such an expression of  $\frac{1}{x-a}$ . It has to be shown that the like considerations apply to any fractional function of  $x$ , and the precise form of the theorem in the case of a rational fractional function of  $x$  which by their intersection determine the nodes, have in common the line  $(x = 0, y = 0)$ , and there is a diminution in the number of their common intersections. I do not pursue the enquiry, but pass to a different question.

(Page 41.) that is,  $0 = \frac{\cos(2n+1)x}{\cos x}$ , the value in question gives  $\sin(2n+1)x = 1$ , and therefore  $\cos(2n+1)x = 0$ ; and it does not give  $\cos x = 0$ ; hence every such factor  $1 - \frac{\sin x}{\sin a}$  is a double factor, viz.

$$1 - \sin(2n+1)x = (1 \pm \sin x) \prod \left(1 - \frac{\sin x}{\sin a}\right)^2.$$

Or the like result may be obtained by considering instead of  $1 - \sin(2n+1)x$ , the more general function

$$\sin(2n+1)a \pm \sin(2n+1)x,$$

and finally assuming  $a = \frac{1}{2}\pi$ .

3. Relates to a theory which is not, but ought to be, treated of in the text books of the University. See Serret's *Algèbre Supérieure*, t. II., Sect. IV.

The substitutions which leave  $ab + cd$  of unaltered are

1 1, that is, the letters remain unchanged,

$\alpha$   $(ab)$ , that is,  $a$  and  $b$  are interchanged,

$\beta$   $(cd)$ , that is,  $c$  and  $d$  are interchanged,

$\gamma$   $(ab)(cd)$ , that is,  $a$  and  $b$  and also  $c$  and  $d$  are simultaneously interchanged,

$\delta$   $(ac)(bd)$ , same with  $a$  and  $d$ ,  $b$  and  $c$ ,

$\epsilon$   $(ad)(bc)$ , same with  $a$  and  $d$ ,  $b$  and  $c$ ,

$\zeta$   $(acbd)$ , that is, we cyclically change  $a$  into  $c$ ,  $c$  into  $b$ ,  $b$  into  $d$ , and  $d$  into  $a$ ,

$\theta$   $(adbc)$ , that is, we cyclically change  $a$  into  $d$ ,  $d$  into  $b$ ,  $b$  into  $c$ , and  $c$  into  $a$ ,

viz. we have eight substitutions 1,  $\alpha$ ,  $\beta$ ,  $\gamma$ ,  $\delta$ ,  $\epsilon$ ,  $\zeta$ ,  $\theta$  forming a group; that is, the product of any two of them, in either order, is a substitution of the group (or, what is the same thing, the same as that of the performance thereon of some other substitution of the group); thus we have  $\alpha^2 = 1$ ,  $\beta^2 = 1$ ,  $\gamma^2 = 1$ ,  $\alpha\beta = \beta\alpha = \gamma$ , &c.; the system of these equations, which verify that the set of substitutions form a group, defines the constitution of the group—thus to take a more simple instance, of 4 may be 1,  $a$ ,  $a^2$  ( $a^3 = 1$ ) or 1,  $a$ ,  $b$ ,  $ab$  ( $a^2 = 1$ ,  $b^2 = 1$ ,  $ab = ba$ ).

4. The expression of the general coefficient is

$$= \frac{1 \cdot 2 \dots 2n}{1 \cdot 2 \dots n - x \cdot 1 \cdot 2 \dots n + x},$$

which can be transformed by the well-known formula

$$1 \cdot 2 \dots u = u^{u+\frac{1}{2}} \sqrt{(\pi)} e^{-u},$$

(Page 42.) viz. the coefficient thus becomes

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} \cdot \frac{1}{\left(1 - \frac{x}{n}\right)^{n-x+\frac{1}{2}} \left(1 + \frac{x}{n}\right)^{n+x+\frac{1}{2}}}.$$

Now  $x$  is supposed small in comparison with  $n$ , and the factors in the denominator have the logarithms

$$(n - x + \tfrac{1}{2}) \log \left(1 - \frac{x}{n}\right) = (n - x + \tfrac{1}{2}) \left(-\frac{x}{n} - \tfrac{1}{2} \frac{x^2}{n^2}\right), \quad = -x + \tfrac{1}{2} \frac{x^2}{n},$$

and

$$(n + x + \tfrac{1}{2}) \log \left(1 + \frac{x}{n}\right) = x + \tfrac{1}{2} \frac{x^2}{n};$$

hence the denominator is  $= e^{\frac{x^2}{n}}$ , and the final approximate value of the coefficient is

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} e^{-\frac{x^2}{n}}.$$

Hence, converting as usual the sum into a definite integral, we have the sum of the  $2x + 1$  coefficients

$$= \frac{2^{2n}}{\sqrt{(n\pi)}} \int_{-x}^x e^{-\frac{x^2}{n}} dx,$$

or, what is the same thing,

$$= \frac{2^{2n}}{\sqrt{(\pi)}} \int_{-\frac{x}{\sqrt{n}}}^{\frac{x}{\sqrt{n}}} e^{-x^2} dx.$$

For the chance that the number of tosses lies between  $n + x$  and  $n - x$ , this has merely to be divided by  $2^{2n}$ ; hence writing  $x = kn$ , the chance that the number may be between  $n(1 + k)$  and  $n(1 - k)$  is

$$= \frac{1}{\sqrt{(\pi)}} \int_{-k\sqrt{n}}^{k\sqrt{n}} e^{-x^2} dx,$$

where observe that the integral, taken with the limits  $\infty, -\infty$  has the value  $\sqrt{(\pi)}$ .

Considering  $k$  as a given fraction however small, by increasing  $n$  we make  $k\sqrt{(n)}$  as large as we please, and therefore the integral, as nearly as we please  $= \sqrt{(\pi)}$ , or the chance as nearly as we please  $= 1$ ; and hence the complete and precise statement of the theorem, viz. by sufficiently increasing the number of tosses, the probability that the deviation from equality shall be any given percentage (as small as we please) of the deviation from equality is unity. The differential equation was considered by the Rev. H. J. Sharpe, who mentioned it to Prof. Stokes.

Further, restoring  $x$  in place of  $kn$ , the chance of a number between  $n + x$  and  $n - x$  is

$$= \frac{1}{\sqrt{(\pi)}} \int_{-\frac{x}{\sqrt{n}}}^{\frac{x}{\sqrt{n}}} e^{-x^2} dx,$$

(Page 43.) which when  $\frac{x}{\sqrt{(n)}}$  is small is  $= \frac{2x}{\sqrt{(n\pi)}}$ , (more accurately  $\frac{2x+1}{\sqrt{(n\pi)}}$ , when  $x$  is small); hence, however large  $x$  is, the chance of a deviation from equality not exceeding  $\pm x$  is as small as we please.

5. The point is represented in the given plane by two points which lie in lines (say  $O$ ) at a fixed point (say  $O$ ) of that plane, the line  $O$  is the intersection of the given plane by the line which joins the two projecting points.

A plane may be represented in the given plane by two lines, viz. these are the projections of the two lines which it makes with the two planes through the projecting points  $O$ .

A plane may be represented in the given plane by two lines, viz. these are the projections of the line in which it joins to projecting points.

6. The solution is represented on the given plane by means of virtual velocities; taking  $a, b, c$  for the lengths of the legs,  $f, g, h$  for the lengths of the strings, and  $z$  for the height of the summit,  $z$  is a known function of  $a, b, c, f, g, h$  ( $z$  is in fact  $= \frac{3V}{\Delta}$ , where  $V$ , the volume of the tetrahedron, is a given function of  $a, b, c, f, g, h$ ; and  $\Delta$ , the area of the base, is a given function of  $a, b, c$ ). Writing then  $F, G, H$  for the tensions, and  $W$  for the weight, and regarding  $z, f, g, h$  as variable, the principle gives

$$Wdz + Fdf + Gdg + Hdh = 0,$$

that is,

$$F, G, H, = -W \frac{dz}{df}, -W \frac{dz}{dg}, -W \frac{dz}{dh},$$

respectively.

7. The ordinary case is when no isoparametric lines has on one side of it larger values, on the other side smaller values, of  $u$ , and  $M$  the general function of the order  $r - m - n$ , then we have a value of  $L + MV = 0$ , where the general number  $L$  of the order  $r$ . Supposing only that  $r > m + n$ , this is the general number  $r$  where  $f$  is the general function of the order  $r$ ,  $L$  that of the order  $r - m$ , and  $M$  that of the order  $r - m - n$ ; the case where  $r - m$  or  $r - m - n$  is negative giving rise to an exception, the case being that no integration is performed by the equation of the surface or surface, of any maximum, parameter, the case excluded.

(Page 44.) The lines in the neighbourhood of a point of maximum, or of minimum, parameter are ovals surrounding this point; in question, each oval being itself surrounded by the consecutive oval. Supposing that there are two points of maximum and one of minimum, parameter, we have round each of them a series of ovals, until at length an oval belonging to the one of them ceases to unite itself with a series of ovals, until of the points of maximum, parameter, into a single point of maximum, parameter; and the remaining curves on the series are then a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

8. The investigation by means of the Potential presents the difficulty that the Potential of the infinite cylinder has no determinate value, as at once appears from the constructive oval. Supposing that there are two points of maximum, and one of minimum parameter, we have round each of them a series of ovals and oval belonging to the one of them ceases to unite itself with a series of ovals; thus the two ovals of maximum, parameter, into a single point of maximum, parameter; and the remaining curves on the series are then a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

The lines in the neighbourhood of a point of maximum, or of minimum, parameter are ovals surrounding this point in question, each oval being itself surrounded by the consecutive oval. Supposing that there are two points of maximum and one of minimum, parameter, we have round each of them a series of ovals, until at length an oval belonging to the one of them ceases to unite itself, with a series of ovals; thus, the two ovals of the points of maximum, parameter; and the remaining curves on the series are then a series of continually diminishing closed curves, and the figure of eight itself surrounded by a series of continually increasing closed curves.

It is necessary, however, before quitting the question, to consider the case where  $\frac{dV}{dx}, \frac{dV}{dy}$  contain only differential coefficients of  $V$ , are determinate. But it is safer to work directly with the Attraction; the Attraction of an infinite line acts in the perpendicular plane through the attracted point, and is inversely proportional to the distance; the problem is thus reduced to the plane

problem of a circle of uniform density, force varying as the distance (distance), attracting a point in its own plane. This is precisely similar to the case of a sphere with the ordinary law of attraction, dividing the circle into rings, each ring is a portion of an interior point, and an attraction as if its mass were collected at its centre, for an exterior point, and as  $r - m - n$ .

10. The numbers are known to be 1, 2, 4, 4, 2, 1, which values are obtained most easily (though not in the way which is theoretically most interesting) by finding for

(Page 45.) the first three cases the equation of the required conic in point-coordinates; and then, by changing these into line-coordinates, we have for points, the equations of any of the required conic in line-coordinates, and the condition of touching a given line gives a quadric for  $\lambda$ ; the number is therefore = 2.

$p = 5$ : 1 point. The equation of the conic is

$$(a, b, c, f, g, h, y, z)^2 = 0,$$

and we have 5 linear equations to determine the ratios of the coefficients; the number is therefore = 1.

$p = 4$ : 4 points and 1 line. Taking  $U = 0$  and  $V = 0$ , the equations of any of two conics each passing through the four points, the equation of the required conic must be of the form  $U + \lambda V = 0$ , and the condition of touching it gives a quadric in  $\lambda$ ; the number is therefore = 2.

$p = 3$ : 3 points and 2 lines. Taking  $U = 0$  and  $V = 0$ , the conics containing the three conics through the three points, the equations are  $\sigma = a, z = c, z = b, z = b$  are the equations of any conic containing these passes through the three points; the conditions of touching the given lines is  $y = \frac{ma}{a^2-1}$ ; and between each pair of conics for  $\sqrt{f}, \sqrt{g}, \sqrt{h}$ ; and therefore each of the eight values of  $\sqrt{f}:\sqrt{g}:\sqrt{h}$ ; that is, the number is = 4.

$p = 2$ : 2 points and 3 lines. Taking  $z = a, z = b, z = c$ , as the equations of the three lines,  $z = b, z = c, \xi a, \eta b, \zeta c$ , respectively. The equation of the conic in question is taken to be

$$\sqrt{\sqrt{\xi} + \sqrt{\eta} + \sqrt{\zeta}} = 0,$$

where each radical multiplied; viz. this is the equation in line-coordinates and adding we have the line equation in line-coordinates of conics touching the three lines  $\xi = 0, \eta = 0, \zeta = 0$ , and the four values of  $\sqrt{f}:\sqrt{g}:\sqrt{h}$ ;

$$a\xi + b\eta + g\zeta = ax + by + gz, \quad \text{etc.,}$$

$$b\xi + c\eta + f\zeta = bx + by + fz, \quad -\lambda\zeta,$$

$$g\xi + f\eta + c\zeta = gx + fy + cz, \quad +\lambda\eta.$$

(Page 46.) where  $\lambda, \mu, \nu$  are arbitrary; viz. multiplying by  $\xi, \eta, \zeta$  and adding we have the equation in question. The two conics determine three values of  $x, y, z$  and the equation in question then becomes three independent variables  $x, y, z$  respectively.

12. The partial differential equation in the form actually equally well have been given here,  $f(x, y, z, p, q, r) = 0$ , the equation then is  $\phi(x, y, z, p, q, r) = 0$ , the general partial differential equation involving the three independent variables  $x, y, z$ , and the derived functions  $p, q, r$  of the dependent variable  $u$ . The question is therefore to find an exact differential; for then writing  $u = \int(pdx + qdy + rdz)$ , we have the solution of the given partial differential equation.

Whatever be the method adopted, it comes out that the solution depends on the integration of the system of ordinary differential equations

$$\frac{dp}{dx} = \frac{dq}{dy} = \frac{dr}{dz} = \frac{dx}{dp} = \frac{dy}{dq} = \frac{dz}{dr},$$

and the answer consists first in showing this and, secondly, in showing how from an integral or integrals of the system we pass to the solution of the partial differential equation.

Considering the partial differential equation in the form actually proposed, and instead of  $\phi$  write  $H$ , where  $H$  will stand for the given function of  $x, y, z, p, q, r$ , which is the arbitrary constant  $H$ ; making this change, and putting the foregoing equal to the arbitrary constant  $H$ ; making this change, and putting the foregoing equations equal to the differential  $dt$  of a new variable, the system of ordinary differential equations is

$$\begin{aligned} \frac{dp}{dt} &= -\frac{dH}{dx}, & \frac{dq}{dt} &= -\frac{dH}{dy}, & \frac{dr}{dt} &= -\frac{dH}{dz}, \\ \frac{dx}{dt} &= \frac{dH}{dp}, & \frac{dy}{dt} &= \frac{dH}{dq}, & \frac{dz}{dt} &= \frac{dH}{dr}, \end{aligned}$$

where  $H$  is a given function of  $x, y, z, p, q, r$ . This is, in fact, the Hamiltonian system of equations; and it was in view to the connexion that the partial differential equation was proposed in its actual form.

## 646.

### ON THE GENERAL EQUATION OF DIFFERENCES OF THE SECOND ORDER.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 23–25.]

(Page 47.) CONSIDER the equation of differences

$$u_s = a_s u_{s-1} + b_{s-1} u_{s-2},$$

viz. we have

$$\begin{aligned} u_1 &= a_1 u_0 + b_0 u_{-1}, \\ u_2 &= a_2 u_1 + b_1 u_0, \\ u_3 &= a_3 u_2 + b_2 u_1, \\ u_4 &= a_4 u_3 + b_3 u_2, \\ u_5 &= a_5 u_4 + b_4 u_3, \\ &\&c., \end{aligned}$$

and thence

$$\begin{aligned} u_2 &= \begin{vmatrix} a_2 a_1 \\ +b_1 \end{vmatrix} u_1 + a_2 b_0 u_0, \\ u_3 &= \begin{vmatrix} a_3 a_2 a_1 \\ +a_3 b_1 \\ +a_1 b_2 \end{vmatrix} u_1 + \begin{vmatrix} a_3 + a_3 a_2 \\ +b_2 \end{vmatrix} b_0 u_0, \\ u_4 &= \begin{vmatrix} a_4 a_3 a_2 a_1 \\ +a_4 a_3 b_1 \\ +a_4 a_1 b_2 \\ +a_2 a_1 b_3 \\ +b_1 b_3 \end{vmatrix} u_1 + \begin{vmatrix} a_4 + a_4 a_3 a_2 \\ +a_4 b_2 \\ +a_2 b_3 \end{vmatrix} b_0 u_0, \end{aligned}$$



(Page 48.)

$$u_5 = \begin{vmatrix} a_5 a_4 a_3 a_2 a_1 \\ +a_5 a_4 a_3 b_1 \\ +a_5 a_4 a_1 b_2 \\ +a_5 a_2 a_1 b_3 \\ +a_3 a_2 a_1 b_4 \\ +a_5 b_1 b_3 \\ +a_3 b_1 b_4 \\ +a_1 b_2 b_4 \end{vmatrix} u_1 + \begin{vmatrix} a_5 a_4 a_3 a_2 \\ +a_5 a_4 b_2 \\ +a_5 a_2 b_3 \\ +a_3 a_2 b_4 \\ +b_2 b_4 \end{vmatrix} b_0 u_0, \\ \&c.$$

It is now easy to see the law; viz. writing for instance

$$u_5 = 54321 \cdot u_1 + 5432 \cdot b_0 u_0,$$

then 54321 has a leading term  $a_5 a_4 a_3 a_2 a_1$ : it has terms derived from this by changing any pair  $a_5 a_4, a_4 a_3$ , into  $b_4, b_3, b_2, b_1$  etc.; it has terms derived by changing two pairs  $a_5 a_4, a_3 a_2$  into  $b_4 b_3, b_2 b_1$ ; and so on; where observe that the expression of two consecutive  $a$ 's.

And, similarly, 5432 has a leading term  $a_5 a_4 a_3 a_2$ ; the other terms being derived from this in the same manner precisely.

The solution of  $u_s = la(au_{s-1} - u_{s-2})$  is included in, and might be deduced from the foregoing, but it is convenient to obtain it separately. Supposing for greater simplicity that  $u_{-1} = 0, u_0 = 1$  (or, what is the same thing,  $u_0 = 1, u_1 = l_1 a$ ), we find

$$\begin{aligned} u_0 &= 1, \\ u_1 &= l_1 a, \\ u_2 &= l_2 l_1 a^2 - l_2, \\ u_3 &= l_3 l_2 l_1 a^3 - \begin{vmatrix} l_3 l_2 \\ +l_3 l_1 \end{vmatrix} a, \\ u_4 &= l_4 l_3 l_2 l_1 a^4 - \begin{vmatrix} l_4 l_3 l_2 \\ +l_4 l_3 l_1 \\ +l_4 l_2 l_1 \end{vmatrix} a^2 + l_4 l_2, \\ u_5 &= l_5 l_4 l_3 l_2 l_1 a^5 - \begin{vmatrix} l_5 l_4 l_3 l_2 \\ +l_5 l_4 l_3 l_1 \\ +l_5 l_4 l_2 l_1 \\ +l_5 l_3 l_2 l_1 \end{vmatrix} a^3 + \begin{vmatrix} l_5 l_3 l_2 \\ +l_5 l_2 l_1 \end{vmatrix} a, \end{aligned}$$

&c.,

(Page 49.) viz. we may for example write

$$u_5 = l_5 \cdot 4321 \cdot a^5 - 4321 \cdot ( ) a^3 + 4321 \cdot ( ) a,$$

where

$$4321 \text{ denotes } l_4 l_3 l_2 l_1;$$

in 4321( ) we omit successively each number, viz. we thus obtain

$$432 + 431 + 421 + 321, \quad = l_4 l_3 l_2 + l_4 l_3 l_1 + l_4 l_2 l_1 + l_3 l_2 l_1;$$

in 4321(·) we omit successively each pair of two non-consecutive numbers, viz. the omitted numbers being 1, 3; 1, 4; 2, 4, we obtain

$$42 + 32 + 31, \quad = l_4 l_2 + l_3 l_2 + l_3 l_1;$$

and so on, the omissions being each four numbers, &c. two of them consecutive; thus in 654321(·) the omissions are 5, 3, 1; 4, 6, 2, or two of them consecutive; or the symbol is

$$642 + 531 + 521 + 421;$$

and so on, the omissions being each four numbers, &c. two of them consecutive;

$$642 + 531 + 21,$$

$$= l_6 l_4 l_2 + l_5 l_3 l_1.$$

As an application, a solution of the differential equation  $\frac{d}{dx} \left( x \frac{dy}{dx} \right) + (x - a)y = 0$  is  $y = u_0 + u_1 x + u_2 x^2 + \&c.$ , where  $s^2 u_s = a u_{s-1} - u_{s-2}$ , and in particular  $1^2 u_1 = a u_0$ , the equation of differences is thus of the form in question, and retaining  $l_s$  in place of its value,  $= s^2$ , the solution is  $u_0 = 1$ ,  $u_1 = l_1 a$ ,  $u_2 = l_2 l_1 a^2 - l_2$ , &c. *ut supra*. The differential equation was considered by the Rev. H. J. Sharpe, who mentioned it to Prof. Stokes.

## 647.

### ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION, SYMMETRICAL DETERMINANT = 0.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 46–52.]

(Page 50.) CONSIDER the equation

$$\nabla = \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} = 0,$$

where for the moment  $(a, b, \dots)$  denote linear functions of the coordinates  $(x, y, z, w)$ . This is a quartic surface having 10 nodes; viz. if we write  $(A, B, \dots)$  for the first minors of the determinant, then the cubic surfaces  $A = 0$ ,  $B = 0$ ,  $\dots$  have in common 10 points which are nodes of the quartic surface.

Suppose that  $(a, b, c, f, g, h)$  are linear functions of the form  $(x, y)$ ; then, observing that every term of  $\nabla$  contains as a factor

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

or one of its first minors, it is clear that the line  $x = 0$ ,  $y = 0$  is a double line on the surface. But the number of nodes is now less than 10; in fact, writing  $(x = 0, y = 0)$ , we make each of the first minors  $A = 0$ ,  $B = 0$ ,  $\dots$  have in common 10 points which are nodes of the quartic surface, which is common to the line  $(x = 0, y = 0)$ , and there is a diminution in the number of their common intersections. I do not pursue the enquiry, but pass to a different question.

# The Collected Mathematical Papers of Arthur Cayley

Vol. X, Pages 51–75

Papers 647 (continuation), 648, 649, 650, 651, 652, 653 (commencement)

Arthur Cayley

Cambridge, 1896

[647] ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION,  
SYMMETRICAL DETERMINANT = 0 (CONTINUED).

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 46–52;  
continued from preceding pages.]

(Page 51.) I, in fact, take the terms  $(a, \dots)$  of the determinant to be homogeneous functions of  $(x, y, z, w)$  of the degrees

$$\begin{array}{cccc} 0, & 1, & 1, & 0 \\ 1, & 2, & 2, & 1 \\ 1, & 2, & 2, & 1 \\ 0, & 1, & 1, & 0 \end{array}$$

respectively, viz.  $a, d, l$  are constants,  $g, h, m, n$  linear functions, and  $b, c, f$  quadric functions of the coordinates;  $\nabla = 0$  still represents a quartic surface; and it appears by a general formula that the number of nodes is = 8. But we can easily show this directly; and further, that the 8 nodes are the intersections of three quadric surfaces; or say that the quartic surface is octadic. For denoting as before the first minors by  $A, \dots$ , then  $B, C, F$  are each of them a quadric function of the coordinates, viz. we have

$$\begin{aligned} B &= d(ac - g^2) - cl^2 - an^2 + 2gln, \\ C &= d(ab - h^2) - am^2 - bl^2 + 2hlm, \\ F &= d(gh - af) + l^2f + mna - nlh - lmg, \end{aligned}$$

and we have identically

$$BC - F^2 = (ad - l^2)\nabla,$$

so that throwing out the constant factor  $ad - l^2$ , the equation of the surface is

$$BC - F^2 = 0,$$

and it has 8 nodes, the intersections of the three quadric surfaces  $B = 0, C = 0, F = 0$ . By equating to zero any other minor of the determinant  $\nabla$ , we have a surface passing through the 8 nodes; we have for instance the quartic surface

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix}^2 = 0.$$

Suppose now (and in all that follows) that, the degrees being as already mentioned, we further assume that  $b, c, f$  are quadric functions of the form  $(x, y)^2$ ,  $g, h$  linear functions of the form  $(x, y)$ ; then since each term of  $\nabla$  contains either

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

or one of its first minors, it is clear that the line  $(x = 0, y = 0)$  is a double line on the surface. But in the present case there is not any diminution in the number of the nodes; in fact, writing  $x = 0, y = 0$ , and therefore  $b, c, f, g, h$  each  $= 0$  (but not  $a = 0$ ), the minors  $B, C, F$  none of them vanish; that is, the line  $x = 0, y = 0$  is not a line on any one of the quadric surfaces, and the quadric surfaces intersect as before in an octad of points.

7—2

(Page 52.) The equation  $\nabla = 0$  thus represents a quartic surface having a double line, and also 8 nodes forming an octad.

We may without loss of generality write  $d = 0$ ; in fact, the determinant is unaltered if we add to the fourth column  $\theta$  times the first column, and then to the fourth line  $\theta$  times the first line; the determinant is thus of the original form, but in place of  $d$  it has  $d + 2\theta l + \theta^2 a$ , which by properly determining  $\theta$  can be made  $= 0$ . And then changing the original  $l, m, n$ , the equation is

$$\nabla = \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & 0 \end{vmatrix} = 0.$$

Or, writing for shortness,

$$K = \begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

and denoting the minors hereof by  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h})$ , then the equation is

$$\nabla = (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid l, m, n)^2 = 0,$$

where the degree of  $K$  is 4, and the degrees of  $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}$  are 4, 2, 2, 2, 3, 3 respectively, those of  $l, m, n$  being 0, 1, 1 respectively.

The nodes are, as before, the intersections of the quadric surfaces  $B = 0, C = 0, F = 0$ , viz. ( $d$  being now  $= 0$ ) the values are

$$\begin{aligned} -B &= cl^2 - 2gln + an^2, \\ -C &= bl^2 - 2hlm + am^2, \\ F &= fl^2 - glm - hln + amn. \end{aligned}$$

But, according to a previous remark, the nodes lie also on the quartic surface  $K^2 = 0$ ; viz. this is a set of four planes intersecting in the line  $x = 0, y = 0$ .

Now, in general, any plane through the line  $x = 0, y = 0$  meets the surface in this line twice and in a conic; if the plane is  $y = \theta x$ , we have

$$(a, b, c, f, g, h \mid x, y)^2 = (a, b'x^2, c'x^2, f'x^2, g'x, h'x),$$

where  $a, b', c', f', g', h'$  are functions of  $\theta$  of the degrees (0, 2, 2, 2, 1, 1) respectively; and thence also

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}) = (\mathfrak{a}', \mathfrak{b}'x^2, \mathfrak{c}'x^2, \mathfrak{f}'x^2, \mathfrak{g}'x^3, \mathfrak{h}'x^3),$$

where  $\mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{f}', \mathbf{g}', \mathbf{h}'$  are functions of  $\theta$  of the degrees 4, 2, 2, 2, 3, 3 respectively; the equation of the surface thus becomes  $(\mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{f}', \mathbf{g}', \mathbf{h}' \mid lx, m, n)^2 = 0$ ; viz. this is a quadric equation which, combined with the equation  $y - \theta x = 0$ , determines the

(Page 53.) conic in question. But for each of the planes  $K = 0$ , we have  $(\mathbf{a}', \mathbf{b}', \mathbf{c}', \mathbf{f}', \mathbf{g}', \mathbf{h}' \mid lx, m, n)^2$ , a perfect square, or the conic a two-fold line; we have thus the 8 nodes lying in pairs on four lines, say the four “rays,” in the four planes  $K = 0$  respectively; each of these rays meets the double line  $x = 0, y = 0$  in a point; and we have thus on the double line 4 points, which are in fact pinch-points of the surface (as to this presently). It has just been stated that for the plane passing through the nodal line and a ray, the conic is a two-fold line (the ray twice) containing upon it a pair of nodes; more properly, the conic is the point-pair composed of the two nodes.

We can find through the nodes four different plane-pairs; in fact, forming the equation

$$B - 2\lambda F + C\lambda^2 = 0,$$

this is

$$l^2(c + 2\lambda f + \lambda^2 b) - 2l(g + \lambda h)(n + \lambda m) + a(n + \lambda m)^2 = 0;$$

or, as this may also be written,

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2 a(b - 2\lambda f + \lambda^2 c) = 0,$$

where  $b, c, f$  and therefore also  $b - 2\lambda f + \lambda^2 c$  are of the form  $(x, y)^2$ ; say that we have

$$b - 2\lambda f + \lambda^2 c = (p, q, r \mid x, y)^2,$$

where  $p, q, r$  are of course quadric functions of  $\lambda$ ; determining  $\lambda$  by the quartic equation  $pr - q^2 = 0$ , we have  $b - 2\lambda f + \lambda^2 c$  a perfect square,  $= (\alpha x + \beta y)^2$  suppose; and we have thus the plane-pair

$$[a(n + \lambda m) - l(g + \lambda h)]^2 - l^2 a(\alpha x + \beta y)^2 = 0$$

containing the eight nodes; viz. there are four such plane-pairs. The two planes of a plane-pair intersect in a line called an “axis”; that is, we have four axes each meeting the nodal line; and we have thus also through the nodal line and the four axes respectively four planes, which are “pinch-planes” of the quartic surface (as to this presently).

It has just been seen that the equation  $B - 2\lambda F + C\lambda^2 = 0$  (where  $\lambda$  is arbitrary) is expressible in the form

$$[a(n + \lambda m) - l(g + \lambda h)]^2 + l^2 a(p, q, r \mid x, y)^2 = 0,$$

viz. this is the equation of a quadric cone having its vertex on the nodal line at the point  $x = 0, y = 0, an - lg + \lambda(am - lh) = 0$ ; this is, in fact, a cone touching the surface, as at once appears by writing the equation of the cone in the form

$$\frac{1}{C}\{BC - F^2 + (\lambda C - F)^2\} = 0,$$

that is,

$$\frac{1}{C}[-l^2 a \nabla + (\lambda C - F)^2] = 0;$$

we thus see that, taking for vertex any point whatever on the nodal line, there is a circumscribed quadric cone.

(Page 54.) For each of the above-mentioned four values of  $\lambda$ , the quadric cone breaks up into a plane-pair; each plane of the plane-pair is thus a “trope” or plane touching the surface along a conic; viz. this is the conic passing through the intersection of the plane (or say of an axis) with the nodal line and through four nodes of the surface. We have thus 8 tropes, intersecting in pairs in the four axes (and the intersection of each axis with the nodal line being a pinch-point). Moreover, joining the nodes in pairs, we have four rays, each meeting the nodal line, the plane through it and the nodal line being a pinch-plane; this is illustrated in the figure.

[Figure: schematic showing nodal line, axis, pinch-point and pinch-planes.]

As to the pinch-planes and pinch-points, remark first that a plane through the nodal line is in general a bitangent plane, its two points of contact being the points where the conic in such plane meets the nodal line. When the two points of contact come to coincide, the plane is a pinch-plane; viz. this happens when the plane passes through a ray, the conic being then the ray twice repeated. And secondly, at a point on the nodal line there are in general two tangent planes, viz. these are the tangent planes to the quadric cone belonging to such point; when the two tangent-planes come to coincide the point is a pinch-point, and this happens when the point is the intersection of the nodal line with an axis, for then (the quadric cone breaking up into the two tropes through the axis) the two tangent planes become the plane through the axis taken twice.

Each section through the nodal line is a conic, and the polar of the nodal line in regard to this conic is a point; the locus of this point (for different sections through the nodal line) is a right line which may be called simply the “polar.” To prove this, considering the section by the plane  $y = \theta x$ , we have to find the pole of the line  $x = 0$  in regard to the conic

$$(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid lx, m, n)^2 = 0;$$

(Page 55.) this is  $lx : m : n = \mathfrak{a}' : \mathfrak{h}' : \mathfrak{g}'$ , viz. if  $\mathfrak{g} = g_0x + g_1y$ ,  $\mathfrak{h} = h_0x + h_1y$ , this is

$$lx : m : n = a : g_0 + g_1\theta : h_0 + h_1\theta,$$

or joining hereto the equation  $y = \theta x$ , we have

$$lx : ly : m : n = a : a\theta : g_0 + g_1\theta : h_0 + h_1\theta,$$

where  $l, a, g_0, g_1, h_0, h_1$  are constants;  $m, n$  are linear functions of the coordinates  $(x, y, z, w)$ . The equations represent, it is clear, a right line which is the polar in question; and they may be written

$$\frac{lx}{a} = \frac{h_1m - g_1n}{h_0g_1 - h_1g_0}, \quad \frac{ly}{a} = \frac{h_0m - g_0n}{h_1g_0 - h_0g_1}.$$

When the plane passes through a ray, the conic becomes, as was stated, the point-pair composed of the two nodes in such ray; the harmonic in regard to these two points of the intersection of the ray with the nodal line is thus a point on the polar: that is, the polar meets the ray; and the two nodes are situate harmonically in regard to the intersections of the ray with the nodal line and the polar respectively.

The polar may be arrived at in a different manner, viz. if instead of a plane through the nodal line we consider a point on the nodal line, this is the vertex of a circumscribed quadric cone; and taking the polar plane of the nodal line in regard to this cone, then considering the point as variable, the different polar planes all pass through a line which is the polar in question. And hence, taking for the point the intersection of the nodal line with an axis, it appears that the axis meets the polar; and, moreover, that the two tropes through the axis are harmonics in regard to the planes through the axis, and the polar and nodal line respectively.

Collecting the foregoing results, we have a quartic surface as follows:

We have two lines, a nodal line and a polar; meeting each of these, four lines called “rays” and four other lines called “axes.” On each ray, harmonically in regard to its intersections with the nodal line and the polar, two nodes of the surface (in all 8 nodes); through each axis, harmonically in regard to the planes through it and the nodal line and the axis respectively, two tropes of the surface (in all 8 tropes). In each trope (or, what is the same thing, in its conic of contact) are 4 nodes; through each node (or, what is the same thing, touching its tangential quadricone) are 4 tropes; the relation of the nodes and tropes may be thus represented, viz. taking the pairs of

nodes to be 1, 2; 3, 4; 5, 6; 7, 8; and those of tropes to be I, II; III, IV; V, VI; VII, VIII; then we have

(Page 56.)

|   | I | II | III | IV | V | VI | VII | VIII |
|---|---|----|-----|----|---|----|-----|------|
| 1 | • |    |     |    | • |    | •   |      |
| 2 |   | •  |     |    |   | •  |     | •    |
| 3 |   |    | •   |    | • |    |     | •    |
| 4 |   |    |     | •  |   | •  | •   |      |
| 5 | • |    | •   |    |   |    |     | •    |
| 6 |   | •  |     | •  |   |    | •   |      |
| 7 | • |    |     | •  |   | •  |     |      |
| 8 |   | •  | •   |    | • |    |     |      |

viz. reading horizontally or vertically, the dots show the tropes through each node, or the nodes in each trope.

The plane through any ray and the nodal line is a pinch-plane of the surface, its point of contact being the intersection of the ray with the nodal line; and the intersection of each axis with the nodal line is a pinch-point of the surface, the tangent plane being the plane through the axis and the nodal line; the surface has thus 4 pinch-planes and 4 pinch-points.

## 648.

### ALGEBRAICAL THEOREM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), p. 53.]

(Page 57.) I WISH to put on record the following theorem, given by me as a Senate-House Problem, January, 1851.

If  $\{\alpha + \beta + \gamma + \dots\}^p$  denote the expansion of  $(\alpha + \beta + \gamma + \dots)^p$ , retaining those terms  $N\alpha^a\beta^b\gamma^c\dots$  only in which

$$b + c + d + \dots > p - 1, \quad c + d + \dots > p - 2, \quad \&c., \quad \&c.,$$

then

$$\begin{aligned} x^n &= (x + a)^n - n\{\alpha\}^1(x + a + \beta)^{n-1} + \frac{1}{2}n(n-1)\{\alpha + \beta\}^2(x + a + \beta + \gamma)^{n-2} \\ &\quad - \frac{1}{6}n(n-1)(n-2)\{\alpha + \beta + \gamma\}^3(x + a + \beta + \gamma + \delta)^{n-3} + \&c. \end{aligned}$$

The theorem, in a somewhat different and imperfectly stated form, is given, Burg, *Crelle*, t. i. (1826), p. 368, as a generalisation of Abel's theorem,

$$\begin{aligned} (x + a)^n &= x^n + na(x + \beta)^{n-1} + \frac{1}{2}n(n-1)a(a-2\beta)(x + 2\beta)^{n-2} \\ &\quad + \frac{1}{6}n(n-1)(n-2)a(a-3\beta)^2(x + 3\beta)^{n-3} + \&c. \end{aligned}$$

## 649.

**ADDITION TO MR GLAISHER'S NOTE ON SYLVESTER'S PAPER,  
"DEVELOPMENT OF AN IDEA OF EISENSTEIN."**

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 83, 84.]

(Page 58.) THE formula (11) [in the Note], under a slightly different form, is demonstrated by me in an addition [263] to Sir J. F. W. Herschel's paper "On the formulae investigated by Dr Brinkley, &c.," *Phil. Trans.*, t. CL., 1860, pp. 321–323. The demonstration is in effect as follows: let  $u$  denote a series of the form  $1 + bx + cx^2 + dx^3 + \dots$ , and let  $u^i$  (where  $i$  is positive or negative, integer or fractional) denote the development of the  $i$ -th power of  $u$ , continued up to the term which involves  $x^n$ , the terms involving higher powers of  $x$  being rejected;  $u^0, u^1, u^2$ , and generally  $u^s$  will denote in like manner the developments of these powers up to the terms involving  $x^n$ , or, what is the same thing, they will be the values of  $u^i$  corresponding to  $i = 0, 1, 2, \dots, s$ . By the formula

$$u^i = 1 + \frac{i}{1}(u-1) + \frac{i(i-1)}{1 \cdot 2}(u-1)^2 + \dots \text{ as far as the term involving } (u-1)^n, \text{ } u^i \text{ is a rational}$$

and integral function of  $i$  of the degree  $n$ , and can therefore be expressed in terms of the values  $u^0, u^1, u^2, \dots, u^n$  which correspond to  $i = 0, 1, 2, \dots, n$ . Let  $s$  have any one of the last-mentioned values, then the expression

$$\frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)},$$

which as regards  $i$  is a rational and integral function of the degree  $n$  (the factor  $i - s$  which occurs in the numerator and denominator being of course omitted), vanishes for each of the values  $i = 0, 1, 2, \dots, n$ , except only for the value  $i = s$ , in which case it becomes equal to unity. The required formula is thus seen to be

$$u^i = \Sigma \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)} u^s \right\},$$

(Page 59.) where the summation extends to the several values  $s = 0, 1, 2, \dots, n$ ; or, what is the same thing, it is

$$u^i = \Sigma \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{(-)^{n-s} 1}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} u^s \right\},$$

or, changing the sign of  $i$ , it is

$$u^{-i} = \Sigma \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^s 1}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} u^s \right\},$$

where, as before,  $s$  has the values  $0, 1, 2, \dots, n$  successively. Or, what is the same thing, we have

$$C_{-i,n} = \Sigma \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^s 1}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} C_{s,n} \right\},$$

where the term corresponding to  $s = 0$ , as containing the factor  $C_{0,n}$  vanishes except in the case  $n = 0$  (for which it is = 1); and omitting this evanescent term, this is in fact the formula (11).



650.

ON A QUARTIC SURFACE WITH TWELVE NODES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 103–106.]

(Page 60.) WRITE for shortness

$$\begin{aligned} a &= \beta - \gamma, & f &= \alpha - \delta, & af &= p, \\ b &= \gamma - \alpha, & g &= \beta - \delta, & bg &= q, \\ c &= \alpha - \beta, & h &= \gamma - \delta, & ch &= r; \end{aligned}$$

then,  $\theta$  being a variable parameter, the surface in question is the envelope of the quadric surface

$$(\alpha + \theta)^2 aghX^2 + (\beta + \theta)^2 bhfY^2 + (\gamma + \theta)^2 cfgZ^2 + (\delta + \theta)^2 abcW^2 = 0;$$

viz. this is

$$\Sigma \theta^2 aghX^2 \cdot \Sigma aghX^2 - \Sigma a aghX^2 = 0.$$

There are no terms in  $X^4$ , &c.; the coefficient of  $Y^2Z^2$  is

$$\gamma^2 cfg \cdot bfh + \beta^2 bfh \cdot cfg - 2\beta\gamma bfh \cdot cfg,$$

which is

$$= bcfgh(\beta - \gamma)^2, = a^2bcf^2gh, = abcfgh \cdot p.$$

Hence the whole equation divides by  $abcfgh$ , and throwing out this factor, the result is

$$p(Y^2Z^2 + X^2W^2) + q(Z^2X^2 + Y^2W^2) + r(X^2Y^2 + Z^2W^2) = 0,$$

or, observing that  $p + q + r = 0$ , this may also be written

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2 = 0,$$

(Page 61.) and also

$$p(YZ - XW)^2 + q(ZX - YW)^2 + r(XY - ZW)^2 = 0.$$

The more general equation

$$(p, q, r, l, m, n \mid YZ + XW, ZX + YW, XY + ZW)^2 = 0$$

represents a quartic surface (octadic) having the 8 nodes

$$\begin{aligned} (1, 0, 0, 0), & \quad (\bar{1}, 1, 1, 1), \\ (0, 1, 0, 0), & \quad (1, \bar{1}, 1, 1), \\ (0, 0, 1, 0), & \quad (1, 1, \bar{1}, 1), \\ (0, 0, 0, 1), & \quad (1, 1, 1, \bar{1}). \end{aligned}$$

We have

|                               |                                |
|-------------------------------|--------------------------------|
| $d_x U =$                     | $d_y U =$                      |
| $p \cdot XW^2 + YZW$          | $p \cdot YZ^2 + XZW$           |
| $q \cdot YW^2 + YZW$          | $q \cdot YW^2 + XZW$           |
| $r \cdot ZW^2 + YZW$          | $r \cdot YX^2 + XZW$           |
| $l \cdot 2XYZ + W(Y^2 + Z^2)$ | $l \cdot 2XYW + Z(W^2 + X^2)$  |
| $m \cdot 2XYW + Z(W^2 + Y^2)$ | $m \cdot 2YZX + W(Z^2 + X^2)$  |
| $n \cdot 2XZW + Y(W^2 + Z^2)$ | $n \cdot 2YZW + X(W^2 + Z^2),$ |

| $d_z U =$                      | $d_w U =$                      |
|--------------------------------|--------------------------------|
| $p \cdot Y^2 Z + XYW$          | $p \cdot X^2 W + XYZ$          |
| $q \cdot X^2 Z + XYW$          | $q \cdot Y^2 W + XYZ$          |
| $r \cdot W^2 Z + XYW$          | $r \cdot Z^2 W + XYZ$          |
| $l \cdot 2ZXW + Y(W^2 + X^2)$  | $l \cdot 2WYZ + X(Y^2 + Z^2)$  |
| $m \cdot 2YZW + X(W^2 + Y^2)$  | $m \cdot 2WZX + Y(Z^2 + X^2)$  |
| $n \cdot 2ZXY + W(X^2 + Y^2),$ | $n \cdot 2WXY + Z(X^2 + Y^2).$ |

Hence there will be a node

$$\begin{aligned}
 1, 1, 1, 1, \quad & \dots \quad p + q + r + 2l + 2m + 2n = 0, \\
 \bar{1}, 1, 1, 1, \quad & \dots \quad p + q + r - 2l + 2m - 2n = 0, \\
 1, \bar{1}, 1, 1, \quad & \dots \quad p + q + r + 2l - 2m + 2n = 0, \\
 1, 1, \bar{1}, 1, \quad & \dots \quad p + q + r - 2l - 2m + 2n = 0;
 \end{aligned}$$

or say there will be

$$\begin{aligned}
 1 \text{ of these nodes } & p + q + r + 2l + 2m + 2n = 0, \\
 2 & p + q + r + 2l = 0, \quad m + n = 0, \\
 3 & p + q + r = 2l = -2m = -2n, \\
 4 & p + q + r = 0, \quad l = 0, \quad m = 0, \quad n = 0;
 \end{aligned}$$

(Page 62.) viz. the surface having the 12 nodes is the original surface

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2,$$

where

$$p + q + r = 0.$$

The Jacobian of the quadrics

$$YZ + XW = 0, \quad ZX + YW = 0, \quad XY + ZW = 0,$$

is

$$\begin{vmatrix} W, & Z, & Y, & X \\ Z, & W, & X, & Y \\ Y, & X, & W, & Z \end{vmatrix} = 0;$$

viz. the equations are

$$\begin{aligned}
 X^3 - X(Y^2 + Z^2 + W^2) + 2YZW &= 0, \\
 Y^3 - Y(Z^2 + X^2 + W^2) + 2ZXW &= 0, \\
 Z^3 - Z(X^2 + Y^2 + W^2) + 2XYW &= 0, \\
 W^3 - W(X^2 + Y^2 + Z^2) + 2XYZ &= 0,
 \end{aligned}$$

each of which is satisfied in virtue of any one of the pairs of equations

$$\begin{aligned}
 (Y - Z = 0, \quad X - W = 0) \quad & (Y + Z = 0, \quad X + W = 0), \\
 (Z - X = 0, \quad Y - W = 0) \quad & (Z + X = 0, \quad Y + W = 0), \\
 (X - Y = 0, \quad Z - W = 0) \quad & (X + Y = 0, \quad Z + W = 0).
 \end{aligned}$$

so that the Jacobian curve is, in fact, the six lines represented by these equations.

Any two of the three tetrads form an octad, the 8 points of intersection of three quadric surfaces: a figure representing the relation of the 12 points to each other may be constructed without difficulty.

Each tetrad is a sibi-conjugate tetrad *quoad* the quadric  $X^2 + Y^2 + Z^2 + W^2 = 0$ . The three tetrads are not on the same quadric surface.

## 651.

## ON A SPECIAL SURFACE OF MINIMUM AREA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 190–196.]

(Page 63.) A VERY remarkable form of the surface of minimum area was obtained by Prof. Schwarz in his memoir “Bestimmung einer speciellen Minimal-fläche,” Berlin, 1871, [*Ges. Werke*, t. I., pp. 6–125], crowned by the Academy of Sciences at Berlin. The equation of the surface is

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where  $\lambda, \mu, \nu$  are functions of  $x, y, z$  respectively, viz.

$$x = - \int \frac{d\theta}{\sqrt{(\frac{1}{4}\theta^4 + \frac{1}{2}\theta^2 + \frac{1}{4})}},$$

and  $y, z$  are the same functions of  $\mu, \nu$  respectively. A direct verification of the theorem that this is a surface of minimum area, satisfying, that is, the differential equation

$$r(1 + q^2) - 2pqs + t(1 + p^2) = 0,$$

is given in the memoir; but the investigation may be conducted in quite a different manner, so as to be at once symmetrical and somewhat more general, viz. we may enquire whether there exists a surface of minimum area

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where the determining equations are

$$\begin{aligned}\lambda'^2 &= a\lambda^4 + b\lambda^2 + c, \\ \mu'^2 &= a\mu^4 + b\mu^2 + c, \\ \nu'^2 &= a\nu^4 + b\nu^2 + c,\end{aligned}$$

(Page 64.) ( $\lambda' = \frac{d\lambda}{dx}$ , &c.). I find that the coefficients  $a, b, c$  must satisfy four homogeneous quadric equations, which, in fact, admit of simultaneous solution, and that in three distinct ways; viz. assuming  $a = 1$ , the solutions are

$$\begin{aligned}a = 1, \quad b = \frac{1}{2}, \quad c = 1, \\ a = 1, \quad b = -2, \quad c = 1, \\ a = 1, \quad b = -\frac{3}{2}, \quad c = -\frac{1}{2};\end{aligned}$$

that is,

$$\lambda'^2 = \lambda^4 + \frac{1}{2}\lambda^2 + 1 \quad \{= \frac{1}{4}(\frac{4}{3}\lambda^4 + \frac{2}{3}\lambda^2 + \frac{4}{3})\},$$

which gives Schwarz's surface;

$$\lambda'^2 = \lambda^4 - 2\lambda^2 + 1 \text{ or } \lambda' = \pm(\lambda^2 - 1),$$

which, it is easy to see, gives only  $x + y + z = \text{const.}$ ; and

$$\lambda'^2 = \lambda^4 - \frac{3}{2}\lambda^2 - \frac{1}{2}, = (\lambda^2 - 1)(\lambda^2 + \frac{1}{2}),$$

which is a surface similar in its nature to Schwarz's surface.

The investigation is as follows: the condition to be satisfied by a surface of minimum area  $U = 0$  is

$$(\mathbf{a} + \mathbf{b} + \mathbf{c})(X^2 + Y^2 + Z^2) - (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid X, Y, Z)^2 = 0,$$

where  $(X, Y, Z)$  are the first derived coefficients and  $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h})$  the second derived coefficients of  $U$  in regard to the coordinates. Considering  $U$  as a function of  $\lambda, \mu, \nu$ , which are functions of  $x, y, z$  respectively, and writing  $(L, M, N)$  and  $(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h})$  for the first and second derived functions of  $U$  in regard to  $\lambda, \mu, \nu$ , also  $\lambda', \lambda''$  for the first and second derived functions of  $\lambda$  in regard to  $x$ , and so for  $\mu, \mu''$  and  $\nu, \nu''$  we have:

$$(X, Y, Z) = (L\lambda', M\mu', N\nu'),$$

$$(\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}) = (\mathbf{a}\lambda'^2 + L\lambda'', \mathbf{b}\mu'^2 + M\mu'', \mathbf{c}\nu'^2 + N\nu'', \mathbf{f}\mu'\nu', \mathbf{g}\nu'\lambda', \mathbf{h}\lambda'\mu'),$$

and for the particular surface  $U = 1 + \mu\nu + \nu\lambda + \lambda\mu = 0$ , the values are

$$(L, M, N, \mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}) = (\mu + \nu, \nu + \lambda, \lambda + \mu, 0, 0, 0, 1, 1, 1).$$

Hence the condition is found to be

$$\begin{aligned} & 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) \\ & + 2\nu'^2\lambda'^2(\mu + \nu)(\mu + \lambda) \\ & + 2\lambda'^2\mu'^2(\nu + \lambda)(\nu + \mu) \\ & - \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} \\ & - \mu''(\nu + \lambda)\{(\mu + \lambda)^2\nu'^2 + (\mu + \nu)^2\lambda'^2\} \\ & - \nu''(\lambda + \mu)\{(\nu + \mu)^2\lambda'^2 + (\nu + \lambda)^2\mu'^2\} = 0, \end{aligned}$$

(Page 65.) or say this is

$$\Sigma 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) - \Sigma \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} = 0.$$

We have to write in this equation  $\lambda'^2 = a\lambda^4 + b\lambda^2 + c$ , and therefore  $\lambda'' = 2a\lambda^3 + b\lambda$ , &c.; the left-hand side, call it  $\Omega$ , is a symmetrical function of  $\lambda, \mu, \nu$ , and is consequently expressible as a rational function of

$$\begin{aligned} p, &= \lambda + \mu + \nu, \\ q, &= \mu\nu + \nu\lambda + \lambda\mu, \\ r, &= \lambda\mu\nu. \end{aligned}$$

We ought to have  $\Omega = 0$ , not identically, but in virtue of the equation  $1 + q = 0$ , that is,  $\Omega$  should divide by  $1 + q$ ; or, what is the same thing,  $\Omega$  should vanish on writing therein  $q = -1$ .

To effect the reduction as easily as possible, observe that we have  $(\lambda + \mu)(\lambda + \nu) = \lambda^2 + q$ ; and therefore

$$\Sigma \mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) = \Sigma \lambda^2\mu'^2\nu'^2 + q\Sigma \mu'^2\nu'^2.$$

Similarly, in the second term,

$$(\mu + \nu)(\lambda + \nu)^2 = (\nu + \lambda)(\nu^2 + q) \text{ and } (\mu + \nu)(\lambda + \mu)^2 = (\mu + \lambda)(\mu^2 + q).$$

The complete value of  $\Omega$  thus is

$$\Omega = 2(Aq + B) - [(C + D)q + E + F],$$

where

$$\begin{aligned} A &= \Sigma \lambda^2 \mu'^2 \nu'^2, & B &= \Sigma \mu'^2 \nu'^2, \\ C &= \Sigma \lambda \lambda'' (\nu \mu'^2 + \mu \nu'^2), & D &= \Sigma \lambda'' (\nu \mu'^2 + \mu \nu'^2), \\ E &= \Sigma \lambda \lambda'' (\mu'^2 + \nu'^2), & F &= \Sigma \lambda'' (\nu \mu'^2 + \mu \nu'^2). \end{aligned}$$

We find without difficulty

$$\begin{aligned} A &= a^2 (q^4 - 4q^2 pr + 2r^2 + 2p^2 r^2) \\ &\quad + ab (-2q^3 + qp^2 + 4qr^2 - 3r^2 - 2p^2 r) \\ &\quad + ac (4q^2 - 3q^2 p^2 + 8pr + 2p^2) \\ &\quad + b^2 (q^2 - 2pr) \\ &\quad + bc (-4q + 2p^2) \\ &\quad + c^2 (3), \end{aligned}$$

$$\begin{aligned} B &= a^2 (q^2 r^2 + 2pr^3) \\ &\quad + ab (-4qr^2 + 2p^2 r^2) \\ &\quad + ac (-2q^3 + q^2 p^2 + 4qpr - 3r^2 - 2p^2 r) \\ &\quad + b^2 (3r^2) \\ &\quad + bc (2q^2 - 4pr) \\ &\quad + c^2 (-2q + 2p^2), \end{aligned}$$

(Page 66.)

$$\begin{aligned} C &= a^2 (4q^2 - 16q^2 pr + 16pr^2 + 8p^2 r^2) \\ &\quad + ab (-6q^3 + 3q^2 p^2 + 12qpr - 9r^2 - 6p^2 r) \\ &\quad + ac (8q^2 - 16qp^2 + 16pr + 4p^4) \\ &\quad + b^2 (2q^2 - 4pr) \\ &\quad + bc (-4q + 2p^2), \end{aligned}$$

$$\begin{aligned} D &= a^2 (4q^2 pr - 2qr^2 - 4p^2 r^2) \\ &\quad + ab (-4qpr + 2p^2 r) \\ &\quad + ac (-4q^2 + 2qp^2 - 2pr) \\ &\quad + b^2 (2pr) \\ &\quad + bc (2q), \end{aligned}$$

$$\begin{aligned} E &= a^2 (4q^2 r - 4pr^2) \\ &\quad + ab (-12qr^2 + 6p^2 r^2) \\ &\quad + ac (-4q^2 + 2qp^2 + 8qpr - 6r^2 - 4p^2 r) \\ &\quad + bc (2q - 4pr), \end{aligned}$$

$$\begin{aligned} F &= a^2 (4pr^2) \\ &\quad + ab ( ) \\ &\quad + ac (4q^2 - 12qpr + 12p^2 r) \\ &\quad + b^2 (qpr - 3r^2) \\ &\quad + bc (-2q^2 + qp^2 - pr), \end{aligned}$$

where in each line the terms are arranged according to their order in  $p, r$ .

Substituting, we find

$$\begin{aligned}\Omega = & a^2 ( -2q^3 + 6q^2pr - 8qr^2 ) \\ & + ab ( 2q^2 - q^2pr + qr^2 - 4r^2 ) \\ & + ac [ -2q^3p^2 + 14qpr - 12r^2 ] \\ & + b^2 ( -3qpr + 3r^2 ) \\ & + bc ( -2q^2 + qp^2 - 3pr ) \\ & + c^2 ( 2q + 2p^2 );\end{aligned}$$

viz. writing  $q = -1$ , this is

$$\begin{aligned}\Omega = & a^2 ( 2 - 6pr - 8r^2 ) \\ & + ab ( 2 + p^2 - pr - 4r^2 ) \\ & + ac ( -2p^2 - 14pr - 12r^2 ) \\ & + b^2 ( 3pr + 3r^2 ) \\ & + bc ( -2 - p^2 - 3pr ) \\ & + c^2 ( -2 - 2p^2 );\end{aligned}$$

(Page 67.) or, what is the same thing, it is

$$\begin{aligned}& ( 2a^2 + 2ab - 2bc - 2c^2 ) \\ & + p^2 ( ab - 2ac - bc + 2c^2 ) \\ & + pr ( -6a^2 - ab - 14ac + 3b^2 - 3bc ) \\ & + r^2 ( -8a^2 - 4ab - 12ac + 3b^2 )\end{aligned}$$

so that, writing for convenience  $a = 1$ , the equations to be satisfied are

$$\begin{aligned}2 - 2c^2 + 2(1 - c)b &= 0, \\ -2c + 2c^2 + (1 - c)b &= 0, \\ -6 - 14c + 3b^2 - (1 + 3c)b &= 0, \\ -8 - 12c + 3b^2 - 4b &= 0.\end{aligned}$$

The first and second are  $(1 - c)(2 + 2c + 2b) = 0$  and  $(1 - c)(-2c + b) = 0$ ; viz. they give  $c = 1$ , or else  $b = -\frac{1}{2}$ ,  $c = \frac{1}{2}$ . In the former case, the third and fourth equations each become  $3b^2 - 4b - 20 = 0$ , that is  $(3b - 10)(b + 2) = 0$ ; in the latter case, they are satisfied identically; hence we have for  $a, b, c$  the three systems of values mentioned at the beginning.

This completes the investigation; but it is interesting to find the values assumed by the other factor of  $\Omega$  on substituting therein for  $a, b, c$  the foregoing several systems of values. We have

in general

$$\begin{aligned}
 \Omega &= -2a^2q^3 + 2abq^2 - 2bcq^2 + 2c^2q \\
 &\quad + p^2(-abq^2 - 2acq^2 + bcq + 2c^2) \\
 &\quad + pr[6a^2q^2 - abq + (14ac - 3b^2)q - 3bc] \\
 &\quad + r^2[-8a^2q + 4abq - 12ac + 3b^2] \\
 &= -2a^2(q^3 - q^2) + 2ab(q^2 - 1) - 2bc(q^2 - 1) + 2c^2(q + 1) \\
 &\quad + p^2[-ab(q^2 + 1) - 2ac(q^2 - 1) + bc(q + 1)] \\
 &\quad + pr[6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)(q + 1)] \\
 &\quad + r^2\{-8a^2(q - 1) + 4ab\} \\
 &= (q + 1)\{-2a^2(q^2 - q + q^2 - q + 1) + 2ab(q^2 - q + q - 1) - 2bc(q - 1) + 2c^2 \\
 &\quad + p^2[-ab(q^2 - q - 1) - 2ac(q - 1) + bc] \\
 &\quad + pr[6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)] \\
 &\quad + r^2\{-8a^2(q - 1) + 4ab\}\}.
 \end{aligned}$$

Hence writing, first,  $a = c = 1$ ,  $b = -\frac{1}{2}$ , we obtain, after some reductions,

$$\Omega = (q + 1)\{-2q(q - 1)(q^2 - \frac{1}{4}q + 1) + p^2(q - 1)(-\frac{1}{2}q - 2) + pr(6q^2 - \frac{1}{4}q - 10) + r^2(-\frac{4}{3} + \frac{1}{2})\};$$

secondly, writing  $a = c = 1$ ,  $b = -2$ , we obtain

$$\Omega = (q + 1)\{-2(q + 1)^2(q^2 + 1) + p^2 \cdot 2(q - 1)^2 + 2pr(3q^2 - 2q + 6) - 8r^2q\};$$

and, thirdly, writing  $a = 1$ ,  $b = -\frac{1}{2}$ ,  $c = -\frac{1}{2}$ , we obtain

$$\Omega = (q + 1)\{(-2q^3 + q^2 - \frac{3}{2}q + \frac{3}{2}q) + p^2(-\frac{1}{2}q^2 + \frac{3}{2}q - \frac{1}{2}q) + pr(6q^2 - \frac{15}{2}q - \frac{17}{4}) + r^2(-8q + \frac{5}{2})\}.$$

9—2

## 652.

### ON A SEXTIC TORSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 229–235.]

(Page 68.) THE torse having for its edge of regression or cuspidal edge the curve defined by the equations  $x = \cos \phi$ ,  $y = \sin \phi$ ,  $z = \cos 2\phi$ , is an interesting and convenient one for the construction of a model, and it is here considered partly from that point of view.

The edge is a quadriquadric curve, the intersection of the cylinder  $x^2 + y^2 = 1$  with the parabolic hyperboloid  $z = x^2 - y^2$ ; the cylinder regarded as a cone having its vertex at infinity on the line  $x = 0$ ,  $y = 0$ , viz. the vertex is on the hyperboloid, or the curve is a nodal quadriquadric (the node being thus an isolated point at infinity on the line in question), and the torse is consequently of the order  $8 - 2$ , = 6, viz. it is a sextic torse.

The edge is a bent oval situate on the cylinder  $x^2 + y^2 = 1$ , such that, regarding  $\phi$  as the azimuth (or angle measured along the circular base from its intersection with the axis of  $x$ ), the altitude  $z$  is given by the equation  $z = \cos 2\phi$ ; viz. there are in the plane  $xz$ , or, say in the planes  $xz$ ,  $x'z$ , two maxima altitudes  $z = 1$ , and in the plane  $yz$ , or, say in the planes  $yz$  and  $y'z$ , two minima altitudes  $z = -1$ . The sections by these principal planes are, as is seen at once, nodal curves on the surface; they are, in fact, the cubic curves  $z = 3 - \frac{2}{x^2}$ , viz. here as  $x$  increases from  $\pm 1$  to  $\pm \infty$ ,  $z$  increases from the before-mentioned value 1 to 3, and  $z = -3 + \frac{2}{y^2}$ , viz. as

$y$  increases from  $\pm 1$  to  $\pm \infty$ ,  $z$  decreases from the before-mentioned value  $-1$  to  $-3$ . The two half-sheets (which meet in the cuspidal edge) intersect each other along these nodal lines, in such wise that the section of the surface by any axial plane (plane through the line  $x = 0, y = 0$ ) is a curve having a cusp on the cuspidal edge, and such that when the axial plane coincides with either of the principal planes  $x = 0, y = 0$ , the

(Page 69.) two half-branches of the curve coincide together with the portions which lie outside the cylinder  $x^2 + y^2 = 1$ , in fact, the portions referred to above, of the nodal curve in the plane in question; the portions which lie inside the cylinder are acnodal or isolated curves without any real sheet through them. It may be added, in the way of general description, that the section of the surface by any cylinder  $x^2 + y^2 = c^2$  ( $c > 1$ ) is a curve of the form  $z = C \cos(2\theta \pm B)$ ,  $\theta$  the angle along the base of the cylinder from the intersection with the axis of  $x$ ;  $C, B$  are functions of  $c$ ; viz. we have for the two half sheets respectively

$$z = C \cos(2\theta + B) \text{ and } z = C \cos(2\theta - B),$$

each curve having thus the two maxima  $+C$ , and the two minima  $-C$ ; and the two curves intersect each other at the four points in the two principal planes respectively; viz. the points for which  $\theta = 0, 90^\circ, 180^\circ, 270^\circ$ , and  $z = C \cos B, -C \cos B, C \cos B, -C \cos B$  accordingly.

Proceeding to discuss the surface analytically, we have for the equations of a generating line

$$\frac{x - \cos \phi}{-\sin \phi} = \frac{y - \sin \phi}{\cos \phi} = \frac{z - \cos 2\phi}{-2 \sin 2\phi} = \rho \text{ suppose,}$$

or say

$$\begin{aligned} x &= \cos \phi - \rho \sin \phi, \\ y &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi, \end{aligned}$$

which equations, considering therein  $\rho, \phi$  as arbitrary parameters, determine the surface.

Writing  $x = 0$ , we find  $y = \frac{1}{\cos \phi}$ , and then  $z = -3 + 2 \sin^2 \phi$ , viz. we have

$$x = 0, \quad z = -3 + \frac{2}{y^2}, \text{ for section in plane } yz,$$

and, similarly, writing  $y = 0$ , we find  $x = \frac{1}{\cos \phi}$ , and then  $z = 3 - 2 \cos^2 \phi$ , viz.

$$y = 0, \quad z = 3 - \frac{2}{x^2}, \text{ for section by plane } xz.$$

By what precedes, these are nodal curves, crunodal for the portions

$$(y = \pm 1 \text{ to } \pm \infty, z = -1 \text{ to } -3) \text{ and } (x = \pm 1 \text{ to } \pm \infty, z = 1 \text{ to } 3)$$

respectively, acnodal for the remaining portions  $y < \pm 1, x < \pm 1$  respectively.

Writing  $x = r \cos \theta, y = r \sin \theta$ , so that the coordinates of a point on the surface are  $r, \theta, z$ , where  $r = \sqrt{(x^2 + y^2)}$  is the projected distance,  $\theta$  is the azimuth from the axis of  $x$ , and  $z$  is the altitude, we have

$$\begin{aligned} r \cos \theta &= \cos \phi - \rho \sin \phi, \\ r \sin \theta &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi. \end{aligned}$$



(Page 70.) We have  $r^2 = 1 + \rho^2$ ; and thence also, if  $\tan \alpha = 2\rho, = \pm 2\sqrt{(r^2 - 1)}$ , that is,

$$\cos \alpha = \frac{1}{\sqrt{(4r^2 - 3)}}, \quad \sin \alpha = \pm \frac{2\sqrt{(r^2 - 1)}}{\sqrt{(4r^2 - 3)}},$$

then

$$z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta),$$

showing that for a given value of  $r$  (or section by the cylinder  $x^2 + y^2 = r^2$ ) the maximum and minimum values of  $z$  are  $z = \pm \sqrt{(4r^2 - 3)}$ .

But proceeding to eliminate  $\phi$ , we find

$$\begin{aligned} r^2 \cos 2\theta &= (1 - \rho^2) \cos 2\phi - 2\rho \sin 2\phi, \\ r^2 \sin 2\theta &= 2\rho \cos 2\phi + (1 - \rho^2) \sin 2\phi \end{aligned}$$

or multiplying these by  $1 + 3\rho^2$  and  $2\rho$  and adding

$$r^2 \{(1 + 3\rho^2) \cos 2\theta + 2\rho^2 \sin 2\theta\} = (1 + \rho^2)^2 (\cos 2\phi - 2\rho \sin 2\phi),$$

that is,

$$r^2 \{(3r^2 - 2) \cos 2\theta \pm 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta\} = r^4 z;$$

or, finally,

$$r^2 z = (3r^2 - 2) \cos 2\theta + 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta,$$

which is the equation of the surface in terms of the coordinates  $r, \theta, z$ .

Observing that  $(3r^2 - 2)^2 + 4(r^2 - 1)^3 = r^4(4r^2 - 3)$ , we may write

$$\begin{aligned} r^2 \sqrt{(4r^2 - 3)} \cos \beta &= 3r^2 - 2, \\ r^2 \sqrt{(4r^2 - 3)} \sin \beta &= 2(r^2 - 1)^{\frac{3}{2}}, \end{aligned}$$

and therefore also

$$\tan \beta = \frac{2(r^2 - 1)^{\frac{3}{2}}}{3r^2 - 2},$$

and the equation thus becomes

$$z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta),$$

where  $z$  is the altitude belonging to the azimuth  $\theta$  in the cylindrical section, radius  $r$ . The maxima and minima altitudes are  $\pm \sqrt{(4r^2 - 3)}$ , and these correspond to the values  $\theta = \pm \frac{1}{2}\beta, \frac{1}{2}\pi + \frac{1}{2}\beta, \pi + \frac{1}{2}\beta, \frac{3}{2}\pi + \frac{1}{2}\beta$ ; it is to be further noticed that when  $r = 1$ , we have  $\beta = 0$ , but as  $r$  increases and becomes ultimately infinite,  $\beta$  increases to  $\frac{1}{2}\pi$ , that is,  $\frac{1}{2}\beta$  increases from 0 to  $\frac{1}{4}\pi$ .

It may be noticed that the surface is a peculiar kind of deformation, obtained by giving proper rotations to the several cylindrical sections of the surface  $z = \sqrt{(4r^2 - 3)} \cos 2\theta$ ; viz. in rectangular coordinates this is  $r^2 z = \sqrt{(4r^2 - 3)}(x^2 - y^2)$ , that is,

$$(x^2 + y^2)^4 z^2 - \{4(x^2 + y^2) - 3\}(x^2 - y^2)^2 = 0.$$

To obtain the equation in rectangular coordinates, we have

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 16(r^2 - 1)^3 \sin^2 2\theta = 0,$$

viz. this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 \left(1 - \frac{4x^2 y^2}{r^4}\right) - 16(r^2 - 1)^3 \frac{4x^2 y^2}{r^4} = 0,$$

(Page 71.) or, what is the same thing, it is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - \frac{4x^2 y^2}{r^4} \{4(r^2 - 1)^3 + (3r^2 - 2)^2\} = 0,$$

viz. the term in  $\{ \}$  being  $r^4(4r^2 - 3)$ , this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 4x^2 y^2(4r^2 - 3) = 0,$$

or say

$$z^2(x^2 + y^2)^2 - 2z(3x^2 + 3y^2 - 2)(x^2 - y^2) + (3x^2 + 3y^2 - 2)^2 - 4x^2 y^2(4x^2 + 4y^2 - 3) = 0.$$

This may also be written

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2 y^2(z^2 - 4x^2 - 4y^2 + 3) = 0,$$

a form which puts in evidence the nodal curves

$$x = 0, \quad zy^2 = -3y^2 + 2, \quad \text{and} \quad y = 0, \quad zx^2 = 3x^2 - 2.$$

It shows also that the quadric cone  $z^2 - 4x^2 - 4y^2 + 3 = 0$  touches the surface along the curve of intersection with the surface  $z(x^2 - y^2) - 3(x^2 + y^2) + 2 = 0$ . This is, in fact, the curve of maxima and minima of the cylindrical sections, viz. reverting to the form  $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$ , or, if for greater clearness, attending only to one sheet of the surface, we write it  $z = \sqrt{(4r^2 - 3)} \cos(2\theta - \beta)$ , we have a maximum,  $z = \sqrt{(4r^2 - 3)}$ , for  $2\theta = \beta$  (or  $2\pi + \beta$ ), giving

$$\cos 2\theta = \cos \beta = \frac{3r^2 - 2}{r^2 \sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

and a minimum,  $z = -\sqrt{(4r^2 - 3)}$ , for  $2\theta = \pi + \beta$  (or  $3\pi + \beta$ ), giving

$$\cos 2\theta = -\cos \beta = -\frac{3r^2 - 2}{r^2 \sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

viz. the locus is  $z^2 = 4(r^2 - 3)$ ,  $z(x^2 - y^2) = 3r^2 - 2$ ; and for  $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$  we find the same locus, viz. the equations of the locus are

$$z^2 - 4x^2 - 4y^2 + 3 = 0, \quad z(x^2 - y^2) - 3x^2 - 3y^2 + 2 = 0,$$

as above.

To put in evidence the cuspidal edge, write for a moment  $\zeta = z - x^2 + y^2$ , the equation becomes  $\{\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 2) - 4x^2 y^2\}^2 + 4x^2 y^2\{\zeta^2 + 2\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 3) - 4x^2 y^2\} = 0$ ;

viz. this is

$$\zeta^2 r^4 + 2\zeta(x^2 - y^2)(r^2 - 1)(r^2 - 2) + (r^2 - 1)^2(r^2 - 2)^2 - 4\zeta^2 x^2 y^2(r^2 - 1)^2 = 0,$$

or writing the last term thereof in the form

$$-[r^4 - (x^2 - y^2)^2](r^2 - 1)^2,$$

and then putting  $r^2 = 1 + U$ , the equation is

$$\zeta^2(1 + 2U + U^2) + 2\zeta U(U - 1)(x^2 - y^2) + U^2(U - 1)^2 - U^2\{(U + 1)^2 - (x^2 - y^2)^2\} = 0;$$

viz. this is

$$\{\zeta - U(x^2 - y^2)\}^2 + 2U\{\zeta^2 + \zeta U(x^2 - y^2) - 2U^2\} + 4U^2 = 0,$$

(Page 72.) showing the cuspidal edge  $\zeta = 0, U = 0$ , viz.  $z = x^2 - y^2, x^2 + y^2 = 1$ . Moreover, along the cuspidal edge the surface is touched by  $\zeta - U(x^2 - y^2) = 0$ , that is, by  $z - (x^2 - y^2) = 0$ ; and at the points where this tangent surface again meets the surface we have  $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$ ; viz. the surface contains upon itself the curve represented by this last equation, and  $z - (x^2 - y^2) = 0$ .

As a verification, in the form

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2y^2(z^2 - 4x^2 - 4y^2 + 3) = 0$$

of the equation of the surface, write  $z = x^2 - y^2$ . If for a moment  $x^2 + y^2 = \lambda, x^2 - y^2 = \mu$ , then the value of  $z$  is  $z = \lambda\mu$ , and the equation becomes

$$(\lambda\mu^2 - 3\lambda + 2)^2 + (\lambda^2 - \mu^2)(\lambda^2\mu^2 - 4\lambda + 3) = 0,$$

that is,

$$\mu^2(\lambda^4 - 6\lambda^2 + 8\lambda - 3) - 4\lambda^3 + 12\lambda^2 - 12\lambda + 4 = 0;$$

or, what is the same thing,

$$(\lambda - 1)^2\{\mu^2(\lambda + 3) - 4\} = 0,$$

so that we have  $(\lambda - 1)^2 = 0$ , or else  $\mu^2(\lambda + 3) - 4 = 0$ ; viz.  $(x^2 + y^2 - 1)^2 = 0$ , or else  $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$ , agreeing with the former result.

In polar coordinates, the surface is touched along the cuspidal edge by the surface  $z = r^2 \cos 2\theta$ , and where this again meets the surface we have  $r^4(r^2 + 3) \cos^2 2\theta - 4 = 0$ .

For the model, taking the unit to be 1 inch, I suppose that for the edge of regression we have

$$x = 2 \cos \phi, \quad y = 2 \sin \phi, \quad z = 5 + (.45) \cos 2\phi,$$

viz. the curve is situate on a cylinder radius 2 inches. And I construct in zinc-plate the cylindric sections, or say the templets, for one sheet of the surface, for the several radii 2, 3, ..., 8 inches; taking the radius as  $k$  inches, the circumference of the cylinder, or entire base of the flattened templet, is  $= 2k\pi$ ; and the altitude, writing  $2\theta$  in place of  $2\theta - \beta$  as above, is given by the formula  $z = 5 + (.45)\sqrt{(k^2 - 3)} \cos 2\theta$ , so that the half altitude of the wave is  $= (.45)\sqrt{(k^2 - 3)}$ ; having this value, the curve is at once constructed geometrically. We have, moreover,  $\cos \beta = \frac{3k^2 - 8}{k^2 \sqrt{(k^2 - 3)}}$ ; the numerical values then are

| $k$ | $2k\pi$ | $(.45)\sqrt{(k^2 - 3)}$ | $\frac{3k^2 - 8}{k^2 \sqrt{(k^2 - 3)}}$ | $\frac{1}{2}\beta$ |
|-----|---------|-------------------------|-----------------------------------------|--------------------|
| 2   | 12.57   | 0.45                    | 1.00                                    | 0°                 |
| 3   | 18.85   | 1.10                    | .86                                     | 15                 |
| 4   | 25.13   | 1.62                    | .69                                     | 23                 |
| 5   | 31.42   | 2.11                    | .57                                     | 27 $\frac{1}{2}$   |
| 6   | 37.70   | 2.59                    | .48                                     | 30 $\frac{1}{2}$   |
| 7   | 43.98   | 3.05                    | .42                                     | 32 $\frac{1}{2}$   |
| 8   | 50.27   | 3.51                    | .36                                     | 34                 |

the altitudes in the successive templets being thus included between the limits  $5 \pm 0.45, 5 \pm 1.10, \dots, 5 \pm 3.51$ .

## 653.

## ON A TORSE DEPENDING ON THE ELLIPTIC FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 235–241.]

(Page 73.) ON attempting to cover with paper one half-sheet of the foregoing sextic torse, [652], I found that the paper assumed approximately the form of a circular annulus of an angle exceeding  $360^\circ$ , and this led me to consider the general theory of the construction of a torse in paper, and, in particular, to consider the torsos such that when developed into a plane the edge of regression becomes a circular arc. It is scarcely necessary to remark that, to construct in paper a circular annulus of an angle exceeding  $360^\circ$ , we have only to take a complete annulus, cut it along a radius, and then insert (gumming it on to the two terminal radii) a portion of an equal circular annulus; drawing from each point of the inner circular boundary a half-tangent, and considering these half-tangents as rigid lines, the paper will bend round them so as to form the half-sheet of a torse having for its edge of regression this inner boundary, which will assume the form of a closed curve with two equal and opposite maxima and two equal and opposite minima, described on a cylinder, and being approximately such as the curve given by the equations

$$x = \cos \theta, \quad y = \sin \theta, \quad z = m \cos 2\theta.$$

Considering, in general, an arc  $PQ$  (without inflexions) of any curve, and drawing at the consecutive points  $P, P', P'', \&c.$  the several half-tangents  $PT, P'T', P''T'', \dots$ , then, considering these as rigid lines and bending the paper round them, we have the half-sheet of a torse, having for its edge of regression the curve in question now bent into a curve of double curvature. It is, moreover, clear that the edge of regression has at each point thereof the same radius of absolute curvature as the original plane curve; in fact, if in the plane curve  $PP' = ds$ , and the angle  $T'PT$  between the consecutive half-tangents  $PT$  and  $P'T'$  be  $= d\phi$ , these quantities  $ds$  and

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(Page 74.)  $d\phi$  remain unaltered in the curve of double curvature; and the radius of absolute curvature is given by the equation  $\rho d\phi = ds$ . In particular when, as above, the arc is a circular one, say of radius  $= a$ , then, however the paper is bent, the edge of regression has at each point thereof the radius of absolute curvature  $= a$ .

Consider on any given surface, at a given point  $P$  thereof, and in a given direction, an element of length  $PP'$ , then (under the restrictions presently mentioned) we can determine the consecutive element  $P'P''$ , such that the curve  $PP'P'' \dots$  shall have at  $P$  a radius of absolute curvature  $= a$ ; in fact,  $r$  being the radius of curvature of the normal section of the surface through the element  $PP'$ , the radius of curvature of the section inclined at an angle  $\theta$  to the normal section is  $= r \cos \theta$ ; so that we have only to take the section at the inclination  $\theta, = \cos^{-1} \frac{a}{r}$  to the normal section, and we have the consecutive element  $P'P''$  such that the radius of absolute curvature of the curve  $PP'P''$  is  $= a$ . The necessary restriction, of course, is that  $r > a$ ; thus, if at the given point  $P$  the two principal radii of curvature are of the same sign (to fix the ideas, let the two principal radii and also  $a$  be each of them positive), then we may on the surface determine a direction  $PQ$ , for which the radius of curvature of the normal section is  $= a$ ; and then the direction of the element  $PP'$  may be any direction between  $PQ$  and the direction  $PR$ , corresponding to the greatest of the two principal radii.

Having obtained the element  $P'P''$ , we may, if the radius of absolute curvature at  $P'$  be given, construct the next element  $P'P'''$ , and so on; that is to say, on a given surface starting from a given point  $P$  and given initial direction  $PP'$ , we can (under a restriction, as above, as to the curvature at the different points of the surface) construct a curve having at the successive points thereof given values of the radius of absolute curvature; viz., the value may be given either as a function of the coordinates of the point on the surface, or as a function of the length of the

curve measured say from the initial point  $P$ ; it is in this last manner that in what follows the value of the radius of absolute curvature is assumed to be given.

We may thus, taking on paper an arc  $PQ$  with its half-tangents, apply it to a given surface, the point  $P$  to a given point, and the infinitesimal arc  $PP'$  to an element  $PP'$  in a given direction from the given point; and we thus obtain the half-sheet of a torse having for its edge of regression a determinate curve upon the surface. In particular, the arc  $PQ$  may be circular of the radius  $a$ , and the surface be a circular cylinder of radius  $\alpha$ ; and we thus obtain the torse having for edge of regression a curve on the cylinder radius  $\alpha$ , and such that the radius of absolute curvature is at each point  $= a$ . There are three cases according as  $\alpha > a$ ,  $\alpha = a$ , or  $\alpha < a$ ; it is to be remarked that if  $\alpha > a$ , then the curve must at each point cut the generating line of the cylinder at an angle not exceeding  $\cos^{-1} \frac{a}{\alpha}$ , but that in the other two cases the angle may have any value whatever; and, further, that in every case when the angle is  $= 0$ , viz. when the curve touches a generating line of the cylinder, then the osculating plane of the curve coincides with the tangent plane of the cylinder.

(Page 75.) The analytical theory is very simple. Taking  $x, y, z$  as functions of the length  $s$ , we have

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1;$$

the condition, which expresses that the radius of absolute curvature is  $= a$ , then is

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = \frac{1}{a^2}.$$

By what precedes, the point  $(x, y, z)$  may be taken to be upon a given surface, say upon the cylinder  $x^2 + y^2 = \alpha^2$ ; we may then write  $x = \alpha \cos \theta$ ,  $y = \alpha \sin \theta$ . Taking instead of  $s$  any independent variable  $u$  whatever, and using accents to denote the derived functions in regard to  $u$ , the equations become

$$\begin{aligned} x'^2 + y'^2 + z'^2 &= s'^2, \\ x''^2 + y''^2 + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \\ x &= \alpha \cos \theta, \quad y = \alpha \sin \theta. \end{aligned}$$

From the last two equations we obtain

$$x'^2 + y'^2 = \alpha^2 \theta'^2, \quad x''^2 + y''^2 = \alpha^2 (\theta''^2 + \theta'^4),$$

and the first two equations thus become

$$\begin{aligned} \alpha^2 \theta'^2 + z'^2 &= s'^2, \\ \alpha^2 (\theta''^2 + \theta'^4) + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \end{aligned}$$

and from the first of these we find

$$s'' = \frac{\alpha^2 \theta' \theta'' + z' z''}{(\alpha^2 \theta'^2 + z'^2)^{\frac{1}{2}}},$$

whence the second equation is

$$\alpha^2 (\theta''^2 + \theta'^4) + z''^2 - \frac{(\alpha^2 \theta' \theta'' + z' z'')^2}{\alpha^2 \theta'^2 + z'^2} = \frac{1}{a^2} (\alpha^2 \theta'^2 + z'^2)^2,$$

or reducing, this is

$$(\alpha^2\theta'^2 + z'^2)(\theta''^2 + \theta'^4) + (z''^2\theta'^2 - 2\theta'\theta''z'z'' - \alpha^2\theta'^2\theta''^2) = \frac{1}{a^2}(\alpha^2\theta'^2 + z'^2)^2.$$

Taking  $\theta$  as the independent variable, we have  $\theta' = 1$ ,  $\theta'' = 0$ , and the equation becomes

$$\alpha^2 + z'^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^2,$$

or, what is the same thing,

$$z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3 - (\alpha^2 + z'^2)^2.$$

Write here

$$\alpha^2 + z'^2 = \Omega^2,$$

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# The Collected Mathematical Papers of Arthur Cayley

Vol. X, Pages 76–100

Papers 647 (conclusion), 648, 649, 650, 651, 652, 653, 654 (commencement)

Arthur Cayley

Cambridge, 1896

[647] ON THE QUARTIC SURFACES REPRESENTED BY THE EQUATION,  
SYMMETRICAL DETERMINANT = 0 (CONCLUDED).

(Page 56.) The following table exhibits, for the surface in question, the incidence of the 8 nodes (numbered 1, ..., 8) with the 8 tropes (numbered I, ..., VIII):

|   | I | II | III | IV | V | VI | VII | VIII |
|---|---|----|-----|----|---|----|-----|------|
| 1 | • | .  | .   | .  | • | .  | •   | .    |
| 2 | . | •  | .   | •  | . | •  | .   | •    |
| 3 | . | .  | •   | .  | • | .  | •   | .    |
| 4 | . | •  | .   | •  | . | •  | .   | •    |
| 5 | • | .  | •   | .  | . | .  | •   | .    |
| 6 | . | •  | .   | •  | . | •  | .   | .    |
| 7 | . | .  | •   | .  | • | .  | .   | •    |
| 8 | . | •  | .   | .  | . | .  | •   | •    |

viz. reading horizontally or vertically, the dots show the tropes through each node, or the nodes in each trope.

The plane through any ray and the nodal line is a pinch-plane of the surface, its point of contact being the intersection of the ray with the nodal line; and the intersection of each axis with the nodal line is a pinch-point of the surface, the tangent plane being the plane through the axis and the nodal line; the surface has thus 4 pinch-planes and 4 pinch-points.

## 648.

## ALGEBRAICAL THEOREM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), p. 53.]

(Page 57.) I wish to put on record the following theorem, given by me as a Senate-House Problem, January, 1851.

If  $\{\alpha + \beta + \gamma + \dots\}^p$  denote the expansion of  $(\alpha + \beta + \gamma + \dots)^p$ , retaining those terms  $N\alpha^a\beta^b\gamma^c\dots$  only in which

$$b + c + d + \dots > p - 1, \quad c + d + \dots > p - 2, \quad \&c., \&c.,$$

then

$$\begin{aligned} x^n = (x + \alpha)^n - n\{\alpha\}^1(x + \alpha + \beta)^{n-1} + \frac{1}{2}n(n-1)\{\alpha + \beta\}^2(x + \alpha + \beta + \gamma)^{n-2} \\ - \frac{1}{6}n(n-1)(n-2)\{\alpha + \beta + \gamma\}^3(x + \alpha + \beta + \gamma + \delta)^{n-3} + \&c. \end{aligned}$$

The theorem, in a somewhat different and imperfectly stated form, is given, Burg, *Crelle*, t. i. (1826), p. 368, as a generalisation of Abel's theorem,

$$\begin{aligned} (x + \alpha)^n = x^n + n\alpha(x + \beta)^{n-1} + \frac{1}{2}n(n-1)\alpha(\alpha - 2\beta)(x + 2\beta)^{n-2} \\ + \frac{1}{6}n(n-1)(n-2)\alpha(\alpha - 3\beta)^2(x + 3\beta)^{n-3} + \&c. \end{aligned}$$

C. X.

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## 649.

**ADDITION TO MR GLAISHER'S NOTE ON SYLVESTER'S PAPER,  
"DEVELOPMENT OF AN IDEA OF EISENSTEIN."**

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 83, 84.]

(Page 58.) The formula (11) [in the Note], under a slightly different form, is demonstrated by me in an addition [263] to Sir J. F. W. Herschel's paper "On the formulae investigated by Dr Brinkley, &c.," *Phil. Trans.*, t. CL., 1860, pp. 321–323. The demonstration is in effect as follows: let  $u$  denote a series of the form  $1 + bx + cx^2 + dx^3 + \dots$ , and let  $u^i$  (where  $i$  is positive or negative, integer or fractional) denote the development of the  $i$ -th power of  $u$ , continued up to the term which involves  $x^n$ , the terms involving higher powers of  $x$  being rejected;  $u^0, u^1, u^2$ , and generally  $u^i$  will denote in like manner the developments of these powers up to the terms involving  $x^n$ , or, what is the same thing, they will be the values of  $u^i$  corresponding to  $i = 0, 1, 2, \dots, s$ . By the formula

$$u^i = 1 + \frac{i}{1}(u-1) + \frac{i \cdot i - 1}{1 \cdot 2}(u-1)^2 + \dots \text{ as far as the term involving } (u-1)^n, \text{ } u^i \text{ is a rational}$$

and integral function of  $i$  of the degree  $n$ , and can therefore be expressed in terms of the values  $u^0, u^1, u^2, \dots, u^n$  which correspond to  $i = 0, 1, 2, \dots, n$ . Let  $s$  have any one of the last-mentioned values, then the expression

$$\frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)},$$

which as regards  $i$  is a rational and integral function of the degree  $n$  (the factor  $i - s$  which occurs in the numerator and denominator being of course omitted), vanishes for each of the values  $i = 0, 1, 2, \dots, n$ , except only for the value  $i = s$ , in which case it becomes equal to unity. The required formula is thus seen to be

$$u^i = \sum \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{1}{s \cdot s - 1 \dots 2 \cdot 1 \cdot -1 \cdot -2 \dots - (n - s)} u^s \right\},$$

(Page 59.) where the summation extends to the several values  $s = 0, 1, 2, \dots, n$ ; or, what is the same thing, it is

$$u^i = \sum \left\{ \frac{i \cdot i - 1 \cdot i - 2 \dots i - n}{i - s} \cdot \frac{(-)^{n-s}}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots (n - s)} u^s \right\},$$

or, changing the sign of  $i$ , it is

$$u^{-i} = \sum \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^n}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} u^s \right\},$$

where, as before,  $s$  has the values  $0, 1, 2, \dots, n$  successively. Or, what is the same thing, we have

$$C_{-i,n} = \sum \left\{ \frac{i \cdot i + 1 \cdot i + 2 \dots i + n}{i + s} \cdot \frac{(-)^n}{1 \cdot 2 \dots s \cdot 1 \cdot 2 \dots n - s} C_{s,n} \right\},$$

where the term corresponding to  $s = 0$ , as containing the factor  $C_{0,n}$  vanishes except in the case  $n = 0$  (for which it is  $= 1$ ); and omitting this evanescent term, this is in fact the formula (11).

650.

ON A QUARTIC SURFACE WITH TWELVE NODES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 103–106.]

(Page 60.) Write for shortness

$$\begin{array}{lll} a = \beta - \gamma, & f = \alpha - \delta, & af = p, \\ b = \gamma - \alpha, & g = \beta - \delta, & bg = q, \\ c = \alpha - \beta, & h = \gamma - \delta, & ch = r; \end{array}$$

then,  $\theta$  being a variable parameter, the surface in question is the envelope of the quadric surface

$$(\alpha + \theta)^2 agh X^2 + (\beta + \theta)^2 bhf Y^2 + (\gamma + \theta)^2 cfg Z^2 + (\delta + \theta)^2 abc W^2 = 0;$$

viz. this is

$$\Sigma \theta^2 agh X^2 \cdot \Sigma agh X^2 - \Sigma \alpha agh X^2 = 0.$$

There are no terms in  $X^4$ , &c.; the coefficient of  $Y^2 Z^2$  is

$$\gamma^2 cfg \cdot bfh + \beta^2 bfh \cdot cfg - 2\beta\gamma bfh \cdot cfg,$$

which is

$$= bc fgh (\beta - \gamma)^2, \quad = a^2 bc f^2 gh, \quad = abc fgh \cdot p.$$

Hence the whole equation divides by  $abc fgh$ , and throwing out this factor, the result is

$$p(Y^2 Z^2 + X^2 W^2) + q(Z^2 X^2 + Y^2 W^2) + r(X^2 Y^2 + Z^2 W^2) = 0,$$

or, observing that  $p + q + r = 0$ , this may also be written

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2 = 0,$$

(Page 61.) and also

$$p(YZ - XW)^2 + q(ZX - YW)^2 + r(XY - ZW)^2 = 0.$$

The more general equation

$$(p, q, r, l, m, n \mid YZ + XW, ZX + YW, XY + ZW)^2 = 0$$

represents a quartic surface (octadic) having the 8 nodes

$$\begin{array}{ll} (1, 0, 0, 0), & (\bar{1}, 1, 1, 1), \\ (0, 1, 0, 0), & (1, \bar{1}, 1, 1), \\ (0, 0, 1, 0), & (1, 1, \bar{1}, 1), \\ (0, 0, 0, 1), & (1, 1, 1, \bar{1}). \end{array}$$

We have

| $d_x U =$                      | $d_y U =$                      |
|--------------------------------|--------------------------------|
| $p. \quad XW^2 + YZW$          | $p. \quad YZ^2 + XZW$          |
| $q. \quad YW^2 + YZW$          | $q. \quad YW^2 + XZW$          |
| $r. \quad ZW^2 + YZW$          | $r. \quad YX^2 + XZW$          |
| $l. \quad 2XYZ + W(Y^2 + Z^2)$ | $l. \quad 2XYW + Z(W^2 + X^2)$ |
| $m. \quad 2XYW + Z(W^2 + Y^2)$ | $m. \quad 2YZX + W(Z^2 + X^2)$ |
| $n. \quad 2XZW + Y(W^2 + Z^2)$ | $n. \quad 2YZW + X(W^2 + Z^2)$ |

| $d_z U =$                      | $d_w U =$                       |
|--------------------------------|---------------------------------|
| $p. \quad Y^2 Z + XYW$         | $p. \quad X^2 W + XYZ$          |
| $q. \quad X^2 Z + XYW$         | $q. \quad Y^2 W + XYZ$          |
| $r. \quad W^2 Z + XYW$         | $r. \quad Z^2 W + XYZ$          |
| $l. \quad 2ZXW + Y(W^2 + X^2)$ | $l. \quad 2WYZ + X(Y^2 + Z^2)$  |
| $m. \quad 2YZW + X(W^2 + Y^2)$ | $m. \quad 2WZX + Y(Z^2 + X^2)$  |
| $n. \quad 2ZXY + W(X^2 + Y^2)$ | $n. \quad 2WXY + Z(X^2 + Y^2).$ |

Hence there will be a node

$$\begin{aligned}
 1, 1, 1, 1, \quad & \dots p + q + r + 2l + 2m + 2n = 0, \\
 \bar{1}, 1, 1, 1, \quad & \dots p + q + r - 2l + 2m - 2n = 0, \\
 1, \bar{1}, 1, 1, \quad & \dots p + q + r - 2l + 2m - 2n = 0, \\
 \bar{1}, \bar{1}, 1, 1, \quad & \dots p + q + r - 2l - 2m + 2n = 0, \\
 1, 1, \bar{1}, 1, \quad & \dots p + q + r + 2l + 2m + 2n = 0;
 \end{aligned}$$

or say there will be

$$\begin{aligned}
 1 \text{ of these nodes if } & p + q + r + 2l + 2m + 2n = 0, \\
 2 \dots & p + q + r + 2l = 0, \quad m + n = 0, \\
 3 \dots & p + q + r = 2l = -2m = -2n, \\
 4 \dots & p + q + r = 0, \quad l = 0, \quad m = 0, \quad n = 0;
 \end{aligned}$$

(Page 62.) viz. the surface having the 12 nodes is the original surface

$$p(YZ + XW)^2 + q(ZX + YW)^2 + r(XY + ZW)^2,$$

where

$$p + q + r = 0.$$

The Jacobian of the quadrics

$$YZ + XW = 0, \quad ZX + YW = 0, \quad XY + ZW = 0,$$

is

$$\begin{vmatrix}
 W, & Z, & Y, & X \\
 Z, & W, & X, & Y \\
 Y, & X, & W, & Z \\
 X, & Y, & Z, & W
 \end{vmatrix} = 0;$$

viz. the equations are

$$\begin{aligned}
 X^3 - X(Y^2 + Z^2 + W^2) + 2YZW &= 0, \\
 Y^3 - Y(Z^2 + X^2 + W^2) + 2ZXW &= 0, \\
 Z^3 - Z(X^2 + Y^2 + W^2) + 2XYW &= 0, \\
 W^3 - W(X^2 + Y^2 + Z^2) + 2XYZ &= 0,
 \end{aligned}$$

each of which is satisfied in virtue of any one of the pairs of equations

$$\begin{aligned}
 (Y - Z = 0, \quad X - W = 0) \quad & (Y + Z = 0, \quad X + W = 0), \\
 (Z - X = 0, \quad Y - W = 0) \quad & (Z + X = 0, \quad Y + W = 0), \\
 (X - Y = 0, \quad Z - W = 0) \quad & (X + Y = 0, \quad Z + W = 0).
 \end{aligned}$$

So that the Jacobian curve is, in fact, the six lines represented by these equations.

Any two of the three tetrads form an octad, the 8 points of intersection of three quadric surfaces: a figure representing the relation of the 12 points to each other may be constructed without difficulty.

Each tetrad is a sibi-conjugate tetrad *quoad* the quadric  $X^2 + Y^2 + Z^2 + W^2 = 0$ . The three tetrads are not on the same quadric surface.

## 651.

## ON A SPECIAL SURFACE OF MINIMUM AREA.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 190–196.]

(Page 63.) A very remarkable form of the surface of minimum area was obtained by Prof. Schwarz in his memoir “Bestimmung einer speciellen Minimal-fläche,” Berlin, 1871, [*Ges. Werke*, t. 1., pp. 6–125], crowned by the Academy of Sciences at Berlin. The equation of the surface is

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where  $\lambda, \mu, \nu$  are functions of  $x, y, z$  respectively, viz.

$$x = - \int_0^\lambda \frac{d\theta}{\sqrt{(\frac{1}{4}\theta^4 + \frac{1}{2}\theta^2 + \frac{1}{4})}},$$

and  $y, z$  are the same functions of  $\mu, \nu$  respectively. A direct verification of the theorem that this is a surface of minimum area, satisfying, that is, the differential equation

$$r(1 + q^2) - 2pqs + t(1 + p^2) = 0,$$

is given in the memoir; but the investigation may be conducted in quite a different manner, so as to be at once symmetrical and somewhat more general, viz. we may enquire whether there exists a surface of minimum area

$$1 + \mu\nu + \nu\lambda + \lambda\mu = 0,$$

where the determining equations are

$$\lambda'^2 = a\lambda^4 + b\lambda^2 + c,$$

$$\mu'^2 = a\mu^4 + b\mu^2 + c,$$

$$\nu'^2 = a\nu^4 + b\nu^2 + c,$$

(Page 64.)  $\left(\lambda' = \frac{d\lambda}{dx}, \text{ \&c.}\right)$ . I find that the coefficients  $a, b, c$  must satisfy four homogeneous quadric equations, which, in fact, admit of simultaneous solution, and that in three distinct ways; viz. assuming  $a = 1$ , the solutions are

$$\begin{aligned} a = 1, \quad b = \frac{1}{2}, \quad c = 1, \\ a = 1, \quad b = -2, \quad c = 1, \\ a = 1, \quad b = -\frac{1}{2}, \quad c = -\frac{1}{2}; \end{aligned}$$

that is,

$$\lambda'^2 = \lambda^4 + \frac{1}{2}\lambda^2 + 1 \left\{ = \frac{1}{4}(4\lambda^4 + 2\lambda^2 + 4) \right\},$$

which gives Schwarz's surface;

$$\lambda'^2 = \lambda^4 - 2\lambda^2 + 1 \text{ or } \lambda' = \pm(\lambda^2 - 1),$$

which, it is easy to see, gives only  $x + y + z = \text{const.}$ ; and

$$\lambda'^2 = \lambda^4 - \frac{1}{2}\lambda^2 - \frac{1}{2}, \quad = (\lambda^2 - 1)(\lambda^2 + \frac{1}{2}),$$

which is a surface similar in its nature to Schwarz's surface.

The investigation is as follows: the condition to be satisfied by a surface of minimum area  $U = 0$  is

$$(\mathfrak{a} + \mathfrak{b} + \mathfrak{c})(X^2 + Y^2 + Z^2) - (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h} \mid X, Y, Z)^2 = 0,$$

where  $(X, Y, Z)$  are the first derived coefficients and  $(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h})$  the second derived coefficients of  $U$  in regard to the coordinates. Considering  $U$  as a function of  $\lambda, \mu, \nu$ , which are functions of  $x, y, z$  respectively, and writing  $(L, M, N)$  and  $(a, b, c, f, g, h)$  for the first and second derived functions of  $U$  in regard to  $\lambda, \mu, \nu$ , also  $\lambda', \lambda''$  for the first and second derived functions of  $\lambda$  in regard to  $x$ , and so for  $\mu, \mu''$  and  $\nu, \nu''$ : we have

$$(X, Y, Z) = (L\lambda', M\mu', N\nu'),$$

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}) = (a\lambda'^2 + L\lambda'', b\mu'^2 + M\mu'', c\nu'^2 + N\nu'', f\mu'\nu', g\nu'\lambda', h\lambda'\mu'),$$

and for the particular surface  $U = 1 + \mu\nu + \nu\lambda + \lambda\mu = 0$ , the values are

$$(L, M, N, a, b, c, f, g, h) = (\mu + \nu, \nu + \lambda, \lambda + \mu, 0, 0, 0, 1, 1, 1).$$

Hence the condition is found to be

$$\begin{aligned} & 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) \\ & + 2\nu'^2\lambda'^2(\mu + \nu)(\mu + \lambda) \\ & + 2\lambda'^2\mu'^2(\nu + \lambda)(\nu + \mu) \\ & - \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} \\ & - \mu''(\nu + \lambda)\{(\mu + \lambda)^2\nu'^2 + (\mu + \nu)^2\lambda'^2\} \\ & - \nu''(\lambda + \mu)\{(\nu + \mu)^2\lambda'^2 + (\nu + \lambda)^2\mu'^2\} = 0, \end{aligned}$$

(Page 65.) or say this is

$$\Sigma 2\mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) - \Sigma \lambda''(\mu + \nu)\{(\lambda + \nu)^2\mu'^2 + (\lambda + \mu)^2\nu'^2\} = 0.$$

We have to write in this equation  $\lambda'^2 = a\lambda^4 + b\lambda^2 + c$ , and therefore  $\lambda'' = 2a\lambda^3 + b\lambda$ , &c.; the left-hand side, call it  $\Omega$ , is a symmetrical function of  $\lambda, \mu, \nu$ , and is consequently expressible as a rational function of

$$\begin{aligned} p &= \lambda + \mu + \nu, \\ q &= \mu\nu + \nu\lambda + \lambda\mu, \\ r &= \lambda\mu\nu. \end{aligned}$$

We ought to have  $\Omega = 0$ , not identically, but in virtue of the equation  $1 + q = 0$ , that is,  $\Omega$  should divide by  $1 + q$ ; or, what is the same thing,  $\Omega$  should vanish on writing therein  $q = -1$ .

To effect the reduction as easily as possible, observe that we have  $(\lambda + \mu)(\lambda + \nu) = \lambda^2 + q$ ; and therefore

$$\Sigma \mu'^2\nu'^2(\lambda + \mu)(\lambda + \nu) = \Sigma \lambda^2\mu'^2\nu'^2 + q\Sigma \mu'^2\nu'^2.$$

Similarly, in the second term,

$$(\mu + \nu)(\lambda + \nu)^2 = (\nu + \lambda)(\nu^2 + q) \quad \text{and} \quad (\mu + \nu)(\lambda + \mu)^2 = (\mu + \lambda)(\mu^2 + q).$$

The complete value of  $\Omega$  thus is

$$\Omega = 2(Aq + B) - [(C + D)q + E + F],$$

where

$$\begin{aligned} A &= \Sigma \lambda^2 \mu'^2 \nu'^2, & B &= \Sigma \mu'^2 \nu'^2, \\ C &= \Sigma \lambda \lambda'' (\nu \mu'^2 + \mu \nu'^2), & D &= \Sigma \lambda'' (\mu^2 \nu'^2 + \nu^2 \mu'^2), \\ E &= \Sigma \lambda \lambda'' (\mu'^2 + \nu'^2), & F &= \Sigma \lambda'' (\nu \mu'^2 + \mu \nu'^2). \end{aligned}$$

We find without difficulty

$$\begin{aligned} A &= a^2 (q^4 - 4q^2 pr + 4qr^2 + 2p^2 r^2) \\ &\quad + ab (-2q^3 + qp^2 + 4qpr - 3r^2 - 2p^2 r) \\ &\quad + ac (4q^2 - 8qp^2 + 8pr + 2p^4) \\ &\quad + b^2 (q^2 - 2pr) \\ &\quad + bc (-4q + 2p^2) \\ &\quad + c^2 (3), \\ B &= a^2 (q^2 r^2 + 2pr^3) \\ &\quad + ab (-4qr^2 + 2p^2 r^2) \\ &\quad + ac (-2q^3 + q^2 p^2 + 4qpr - 3r^2 - 2p^2 r) \\ &\quad + b^2 (3r^2) \\ &\quad + bc (2q^2 - 4pr) \\ &\quad + c^2 (-2q + 2p^2), \end{aligned}$$

(Page 66.)

$$\begin{aligned} C &= a^2 ( ) \\ &\quad + ab (-6q^3 + 3q^2 p^2 + 12qpr - 9r^2 - 6p^2 r) \\ &\quad + ac (8q^2 - 16qp^2 + 16pr + 4p^4) \\ &\quad + b^2 (2q^2 - 4pr) \\ &\quad + bc (4q + 2p^2), \\ D &= a^2 (4q^2 pr - 2qr^2 - 4p^2 r^2) \\ &\quad + ab (-4qpr + 2p^2 r) \\ &\quad + ac (-4q^2 + 2qp^2 - 2pr) \\ &\quad + b^2 (2pr) \\ &\quad + bc (2q), \\ E &= a^2 (4q^2 r^2 - 4pr^3) \\ &\quad + ab (-12qr^2 + 6p^2 r^2) \\ &\quad + ac (-4q^3 + 2q^2 p^2 + 8qpr - 6r^2 - 4p^2 r) \\ &\quad + b^2 ( ) \\ &\quad + bc (2q^2 - 4pr), \\ F &= a^2 ( ) \\ &\quad + ab ( ) \\ &\quad + ac (4q^3 - 12qpr + 12p^2 r) \\ &\quad + b^2 (qpr - 3r^2) \\ &\quad + bc (-2q^2 + qp^2 - pr), \end{aligned}$$

where in each line the terms are arranged according to their order in  $p, r$ .

Substituting, we find

$$\begin{aligned}\Omega = & a^2(-2q^3 + 6q^2pr - 8qr^2) \\ & + ab(2q^2 - q^3pr - 4qr^2) \\ & + ac(-2q^2p^2 + 14qpr - 12r^2) \\ & + b^2(-3qpr + 3r^2) \\ & + bc(-2q^2 + qp^2 - 3pr) \\ & + c^2(2q + 2p^2); \end{aligned}$$

viz. writing  $q = -1$ , this is

$$\begin{aligned}\Omega = & a^2(2 - 6pr - 8r^2) \\ & + ab(2 + p^2 - pr - 4r^2) \\ & + ac(-2p^2 - 14pr - 12r^2) \\ & + b^2(3pr + 3r^2) \\ & + bc(-2 - p^2 - 3pr) \\ & + c^2(-2 - 2p^2); \end{aligned}$$

(Page 67.) or, what is the same thing, it is

$$\begin{aligned}\Omega = & (2a^2 + 2ab - 2bc - 2c^2) \\ & + p^2(ab - 2ac - bc + 2c^2) \\ & + pr(-6a^2 - ab - 14ac + 3b^2 - 3bc) \\ & + r^2(-8a^2 - 4ab - 12ac + 3b^2); \end{aligned}$$

so that, writing for convenience  $a = 1$ , the equations to be satisfied are

$$\begin{aligned}2 - 2c^2 + 2(1 - c)b &= 0, \\ -2c + 2c^2 + (1 - c)b &= 0, \\ -6 - 14c + 3b^2 - (1 + 3c)b &= 0, \\ -8 - 12c + 3b^2 - 4b &= 0. \end{aligned}$$

The first and second are  $(1 - c)(2 + 2c + 2b) = 0$  and  $(1 - c)(-2c + b) = 0$ ; viz. they give  $c = 1$ , or else  $b = -2$ ,  $c = -\frac{1}{2}$ . In the former case, the third and fourth equations each become  $3b^2 - 4b - 20 = 0$ , that is  $(3b - 10)(b - 2) = 0$ ; in the latter case, they are satisfied identically; hence we have for  $a, b, c$  the three systems of values mentioned at the beginning.

This completes the investigation; but it is interesting to find the values assumed by the other factor of  $\Omega$  on substituting therein for  $a, b, c$  the foregoing several systems of values. We have



in general

$$\begin{aligned}
\Omega &= -2a^2q^3 + 2abq^2 - 2bcq + 2c^2q \\
&\quad + p^2(-abq^2 - 2acq^2 + bcq + 2c^2) \\
&\quad + pr(6a^2q^2 - abq + 14acq - 3b^2q - 3bc) \\
&\quad + r^2(-8a^2q + 4abq - 12ac + 3b^2) \\
&= -2a^2(q^3 - 1) + 2ab(q^2 - 1) - 2bc(q - 1) + 2c^2(q + 1) \\
&\quad + p^2\{-ab(q^2 + 1) - 2ac(q^2 - 1) + bc(q + 1)\} \\
&\quad + pr\{6a^2(q^2 + 1) - ab(q - 1) + (14ac - 3b^2)(q + 1)\} \\
&\quad + r^2\{-8a^2(q - 1) + 4ab\} \\
&= (q + 1)\{-2a^2(q^2 - q + 1) + 2ab(q - 1) - 2bc + 2c^2 \\
&\quad + p^2[-ab(q^2 - q + 1) - 2ac(q - 1) + bc] \\
&\quad + pr[6a^2(q^2 + q + 1) - ab(q - 1) + (14ac - 3b^2)] \\
&\quad + r^2[-8a^2(q - 1) + 4ab]\}.
\end{aligned}$$

Hence writing, first,  $a = c = 1$ ,  $b = \frac{1}{2}$ , we obtain, after some reductions,

$$\Omega = (q + 1)\{-2q(q - 1)(q^2 - \frac{1}{2}q + 1) + p^2(q - 1)(-\frac{1}{2}q - 2) + pr(6q^2 - \frac{1}{2}q - \frac{19}{2}) + r^2(-8q + \frac{9}{2})\};$$

secondly, writing  $a = c = 1$ ,  $b = -2$ , we obtain

$$\Omega = (q + 1)\{-2(q + 1)^2(q^2 - q + 1) + p^2 \cdot 2(q - 1)^2 + 2pr(3q^2 - 2q + 6) - 8r^2q\};$$

and, thirdly, writing  $a = 1$ ,  $b = -\frac{1}{2}$ ,  $c = -\frac{1}{2}$ , we obtain

$$\Omega = (q + 1)\{(-2q^3 + q^2 - \frac{1}{2}q + \frac{1}{2}) + p^2(-\frac{1}{2}q^2 + \frac{1}{2}q - \frac{1}{2}) + pr(6q^2 - \frac{19}{2}q - \frac{27}{4}) + r^2(-8q + \frac{1}{2})\}.$$

## 652.

## ON A SEXTIC TORSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 229–235.]

(Page 68.) The torse having for its edge of regression or cuspidal edge the curve defined by the equations  $x = \cos \phi$ ,  $y = \sin \phi$ ,  $z = \cos 2\phi$ , is an interesting and convenient one for the construction of a model, and it is here considered partly from that point of view.

The edge is a quadriquadric curve, the intersection of the cylinder  $x^2 + y^2 = 1$  with the parabolic hyperboloid  $z = x^2 - y^2$ ; the cylinder regarded as a cone having its vertex at infinity on the line  $x = 0$ ,  $y = 0$ , viz. the vertex is on the hyperboloid, or the curve is a nodal quadriquadric (the node being thus an isolated point at infinity on the line in question), and the torse is consequently of the order  $8 - 2$ , = 6, viz. it is a sextic torse.

The edge is a bent oval situate on the cylinder  $x^2 + y^2 = 1$ , such that, regarding  $\phi$  as the azimuth (or angle measured along the circular base from its intersection with the axis of  $x$ ), the altitude  $z$  is given by the equation  $z = \cos 2\phi$ ; viz. there are in the plane  $xz$ , or, say in the planes  $xz$ ,  $x'z$ , two maxima altitudes  $z = 1$ , and in the plane  $yz$ , or, say in the planes  $yz$  and  $y'z$ , two minima altitudes  $z = -1$ . The sections by these principal planes are, as is seen at once, nodal curves on the surface; they are, in fact, the cubic curves  $z = 3 - \frac{2}{x^2}$ , viz. here as  $x$  increases from  $\pm 1$  to  $\pm\infty$   $z$  increases from the before-mentioned value 1 to 3, and  $z = -3 + \frac{2}{y^2}$ , viz. as  $y$  increases from  $\pm 1$  to  $\pm\infty$ ,  $z$  decreases from the before-mentioned value  $-1$  to  $-3$ . The two half-sheets (which meet in the cuspidal edge) intersect each other along these nodal lines, in such wise that the section of the surface by any axial plane (plane through the line  $x = 0$ ,  $y = 0$ ) is a curve having a cusp on the cuspidal edge, and such that when the axial plane coincides with either of the principal planes  $x = 0$ ,  $y = 0$ , the

(Page 69.) two half-branches of the curve coincide together with the portions which lie outside the cylinder  $x^2 + y^2 = 1$ , in fact, the portions referred to above, of the nodal curve in the plane in question; the portions which lie inside the cylinder are acnodal or isolated curves without any real sheet through them. It may be added, in the way of general description, that the section of the surface by any cylinder  $x^2 + y^2 = c^2$  ( $c > 1$ ) is a curve of the form  $z = C \cos(2\theta \pm B)$ ,  $\theta$  the angle along the base of the cylinder from the intersection with the axis of  $x$ ;  $C$ ,  $B$  are functions of  $c$ ; viz. we have for the two half sheets respectively

$$z = C \cos(2\theta + B) \quad \text{and} \quad z = C \cos(2\theta - B),$$

each curve having thus the two maxima  $+C$ , and the two minima  $-C$ ; and the two curves intersect each other at the four points in the two principal planes respectively; viz. the points for which  $\theta = 0, 90^\circ, 180^\circ, 270^\circ$ , and  $z = C \cos B, -C \cos B, C \cos B, -C \cos B$  accordingly.

Proceeding to discuss the surface analytically, we have for the equations of a generating line

$$\frac{x - \cos \phi}{-\sin \phi} = \frac{y - \sin \phi}{\cos \phi} = \frac{z - \cos 2\phi}{-2 \sin 2\phi} = \rho \text{ suppose,}$$

or say

$$\begin{aligned} x &= \cos \phi - \rho \sin \phi, \\ y &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi, \end{aligned}$$

which equations, considering therein  $\rho$ ,  $\phi$  as arbitrary parameters, determine the surface.

Writing  $x = 0$ , we find  $y = \frac{1}{\cos \phi}$ , and then  $z = -3 + 2 \sin^2 \phi$ , viz. we have

$$x = 0, \quad z = -3 + \frac{2}{y^2}, \text{ for section in plane } yz,$$

and, similarly, writing  $y = 0$ , we find  $x = \frac{1}{\cos \phi}$ , and then  $z = 3 - 2 \cos^2 \phi$ , viz.

$$y = 0, \quad z = 3 - \frac{2}{x^2}, \text{ for section by plane } xz.$$

By what precedes, these are nodal curves, crunodal for the portions

$$(y = \pm 1 \text{ to } \pm \infty, z = -1 \text{ to } -3) \text{ and } (x = \pm 1 \text{ to } \pm \infty, z = 1 \text{ to } 3)$$

respectively, acnodal for the remaining portions  $y < \pm 1, x < \pm 1$  respectively.

Writing  $x = r \cos \theta, y = r \sin \theta$ , so that the coordinates of a point on the surface are  $r, \theta, z$ , where  $r = \sqrt{x^2 + y^2}$  is the projected distance,  $\theta$  is the azimuth from the axis of  $x$ , and  $z$  is the altitude, we have

$$\begin{aligned} r \cos \theta &= \cos \phi - \rho \sin \phi, \\ r \sin \theta &= \sin \phi + \rho \cos \phi, \\ z &= \cos 2\phi - 2\rho \sin 2\phi. \end{aligned}$$

(Page 70.) We have  $r^2 = 1 + \rho^2$ ; and thence also, if  $\tan \alpha = 2\rho, = \pm 2\sqrt{r^2 - 1}$ , that is,

$$\cos \alpha = \frac{1}{\sqrt{4r^2 - 3}}, \quad \sin \alpha = \pm \frac{2\sqrt{r^2 - 1}}{\sqrt{4r^2 - 3}},$$

then

$$z = \sqrt{4r^2 - 3} \cos(2\phi + \alpha),$$

showing that for a given value of  $r$  (or section by the cylinder  $x^2 + y^2 = r^2$ ) the maximum and minimum values of  $z$  are  $z = \pm \sqrt{4r^2 - 3}$ .

But proceeding to eliminate  $\phi$ , we find

$$\begin{aligned} r^2 \cos 2\theta &= (1 - \rho^2) \cos 2\phi - 2\rho \sin 2\phi, \\ r^2 \sin 2\theta &= 2\rho \cos 2\phi + (1 - \rho^2) \sin 2\phi, \end{aligned}$$

or multiplying these by  $1 + \rho^2$  and  $2\rho$  and adding

$$r^2 \{(1 + 3\rho^2) \cos 2\theta + 2\rho^2 \sin 2\theta\} = (1 + \rho^2)^2 (\cos 2\phi - 2\rho \sin 2\phi),$$

that is,

$$r^2 \{(3r^2 - 2) \cos 2\theta \pm 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta\} = r^4 z;$$

or, finally,

$$r^2 z = (3r^2 - 2) \cos 2\theta + 2(r^2 - 1)^{\frac{1}{2}} \sin 2\theta,$$

which is the equation of the surface in terms of the coordinates  $r, \theta, z$ .

Observing that  $(3r^2 - 2)^2 + 4(r^2 - 1)^3 = r^4(4r^2 - 3)$ , we may write

$$\begin{aligned} r^2 \sqrt{4r^2 - 3} \cos \beta &= 3r^2 - 2, \\ r^2 \sqrt{4r^2 - 3} \sin \beta &= 2(r^2 - 1)^{\frac{3}{2}}, \end{aligned}$$

and therefore also

$$\tan \beta = \frac{2(r^2 - 1)^{\frac{3}{2}}}{3r^2 - 2},$$

and the equation thus becomes

$$z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta),$$

where  $z$  is the altitude belonging to the azimuth  $\theta$  in the cylindrical section, radius  $r$ . The maxima and minima altitudes are  $\pm\sqrt{(4r^2 - 3)}$ , and these correspond to the values  $\theta = \pm\frac{1}{2}\beta$ ,  $\frac{1}{2}\pi + \frac{1}{2}\beta$ ,  $\pi + \frac{1}{2}\beta$ ,  $\frac{3}{2}\pi + \frac{1}{2}\beta$ ; it is to be further noticed that when  $r = 1$ , we have  $\beta = 0$ , but as  $r$  increases and becomes ultimately infinite,  $\beta$  increases to  $\frac{1}{2}\pi$ , that is,  $\frac{1}{2}\beta$  increases from 0 to  $\frac{1}{4}\pi$ .

It may be noticed that the surface is a peculiar kind of deformation, obtained by giving proper rotations to the several cylindrical sections of the surface  $z = \sqrt{(4r^2 - 3)} \cos 2\theta$ ; viz. in rectangular coordinates this is  $r^2 z = \sqrt{(4r^2 - 3)}(x^2 - y^2)$ , that is,

$$(x^2 + y^2)^4 z^2 - \{4(x^2 + y^2) - 3\}(x^2 - y^2)^2 = 0.$$

To obtain the equation in rectangular coordinates, we have

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 16(r^2 - 1)^3 \sin^2 2\theta = 0,$$

viz. this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 \left(1 - \frac{4x^2 y^2}{r^4}\right) - 16(r^2 - 1)^3 \frac{4x^2 y^2}{r^4} = 0,$$

(Page 71.) or, what is the same thing, it is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - \frac{4x^2 y^2}{r^4} \{4(r^2 - 1)^3 + (3r^2 - 2)^2\} = 0,$$

viz. the term in  $\{ \}$  being  $r^4(4r^2 - 3)$ , this is

$$r^4 z^2 - 2z(3r^2 - 2)(x^2 - y^2) + (3r^2 - 2)^2 - 4x^2 y^2(4r^2 - 3) = 0,$$

or say

$$z^2(x^2 + y^2)^2 - 2z(3x^2 + 3y^2 - 2)(x^2 - y^2) + (3x^2 + 3y^2 - 2)^2 - 4x^2 y^2(4x^2 + 4y^2 - 3) = 0.$$

This may also be written

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2 y^2(z^2 - 4x^2 - 4y^2 + 3) = 0,$$

a form which puts in evidence the nodal curves

$$x = 0, \quad zy^2 = -3y^2 + 2, \quad \text{and} \quad y = 0, \quad zx^2 = 3x^2 - 2.$$

It shows also that the quadric cone  $z^2 - 4x^2 - 4y^2 + 3 = 0$  touches the surface along the curve of intersection with the surface  $z(x^2 - y^2) - 3(x^2 + y^2) + 2 = 0$ . This is, in fact, the curve of maxima and minima of the cylindrical sections, viz. reverting to the form  $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$ , or, if for greater clearness, attending only to one sheet of the surface, we write it  $z = \sqrt{(4r^2 - 3)} \cos(2\theta - \beta)$ , we have a maximum,  $z = \sqrt{(4r^2 - 3)}$ , for  $2\theta = \beta$  (or  $2\pi + \beta$ ), giving

$$\cos 2\theta = \cos \beta = \frac{3r^2 - 2}{r^2 \sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

and a minimum,  $z = -\sqrt{(4r^2 - 3)}$ , for  $2\theta = \pi + \beta$  (or  $3\pi + \beta$ ), giving

$$\cos 2\theta = -\cos \beta = -\frac{3r^2 - 2}{r^2\sqrt{(4r^2 - 3)}} = \frac{3r^2 - 2}{r^2 z},$$

viz. the locus is  $z^2 = 4(r^2 - 3)$ ,  $z(x^2 - y^2) = 3r^2 - 2$ ; and for  $z = \sqrt{(4r^2 - 3)} \cos(2\theta + \beta)$  we find the same locus, viz. the equations of the locus are

$$z^2 - 4x^2 - 4y^2 + 3 = 0, \quad z(x^2 - y^2) - 3x^2 - 3y^2 + 2 = 0,$$

as above.

To put in evidence the cuspidal edge, write for a moment  $\zeta = z - x^2 + y^2$ , the equation becomes  $\{\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 2) - 4x^2y^2\}^2 + 4x^2y^2\{\zeta^2 + 2\zeta(x^2 - y^2) + (r^2 - 1)(r^2 - 3) - 4x^2y^2\} = 0$ ; viz. this is

$$\zeta^2 r^4 + 2\zeta(x^2 - y^2)(r^2 - 1)(r^2 - 2) + (r^2 - 1)^2(r^2 - 2)^2 - 4\zeta^2 x^2 y^2 (r^2 - 1)^2 = 0,$$

or writing the last term thereof in the form

$$-[r^4 - (x^2 - y^2)^2](r^2 - 1)^2,$$

and then putting  $r^2 = 1 + U$ , the equation is

$$\zeta^2(1 + 2U + U^2) + 2\zeta U(U - 1)(x^2 - y^2) + U^2(U - 1)^2 - U^2\{(U + 1)^2 - (x^2 - y^2)^2\} = 0;$$

viz. this is

$$\{\zeta - U(x^2 - y^2)\}^2 + 2U\{\zeta^2 + \zeta U(x^2 - y^2) - 2U^2\} + 4U^2 = 0,$$

(Page 72.) showing the cuspidal edge  $\zeta = 0$ ,  $U = 0$ , viz.  $z = x^2 - y^2$ ,  $x^2 + y^2 = 1$ . Moreover, along the cuspidal edge the surface is touched by  $\zeta - U(x^2 - y^2) = 0$ , that is, by  $z - (x^2 - y^2) = 0$ ; and at the points where this tangent surface again meets the surface we have  $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$ ; viz. the surface contains upon itself the curve represented by this last equation, and  $z - (x^2 - y^2) = 0$ .

As a verification, in the form

$$\{z(x^2 - y^2) - 3x^2 - 3y^2 + 2\}^2 + 4x^2y^2(z^2 - 4x^2 - 4y^2 + 3) = 0$$

of the equation of the surface, write  $z = x^2 - y^2$ . If for a moment  $x^2 + y^2 = \lambda$ ,  $x^2 - y^2 = \mu$ , then the value of  $z$  is  $z = \lambda\mu$ , and the equation becomes

$$(\lambda\mu^2 - 3\lambda + 2)^2 + (\lambda^2 - \mu^2)(\lambda^2\mu^2 - 4\lambda + 3) = 0,$$

that is,

$$\mu^2(\lambda^3 - 6\lambda^2 + 8\lambda - 3) - 4\lambda^3 + 12\lambda^2 - 12\lambda + 4 = 0;$$

or, what is the same thing,

$$(\lambda - 1)^2\{\mu^2(\lambda + 3) - 4\} = 0,$$

so that we have  $(\lambda - 1)^2 = 0$ , or else  $\mu^2(\lambda + 3) - 4 = 0$ ; viz.  $(x^2 + y^2 - 1)^2 = 0$ , or else  $(x^2 - y^2)^2(x^2 + y^2 + 3) - 4 = 0$ , agreeing with the former result.

In polar coordinates, the surface is touched along the cuspidal edge by the surface  $z = r^2 \cos 2\theta$ , and where this again meets the surface we have  $r^4(r^2 + 3) \cos^2 2\theta - 4 = 0$ .

For the model, taking the unit to be 1 inch, I suppose that for the edge of regression we have

$$x = 2 \cos \phi, \quad y = 2 \sin \phi, \quad z = 5 + (.45) \cos 2\phi$$

viz. the curve is situate on a cylinder radius 2 inches. And I construct in zinc-plate the cylindric sections, or say the templets, for one sheet of the surface, for the several radii 2, 3, ..., 8 inches; taking the radius as  $k$  inches, the circumference of the cylinder, or entire base of the flattened templet, is  $= 2k\pi$ ; and the altitude, writing  $2\theta$  in place of  $2\theta - \beta$  as above, is given by the formula  $z = 5 + (.45)\sqrt{(k^2 - 3)}\cos 2\theta$ , so that the half altitude of the wave is  $= (.45)\sqrt{(k^2 - 3)}$ ; having this value, the curve is at once constructed geometrically. We have, moreover,  $\cos \beta = \frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$ ; the numerical values then are

| $k$ | $2k\pi$ | $(.45)\sqrt{(k^2 - 3)}$ | $\frac{3k^2 - 8}{k^2\sqrt{(k^2 - 3)}}$ | $\frac{1}{2}\beta$ |
|-----|---------|-------------------------|----------------------------------------|--------------------|
| 2   | 12.57   | 0.45                    | 1.00                                   | 0°                 |
| 3   | 18.85   | 1.10                    | .86                                    | 15                 |
| 4   | 25.13   | 1.62                    | .69                                    | 23                 |
| 5   | 31.42   | 2.11                    | .57                                    | 27 $\frac{1}{2}$   |
| 6   | 37.70   | 2.59                    | .48                                    | 30 $\frac{1}{2}$   |
| 7   | 43.98   | 3.05                    | .42                                    | 32 $\frac{1}{2}$   |
| 8   | 50.27   | 3.51                    | .36                                    | 34                 |

the altitudes in the successive templets being thus included between the limits  $5 \pm 0.45$ ,  $5 \pm 1.10$ , ...,  $5 \pm 3.51$ .

## 653.

## ON A TORSE DEPENDING ON THE ELLIPTIC FUNCTIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 235–241.]

(Page 73.) On attempting to cover with paper one half-sheet of the foregoing sextic torse, [652], I found that the paper assumed approximately the form of a circular annulus of an angle exceeding  $360^\circ$ , and this led me to consider the general theory of the construction of a torse in paper, and, in particular, to consider the torsos such that when developed into a plane the edge of regression becomes a circular arc. It is scarcely necessary to remark that, to construct in paper a circular annulus of an angle exceeding  $360^\circ$ , we have only to take a complete annulus, cut it along a radius, and then insert (gumming it on to the two terminal radii) a portion of an equal circular annulus; drawing from each point of the inner circular boundary a half-tangent, and considering these half-tangents as rigid lines, the paper will bend round them so as to form the half-sheet of a torse having for its edge of regression this inner boundary, which will assume the form of a closed curve with two equal and opposite maxima and two equal and opposite minima, described on a cylinder, and being approximately such as the curve given by the equations

$$x = \cos \theta, \quad y = \sin \theta, \quad z = m \cos 2\theta.$$

Considering, in general, an arc  $PQ$  (without inflexions) of any curve, and drawing at the consecutive points  $P, P', P'', \&c.$  the several half-tangents  $PT, P'T', P''T'', \dots$ , then, considering these as rigid lines and bending the paper round them, we have the half-sheet of a torse, having for its edge of regression the curve in question now bent into a double curvature. It is, moreover, clear that the edge of regression has at each point thereof the same radius of absolute curvature as the original plane curve; in fact, if in the plane curve  $PP' = ds$ , and the angle  $T'PT$  between the consecutive half-tangents  $PT$  and  $P'T'$  be  $= d\phi$ , these quantities  $ds$  and

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(Page 74.)  $d\phi$  remain unaltered in the curve of double curvature; and the radius of absolute curvature is given by the equation  $\rho d\phi = ds$ . In particular when, as above, the arc is a circular one, say of radius  $= a$ , then, however the paper is bent, the edge of regression has at each point thereof the radius of absolute curvature  $= a$ .

Consider on any given surface, at a given point  $P$  thereof, and in a given direction, an element of length  $PP'$ , then (under the restrictions presently mentioned) we can determine the consecutive element  $P'P''$ , such that the curve  $PP'P'' \dots$  shall have at  $P$  a radius of absolute curvature  $= a$ ; in fact,  $r$  being the radius of curvature of the normal section of the surface through the element  $PP'$ , the radius of curvature of the section inclined at an angle  $\theta$  to the normal section is  $= r \cos \theta$ ; so that we have only to take the section at the inclination  $\theta, = \cos^{-1} \frac{a}{r}$ , to the normal section, and we have the consecutive element  $P'P''$  such that the radius of absolute curvature of the curve  $PP'P''$  is  $= a$ . The necessary restriction, of course, is that  $r > a$ ; thus, if at the given point  $P$  the two principal radii of curvature are of the same sign (to fix the ideas, let the two principal radii and also  $a$  be each of them positive), then we may on the surface determine a direction  $PQ$ , for which the radius of curvature of the normal section is  $= a$ ; and then the direction of the element  $PP'$  may be any direction between  $PQ$  and the direction  $PR$ , corresponding to the greatest of the two principal radii.

Having obtained the element  $P'P''$ , we may, if the radius of absolute curvature at  $P'$  be given, construct the next element  $P''P'''$ , and so on; that is to say, on a given surface starting from a given point  $P$  and given initial direction  $PP'$ , we can (under a restriction, as above, as to the curvature at the different points of the surface) construct a curve having at the successive points thereof given values of the radius of absolute curvature; viz., the value may be given either as

a function of the coordinates of the point on the surface, or as a function of the length of the curve measured say from the initial point  $P$ ; it is in this last manner that in what follows the value of the radius of absolute curvature is assumed to be given.

We may thus, taking on paper an arc  $PQ$  with its half-tangents, apply it to a given surface, the point  $P$  to a given point, and the infinitesimal arc  $PP'$  to an element  $PP'$  in a given direction from the given point; and we thus obtain the half-sheet of a torse having for its edge of regression a determinate curve upon the surface. In particular, the arc  $PQ$  may be circular of the radius  $a$ , and the surface be a circular cylinder of radius  $\alpha$ ; and we thus obtain the torse having for edge of regression a curve on the cylinder radius  $\alpha$ , and such that the radius of absolute curvature is at each point  $= a$ . There are three cases according as  $a > \alpha$ ,  $a = \alpha$ , or  $a < \alpha$ ; it is to be remarked that if  $a > \alpha$ , then the curve must at each point cut the generating line of the cylinder at an angle not exceeding  $\cos^{-1} \frac{\alpha}{a}$ , but that in the other two cases the angle may have any value whatever; and, further, that in every case when the angle is  $= 0$ , viz. when the curve touches a generating line of the cylinder, then the osculating plane of the curve coincides with the tangent plane of the cylinder.

(Page 75.) The analytical theory is very simple. Taking  $x, y, z$  as functions of the length  $s$ , we have

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1;$$

the condition, which expresses that the radius of absolute curvature is  $= a$ , then is

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = \frac{1}{a^2}.$$

By what precedes, the point  $(x, y, z)$  may be taken to be upon a given surface, say upon the cylinder  $x^2 + y^2 = \alpha^2$ ; we may then write  $x = \alpha \cos \theta$ ,  $y = \alpha \sin \theta$ . Taking instead of  $s$  any independent variable  $u$  whatever, and using accents to denote the derived functions in regard to  $u$ , the equations become

$$\begin{aligned} x'^2 + y'^2 + z'^2 &= s'^2, \\ x''^2 + y''^2 + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \\ x &= \alpha \cos \theta, \quad y = \alpha \sin \theta. \end{aligned}$$

From the last two equations we obtain

$$x'^2 + y'^2 = \alpha^2 \theta'^2, \quad x''^2 + y''^2 = \alpha^2 (\theta'^4 + \theta''^2),$$

and the first two equations thus become

$$\begin{aligned} \alpha^2 \theta'^2 + z'^2 &= s'^2, \\ \alpha^2 \theta'^4 + \alpha^2 \theta''^2 + z''^2 - s''^2 &= \frac{1}{a^2} s'^4, \end{aligned}$$

and from the first of these we find

$$s'' = \frac{\alpha^2 \theta' \theta'' + z' z''}{(\alpha^2 \theta'^2 + z'^2)^{\frac{1}{2}}},$$

whence the second equation is

$$\alpha^2 (\theta'^4 + \theta''^2) + z''^2 - \frac{(\alpha^2 \theta' \theta'' + z' z'')^2}{\alpha^2 \theta'^2 + z'^2} = \frac{1}{a^2} (\alpha^2 \theta'^2 + z'^2)^2,$$



or reducing, this is

$$(\alpha^2 \theta'^2 + z'^2)(\theta''^2 + \theta'^4) + (z''^2 \theta'^2 - 2\theta' \theta'' z' z'' - \alpha^2 \theta'^2 \theta''^2) = \frac{1}{a^2}(\alpha^2 \theta'^2 + z'^2)^3.$$

Taking  $\theta$  as the independent variable, we have  $\theta' = 1$ ,  $\theta'' = 0$ , and the equation becomes

$$(\alpha^2 + z'^2) + z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3,$$

or, what is the same thing,

$$z''^2 = \frac{1}{a^2}(\alpha^2 + z'^2)^3 - (\alpha^2 + z'^2)^2.$$

Write here

$$\alpha^2 + z'^2 = \Omega^2,$$

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(Page 76.) then

$$z'' = \frac{\Omega \Omega'}{\sqrt{(\Omega^2 - \alpha^2)}},$$

and the equation becomes

$$\frac{\Omega'^2}{\Omega^2 - \alpha^2} = \frac{\Omega^4}{a^2 \alpha^2} - 1,$$

or say

$$\frac{a\alpha d\Omega}{\sqrt{(\Omega^2 - \alpha^2) \cdot \Omega^4 - a^2 \alpha^2}} = d\theta,$$

viz. this equation determines  $\Omega$  as a function of  $\theta$ , and we then have

$$\begin{cases} ds = \Omega d\theta, \\ dz = \sqrt{(\Omega^2 - \alpha^2)} d\theta, \\ x = \alpha \cos \theta, \\ y = \alpha \sin \theta, \end{cases}$$

equations which determine  $x, y, z, s$  as functions of the parameter  $\theta$ , and give thus the edge of regression of the torse in question.

It is clear that the formulae are very much simplified in the case  $a = \alpha$ , where the radius of absolute curvature  $a$  is equal to the radius  $\alpha$  of the cylinder; but it is worth while to develop the general case somewhat further.

Considering the elliptic functions  $\text{sn } u, \text{cn } u, \text{dn } u$ , to the modulus  $k (= k') = \frac{1}{\sqrt{2}}$ , assume

$$\Omega = \frac{\sqrt{(a\alpha)} \text{dn } u}{k \text{sn } u},$$

then

$$\begin{aligned} d\Omega &= -\frac{\sqrt{(a\alpha)} \text{cn } u \text{dn } u}{k \text{sn}^2 u}, \\ \Omega^2 - \alpha^2 &= \frac{a\alpha}{k^2 \text{sn}^2 u} \left( \text{dn}^2 u - \frac{\alpha}{a} k^2 \text{sn}^2 u \right), \\ &= \frac{a\alpha}{k^2 \text{sn}^2 u} \left\{ 1 - \left( 1 + \frac{\alpha}{a} \right) k^2 \text{sn}^2 u \right\}, \\ \Omega^4 - a^2 \alpha^2 &= \frac{a^2 \alpha^2}{k^4 \text{sn}^4 u} (\text{dn}^4 u - k^4 \text{sn}^4 u), \end{aligned}$$

$$= \frac{a^2 \alpha^2}{k^4 \operatorname{sn}^4 u} (1 - 2k^2 \operatorname{sn}^2 u), \quad = \frac{a^2 \alpha^2}{k^4 \operatorname{sn}^4 u} \operatorname{cn}^2 u,$$

and hence

$$d\theta = \frac{k^2 \operatorname{sn} u \, du}{\sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \operatorname{sn}^2 u\}}},$$

$$dz = \frac{k \sqrt{(a\alpha)} \, \operatorname{dn} u \, du}{\sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \operatorname{sn}^2 u\}}},$$

(Page 77.) We have thus  $z = k\sqrt{(a\alpha)}u$ , no constant of integration being required, viz.  $u$  is a mere constant multiple of  $z$ : and the first and second equations then give  $s$  and  $\theta$  as functions of  $u$ , that is, of  $z$ ; but it is obviously convenient to retain  $u$  instead of expressing it in terms of  $z$ . As regards the form of these integrals observe that, writing  $\operatorname{sn} u = \lambda$ , we have

$$du = \frac{d\lambda}{\operatorname{dn} u \cdot \operatorname{cn} u},$$

and thence

$$d\theta = \frac{k^2 \lambda \, d\lambda}{\operatorname{dn} u \cdot \operatorname{cn} u \cdot \sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \lambda^2\}}},$$

$$ds = \frac{k \sqrt{(a\alpha)} \, d\lambda}{\operatorname{cn} u \cdot \sqrt{\{1 - (1 + \frac{\alpha}{a})k^2 \lambda^2\}}},$$

each of which is in fact reducible to elliptic integrals, but I do not further pursue this general case.

In the particular case  $a = \alpha$ , we have

$$1 - \left(1 + \frac{\alpha}{a}\right) k^2 \operatorname{sn}^2 u = \operatorname{cn}^2 u,$$

and the equations become

$$d\theta = \frac{k^2 \operatorname{sn} u \, du}{\operatorname{cn} u}, \quad ds = \frac{k\alpha \operatorname{dn} u \, du}{\operatorname{cn} u},$$

which admit of immediate integration; viz. we have

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'},$$

or determining the constant so that  $\theta$  may vanish for  $u = 0$ , say

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right),$$

and

$$s = \frac{1}{2} k\alpha \log \left( \frac{1 + \operatorname{sn} u}{1 - \operatorname{sn} u} \right);$$

viz. to verify these results we have

$$\frac{d\theta}{du} = \frac{1}{2} \frac{k^2}{k'} \cdot \frac{[\operatorname{dn} u + k']'(\operatorname{dn} u - k') - (\operatorname{dn} u + k')[\operatorname{dn} u - k']'}{\operatorname{dn}^2 u - k'^2} = -\frac{k^2 \operatorname{sn} u \operatorname{cn} u}{\operatorname{dn}^2 u - k'^2} = k^2 \frac{\operatorname{sn} u}{\operatorname{cn} u},$$

and

$$\frac{ds}{du} = \frac{1}{2} k\alpha \left( \frac{1}{1 + \operatorname{sn} u} + \frac{1}{1 - \operatorname{sn} u} \right) \cdot \operatorname{cn} u \operatorname{dn} u = \frac{k\alpha \operatorname{cn} u \operatorname{dn} u}{1 - \operatorname{sn}^2 u} = \frac{k\alpha \operatorname{dn} u}{\operatorname{cn} u}.$$

(Page 78.) Hence, recurring to the original equations, and writing for convenience  $a = \alpha = 1$ , we see that a solution of the simultaneous equations

$$\left(\frac{dx}{ds}\right)^2 + \left(\frac{dy}{ds}\right)^2 + \left(\frac{dz}{ds}\right)^2 = 1,$$

$$\left(\frac{d^2x}{ds^2}\right)^2 + \left(\frac{d^2y}{ds^2}\right)^2 + \left(\frac{d^2z}{ds^2}\right)^2 = 1,$$

is

$$x = \cos \theta, \quad y = \sin \theta, \quad z = ku,$$

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right), \quad s = \frac{1}{2} k \log \left( \frac{1 + \operatorname{sn} u}{1 - \operatorname{sn} u} \right),$$

where, as before,  $k = k' = \frac{1}{\sqrt{2}}$ .

Restoring the radius  $\alpha$ , and writing the system in the form

$$x = \alpha \cos \theta, \quad y = \alpha \sin \theta, \quad z = k\alpha u,$$

$$\theta = \frac{1}{2} \frac{k^2}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right), \quad s = \frac{1}{2} k\alpha \log \left( \frac{1 + \operatorname{sn} u}{1 - \operatorname{sn} u} \right),$$

we see that, as  $u$  passes from  $u = 0$  to  $u = K$ , and therefore  $z$  from  $z = 0$  to  $z = k\alpha K$  ( $K$  the complete function  $\int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}$ ), then  $\theta$  and  $s$  each pass from 0 to  $\infty$ ; and, similarly, as  $u$  passes from  $u = 0$  to  $u = -K$ , that is, as  $z$  passes from 0 to  $-k\alpha K$ , then  $\theta$  passes from 0 to  $\infty$ , and  $s$  from  $s = 0$  to  $s = -\infty$ ; viz. the curve makes in each direction an infinity of revolutions about the cylinder. Developing the cylinder,  $\alpha\theta$  becomes an  $x$ -coordinate; viz. we have thus the plane curve

$$z = k\alpha u, \quad x = \frac{1}{2} \frac{k^2 \alpha}{k'} \log \left( \frac{\operatorname{dn} u + k'}{\operatorname{dn} u - k'} \cdot \frac{1 - k'}{1 + k'} \right),$$

which is a curve extending from the origin in the direction  $x$  positive, to touch at infinity the two parallel asymptotes  $z = \pm k\alpha K$ ; and conversely, when such a plane curve is wound about the cylinder, there will be in each direction an infinity of revolutions round the cylinder.

## 654.

## ON CERTAIN OCTIC SURFACES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 249–264.]

(Page 79.) I consider the torse generated by the tangents of the quadriquadric curve, the intersection of the two quadric surfaces

$$\begin{aligned} ax^2 + by^2 + cz^2 + dw^2 &= 0, \\ a'x^2 + b'y^2 + c'z^2 + d'w^2 &= 0; \end{aligned}$$

then, writing

$$\begin{aligned} bc' - b'c &= a', & ad' - a'd &= f', \\ ca' - c'a &= b', & bd' - b'd &= g', \\ ab' - a'b &= c', & cd' - c'd &= h', \end{aligned}$$

and therefore

$$a'f' + b'g' + c'h' = 0,$$

the equation of the torse, writing for greater convenience  $(a, b, c, f, g, h)$  in place of  $(a', b', c', f', g', h')$ , but understanding these letters as signifying the accented letters  $(a', b', c', f', g', h')$ , is

$$\begin{aligned} &a^2 f^2 y^2 z^4 + b^2 g^2 z^2 x^4 + c^2 h^2 x^2 y^4 \\ &+ a^2 f^2 z^2 w^4 + b^2 g^2 x^2 w^4 + c^2 h^2 y^2 w^4 \\ &+ 2b^2 d' g h x^2 y^2 z^2 - 2c^2 f^2 a h x^4 y^2 w^2 + b^2 f^2 a g x^4 z^2 w^2 \\ &+ 2a^2 h^2 f g z^2 w^2 x^2 - 2a^2 h f b f y^2 z^2 w^2 + 2c^2 g^2 b h f x^2 w^2 \\ &+ 2a^2 g^2 f g z^2 w^2 x^2 - 2b^2 h^2 c g z^2 x^2 w^2 + 2a^2 h^2 c f z^2 y^2 w^2 \\ &- 2b c g^2 h^2 z^2 w^2 x^2 - 2c a h^2 f^2 z^2 w^2 x^2 - 2a b f^2 g^2 w^4 x^2 y^2 \\ &+ 2(bg - ch)(ch - af)(af - bg)x^2 y^2 z^2 w^2 = 0. \end{aligned}$$

(Page 80.) If in this equation we write

$$\begin{aligned} a^2 f^2 &= a, & b^2 g^2 &= b, & c^2 h^2 &= c, \\ a'^2 f'^2 &= f, & b'^2 g'^2 &= g, & c'^2 h'^2 &= h; \end{aligned}$$

and therefore

$$\begin{aligned} bc' - b'c &= \frac{a}{\sqrt{af}}, & ad' - a'd &= \frac{f}{\sqrt{af}}, \\ ca' - c'a &= \frac{b}{\sqrt{bg}}, & bd' - b'd &= \frac{g}{\sqrt{bg}}, \\ ab' - a'b &= \frac{c}{\sqrt{ch}}, & cd' - c'd &= \frac{h}{\sqrt{ch}}, \end{aligned}$$

and consequently

$$(af)^{\frac{1}{2}} + (bg)^{\frac{1}{2}} + (ch)^{\frac{1}{2}} = 0;$$

then the equation becomes

$$\begin{aligned}
 & a^2y^2z^4 + b^2z^2x^4 + c^2x^2y^4 \\
 & + f^2x^2w^4 + g^2y^2w^4 + h^2z^2w^4 \\
 & + 2bcx^4y^2z^2 - 2cfx^4y^2w^2 + 2bf x^4z^2w^2 \\
 & + 2cay^4z^2x^2 - 2agy^4z^2w^2 + 2cgy^4x^2w^2 \\
 & + 2ahz^4x^2y^2 - 2bh z^4x^2w^2 + 2ahz^4y^2w^2 \\
 & - 2ghw^4y^2z^2 - 2hfw^4z^2x^2 - 2fgw^4x^2y^2 \\
 & + 2\{(bg)^{\frac{1}{2}} - (ch)^{\frac{1}{2}}\}\{(ch)^{\frac{1}{2}} - (af)^{\frac{1}{2}}\}\{(af)^{\frac{1}{2}} - (ch)^{\frac{1}{2}}\}x^2y^2z^2w^2 = 0.
 \end{aligned}$$

This same equation, without the relation

$$(af)^{\frac{1}{2}} + (bg)^{\frac{1}{2}} + (ch)^{\frac{1}{2}} = 0,$$

and with an arbitrary coefficient for  $x^2y^2z^2w^2$ ; or say, the equation

$$\begin{aligned}
 & a^2y^2z^4 + b^2z^2x^4 + c^2x^2y^4 \\
 & + f^2x^2w^4 + g^2y^2w^4 + h^2z^2w^4 \\
 & + 2bcx^4y^2z^2 - 2cfx^4y^2w^2 + 2bf x^4z^2w^2 \\
 & + 2cay^4z^2x^2 - 2agy^4z^2w^2 + 2cgy^4x^2w^2 \\
 & + 2ahz^4x^2y^2 - 2bh z^4x^2w^2 + 2ahz^4y^2w^2 \\
 & - 2ghw^4y^2z^2 - 2hfw^4z^2x^2 - 2fgw^4x^2y^2 \\
 & + 2k x^2y^2z^2w^2 = 0,
 \end{aligned}$$

where  $a, b, c, f, g, h, k$  are arbitrary coefficients, is the general equation of an octic surface having the four nodal curves

$$\begin{aligned}
 x = 0, & \quad hz^2w^2 - gw^2y^2 + ay^2z^2 = 0, \\
 y = 0, & \quad -hz^2w^2 + fw^2x^2 + bz^2x^2 = 0, \\
 z = 0, & \quad gw^2x^2 - fw^2y^2 + cx^2y^2 = 0, \\
 w = 0, & \quad -ay^2z^2 - bz^2x^2 - cx^2y^2 = 0.
 \end{aligned}$$

# The Collected Mathematical Papers of Arthur Cayley

Vol. X, Pages 101–125

Papers 654 (conclusion), 655 (commencement)

Arthur Cayley

Cambridge, 1896

## [654] ON CERTAIN OCTIC SURFACES (CONCLUDED).

(Page 81.) In fact, the equation of the surface may be written in the form

$$\begin{aligned} w^8 + \{f^2x^4 + g^2y^4 + h^2z^4 - 2ghy^2z^2 - 2hfx^2z^2 - 2fgx^2y^2\} w^6 \\ + 2w^4 \left\{ \begin{aligned} &-cfx^2y^2 - agy^2z^2 - bhx^2z^2 + 2kx^2y^2z^2 \\ &+ bfx^2z^2 + cgy^2x^2 + ahx^2y^2 \end{aligned} \right\} \\ + (ay^2z^2 + bz^2x^2 + cx^2y^2)^2 = 0, \end{aligned}$$

which puts in evidence the nodal curve

$$w = 0, \quad -ay^2z^2 - bz^2x^2 - cx^2y^2 = 0 :$$

there are three similar forms which put in evidence the other three nodal curves.

The four curves are so related to each other that every line which meets three of them meets also the fourth curve; there is consequently a singly infinite series of lines meeting each of the four curves; these break up into four series of lines each forming an octic scroll, and each scroll has the four curves for nodal curves respectively; that is, each scroll is a surface included under the foregoing general equation, and derived from it by assigning a proper value to the constant  $k$ . To determine these values, write

$$\begin{aligned} \lambda + \mu + \nu &= 0, \\ \frac{\lambda}{af} + \frac{\mu}{bg} + \frac{\nu}{ch} &= 0, \\ \frac{k - \mu}{af} + \frac{\nu - \lambda}{bg} + \frac{\mu - \nu}{ch} &= 0, \end{aligned}$$

equations which give four systems of values for the ratios  $(\lambda : \mu : \nu)$ . We have then

$$k = af \cdot \frac{\nu - \mu}{\lambda} + bg \cdot \frac{\lambda - \nu}{\mu} + ch \cdot \frac{\mu - \lambda}{\nu},$$

viz.  $k$  has four values corresponding to the several values of  $(\lambda : \mu : \nu)$ .

The scroll in question is M. De La Gournerie's scroll  $\Sigma_1$ ; the equation of the scroll  $\Sigma_1$  is consequently obtained from the octic equation by writing therein the last-mentioned value of  $k$ .

It is to be noticed that  $k$  is, in effect, determined by a quartic equation; and, that, for a certain relation between the coefficients, this equation will have a twofold root. Assuming that this relation is satisfied, and assigning to  $k$  its twofold value, the resulting scroll becomes a torse; that is, two of the four scrolls coincide together

and degenerate into a torse; corresponding to the remaining two values of  $k$  we have two scrolls, companions of the torse. In order to a twofold value of  $k$ , we must have

$$\frac{af}{\lambda^2} = \frac{bg}{\mu^2} = \frac{ch}{\nu^2},$$

and thence

$$(af)^{\frac{1}{3}} + (bg)^{\frac{1}{3}} + (ch)^{\frac{1}{3}} = 0;$$

or, what is the same thing,

$$(af + bg + ch)^3 - 27abcfgh = 0.$$

If for a moment we write  $af = \alpha^3$ ,  $bg = \beta^3$ ,  $ch = \gamma^3$ , and, therefore,  $\alpha + \beta + \gamma = 0$ ; then for the twofold root, we have  $\lambda : \mu : \nu = \alpha : \beta : \gamma$ , and consequently

$$k = \alpha^2(\gamma - \beta) + \beta^2(\alpha - \gamma) + \gamma^2(\beta - \alpha) = (\alpha - \beta)(\beta - \gamma)(\gamma - \alpha),$$

that is,

$$k = \left\{ (af)^{\frac{1}{3}} - (bg)^{\frac{1}{3}} \right\} \left\{ (bg)^{\frac{1}{3}} - (ch)^{\frac{1}{3}} \right\} \left\{ (ch)^{\frac{1}{3}} - (af)^{\frac{1}{3}} \right\},$$

which agrees with the result in regard to the octic torse.

If in the octic equation we write  $(x, y, z, w)$  in place of  $(x^2, y^2, z^2, w^2)$ , then we have the quartic equation

$$\begin{aligned} & f^2x^2w^2 + g^2y^2w^2 + h^2z^2w^2 \\ & + 2hcx^2yz - 2cfx^2yw + 2bf x^2zw \\ & + 2ahy^2zx - 2agy^2zw + 2cgy^2xw \\ & + 2abz^2xy - 2bh z^2xw + 2ahz^2yw \\ & - 2ghw^2yz - 2hf w^2zx - 2fgw^2xy \\ & + 2kxyzw = 0, \end{aligned}$$

which is the equation of a quartic surface touched by the planes  $x = 0$ ,  $y = 0$ ,  $z = 0$ ,  $w = 0$ , in the four conics

$$\begin{aligned} x = 0, & \quad hzw - gwy + ayz = 0, \\ y = 0, & \quad -hzw + fwx + bzx = 0, \\ z = 0, & \quad gyw - fwx + cxy = 0, \\ w = 0, & \quad -ayz - bzx - cxy = 0, \end{aligned}$$

respectively.

## II. The octic surface

$$\begin{aligned} U = & b^2c^2f^2x^4 + c^2a^2g^2y^4 + a^2b^2h^2z^4 + f^2g^2h^2w^4 \\ & - 2a^2cg(bg - ch)y^2z^2 - 2b^2ah(ch - af)z^2x^2 - 2c^2bf(af - bg)x^2y^2 \\ & + 2a^2bh(\cdot)y^2z^2 + 2b^2cf(\cdot)z^2x^2 + 2c^2ag(\cdot)x^2y^2 \\ & - 2f^2bc(\cdot)x^2w^2 - 2g^2ca(\cdot)y^2w^2 - 2h^2ab(\cdot)z^2w^2 \\ & + 2f^2gh(\cdot)x^2w^2 + 2g^2hf(\cdot)y^2w^2 + 2h^2fg(\cdot)z^2w^2 \\ & + f^2(b^2g^2 + c^2h^2 - 4bgch)w^2x^2 + g^2(c^2h^2 + a^2f^2 - 4chaf)w^2y^2 + h^2(a^2f^2 + b^2g^2 - 4afbg)w^2z^2 \\ & + a^2(\cdot)y^2z^2 + b^2(\cdot)z^2x^2 + c^2(\cdot)x^2y^2 \\ & - 2gh(bcgh - a^2f^2 - 2afbg - 2afch)w^2y^2z^2 \\ & - 2bh(\cdot)z^2x^2 \\ & + 2cg(\cdot)y^2x^2w^2 \\ & + 2bc(\cdot)x^2y^2z^2 \end{aligned}$$

$$\begin{aligned}
 & -2hf(cahf - b^2g^2 - 2bgaf - 2bgch)w^2z^2x^2 \\
 & -2cf(\cdot)x^2y^2 \\
 & +2ah(\cdot)z^2y^2w^2 \\
 & +2ca(\cdot)y^2x^2z^2 \\
 & -2fg(abfg - c^2h^2 - 2chaf - 2chbg)w^2x^2y^2 \\
 & +2ag(\cdot)y^2z^2w^2 \\
 & +2bf(\cdot)x^2z^2w^2 \\
 & +2ab(\cdot)z^2x^2y^2 \\
 & +2\Omega x^2y^2z^2w^2 = 0,
 \end{aligned}$$

where the values of the coefficients indicated by  $(\cdot)$  are at once obtained by the proper interchanges of the letters, and where  $\Omega$  is an arbitrary coefficient, is a surface having the four nodal conics

$$\begin{aligned}
 x = 0, \quad & cy^2 - bz^2 + fw^2 = 0, \\
 y = 0, \quad & -cx^2 + az^2 + gw^2 = 0, \\
 z = 0, \quad & bx^2 - ay^2 + hw^2 = 0, \\
 w = 0, \quad & -fx^2 - gy^2 - hz^2 = 0.
 \end{aligned}$$

In fact, writing the equation under the form

$$w^4\Theta + (fx^2 + gy^2 + hz^2)^2 \times (b^2c^2x^4 + c^2a^2y^4 + a^2b^2z^4 - 2a^2bcy^2z^2 - 2b^2caz^2x^2 - 2c^2abx^2y^2) = 0,$$

we put in evidence the nodal conic  $w = 0, fx^2 + gy^2 + hz^2 = 0$ : and similarly for the other nodal conics.

It is to be observed, that the complete section by the plane  $w = 0$  is the conic  $fx^2 + gy^2 + hz^2 = 0$ , twice repeated, and the quartic

$$b^2c^2x^4 + c^2a^2y^4 + a^2b^2z^4 - 2a^2bcy^2z^2 - 2ab^2cz^2x^2 - 2abc^2x^2y^2 = 0 :$$

the latter being the system of four lines

$$\frac{x}{\sqrt{(a)}} \pm \frac{y}{\sqrt{(b)}} \pm \frac{z}{\sqrt{(c)}} = 0.$$

The plane in question,  $w = 0$ , meets the other nodal conics in the six points

$$(x = 0, bz^2 - cy^2 = 0), \quad (y = 0, cx^2 - az^2 = 0), \quad (z = 0, ay^2 - bx^2 = 0),$$

which six points are the angles of the quadrilateral formed by the above-mentioned four lines.

The four conics are such, that every line meeting three of these conics meets also the fourth conic. The lines in question form a double system: each of these



systems has, in reference to any pair of nodal conics, a homographic property as follows: viz. considering for example the two conics in the planes  $z = 0$  and  $w = 0$  respectively, if a line meets these conics in the points  $P$  and  $Q$  respectively, and through these points respectively and the line  $x = 0, y = 0$  we draw planes, then the system of the  $P$  planes and the system of the  $Q$  planes correspond homographically to each other, the coincident planes of the two systems being the planes  $x = 0$  and  $y = 0$  respectively.

Conversely, if through the line  $(x = 0, y = 0)$  we draw the two homographically related planes meeting the two conics in the points  $P$  and  $Q$  respectively, then, for a proper value (determined by a quadratic equation) of the constant  $k (= \theta' \div \theta)$  which determines the homographic relation, the line  $PQ$  will be a line meeting each of the four conics, and will belong to one or other of the above-mentioned two systems, as  $k$  is equal to one or the other of the two roots of the quadratic equation. The scroll generated by the lines meeting each of the four conics, or what is the same thing, any three of these conics, is *primâ facie* a scroll of the order 16; but by what precedes, it appears that this scroll breaks up into two scrolls, which will be each of the order 8. Moreover, each scroll has the four conics for nodal curves; and since the equation  $U = 0$  is the general equation of an octic surface having the four conics for nodal curves, it follows, that the equation of the scroll is derived from that of the octic surface  $U = 0$ , by assigning a proper value to the indeterminate coefficient  $\Omega$ ; so that there are in fact two values of  $\Omega$ , for each of which the surface  $U = 0$  becomes a scroll.

To sustain the foregoing conclusions, take  $x = \theta y, x = \theta' y$  for the equations of the two planes through the line  $(x = 0, y = 0)$ , which meet the  $z$ -conic and  $w$ -conic in the points  $P$  and  $Q$  respectively; then the equations of the line  $PQ$  are

$$\begin{aligned} \sqrt{(f\theta^2 + g)}(x - \theta'y) + \sqrt{(-h)}(\theta' - \theta)z &= 0, \\ -\sqrt{(b\theta'^2 - a)}(x - \theta y) + \sqrt{(-h)}(\theta' - \theta)w &= 0, \end{aligned}$$

or, writing therein  $\theta' = k\theta$ , the equations are

$$\begin{aligned} \sqrt{(f\theta^2 + g)}(x - k\theta y) + \sqrt{(-h)}(k - 1)\theta z &= 0, \\ -\sqrt{(bk^2\theta^2 - a)}(x - \theta y) + \sqrt{(-h)}(k - 1)\theta w &= 0. \end{aligned}$$

To find where the line in question meets the plane  $y = 0$ , we have

$$\begin{aligned} \sqrt{(f\theta^2 + g)}x + \sqrt{(-h)}(k - 1)\theta z &= 0, \\ -\sqrt{(bk^2\theta^2 - a)}x + \sqrt{(-h)}(k - 1)\theta w &= 0, \end{aligned}$$

and thence

$$\begin{aligned} (f\theta^2 + g)x^2 + h(k - 1)^2\theta^2 z^2 &= 0, \\ (bk^2\theta^2 - a)x^2 + h(k - 1)^2\theta^2 w^2 &= 0, \end{aligned}$$

or multiplying by  $a, g$  and adding

$$(af + bgk^2)x^2 + h(k - 1)^2(az^2 + gw^2) = 0,$$

or assuming

$$af + bgk^2 + ch(k - 1)^2 = 0,$$

the equation is

$$-cx^2 + az^2 + gw^2 = 0.$$

That is,  $k$  being determined by the quadric equation  $af + bgk^2 + ch(k-1)^2 = 0$ , the line  $PQ$  meets the  $y$ -conic  $y = 0$ ,  $-cx^2 + az^2 + gw^2 = 0$ ; and, in a similar manner, it appears that the line  $PQ$  also meets the  $x$ -conic  $x = 0$ ,  $cy^2 - bz^2 + fw^2 = 0$ .

Writing for greater symmetry  $1 : -k : k-1 = \lambda : \mu : \nu$ , we have

$$\lambda + \mu + \nu = 0,$$

$$af\lambda^2 + bg\mu^2 + ch\nu^2 = 0,$$

so that there are two systems of values of  $(\lambda, \mu, \nu)$  corresponding to, and which may be used in place of, the two values of  $k$  respectively.

Starting now from the equations

$$(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2 = 0,$$

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 = 0,$$

the elimination of  $\theta$  from these equations leads to an equation  $U = 0$ , of the above-mentioned form but with a determinate value of the coefficient.

The process, although a long one, is interesting and I give it in some detail.

*Elimination of  $\theta$  from the foregoing equations.*

We have

$$U = M \Pi [(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2],$$

where  $\Pi$  denotes the product of the expressions corresponding to the four roots of the equation

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 = 0.$$

Observing that this equation does not contain  $z$ , and that the expression under the sign  $\Pi$  does not contain  $w$ , it is at once seen that the product  $\Pi$  is in regard to  $(z, w)$  a rational and integral function of the form  $(z^2, w^2)^4$ ; and since, in regard to  $(z, w)$ ,  $U$  is also a rational and integral function of the same form  $(z^2, w^2)^4$ , it is clear that the factor  $M$  does not contain  $z$  or  $w$ , but is a function of only  $(x, y)$ . To determine it we may write  $z = 0$ ,  $w = 0$ : this gives

$$c^4(bk^2\theta^2 - ay^2)^4(fx^2 + gy^2)^4 = M h^4(f\theta^2 + g)^4(k\theta y - x)^8,$$

where

$$(bk^2\theta^2 - a)(\theta y - x)^2 = 0,$$

and the values of  $\theta$  are therefore  $+\frac{x}{y}$ ,  $-\frac{x}{y}$ ,  $+\frac{\sqrt{(a)}}{k\sqrt{(b)}}$ ,  $-\frac{\sqrt{(a)}}{k\sqrt{(b)}}$ . Hence substituting and observing that

$$ch^2(k-1)^2 = (af + bgk^2)^2,$$

it is easy to find

$$M = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4},$$

that is, we have

$$U = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4} \Pi [(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2z^2],$$

where

$$(bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2w^2 = 0.$$

If for greater convenience we write  $\theta = \frac{x}{y}\phi$ , then this formula becomes

$$U = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4} \Pi [(fx^2\phi^2 + gy^2)(k\phi - 1)^2 + h(k-1)^2\phi^2z^2],$$

where

$$(bk^2x^2\phi^2 - ay^2)(\phi - 1)^2 + h(k-1)^2\phi^2w^2 = 0,$$

or, what is the same thing,

$$(bk^2x^2\phi^2 - ay^2)(\phi - 1)^2 + h(k-1)^2\phi^2w^2 = 0.$$

Suppose that the terms in  $U$  which contain  $z^2$  are  $= \Theta z^2$ ; then we have

$$\Theta z^2 = \frac{x^4}{y^4} \cdot \frac{h^4(k-1)^2}{b^4k^4} \Sigma' [\Pi' \{ (fx^2\phi^2 + gy^2)(k\phi - 1)^2 \} \cdot h(k-1)^2\phi^2z^2],$$

or, what is the same thing,

$$\Theta = \frac{x^4}{y^4} \cdot \frac{h^5(k-1)^4}{b^4k^4} \Sigma [\phi^2 \Pi' \{ (fx^2\phi^2 + gy^2)(k\phi - 1)^2 \}],$$

where  $\Pi'$  refers to the remaining three roots  $\phi_2, \phi_3, \phi_4$ ; this may also be written

$$\Theta = \frac{x^4}{y^4} \cdot \frac{h^5(k-1)^4}{b^4k^4} \Sigma \left\{ \frac{\phi^2}{(fx^2\phi^2 + gy^2)(k\phi - 1)^2} \right\} \cdot \Pi \{ (fx^2\phi^2 + gy^2)(k\phi - 1)^2 \}.$$

Hence, observing that we have identically

$$fx^2\phi^2 + gy^2 = \{\phi x\sqrt{(f)} + iy\sqrt{(g)}\}\{\phi x\sqrt{(f)} - iy\sqrt{(g)}\},$$

and writing  $\phi = \pm \frac{iy\sqrt{(g)}}{x\sqrt{(f)}}$ ,  $\{i = \sqrt{(-1)} \text{ as usual}\}$ , we find

$$\Pi\{\phi x\sqrt{(f)} \pm iy\sqrt{(g)}\} = -\frac{1}{b^2k^4x^4} [c\{x\sqrt{(f)} \pm iy\sqrt{(g)}\}^2 - fgv^2] y^2,$$

$$\Pi(k\phi - 1) = \frac{1}{k^2b^2x^4}(bx^2 - ay^2 + hw^2);$$

whence, writing for shortness

$$\begin{aligned} \Delta &= [c\{x\sqrt{(f)} + iy\sqrt{(g)}\}^2 - fgw^2] [c\{x\sqrt{(f)} - iy\sqrt{(g)}\}^2 - fgw^2] \\ &= c^2f^2x^4 + c^2g^2y^4 + f^2g^2w^4 + 2c f g^2 y^2 w^2 - 2c f g x^2 w^2 + 2c^2 f g x^2 y^2, \end{aligned}$$

we find

$$\begin{aligned}\Pi(fx^2\phi^2 + gy^2) &= \frac{h^2(k-1)^4}{b^4k^8} \cdot \frac{y^4}{x^8} \Delta, \\ \Pi(k\phi - 1)^2 &= \frac{1}{b^4k^8x^8} (bx^2 - ay^2 + hw^2)^2,\end{aligned}$$

and thence

$$\Pi(fx^2\phi^2 + gy^2)(k\phi - 1)^2 = \frac{h^2(k-1)^4}{b^8k^{16}} \Delta (bx^2 - ay^2 + hw^2)^2 \frac{y^4}{x^{16}},$$

and consequently

$$\Theta = \frac{h^7(k-1)^8}{b^{12}k^{20}} \cdot \frac{1}{x^{12}} \Delta (bx^2 - ay^2 + hw^2)^2 \Sigma \left\{ \frac{\phi^2}{(fx^2\phi^2 + gy^2)(k\phi - 1)^2} \right\}.$$

Hence, writing

$$\Sigma \left\{ \frac{\phi^2}{(fx^2\phi^2 + gy^2)(k\phi - 1)^2} \right\} = \frac{A}{(k\phi - 1)^2} + \frac{B}{k\phi - 1} + \frac{C}{x\phi\sqrt{(f)} + iy\sqrt{(g)}} + \frac{D}{x\phi\sqrt{(f)} - iy\sqrt{(g)}},$$

we may calculate separately the terms

$$\Sigma \left\{ \frac{A}{(k\phi - 1)^2} + \frac{B}{k\phi - 1} \right\}$$

and

$$\Sigma \left\{ \frac{C}{x\phi\sqrt{(f)} + iy\sqrt{(g)}} + \frac{D}{x\phi\sqrt{(f)} - iy\sqrt{(g)}} \right\}.$$

The first of these is

$$= \frac{1}{h(k-1)^2(fx^2 + k^2gy^2)^2} (bx^2 - ay^2 + hw^2)^2 \cdot \{x, y, w\}^2,$$

if for shortness

$$\begin{aligned}\{x, y, w\}^2 &= (fx^2 + k^2gy^2) [4\{(2-k)bx^2 - ay^2 + h(1-k)w^2\}^2 \\ &\quad - 2(bx^2 - ay^2 + hw^2)\{(6-6k+k^2)bx^2 - ay^2 + h(1-k)^2w^2\}] \\ &\quad + 4k^2(k-1)gy^2 - (bx^2 - ay^2 + hw^2)\{(2-k)bx^2 - ay^2 + h(1-k)w^2\};\end{aligned}$$

the second is

$$= \frac{2}{(k-1)^2(fx^2 + k^2gy^2)^2} \cdot (x, y, w)^2,$$

if for shortness

$$\begin{aligned}(x, y, w)^2 &= [(fx^2 - k^2gy^2)(cfx^2 - cgy^2 - fgw^2) - 4ckfgx^2y^2] \\ &\quad \times [fgbk^2x^2 + \{2ch(k-1)^2 - af\}gy^2 + fgh(k-1)^2w^2] \\ &\quad + 2[k^2bg - ch(k-1)^2] fgx^2y^2 [c(k+1)(fx^2 - kgy^2) - kfgw^2];\end{aligned}$$

and hence

$$\Theta = \frac{h^6(k-1)^6}{b^{12}k^{20}(fx^2 + k^2gy^2)^2} \Delta [\{x, y, w\}^2 + 2(bx^2 - ay^2 + hw^2)^2 (x, y, w)^2],$$

which must be a rational and integral function of  $(x, y, w)$ .

In partial verification of this, observe that, because  $U$  contains the terms

$$2h^2cf(ch - af)x^2z^2 + 2\Omega x^2y^2z^2w^2,$$

$\Theta$  should contain the terms

$$2h^2cf(ch - af)x^2 + 2\Omega x^2y^2w^2,$$

viz. in the term in  $x^4$  should be  $= 2h^2cf(ch - af)x^4$ .

Now writing  $y = 0$ ,  $w = 0$ , we have

$$\Delta = c^2f^2x^4,$$

$$\{x, y, w\}^2 = bx^2\{4(2 - k)^2 - 2(6 - 6k + k^2)\}x^4 = bx^2(4 - 4k + 2k^2)x^4,$$

$$(x, y, w)^2 = bcfgk^2x^4;$$

and hence the required term of  $\Theta$  is  $x^4$  multiplied by

$$b^3c^3fh(4 - 4k + 2k^2) + 2hb^2cfgk^2,$$

viz. the coefficient is

$$= 2b^2cf[c^2h(2 - 2k + k^2) + bgk^2] = 2b^2cf[ch + ch(1 - k)^2 + bgk^2],$$

which in virtue of the relation  $af + bgk^2 + ch(1 - k)^2 = 0$  becomes, as it should do,

$$= 2b^2cf(ch - af).$$

The actual division by  $(fx^2 + k^2gy^2)^2$  would, however, be a very tedious process, and it is to be observed, that we only require to know the term  $2\Omega x^2y^2w^2$  of  $\Theta$ . We may therefore adopt a more simple course as follows: the terms of  $\Theta$  which contain  $w^2$  are  $= (Ax^2 + 2\Omega x^2y^2 + By^4)w^2$ , hence writing for a moment

$$\{x, y, w\}^2 = P + Qw^2, \quad (x, y, w)^2 = R + Sw^2,$$

and observing that we have

$$\Delta = c^2(fx^2 + gy^2)^2 - 2cfg(fx^2 - gy^2)w^2 + \&c.,$$

$$(bx^2 - ay^2 + hw^2)^2 = (bx^2 - ay^2)^2 + 2h(bx^2 - ay^2)w^2 + \&c.,$$

we have

$$(fx^2 + k^2gy^2)^2(Ax^2 + 2\Omega x^2y^2 + By^4) = c^2h(fx^2 + gy^2)^2Q - 2c^2fgh(fx^2 - gy^2)P + (bx^2 - ay^2)^2S + 2h(bx^2 - ay^2)R.$$

But in this identical equation we may write  $x^2 = a$ ,  $y^2 = b$ , which gives

$$(af + k^2bg)^2(Aa^2 + 2\Omega ab + Bb^2) = c^2h(af + bg)^2Q - 2c^2fgh(af - bg)P;$$

and from the equation

$$\{x, y, w\}^2 = P + Qw^2,$$

we have

$$\begin{aligned} P + Qw^2 &= (af + bgk^2)[4\{(2 - k)ab - ab + (1 - k)hw^2\}^2 \\ &\quad - 2(ab + hw^2)\{(6 - 6k + k^2)ab - ab + (1 - k)^2hw^2\}] \\ &\quad + 4k^2(k - 1)bghw^2(1 - k)ab \\ &= -ch(k - 1)^2[4(k - 1)^2(a^2b^2 + 2w^2abh) - 2(6 - 6k + k^2)hw^2] \\ &\quad - 4k^2(k - 1)^2ab^2ghw^2, \end{aligned}$$

that is,

$$Q = (k-1)^2 abh [ch(-6k^2 + 4k + 2) - 4k^2 bg],$$

whence

$$(k-1)^2 (Aa^2 + 2\Omega ab + Bb^2) = ab(af+bg)^2 [ch(-6k^2 + 4k + 2) - 4k^2 bg] + 8fgh(af-bg)(k-1)^2 (ab)^2.$$

But we have

$$Aa^2 + Bb^2 = -2ab(af-bg)(-afbg + c^2h^2 + 2chaf + 2chbg),$$

and thence

$$2(k-1)^2 \Omega = (af+bg)^2 [ch(-6k^2 + 4k + 2) - 4k^2 bg] + (k-1)^2 (af-bg) [-2afbg + 2c^2h^2 + 4chaf + 4chbg],$$

or

$$(k-1)^2 \Omega = (af+bg)^2 [ch(-3k^2 + 2k + 1) - 2k^2 bg] + (k-1)^2 (af-bg) [3afbg + c^2h^2 + 2chaf + 2chbg].$$

Writing  $-3k^2 + 2k + 1 = -3(k-1)^2 - 4(k-1)$ , this is

$$\Omega = (af+bg)^2 \left[ ch \left( -3 - \frac{4}{k-1} \right) - 2 \frac{k^2}{(k-1)^2} bg \right] + (af-bg) [3afbg + c^2h^2 + 2chaf + 2chbg],$$

or since  $1 : -k : k-1 = \lambda : \mu : \nu$ ; and writing for shortness  $(af, bg, ch) = (\alpha, \beta, \gamma)$ , this is

$$\Omega = (\alpha + \beta)^2 \left( -3\gamma - \frac{4\gamma\lambda}{\nu} - \frac{2\mu^2}{\nu^2} \beta \right) + (\alpha - \beta) \{3\alpha\beta + \gamma^2 + 2\gamma\alpha + 2\gamma\beta\},$$

which is the value of  $\Omega$ : viz. the conclusion arrived at is that, eliminating  $\theta$  from the equations

$$(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2 \theta^2 z^2 = 0,$$

$$(bk^2 \theta^2 - a)(\theta y - x)^2 + h(k-1)^2 \theta^2 w^2 = 0,$$

where  $k$  denotes a determinate function of  $af, bg, ch$ , viz. writing  $af, bg, ch = \alpha, \beta, \gamma$  and  $1 : -k : k-1 = \lambda : \mu : \nu$ , we have

$$\lambda + \mu + \nu = 0,$$

$$\alpha\lambda^2 + \beta\mu^2 + \gamma\nu^2 = 0,$$

equations which serve to determine  $k$ : the result of the elimination is the octic equation

$$b^2 c^2 f^2 x^4 + \dots + 2\Omega xyz^2 w^2 = 0,$$

where  $\Omega$  has the last-mentioned value.

The value of  $\Omega$  is unsymmetrical in its form, and there are apparently six values; viz. writing

$$\begin{aligned} A &= (\beta + \gamma)^2 \left\{ \alpha \left( -3 - \frac{4\mu}{\nu} \right) - \frac{2\mu^2}{\nu^2} \gamma \right\} + (\beta - \gamma)(S + \beta\gamma + \alpha^2), \\ B &= (\gamma + \alpha)^2 \left\{ \beta \left( -3 - \frac{4\nu}{\lambda} \right) - \frac{2\nu^2}{\lambda^2} \alpha \right\} + (\gamma - \alpha)(S + \gamma\alpha + \beta^2), \\ C &= (\alpha + \beta)^2 \left\{ \gamma \left( -3 - \frac{4\lambda}{\mu} \right) - \frac{2\lambda^2}{\mu^2} \beta \right\} + (\alpha - \beta)(S + \alpha\beta + \gamma^2), \\ A_1 &= (\beta + \gamma)^2 \left\{ \alpha \left( -3 - \frac{4\nu}{\mu} \right) - \frac{2\nu^2}{\mu^2} \beta \right\} - (\beta - \gamma)(S + \beta\gamma + \alpha^2), \\ B_1 &= (\gamma + \alpha)^2 \left\{ \beta \left( -3 - \frac{4\lambda}{\nu} \right) - \frac{2\lambda^2}{\nu^2} \gamma \right\} - (\gamma - \alpha)(S + \gamma\alpha + \beta^2), \\ C_1 &= (\alpha + \beta)^2 \left\{ \gamma \left( -3 - \frac{4\mu}{\lambda} \right) - \frac{2\mu^2}{\lambda^2} \alpha \right\} - (\alpha - \beta)(S + \alpha\beta + \gamma^2), \end{aligned}$$

where for shortness  $S = 2(\beta\gamma + \gamma\alpha + \alpha\beta)$ , the six values would be  $A, B, C, A_1, B_1, C_1$ .

But we have really

$$A = B = C = -A_1 = -B_1 = -C_1,$$

so that  $\Omega$  has really only two values, equal and of opposite signs, or, what is the same thing,  $\Omega^2$  has a unique value. In fact, writing for shortness

$$\lambda + \mu + \nu = P, \quad \alpha\lambda^2 + \beta\mu^2 + \gamma\nu^2 = X,$$

we find at once the identity

$$\lambda^2\nu^2(A + A_1) = (\beta + \gamma)^2(-2X - 4X\alpha P),$$

so that  $A = -A_1$ , in value of  $P = 0, X = 0$ . And similarly  $B = -B_1, C = -C_1$ .

But the demonstration of the equation  $A = B$  is more complicated. We have

$$\begin{aligned} A - B &= -3\alpha(\beta + \gamma)^2 - 4\alpha(\beta + \gamma)^2 \frac{\mu}{\nu} - 2\gamma(\beta + \gamma)^2 \frac{\mu^2}{\nu^2} + (\beta - \gamma)(S + \beta\gamma + \alpha^2) \\ &\quad + 3\beta(\gamma + \alpha)^2 + 4\beta(\gamma + \alpha)^2 \frac{\nu}{\lambda} + 2\alpha(\gamma + \alpha)^2 \frac{\nu^2}{\lambda^2} - (\gamma - \alpha)(S + \gamma\alpha + \beta^2), \end{aligned}$$

that is,

$$\begin{aligned} \lambda^2\nu^2(A - B) &= \{-3\alpha(\beta + \gamma)^2 + 3\beta(\gamma + \alpha)^2 + (\beta - \gamma)(S + \beta\gamma + \alpha^2) - (\gamma - \alpha)(S + \gamma\alpha + \beta^2)\} \lambda^2\nu^2 \\ &\quad - 4\alpha(\beta + \gamma)^2 \lambda^2\mu\nu \\ &\quad - 2\gamma(\beta + \gamma)^2 \lambda^2\mu^2 \\ &\quad + 4\beta(\gamma + \alpha)^2 \nu^3\lambda \\ &\quad + 2\alpha(\gamma + \alpha)^2 \nu^4, \end{aligned}$$

or, denoting for a moment the coefficient of  $\lambda^2\nu^2$  by  $K$ , and writing also  $\gamma\nu^2 = X - \alpha\lambda^2 - \beta\mu^2$ ,  $\nu = P - \lambda - \mu$ , this is

$$\begin{aligned}
 &= K\lambda^2\nu^2 \\
 &\quad - 4\alpha(\beta + \gamma)^2\lambda^2\mu\nu \\
 &\quad - 2(\beta + \gamma)^2\mu^2(X - \alpha\lambda^2 - \beta\mu^2) \\
 &\quad + 4\beta(\gamma + \alpha)^2\lambda(P - \lambda - \mu)\nu^2 \\
 &\quad + 2\alpha(\gamma + \alpha)^2\nu^4, \\
 &= -2(\beta + \gamma)^2\mu^2X + 4\beta(\gamma + \alpha)^2\lambda\nu^2P \\
 &\quad + 2\alpha(\gamma + \alpha)^2\nu^4 \\
 &\quad - 4\beta(\gamma + \alpha)^2\lambda^2\nu^2 \\
 &\quad + \{-4\beta(\gamma + \alpha)^2 + K + 2\alpha(\beta + \gamma)^2\}\lambda^2\nu^2 \\
 &\quad - 4\alpha(\beta + \gamma)^2\lambda^2\mu\nu \\
 &\quad + 2\beta(\beta + \gamma)^2\mu^4,
 \end{aligned}$$

and here the coefficient of  $\lambda^2\nu^2$  is found to be

$$= 2\{\alpha\beta(\alpha + \beta) + \gamma(\alpha - \beta)^2 - 3\gamma^2(\alpha + \beta)\}.$$

Hence, the terms without  $X$  or  $P$  are  $= 2V$ , where

$$\begin{aligned}
 V &= \alpha(\gamma + \alpha)^2\nu^4 \\
 &\quad - 2\beta(\gamma + \alpha)^2\lambda^2\nu^2 \\
 &\quad + \{\alpha\beta(\alpha + \beta) + \gamma(\alpha - \beta)^2 - 3\gamma^2(\alpha + \beta)\}\lambda^2\nu^2 \\
 &\quad - 2\alpha(\beta + \gamma)^2\lambda^2\mu\nu \\
 &\quad + \beta(\beta + \gamma)^2\mu^4,
 \end{aligned}$$

and this is identically

$$\begin{aligned}
 &= \{(\alpha + \gamma)\lambda^2 + (\beta + \gamma)\mu^2 + 2\gamma\lambda\mu\} \\
 &\quad \times \{\alpha(\gamma + \alpha)\nu^2 - 2(\beta\gamma + \gamma\alpha + \alpha\beta)\lambda\mu + \beta(\beta + \gamma)\mu^2\};
 \end{aligned}$$

where observing that

$$\alpha\lambda^2 + \beta\mu^2 + \gamma(P - \lambda - \mu)^2 = X,$$

we have the first factor

$$(\alpha + \gamma)\lambda^2 + (\beta + \gamma)\mu^2 + 2\gamma\lambda\mu = X - \gamma P^2 + 2\gamma P(\lambda + \mu),$$

and consequently

$$\begin{aligned}
 \lambda^2\nu^2(A - B) &= -2(\beta + \gamma)^2\mu^2X + 4\beta(\gamma + \alpha)^2\lambda\nu^2P \\
 &\quad + 2\{X - \gamma P^2 + 2\gamma P(\lambda + \mu)\}\{\alpha(\gamma + \alpha)\nu^2 - 2(\beta\gamma + \gamma\alpha + \alpha\beta)\lambda\mu + \beta(\beta + \gamma)\mu^2\};
 \end{aligned}$$

viz. in virtue of  $P = 0$ ,  $X = 0$ , we have  $A = B$ . And thus

$$A = B = C = -A_1 = -B_1 = -C_1;$$

so that the only values of  $\Omega$  are, say,  $A$  and  $-A$ .



Reverting to the original equations

$$\begin{aligned}(f\theta^2 + g)(k\theta y - x)^2 + h(k-1)^2\theta^2 z^2 &= 0, \\ (bk^2\theta^2 - a)(\theta y - x)^2 + h(k-1)^2\theta^2 w^2 &= 0,\end{aligned}$$

say these are

$$\begin{aligned}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e})(\theta, 1)^4 &= 0, \\ (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{d}', \mathfrak{e}')(\theta, 1)^4 &= 0,\end{aligned}$$

then the coefficients in the two equations have the values

$$\begin{array}{ll}fk y^2, & bk^2 y^2, \\ -2k fxy, & -2bk^2 xy, \\ fx^2 + gky^2 + h(k-1)^2 z^2, & bk^2 x^2 - ay^2 + h(k-1)^2 w^2, \\ -2gkxy, & 2axy, \\ gx^2, & -ax^2,\end{array}$$

where observe that only  $\mathfrak{c}$  contains  $z^2$ , and only  $\mathfrak{c}'$  contains  $w^2$ . The result of the elimination is

$$\begin{vmatrix} \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} & \cdot & \cdot & \cdot \\ \cdot & \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} & \cdot & \cdot \\ \cdot & \cdot & \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} & \cdot \\ \cdot & \cdot & \cdot & \mathfrak{a} & \mathfrak{b} & \mathfrak{c} & \mathfrak{d} & \mathfrak{e} \\ \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' & \cdot & \cdot & \cdot \\ \cdot & \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' & \cdot & \cdot \\ \cdot & \cdot & \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' & \cdot \\ \cdot & \cdot & \cdot & \mathfrak{a}' & \mathfrak{b}' & \mathfrak{c}' & \mathfrak{d}' & \mathfrak{e}' \end{vmatrix} = 0;$$

viz. here the only terms which contain  $z^2$  and  $w^2$  are

$$\mathfrak{c}^4 z^8 \cdot \mathfrak{c}'^4 w^8,$$

and hence the terms in  $z^8$  and  $w^8$  are

$$h^4(k-1)^8 z^8 \cdot a^4 b^4 k^8 x^8 y^8 + h^4(k-1)^8 w^8 \cdot f^4 g^4 k^4 x^8 y^8,$$

viz. these are

$$= h^4 k^8 (k-1)^8 x^8 y^8 (a^4 b^4 z^8 + f^4 g^4 w^8),$$

or assuming that the determinant contains as a factor the function  $b^2 c^2 f^2 x^4 + \dots + 2\Omega x^2 y^2 z^2 w^2$ , with a properly determined value of  $\Omega$ , we see that the other factor is  $= h^4 k^8 (k-1)^8 x^4 y^4$ , which agrees with a preceding result.

## [655] A MEMOIR ON DIFFERENTIAL EQUATIONS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. xiv. (1877), pp. 292–339.]

We have to do with a set of variables, which is either unipartite  $(x, y, z, \dots)$ , or else bipartite  $(x, y, z, \dots; p, q, r, \dots)$ , the variables in the latter case corresponding in pairs  $x$  and  $p$ ,  $y$  and  $q$ , &c.

A letter not otherwise explained denotes a function of the variables. Any such letter may be put  $= \text{const.}$ , viz. we thereby establish a relation between the variables; and when this is so, we use the same letter to denote the constant value of the function. Thus the set being  $(x, y, z; p, q, r)$ ,  $H$  may denote a given function  $pqr - xyz$ ; and then, if  $H = \text{const.}$ , we have  $pqr - xyz = H$  (a constant). This notation, when once clearly understood, is I think a very convenient one.

The present memoir relates chiefly to the following subjects:

A. *Unipartite set*  $(x, y, z, \dots)$ . The differential system

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \dots,$$

and connected therewith the linear partial differential equation

$$X \frac{d\theta}{dx} + Y \frac{d\theta}{dy} + Z \frac{d\theta}{dz} + \dots = 0 :$$

also the lineo-differential

$$X dx + Y dy + Z dz + \dots$$

B. *Bipartite set*  $(x, y, z, \dots; p, q, r, \dots)$ . The Hamiltonian system

$$\frac{dx}{\frac{dH}{dp}} = \frac{dy}{\frac{dH}{dq}} = \frac{dz}{\frac{dH}{dr}} = \dots = \frac{dp}{-\frac{dH}{dx}} = \frac{dq}{-\frac{dH}{dy}} = \frac{dr}{-\frac{dH}{dz}} = \dots$$

and connected therewith the linear partial differential equation

$$\frac{dH}{dp} \frac{d\theta}{dx} - \frac{dH}{dx} \frac{d\theta}{dp} + \frac{dH}{dq} \frac{d\theta}{dy} - \frac{dH}{dy} \frac{d\theta}{dq} + \dots = 0,$$

otherwise written

$$\frac{d(H, \theta)}{d(p, x)} + \frac{d(H, \theta)}{d(q, y)} + \dots = 0,$$

where  $H$  denotes a given function of the variables: also the Hamiltonian system as augmented by an equality  $= dt$ , and as augmented by this and another equality

$$= dV \div \left( p \frac{dH}{dp} + q \frac{dH}{dq} + r \frac{dH}{dr} + \dots \right).$$

C. *Bipartite set*  $(x, y, z, \dots; p, q, r, \dots)$ . The partial differential equation  $H = \text{const.}$ , where, as before,  $H$  is a given function of the variables, but  $p, q, r, \dots$  are now the differential coefficients in regard to  $x, y, z, \dots$  respectively of a function  $V$  of these variables, or, what is the same thing, there exists a function

$$V = \int (p dx + q dy + r dz + \dots),$$

of the variables  $x, y, z, \dots$ .

In what precedes, I have written  $(x, y, z, \dots)$  to denote a set of any number  $n$  of variables, and  $(x, y, z, \dots; p, q, r, \dots)$  to denote a set of any even number  $2n$  of variables, and the investigations are for the most part applicable to these general cases. But for greater clearness and facility of expression, I usually consider the case of a set  $(x, y, z, w)$ , or  $(x, y, z; p, q, r)$ , &c., as the case may be, consisting of a definite number of variables.

The greater part of the theory is not new, but I think that I have presented it in a more compact and intelligible form than has hitherto been done, and I have added some new results.

### *Introductory Remarks.* Art. Nos. 1 to 3.

1. As already noticed, a letter not otherwise explained is considered as denoting a function of the variables of the set; but when necessary we indicate the variables by a notation such as  $z = z(x, y)$ ;  $z$  is here a function (known or unknown as the case may be) of the variables  $x, y$ , the  $z$  on the right-hand side being in fact a functional symbol. And thus also  $z = z(x, y) = \text{const.}$ , denotes that the function  $z(x, y)$  of the variables  $x, y$  has a constant value, which constant value is  $= z$ , viz. we thus indicate a relation between the variables  $x, y$ .

2. The variables  $x, y$ , &c., may have infinitesimal increments  $dx, dy$ , &c.; and the equations of connexion between the variables then give rise to linear relations between these increments, the coefficients therein being differential coefficients and,

as such, represented in the usual notation; thus if  $z = z(x, y)$ , we have  $dz = \frac{dz}{dx} dx + \frac{dz}{dy} dy$ , where  $\frac{dz}{dx}$ ,  $\frac{dz}{dy}$  are the so-called partial differential coefficients of  $z$  in regard to  $x$ ,  $y$  respectively.

If we have  $y = y(x)$ , then also  $dy = \frac{dy}{dx} dx$ , and the foregoing equation becomes

$$dz = \left( \frac{dz}{dx} + \frac{dz}{dy} \frac{dy}{dx} \right) dx;$$

but considering the two equations  $z = z(x, y)$  and  $y = y(x)$  as determining  $z$  as a function of  $x$ , say  $z = z(x)$ , we have  $dz = \frac{d(z)}{dx} dx$ ; whence comparing the two formulae

$$\frac{d(z)}{dx} = \frac{dz}{dx} + \frac{dz}{dy} \frac{dy}{dx},$$

where  $\frac{d(z)}{dx}$  is the so-called total differential coefficient of  $z$  in regard to  $x$ . The distinction is best made, not by any difference of notation  $\left( \frac{d(z)}{dx}, \frac{dz}{dx} \right)$ , but by appending in any case of doubt the equations or equation used in the differentiation. Thus we have  $\frac{dz}{dx}$ , where  $z = z(x, y)$ : or, as the case may be,  $\frac{dz}{dx}$  where  $z = z(x, y)$  and  $y = y(x)$ .

3. A relation between increments is always really a relation between differential coefficients; but we use the increments for symmetry and conciseness, as in the case of a differential system  $\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z}$ , or in a question relating to the lineo-differential  $X dx + Y dy + Z dz$ , for instance in the question whether this can be put  $= du$ .

*Notations.* Art. Nos. 4 to 6.

4. *Functional determinants.* If  $a, b, c, \dots$  are functions of the variables  $x, y, z, w, \dots$ , then the determinants

$$\begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} \\ \frac{db}{dx} & \frac{db}{dy} \end{vmatrix}, \quad \begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} & \frac{da}{dz} \\ \frac{db}{dx} & \frac{db}{dy} & \frac{db}{dz} \\ \frac{dc}{dx} & \frac{dc}{dy} & \frac{dc}{dz} \end{vmatrix}, \quad \&c.,$$

are for shortness represented by

$$\frac{d(a, b)}{d(x, y)}, \quad \frac{d(a, b, c)}{d(x, y, z)}, \quad \&c.,$$

the notation being especially used in the first-mentioned case where the symbol is  $\frac{d(a,b)}{d(x,y)} = 0$ .

It is sometimes convenient to extend this notation, and for instance use  $\frac{d(a,b)}{d(x,y,z)}$  to denote the series of determinants

$$\begin{vmatrix} \frac{da}{dx} & \frac{da}{dy} & \frac{da}{dz} \\ \frac{db}{dx} & \frac{db}{dy} & \frac{db}{dz} \end{vmatrix}$$

which can be formed by selecting in every way two columns to form thereout a determinant; the equation

$$\frac{d(a,b)}{d(x,y,z)} = 0$$

will then denote that each of these determinants is  $= 0$ .

The analogous notation

$$\frac{d(a,b,c)}{d(x,y)}$$

would denote non-existent determinants, viz. there are here not columns enough to form with them a determinant: and the notation is not required.

5. In the case of a bipartite set  $(x, y, z, \dots; p, q, r, \dots)$ , if  $a, b$  are any functions of these variables, we consider the derivative

$$(a, b) = \frac{d(a, b)}{d(p, x)} + \frac{d(a, b)}{d(q, y)} + \frac{d(a, b)}{d(r, z)} + \dots,$$

viz.  $(a, b)$  is used to denote the sum of the functional determinants on the right hand.

6. Taking again  $(x, y, z, w, \dots)$  as the variables, then in the theory of the lineo-differential  $X dx + Y dy + Z dz + W dw + \dots$ , we use certain derivative functions analogous to Pfaffians. They may be thus defined; viz. considering the numbers  $1, 2, 3, 4, \dots$  as corresponding to the variables  $x, y, z, w, \dots$  respectively, we have

$$\begin{aligned} 1 &= X, \quad 2 = Y, \quad 3 = Z, \quad 4 = W, \quad \&c., \\ 12 &= \frac{dY}{dx} - \frac{dX}{dy}, \quad 13 = \frac{dZ}{dx} - \frac{dX}{dz}, \quad \&c., \\ 123 &= 1 \cdot 23 + 2 \cdot 31 + 3 \cdot 12 \\ &= X \left( \frac{dZ}{dy} - \frac{dY}{dz} \right) + Y \left( \frac{dX}{dz} - \frac{dZ}{dx} \right) + Z \left( \frac{dY}{dx} - \frac{dX}{dy} \right), \\ 1234 &= 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23 \\ &= \left( \frac{dY}{dx} - \frac{dX}{dy} \right) \left( \frac{dW}{dz} - \frac{dZ}{dw} \right) + \left( \frac{dZ}{dx} - \frac{dX}{dz} \right) \left( \frac{dY}{dw} - \frac{dW}{dy} \right) + \left( \frac{dW}{dx} - \frac{dX}{dw} \right) \left( \frac{dZ}{dy} - \frac{dY}{dz} \right), \end{aligned}$$

and, adding for greater distinctness the next following cases,

$$12345 = 1 \cdot 2345 + 2 \cdot 3451 + 3 \cdot 4512 + 4 \cdot 5123 + 5 \cdot 1234,$$

$$123456 = 12 \cdot 3456 + 13 \cdot 4561 + 14 \cdot 5612 + 15 \cdot 6123 + 16 \cdot 2345,$$

where of course 2345, &c., have the significations mentioned above.

*Dependency of Functions.* Art. Nos. 7 and 8.

7. Two or more functions of the same variables may be independent, or else dependent or connected; viz. in the latter case any one of the functions is a function of the others:  $a = a(x)$ ,  $b = b(x)$ , the functions  $a, b$  are dependent, but if

$$a = a(x, y), \quad b = b(x, y),$$

then the condition of dependency is

$$\frac{d(a, b)}{d(x, y)} = 0,$$

and, similarly, if  $a = a(x, y, z)$ ,  $b = b(x, y, z)$ , then the conditions of dependency are

$$\frac{d(a, b)}{d(x, y, z)} = 0,$$

viz. if the equations thus represented are all of them satisfied, the functions are dependent, but if not, then they are independent.

Observe that, when  $a = a(x, y, z)$ ,  $b = b(x, y, z)$  as above, if we choose to attend only to the variables  $x, y$ , treating  $z$  as a mere constant, there is then a single condition of dependency  $\frac{d(a, b)}{d(x, y)} = 0$ , and so if we attend only to the variable  $x$ , treating  $y, z$  as mere constants, then  $a$  and  $b$  are dependent. Thus when  $a = x$ ,  $b = x^2 + y$ , the functions  $a, b$  are independent if we attend to both the variables  $x, y$ ; dependent if  $y$  be regarded as a constant.

8. Further when  $a = a(x, y)$ ,  $b = b(x, y)$ ,  $c = c(x, y)$ , the functions  $a, b, c$  are dependent; but when  $a = a(x, y, z)$ ,  $b = b(x, y, z)$ ,  $c = c(x, y, z)$ , the condition of dependency is

$$\frac{d(a, b, c)}{d(x, y, z)} = 0,$$

and so when  $a = a(x, y, z, w)$ ,  $b = b(x, y, z, w)$ ,  $c = c(x, y, z, w)$ , the conditions of dependency are

$$\frac{d(a, b, c)}{d(x, y, z, w)} = 0,$$

viz. if all the equations thus represented are satisfied, the functions are dependent; but if not, then they are independent. And so in other cases.

*The General Differential System.* Art. Nos. 9 to 22.

9. Taking the set of variables to be  $(x, y, z, w)$ , the system is

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \frac{dw}{W},$$

and we associate with this the linear partial differential equation

$$X \frac{d\theta}{dx} + Y \frac{d\theta}{dy} + Z \frac{d\theta}{dz} + W \frac{d\theta}{dw} = 0.$$

10. It is tolerably evident that the differential equations establish between  $x, y, z, w$  a threefold relation depending upon three arbitrary constants; in fact, regarding  $(x, y, z, w)$  as the coordinates of a point in four-dimensional space, and starting from any given point, the differential equations determine the ratios of the increments  $dx, dy, dz, dw$ , that is, the direction of passage to a consecutive point; and then again taking for  $x, y, z, w$  the coordinates of this point, the same equations give the direction of passage to the next consecutive point, and so on. The locus of the point is therefore a curve, or we have between the coordinates a threefold relation, and (the initial point being arbitrary) we have a curve of the system through each point of the four-dimensional space, viz. the relation must involve three arbitrary constants. But this being so, the constants will be expressible as functions of the coordinates, viz. the threefold relation involving the three constants will be expressible in the form  $a = \text{const.}$ ,  $b = \text{const.}$ ,  $c = \text{const.}$ , where  $a, b, c$  denote respectively functions of the coordinates  $(x, y, z, w)$ .

11. Supposing that one of the relations is  $a = \text{const.}$ , it is clear that the increment

$$da = \frac{da}{dx} dx + \frac{da}{dy} dy + \frac{da}{dz} dz + \frac{da}{dw} dw,$$

must become  $= 0$ , on substituting therein for  $dx, dy, dz, dw$ , the values  $X, Y, Z, W$  to which by virtue of the differential equations they are proportional, viz. that we must have identically

$$X \frac{da}{dx} + Y \frac{da}{dy} + Z \frac{da}{dz} + W \frac{da}{dw} = 0.$$

Conversely, when this is so, we have  $da = 0$ , by virtue of the differential equation.

We say that  $a$  is a solution of the partial differential equation, and an integral of the differential equations, viz. any solution of the partial differential equation is an integral of the differential equations, and any integral of the differential equations is a solution of the partial differential equation, or, this being so, we may in general without risk of ambiguity, say simply  $a$  is an integral<sup>1</sup>; similarly  $b$  and  $c$  are integrals, and, by what precedes, there are three integrals  $a, b, c$ .

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<sup>1</sup>Viz. we use indifferently, in regard to the differential equations and to the partial differential equation, the term integral, which is appropriate to the differential equations; the appropriate term in regard to the partial differential equation would be solution.

Observe that, in speaking of an integral  $a$ , we mean a function of the variables; the differential equations give between the variables the relation  $a = \text{const.}$ , and when this is so, we use the same letter  $a$  to denote the constant value of this function.

12. In speaking of the three integrals  $a, b, c$  we mean independent integrals; any function whatever  $\phi a$  of an integral  $a$ , or any function whatever  $\phi(a, b)$  of two integrals  $a, b$ , is an integral (viz. it is an integral of the differential equations, and also a solution of the partial differential equation), but such dependent integrals give nothing new, and we require a third independent integral  $c$ , viz. we need this to express the threefold relation between the variables, given by the differential equations, and also to express the general solution  $\phi(a, b, c)$  of the partial differential equation.

13. By what precedes the analytical condition, in order that the integrals  $a, b, c$  may be independent, is that they are such as not to satisfy the relations

$$\frac{d(a, b, c)}{d(x, y, z, w)} = 0.$$

14. We moreover see *à posteriori*, that there cannot be more than three independent integrals; in fact, if  $a, b, c, d$  are integrals, then, considering them as solutions of the partial differential equation, we have four equations which by the elimination therefrom of  $X, Y, Z, W$ , give

$$\frac{d(a, b, c, d)}{d(x, y, z, w)} = 0,$$

and this is the very equation which expresses that  $a, b, c, d$  are not independent.

15. The notion of the integrals may be arrived at somewhat differently thus: take  $a, b, c, d$  any functions of the variables, and write

$$A = X \frac{da}{dx} + Y \frac{da}{dy} + Z \frac{da}{dz} + W \frac{da}{dw},$$

and the like for  $B, C, D$ : then replacing the original variables  $x, y, z, w$  by the new variables  $a, b, c, d$ , the differential equations become

$$\frac{da}{A} = \frac{db}{B} = \frac{dc}{C} = \frac{dd}{D},$$

where  $A, B, C, D$  are to be (by means of the given values of  $a, b, c, d$  as functions of  $x, y, z, w$ ) expressed as functions of  $a, b, c, d$ . If then  $A = 0, B = 0, C = 0$ , the differential equations become

$$\frac{da}{0} = \frac{db}{0} = \frac{dc}{0} = \frac{dd}{D},$$

viz. we have  $da = 0, db = 0, dc = 0$ , and therefore  $a = \text{const.}, b = \text{const.}, c = \text{const.}$ , that is, we have the integrals  $a, b, c$  as before.



16. There is no general process for obtaining an integral  $a$  of the differential equations. Supposing such integral known, we can introduce it as a variable, in place of one of the original variables, say  $w$ , viz. we thus reduce the system to

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \frac{da}{0},$$

where  $X, Y, Z$  now denote the values assumed by these functions upon expressing therein  $w$  as a function of  $x, y, z, a$ , viz. they are now functions of  $x, y, z, a$ . The system thus breaks up into  $da = 0$  and the system

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z},$$

in which last (by virtue of the first equation, or  $a = \text{const.}$ )  $a$  is to be regarded as a constant; the original system of three equations between four variables is thus reduced to a system of two equations between three variables. Supposing  $b$  to be an integral of this reduced system,  $b$  is given as a function of  $x, y, z, a$ , but upon substituting herein for  $a$  its value as a function of  $x, y, z, w$ , we have  $b$  a function of the original variables  $x, y, z, w$ , and  $b$  is then a second integral of the original system.

17. In like manner supposing  $a$  and  $b$  to be known, we reduce the system to the single equation

$$\frac{dx}{X} = \frac{dy}{Y},$$

where  $X, Y$  are now functions of  $x, y, a, b$ ; supposing an integral hereof to be  $c$ , we have  $c$  a function of  $x, y, a, b$ ; but upon substituting herein for  $a, b$  their values as functions of  $x, y, z, w$ , we have  $c$  a function of  $x, y, z, w$ , and as such it is the third integral of the original system.

18. It may be remarked that if, to the original system, we join on an equality  $= dt$ , viz. if we consider the system

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z} = \frac{dw}{W} = dt,$$

where  $X, Y, Z, W$  are as before functions of the variables  $(x, y, z, w)$ , then the integrals  $a, b, c$  of the original system being known, we can by means of them express for instance  $X$  as a function of  $x, a, b, c$ , and we have then,  $\text{const.} = t - \int \frac{dx}{X}$ , where the integration is to be performed regarding  $a, b, c$  as constants; writing  $\int \frac{dx}{X} = \tau$ , but after the integration replacing  $a, b, c$  by their values as functions of  $x, y, z, w$ , we have  $\tau$  a function of  $x, y, z, w$ ; and we say that  $t - \tau$  is an integral; putting it  $= \text{const.}$ , we use also  $t$  to denote the constant value of the integral  $t - \tau$  in question. Observe that here, the integrals  $a, b, c$  being known, the last integral  $t - \tau$  is obtained by a quadrature.

19. The result would have been similar, if the adjoined equality had been  $= \frac{dt}{T}$ , ( $T$  a function of  $x, y, z, w$ ), but in reference to subsequent matter, I retain the equality  $= dt$ , and adjoin a second equality  $= \frac{dV}{\Omega}$  ( $\Omega$  a function of  $x, y, z, w$ ); we have then the integral  $t - \tau$  as before, and another integral  $V - \int \frac{\Omega dx}{X}$ , where  $\Omega, X$  are first expressed as functions of  $x, a, b, c$ , but after the integration  $a, b, c$  are replaced by their values as functions of  $(x, y, z, w)$ , say this is the integral  $V - \lambda$ ; this, when the integrals  $a, b, c$  are known, is (like  $t - \tau$ ) obtained by a quadrature.

20. Attending only to the adjoined equality  $= dt$ , we can by means of the four integrals express each of the variables  $x, y, z, w$  as a function of  $a, b, c, t - \tau$ ; viz. these four equations, regarding therein  $t - \tau$  as a variable parameter, are in fact equivalent to the equations  $a = \text{const.}$ ,  $b = \text{const.}$ ,  $c = \text{const.}$ , which connect the variables  $x, y, z, w$  with the integrals  $a, b, c$  regarded as constants.

21. All that precedes is of course applicable to a system of  $n - 1$  equations between  $n$  variables, the number of independent integrals being  $= n - 1$ .

22. I take an example with the three variables  $x, y, z$ ; the differential equations being

$$\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)},$$

and therefore the partial differential equation

$$x(y-z)\frac{d\theta}{dx} + y(z-x)\frac{d\theta}{dy} + z(x-y)\frac{d\theta}{dz} = 0.$$

The integrals are  $a = x + y + z$ ,  $b = xyz$ ; and it will be shown how either of these integrals being known, the system is reduced to a single equation between two variables, say  $x, y$ .

First,  $a$  being known,  $= x + y + z$  as before, we have

$$x(y-z) = x(x+2y-a), \quad y(z-x) = y(a-2x-y),$$

and the system is

$$\frac{dx}{x(x+2y-a)} = \frac{dy}{y(a-2x-y)},$$

which has the integral  $b = xy(a-x-y)$ ; observe that this is a solution of the partial differential equation

$$x(x+2y-a)\frac{d\theta}{dx} + y(a-2x-y)\frac{d\theta}{dy} = 0.$$

For  $a$  putting its value we find  $b = xyz$ .

Secondly,  $b$  being known,  $= xyz$  as before, we have

$$x(y - z) = xy - \frac{b}{z}, \quad y(z - x) = \frac{b}{x} - xy,$$

and the system is

$$\frac{dx}{xy - \frac{b}{y}} = \frac{dy}{\frac{b}{x} - xy},$$

which has the integral  $a = x + y + \frac{b}{xy}$ ; observe that this is a solution of the partial differential equation

$$\left(xy - \frac{b}{y}\right) \frac{d\theta}{dx} + \left(\frac{b}{x} - xy\right) \frac{d\theta}{dy} = 0.$$

For  $b$  putting its value, we find  $a = x + y + z$ .

*The Multiplier.* Art. Nos. 23 to 29.

23. First, if there are only two variables  $(x, y)$ , the system consists of the single equation

$$\frac{dx}{X} = \frac{dy}{Y},$$

which may be written

$$Y dx - X dy = 0.$$

Hence, if  $a$  be an integral, we have

$$\frac{da}{dx} dx + \frac{da}{dy} dy = 0;$$

the two will agree if there exists a function  $M$  such that

$$\frac{da}{dx} = MY, \quad \frac{da}{dy} = -MX,$$

and thence, in virtue of the identity

$$\frac{d}{dy} \frac{da}{dx} = \frac{d}{dx} \frac{da}{dy},$$

we find

$$\frac{d(MX)}{dx} + \frac{d(MY)}{dy} = 0,$$

or, as this may also be written,

$$X \frac{dM}{dx} + Y \frac{dM}{dy} + M \left( \frac{dX}{dx} + \frac{dY}{dy} \right) = 0,$$

as the condition to determine the multiplier  $M$ . Supposing  $M$  known, we have  $M(Y dx - X dy) = da$ , or say  $a = \int M(Y dx - X dy)$ , viz. the integral  $a$  is determined by a quadrature.

24. In the case of three variables  $(x, y, z)$ , the system is

$$\frac{dx}{X} = \frac{dy}{Y} = \frac{dz}{Z},$$

or, writing these in the form

$$Y dz - Z dy = 0, \quad Z dx - X dz = 0, \quad X dy - Y dx = 0,$$

the course which immediately suggests itself is to seek for factors  $L, M, N$ , such that,  $a$  being an integral, we may have

$$L(Y dz - Z dy) + M(Z dx - X dz) + N(X dy - Y dx) = da,$$

but this does not lead to any result. The course taken by Jacobi is quite a different one: he, in fact, determines a multiplier  $M$  connected with two integrals  $a, b$ .

25. Supposing that  $a, b$  are independent integrals, we have

$$X \frac{da}{dx} + Y \frac{da}{dy} + Z \frac{da}{dz} = 0,$$

$$X \frac{db}{dx} + Y \frac{db}{dy} + Z \frac{db}{dz} = 0;$$

and determining from these equations the ratios of the quantities  $X, Y, Z$ , we may, it is clear, write

$$MX, MY, MZ = \frac{d(a, b)}{d(y, z)}, \frac{d(a, b)}{d(z, x)}, \frac{d(a, b)}{d(x, y)}.$$

It may be shown that we have identically

$$\frac{d}{dx} \frac{d(a, b)}{d(y, z)} + \frac{d}{dy} \frac{d(a, b)}{d(z, x)} + \frac{d}{dz} \frac{d(a, b)}{d(x, y)} = 0,$$

and we thence deduce

$$\frac{d(MX)}{dx} + \frac{d(MY)}{dy} + \frac{d(MZ)}{dz} = 0,$$

or, what is the same thing,

$$X \frac{dM}{dx} + Y \frac{dM}{dy} + Z \frac{dM}{dz} + M \left( \frac{dX}{dx} + \frac{dY}{dy} + \frac{dZ}{dz} \right) = 0,$$

as the condition for determining the multiplier  $M$ .

26. The use is as follows: supposing that  $M$  is known, and supposing also that one integral  $a$  of the system is known, we can then by a quadrature determine the other integral  $b$ . Thus, supposing that we know the integral  $a, = a(x, y, z)$ , we can by means of this integral express  $z$  in terms of  $x, y, a$ ; and hence we may regard the unknown integral  $b$  as expressed in the like form,  $b = b(x, y, a)$ . The original values of  $\frac{db}{dx}, \frac{db}{dy}, \frac{db}{dz}$  become on this supposition

$$\frac{db}{dx} + \frac{db}{da} \frac{da}{dx}, \quad \frac{db}{dy} + \frac{db}{da} \frac{da}{dy}, \quad \frac{db}{da} \frac{da}{dz},$$

and we thence find

$$\frac{d(a, b)}{d(y, z)}, \quad \frac{d(a, b)}{d(z, x)}, \quad \frac{d(a, b)}{d(x, y)} = -\frac{da}{dz} \frac{db}{dy}, \quad \frac{da}{dz} \frac{db}{dx}, \quad \frac{d(a, b)}{d(x, y)}.$$

We have therefore

$$MX, MY = -\frac{da}{dz} \frac{db}{dy}, \quad \frac{da}{dz} \frac{db}{dx},$$

and, consequently,

$$db = \frac{db}{dx} dx + \frac{db}{dy} dy = \frac{M}{\frac{da}{dz}} (Y dx - X dy);$$

viz.  $M, \frac{da}{dz}, Y, X$  being all of them expressed as functions of  $x, y, a$ , the expression on the right-hand is a complete differential, and we have

$$b = \int \frac{M}{\frac{da}{dz}} (Y dx - X dy);$$

that is, the integral  $b$  is determined by a quadrature.

27. Thus, in the example No. 22,

$$\frac{dX}{dx} + \frac{dY}{dy} + \frac{dZ}{dz} = 0,$$

and a value of the multiplier is  $= 1$ . Supposing that the given integral is  $a = x + y + z$ , then  $\frac{da}{dz} = 1$ , and we have accordingly 1 as the multiplier of the equation

$$y(a - 2x - y) dx + x(a - x - 2y) dy = 0,$$

viz. this equation is integrable *per se*. Supposing the given integral to be  $b = xyz$ , then  $\frac{db}{dz} = xy$ ,

viz. we have  $\frac{1}{xy}$  as the multiplier of the equation

$$\left(\frac{b}{x} - xy\right) dx + \left(xy - \frac{b}{y}\right) dy = 0,$$

and we thus in each case obtain the other integral as before.

28. The foregoing result may be presented in a more symmetrical form by taking in place of  $x, y$  any two variables  $u = u(x, y, z), v = v(x, y, z)$ .

Supposing the integral  $a$  known as before, the system then is

$$\frac{du}{U} = \frac{dv}{V} = \frac{da}{0},$$

where  $U, V = X \frac{du}{dx} + Y \frac{du}{dy} + Z \frac{du}{dz}, X \frac{dv}{dx} + Y \frac{dv}{dy} + Z \frac{dv}{dz}$ , these being expressed as functions of  $u, v, a$ ; or, what is the same thing, we have  $V du - U dv = 0$ ,  $a$  being in this equation regarded as a constant.

From the foregoing values of  $MX, MY, MZ$ , we deduce

$$MU, MV = \frac{d(u, a, b)}{d(x, y, z)}, \frac{d(v, a, b)}{d(x, y, z)}.$$

But forming the values of  $du, dv, da, db$ , we have an equation, determinant = 0, which equation may be written

$$\frac{d(v, a, b)}{d(x, y, z)} du - \frac{d(u, a, b)}{d(x, y, z)} dv + \frac{d(u, v, b)}{d(x, y, z)} da - \frac{d(u, v, a)}{d(x, y, z)} db = 0,$$

or, writing herein  $da = 0$ , this is

$$\frac{d(v, a, b)}{d(x, y, z)} du - \frac{d(u, a, b)}{d(x, y, z)} dv = \frac{d(u, v, a)}{d(x, y, z)} db,$$

viz. this is

$$M(V du - U dv) = db \frac{d(u, v, a)}{d(x, y, z)},$$

or say

$$db = M(V du - U dv) \div \frac{d(u, v, a)}{d(x, y, z)},$$

where, on the right-hand side, everything must be expressed in terms of  $u, v, a$ . It thus appears that on expressing the final equation as a relation  $V du - U dv = 0$  between the variables  $u$  and  $v$ , the multiplier hereof is  $M \div \frac{d(u, v, a)}{d(x, y, z)}$ . If  $u, v = x, y$ , this agrees with a foregoing result.

29. The theory is precisely the same for any number of variables. Thus, if there are four variables  $x, y, z, w$ , we have

$$MX, MY, MZ, MW = \frac{d(a, b, c)}{d(y, z, w)}, \frac{d(a, b, c)}{d(z, w, x)}, \frac{d(a, b, c)}{d(w, x, y)}, \frac{d(a, b, c)}{d(x, y, z)},$$

and, we have between the functions on the right-hand an identical relation, in virtue of which

$$\frac{d(MX)}{dx} + \frac{d(MY)}{dy} + \frac{d(MZ)}{dz} + \frac{d(MW)}{dw} = 0;$$

then, supposing that a value of  $M$  is known, and also any two integrals  $a, b$ , and that by means of these the equation to be finally integrated is expressed as a relation  $V du - U dv = 0$  between any two variables  $u$  and  $v$ , the multiplier of this is

$$= M \div \frac{d(u, v, a, b)}{d(x, y, z, w)},$$

where  $U, V$  and this multiplier are to be expressed in terms of  $u, v, a, b$ .

The general result is that, given a value of the multiplier, and also all but one of the integrals, the final integral is expressible by a quadrature.

## The Collected Mathematical Papers of Arthur Cayley

Vol. X, Pages 126–150 (PDF)

Paper 655 (cont.): A Memoir on Differential Equations

Arthur Cayley

Cambridge, 1896

*Pfaffian Theorem.* Art. No. 30.

30. According as

the variables are we have

$$\begin{array}{ll}
x, & X dx = du, \\
x, y, & X dx + Y dy = \lambda du, \\
x, y, z, & X dx + Y dy + Z dz = \lambda du + dv, \\
x, y, z, w, & X dx + Y dy + Z dz + W dw = \lambda du + \mu dv,
\end{array}$$

and so on; viz. the theorem is that, taking for instance two variables, a given lineo-differential  $X dx + Y dy$  is  $\lambda du$ , that is, there exist  $\lambda, u$  functions of  $x, y$ , which verify this identity, or, what is the same thing, such that we have

$$X, Y = \lambda \frac{du}{dx}, \quad \lambda \frac{du}{dy};$$

and so, in the case of three variables, there exist  $\lambda, u, v$  functions of  $x, y, z$ , such that

$$X, Y, Z = \lambda \frac{du}{dx} + \frac{dv}{dx}, \quad \lambda \frac{du}{dy} + \frac{dv}{dy}, \quad \lambda \frac{du}{dz} + \frac{dv}{dz}.$$

The problem of determining the functions on the right-hand side is known as the Pfaffian Problem; this I do not at present consider, but only assume that there exist such functions.

*The Hamiltonian System, its derivation from the general System.* Art. Nos. 31 to 34.

31. Considering a bipartite set  $(x, y, z; p, q, r)$ , the general system of differential equations may be written

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{dp}{-X} = \frac{dq}{-Y} = \frac{dr}{-Z}.$$

But by the Pfaffian theorem we may write

$$X dx + Y dy + Z dz + P dp + Q dq + R dr = \xi d\rho + \eta d\sigma + \zeta d\tau,$$

viz. there exist  $\xi, \eta, \zeta, \rho, \sigma, \tau$  functions of the variables  $x, y, z, p, q, r$ , such that we have

$$X = \xi \frac{d\rho}{dx} + \eta \frac{d\sigma}{dx} + \zeta \frac{d\tau}{dx}, \quad \dots, \quad P = \xi \frac{d\rho}{dp} + \eta \frac{d\sigma}{dp} + \zeta \frac{d\tau}{dp}, \quad \dots;$$

and we have the foregoing general system expressed by means of these given functions  $\xi, \eta, \zeta, \rho, \sigma, \tau$  of the variables.

32. But the lineo-differential

$$X dx + Y dy + Z dz + P dp + Q dq + R dr$$

may be of a more special form; for instance, it may be a sum of two terms  $= \xi d\rho + \eta d\sigma$ ; or, finally, it may be a single term  $= \xi d\rho$ , and in this case we have the Hamiltonian system, viz. writing  $H$  in place of  $\rho$ , if we have

$$X dx + Y dy + Z dz + P dp + Q dq + R dr = \xi dH,$$

where  $H$  is a given function of the variables, then the system is

$$\frac{dx}{\frac{dH}{dp}} = \frac{dy}{\frac{dH}{dq}} = \frac{dz}{\frac{dH}{dr}} = -\frac{dp}{\frac{dH}{dx}} = -\frac{dq}{\frac{dH}{dy}} = -\frac{dr}{\frac{dH}{dz}},$$

which is the system in question.

33. Any integral  $\theta$  of the system is a solution of

$$\frac{dH}{dp} \frac{d\theta}{dx} + \frac{dH}{dq} \frac{d\theta}{dy} + \frac{dH}{dr} \frac{d\theta}{dz} - \frac{dH}{dx} \frac{d\theta}{dp} - \frac{dH}{dy} \frac{d\theta}{dq} - \frac{dH}{dz} \frac{d\theta}{dr} = 0;$$

viz. writing, as above,

$$(H, \theta) = \frac{d(H, \theta)}{d(p, x)} + \frac{d(H, \theta)}{d(q, y)} + \frac{d(H, \theta)}{d(r, z)}$$

= last-mentioned expression, the partial differential equation is  $(H, \theta) = 0$ ; and, conversely, any solution of this equation is an integral of the differential equations.

34. It is obvious that a solution of  $(H, \theta) = 0$  is  $H$ ; hence the entire system of independent solutions may be taken to be  $H, a, b, c, d$ ; or, if we choose to consider a set of five independent solutions  $a, b, c, d, e$ , then we have  $H$  a function of these solutions.

*An Identity in regard to the Functions  $(H, \theta)$ . Art. Nos. 35 and 36.*

35. Taking the variables to be  $(x, y, z, p, q, r)$ , and  $H, a, b$  to be any functions of these variables, we have the identity

$$(H, (a, b)) + (a, (b, H)) + (b, (H, a)) = 0,$$

which is now to be proved. For this purpose we write it in the slightly different form

$$((a, b), H) = (a, (b, H)) - (b, (a, H)).$$

The first term on the right-hand side is

$$\left( \frac{da}{dp} \frac{d}{dx} + \frac{da}{dq} \frac{d}{dy} + \frac{da}{dr} \frac{d}{dz} - \frac{da}{dx} \frac{d}{dp} - \frac{da}{dy} \frac{d}{dq} - \frac{da}{dz} \frac{d}{dr} \right)$$

operating upon

$$\left( \frac{db}{dp} \frac{dH}{dx} + \frac{db}{dq} \frac{dH}{dy} + \frac{db}{dr} \frac{dH}{dz} - \frac{db}{dx} \frac{dH}{dp} - \frac{db}{dy} \frac{dH}{dq} - \frac{db}{dz} \frac{dH}{dr} \right);$$



and if we herein attend to the terms which contain the second differential coefficients of  $H$ , these are symmetrical functions of  $a, b$ . For instance,

$$\begin{aligned} \frac{d^2 H}{dx^2} \text{ coefficient is } & \frac{da}{dp} \frac{db}{dp}, \\ \frac{d^2 H}{dx dy} & , \quad \frac{da}{dp} \frac{db}{dq} + \frac{da}{dq} \frac{db}{dp}, \\ \frac{d^2 H}{dx dp} & , \quad -\frac{da}{dp} \frac{db}{dx} - \frac{da}{dx} \frac{db}{dp}, \\ \frac{d^2 H}{dx dq} & , \quad -\frac{da}{dp} \frac{db}{dy} - \frac{da}{dy} \frac{db}{dp}. \end{aligned}$$

Hence, forming the like terms of the second term  $(b, (a, H))$  and subtracting, the terms in question all vanish: and we thus see that  $(a, (b, H)) - (b, (a, H))$  is a linear function of the differential coefficients

$$\frac{dH}{dx}, \quad \frac{dH}{dy}, \quad \frac{dH}{dz}, \quad \frac{dH}{dp}, \quad \frac{dH}{dq}, \quad \frac{dH}{dr}.$$

36. Attending to any one of these, suppose  $\frac{dH}{dx}$ , the coefficient of this

$$\text{in } (a, (b, H)) \text{ is } = \left( a, \frac{db}{dp} \right),$$

$$\text{in } (b, (a, H)) \quad , \quad \left( b, \frac{da}{dp} \right), = - \left( \frac{da}{dp}, b \right),$$

wherefore, in the difference of these, it is

$$\left( a, \frac{db}{dp} \right) + \left( \frac{da}{dp}, b \right) = \frac{d}{dp}(a, b).$$

Hence, for the several terms

$$\frac{dH}{dx}, \quad \frac{dH}{dy}, \quad \frac{dH}{dz}, \quad \frac{dH}{dp}, \quad \frac{dH}{dq}, \quad \frac{dH}{dr},$$

the coefficients are

$$\frac{d}{dp}(a, b), \quad \frac{d}{dq}(a, b), \quad \frac{d}{dr}(a, b), \quad -\frac{d}{dx}(a, b), \quad -\frac{d}{dy}(a, b), \quad -\frac{d}{dz}(a, b);$$

or, what is the same thing, we have

$$(a, (b, H)) - (b, (a, H)) = ((a, b), H),$$

the identity in question.

*The Poisson-Jacobi Theorem.* Art. Nos. 37 to 39.

37. The foregoing identity shows that if  $(H, a) = 0$ , and  $(H, b) = 0$ , then also  $(H, (a, b)) = 0$ ; or, what is the same thing, if  $a$  and  $b$  are solutions of the partial differential equation  $(H, \theta) = 0$ , then also  $(a, b)$  is a solution; or, say, if  $a, b$  are integrals, then also  $(a, b)$  is an integral.

Supposing that the set is  $(x, y, z, p, q, r)$ , so that there are in all five integrals  $a, b, c, d, e$ , then the theorem may be otherwise stated, we have  $(a, b)$  a function of the integrals  $a, b, c, d, e$ .

Observe that, knowing only the integrals  $a$  and  $b$ , we find  $(a, b)$  as a function of  $x, y, z, p, q, r$ ; this may be  $= 0$ , or a determinate constant, or it may be such a function that by virtue of the given values of  $a$  and  $b$  it reduces itself to a function of  $a$  and  $b$ ; in any of these cases the theorem does not determine a new integral. But if contrariwise the value of  $(a, b)$ , obtained as above as a function of the variables, is not a function of  $a, b$ , then it is a new integral which may be called  $c$ .

38. To obtain in this way a new integral, we require two integrals  $a, b$  other than  $H$ ; for knowing only the integrals  $a, H$ , the theorem gives only  $(a, H)$  an integral, and we have of course  $(a, H) = 0$ , viz. we do not obtain a new integral.

But starting from two integrals  $a, b$  other than  $H$ , we may obtain as above a new integral  $c$ ; and then again  $(a, c)$  and  $(b, c)$  will be integrals, one or both of which may be new. And it may therefore happen that in this way we obtain all the independent integrals  $a, b, c, d, e$ ; or the process may on the other hand terminate, without giving all the independent integrals.

The theory is obviously applicable throughout to the case of a bipartite set  $(x, y, z, \dots, p, q, r, \dots)$  of  $2n$  variables.

39. It may be remarked here that, in the Hamiltonian system, a value of the multiplier is  $M = 1$ ; and consequently, if in any way all but one of the integrals, that is,  $2n - 2$  integrals, be known, the remaining integral can be found by a quadrature.

It is further to be noticed that, if we adjoin a new variable  $t$  and a term  $= dt$  to the system of equations; then the  $2n - 1$  integrals of the original system being known, all the original variables can be expressed in terms of the  $2n - 1$  integrals regarded as constants and of one of the variables, say  $x$ : we then have

$$dt = dx \div \frac{dH}{dp},$$

or

$$t - \epsilon = \int dx \div \frac{dH}{dp},$$

or say

$$\epsilon = t - \int dx \div \frac{dH}{dp},$$

viz. if after the integration we suppose the  $2n - 1$  integrals replaced each of them by its value, we have

$$\epsilon = t - \phi(x, y, z, \dots, p, q, r, \dots),$$

which is the remaining or  $2n$ th integral of the original system as augmented by the term  $= dt$ .

*The Poisson-Jacobi theorem peculiar to the Hamiltonian Form.* Art. Nos. 40 to 45.

40. Taking for greater simplicity the set  $(x, y, p, q)$ , and writing

$$X dx + Y dy + P dp + Q dq = \xi d\rho + \eta d\sigma,$$

then the general system

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dp}{-X} = \frac{dq}{-Y}$$

becomes

$$\frac{\frac{dx}{\xi \frac{d\rho}{dp} + \eta \frac{d\sigma}{dp}}}{\frac{dy}{\xi \frac{d\rho}{dq} + \eta \frac{d\sigma}{dq}}} = \frac{\frac{dp}{-\left(\xi \frac{d\rho}{dx} + \eta \frac{d\sigma}{dx}\right)}}{\frac{dq}{-\left(\xi \frac{d\rho}{dy} + \eta \frac{d\sigma}{dy}\right)}},$$

and the corresponding partial differential equation ( $\theta$  the independent variable) is

$$\xi(\rho, \theta) + \eta(\sigma, \theta) = 0,$$

where

$$(\rho, \theta) = \frac{d(\rho, \theta)}{d(p, x)} + \frac{d(\rho, \theta)}{d(q, y)}, \quad (\sigma, \theta) = \frac{d(\sigma, \theta)}{d(p, x)} + \frac{d(\sigma, \theta)}{d(q, y)}.$$

It is to be shown that if  $a, b$  are solutions, viz. if we have

$$\xi(\rho, a) + \eta(\sigma, a) = 0,$$

$$\xi(\rho, b) + \eta(\sigma, b) = 0,$$

implying of course

$$(\rho, a)(\sigma, b) - (\rho, b)(\sigma, a) = 0,$$

then it is *not* in general true that we have  $(a, b)$  a solution; that is, not in general true that

$$\xi(\rho, (a, b)) + \eta(\sigma, (a, b)) = 0;$$

the condition for the truth of this equation is in fact  $\frac{\eta}{\xi}$  a function of  $\rho, \sigma$ , but when this is so,  $\xi d\rho + \eta d\sigma$  is  $\lambda dH$ , viz. there exist  $\lambda, H$  functions of  $\rho, \sigma$  (and therefore ultimately of  $x, y, p, q$ ) satisfying this equation, and the system is really Hamiltonian.

41. We consider whether it is true that

$$\xi(\rho, (a, b)) + \eta(\sigma, (a, b)) = 0.$$

We have identically

$$((a, b), \rho) + ((b, \rho), a) + ((\rho, a), b) = 0,$$

$$((a, b), \sigma) + ((b, \sigma), a) + ((\sigma, a), b) = 0,$$

so that multiplying by  $\xi, \eta$ , and adding, the equation in question is

$$\xi [((b, \rho), a) + ((\rho, a), b)] + \eta [((b, \sigma), a) + ((\sigma, a), b)] = 0.$$

But in virtue of the equations satisfied by  $a, b$ , we may write

$$(\rho, a) = l\eta, \quad (b, \rho) = -(\rho, b) = m\eta,$$

$$(\sigma, a) = -l\xi, \quad (b, \sigma) = -(\sigma, b) = -m\xi,$$

where  $l, m$  are indeterminate functions of  $x, y, p, q$ ; and the equation in question now becomes

$$\xi [-(m\eta, a) + (l\eta, b)] + \eta [(m\xi, a) - (l\xi, b)] = 0,$$

that is,

$$\begin{aligned} & \xi [-\eta (m, a) - m (\eta, a) + \eta (l, b) + l (\eta, b)] \\ & + \eta [\xi (m, a) + m (\xi, a) - \xi (l, b) - l (\xi, b)] = 0; \end{aligned}$$

viz. omitting the terms which destroy each other, this is

$$-m\xi (\eta, a) + l\xi (\eta, b) + m\eta (\xi, a) - l\eta (\xi, b) = 0.$$

Substituting for  $m\xi$ , &c., their values, we have

$$(\sigma, a)(\eta, a) - (\sigma, a)(\eta, b) + (\rho, b)(\xi, a) - (\rho, a)(\xi, b) = 0;$$

and the question is whether this is implied in the equations

$$\xi (\rho, a) + \eta (\sigma, a) = 0,$$

$$\xi (\rho, b) + \eta (\sigma, b) = 0.$$

42. Write  $\eta = \kappa\xi$ , the equation in question is

$$(\sigma, b)(\kappa\xi, a) - (\sigma, a)(\kappa\xi, b) + (\rho, b)(\xi, a) - (\rho, a)(\xi, b) = 0;$$

that is,

$$(\xi, a)[(\rho, b) + \kappa (\sigma, b)] - (\xi, b)[(\rho, a) + \kappa (\sigma, a)] + \xi [(\sigma, b)(\kappa, a) - (\sigma, a)(\kappa, b)] = 0;$$

viz. and we wish to see whether this is implied in

$$(\rho, a) + \kappa (\sigma, a) = 0,$$

$$(\rho, b) + \kappa (\sigma, b) = 0,$$

which give

$$(\sigma, b)(\rho, a) - (\sigma, a)(\rho, b) = 0;$$

or, what is the same thing, whether these last equations imply

$$(\sigma, b)(\kappa, a) - (\sigma, a)(\kappa, b) = 0.$$

Suppose  $\kappa$  is a function of  $\rho, \sigma$ , then, as is at once seen,

$$(\kappa, a) = \frac{d\kappa}{d\rho} (\rho, a) + \frac{d\kappa}{d\sigma} (\sigma, a),$$

and thence

$$(\sigma, b)(\kappa, a) - (\sigma, a)(\kappa, b) = \frac{d\kappa}{d\rho} [(\sigma, b)(\rho, a) - (\sigma, a)(\rho, b)];$$

viz.  $\kappa$  being a function of  $\rho$  and  $\sigma$ , the two equations imply the third.

43. But we wish to prove the converse, viz. that, if the two equations imply the third, then  $\kappa$  is a function of  $\rho, \sigma$ .

Now the equations

$$(\sigma, b)(\kappa, a) - (\sigma, a)(\kappa, b) = 0, \quad (\sigma, b)(\rho, a) - (\sigma, a)(\rho, b) = 0,$$

are transformable into

$$\begin{array}{ll} \frac{d(\sigma, \kappa)}{d(p, x)} \frac{d(b, a)}{d(p, x)} = 0, & \frac{d(\sigma, \rho)}{d(p, x)} \frac{d(b, a)}{d(p, x)} = 0, \\ + q, y, & + q, y, \\ + p, q, & + p, q, \\ + p, y, & + p, y, \\ + x, q, & + x, q, \\ + x, y, & + x, y, \end{array}$$

the lines after the first being the corresponding terms with  $q, y$ , &c. instead of  $p, x$ . And if independently of the values of  $a, b$ , one of these equations implies the other, we must have

$$\frac{d(\sigma, \kappa)}{d(p, x)}, \frac{d(\sigma, \kappa)}{d(q, y)}, \frac{d(\sigma, \kappa)}{d(p, q)}, \frac{d(\sigma, \kappa)}{d(p, y)}, \frac{d(\sigma, \kappa)}{d(x, q)}, \frac{d(\sigma, \kappa)}{d(x, y)},$$

proportional to the like expressions with  $\sigma, \rho$  instead of  $\sigma, \kappa$ ; say these are

$$\frac{d(\sigma, \kappa)}{d(p, x)} = \Lambda \frac{d(\sigma, \rho)}{d(p, x)}, \quad \&c.$$

44. Assume  $\kappa$  a function of  $\rho, \sigma, x, y$ : we have

$$\frac{d\kappa}{dp} = \frac{d\kappa}{d\rho} \frac{d\rho}{dp} + \frac{d\kappa}{d\sigma} \frac{d\sigma}{dp}, \quad \frac{d\kappa}{dx} = \frac{d\kappa}{d\rho} \frac{d\rho}{dx} + \frac{d\kappa}{d\sigma} \frac{d\sigma}{dx} + \frac{d\kappa}{dx};$$

the condition  $\frac{d(\sigma, \kappa)}{d(p, x)}$  becomes

$$\frac{d(\sigma, \kappa)}{d(p, x)} = \frac{d\kappa}{d\rho} \frac{d(\sigma, \rho)}{d(p, x)} + \frac{d\kappa}{dx} \frac{d\sigma}{dp},$$

and similarly

$$\frac{d(\sigma, \kappa)}{d(q, y)} = \frac{d\kappa}{d\rho} \frac{d(\sigma, \rho)}{d(q, y)} + \frac{d\kappa}{dy} \frac{d\sigma}{dq},$$

$$\frac{d(\sigma, \kappa)}{d(p, q)} = \frac{d\kappa}{d\rho} \frac{d(\sigma, \rho)}{d(p, q)},$$

$$\frac{d(\sigma, \kappa)}{d(p, y)} = \frac{d\kappa}{d\rho} \frac{d(\sigma, \rho)}{d(p, y)} + \frac{d\kappa}{dy} \frac{d\sigma}{dp},$$

$$\frac{d(\sigma, \kappa)}{d(x, q)} = \frac{d\kappa}{d\rho} \frac{d(\sigma, \rho)}{d(x, q)} - \frac{d\kappa}{dx} \frac{d\sigma}{dq},$$

$$\frac{d(\sigma, \kappa)}{d(x, y)} = \frac{d\kappa}{d\rho} \frac{d(\sigma, \rho)}{d(x, y)}.$$

Hence, unless  $\frac{d(\sigma, \rho)}{d(p, q)} = 0$ , we have  $\Lambda = \frac{d\kappa}{d\rho}$ . The remaining five equations then are

$$\frac{d\kappa}{dx} \frac{d\sigma}{dp} = 0, \quad \frac{d\kappa}{dy} \frac{d\sigma}{dq} = 0,$$

$$\frac{d\kappa}{dy} \frac{d\sigma}{dp} = 0, \quad \frac{d\kappa}{dx} \frac{d\sigma}{dq} = 0, \quad \frac{d\kappa}{dx} \frac{d\sigma}{dy} - \frac{d\kappa}{dy} \frac{d\sigma}{dx} = 0,$$

which give, and are satisfied by  $\frac{d\kappa}{dx} = 0$ ,  $\frac{d\kappa}{dy} = 0$ , viz. we then have  $\kappa$  a function of  $\rho, \sigma$  without  $x, y$ , which is the theorem in question.

45. The proof fails if  $\frac{d(\sigma, \rho)}{d(p, q)} = 0$ . But here, unless also  $\frac{d(\sigma, \rho)}{d(x, y)} = 0$ , we can, by assuming in the first instance  $\kappa$  a function of  $\rho, \sigma, p, q$ , prove in like manner that  $\kappa$  is a function of only  $\rho$  and  $\sigma$ : if however we have as well  $\frac{d(\sigma, \rho)}{d(p, q)} = 0$  and  $\frac{d(\sigma, \rho)}{d(x, y)} = 0$ , the last-mentioned process would also fail, but it can be shown the conclusion holds good in this case also; hence the conclusion that the Poisson-Jacobi theorem holds good only for a Hamiltonian system.

*Conjugate Integrals of the Hamiltonian System.* Art. Nos. 46 to 51.

46. For greater clearness, let  $n = 4$ , or let the variables be  $x, y, z, w, p, q, r, s$ ; the system of differential equations therefore is

$$\frac{dx}{dH} = \frac{dy}{dH} = \frac{dz}{dH} = \frac{dw}{dH} = \frac{dp}{dH} = \frac{dq}{dH} = \frac{dr}{dH} = \frac{ds}{dH};$$

and any integral hereof is as before a solution of  $(H, \theta) = 0$ . Assume that the integrals are  $H, a, b, c, d, e, f$ , so that

$$(H, a) = 0, \quad (H, b) = 0, \quad (H, c) = 0, \quad (H, d) = 0, \quad (H, e) = 0, \quad (H, f) = 0.$$

Considering here  $a$  as denoting any integral whatever, that is, any solution whatever of the partial differential equation  $(H, \theta) = 0$ , it is to be shown that it is possible to determine  $\theta$  so as to satisfy as well this equation  $(H, \theta) = 0$ , as also the new equation  $(a, \theta) = 0$ .

47. We, in fact, satisfy the first equation by taking

$$\theta = \theta(H, a, b, c, d, e, f),$$

any function whatever of the seven integrals. But, having this value, we find

$$(a, \theta) = (a, H) \frac{d\theta}{dH} + (a, a) \frac{d\theta}{da} + (a, b) \frac{d\theta}{db} + (a, c) \frac{d\theta}{dc} + (a, d) \frac{d\theta}{dd} + (a, e) \frac{d\theta}{de} + (a, f) \frac{d\theta}{df};$$

or, since the first two terms on the right-hand vanish, the equation  $(a, \theta) = 0$  thus becomes

$$(a, b) \frac{d\theta}{db} + (a, c) \frac{d\theta}{dc} + (a, d) \frac{d\theta}{dd} + (a, e) \frac{d\theta}{de} + (a, f) \frac{d\theta}{df} = 0.$$

But by the Poisson-Jacobi theorem  $(a, b)$ , &c., are each of them a solution of  $(H, \theta) = 0$ , viz. they are each of them a function of  $H, a, b, c, d, e, f$ . This is then a linear partial differential equation wherein the variables are  $H, a, b, c, d, e, f$ ; or, since there are no terms in  $\frac{d\theta}{dH}$ ,  $\frac{d\theta}{da}$ , we may regard  $H, a$ , as constants, and treat it as a linear partial differential equation in  $b, c, d, e, f$ , the solutions of the equation being in fact the integrals, or any functions of the integrals, of

$$\frac{db}{(a, b)} = \frac{dc}{(a, c)} = \frac{dd}{(a, d)} = \frac{de}{(a, e)} = \frac{df}{(a, f)}.$$

48. Suppose any four integrals are  $b', c', d', e'$ , so that a general integral is  $\theta(H, a, b', c', d', e')$ , then  $b', c', d', e'$  qua functions of  $H, a, b, c, d, e, f$  are integrals of the original equation  $(H, \theta) = 0$ ; hence changing the notation and writing  $b, c, d, e$  in place of these accented letters we have  $\theta(H, a, b, c, d, e)$  as solutions of the two equations  $(H, \theta) = 0$ ,  $(a, \theta) = 0$ ; viz.  $a$  being any integral of the first of these equations, we see how to find four other integrals  $(b, c, d, e)$  which are such that

$$\begin{aligned} (H, a) &= 0, & (H, b) &= 0, & (H, c) &= 0, & (H, d) &= 0, & (H, e) &= 0, \\ (a, b) &= 0, & (a, c) &= 0, & (a, d) &= 0, & (a, e) &= 0. \end{aligned}$$

49. We proceed in the same course and endeavour to find  $\theta$ , so that not only  $(H, \theta) = 0$ ,  $(a, \theta) = 0$ , but also  $(b, \theta) = 0$ . Assuming here  $\theta(H, a, b, c, d, e)$  an arbitrary function of the integrals, the first and second equations are satisfied; for the third equation, we have

$$(b, \theta) = (b, H) \frac{d\theta}{dH} + (b, a) \frac{d\theta}{da} + (b, b) \frac{d\theta}{db} + (b, c) \frac{d\theta}{dc} + (b, d) \frac{d\theta}{dd} + (b, e) \frac{d\theta}{de};$$

viz. the first three terms here vanish, and the equation  $(b, \theta) = 0$  becomes

$$(b, c) \frac{d\theta}{dc} + (b, d) \frac{d\theta}{dd} + (b, e) \frac{d\theta}{de} = 0,$$

where,  $b, c, d, e$  being solutions as well of  $(H, \theta) = 0$  as of  $(a, \theta) = 0$ , we have  $(b, c)$  a solution of these two equations, and as such a function of  $H, a, b, c, d, e$ ; and so  $(b, d)$  and  $(b, e)$  are each of them a function of the same variables. The above is therefore a linear partial differential equation wherein the variables are  $H, a, b, c, d, e$ , but as the equation does not contain  $\frac{d\theta}{dH}$ ,  $\frac{d\theta}{da}$ , or  $\frac{d\theta}{db}$ , we may regard  $H, a, b$ , as constants; and the solutions of the equation are, in fact, the integrals of

$$\frac{dc}{(b, c)} = \frac{dd}{(b, d)} = \frac{de}{(b, e)}.$$

50. Supposing that any two integrals are  $c', d'$ , so that a general integral is  $\theta(H, a, b, c', d')$ , then  $c', d'$  qua functions of  $H, a, b, c, d, e$  are integrals of the former equations  $(H, \theta) = 0, (a, \theta) = 0$ , so that again changing the notation, and writing  $c, d$  instead of the accented letters, we have  $\theta(H, a, b, c, d)$  as solutions of the three equations  $(H, \theta) = 0, (a, \theta) = 0, (b, \theta) = 0$ , viz.  $a$  being any solution of the first equation, and  $b$  any solution of the first and second equations, we see how to find two others  $c, d$ , of the same two equations, which are such that

$$\begin{aligned}(H, a) &= 0, \quad (H, b) = 0, \quad (H, c) = 0, \quad (H, d) = 0, \\ (a, b) &= 0, \quad (a, c) = 0, \quad (a, d) = 0, \\ (b, c) &= 0, \quad (b, d) = 0;\end{aligned}$$

or, attending only to the integrals  $H, a, b, c$ , these are integrals of the equations  $(H, \theta) = 0, (a, \theta) = 0, (b, \theta) = 0$ , such that

$$(H, a) = 0, \quad (H, b) = 0, \quad (H, c) = 0, \quad (a, b) = 0, \quad (a, c) = 0, \quad (b, c) = 0.$$

We here say that  $H, a, b, c$  are a system of conjugate solutions. Attempting to continue the process, it would appear that there is not any new independent integral  $d$ , such that  $(H, d) = 0, (a, d) = 0, (b, d) = 0, (c, d) = 0$  (the first three of these are satisfied by the integral  $d$  found above, but the last of them is not); we may, however, taking  $d$  an arbitrary function of  $H, a, b, c$ , replace  $H$  by  $d$ ; viz. we thus have the four integrals  $a, b, c, d$ , such that

$$(a, b) = 0, \quad (a, c) = 0, \quad (a, d) = 0, \quad (b, c) = 0, \quad (b, d) = 0, \quad (c, d) = 0,$$

and which are consequently said to form a conjugate system.

51. The process is of course general, and it shows how, in the case of a Hamiltonian system of  $2n$  variables, it is possible to find a system  $H, a, b, \dots, f$ , consisting of  $H$  and  $n - 1$  other integrals, or, if we please, a system of  $n$  integrals  $a, b, \dots, f, g$ , such that the derivative of any two integrals whatever of the system is  $= 0$ ; any such system is termed a conjugate system.

*Hamiltonian System — the function  $V$ . Art. Nos. 52 to 58.*

52. Taking a Hamiltonian system with the original variables  $x, y, z, p, q, r$ , we adjoin the two new variables  $t, V$ , forming the extended system

$$\frac{dx}{dp} = \frac{dy}{dq} = \frac{dz}{dr} = -\frac{dp}{dx} = -\frac{dq}{dy} = -\frac{dr}{dz} = dt = \frac{dV}{p \frac{dH}{dp} + q \frac{dH}{dq} + r \frac{dH}{dr}}.$$

Supposing the integrals of the original system to be  $a, b, c, d, e$ , we have  $H = H(a, b, c, d, e)$  a determinate function of these integrals; also an integral  $\tau = t - \phi(x, y, z, p, q, r)$  and an integral  $\lambda = V - \psi(x, y, z, p, q, r)$ ; these integrals, exclusive of the last of them, serve to express  $x, y, z, p, q, r$  as functions of  $a, b, c, d, e, t - \tau$ ; and the last integral then gives  $V = \lambda + a$  function of the last-mentioned quantities.



53. We consider the differential expression

$$dV - p dx - q dy - r dz,$$

which, treating the integrals as constants, that is, in the expressions of  $V, x, y, z$ , regarding  $t$  as the only variable, is at once seen to be  $= 0$ ; hence, if we regard all the integrals as variables, the value is

$$= d\lambda + A da + B db + C dc + D dd + E de,$$

without any term in  $d\tau$ , since this enters originally in the form  $dt - d\tau$ , and therefore disappears with  $dt$ .

The coefficients  $A, B, C, D, E$  are of course functions of  $a, b, c, d, e, t - \tau$ ; it is to be shown that they contain  $t - \tau$  linearly, viz. that in these coefficients respectively the coefficients of  $t - \tau$  are

$$\frac{dH}{da}, \quad \frac{dH}{db}, \quad \frac{dH}{dc}, \quad \frac{dH}{dd}, \quad \frac{dH}{de},$$

where  $H$  is expressed as above in the form  $H(a, b, c, d, e)$ ; this being so, the entire term in  $t - \tau$  will be  $(t - \tau) dH$ ; each coefficient, for instance  $A$ , has besides a part  $A'$ , which is a function of  $a, b, c, d, e$  without  $t - \tau$ , or changing the notation and writing the unaccented letters to denote these parts of the original coefficients, the final result is

$$dV - p dx - q dy - r dz = (t - \tau) dH + d\lambda + A da + B db + C dc + D dd + E de,$$

where  $H$  stands for its value  $H(a, b, c, d, e)$ , and  $A, B, C, D, E$  are functions of  $a, b, c, d, e$  without  $t - \tau$ .

54. To prove the theorem, we have

$$A = \frac{dV}{da} - p \frac{dx}{da} - q \frac{dy}{da} - r \frac{dz}{da},$$

and thence

$$\begin{aligned} \frac{dA}{dt} &= \frac{d^2V}{da dt} - \frac{dp}{dt} \frac{dx}{da} - \frac{dq}{dt} \frac{dy}{da} - \frac{dr}{dt} \frac{dz}{da} - p \frac{d^2x}{da dt} - q \frac{d^2y}{da dt} - r \frac{d^2z}{da dt} \\ &= \frac{d}{da} \left( \frac{dV}{dt} - p \frac{dx}{dt} - q \frac{dy}{dt} - r \frac{dz}{dt} \right) \\ &\quad + \frac{dp}{da} \frac{dx}{dt} + \frac{dq}{da} \frac{dy}{dt} + \frac{dr}{da} \frac{dz}{dt} - \frac{dp}{dt} \frac{dx}{da} - \frac{dq}{dt} \frac{dy}{da} - \frac{dr}{dt} \frac{dz}{da}; \end{aligned}$$

and then substituting for  $\frac{dV}{dt}$ , &c., their values from the system of differential equations, the first line vanishes, and the second line becomes

$$\begin{aligned} \frac{dH}{dp} \frac{dp}{da} + \frac{dH}{dq} \frac{dq}{da} + \frac{dH}{dr} \frac{dr}{da} - \frac{dH}{dx} \frac{dx}{da} - \frac{dH}{dy} \frac{dy}{da} - \frac{dH}{dz} \frac{dz}{da}, \\ = \frac{dH}{da}; \end{aligned}$$

and hence  $A = (t - \tau) \frac{dH}{da} + A'$ , and the like for the other coefficients  $B, C, D, E$ , which is the theorem in question.

55. We may have between two coefficients of the formula, for instance,  $D$  and  $E$ , a relation  $\frac{dD}{de} = \frac{dE}{dd}$ , and I will for the present assume, without proving it, the theorem that if  $a, b, c$  are conjugate integrals, then this relation  $\frac{dD}{de} - \frac{dE}{dd} = 0$ , holds good, merely mentioning that the proof depends on the consideration of certain symbols  $[a, b]$ , which are the converses, so to speak, of the symbols  $(a, b)$ , viz. considering the variables  $x, y, z, p, q, r$  as given functions of  $a, b, c, d, e, t - \tau$ , then we have

$$[a, b] = \frac{d(p, x)}{d(a, b)} + \frac{d(q, y)}{d(a, b)} + \frac{d(r, z)}{d(a, b)}.$$

The assumption is used only in the two following Nos. 56 and 57.

56. Supposing then that  $a, b, c$  are conjugate integrals, we have  $\frac{dD}{de} - \frac{dE}{dd} = 0$ , and there exists therefore  $\phi$ , a function of  $a, b, c, d, e$ , such that

$$d\phi = A' da + B' db + C' dc + D dd + E de,$$

( $A', B', C'$  functions of the same quantities  $a, b, c, d, e$ ), we have therefore

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH + d\phi + (A - A') da + (B - B') db + (C - C') dc.$$

Taking as above  $a, b$  conjugate integrals  $(a, b) = 0$ , and  $c$  any function whatever of  $a, b, H$ , then  $a, b, c$  are conjugate integrals, and the formula holds good. Suppose further that  $a, b, H$  are absolute constants, then  $dH = 0$ ,  $da = 0$ ,  $db = 0$ ,  $dc = 0$ , and the formula becomes

$$dV - p dx - q dy - r dz = d\lambda + d\phi;$$

or, writing this under the form,

$$p dx + q dy + r dz = dV - d\lambda - d\phi,$$

it follows that  $p dx + q dy + r dz$  is an exact differential, a theorem which may be stated as follows: viz. if  $a, b$  are conjugate integrals of the Hamiltonian system, and if from the equations  $H = \text{const.}$ ,  $a = \text{const.}$ ,  $b = \text{const.}$ , we express  $p, q, r$  as functions of  $x, y, z$ , then  $p dx + q dy + r dz$  is an exact differential; or, what is the same thing,  $p, q, r$  are the differential coefficients  $\frac{dU}{dx}, \frac{dU}{dy}, \frac{dU}{dz}$  of a function of  $x, y, z$ .

This is, in fact, a fundamental theorem in regard to the partial differential equation  $H = \text{const.}$ , and it will presently be proved in a different manner.

57. If, as before,  $a$  and  $b$  are conjugate integrals, then, writing as we may do  $V$  in place of  $\lambda + \phi$ , and finding  $H$  as a function of  $x, y, z, a, b$ , from the equation

$$V = \lambda + \int (p dx + q dy + r dz),$$

and again treating  $a, b, H$  as variable, we have

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH + A da + B db,$$

where  $A, B$  are functions of the integrals  $a, b, c, d, e$ , that is, they are themselves integrals, which may be taken for the integrals  $d, e$ , or we have

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH + d da + e db;$$

we have therefore

$$\frac{dV}{da} = d, \quad \frac{dV}{db} = e,$$

equations which, on substituting therein for  $a, b, H$  their values as functions of  $x, y, z, p, q, r$ , determine the integrals  $d, e$ , which with  $a, b, H$  or  $c$ , are the remaining integrals of the Hamiltonian system; and further

$$\frac{dV}{dH} = t - \tau,$$

which, when in like manner, we substitute therein for  $a, b, H$ , their values as functions of  $x, y, z, p, q, r$ , determines  $\tau$ , the remaining integral of the system as increased by the equality  $= dt$ .

58. Reverting to the general theorem No. 52, let  $x_0, y_0, z_0, p_0, q_0, r_0, t_0$  be corresponding values of the variables  $x, y, z, p, q, r, t$ ; and let  $a_0, \&c., \dots, V_0$  be the same functions of  $x_0, y_0, z_0, p_0, q_0, r_0, t_0$  that  $a, \&c., \dots, V$  are of the variables; we have  $a = a_0, \dots, e = e_0$ , and corresponding to the equation

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH + A da + \dots + E de,$$

the like equation

$$dV_0 - p_0 dx_0 - q_0 dy_0 - r_0 dz_0 = d\lambda + (t_0 - \tau) dH + A da + \dots + E de.$$

Hence, subtracting

$$dV - dV_0 = (t - t_0) dH + p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0,$$

or, considering only  $H$  as an absolute constant,

$$dV - dV_0 = p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0;$$

viz. if from the equations  $H = \text{const.}$ ,  $a = a_0, b = b_0, c = c_0, d = d_0, e = e_0$ , we express  $p, q, r, p_0, q_0, r_0$  as functions of  $x, y, z, x_0, y_0, z_0, H$ , then

$$p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0$$

will be an exact differential. And in particular regarding  $x_0, y_0, z_0$  as constants, then  $p dx + q dy + r dz$  is an exact differential, viz. there exists a function

$$V = \lambda + \int (p dx + q dy + r dz).$$

We have thus again arrived at a solution of the partial differential equation  $H = \text{const.}$

*The Partial Differential Equation  $H = \text{const.}$  Art. Nos. 59 to 70.*

59. In what just precedes we have, in fact, brought the theory of the Hamiltonian system into connexion with a partial differential equation, viz. we have determined the variables  $p, q, r$  as functions of  $x, y, z$  such that  $p dx + q dy + r dz$  is an exact differential  $= dV$ ; but we now consider the subject in a more regular manner.

The partial differential equation is  $H = \text{const.}$ , viz. here  $H$  denotes, in the first instance, a given function of  $p, q, r, x, y, z$ , where  $p, q, r$  are the differential coefficients of a function  $V$  of  $x, y, z$ , or, what is the same thing, there exists a function  $V$  of  $x, y, z$  such that  $p dx + q dy + r dz = dV$ ; and then, this function  $H$  being constant, we use the same letter  $H$  to denote the constant value of the function. The equation  $H = \text{const.}$  is the most general form of a partial differential equation of the first order which contains the independent variable only through its differential coefficients  $p, q, r$ , and it is for convenience put in a form containing the arbitrary constant  $H$ , which constant might without loss of generality be put  $= 0$  or  $=$  any other determinate value.

60. We seek to determine  $p, q, r$  as functions of  $x, y, z$ , satisfying the given equation  $H = \text{const.}$ , and such that we have  $p dx + q dy + r dz$  an exact differential  $= dV$ ; this would be done if we can find two other equations  $K = \text{const.}$  and  $L = \text{const.}$ , such that the values of  $p, q, r$  obtained from the three equations give  $p, q, r$  functions having the property in question. Attending to only two of the equations, say  $H = \text{const.}$  and  $K = \text{const.}$ , we have here  $p, q, r$  functions of  $x, y, z$ , such that  $p dx + q dy + r dz$  is an exact differential, and two of the equations which serve to determine  $p, q, r$  as functions of  $x, y, z$  are  $H = \text{const.}$ ,  $K = \text{const.}$  We have to prove the following fundamental theorem, viz. that  $(H, K) = 0$ .

61. In fact, from the equations  $H = \text{const.}$ ,  $K = \text{const.}$ , treating  $x, y, z$  as independent variables, we have

$$\begin{aligned}\frac{dH}{dx} + \frac{dH}{dp} \frac{dp}{dx} + \frac{dH}{dq} \frac{dq}{dx} + \frac{dH}{dr} \frac{dr}{dx} &= 0, \\ \frac{dK}{dx} + \frac{dK}{dp} \frac{dp}{dx} + \frac{dK}{dq} \frac{dq}{dx} + \frac{dK}{dr} \frac{dr}{dx} &= 0,\end{aligned}$$

and if from these equations in order to eliminate  $\frac{dp}{dx}$  we multiply by  $\frac{dK}{dp}$ ,  $-\frac{dH}{dp}$ , and add, we find

$$\frac{d(K, H)}{d(p, x)} + \frac{d(K, H)}{d(p, q)} \frac{dq}{dx} + \frac{d(K, H)}{d(p, r)} \frac{dr}{dx} = 0,$$

and, in precisely the same way,

$$\begin{aligned}\frac{d(K, H)}{d(q, y)} + \frac{d(K, H)}{d(q, p)} \frac{dp}{dy} + \frac{d(K, H)}{d(q, r)} \frac{dr}{dy} &= 0, \\ \frac{d(K, H)}{d(r, z)} + \frac{d(K, H)}{d(r, p)} \frac{dp}{dz} + \frac{d(K, H)}{d(r, q)} \frac{dq}{dz} &= 0.\end{aligned}$$

Adding these together, we have

$$(K, H) + \frac{d(K, H)}{d(q, r)} \left( \frac{dr}{dy} - \frac{dq}{dz} \right) + \frac{d(K, H)}{d(r, p)} \left( \frac{dp}{dz} - \frac{dr}{dx} \right) + \frac{d(K, H)}{d(p, q)} \left( \frac{dq}{dx} - \frac{dp}{dy} \right) = 0;$$

viz. if  $p dx + q dy + r dz$  be an exact differential, then  $(H, K) = 0$ , which is the theorem in question.

62. In the case where the variables are  $(x, y, p, q)$ , we have simply

$$(K, H) + \frac{d(K, H)}{d(p, q)} \left( \frac{dq}{dx} - \frac{dp}{dy} \right) = 0;$$

viz.  $p dx + q dy$  being a complete differential,  $(K, H) = 0$ . Conversely, if  $(K, H) = 0$ , then  $\frac{dq}{dx} - \frac{dp}{dy} = 0$ , and  $p dx + q dy$  is an exact differential; viz. this is so unless  $\frac{d(K, H)}{d(p, q)} = 0$ ; this equation would imply that  $K, H$ , considered as functions of  $p, q$ , are functions one of the other: and, supposing it to hold good, we could not from the equations  $H = 0, K = 0$  determine  $p, q$  as functions of  $x, y$ , for, eliminating one of the variables  $p, q$ , the other would disappear of itself. We hence obtain the complete statement of the converse theorem, viz. the functions  $H, K$  being such that it is possible from the equations  $H = 0, K = 0$  to express  $p, q$  as functions of  $x, y$ , then, if  $(H, K) = 0$ , we have  $p dx + q dy$  an exact differential.

63. Returning to the case of the variables  $(x, y, z, p, q, r)$ , if  $p, q, r$  are determined as functions of  $x, y, z$  by the three equations  $H = 0, K = 0, L = 0$ , then, by what precedes, in order that  $p dx + q dy + r dz$  may be a complete differential, we must have  $(H, K) = 0, (H, L) = 0, (K, L) = 0$ ; and it further appears that if these equations are satisfied, then we have, conversely,

$$\frac{dr}{dy} - \frac{dq}{dz} = 0, \quad \frac{dp}{dz} - \frac{dr}{dx} = 0, \quad \frac{dq}{dx} - \frac{dp}{dy} = 0,$$

that is,  $p dx + q dy + r dz$  is an exact differential; viz. this is the case unless we have between  $H, K, L$  the relation

$$\begin{vmatrix} \frac{d(H, K)}{d(q, r)}, & \frac{d(H, K)}{d(r, p)}, & \frac{d(H, K)}{d(p, q)} \\ H, L, & H, L, & H, L \\ K, L, & K, L, & K, L \end{vmatrix} = 0,$$

where in the determinant the second and third lines are the same functions of  $H, L$  and  $K, L$  respectively that the first line is of  $H, K$ .

The determinant is, in fact, equal to the square of

$$\frac{d(H, K, L)}{d(p, q, r)},$$

and, if it vanish, it is impossible, by means of the equations  $H = 0$ ,  $K = 0$ ,  $L = 0$ , to determine  $p, q, r$  as functions of  $x, y, z$ . Hence, if the last-mentioned equations are such that by means of them it is possible to effect the determination, and if, moreover,  $(H, K) = 0$ ,  $(H, L) = 0$ ,  $(K, L) = 0$ , then  $p dx + q dy + r dz$  will be an exact differential.

64. Considering  $H$  as given, we have, by what precedes,  $K, L$  solutions of the linear partial differential equation  $(H, \theta) = 0$ ; and since also  $K, L$  must be such that  $(K, L) = 0$ , they are conjugate solutions; or in conformity with what precedes, using the small letters  $a, b$  instead of  $K, L$ , we have the following theorem for the integration of the partial differential equation  $H = \text{const.}$ , where as before  $H$  is a given function of  $x, y, z, p, q, r$ .

Find  $a$  and  $b$ , such that  $H, a, b$  are a system of conjugate solutions of the linear partial differential equation  $(H, \theta) = 0$ : then from the equations  $H = \text{const.}$ ,  $a = \text{const.}$ ,  $b = \text{const.}$ , determining  $p, q, r$  as functions of  $a, b, H$ , and in the result treating these quantities as constants, we have  $p dx + q dy + r dz$  an exact differential  $= dV$ , and thence

$$V = \lambda + \int (p dx + q dy + r dz),$$

an expression for  $V$  containing the three arbitrary constants  $\lambda, a, b$ , and therefore a complete solution of the given partial differential equation  $H = \text{const.}$

The theorem applies to the case where  $n$  has any value whatever, viz. if there are  $n$  variables  $x, y, z, \dots$ , then we have to find the  $n - 1$  integrals  $a, b, c, \dots$  constituting with  $H$  a system of conjugate integrals; and the theorem holds good.

In particular, if  $n = 2$ , or the independent variables are  $x$  and  $y$ , then we find any solution  $a$  of the partial differential equation  $(H, \theta) = 0$ ; the values  $p, q$  derived from the equations  $H = \text{const.}$ ,  $a = \text{const.}$ , give  $V = \lambda + \int (p dx + q dy)$ , a complete solution.

65. But there is a different solution depending on the consideration of corresponding values; viz. if the independent variables be as before  $x, y, z, p, q, r$ , and if  $x_0, y_0, z_0, p_0, q_0, r_0$  are corresponding values of  $x, y, z, p, q, r$ , then, taking  $a, b, c, d, e$  to be integrals of  $(H, \theta) = 0$ : so that  $H$  is here a given function of  $a, b, c, d, e$ , since the number of independent variables is  $= 5$ : and representing by  $a_0, b_0, c_0, d_0, e_0$  the like functions of  $x_0, y_0, z_0, p_0, q_0, r_0$ , we form the equations

$$H = \text{const.}, \quad a = a_0, \quad b = b_0, \quad c = c_0, \quad d = d_0, \quad e = e_0.$$

We have the theorem that, expressing by means of these equations  $p, q, r$ , as functions of  $x, y, z, x_0, y_0, z_0, H$ , and regarding therein  $x_0, y_0, z_0$  as constants, we have  $p dx + q dy + r dz$  an exact differential, and therefore

$$V = \lambda + \int (p dx + q dy + r dz),$$

a solution of the equation  $H = \text{const.}$  involving the arbitrary constants  $\lambda, x_0, y_0, z_0$  (one more than required for a complete solution).

The theorem is here stated in the form proper for the solution of the partial differential equation  $H = \text{const.}$ ; a more general statement will be given further on.

66. I take first  $n = 2$ , or the independent variables to be  $x, y$ ; here  $p, q$  are determined by the equations  $H = \text{const.}$ ,  $a = a_0, b = b_0, c = c_0$ , and it is to be shown that  $p dx + q dy = dV$ .

Considering  $p, q, p_0, q_0$  as functions of the independent variables  $x, y$ , then differentiating in regard to  $x$ , and eliminating  $\frac{dp_0}{dx}, \frac{dq_0}{dx}$ , we find

$$\begin{vmatrix} \frac{da}{dx} + \frac{da}{dq} \frac{dq}{dx}, & \frac{da}{db}, & \frac{da}{dp_0}, & \frac{da}{dq_0} \\ \frac{db}{dx} + \frac{db}{dq} \frac{dq}{dx}, & \frac{db}{dp}, & \frac{db}{dp_0}, & \frac{db}{dq_0} \\ \frac{dc}{dx} + \frac{dc}{dq} \frac{dq}{dx}, & \frac{dc}{dp}, & \frac{dc}{dp_0}, & \frac{dc}{dq_0} \\ \frac{dH}{dx} + \frac{dH}{dq} \frac{dq}{dx}, & \frac{dH}{dp}, & 0, & 0 \end{vmatrix} = 0,$$

viz. this is

$$\frac{d(a_0, b_0)}{d(p_0, q_0)} \left\{ \frac{d(H, c)}{d(p, x)} + \frac{d(H, c)}{d(p, q)} \frac{dq}{dx} \right\} + \&c. = 0.$$

But in the same way

$$\frac{d(a_0, b_0)}{d(p_0, q_0)} \left\{ \frac{d(H, c)}{d(q, y)} + \frac{d(H, c)}{d(q, p)} \frac{dp}{dy} \right\} + \&c. = 0;$$

adding these two equations we have

$$\frac{d(a_0, b_0)}{d(p_0, q_0)} \left\{ (H, c) + \frac{d(H, c)}{d(p, q)} \left( \frac{dq}{dx} - \frac{dp}{dy} \right) \right\} + \&c. = 0,$$

the terms denoted by the  $\&c.$  being the like terms with  $b, c, a$  and  $c, a, b$  in place of  $a, b, c$ . We have  $(H, a) = 0, (H, b) = 0, (H, c) = 0$ , and the equation, in fact, is

$$\left\{ \Sigma \frac{d(a_0, b_0)}{d(p_0, q_0)} \frac{d(H, c)}{d(p, q)} \right\} \left( \frac{dq}{dx} - \frac{dp}{dy} \right) = 0;$$

viz. we have  $\frac{dq}{dx} - \frac{dp}{dy} = 0$ , the condition for an exact differential.

67. Coming now to the case where the independent variables are  $x, y, z$ , we proceed in the same way with the equations  $H = \text{const.}$ ,  $a = a_0, b = b_0, c = c_0, d = d_0, e = e_0$ . Differentiating in regard to  $x$ , and eliminating

$$\frac{dp}{dx}, \frac{dq}{dx}, \frac{dp_0}{dx}, \frac{dq_0}{dx}, \frac{dr_0}{dx},$$

we find for  $\frac{dr}{dx}$  the equation

$$\frac{d(c_0, d_0, e_0)}{d(p_0, q_0, r_0)} \left\{ \frac{dr}{dx} \frac{d(a, b, H)}{d(r, p, q)} + \frac{d(a, b, H)}{d(x, p, q)} \right\} + \&c. = 0.$$

We have in the same way for  $\frac{dp}{dz}$  the equation

$$\frac{d(c_0, d_0, e_0)}{d(p_0, q_0, r_0)} \left\{ \frac{dp}{dz} \frac{d(a, b, H)}{d(p, r, q)} + \frac{d(a, b, H)}{d(z, r, q)} \right\} + \&c. = 0;$$

whence, adding, we obtain

$$\left( \frac{dr}{dx} - \frac{dp}{dz} \right) \frac{d(a, b, H)}{d(r, p, q)} + \frac{d(a, b, H)}{d(x, p, q)} + \frac{d(a, b, H)}{d(r, z, q)} + \&c. = 0,$$

where the terms denoted by the &c. are the like terms corresponding to the different permutations of the letters  $a, b, c, d, e$ .

The equation may be simplified; we have identically

$$\frac{d}{dq}(a, b) - \frac{dH}{dq}(H, a) - \frac{dH}{dq}(a, b) = \frac{d(a, b, H)}{d(x, p, q)} + \frac{d(a, b, H)}{d(z, r, q)};$$

or, since  $(H, a) = 0$ ,  $(H, b) = 0$ , the left-hand side is simply  $\frac{dH}{dq}(a, b)$ , and the equation becomes

$$\frac{d(c_0, d_0, e_0)}{d(p_0, q_0, r_0)} \left\{ \left( \frac{dr}{dx} - \frac{dp}{dz} \right) \frac{d(a, b, H)}{d(r, p, q)} + \frac{dH}{dq}(a, b) \right\} + \&c. = 0.$$

68. This ought to give  $\frac{dr}{dx} = \frac{dp}{dz}$ ; it will, if only

$$\Sigma \left\{ \frac{d(c_0, d_0, e_0)}{d(p_0, q_0, r_0)} (a, b) \right\} = 0,$$

which is thus the condition which has to be proved. By the Poisson-Jacobi theorem,  $(a, b)$  is a function of  $a, b, c, d, e$ : if we write

$$(a_0, b_0) = \frac{d(a_0, b_0)}{d(p_0, x_0)} + \frac{d(a_0, b_0)}{d(q_0, y_0)} + \frac{d(a_0, b_0)}{d(r_0, z_0)},$$

then  $(a_0, b_0)$  is the same function of  $a_0, b_0, c_0, d_0, e_0$ ; but these are equal to  $a, b, c, d, e$  respectively, and we then have  $(a, b) = (a_0, b_0)$ , and the theorem to be proved is

$$\Sigma \left\{ \frac{d(c_0, d_0, e_0)}{d(p_0, q_0, r_0)} (a_0, b_0) \right\} = 0.$$

But, substituting for  $(a_0, b_0)$  its value, the function on the left-hand side is, it is easy to see, the sum of the three functional determinants

$$\frac{d(a_0, b_0, c_0, d_0, e_0)}{d(p_0, q_0, r_0, p_0, x_0)}, \quad \frac{d(a_0, b_0, c_0, d_0, e_0)}{d(p_0, q_0, r_0, q_0, y_0)}, \quad \frac{d(a_0, b_0, c_0, d_0, e_0)}{d(p_0, q_0, r_0, r_0, z_0)},$$



and each of these, as containing the same letter twice in the denominator, that is, as having two identical columns, is = 0; the theorem is thus proved. And in the same way  $\frac{dp}{dy} - \frac{dq}{dx}$ ,  $\frac{dq}{dz} - \frac{dr}{dy}$  are each = 0; that is,  $p dx + q dy + r dz = dV$ .

69. The proof would fail if the factors multiplying  $\frac{dr}{dx} - \frac{dp}{dz}$ , &c., or any one of these factors, were = 0. I have not particularly examined this, but the meaning must be that here the equations  $a = a_0$ , &c.,  $H = \text{const.}$ , fail to give for  $p, q, r$  expressions as functions of  $x, y, z, x_0, y_0, z_0, H$ ; whenever such expressions are obtainable, we have

$$p dx + q dy + r dz = dV.$$

The proof in the case of a greater number of variables, say in the next case where the independent variables are  $x, y, z, w$ , would probably present greater difficulty, but I have not examined this.

70. Taking the independent variables to be  $x$  and  $y$ , we may from the equations  $a = a_0, b = b_0, c = c_0, H = \text{const.}$  (which last equation may also be written  $H_0 = \text{const.}$ ) find  $p, q, p_0, q_0$  as functions of  $x, y, x_0, y_0, H$ ; and we have then the theorem that, considering only  $H$  as a constant,

$$p dx + q dy - p_0 dx_0 - q_0 dy_0 = dV.$$

To show this, we have to prove the further equations  $\frac{dp}{dx_0} + \frac{dp_0}{dx} = 0$ , &c.; we find

$$\begin{aligned} \frac{dp}{dx_0} \left\{ \frac{d(b_0, c_0)}{d(p_0, q_0)} \frac{d(a, H)}{d(p, q)} \right\} - \frac{dH}{dp} \frac{d(a_0, b_0, c_0)}{d(x_0, p_0, q_0)} &= 0, \\ \frac{dp_0}{dx} \left\{ \frac{d(b, c)}{d(p, q)} \frac{d(a_0, H_0)}{d(p_0, q_0)} \right\} - \frac{dH_0}{dq_0} \frac{d(a, b, c)}{d(x, p, q)} &= 0, \end{aligned}$$

and it is to be shown that the coefficients of  $\frac{dp}{dx_0}, \frac{dp_0}{dx}$  are equal and of opposite signs, and that the other two terms are equal; viz. this being so, subtracting the two equations, we have the required relation  $\frac{dp}{dx_0} + \frac{dp_0}{dx} = 0$ . Now  $H, H_0$  are the same functions of  $a, b, c$  and of  $a_0, b_0, c_0$ , and there is no real loss of generality in assuming  $c = H, c_0 = H_0$ ; but this being so, the first coefficient is

$$\frac{d(b_0, H_0)}{d(p_0, q_0)} \frac{d(a, H)}{d(p, q)} = \frac{d(H_0, a_0)}{d(p_0, q_0)} \frac{d(b, H)}{d(p, q)},$$

and the second is

$$\frac{d(b, H)}{d(p, q)} \frac{d(a_0, H_0)}{d(p_0, q_0)} = \frac{d(H, a)}{d(p, q)} \frac{d(b_0, H_0)}{d(p_0, q_0)},$$

which only differ by their signs. As regards the other two terms, we have identically

$$\frac{dH}{dp} (b, H) + \frac{dH}{dq} (H, a) + \frac{dH}{dr} (a, b) = \frac{d(a, b, H)}{d(x, p, q)},$$

which, in virtue of  $(a, H) = 0$ ,  $(b, H) = 0$ , becomes

$$\frac{dH}{dr}(a, b) = \frac{d(a, b, H)}{d(x, p, q)};$$

similarly,

$$\frac{dH_0}{dr_0}(a_0, b_0) = \frac{d(a_0, b_0, H_0)}{d(x_0, p_0, q_0)}.$$

Hence the terms in question are

$$-\frac{dH}{dr}(a, b), \quad -\frac{dH_0}{dr_0}(a_0, b_0),$$

which are equal in virtue of  $(a, b) = (a_0, b_0)$ ; and, similarly, the other conditions might be proved. But the proof would be more difficult in the case of a greater number of variables.

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71. The variables are taken to be  $x, y, z, p, q, r$ . As a first example, which will serve as an illustration of most of the preceding theorems, suppose  $pqr - 1 = H$ ; the Hamiltonian system, with the adjoined equalities, is here

$$\frac{dx}{qr} = \frac{dy}{rp} = \frac{dz}{pq} = \frac{dp}{0} = \frac{dq}{0} = \frac{dr}{0} = dt = \frac{dV}{3pqr}.$$

The integrals of the original system may be taken to be

$$\begin{aligned} a &= p, \\ b &= q, \\ c &= r, \\ d &= qy - px, \\ e &= rz - px, \end{aligned}$$

and there is of course the integral  $H = pqr - 1$ , which is connected with the foregoing five integrals by the relation  $H = abc - 1$ .

We form at once the equations

$$\begin{aligned} (a, b) &= 0, \quad (a, c) = 0, \quad (a, d) = -a, \quad (a, e) = -a, \\ (b, c) &= 0, \quad (b, d) = b, \quad (b, e) = 0, \\ (c, d) &= 0, \quad (c, e) = c, \\ (d, e) &= 0; \end{aligned}$$

hence it happens that no two of these integrals  $a, b, c, d, e$  give by the Poisson-Jacobi theorem a new integral. To show how the theorem might have given a new integral, suppose that the known integrals had been  $a = p + q$ , and  $e = rz - px$ , then  $(a, e) = -p$ : or the theorem gives the new integral  $a = p$ .

We have as a conjugate system  $a, b, c$ ; also the conjugate systems  $H, a, b$ ;  $H, a, c$ ;  $H, b, c$ ;  $H, b, e$ ;  $H, c, d$ ;  $H, d, e$ ; but the first three of these, considering therein  $H$  as standing for its value  $abc - 1$ , are substantially equivalent to the first-mentioned system  $(a, b, c)$ .

72. Postponing the consideration of the augmented system, we now consider the partial differential equation  $pqr = 1 + H$ , where  $H$  is a given constant and  $p, q, r$  denote the differential coefficients of a function  $V$ . The most simple solution is that given by the conjugate system  $H, a, b$ , viz. here  $p, q, r$  are determined by the equations  $p = a$ ,  $q = b$ ,  $pqr = 1 + H$ , that is,  $r = \frac{1+H}{ab}$ ; or, introducing for symmetry the constant  $c$ , where  $abc = 1 + H$  as before, then  $r = c$ , and we have

$$V = \lambda + \int (a dx + b dy + c dz) = \lambda + ax + by + cz,$$

where  $a, b, c$  are connected by the just-mentioned equation  $abc = 1 + H$ . This is therefore a solution containing say the arbitrary constants  $\lambda, a, b$ , and, as such, is a complete solution.

But any other conjugate system gives a complete solution, and a very elegant one is obtained from the system  $H, d, e$ . Writing for symmetry  $\beta - \alpha$ ,  $\gamma - \alpha$  in place of  $d, e$ , we have here to find  $p, q, r$  from the equations

$$H = pqr - 1, \quad qy - px = \beta - \alpha, \quad rz - px = \gamma - \alpha;$$

or, if we assume  $\theta = px - \alpha$ , then

$$H = pqr - 1; \quad px, qy, rz = \theta + \alpha, \theta + \beta, \theta + \gamma,$$

respectively, whence

$$(1 + H)xyz = (\theta + \alpha)(\theta + \beta)(\theta + \gamma),$$

which equation determines  $\theta$  as a function of  $x, y, z$  (in fact, it is a function of the product  $xyz$ ), and then

$$p, q, r = \frac{\theta + \alpha}{x}, \quad \frac{\theta + \beta}{y}, \quad \frac{\theta + \gamma}{z},$$

and we have

$$V = \lambda + \int \left( \frac{\theta + \alpha}{x} dx + \frac{\theta + \beta}{y} dy + \frac{\theta + \gamma}{z} dz \right).$$

There is no difficulty in effecting the integration directly by introducing  $\theta$  as a new variable, and we find

$$V = \lambda + 3\theta - \alpha \log \frac{\theta + \alpha}{x} - \beta \log \frac{\theta + \beta}{y} - \gamma \log \frac{\theta + \gamma}{z}.$$

Or, starting from this form, we may verify it by differentiation; the value of  $dV$  is

$$3 d\theta - \left( \frac{\alpha}{\theta + \alpha} + \frac{\beta}{\theta + \beta} + \frac{\gamma}{\theta + \gamma} \right) d\theta + \frac{\alpha dx}{x} + \frac{\beta dy}{y} + \frac{\gamma dz}{z},$$

where the term in  $d\theta$  is

$$= \theta d\theta \left( \frac{1}{\theta + \alpha} + \frac{1}{\theta + \beta} + \frac{1}{\theta + \gamma} \right),$$

which, from the equation which determines  $\theta$ , is

$$= \theta \left( \frac{dx}{x} + \frac{dy}{y} + \frac{dz}{z} \right),$$

and the value of  $dV$  is thus

$$= \frac{\theta + \alpha}{x} dx + \frac{\theta + \beta}{y} dy + \frac{\theta + \gamma}{z} dz.$$

The solution contains apparently the four constants  $\lambda, \alpha, \beta, \gamma$ , but there is no loss of generality in writing, for instance,  $\alpha = 0$ , and the number of constants really contained in the solution may be regarded as 3.

73. To show how the equations  $H = \text{const.}$ ,  $a = a_0$ ,  $b = b_0$ ,  $c = c_0$ ,  $d = d_0$ ,  $e = e_0$  give a solution; remarking that these equations are  $pqr - 1 = H$ ,  $p = p_0$ ,  $q = q_0$ ,  $r = r_0$ ,  $qy - px = q_0y_0 - p_0x_0$ ,  $rz - px = r_0z_0 - p_0x_0$ , we find

$$p(x - x_0) = q(y - y_0) = r(z - z_0),$$

and consequently  $p, q, r =$

$$\frac{\sqrt[3]{(1+H)(y-y_0)(z-z_0)}}{(x-x_0)^{\frac{2}{3}}}, \quad \frac{\sqrt[3]{(1+H)(z-z_0)(x-x_0)}}{(y-y_0)^{\frac{2}{3}}}, \quad \frac{\sqrt[3]{(1+H)(x-x_0)(y-y_0)}}{(z-z_0)^{\frac{2}{3}}},$$

respectively: whence

$$\begin{aligned} V &= \lambda + \int (p dx + q dy + r dz), \\ &= \lambda + 3 \sqrt[3]{(1+H)(x-x_0)(y-y_0)(z-z_0)}, \end{aligned}$$

which is the solution involving the four constants  $\lambda, x_0, y_0, z_0$ .

If in the foregoing value of  $V$  we consider  $x_0, y_0, z_0$  as variables, then  $p, q, r$  having the values just mentioned, and  $p_0, q_0, r_0$  being equal to these respectively, we obviously have

$$dV = p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0.$$

74. Considering now the augmented Hamiltonian system, we join to the foregoing integrals  $a, b, c, d, e$ , the new integrals  $\epsilon = t - \frac{V}{3px}$  and  $\lambda = V - 3px$ . And then expressing all the quantities in terms of  $t - \tau$ ,

$$\begin{aligned} x &= bc(t - \tau), \\ y &= ca(t - \tau) + \frac{d}{b}, \\ z &= ab(t - \tau) + \frac{e}{c}, \\ p &= a, \quad q = b, \quad r = c, \quad H = abc - 1, \\ V &= \lambda + 3abc(t - \tau). \end{aligned}$$

Forming from these the expression for  $dV - p dx - q dy - r dz$ , the term in  $dt - d\tau$  disappears; there is a term in  $t - \tau$ , the coefficient of which is

$$3d.abc - ad.bc - bd.ca - cd.ab,$$

which is  $= d.abc$ , or the term is  $(t - \tau) dH$ ; and we have, finally,

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH - bd \frac{d}{b} - cd \frac{e}{c},$$

viz.  $t$  enters only in the combination  $(t - \tau) dH$ , which is the fundamental theorem. Considering  $H$  as a determinate constant, this term disappears.

We may show how this formula leads to the solution of the partial differential equation  $pqr = 1 + H$ ; treating  $H$  as a definite constant, then in order that the formula may give  $dV - p dx - q dy - r dz = d\lambda$ , or  $V = \lambda + \int (p dx + q dy + r dz)$ , as before, the last two terms of the formula must disappear; this will be the case if  $\frac{d}{b}, \frac{e}{c}$  are constants, or, say,  $d = b\beta, e = c\gamma$ ,  $\beta$  and  $\gamma$  being constants. But, this being so, we have  $q\beta = qy - px, \gamma r = rz - px$ , that is,  $px = q(y - \beta) = r(z - \gamma)$ ,  $pqr = 1 + H$ , giving the values of  $p, q, r$ ; and then

$$V = \lambda + \int (p dx + q dy + r dz), = \lambda + 3\sqrt[3]{(1 + H) x^{\frac{1}{3}}(y - \beta)^{\frac{1}{3}}(z - \gamma)^{\frac{1}{3}}},$$

which is substantially the same solution as is obtained above by a different process.

Or, again, observing that we have

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH - dd - de + \frac{d}{b} db + \frac{e}{c} dc,$$

then, taking  $H, b, c$  constants, we have

$$dV - p dx - q dy - r dz = d\lambda - dd - de,$$

which, changing the value of  $\lambda$ , gives the before-mentioned solution

$$V = \lambda + ax + by + cz, \quad (abc = 1 + H).$$

75. As a second example, suppose

$$\frac{1}{2}(p^2 + q^2 + r^2 - x^2 - y^2 - z^2) = H;$$

the augmented system is

$$\frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r} = \frac{dp}{x} = \frac{dq}{y} = \frac{dr}{z} = dt = \frac{dV}{p^2 + q^2 + r^2},$$

corresponding to the dynamical problem of the motion of a particle acted upon by a repulsive central force equal to the distance.

The integrals of the original system may be expressed in various forms, viz. the quotient of any two of the expressions  $x + p, y + q, z + r$ , or of any two of the

expressions  $x - p, y - q, z - r$  is an integral, or again the product of any expression of the first set into any expression of the second set is an integral: we may take as integrals

$$\alpha = x^2 - p^2, \quad \beta = y^2 - q^2, \quad \gamma = z^2 - r^2, \quad \delta = \frac{y + q}{x + p}, \quad \epsilon = \frac{z + r}{x + p}.$$

We have then

$$dt = \frac{dx}{\sqrt{(x^2 - \alpha)}}, \quad \text{that is,} \quad t - \tau = \log \left[ x + \sqrt{(x^2 - \alpha)} \right] = \log(x + p),$$

giving  $x + p = e^{t-\tau}$ , and thence the other quantities  $x - p, y + q$ , &c. For greater symmetry, I introduce a new set of constants  $a, b, c, a', b', c'$ , and I write also  $e^{t-\tau} = T, e^{-t+\tau} = T'$  (where  $TT' = 1$ ). We then have

$$\begin{aligned} x &= aT + a'T', & p &= aT - a'T', \\ y &= bT + b'T', & q &= bT - b'T', \\ z &= cT + c'T', & r &= cT - c'T'; \end{aligned}$$

also, comparing with the values obtained as above,

$$\alpha = \frac{1}{4}, \quad b = \frac{1}{4} \frac{\beta}{\delta}, \quad c = \frac{1}{4} \frac{\gamma}{\epsilon}.$$

We have, moreover,

$$H = -2(aa' + bb' + cc') = -\frac{1}{2}(1 + \beta + \gamma).$$

76. We find

$$p^2 + q^2 + r^2 = H + (a^2 + b^2 + c^2) T^2 + (a'^2 + b'^2 + c'^2) T'^2,$$

and thence

$$\begin{aligned} V &= \lambda + \int (p^2 + q^2 + r^2) dt \\ &= \lambda + H(t - \tau) - \frac{1}{2}(a^2 + b^2 + c^2) T^2 - \frac{1}{2}(a'^2 + b'^2 + c'^2) T'^2. \end{aligned}$$

We may from this obtain the expression for

$$dV - p dx - q dy - r dz,$$

when everything is variable. The terms in  $(dt - d\tau)$ , as is obvious, disappear; omitting these from the beginning, we have

$$dV = d\lambda + (t - \tau) dH + (a da + b db + c dc) T^2 - (a' da' + b' db' + c' dc') T'^2;$$

also

$$\begin{aligned} p dx &= (aT - a'T')(T da + T' da'), \\ &= da(aT^2 - a') + da'(-aT'^2 + a); \end{aligned}$$

thence forming the analogous expressions for  $q dy$  and  $r dz$ , we have

$$\begin{aligned} p dx + q dy + r dz &= (a da + b db + c dc) T^2 - (a' da' + b' db' + c' dc') T'^2 \\ &\quad - (a' da + b' db + c' dc) + (a da' + b db' + c dc'); \end{aligned}$$

whence

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH + a' da + b' db + c' dc - a da' - b db' - c dc';$$

or, in place of  $a, b, c, a', b', c'$ , introducing  $\alpha, \beta, \gamma, \delta, \epsilon$ , and attending to the value of  $H$ ,

$$dV - p dx - q dy - r dz = d\lambda + (t - \tau) dH + \frac{1}{2} dH + \frac{1}{2} \frac{\beta}{\delta} d\delta + \frac{1}{2} \frac{\gamma}{\epsilon} d\epsilon.$$

77. Suppose  $H, \delta, \epsilon$  absolute constants, this becomes

$$d(V - \lambda) = p dx + q dy + r dz,$$

or

$$V = \lambda + \int (p dx + q dy + r dz),$$

and we have thus a solution of the partial differential equation

$$p^2 + q^2 + r^2 = x^2 + y^2 + z^2 + 2H;$$

viz.  $p, q, r$  are here to be determined as functions of  $x, y, z$  by the equations

$$\begin{aligned} p^2 + q^2 + r^2 &= x^2 + y^2 + z^2 + 2H, \\ y + q &= \delta(x + p), \\ z + r &= \epsilon(x + p). \end{aligned}$$

We have

$$2H + x^2 + y^2 + z^2 = p^2 + [y - \delta(x + p)]^2 + [z - \epsilon(x + p)]^2;$$

or, on the right-hand side, writing  $p^2 = (x + p)^2 - 2x(x + p) + x^2$ ,

$$\text{left } , , , \quad x^2 = (x - p)^2 - 2x(x + p) + p^2,$$

the equation is

$$(1 + \delta^2 + \epsilon^2)(x + p)^2 - 2(x + \delta y + \epsilon z)(x + p) - 2H = 0,$$

which gives  $p$  as a function of  $x, y, z$ . But the result is a complicated one, except in the case  $H = 0$ ; we then have

$$\begin{aligned} x + p &= \frac{2(x + \delta y + \epsilon z)}{1 + \delta^2 + \epsilon^2}, \\ y + q &= \frac{2\delta(x + \delta y + \epsilon z)}{1 + \delta^2 + \epsilon^2}, \\ z + r &= \frac{2\epsilon(x + \delta y + \epsilon z)}{1 + \delta^2 + \epsilon^2}, \end{aligned}$$

and thence

$$V = \lambda - \frac{1}{2}(x^2 + y^2 + z^2) + \frac{(x + \delta y + \epsilon z)^2}{1 + \delta^2 + \epsilon^2},$$

a complete solution of the partial differential equation

$$p^2 + q^2 + r^2 = x^2 + y^2 + z^2.$$

# The Collected Mathematical Papers of Arthur Cayley

Vol. X, Pages 151–175 (PDF)

Papers 655 (concl.), 656, 657, 658, 659, 660, 661 (begin)

Arthur Cayley

Cambridge, 1896

## 655 (continued): A MEMOIR ON DIFFERENTIAL EQUATIONS.

More symmetrically, we have the solution

$$V = \lambda - \frac{1}{2}(x^2 + y^2 + z^2) + \frac{(ax + by + cz)^2}{a^2 + b^2 + c^2},$$

as can be at once verified.

78. In the same particular case  $H = 0$ , introducing the corresponding values  $V_0, p_0, q_0, r_0, x_0, y_0, z_0$ , we find a very simple expression for  $V_0$ , as a function of  $x, y, z, x_0, y_0, z_0$ . We have, writing  $T = e^{\sigma+\sigma'}$ ,  $T_0 = e^{-\sigma+\sigma'}$ , and therefore  $T_0 T'_0 = 1$ ,

$$\begin{aligned} x - x_0 &= a(T - T_0) + a'(T' - T'_0), \\ y - y_0 &= b(T - T_0) + b'(T' - T'_0), \\ z - z_0 &= c(T - T_0) + c'(T' - T'_0). \end{aligned}$$

Forming the analogous quantities  $y - y_0$ , &c., we deduce

$$\begin{aligned} (x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 &= (T - T_0)^2 \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2) \frac{1}{T^2 T_0^2} \right\}, \\ (x + x_0)^2 + (y + y_0)^2 + (z + z_0)^2 &= (T + T_0)^2 \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2) \frac{1}{T^2 T_0^2} \right\}. \end{aligned}$$

But we have

$$\begin{aligned} V - V_0 &= \frac{1}{4} \{ (a^2 + b^2 + c^2)(T^2 - T_0^2) + (a'^2 + b'^2 + c'^2)(T'^2 - T_0'^2) \}, \\ T^2 - T_0^2 &= \frac{1}{4} (T^2 - T_0^2) \left\{ a^2 + b^2 + c^2 + (a'^2 + b'^2 + c'^2) \frac{1}{T^2 T_0^2} \right\}, \end{aligned}$$

and hence the required formula

$$V - V_0 = \frac{1}{4} \sqrt{(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2} \sqrt{(x + x_0)^2 + (y + y_0)^2 + (z + z_0)^2},$$

or, say, for shortness,

$$V - V_0 = \frac{1}{4} \sqrt{(R)} \sqrt{(S)}.$$

79. We ought, therefore, to have

$$\frac{1}{4} d \left[ \sqrt{(R)} \sqrt{(S)} \right] = p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0,$$

where  $p, q, r, p_0, q_0, r_0$  denote as above, and consequently

$$p^2 + q^2 + r^2 = x^2 + y^2 + z^2, \quad p_0^2 + q_0^2 + r_0^2 = x_0^2 + y_0^2 + z_0^2.$$



We have in fact

$$p = \frac{1}{4} \left\{ \frac{\sqrt{(R)}}{\sqrt{(S)}}(x - x_0) + \frac{\sqrt{(S)}}{\sqrt{(R)}}(x + x_0) \right\}, \quad \&c.,$$

$$p_0 = \frac{1}{4} \left\{ -\frac{\sqrt{(R)}}{\sqrt{(S)}}(x - x_0) + \frac{\sqrt{(S)}}{\sqrt{(R)}}(x + x_0) \right\}, \quad \&c.;$$

and thence

$$p^2 + q^2 + r^2 = \frac{1}{16} \{ R + S + 2(x^2 + y^2 + z^2 - x_0^2 - y_0^2 - z_0^2) \} = x^2 + y^2 + z^2,$$

$$p_0^2 + q_0^2 + r_0^2 = \frac{1}{16} \{ R + S - 2(x^2 + y^2 + z^2 - x_0^2 - y_0^2 - z_0^2) \} = x_0^2 + y_0^2 + z_0^2.$$

or the last-mentioned results are thus verified.

*Partial Differential Equation containing the Dependent Variable: Reduction to Standard Form.*

Art. Nos. 80, 81.

80. The equation  $H = \text{const.}$  is the most general form of a partial differential equation not containing the dependent variable  $V$ ; but if a partial differential equation does contain the independent variable, we can, by regarding this as one of the dependent variables, and in place of it introducing a new independent variable, exhibit the equation in the standard form  $H = \text{const.}$ ,  $H$  being here a homogeneous function of the order zero in the differential coefficients. Thus, if the independent variables are  $x, y$ , the dependent variable  $z$ , and its differential coefficients  $p, q$ , then the given partial differential equation may be  $H_1 = (p, q, x, y, z) = \text{const.}$  But we may determine  $z$  as a function of  $x, y$  by an equation  $V = \text{const.}$ ,  $V$  being a desired function of  $x, y, z$ ; and then writing  $p, q, r$  for the differential coefficients  $\frac{dV}{dx}, \frac{dV}{dy}, \frac{dV}{dz}$ , we have  $p = -\frac{p}{r}, q = -\frac{q}{r}$ , and the proposed partial differential equation becomes

$$H = H_1 \left( -\frac{p}{r}, -\frac{q}{r}, x, y, z \right) = \text{const.}$$

viz. this is an equation containing only the differential coefficients  $p, q, r$  of the dependent variable  $V$ , a function of  $x, y, z$ . And, moreover,  $H$  is homogeneous of the order zero in  $p, q, r$ ; consequently

$$p \frac{dH}{dp} + q \frac{dH}{dq} + r \frac{dH}{dr} = 0,$$

or, in the augmented Hamiltonian system, the last equality is  $= -\frac{dV}{dt}$ , so that an integral is  $V = \text{const.}$ ; as already stated, this is the equation by which  $z$  is determined as a function of  $x, y$ .

81. Thus, if the given partial differential equation be  $pq = z$ , we here consider the equation  $pq - z = H$ . The Hamiltonian system is

$$\frac{r dx}{q} = \frac{r dy}{p} = \frac{-r^2 dz}{2pq} = \frac{dp}{0} = \frac{dq}{0} = \frac{dr}{r} \left( = -\frac{dV}{0} \right)$$

having the integrals

$$\begin{aligned} a &= p, \\ b &= q, \\ c &= px - qy, \\ d &= \frac{r}{q} - \frac{1}{p}, \\ e &= \frac{x}{pq} - \frac{1}{r^2}, \end{aligned}$$

(whence  $H = -abe$ ). We have  $H, a, b$  a system of conjugate integrals and, in terms of these,

$$p = a, \quad q = b, \quad r = \sqrt{(ab/(z + H))};$$

hence, writing  $\lambda$  for the constant value of  $V$ , we have

$$\lambda = \int \left\{ a \, dx + b \, dy + \sqrt{\frac{ab}{z + H}} \, dz \right\},$$

that is,

$$\lambda = ax + by + 2\sqrt{[ab(z + H)]},$$

or say,

$$4ab(z + H) = (ax + by - \lambda)^2,$$

a solution containing really the two constants  $\lambda$  and  $H$ , and thus a complete solution of the given equation  $pq - z = H$ . We have, in fact,

$$2abp = a(ax + by - \lambda),$$

$$2abq = b(ax + by - \lambda),$$

that is,

$$4a^2b^2pq = ab(ax + by - \lambda)^2, = 4a^2b^2(z + H),$$

or

$$pq = z + H,$$

as it should be.

## 656.

## ON THE THEORY OF PARTIAL DIFFERENTIAL EQUATIONS.

[From the *Mathematische Annalen*, t. XI. (1877), pp. 194–198.]

In what follows, any letter not otherwise explained denotes a function of certain variables  $(x, y, p, q)$ , or  $(x, y, z, p, q, r)$ , &c., as will be stated in each particular case.

An equation  $a = \text{const.}$  denotes that the function  $a$  of the variables is, in fact, a constant (viz. by such equation we establish a relation between the variables): and when this is so, we use the same letter  $a$  to denote the constant value of the function in question; I find this a very convenient notation.

Thus if the variables are  $x, y, z, p, q, r$  and if  $p, q, r$  are the differential coefficients in regard to  $x, y, z$  respectively of a function  $V$  of  $x, y, z$ , then  $H$  (as a letter not otherwise explained) denotes a function of  $x, y, z, p, q, r$  and considering it as a given function,

$$H = \text{const.}$$

will be a partial differential equation containing the constant  $H$ . For instance, if  $H$  denote the function  $pqr - xyz$ ,  $H = \text{const.}$  is the partial differential equation,  $pqr - xyz = \text{const.}$  (a given constant).

The integration of the partial differential equation,  $H = \text{const.}$ , depends upon that of the linear partial differential equation

$$(H, \theta) = 0,$$

where as usual  $(H, \theta)$  signifies

$$\frac{\partial(H, \theta)}{\partial(p, x)} + \frac{\partial(H, \theta)}{\partial(q, y)} + \frac{\partial(H, \theta)}{\partial(r, z)}.$$

It can be effected if we know two conjugate solutions  $a, b$  of the equation  $(H, \theta) = 0$ , viz.  $a, b$  as solutions are such that  $(H, a) = 0$ ,  $(H, b) = 0$ , and (as conjugate solutions) are also such that  $(a, b) = 0$ ; in this case if from the equations

$$H = \text{const.}, \quad a = \text{const.}, \quad b = \text{const.}$$

we determine  $p, q, r$  as functions of  $x, y, z$ , the resulting value of  $p dx + q dy + r dz$  is an exact differential, and we have

$$V = \lambda + \int (p dx + q dy + r dz),$$

a solution containing three arbitrary constants,  $\lambda, a, b$ , and therefore a complete solution of the proposed partial differential equation  $H = \text{const.}$

But (as is known) there is a different process of integration, for which the conjugate solutions are not required, and which has reference to a system of initial values  $x_0, y_0, z_0, p_0, q_0, r_0$ : viz. if the independent solutions of  $(H, \theta) = 0$  are  $a, b, c, d, e$ , and if  $a_0, b_0, c_0, d_0, e_0$  denote respectively the same functions of the initial variables that  $a, b, c, d, e$  are of  $x, y, z, p, q, r$ , then if from the equations

$$a = a_0, \quad b = b_0, \quad c = c_0, \quad d = d_0, \quad e = e_0, \quad H = \text{const.}$$

we express  $p, q, r$  as functions of  $x, y, z$  and of  $x_0, y_0, z_0, H$ , these last being regarded as constants, we have  $p dx + q dy + r dz$  an exact differential, and

$$V = \lambda + \int (p dx + q dy + r dz),$$

a solution containing the constants  $\lambda, x_0, y_0, z_0$  (that is, one supernumerary constant), and as such a complete solution.

It is interesting to prove directly that  $p dx + q dy + r dz$  is an exact differential.

I consider first the more simple case where the variables are  $p, q, x, y$ . Here  $p, q$  are to be found from the equations

$$a = a_0, \quad b = b_0, \quad H = \text{const.}$$

and it is to be shown that  $p dx + q dy$  is an exact differential.

Considering  $p, q, p_0, q_0$  as functions of the independent variables  $x, y$ , then differentiating in regard to  $x$ , and eliminating  $\frac{dp_0}{dx}, \frac{dq_0}{dx}, \frac{dp}{dx}$ , we have

$$\left| \begin{array}{ccc} \frac{da}{dx} + \frac{da}{dq} \frac{dq}{dx}, & \frac{da}{dp}, & \frac{da}{dp_0}, & \frac{da}{dq_0} \\ \frac{db}{dx} + \frac{db}{dq} \frac{dq}{dx}, & \frac{db}{dp}, & \frac{db}{dp_0}, & \frac{db}{dq_0} \\ \frac{dc}{dx} + \frac{dc}{dq} \frac{dq}{dx}, & \frac{dc}{dp}, & \frac{dc}{dp_0}, & \frac{dc}{dq_0} \\ \frac{dx}{dH} + \frac{dq}{dH} \frac{dq}{dx}, & \frac{dp}{dH}, & 0, & 0 \end{array} \right| = 0,$$

or introducing a well-known notation for functional determinants, and expanding the determinant, this is

$$\frac{\partial(a_0, b_0)}{\partial(p_0, q_0)} \left\{ \frac{\partial(H, c)}{\partial(p, x)} + \frac{\partial(H, c)}{\partial(p, q)} \frac{dq}{dx} \right\} + \&c. = 0.$$

But in the same way

$$\frac{\partial(a_0, b_0)}{\partial(p_0, q_0)} \left\{ \frac{\partial(H, c)}{\partial(q, y)} + \frac{\partial(H, c)}{\partial(q, p)} \frac{dp}{dy} \right\} + \&c. = 0;$$

or adding these, attending to the value of  $(H, c)$ , and observing that  $\frac{\partial(H, c)}{\partial(p, q)} = -\frac{\partial(H, c)}{\partial(q, p)}$ , we have

$$\frac{\partial(a_0, b_0)}{\partial(p_0, q_0)} \left\{ (H, c) - \frac{\partial(H, c)}{\partial(p, q)} \left( \frac{dq}{dx} - \frac{dp}{dy} \right) \right\} + \&c. = 0,$$

the terms denoted by the  $\&c.$  being the like terms with  $b, c, a$  and  $c, a, b$  in place of  $a, b, c$ . We have  $(H, a) = 0$ ,  $(H, b) = 0$ ,  $(H, c) = 0$ , and the equation in fact is

$$\left\{ \frac{\partial(a_0, b_0)}{\partial(p_0, q_0)} \frac{\partial(H, c)}{\partial(p, q)} + \&c. \right\} \left( \frac{dq}{dx} - \frac{dp}{dy} \right) = 0,$$

viz. we have  $\frac{dq}{dx} - \frac{dp}{dy} = 0$ , the condition for the exact differential.

Coming now to the case where the variables are  $x, y, z, p, q, r$ , and in the six equations treating  $p, q, r, p_0, q_0, r_0$  as functions of the independent variables  $x, y, z$ , then differentiating with regard to  $x$  and proceeding as before, we find for  $\frac{dr}{dx}$  the equation

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left\{ \frac{dr}{dx} \frac{\partial(a, b, H)}{\partial(r, p, q)} + \frac{\partial(a, b, H)}{\partial(x, p, q)} \right\} + \&c. = 0.$$

We have, in the same way, for  $\frac{dp}{dz}$  the equation

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left\{ \frac{dp}{dz} \frac{\partial(a, b, H)}{\partial(p, r, q)} + \frac{\partial(a, b, H)}{\partial(z, r, q)} \right\} + \&c. = 0;$$

or, adding the two equations,

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left\{ \left( \frac{dr}{dx} - \frac{dp}{dz} \right) \frac{\partial(a, b, H)}{\partial(r, p, q)} + \frac{\partial(a, b, H)}{\partial(x, p, q)} - \frac{\partial(a, b, H)}{\partial(z, r, q)} \right\} + \&c. = 0,$$

where the terms denoted by the  $\&c.$  indicate the like terms corresponding to the different partitions of the letters  $a, b, c, d, e$ .

The equation may be simplified; we have identically

$$-H \frac{da_0}{dq_0} \frac{db_0}{dq_0} + H \frac{da_0}{dq_0} \frac{dH_0}{dq_0} + \&c. = \frac{\partial(a_0, b, H)}{\partial(x, p, q)} - \frac{\partial(a, b, H)}{\partial(z, r, q)},$$

or since  $(H, a) = 0$ ,  $(b, H) = 0$ , the left-hand side is simply  $-\frac{dH}{dq_0}(a, b)$ , and the equation becomes

$$\frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \left\{ \left( \frac{dr}{dx} - \frac{dp}{dz} \right) \frac{\partial(a, b, H)}{\partial(r, p, q)} - \frac{dH}{dq_0}(a, b) \right\} + \&c. = 0.$$

This ought to give  $\frac{dr}{dx} - \frac{dp}{dz} = 0$ , and we shall in fact have

$$\left\{ \frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \frac{dH}{dq_0}(a, b) + \&c. \right\} = 0;$$

this is then the equation which has to be proved. By the Poisson-Jacobi theorem,  $(a, b)$  is a function of  $a, b, c, d, e$ : if we write

$$(a, b) = f(a, b, c, d, e),$$

then  $(a_0, b_0)$  is the same function of  $a_0, b_0, c_0, d_0, e_0$ ; but these are  $= a, b, c, d, e$  respectively, and we thence have  $(a, b) = (a_0, b_0)$ , and the theorem to be proved is

$$\left\{ \frac{\partial(c_0, d_0, e_0)}{\partial(p_0, q_0, r_0)} \frac{dH}{dq_0}(a_0, b_0) + \&c. \right\} = 0.$$

But substituting for  $(a_0, b_0)$  its value, the function on the left-hand is (it is easy to see) the sum of the three functional determinants

$$\frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, q_0, r_0)}, \quad \frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, q_0, r_0)}, \quad \frac{\partial(a_0, b_0, c_0, d_0, e_0)}{\partial(p_0, q_0, r_0, r_0, q_0)},$$

each of which vanishes as containing the same letter twice in the denominator, that is, as having two identical columns; and the theorem in question is thus proved. And in the same way  $\frac{dp}{dy} - \frac{dq}{dx}$ ,  $\frac{dq}{dz} - \frac{dr}{dy}$  are each  $= 0$ : or we have  $p dx + q dy + r dz$  an exact differential.

The proof would fail if the factors multiplying  $\frac{dr}{dx} - \frac{dp}{dz}$ , &c., or if any one of these factors, were  $= 0$ ; I have not particularly examined this, but the meaning would be, that here the equations in question  $a = a_0$ , &c.,  $H = \text{const.}$ , are such as not to give rise to expressions for  $p, q, r$  as functions of  $x, y, z, x_0, y_0, z_0, H$ , as assumed in the theorem; whenever such expressions are obtainable, then we have  $p dx + q dy + r dz$  an exact differential.

The proof in the case of a greater number of variables, say in the next case where the variables are  $x, y, z, w, p, q, r, s$ , would present more difficulty; but I have not proceeded further in the question.

It is worth while to put the two processes into connexion with each other: taking in each case the variables to be  $x, y, z, p, q, r$ , and the partial differential equation to be  $H = \text{const.}$

In the one case,  $a, b$  being conjugate solutions of  $(H, \theta) = 0$ ,

from the equations  $H = \text{const.}$ ,  $a = \text{const.}$ ,  $b = \text{const.}$ ,  
we find  $p, q, r$  functions of  $x, y, z, H, a, b$   
and then  $p dx + q dy + r dz$  is an exact differential.

In the other case,  $a, b, c, d, e$  being the solutions of  $(H, \theta) = 0$ ,

from the equations  $H = \text{const.}$ ,  $a = a_0$ ,  $b = b_0$ ,  $c = c_0$ ,  $d = d_0$ ,  $e = e_0$ ,  
we find  $p, q, r$  functions of  $x, y, z, x_0, y_0, z_0, H$ :  
and then  $p dx + q dy + r dz$  is an exact differential.

It may be added that, if from the last mentioned equations we determine also  $p_0, q_0, r_0$  as functions of  $x, y, z, x_0, y_0, z_0$ , then considering only  $H$  as a constant, we ought to have  $p dx + q dy + r dz - p_0 dx_0 - q_0 dy_0 - r_0 dz_0$  an exact differential; I have not examined the direct proof.

*Cambridge, 28 Nov., 1876.*

## 657.

## NOTE ON THE THEORY OF ELLIPTIC INTEGRALS.

[From the *Mathematische Annalen*, t. XII. (1877), pp. 143–146.]

The equation

$$\frac{M dy}{\sqrt{1 - y^2 \cdot 1 - k^2 y^2}} = \frac{dx}{\sqrt{1 - x^2 \cdot 1 - k^2 x^2}}$$

is integrable algebraically when  $M$  is rational: and so long as the modulus is arbitrary, then conversely, in order that the equation may be integrable algebraically,  $M$  must be rational. For particular values however of the modulus, the equation is integrable algebraically for values of the form  $M$ , or (what is the same thing)  $\frac{1}{M}$ , = a rational quantity + square root of a negative rational quantity, say =  $\frac{1}{p}(l + m\sqrt{-n})$ , where  $l, m, n, p$  are integral and  $n$  is positive; we may for shortness call this a half-rational numerical value. The theory is considered by Abel in two Memoirs in the *Astr. Nach.* Nos. 138 & 147 (1828), being the Memoirs<sup>1</sup> XIII & XIV in the *Œuvres Complètes* (Christiania 1839). I here reproduce the investigation in a somewhat altered (and, as it appears to me, improved) form.

Putting the two differentials each =  $du$ , we have  $x = \operatorname{sn}(u + \alpha)$ ,  $y = \operatorname{sn}\left(\frac{u}{M} + \beta\right)$ ; and the question is whether there exists an algebraical relation between these functions, or, what is the same thing, an algebraical relation between the functions  $x = \operatorname{sn} u$  and  $y = \operatorname{sn} \frac{u}{M}$ .

Suppose that  $A$  and  $B$  are independent periods of  $\operatorname{sn} u$ ; so that  $\operatorname{sn}(u + A) = \operatorname{sn} u$ ,  $\operatorname{sn}(u + B) = \operatorname{sn} u$ , and that every other period is =  $mA + nB$ , where  $m$  and  $n$  are integers. Then if  $u$  has successively the values  $u, u + A, u + 2A$ , etc., the value of  $x$

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<sup>1</sup>They are the Memoirs xix. and xx. in the *Œuvres Complètes*, t. i., Christiania, 1881.



remains always the same, and if  $x$  and  $y$  are algebraically connected,  $y$  can have only a finite number of values: there are consequently integer values  $p', p''$  for which  $\operatorname{sn} \frac{1}{M}(u+p'A) = \operatorname{sn} \frac{1}{M}(u+p''A)$ : or writing  $u - p'A$  for  $u$  and putting  $p'' - p' = p$ , there is an integer value  $p$  for which  $\operatorname{sn} \frac{1}{M}(u + pA) = \operatorname{sn} \frac{1}{M}u$ .

Similarly there is an integer value  $q$  for which  $\operatorname{sn} \frac{1}{M}(u + qB) = \operatorname{sn} \frac{1}{M}u$ ; and we are at liberty to assume  $q = p$ ; for if the original values are unequal, we have only in the place of each of them to substitute their least common multiple.

We have thus an integer  $p$ , for which

$$\operatorname{sn} \frac{1}{M}(u + pA) = \operatorname{sn} \frac{1}{M}u,$$

$$\operatorname{sn} \frac{1}{M}(u + pB) = \operatorname{sn} \frac{1}{M}u.$$

There are consequently integers  $m, n, r, s$  such that

$$\frac{pA}{M} = mA + nB,$$

$$\frac{pB}{M} = rA + sB,$$

equations which will constitute a single relation  $\frac{p}{M} = m$ , if  $m = s, r = n = 0$ ; but in every other case will be two independent relations. In the case first referred to, the modulus is arbitrary and  $M$  is rational.

But excluding this case, the equations give

$$B(mA + nB) = A(rA + sB),$$

or, what is the same thing,

$$rA^2 - (m - s)AB - nB^2 = 0,$$

an equation which implies that the modulus has some one value out of a set of given values. The ratio  $A : B$  of the two periods is of necessity imaginary, and hence the integers  $m, n, r, s$  must be such that  $(m + s)^2 - 4nr$  is negative.

The foregoing equations may be written

$$\left(m - \frac{p}{M}\right)A + nB = 0,$$

$$rA + \left(s - \frac{p}{M}\right)B = 0,$$

whence eliminating  $A$  and  $B$  we have

$$\left(m - \frac{p}{M}\right)\left(s - \frac{p}{M}\right) - nr = 0,$$

that is,

$$\left(\frac{p}{M}\right)^2 - (m+s)\frac{p}{M} + ms - nr = 0,$$

and consequently

$$\frac{p}{M} = \frac{1}{2}(m+s) \pm \frac{1}{2}\sqrt{(m-s)^2 + 4nr},$$

where, by what precedes, the integer under the radical sign is negative: and we have thus the above mentioned theorem.

As a very general example, consider the two rational transformations

$$z = z(x, u, v); \text{ mod. eq. } Q(u, v) = 0; \quad \frac{N dz}{\sqrt{1-z^2} \cdot 1-v^4 z^2} = \frac{dx}{\sqrt{1-x^2} \cdot 1-u^4 x^2},$$

$$y = y(z, v, w); \text{ mod. eq. } P(v, w) = 0; \quad \frac{M dy}{\sqrt{1-y^2} \cdot 1-w^4 y^2} = \frac{dz}{\sqrt{1-z^2} \cdot 1-v^4 z^2};$$

viz.  $z$  is taken to be a rational function of  $x$ , and of the modular fourth roots  $u, v$ ; and  $y$  to be a rational function of  $z$ , and of the modular fourth roots  $v, w$ ; the transformations being (to fix the ideas) of different orders. We have  $y$  a rational function of  $x$ , corresponding to the differential relation

$$\frac{MN dy}{\sqrt{1-y^2} \cdot 1-w^4 y^2} = \frac{dx}{\sqrt{1-x^2} \cdot 1-u^4 x^2}.$$

Suppose here  $w^4 = u^4$ , or say  $w = \theta u$ ,  $\theta$  being an eighth root of unity: we then have  $Q(u, v) = 0$ ,  $P(v, \theta u) = 0$ , equations which determine  $u$ . The differential equation is then

$$\frac{MN dy}{\sqrt{1-y^2} \cdot 1-\theta^4 u^4 y^2} = \frac{dx}{\sqrt{1-x^2} \cdot 1-u^4 x^2},$$

an equation the algebraical integral of which is  $y =$  a rational function of  $x$  as above; hence, by what precedes, we have  $MN =$  a half-rational numerical value, as above.

To explain what the algebraical theorem implied herein is, observe that the equations  $Q(u, v) = 0$ ,  $P(v, \theta u) = 0$ , give for  $u$  an algebraical equation. Admitting  $\theta$  as an adjoint radical, suppose that an irreducible factor is  $\phi(u)$ , and take  $u$  to be determined by the equation  $\phi u = 0$ ; then  $v$ , and consequently also any rational function  $\frac{1}{MN}$  of  $u, v$ , can be expressed as a rational integral function of  $u$ , of a degree which is at most equal to the degree of the function  $\phi u$  less unity. The theorem is that, in virtue of the equation  $\phi u = 0$ , this rational function of  $u$  becomes equal to a half-rational numerical value as above. Thus in a simple case, which actually presented itself, the equation  $\phi u = 0$  was  $u^2 - 4u + 1 = 0$ ; and  $\frac{1}{MN}$  had the value  $u - 2$ , which in virtue of this equation becomes  $= \pm\sqrt{-3}$ .

Thus if the second transformation be the identity  $z = y$ ,  $M = v$ ,  $w = v$ : we have  $v = \theta u$ ; and the equations are

$$\frac{N dy}{\sqrt{1 - y^2 \cdot 1 - \theta^4 u^4 y^2}} = \frac{dx}{\sqrt{1 - x^2 \cdot 1 - u^4 x^2}}, \quad Q(x, u, \theta u) = 0, \quad Q(u, \theta u) = 0.$$

In particular, if the relation between  $y, x$  be given by the cubic transformation

$$y = x \frac{-v^4 x^2 + 2u^2 v^2 + v^4}{x^2 - 2u^2 v^2 x^2 + u^4 v^4},$$

so that the modular equation  $Q(u, v) = 0$  is  $u^4 - v^4 + 2uv(1 - u^2 v^2) = 0$ ; then, writing herein  $v = \theta u$ , and taking  $\theta$  a prime eighth root of unity, that is, a root of  $\theta^4 + 1 = 0$ , we have

$$Q(u, \theta u) = -u^4 + \theta^4 + 2\theta u^2(1 - \theta^2 u^4)$$

viz. disregarding the factor  $u^2$ , the equation for  $u$  is  $u^4 + \theta u^2 + \theta^2 = 0$ ; or, if  $\omega$  be an imaginary cube root of unity ( $\omega^2 + \omega + 1 = 0$ ), this is  $(u^2 - \omega\theta)(u^2 - \omega^2\theta) = 0$ ; so that a value of  $u^2$  is  $= -\omega\theta$ .

Assuming then  $\theta^4 + 1 = 0$ ,  $v = \theta u$  and  $u^2 = -\omega\theta$ , we have  $v + 2u^2 = v + \theta u^2(1 + 2\omega)$ ,  $= -\theta^2\omega(\omega^2 - \omega) := \theta^2\omega'$ ;  $-\omega^2 := -\omega^2$ ,  $(v + 2u^2)vu^2 = -\omega^2(\omega - \omega^2)$ ,  $u^4 = -\omega^2\theta^2$ ; and the formula becomes

$$y = x \frac{1 - \omega^2(\omega - \omega^2)\theta^2 x^2}{1 - \omega(\omega - \omega^2)\theta^2 x^2},$$

giving

$$\frac{dy}{\sqrt{1 - y^2 \cdot 1 + \omega\theta^2 y^2}} = \frac{(\omega - \omega^2) dx}{\sqrt{1 - x^2 \cdot 1 + \omega^2\theta^2 x^2}},$$

where as before  $\omega^2 + \omega + 1 = 0$ , a result which can be at once verified. We have  $\omega - \omega^2 = \sqrt{-3}$ , or the coefficient  $\omega - \omega^2$  in the differential equation is  $= \sqrt{-3}$ , which is of the form mentioned in the general theorem.

We might, instead of  $z = y$ , have assumed between  $y$  and  $z$  the relation corresponding to any other of the six linear transformations of an elliptic integral, and thus have obtained in each case, for a properly determined value of the modulus, a cubic transformation to the same modulus.

*Cambridge, 10 April, 1877.*

## 658.

## ON SOME FORMULÆ IN ELLIPTIC INTEGRALS.

[From the *Mathematische Annalen*, t. XII. (1877), pp. 369–374.]

I reproduce in a modified form an investigation contained in the memoir, Zolotareff, “Sur la méthode d’intégration de M. Tchébychef,” *Mathematische Annalen*, t. V. (1872), pp. 560–580.

Starting from the quartic

$$(a, b, c, d, e)(x, 1)^4 = a \cdot \overline{x - \alpha} \cdot \overline{x - \beta} \cdot \overline{x - \gamma} \cdot \overline{x - \delta},$$

we derive from it the quartic

$$(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4 = a_1 \cdot \overline{x_1 - \alpha_1} \cdot \overline{x_1 - \beta_1} \cdot \overline{x_1 - \gamma_1} \cdot \overline{x_1 - \delta_1},$$

where, writing for shortness

$$\lambda = -\alpha + \beta + \gamma - \delta,$$

the roots of the new quartic are

$$\alpha_1 = \theta + \frac{\mu\nu}{2\lambda},$$

$$\beta_1 = \theta + \frac{\nu\lambda}{2\mu},$$

$$\gamma_1 = \theta + \frac{\lambda\mu}{2\nu},$$

$$\delta_1 = \theta,$$

where

$$\mu = \alpha - \beta + \gamma - \delta, \quad \nu = \alpha + \beta - \gamma - \delta,$$

$\theta$  being arbitrary: the differences of the roots  $\alpha_1, \beta_1, \gamma_1, \delta_1$  are, it will be observed, functions of the differences of the roots  $\alpha, \beta, \gamma, \delta$ .

We assume  $a_1 = a = 1$ , nevertheless retaining in the formulae  $a$ , or  $a_1$  (each meaning 1), whenever, for the sake of homogeneity, it is convenient to do so. The relations between the remaining coefficients  $b_1, c_1, d_1, e_1$ , and  $b, c, d, e$ , are of course to be calculated from the formulae  $-4b = \Sigma\alpha$ ,  $6c = \Sigma\alpha\beta$ , &c., and the like formulae  $-4b_1 = \Sigma\alpha_1$ ,  $6c_1 = \Sigma\alpha_1\beta_1$ , &c. We thus have

$$-4b_1 = 4\theta + \frac{1}{2}\Sigma\frac{\mu\nu}{\lambda},$$

$$-4d_1 = 4\theta^3 + \frac{3}{2}\theta^2\Sigma\frac{\mu\nu}{\lambda} + \frac{1}{4}\theta\Sigma\lambda^2 + \frac{1}{8}\lambda\mu\nu,$$

where  $\Sigma\frac{\mu\nu}{\lambda} = \frac{1}{\lambda\mu\nu}\Sigma\lambda^2\mu^2$ .

Writing, for shortness,

$$\begin{aligned} C &= ac - b^2, \\ D &= a^2d - 3abc + 2b^3, \\ E &= a^3e - 4a^2bd + 6ab^2c - 3b^4 = a^2I - 3C^2, \\ I &= ae - 4bd + 3c^2, \\ J &= ace - ad^2 - b^2e + 2bcd - c^3, \end{aligned}$$

we have

$$\begin{aligned} a^2E &= -a^2I + 12C^2, \\ \Sigma\lambda^2 &= -4(b + B), \\ \Sigma\lambda^4 &= -48C, \\ \Sigma\lambda^2\mu^2 &= 24a^2c - 8(b + B)^2, \\ \Sigma\lambda\mu\nu &= 32D, \\ \Sigma\lambda^2\mu^2\nu^2 &= 64(-a^2I + 12C^2), \end{aligned}$$

where the last equation may be verified by means of the formula

$$(\Sigma\lambda^2)^2 = \Sigma\lambda^4 + 2\Sigma\lambda^2\mu^2.$$

And we hence obtain

$$\begin{aligned} a_1 &= 1, \\ b_1 &= -\theta - B, \\ c_1 &= \theta^2 + 2B\theta - 2C, \\ d_1 &= -\theta^3 - 3B\theta^2 + 6C\theta - D, \\ e_1 &= \theta^4 + 4B\theta^3 - 12C\theta^2 + 4D\theta. \end{aligned}$$

And consequently

$$(a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4 = (1, -B, -2C, -D, 0)(x_1 - \theta, 1)^4.$$

Hence also

$$\begin{aligned} I_1 &= a_1 e_1 - 4b_1 d_1 + 3c_1^2 = -4BD + 12C^2 = a^2 I, \\ J_1 &= a_1 c_1 e_1 - a_1 d_1^2 - b_1^2 e_1 + 2b_1 c_1 d_1 - c_1^3 = -D^2 + 8C^3 - 4BCD \\ &= -D^2 + 8C^3 + C(a^2 I - 12C^2) \\ &= a^2 CI - 4C^3 - D^2 \\ &= a^2 J; \end{aligned}$$

where, as regards this last equation  $a^2 CI - 4C^3 - D^2 = a^2 J$ , observe that  $C$  and  $D$  are the leading coefficients of the Hessian  $H$  and the cubicovariant  $\Phi$  of the quartic function  $U$ , and hence that the identity  $-\Phi^2 = JU^3 - IU^2H + 4H^3$ , attending only to the term in  $a^6$ , becomes  $-D^2 = a^4 J - a^2 CI + 4C^3$ , which is the equation in question.

We thus have  $I_1 = I, J_1 = J$ ; viz. the functions  $(a, b, c, d, e)(x, 1)^4, (a_1, b_1, c_1, d_1, e_1)(x_1, 1)^4$ , are linearly transformable the one into the other, and that by a unimodular substitution  $x_1 = \rho x + \sigma, x = \rho' x_1 + \sigma'$ , where  $\rho\sigma' - \rho'\sigma = 1$ . It may be remarked that we have  $(a, b, c, d, e)(x, 1)^4 = (1, 0, C, D, E)(x + b, 1)^4$ ; and hence the theorem may be stated in the form: the quartic functions  $(1, 0, C, D, E)(x, 1)^4$ , and  $(1, -B, -2C, -D, 0)(x_1, 1)^4$  are transformable the one into the other by a unimodular substitution: or again, substituting for  $E$  its value  $a^2 I - 3C^2, = -4BD + 9C^2$ , the quartic functions

$$(1, 0, C, D, -4BD + 9C^2)(x, 1)^4, \text{ and } (1, -B, -2C, -D, 0)(x_1, 1)^4$$

are linearly transformable the one into the other by a unimodular substitution. In this last form  $B, C, D$  are arbitrary quantities; it is at once verified that the invariants  $I, J$  have the same values for the two functions respectively; and the theorem is thus self-evident.

Reverting to the expressions for  $\alpha_1, \beta_1, \gamma_1, \delta_1$  we obtain

$$\begin{aligned} \alpha_1 - \delta_1 &= \frac{\mu\nu}{2\lambda} = \frac{1}{2\lambda}(\alpha - \delta)(\beta - \gamma), \\ \beta_1 - \delta_1 &= \frac{\nu\lambda}{2\mu} = \frac{1}{2\mu}(\beta - \delta)(\gamma - \alpha), \\ \gamma_1 - \delta_1 &= \frac{\lambda\mu}{2\nu} = \frac{1}{2\nu}(\gamma - \delta)(\alpha - \beta), \end{aligned}$$

$$\beta_1 - \gamma_1 = \frac{1}{2\lambda}(\overline{\alpha - \delta} \cdot \overline{\beta - \gamma}), \quad \gamma_1 - \alpha_1 = \frac{1}{2\mu}(\overline{\beta - \delta} \cdot \overline{\gamma - \alpha}), \quad \alpha_1 - \beta_1 = \frac{1}{2\nu}(\overline{\gamma - \delta} \cdot \overline{\alpha - \beta}).$$

Hence also

$$\begin{aligned} &\overline{\alpha - \delta} \cdot \overline{\beta - \gamma} \cdot \overline{\beta - \delta} \cdot \overline{\gamma - \alpha} \cdot \overline{\gamma - \delta} \cdot \overline{\alpha - \beta} \\ &= \overline{\alpha_1 - \delta_1} \cdot \overline{\beta_1 - \gamma_1} \cdot \overline{\beta_1 - \delta_1} \cdot \overline{\gamma_1 - \alpha_1} \cdot \overline{\gamma_1 - \delta_1} \cdot \overline{\alpha_1 - \beta_1}, \end{aligned}$$

which agrees with the foregoing equations  $I_1 = I$  and  $J_1 = J$ , since  $I, J$  are functions of the first set of quantities and  $I_1, J_1$  the like functions of the second set; in fact,  $I = \frac{1}{3}(P^2 + Q^2 + R^2)$ , and  $J = \frac{1}{27}(Q - R)(R - P)(P - Q)$ , if for a moment the quantities are called  $P, Q, R$ .

We consider now the differential expression  $\frac{dx}{\sqrt{x-\alpha} \cdot \overline{x-\beta} \cdot \overline{x-\gamma} \cdot \overline{x-\delta}}$ ; to transform this into the elliptic form, assume

$$k^2 = \frac{\alpha - \beta}{\gamma - \alpha} \cdot \frac{\gamma - \delta}{\beta - \delta}, \quad \text{sn}^2 \alpha = \frac{\gamma - \alpha}{\gamma - \delta},$$

(where  $\alpha$  is of course not the coefficient,  $= 1$ , heretofore represented by that letter: as  $\alpha$  will only occur under the functional signs sn, cn, dn, there is no risk of ambiguity). And then further

$$x = \frac{-\alpha \text{sn}^2 u - \delta \text{sn}^2 \alpha - u}{\text{sn}^2 \alpha - \text{sn}^2 u}.$$

Forming the equations

$$x - \alpha = \frac{\alpha - \beta}{\gamma - \alpha} \cdot \&c., \quad \&c.,$$

we deduce without difficulty

$$\begin{aligned} \text{sn}^2 \alpha &= \frac{\gamma - \alpha}{\gamma - \delta}, & \frac{\text{sn}^2 u}{\text{sn}^2 \alpha} &= \frac{x - \delta}{x - \alpha}, \\ \text{cn}^2 \alpha &= \frac{\alpha - \delta}{\gamma - \delta}, & \frac{\text{cn}^2 u}{\text{cn}^2 \alpha} &= \frac{x - \gamma}{x - \alpha}, \\ \text{dn}^2 \alpha &= \frac{\alpha - \beta}{\gamma - \beta}, & \frac{\text{dn}^2 u}{\text{dn}^2 \alpha} &= \frac{x - \beta}{x - \alpha}, \\ 1 - k^2 \text{sn}^4 \alpha &= \frac{(\alpha - \delta)(-\alpha + \beta + \gamma - \delta)}{\beta - \delta \cdot \gamma - \delta} = \frac{\lambda(\alpha - \delta)}{\beta - \delta \cdot \gamma - \delta}, \end{aligned}$$

the use of which last equation will presently appear.

We hence obtain

$$\begin{aligned} 2 \text{sn} u \text{cn} u \text{dn} u \, du &= -(\alpha - \delta) \text{sn}^2 \alpha \frac{dx}{(x - \alpha)^2}, \\ \text{sn} u \text{cn} u \text{dn} u &= \frac{1}{\text{sn} \alpha \text{cn} \alpha \text{dn} \alpha} \frac{\sqrt{x - \alpha} \cdot \overline{x - \beta} \cdot \overline{x - \gamma} \cdot \overline{x - \delta}}{(x - \alpha)^2}, \end{aligned}$$

and consequently

$$2 \, du = -\frac{(\alpha - \delta) \text{sn} \alpha}{\text{cn} \alpha \text{dn} \alpha} \frac{dx}{\sqrt{x - \alpha} \cdot \overline{x - \beta} \cdot \overline{x - \gamma} \cdot \overline{x - \delta}},$$

or, reducing the coefficient,

$$\frac{dx}{\sqrt{x - \alpha} \cdot \overline{x - \beta} \cdot \overline{x - \gamma} \cdot \overline{x - \delta}} = \frac{-2}{\sqrt{\gamma - \alpha} \cdot \overline{\beta - \delta}} \, du,$$

which is the required formula.

We next have

$$k^2 = \frac{4 \text{sn}^2 \alpha \text{cn}^2 \alpha \text{dn}^2 \alpha}{4} = \frac{\overline{\beta - \delta} \cdot \overline{\gamma - \alpha}}{\gamma_1 - \alpha_1} \cdot \&c.$$

in virtue of the foregoing values

$$\gamma_1 - \alpha_1 = \frac{1}{2\mu}(\beta - \delta)(\gamma - \alpha) \quad \text{and} \quad \gamma_1 - \delta_1 = \frac{\lambda\mu}{2\nu}.$$

Moreover

$$k_1^2 = -\frac{\overline{\alpha - \beta} \cdot \overline{\gamma - \delta}}{\overline{\gamma - \alpha} \cdot \overline{\beta - \delta}} = \frac{\overline{\alpha_1 - \beta_1} \cdot \overline{\gamma_1 - \delta_1}}{\overline{\gamma_1 - \alpha_1} \cdot \overline{\beta_1 - \delta_1}}.$$

Hence the like formulæ with the same value of  $k^2$ , and with  $2\alpha$  in place of  $\alpha$ , will be applicable to the like differential expression in  $x_1$ : viz. assuming

$$x_1 = \frac{-\alpha_1 \operatorname{sn}^2 u_1 - \delta_1 \operatorname{sn}^2 2\alpha}{\operatorname{sn}^2 u_1 - \operatorname{sn}^2 2\alpha},$$

we have

$$\frac{dx_1}{\sqrt{\overline{x_1 - \alpha_1} \cdot \overline{x_1 - \beta_1} \cdot \overline{x_1 - \gamma_1} \cdot \overline{x_1 - \delta_1}}} = \frac{-2}{\sqrt{\overline{\gamma_1 - \alpha_1} \cdot \overline{\beta_1 - \delta_1}}} du_1.$$

We have thus the integral of the differential equation

$$\frac{dx_1}{\sqrt{\overline{x_1 - \alpha_1} \cdot \overline{x_1 - \beta_1} \cdot \overline{x_1 - \gamma_1} \cdot \overline{x_1 - \delta_1}}} = \frac{dx}{\sqrt{\overline{x - \alpha} \cdot \overline{x - \beta} \cdot \overline{x - \gamma} \cdot \overline{x - \delta}}}$$

(the two quartic functions being of course connected as before); viz. assuming  $x, x_1$  functions of  $u, u_1$  respectively as above and recollecting that  $\overline{\gamma_1 - \alpha_1} \cdot \overline{\beta_1 - \delta_1} = \overline{\gamma - \alpha} \cdot \overline{\beta - \delta}$ ,

$du_1 = du$ ; and therefore  $u_1 = u + f$  ( $f$  an arbitrary constant); the required

integral is thus given by the equations

$$\frac{x - \delta}{x - \alpha} = \frac{\operatorname{sn}^2 u}{\operatorname{sn}^2 \alpha}, \quad \frac{x_1 - \delta_1}{x_1 - \alpha_1} = \frac{\operatorname{sn}^2(u + f)}{\operatorname{sn}^2 2\alpha}, \quad (f \text{ constant of integration}).$$

Using the formula

$$\operatorname{sn}^2(u + f) - \operatorname{sn}^2 f = \frac{\operatorname{sn} u \operatorname{cn} u \operatorname{dn} u \operatorname{sn} f \operatorname{cn} f \operatorname{dn} f}{1}$$

we obtain

$$\begin{aligned} & \{(x - \delta) \operatorname{sn}^2 \alpha - (x - \alpha) \operatorname{sn}^2 f\}^2 \\ & \times \frac{x_1 - \delta_1}{\operatorname{sn}^2 2\alpha} = \{\sqrt{\overline{x - \alpha} \cdot \overline{x - \delta}} \operatorname{sn} \alpha \operatorname{cn} f \operatorname{dn} f - \sqrt{\overline{x - \beta} \cdot \overline{x - \gamma}} \operatorname{sn} f \operatorname{cn} \alpha \operatorname{dn} \alpha\}^2, \end{aligned}$$

which is the general integral.

We obtain a particular integral of a very simple form by assuming  $f = \alpha$ , viz. this is

$$\frac{x_1 - \alpha_1}{x_1 - \delta_1} \frac{1}{\operatorname{sn}^2 2\alpha} = \operatorname{cn}^2 \alpha \operatorname{dn}^2 \alpha \{\sqrt{\overline{x - \alpha} \cdot \overline{x - \delta}} - \sqrt{\overline{x - \beta} \cdot \overline{x - \gamma}}\}^2,$$

this is

$$\frac{x_1 - \delta_1 \cdot \gamma_1 - \beta_1}{x_1 - \alpha_1 \cdot \gamma_1 - \delta_1} = \frac{\overline{\gamma - \alpha} \cdot \overline{\beta - \delta}}{\{\sqrt{\overline{x - \alpha} \cdot \overline{x - \delta}} - \sqrt{\overline{x - \beta} \cdot \overline{x - \gamma}}\}^2}.$$



or writing  $\overline{\gamma - \alpha} \cdot \overline{\beta - \delta} = \overline{\gamma_1 - \alpha_1} \cdot \overline{\beta_1 - \delta_1}$ , reducing and inverting, we have

$$\frac{\alpha_1 - \beta_1 \cdot \beta_1 - \delta_1}{x_1 - \delta_1} = \frac{1}{\beta_1 - \delta_1 \cdot \alpha_1 - \delta_1} \{ \sqrt{x - \alpha \cdot x - \delta} - \sqrt{x - \beta \cdot x - \gamma} \}^2.$$

which may also be written in the equivalent forms

$$\begin{aligned} \frac{x_1 - \beta_1}{x_1 - \alpha_1 \cdot \beta_1 - \delta_1} &= \frac{1}{2} \{ \sqrt{x - \gamma \cdot x - \delta} - \sqrt{x - \alpha \cdot x - \beta} \}^2, \\ \frac{x_1 - \delta_1}{x_1 - \beta_1 \cdot \alpha_1 - \delta_1} &= \frac{1}{2} \{ \sqrt{x - \gamma \cdot x - \delta} - \sqrt{x - \alpha \cdot x - \beta} \}^2. \end{aligned}$$

In fact, from the first equation we have

$$\frac{x_1 - \alpha_1 \cdot \beta_1 - \delta_1 \cdot \gamma_1 - \delta_1}{x_1 - \delta_1} = (x - \alpha)(x - \delta) - \{ \sqrt{x - \alpha \cdot x - \delta} - \sqrt{x - \beta \cdot x - \gamma} \}^2,$$

where the expression on the right-hand side is

$$\delta_1^2 - \delta_1(\alpha_1 + \beta_1 + \gamma_1) + \alpha_1\beta_1 + \beta_1\gamma_1 - 2\alpha_1 + \delta_1^2(\alpha + \beta + \gamma + \delta) - \alpha\delta - \beta\gamma + 2\sqrt{X},$$

$X$  having here the value

$$X = \overline{x - \alpha} \cdot \overline{x - \beta} \cdot \overline{x - \gamma} \cdot \overline{x - \delta}.$$

Writing for a moment

$$\begin{aligned} P &= \alpha\delta + \beta\gamma, & P_1 &= \alpha_1\delta_1 + \beta_1\gamma_1, \\ Q &= \beta\delta + \gamma\alpha, & Q_1 &= \beta_1\delta_1 + \gamma_1\alpha_1, \\ R &= \gamma\delta + \alpha\beta, & R_1 &= \gamma_1\delta_1 + \alpha_1\beta_1, \end{aligned}$$

then, by what precedes,  $Q_1 - R_1$ ,  $R_1 - P_1$ ,  $P_1 - Q_1$  are equal to  $Q - R$ ,  $R - P$ ,  $P - Q$  respectively; that is,  $P_1 - P = Q_1 - Q = R_1 - R =$  (suppose)  $\Omega$ , a function symmetrical in regard to  $\alpha_1, \beta_1, \gamma_1; \alpha, \beta, \gamma$ : the equation therefore is

$$\frac{x_1 - \alpha_1 \cdot \beta_1 - \delta_1 \cdot \gamma_1 - \delta_1}{x_1 - \delta_1} = \alpha_1(\beta_1 - \alpha_1 - \beta_1 - \gamma_1) - 2\alpha + \delta_1(\alpha + \beta + \gamma + \delta) + 2\sqrt{X} + \Omega,$$

or the relation is symmetrical in regard to  $\alpha_1, \beta_1, \gamma_1; \alpha, \beta, \gamma$ : and the first form implies therefore each of the other two forms.

*Cambridge, 8 May, 1877.*

## 659.

## A THEOREM ON GROUPS.

[From the *Mathematische Annalen*, t. XIII. (1878), pp. 561–565.]

The following theorem is very simple; but it seems to belong to a class of theorems, the investigation of which is desirable.

I consider a substitution-group of a given order upon a given number of letters; and I seek to double the group, that is to derive from it a group of twice the order upon twice the number of letters. This can be effected for any group, in a manner which is self-evident and in nowise interesting; but in a different manner for a commutative group (or group such that any two of its substitutions satisfy the condition  $AB = BA$ ): it is to be observed that the double group is not in general commutative.

Let the letters of the original group be  $abcde \dots$ , we may for shortness write  $U = abcde \dots$ ; and take  $U$  as the primitive arrangement: and let the group then be  $1, A, B, \dots$  where  $A, B, \dots$  represent substitutions: the corresponding arrangements are  $U, AU, BU, \dots$  and these may for shortness be represented by  $1, A, B, \dots$ ; viz.  $1, A, B, \dots$  represent, properly and in the first instance, substitutions; but when it is explained that they represent arrangements, then they represent the arrangements  $U, AU, BU, \dots$ .

For the double group the letters are taken to be  $a_1b_1c_1d_1e_1 \dots$  and  $a_2b_2c_2d_2e_2 \dots = U_1$  and  $U_2$  suppose, and  $U_1U_2$  is regarded as the primitive arrangement;  $A_1$  and  $A_2$  denote the same substitutions in regard to  $U_1$  and  $U_2$  respectively, that  $A$  denotes in regard to  $U$ : and so for  $B_1, B_2$ , etc.; moreover  $12$  denotes the substitution  $(a_1a_2)(b_1b_2)(c_1c_2)(d_1d_2)(e_1e_2) \dots$ , or interchange of the suffixes 1 and 2. The substitutions  $A_1^i, A_2^j$ , or any powers of these  $A_1^i, A_2^j$ , are obviously commutative; applying them to the primitive arrangement  $U_1U_2$ , we have  $A_1^i A_2^j U_1U_2$  and  $A_2^j A_1^i U_1U_2$ , each  $= A_1^i U_1 A_2^j U_2$ . But  $A_1^i, A_2^j$  are not commutative with  $12$ : we have for instance  $12 A_1^i \cdot U_1U_2 = 12 A_1^i U_1 \cdot U_2 = A_2^i U_2 \cdot U_1$  but  $A_1^i 12 U_1U_2 = A_1^i \cdot U_2 U_1 = U_2 \cdot A_1^i U_1$ . If instead of the substitutions we consider the arrangements obtained by operating upon  $U_1U_2$ , then we

may for shortness consider for instance  $A_1A_2$  as denoting the arrangement  $A_1U_1.A_2U_2$ . But observe that in this use of the symbols the  $A_1, A_2$  are not commutative,  $A_2A_1$  would denote the different arrangement  $A_1U_2.A_2U_1$ : in this use of the symbols, 1 would denote  $U_1U_2$ , and 12 would denote  $U_2U_1$ , but it would be clearer to use 12, 21 as denoting  $U_1U_2$  and  $U_2U_1$  respectively.

These explanations having been given, I remark that in every case the substitution-group  $1, A, B, \dots$  gives the double group

$$\begin{aligned} &1, A_1A_2, B_1B_2, \dots \\ &12, 12A_1A_2, 12B_1B_2, \dots \end{aligned}$$

as is at once seen to be true: but further when the original group  $1, A, B, \dots$  is commutative, then if  $m$  be any integer number, such that  $m^2 = 1 \pmod{\mu}$  (mod the order of the original group), we have also the double group

$$\begin{aligned} &1, A_1A_2^m, B_1B_2^m, \dots \\ &12, 12A_1A_2^m, 12B_1B_2^m, \dots \end{aligned}$$

where of course if the order of the original group ( $= \mu$  suppose) be prime, we have  $m^2 = 1 \pmod{\mu}$ , say  $m = 1$  or else  $m = \mu - 1$ ; but if the order  $\mu$  be composite, then the number of solutions may be greater.

The condition in order to the existence of the double group of course is that, in the system of substitutions just written down, the combination of any two substitutions may give a substitution of the system. And this is in fact the case in virtue of the formulae

$$\begin{aligned} 1^\circ. & A_1A_2^m.B_1B_2^m = A_1B_1(A_2B_2)^m, \\ 2^\circ. & A_1A_2^m.12B_1B_2^m = 12A_1^mB_1(A_2^mB_2)^m, \\ 3^\circ. & 12A_1A_2^m.B_1B_2^m = 12(A_1B_2)(A_2B_1)^m, \\ 4^\circ. & 12A_1A_2^m.12B_1B_2^m = A_1^mB_1(A_2^mB_2)^m, \end{aligned}$$

inasmuch as  $1, A, B, \dots$  being a group,  $AB$  and  $A^mB$  are each of them a substitution of the group,  $= C$  suppose; we have of course in like manner  $A_1B_1 = C_1, A_2B_2 = C_2$ , etc., and the right-hand sides of the four formulæ are thus of the forms  $C_1C_2^m, 12C_1C_2^m, 12C_1C_2^m, C_1C_2^m$  respectively, viz. these are substitutions of the system.

To prove for instance the formula  $2^\circ$ , considering the arrangements obtained by operating upon  $U_1U_2$ , we have

$$B_1B_2^mU_1U_2 = B_1B_2^m, \quad 12B_1B_2^mU_1U_2 = B_2^mB_1, \quad A_1A_2^m.12B_1B_2^mU_1U_2 = A_1B_2^m.A_2^mB_1.$$

where of course the expressions on the right-hand side denote arrangements. By reason that the original group is commutative  $(A^{m-1}B)^m$  is  $= A^{m^2-m}B^m$  or since  $m^2 = 1 \pmod{\mu}$  this is  $= A^{1-m}B^m$ ; hence also  $(A_2^{m-1}B_2)^m = A_2^{1-m}B_2^m$ : hence, considering as before the arrangements obtained by operating on  $U_1U_2$ , we have

$$(A_2^{m-1}B_2)^m U_1 U_2 = 1 \cdot A_2 B_2^m; \quad A_1^m B_1 (A_2^{m-1}B_2)^m U_1 U_2 = A_1^m B_1 A_2 B_2^m,$$

and

$$12A_1^m B_1 (A_2^{m-1}B_2)^m U_1 U_2 = A_2^m B_2 A_1 B_1^m,$$

where of course the right-hand sides denote arrangements. Hence in the formula  $2^\circ$ , the two substitutions operating on  $U_1 U_2$  give each of them the same arrangement  $A_2^m B_2 A_1 B_1^m$ , that is, the two substitutions are equal. And similarly the other formulae  $1^\circ, 3^\circ, 4^\circ$  may be proved.

By interchanging  $A$  and  $B$ , in the formulae I obtain

$$1^\circ. \quad A_1 A_2^m \cdot B_1 B_2^m = A_1 B_1 (A_2 B_2)^m;$$

$$B_1 B_2^m \cdot A_1 A_2^m = B_1 A_1 (B_2 A_2)^m = A_1 B_1 (A_2 B_2)^m,$$

which is

$$= A_1 A_2^m \cdot B_1 B_2^m.$$

$$2^\circ \text{ and } 3^\circ. \quad A_1 A_2^m \cdot 12B_1 B_2^m = 12A_1^m B_1 (A_2^{m-1}B_2)^m;$$

$$12B_1 B_2^m \cdot A_1 A_2^m = 12B_1 A_1 (B_2 A_2)^m = 12A_1 B_1 (A_2 B_2)^m,$$

which is not

$$= A_1 A_2^m \cdot 12B_1 B_2^m.$$

$$3^\circ \text{ and } 2^\circ. \quad 12A_1 A_2^m \cdot B_1 B_2^m = 12A_1 B_2 (A_2 B_1)^m;$$

$$B_1 B_2^m \cdot 12A_1 A_2^m = 12A_1 B_2^m (A_2 B_1^m)^m = 12A_1 B_2^m A_2^m B_1,$$

which is not

$$= 12A_1 A_2^m \cdot B_1 B_2^m.$$

$$4^\circ. \quad 12A_1 A_2^m \cdot 12B_1 B_2^m = A_1^m B_1 (A_2^{m-1}B_2)^m;$$

which is not

$$\begin{aligned} 12B_1 B_2^m \cdot 12A_1 A_2^m &= A_1 B_1^m (A_2 B_2^m)^m = (A_1^m B_1)^m A_2^m B_2, \\ &= 12A_1 A_2^m \cdot 12B_1 B_2^m. \end{aligned}$$

That is, in the double group any two substitutions of the form  $A_1 A_2^m$  are commutative, but a substitution of this form is not in general commutative with a substitution of the form  $12B_1 B_2^m$ , nor are two substitutions of the last-mentioned form  $12A_1 A_2^m$  in general commutative with each other; hence the double group is not in general commutative.

In the formula  $4^\circ$ , writing  $B = A$ , we have

$$(12A_1 A_2^m)^2 = A_1^{m+1} A_2^{m^2+m} = A_1^{m+1} A_2^{m+1},$$

hence, if  $\lambda$  is the least integer value such that

$$\lambda(m+1) \equiv 0 \pmod{\mu},$$

we have  $(12A_1 A_2^m)^\lambda = 1$ , viz. in the double group the substitutions of the second row are each of them of an order not exceeding  $2\lambda$ , the substitution  $12$  being of course of the order  $2$ . In particular, if  $m = \mu - 1$ , then  $\lambda = 1$ : and the substitutions of the second row are each of them of the order  $2$ .

As the most simple instance of the theorem, suppose that the original group is the group  $1, (abc), (acb)$ , or say  $1, \Theta, \Theta^2$ , of the cyclical substitutions upon the 3 letters  $abc$ . Here  $m^2 = 1 \pmod{3}$  or except  $m = 1$  the only solution is  $m = 2$ , and thence  $\lambda = 1$ . The double group is a group of the order 6 on the letters  $a_1b_1c_1a_2b_2c_2$ : viz. writing  $\Theta = (abc)$ , and therefore  $\Theta_1 = (a_1b_1c_1)$ ,  $\Theta_1^2 = (a_1c_1b_1)$ ,  $\Theta_2 = (a_2b_2c_2)$ ,  $\Theta_2^2 = (a_2c_2b_2)$ , also writing  $12 = \alpha$ , the substitutions are

$$\begin{aligned} &1, \Theta_1\Theta_2^2, \Theta_1^2\Theta_2, \\ &\alpha, \alpha\Theta_1\Theta_2^2, \alpha\Theta_1^2\Theta_2, \end{aligned}$$

the arrangements corresponding to the second row of substitutions are  $a_2b_2c_2a_1b_1c_1, b_2c_2a_2c_1a_1b_1, c_2a_2b_2b_1c_1a_1$ , viz. the substitutions are  $(a_1a_2)(b_1b_2)(c_1c_2)$ ,  $(a_1b_2)(b_1c_2)(c_1a_2)$ ,  $(a_1c_2)(b_1a_2)(c_1b_2)$ , each of them of the second order as they should be.

I take the opportunity of mentioning a further theorem. Let  $\mu$  be the order of the group, and  $\alpha$  the order of any term  $A$  thereof,  $\alpha$  being of course a submultiple of  $\mu$ : and let the term  $A$  be called quasi-positive when  $\mu\left(1 - \frac{1}{\alpha}\right)$  is even, quasi-negative when  $\mu\left(1 - \frac{1}{\alpha}\right)$  is odd. The theorem is that the product of two quasi-positive terms, or of two quasi-negative terms, is quasi-positive; but the product of a quasi-positive term and a quasi-negative term is quasi-negative. And it follows hence that, either the terms of a group are all quasi-positive, or else one half of them are quasi-positive and the other half of them are quasi-negative.

The proof is very simple: a term  $A$  of the group operating on the  $\mu$  terms  $(1, A, B, C, \dots)$  of the group, gives these same terms in a different order, or it may be regarded as a substitution upon the  $\mu$  symbols  $1, A, B, C, \dots$ ; so regarded it is a regular substitution (this is a fundamental theorem, which I do not stop to prove), and hence since it must be of the order  $\alpha$  it is a substitution composed of  $\frac{\mu}{\alpha}$  cycles, each of  $\alpha$  letters. But in general a substitution is positive or negative according as it is equivalent to an even or an odd number of inversions; a cyclical substitution upon  $\alpha$  letters is positive or negative according as  $\alpha - 1$  is even or odd; and the substitution composed of the  $\frac{\mu}{\alpha}$  cycles is positive or negative according as  $\frac{\mu}{\alpha}(\alpha - 1)$ , that is,  $\mu\left(1 - \frac{1}{\alpha}\right)$ , is even or odd. Hence by the foregoing definition, the term  $A$ , according as it is quasi-positive or quasi-negative, corresponds to a positive substitution or to a negative substitution; and such terms combine together in like manner with positive and negative substitutions.

*Cambridge, 3rd April, 1878.*

## 660.

## ON THE CORRESPONDENCE OF HOMOGRAPHIES AND ROTATIONS.

[From the *Mathematische Annalen*, t. XV. (1879), pp. 238–240.]

It is a fundamental notion in Prof. Klein's theory of the "Icosahedron" that homographies correspond to rotations (of a solid body about a fixed point): in such wise that, considering the homographies which correspond to two given rotations, the homography compounded of these corresponds to the rotation compounded of the two rotations.

Say the two homographies are  $A + Bp + Cq + Dpq = 0$ ,  $A_1 + B_1q + C_1r + D_1qr = 0$ , then, eliminating  $q$ , the compound homography is  $A_2 + B_2p + C_2r + D_2pr = 0$ , where

$$A_2, B_2, C_2, D_2 = B_1A - A_1C, \quad B_1B - A_1D, \quad D_1A - C_1C, \quad D_1B - C_1D;$$

and the theorem is that, corresponding to these, we have rotations depending on the parameters  $(\lambda, \mu, \nu)$ ,  $(\lambda_1, \mu_1, \nu_1)$ ,  $(\lambda_2, \mu_2, \nu_2)$  respectively, such that the third rotation is that compounded of the first and second rotations. The question arises to find the expression for the parameters of the homography in terms of the parameters of the corresponding rotation.

The rotation  $(\lambda, \mu, \nu)$  is taken to denote a rotation through an angle  $\vartheta$  about an axis the inclinations of which to the axes of coordinates are  $f, g, h$ , the values of  $\lambda, \mu, \nu$  then being  $\tan \frac{1}{2}\vartheta \cos f$ ,  $\tan \frac{1}{2}\vartheta \cos g$ ,  $\tan \frac{1}{2}\vartheta \cos h$  respectively:  $(\lambda_1, \mu_1, \nu_1)$  and  $(\lambda_2, \mu_2, \nu_2)$  have of course the like significations; and then, if  $(\lambda, \mu, \nu)$  refer to the first rotation, and  $(\lambda_1, \mu_1, \nu_1)$  to the second rotation, the values of  $(\lambda_2, \mu_2, \nu_2)$  for the rotation compounded of these are taken to be<sup>2</sup>

$$\begin{aligned}\lambda_2 &= \lambda + \lambda_1 + \mu\nu_1 - \mu_1\nu, \\ \mu_2 &= \mu + \mu_1 + \nu\lambda_1 - \nu_1\lambda, \\ \nu_2 &= \nu + \nu_1 + \lambda\mu_1 - \lambda_1\mu,\end{aligned}$$

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<sup>2</sup>The numerators might equally well have been  $\lambda + \lambda_1 - (\mu\nu_1 - \mu_1\nu)$ , etc., but there is no essential difference: we pass from one set of formulæ to the other by reversing the signs of all the symbols: and hence, by properly fixing the sense of the rotations, the signs may be made to be  $+$  as in the text. Assuming this to be so, if we then reverse the order of the component rotations, we have for the new compound rotation the like formulæ with the signs  $-$  instead of  $+$ ; but this in passing. The formulæ, virtually due to Rodrigues, are given in my paper "On the motion of rotation of a solid body," *Camb. Math. Journal*, t. III. (1843), [6].

each divided by

$$1 - \lambda\lambda_1 - \mu\mu_1 - \nu\nu_1;$$

and if we then write for  $\lambda, \mu, \nu$ , the quotients  $x, y, z$  each divided by  $w$ , and in like manner for  $\lambda_1, \mu_1, \nu_1$  and  $\lambda_2, \mu_2, \nu_2$ , the quotients  $x_1, y_1, z_1$  each divided by  $w_1$ , and  $x_2, y_2, z_2$  each divided by  $w_2$ , the formulae for the composition of the rotations are

$$\begin{aligned} x_2 &= xw_1 + x_1w + yz_1 - y_1z, \\ y_2 &= yw_1 + y_1w + zx_1 - z_1x, \\ z_2 &= zw_1 + z_1w + xy_1 - x_1y, \\ w_2 &= ww_1 - xx_1 - yy_1 - zz_1; \end{aligned}$$

and the question is to express  $A, B, C, D$  as functions of  $(x, y, z, w)$ , such that  $A_1, B_1, C_1, D_1$  denoting the like functions of  $(x_1, y_1, z_1, w_1)$ ,  $A_2, B_2, C_2, D_2$  shall denote the like functions of  $(x_2, y_2, z_2, w_2)$ .

It is found that the required conditions are satisfied by assuming

$$A, B, C, D = ix - y, -iz + w, -iz - w, -ix - y,$$

(where  $i = \sqrt{-1}$  as usual): in fact, we then have

$$\begin{aligned} A_2 &= B_1A - A_1C \\ &= (-iz_1 + w_1)(ix - y) - (ix_1 - y_1)(-iz - w) \\ &= i(xw_1 + x_1w + yz_1 - y_1z) - (yw_1 + y_1w + zx_1 - z_1x) \end{aligned}$$

as it should be: and we verify in like manner the values of  $B_2, C_2$  and  $D_2$  respectively.

The result consequently is that we have the homography

$$(ix - y) + (-iz + w)p + (-iz - w)q + (-ix - y)pq = 0$$

corresponding to the rotation  $\left(\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right)$ : where  $\frac{x}{w}, \frac{y}{w}, \frac{z}{w}$  are the parameters of rotation,  $\tan \frac{1}{2}\vartheta \cos f, \tan \frac{1}{2}\vartheta \cos g, \tan \frac{1}{2}\vartheta \cos h$ .

I remark as regards the geometrical theory that, if we consider two lines  $J$  and  $K$  fixed in space, and a third line  $L$  fixed in the solid body and moveable with it: then, for any given position of the solid body, the three lines  $J, K, L$  are directrices of a hyperboloid, the generatrices whereof meet each of the three lines: and these generatrices determine, say on the fixed lines  $J$  and  $K$ , two series of points corresponding homographically to each other: that is, corresponding to any given position of the solid body we have a homography. But it is not immediately obvious how we can thence obtain the foregoing analytical formula.

*Cambridge, 3 April, 1879.*

## 661.

ON THE DOUBLE  $\vartheta$ -FUNCTIONS.

[From the *Proceedings of the London Mathematical Society*, vol. IX. (1878), pp. 29, 30.]

Prof. Cayley gave an account of researches<sup>3</sup> on which he is engaged upon the double  $\vartheta$ -functions. In regard to these, he establishes in a strictly analogous manner the theory of the single  $\vartheta$ -functions, the process for the single functions being in fact as follows: Considering  $u, x$  as connected by the differential relation

$$du = \frac{dx}{\sqrt{a-x} \cdot \sqrt{b-x} \cdot \sqrt{c-x} \cdot \sqrt{d-x}},$$

then, if  $A, B, C, D, \Omega$  denote functions of  $u$ , viz. for shortness, the single letters are used, instead of writing them as functional symbols,  $A(u), B(u)$ , &c., then, by way of definition of these functions (called, the first four of them  $\vartheta$ -functions, and the last an  $\omega$ -function), we assume

$$A, B, C, D = \Omega\sqrt{a-x}, \Omega\sqrt{b-x}, \Omega\sqrt{c-x}, \Omega\sqrt{d-x}$$

respectively, together with one other equation, as presently mentioned. Without in any wise defining the meaning of  $\Omega$ , we then obtain a set of equations of the form

$$A\delta B - B\delta A = \&c. \sqrt{c-x} \cdot \sqrt{d-x} \delta u,$$

(mere constant coefficients are omitted), or, what is the same thing,

$$A\delta B - B\delta A = CD \delta u,$$

which are differential equations defining the nature of the ratio-functions  $A : B : C : D$ . If, proceeding to second differential coefficients, we attempt to form the expressions for  $A\delta^2 A - (\delta A)^2$ , &c., these involve multiples of  $\Omega\delta^2\Omega - (\delta\Omega)^2$ ; in order to obtain a con-

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<sup>3</sup>See paper, number 665.



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Arthur Cayley

Cambridge, 1896

**661 (continued): ON THE DOUBLE  $\vartheta$ -FUNCTIONS.**

venient form, we assume  $\Omega\delta^2\Omega - (\delta\Omega)^2 = \Omega^2 M (\delta u)^2$ , where  $M$  is a function of  $x$ . We thus obtain an equation  $A\delta^2 A - (\delta A)^2 = \Omega^2 \mathfrak{A} (\delta u)^2$ , where the value of  $\mathfrak{A}$  depends upon that of  $M$ . The value of  $M$  has to be taken so as to simplify as much as may be the expression of  $\mathfrak{A}$ , but so that  $\mathfrak{A}$  shall be a symmetrical function of the constants  $a, b, c, d$ : this last condition is assigned in order that the like simplification may present itself in the analogous relations  $B\delta^2 B - (\delta B)^2 = \Omega^2 \mathfrak{B} (\delta u)^2$ , &c. The proper expression of  $M$  is found to be

$$M = -2x^2 + x(a + b + c + d) + a^2 + b^2 + c^2 + d^2 - 2bc - 2ca - 2ab - 2ad - 2bd - 2cd,$$

viz.  $M$  having this value, the one other equation above referred to is

$$\Omega\delta^2\Omega - (\delta\Omega)^2 = \Omega^2 M (\delta u)^2;$$

and we then have a system of four equations such as

$$A\delta^2 A - (\delta A)^2 = \Omega^2 \mathfrak{A} (\delta u)^2,$$

where  $\mathfrak{A}$  is a linear function of  $x$ , and where consequently  $\Omega^2 \mathfrak{A}$  can be expressed as a linear function of any two of the four squares  $A^2, B^2, C^2, D^2$ .

To establish the connexion with the Jacobian  $\Theta$  and  $H$  functions, the differential relation between  $u, x$  may be taken to be

$$\delta u = \frac{\delta x}{\sqrt{x \cdot 1 - x \cdot 1 - k^2 x}},$$

viz. we have here  $d = \infty$ , and to adapt the formulae to this value it is necessary to write  $\frac{u}{\sqrt{d}}$  instead of  $u$ , and make other easy changes. The result is that  $\Omega$  differs from  $D$  by a constant factor only, so that, instead of the five functions  $A, B, C, D, \Omega$ , we have only the four functions  $A, B, C, D$ . The equation  $\Omega\delta^2\Omega - (\delta\Omega)^2 = \Omega^2 \mathfrak{A} (\delta u)^2$  is replaced by an equation  $D\delta^2 D - (\delta D)^2 = D^2 \mathfrak{D} (\delta u)^2$ , or say  $= \delta^2(\log D) D^2 (\delta u)^2$ , which gives a result of the form

$$D = \Theta e^{\alpha u^2},$$

showing that  $D$  differs from Jacobi's  $\Theta(u)$  only by an exponential factor of the form  $\Theta e^{\alpha u^2}$ . And it then further appears that  $A, B, C$  differ only by factors of the like form from the three numerator functions  $H(u), H(u + K), \Theta(u + K)$ , so that, neglecting constant factors, the functions

$$\frac{A}{D}, \quad \frac{B}{D}, \quad \frac{C}{D}$$

are equal to  $\frac{H(u)}{\Theta(u)}, \frac{H(u + K)}{\Theta(u)}, \frac{\Theta(u + K)}{\Theta(u)}$ , that is, to the elliptic functions  $\text{sn } u, \text{cn } u, \text{dn } u$ .

## 662.

ON THE DOUBLE  $\vartheta$ -FUNCTIONS IN CONNEXION WITH A 16-NODAL QUARTIC SURFACE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXIII. (1877), pp. 210–219.]

I have before me Göpel's memoir, "Theoriae transcendentium Abelianarum primi ordinis ad-umbratio levis," Crelle's Journal, t. XXXV. (1847), pp. 277–312. Writing  $P_1, P_2, P_3, P_4$ , etc., in place of his  $P', P'', P'''$ , etc., also  $\alpha, \beta, \gamma, \delta, X', Y', Z', W'$ , in place of his  $t, u, v, w, T, U, V, W$ , the system of 16 equations (given p. 287) is

$$\begin{aligned}
 (1) \quad P^2 &= (\alpha, -\beta, -\gamma, \delta)(X', Y', Z', W'), \\
 (4) \quad P_1^2 &= (\alpha, \beta, -\gamma, -\delta)(X', Y', Z', W'), \\
 (9) \quad P_2^2 &= (\alpha, -\beta, \gamma, -\delta)(X', Y', Z', W'), \\
 (12) \quad P_3^2 &= (\alpha, \beta, \gamma, \delta)(X', Y', Z', W'); \\
 (3) \quad Q^2 &= (\beta, -\alpha, -\delta, \gamma)(X', Y', Z', W'), \\
 (2) \quad Q_1^2 &= (\beta, \alpha, -\delta, -\gamma)(X', Y', Z', W'), \\
 (11) \quad Q_2^2 &= (\beta, -\alpha, \delta, -\gamma)(X', Y', Z', W'), \\
 (10) \quad Q_3^2 &= (\beta, \alpha, \delta, \gamma)(X', Y', Z', W'); \\
 (13) \quad R^2 &= (\gamma, -\delta, -\alpha, \beta)(X', Y', Z', W'), \\
 (16) \quad R_1^2 &= (\gamma, \delta, -\alpha, -\beta)(X', Y', Z', W'), \\
 (5) \quad R_2^2 &= (\gamma, -\delta, \alpha, -\beta)(X', Y', Z', W'), \\
 (8) \quad R_3^2 &= (\gamma, \delta, \alpha, \beta)(X', Y', Z', W'); \\
 (15) \quad S^2 &= (\delta, -\gamma, -\beta, \alpha)(X', Y', Z', W'), \\
 (14) \quad S_1^2 &= (\delta, \gamma, -\beta, -\alpha)(X', Y', Z', W'), \\
 (7) \quad S_2^2 &= (\delta, -\gamma, \beta, -\alpha)(X', Y', Z', W'), \\
 (6) \quad S_3^2 &= (\delta, \gamma, \beta, \alpha)(X', Y', Z', W');
 \end{aligned}$$

viz. we have  $P^2 = \alpha X' - \beta Y' - \gamma Z' + \delta W'$ , etc. The reason for the apparently arbitrary manner in which I have numbered these equations, will appear further on. I recall that the sixteen double  $\vartheta$ -functions, that is,  $\vartheta$ -functions of two arguments  $u, u'$ , are<sup>1</sup>

$$\begin{array}{cccc} P, & P_1, & P_2, & P_3, \\ iQ, & Q_1, & iQ_2, & Q_3, \\ iR, & iR_1, & R_2, & R_3, \\ S, & iS_1, & iS_2, & S_3, \end{array}$$

the factor  $i, = \sqrt{-1}$ , being introduced in regard to the six functions which are odd functions of the arguments. But disregarding the  $i$  sign, I speak of  $P^2, P_1^2, \dots, Q^2$ , etc., as the squared functions, or simply as the squares;  $\alpha, \beta, \gamma, \delta$  are constants, depending of course on the parameters of the  $\vartheta$ -functions;  $X', Y', Z', W'$ , which are however to be eliminated, are themselves  $\vartheta$ -functions to a different set of parameters: the 16 equations express that the squared functions  $P^2, P_1^2$ , etc., are linear functions of  $X', Y', Z', W'$ , and they consequently serve to obtain linear relations between the squared functions: viz. by means of them, Göpel expresses the remaining 12 squares, each in terms of the selected four squares  $P_1^2, P_3^2, S_1^2, S_3^2$ , which are linearly independent: that is, he obtains linear relations between five squares, and he seems to have assumed that there were not any linear relations between fewer than five squares.

It appears however by Rosenhain's "Mémoire sur les fonctions de deux variables et à quatre périodes etc.", Mém. Sav. Etrangers, t. XI. (1851), pp. 364–468, that there are, in fact, linear relations between four squares, viz. that there exist sixes of squares such that, selecting at pleasure any three out of the six, each of the remaining three squares can be expressed as a linear function of these three squares. Knowing this result, it is easy to verify it by means of the sixteen equations, and moreover to show that there are in all 16 such sixes: these are shown by the following scheme which I copy from Kummer's memoir "Ueber die algebraischen Strahlensysteme u. s. w." Berlin. Abh. (1866), p. 66: viz. the scheme is

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |     |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16  |
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8   |
| 13 | 14 | 15 | 16 | 9  | 10 | 11 | 12 | 5  | 6  | 7  | 8  | 1  | 2  | 3  | 4   |
| 8  | 7  | 6  | 5  | 4  | 3  | 2  | 1  | 16 | 15 | 14 | 13 | 12 | 11 | 10 | 9   |
| 7  | 8  | 5  | 6  | 3  | 4  | 1  | 2  | 15 | 16 | 13 | 14 | 11 | 12 | 9  | 10  |
| 6  | 5  | 8  | 7  | 2  | 1  | 4  | 3  | 14 | 13 | 16 | 15 | 10 | 9  | 12 | 11. |

<sup>1</sup>The same functions in Rosenhain's notation are

$$\begin{array}{l} 00, 02, 20, 22, \\ 01, 03, 21, 23, \\ 10, 12, 30, 32, \\ 11, 13, 31, 33; \end{array}$$

viz. the figures here written down are the suffixes of his  $\vartheta$ -functions,  $00 = \vartheta_{0,0}(v, w)$ , etc.

In fact, to show that any four of the squares, for instance 1, 9, 13, 8, that is,  $P^2, P_2^2, R^2, R_3^2$ , are linearly connected, it is only necessary to show that the determinant of coefficients

$$\begin{vmatrix} \alpha, & -\beta, & -\gamma, & \delta \\ \alpha, & -\beta, & \gamma, & -\delta \\ \gamma, & -\delta, & -\alpha, & \beta \\ \gamma, & \delta, & \alpha, & \beta \end{vmatrix}$$

is  $= 0$ , or what is the same thing, that there exists a linear function of the new variables  $(X, Y, Z, W)$ , which will become  $= 0$  on putting for these variables the values in any line of this determinant: we have such a function, viz. this is

$$\beta X + \alpha Y - \delta Z - \gamma W,$$

or say

$$[1] (\beta, \alpha, -\delta, -\gamma)(X, Y, Z, W).$$

This function also vanishes if for  $(X, Y, Z, W)$  we substitute the values

$$\delta, -\gamma, -\beta, -\alpha,$$

$$\delta, \gamma, \beta, \alpha,$$

which belong to 7, 6, that is,  $S_2^2$  and  $S_3^2$  respectively. It thus appears that 1, 9, 13, 8, 7, 6, that is,  $P^2, P_2^2, R^2, R_3^2, S_2^2, S_3^2$ , are a set of six squares having the property in question. I remark that the process of forming the linear functions is a very simple one; we write down six lines, and thence directly obtain the result, thus

$$\begin{array}{rcll} 1 & \alpha, & -\beta, & -\gamma, & \delta \\ 9 & \alpha, & -\beta, & \gamma, & -\delta \\ 13 & \gamma, & -\delta, & -\alpha, & \beta \\ 8 & \gamma, & \delta, & \alpha, & \beta \\ 7 & \delta, & -\gamma, & \beta, & -\alpha \\ 6 & \delta, & \gamma, & \beta, & \alpha \\ & \beta, & \alpha, & -\delta, & -\gamma : \end{array}$$

viz.  $\beta, \alpha, \delta, \gamma$  are the letters not previously occurring in the four columns respectively: the first letter  $\beta$  is taken to have the sign  $+$ , and then the remaining signs are determined by the condition that, combining the last line with any line above it (e.g. with the line next above it  $\beta\delta + \alpha\gamma - \delta\beta - \gamma\alpha$ ), the sum must be zero.

We find in this way, as the conditions for the existence of the 16 sixes respectively,

$$\begin{array}{ll} [1] & (\beta, \alpha, -\delta, -\gamma)(X, Y, Z, W) = 0, \\ [2] & (\alpha, -\beta, -\gamma, \delta)(X, Y, Z, W) = 0, \\ [3] & (\alpha, \beta, -\gamma, -\delta)(X, Y, Z, W) = 0, \\ [4] & (\beta, -\alpha, -\delta, \gamma)(X, Y, Z, W) = 0, \end{array}$$

- [5]  $(\delta, \gamma, \beta, \alpha)(X, Y, Z, W) = 0,$
- [6]  $(\gamma, -\delta, \alpha, -\beta)(X, Y, Z, W) = 0,$
- [7]  $(\gamma, \delta, \alpha, \beta)(X, Y, Z, W) = 0,$
- [8]  $(\delta, -\gamma, \beta, -\alpha)(X, Y, Z, W) = 0,$
- [9]  $(\beta, \alpha, \delta, \gamma)(X, Y, Z, W) = 0,$
- [10]  $(\alpha, -\beta, \gamma, -\delta)(X, Y, Z, W) = 0,$
- [11]  $(\alpha, \beta, \gamma, \delta)(X, Y, Z, W) = 0,$
- [12]  $(\beta, -\alpha, \delta, -\gamma)(X, Y, Z, W) = 0,$
- [13]  $(\delta, -\gamma, -\beta, \alpha)(X, Y, Z, W) = 0,$
- [14]  $(\gamma, -\delta, -\alpha, \beta)(X, Y, Z, W) = 0,$
- [15]  $(\gamma, \delta, -\alpha, -\beta)(X, Y, Z, W) = 0,$
- [16]  $(\delta, \gamma, -\beta, -\alpha)(X, Y, Z, W) = 0.$

I repeat in a new order the sets of coefficients which belong to the several squares, viz. these are

- (1)  $P^2 = (\alpha, -\beta, -\gamma, \delta),$
- (2)  $Q_1^2 = (\beta, \alpha, -\delta, -\gamma),$
- (3)  $Q^2 = (\beta, -\alpha, -\delta, \gamma),$
- (4)  $P_1^2 = (\alpha, \beta, -\gamma, -\delta),$
- (5)  $R_2^2 = (\gamma, -\delta, \alpha, -\beta),$
- (6)  $S_3^2 = (\delta, \gamma, \beta, \alpha),$
- (7)  $S_2^2 = (\delta, -\gamma, \beta, -\alpha),$
- (8)  $R_3^2 = (\gamma, \delta, \alpha, \beta),$
- (9)  $P_2^2 = (\alpha, -\beta, \gamma, -\delta),$
- (10)  $Q_3^2 = (\beta, \alpha, \delta, \gamma),$
- (11)  $Q_2^2 = (\beta, -\alpha, \delta, -\gamma),$
- (12)  $P_3^2 = (\alpha, \beta, \gamma, \delta),$
- (13)  $R^2 = (\gamma, -\delta, -\alpha, \beta),$
- (14)  $S_1^2 = (\delta, \gamma, -\beta, -\alpha),$
- (15)  $S^2 = (\delta, -\gamma, -\beta, \alpha),$
- (16)  $R_1^2 = (\gamma, \delta, -\alpha, -\beta).$

And I remark that, if we connect these with the multipliers  $(Y, -X, W, -Z)$ , we obtain, except that there is sometimes a reversal of all the signs, the same linear functions of  $(X, Y, Z, W)$  as are written down under the same numbers in square brackets above: thus (1) gives

$$(\alpha, -\beta, -\gamma, \delta)(Y, -X, W, -Z), \text{ which is } (\beta, \alpha, -\delta, -\gamma)(X, Y, Z, W), = [1];$$

and so (2) gives

$$(\beta, \alpha, -\delta, -\gamma)(Y, -X, W, -Z), \text{ which is } (-\alpha, \beta, \gamma, -\delta)(X, Y, Z, W),$$

or, reversing the signs,

$$(\alpha, -\beta, -\gamma, \delta)(X, Y, Z, W), = [2].$$

Comparing with the geometrical theory in Kummer's Memoir, it appears that the several systems of values (1), (2), . . . , (16) are the coordinates of the nodes of a 16-nodal quartic surface, which nodes lie by sixes in the singular tangent planes, in the manner expressed by the foregoing scheme, wherein each top number may refer to a singular tangent plane, and then the numbers below it show the nodes in this plane: or else the top number may refer to a node, and then the numbers below it show the singular planes through this node.

And, from what precedes, we have the general result: the 16 squared double  $\vartheta$ -functions correspond (one to one) to the nodes of a 16-nodal quartic surface, in such wise that linearly connected squared functions correspond to nodes in the same singular tangent plane.

The question arises, to find the equation of the 16-nodal quartic surface, having the foregoing nodes and singular tangent planes. Starting from one of the irrational forms, say

$$\sqrt{A} [1][5] + \sqrt{B} [2][6] + \sqrt{C} [3][7] = 0,$$

the coefficients  $A, B, C$  are readily determined; and the result written at full length is

$$\begin{aligned} & \sqrt{2}(\alpha\beta - \gamma\delta)(\alpha\delta + \beta\gamma)(\beta X + \alpha Y - \delta Z - \gamma W)(\delta X + \gamma Y + \beta Z + \alpha W) \\ & + \sqrt{(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(\alpha\gamma - \beta\delta)}(\alpha X - \beta Y - \gamma Z + \delta W)(\gamma X - \delta Y - \alpha Z + \beta W) \\ & + \sqrt{(\alpha^2 - \beta^2 - \gamma^2 + \delta^2)(\alpha\gamma + \beta\delta)}(\alpha X + \beta Y - \gamma Z - \delta W)(\gamma X + \delta Y + \alpha Z + \beta W) = 0. \end{aligned}$$

It is a somewhat long, but nevertheless interesting, piece of algebraical work to rationalise the foregoing equation: the result is

$$\begin{aligned} & (\gamma^2\delta^2 - \alpha^2\beta^2)(\beta^2\gamma^2 - \alpha^2\delta^2)(\alpha^2\beta^2 - \gamma^2\delta^2)(X^4 + Y^4 + Z^4 + W^4) \\ & + (\gamma^2\delta^2 - \alpha^2\beta^2)(\alpha^2\beta^2 - \gamma^2\delta^2)(\alpha^2 + \delta^2 - \beta^2 - \gamma^2)(Y^2Z^2 + X^2W^2) \\ & + (\alpha^2\beta^2 - \gamma^2\delta^2)(\beta^2\gamma^2 - \alpha^2\delta^2)(\beta^2 + \delta^2 - \gamma^2 - \alpha^2)(Z^2X^2 + Y^2W^2) \\ & + (\beta^2\gamma^2 - \alpha^2\delta^2)(\gamma^2\delta^2 - \alpha^2\beta^2)(\gamma^2 + \delta^2 - \alpha^2 - \beta^2)(X^2Y^2 + Z^2W^2) \\ & + 2\alpha\beta\gamma\delta(\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(\alpha^2 - \beta^2 - \gamma^2 + \delta^2)XYZW = 0; \end{aligned}$$

or, if we write for shortness

$$\begin{aligned} L &= \beta^2\gamma^2 - \alpha^2\delta^2, & F &= \alpha^2 + \beta^2 - \gamma^2 - \delta^2, \\ M &= \gamma^2\alpha^2 - \beta^2\delta^2, & G &= \alpha^2 - \beta^2 + \gamma^2 - \delta^2, \\ N &= \alpha^2\beta^2 - \gamma^2\delta^2, & H &= \alpha^2 - \beta^2 - \gamma^2 + \delta^2, \\ & & \Delta &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2, \end{aligned}$$

then the result is

$$\begin{aligned} & LMN(X^4 + Y^4 + Z^4 + W^4) \\ & + MN(F\Delta + 2L)(Y^2Z^2 + X^2W^2) \\ & + NL(G\Delta + 2M)(Z^2X^2 + Y^2W^2) \\ & + LM(H\Delta + 2N)(X^2Y^2 + Z^2W^2) \\ & - 2\alpha\beta\gamma\delta FGH\Delta XYZW = 0. \end{aligned}$$

It may be easily verified that any one of the sixteen points, for instance  $(\alpha, \beta, \gamma, \delta)$ , is a node of the surface. Thus to show that the derived function in respect to  $X$ , vanishes for  $X, Y, Z, W = \alpha, \beta, \gamma, \delta$ ; the derived function here divides by  $2\alpha$ , and omitting this factor, the equation to be verified is

$$LMN \cdot 2\alpha^2 + MN(F\Delta + 2L)\delta^2 + NL(G\Delta + 2M)\gamma^2 + LM(H\Delta + 2N)\beta^2 - \beta^2\gamma^2\delta^2 FGH\Delta = 0,$$

viz. the whole coefficient of  $LMN$  is  $2(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)$ ,  $= 2\Delta$ ; hence throwing out the factor  $\Delta$ , the equation becomes

$$2LMN + MNF\delta^2 + NLG\gamma^2 + LMH\beta^2 - \beta^2\gamma^2\delta^2 FGH = 0.$$

Writing this in the form

$$L(2MN + NG\gamma^2 + MH\beta^2) = \beta^2\delta^2(\gamma^2\delta^2 FGH - MNF),$$

we find without difficulty  $GH\gamma^2\delta^2 - MN = -L(\delta^2 - \gamma^2)^2$ ; hence throwing out the factor  $L$ , the equation becomes

$$G(2M + G\gamma^2) + MH\beta^2 + F\delta^2(\delta^2 - \gamma^2)^2 = 0;$$

we find

$$\begin{aligned} MH\beta^2 + F\delta^2(\delta^2 - \gamma^2)^2 &= (\alpha^2\beta^2 - \gamma^2\delta^2)(2\delta^2\beta^2 - \gamma^2(\alpha^2 + \beta^2 + \delta^2) + \gamma^4) \\ &= -N(2\beta^2\delta^2 - \gamma^2(\alpha^2 + \beta^2 + \delta^2) + \gamma^4), \end{aligned}$$

or throwing out the factor  $N$ , the equation becomes

$$2M + G\gamma^2 - 2\beta^2\delta^2 + \gamma^2(\alpha^2 + \beta^2 + \delta^2) - \gamma^4 = 0,$$

which is at once verified: and similarly it can be shown that the other derived functions vanish, and the point  $(\alpha, \beta, \gamma, \delta)$  is thus a node.

The surface seems to be the general 16-nodal surface, viz. replacing  $X, Y, Z, W$  by any linear functions of four coordinates, we have thus  $4 \cdot 4 - 1 = 15$  constants, and the equation contains besides the three ratios  $\alpha : \beta : \gamma : \delta$ , that is, in all 18 constants: the general quartic surface has 34 constants, and therefore the general 16-nodal surface  $34 - 16 = 18$  constants: but the conclusion requires further examination.

Göpel and Rosenhain each connect the theory with that of the ultra-elliptic functions involving the radical  $\sqrt{X}$ ,  $= \sqrt{x \cdot 1 - x \cdot 1 - lx \cdot 1 - mx \cdot 1 - nx}$ ; viz. it appears by their formulae (more completely by those of Rosenhain) that the ratios of the 16 squares can be expressed rationally in terms of the two variables  $x, x'$ , and the radicals

$\sqrt{X}, \sqrt{X'}$ ,  $X'$  being the same function of  $x'$  that  $X$  is of  $x$ . We may instead of the preceding form take  $X$  to be the general quintic function, or what is better take it to be the sextic function  $\overline{a-x} \cdot \overline{b-x} \cdot \overline{c-x} \cdot \overline{d-x} \cdot \overline{e-x} \cdot \overline{f-x}$ ; and we thus obtain a remarkable algebraical theorem: viz. I say that the 16 squares, each divided by a proper constant factor, are proportional to six functions of the form

$$\overline{a-x} \cdot \overline{a-x'},$$

and ten functions of the form

$$\frac{1}{(x-x')^2} \left\{ \sqrt{\overline{a-x} \cdot \overline{b-x} \cdot \overline{c-x} \cdot \overline{d-x'} \cdot \overline{e-x'} \cdot \overline{f-x'}} - \sqrt{\overline{a-x'} \cdot \overline{b-x'} \cdot \overline{c-x'} \cdot \overline{d-x} \cdot \overline{e-x} \cdot \overline{f-x}} \right\}^2,$$

and consequently that these 16 algebraical functions of  $x, x'$  are linearly connected in the manner of the 16 squares; viz. there exist 16 sixes such that, in each six, the remaining three functions can be linearly expressed in terms of any three of them.

To further develop the theory, I remark that the six functions may be represented by  $A, B, C, D, E, F$  respectively: any one of the ten functions would be properly represented by  $\frac{ABC}{DEF}$ , but isolating one letter  $F$ , and writing  $DEF$  to denote this function, may be represented simply as  $DE$ ; and the ten functions thus are  $AB, AC, AD, AE, BC, BD, BE, CD, CE, DE$ .

Writing for shortness  $a, b, c, d, e, f$ , to denote  $a-x, b-x$ , etc., and similarly  $a', b', c', d', e', f'$ , to denote  $a-x', b-x'$ , etc., we thus have

$$\begin{aligned} (13) \quad A &= aa', \quad (= E), \\ (9) \quad B &= bb', \\ (7) \quad C &= cc', \quad (= F), \\ (8) \quad D &= dd', \\ (6) \quad E &= ee', \\ (1) \quad F &= ff', \\ (3) \quad DE &= \frac{1}{(x-x')^2} \{ \sqrt{abcd'e'f'} - \sqrt{a'b'c'def} \}^2, \quad (= D), \\ (4) \quad CE &= \frac{1}{(x-x')^2} \{ \sqrt{abdc'e'f'} - \sqrt{a'b'd'cef} \}^2, \quad (= E), \\ (2) \quad CD &= \frac{1}{(x-x')^2} \{ \sqrt{abec'd'f'} - \sqrt{a'b'e'cdf} \}^2, \\ (14) \quad BE &= \frac{1}{(x-x')^2} \{ \sqrt{acdb'e'f'} - \sqrt{a'c'd'bef} \}^2, \quad (= B), \\ (16) \quad BD &= \frac{1}{(x-x')^2} \{ \sqrt{aceb'd'f'} - \sqrt{a'c'e'bdf} \}^2, \\ (15) \quad BC &= \frac{1}{(x-x')^2} \{ \sqrt{adeb'c'f'} - \sqrt{a'd'e'bcf} \}^2, \\ (10) \quad AE &= \frac{1}{(x-x')^2} \{ \sqrt{bcda'e'f'} - \sqrt{b'c'd'ae'f} \}^2, \quad (= I), \end{aligned}$$



$$\begin{aligned}
 (12) \quad AD &= \frac{1}{(x-x')^2} \{ \sqrt{bcea'd'f'} - \sqrt{b'c'e'adf} \}^2, \\
 (11) \quad AC &= \frac{1}{(x-x')^2} \{ \sqrt{bdea'c'f'} - \sqrt{b'd'e'acf} \}^2, \\
 (5) \quad AB &= \frac{1}{(x-x')^2} \{ \sqrt{cdea'b'f'} - \sqrt{c'd'e'abf} \}^2,
 \end{aligned}$$

where the numbers are in accordance with the foregoing scheme; viz. the scheme becomes

| (1)      | (2)       | (3)       | (4)       | (5)       | (6)       | (7)       | (8)       | (9)       | (10)      | (11)      | (12)      | (13)      | (14)      | (15)      | (16)      |
|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|
| <i>F</i> | <i>CD</i> | <i>DE</i> | <i>CE</i> | <i>AB</i> | <i>E</i>  | <i>C</i>  | <i>D</i>  | <i>B</i>  | <i>AE</i> | <i>AC</i> | <i>AD</i> | <i>A</i>  | <i>BE</i> | <i>BC</i> | <i>BD</i> |
| <i>B</i> | <i>AE</i> | <i>AC</i> | <i>AD</i> | <i>A</i>  | <i>BE</i> | <i>BC</i> | <i>BD</i> | <i>F</i>  | <i>CD</i> | <i>DE</i> | <i>CE</i> | <i>AB</i> | <i>E</i>  | <i>C</i>  | <i>D</i>  |
| <i>C</i> | <i>D</i>  | <i>A</i>  | <i>BE</i> | <i>BC</i> | <i>BD</i> | <i>B</i>  | <i>AE</i> | <i>AC</i> | <i>AD</i> | <i>AB</i> | <i>E</i>  | <i>C</i>  | <i>D</i>  | <i>F</i>  | <i>CD</i> |
|          | <i>E</i>  | <i>AB</i> | <i>CE</i> | <i>DE</i> | <i>CD</i> | <i>F</i>  | <i>BD</i> | <i>BC</i> | <i>BE</i> | <i>A</i>  | <i>AD</i> | <i>AC</i> | <i>AE</i> | <i>B</i>  | <i>DE</i> |
| <i>D</i> | <i>AB</i> | <i>E</i>  | <i>C</i>  | <i>DE</i> | <i>CE</i> | <i>F</i>  | <i>CD</i> | <i>BC</i> | <i>BD</i> | <i>A</i>  | <i>BE</i> | <i>AC</i> | <i>AD</i> | <i>B</i>  | <i>AE</i> |
| <i>E</i> | <i>AB</i> | <i>D</i>  | <i>C</i>  | <i>CD</i> | <i>F</i>  | <i>CE</i> | <i>DE</i> | <i>DE</i> | <i>A</i>  | <i>BD</i> | <i>BC</i> | <i>AE</i> | <i>B</i>  | <i>AD</i> | <i>AC</i> |

There is of course the six  $A, B, C, D, E, F$ ; for each of these is a linear function of  $1, x+x', xx'$ , and there is thus a linear relation between any four of them. It would at first sight appear that the remaining sixes were of two different forms,  $A, B, AB, CE, CD, DE$ , and  $F, A, AB, AC, AD, AE$ ; but these are really identical, for taking any two letters  $E, F$ , the six is  $E, F, AE, BE, CE, DE$ , or, as this might be written,  $E, F, AEF, BEF, CEF, DEF$ , where  $AEF$  means  $BCD \cdot AEF$ , etc.; and we thus obtain each of the remaining fifteen sixes. The six just referred to, viz.  $E, F, AE, BE, CE, DE$ , or changing the notation say  $E, F, A, B, C, D$  as indicated in the table, thus represents any one of the sixes other than the rational six  $A, B, C, D, E, F$ : and there is no difficulty in actually finding each of the fifteen relations between four functions of the six in question,  $E, F, A, B, C, D$ . It is to be observed that every such function as  $A$  contains the same irrational part

$$\frac{2}{(x-x')^2} \sqrt{abcdef a'b'c'd'e'f'},$$

and that the linear relations involve therefore only the differences  $A-B, A-C$ , etc., which are rational. Proceeding to calculate these differences, we have for instance

$$C-D = \frac{1}{(x-x')^2} (cefa'b'd' + c'e'fabd - defa'b'c' - d'e'f'abc) = \frac{1}{(x-x')^2} (cd' - c'd)(efa'b' - e'f'ab);$$

or, substituting for  $a, a'$ , etc. their values  $a-x, a-x'$ , etc., we have

$$\begin{aligned}
 cd' - c'd &= (x-x')(c-d), \\
 efa'b' - e'f'ab &= -(x-x') \begin{vmatrix} 1, & x+x', & xx' \\ 1, & a+b, & ab \\ 1, & e+f, & ef \end{vmatrix} = (x-x')[xx'abef],
 \end{aligned}$$

or say for shortness.

We have therefore

$$C - D = (c - d)[xx'abef];$$

and in like manner we obtain the equations

$$\begin{aligned} A - B &= (a - b)[xx'cdef], & C - D &= (c - d)[xx'abef], \\ A - C &= (a - c)[xx'bdef], & B - D &= (b - d)[xx'caef], \\ A - D &= (a - d)[xx'bcef], & B - C &= (b - c)[xx'ade f]. \end{aligned}$$

It is now easy to form the system of formulae

| <i>E</i>                | <i>F</i>                 | <i>A</i>                 | <i>B</i>                 | <i>C</i>       | <i>D</i> |
|-------------------------|--------------------------|--------------------------|--------------------------|----------------|----------|
| $ae \cdot af \cdot bcd$ | $-be \cdot bf \cdot cda$ | $+ce \cdot cf \cdot dab$ | $-de \cdot df \cdot abc$ |                | $= 0,$   |
| $ad \cdot bf \cdot cf$  | $-ad \cdot be \cdot ce$  | $+ef$                    | $-ef$                    |                | $= 0,$   |
| $bd \cdot cf \cdot af$  | $-bd \cdot ce \cdot ae$  | $+ef$                    | $-ef$                    |                | $= 0,$   |
| $cd \cdot af \cdot bf$  | $-cd \cdot ae \cdot be$  | $+ef$                    | $-ef$                    |                | $= 0,$   |
| $bc \cdot af \cdot df$  | $-bc \cdot ae \cdot de$  | $+ef$                    | $-ef$                    |                | $= 0,$   |
| $ca \cdot bf \cdot df$  | $-ca \cdot be \cdot de$  | $+ef$                    | $-ef$                    |                | $= 0,$   |
| $ab \cdot cf \cdot df$  | $-ab \cdot ce \cdot de$  | $+ef$                    | $-ef$                    |                | $= 0,$   |
| $-af \cdot bcd$         |                          | $+be \cdot cd$           | $+ce \cdot db$           | $+de \cdot bc$ | $= 0,$   |
| $-bf \cdot cda$         |                          | $+ae \cdot cd$           | $+ce \cdot da$           | $+de \cdot ac$ | $= 0,$   |
| $-cf \cdot dab$         | $+ae \cdot bd$           | $+be \cdot ca$           | $+ce \cdot ab$           |                | $= 0,$   |
| $-df \cdot abc$         | $+ae \cdot bc$           |                          |                          | $+de \cdot ab$ | $= 0,$   |
| $-ae \cdot bcd$         |                          | $+bf \cdot cd$           | $+cf \cdot db$           | $+df \cdot bc$ | $= 0,$   |
| $-be \cdot cda$         |                          | $+af \cdot cd$           | $+cf \cdot da$           | $+df \cdot ac$ | $= 0,$   |
| $-ce \cdot dab$         | $+af \cdot bd$           | $+bf \cdot ca$           | $+cf \cdot ab$           |                | $= 0,$   |
| $-de \cdot abc$         | $+af \cdot bc$           |                          |                          | $+df \cdot ab$ | $= 0,$   |

where for shortness  $ab, ac$ , etc., are written to denote  $a - b, a - c$ , etc.; also  $abc$ , etc., to denote  $(b - c)(c - a)(a - b)$ , etc.: the equations contain all of them only the differences of  $A, B, C, D$ ; thus the first equation is equivalent to

$$ae \cdot af \cdot bcd (A - D) - be \cdot bf \cdot cde (B - D) + ce \cdot cf \cdot dab (C - D) = 0,$$

and so in other cases.

Cambridge, 14 March, 1877.

## 663.

FURTHER INVESTIGATIONS ON THE DOUBLE  $\vartheta$ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXIII. (1877), pp. 220–233.]

I consider six letters

$$a, b, c, d, e, f;$$

a duad  $ab$  not containing  $f$  may be completed into the triad  $abf$ , and then into the double triad  $abf, cde$ ; there are in all ten double triads, represented by the duads

$$ab, ac, ad, ae, bc, bd, be, cd, ce, de,$$

and the whole number of letters and of double triads is = 16.

Taking  $x, x'$  as variables, I form sixteen functions; viz. these are

$$[a] = \overline{a-x} \cdot \overline{a-x'},$$

$$[ab] = \frac{1}{(x-x')^2} \left\{ \sqrt{\frac{\overline{a-x} \cdot \overline{b-x} \cdot \overline{f-x}}{\overline{c-x'} \cdot \overline{d-x'} \cdot \overline{e-x'}}} \pm \sqrt{\frac{\overline{a-x'} \cdot \overline{b-x'} \cdot \overline{f-x'}}{\overline{c-x} \cdot \overline{d-x} \cdot \overline{e-x}}} \right\}^2,$$

where the function under each radical sign is the product of six factors, the arrangement in two lines being for convenience only: the sign  $\pm$  has the same value in all the functions, and it will be observed that the irrational part is

$$\pm \frac{2}{(x-x')^2} \sqrt{\frac{\overline{a-x} \cdot \overline{b-x} \cdot \overline{c-x} \cdot \overline{d-x} \cdot \overline{e-x} \cdot \overline{f-x}}{\overline{a-x'} \cdot \overline{b-x'} \cdot \overline{c-x'} \cdot \overline{d-x'} \cdot \overline{e-x'} \cdot \overline{f-x'}}},$$

viz. this has the same value in all the functions.

The general property of the double  $\vartheta$ -functions is that the squares of the sixteen functions are proportional to constant multiples of the sixteen functions  $[a], [ab]$ ; but this theorem may be presented in a much more definite form, viz. we can determine, and

that very simply, the actual expressions for the constant factors; and so we can enunciate the theorem as follows; the squares of the sixteen double  $\vartheta$ -functions are proportional to sixteen functions  $\{a\}, \{ab\}$ , where, in a notation about to be explained,

$$\{a\} = \sqrt{a} [a], \quad \{ab\} = \sqrt{ab} [ab].$$

Here in the radical  $\sqrt{a}$ ,  $a$  is to be considered as standing in the first place for the pentad  $bcdef$ , which is to be interpreted as a product of differences,

$$= bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef,$$

(where  $bc, bd$ , etc., denote the differences  $b - c, b - d$ , etc.). Similarly, in the radical  $\sqrt{ab}$ ,  $ab$  is to be considered as standing in the first instance for the double triad  $abf \cdot cde$ , which is to be interpreted as a product of differences,  $= ab \cdot af \cdot bf \cdot cd \cdot ce \cdot de$ , (where  $ab, af$ , etc., denote the differences  $a - b, a - f$ , etc.).

It is convenient to consider  $a, b, c, d, e, f$  as denoting real magnitudes taken in decreasing order: in all the products  $bcdef$ , etc., and in each term  $abf$  or  $cde$  of a product  $abf \cdot cde$ , the letters are to be written in alphabetical order; the differences  $bc, bd$ , etc.,  $ab, af$ , etc., which present themselves in the several products, are thus all of them positive; and the radicals, being all of them the roots of positive quantities, may themselves be taken to be positive.

We have to consider the values of the functions  $[a], [ab]$ , or  $\{a\}, \{ab\}$ , in the case where the variables  $x, x'$  become equal to any two of the letters  $a, b, c, d, e, f$ ; it is clearly the same thing whether we have for instance  $x = b, x' = c$ , or  $x = c, x' = b$ , etc.; we have therefore to consider for  $x, x'$  the fifteen values  $ab, ac, \dots, af, \dots, ef$ ; there is besides a sixteenth set of values  $x, x'$  each infinite, without any relation between the infinite values.

Taking this case first,  $x, x'$  each infinite, and in  $[ab]$ , etc., the sign  $\pm$  to be  $+$ , we have

$$[a] = xx', \quad [ab] = \frac{4x^3 x'^3}{(x - x')^2},$$

or, attending only to the ratios of these values,

$$[a] = 1, \quad [ab] = \frac{4x^3 x'^3}{(x - x')^2},$$

where  $\frac{4x^3 x'^3}{(x - x')^2}$  is infinite, and the values may finally be written

$$[a] = 0, \quad [ab] = 1;$$

whence also, for  $x, x'$  infinite,

$$\{a\} = 0, \quad \{ab\} = \sqrt{ab},$$

the radical  $\sqrt{ab}$  being understood as before.

Suppose next that  $x, x'$  denote any two of the letters, for instance  $a, b$ ; then two of the functions  $[a]$  vanish, viz. these are  $[a], [b]$ , but the remaining four functions acquire determinate values; and moreover four of the functions  $[ab]$  vanish, viz. these are  $[ab], [cd], [ce], [de]$ , for each of which the  $xx'$  letters  $a, b$  occur in the same triad (the

double triads for the four functions are, in fact,  $abf \cdot cde, cdf \cdot abe, cef \cdot abd, def \cdot abc$ ); but the other six functions  $[ab]$ , for which the letters  $a, b$  occur in separate triads, acquire determinate values.

It is important to attend to the signs: for example, if  $x, x' = b, c$ , we have

$$[c] = ce \cdot cb = -bc \cdot ce,$$

$$[ce] = \frac{cb \cdot eb \cdot fb}{ae \cdot be \cdot de} = \frac{cb \cdot fb}{ae \cdot de} = -\frac{bc \cdot bf}{ae \cdot de}.$$

TABLE I. OF THE VALUES OF  $[a], [ab]$ , ETC., FOR THE SIXTEEN SPECIAL VALUES OF  $x, x'$ .

[The original printed table is a large  $16 \times 16$  array giving, for each of the sixteen special pairs  $x, x' = \infty\infty, ab, ac, ad, ae, af, bc, bd, be, bf, cd, ce, cf, de, df, ef$  (across the top), the corresponding values (with proper signs) of the sixteen functions

$$[a], [b], [c], [d], [e], [f], [ab], [ac], [ad], [ae], [bc], [bd], [be], [cd], [ce], [de]$$

(down the side). Each non-zero entry is a product of two differences such as  $+ab \cdot ac, -bc \cdot ef$ , etc.; the diagonal entries  $[a]$  for  $x, x' = ab$  vanish (since one of  $x, x'$  equals  $a$ ), as do entries  $[ab]$  when  $a, b$  occur in the same triad of the double triad. The full table occupies one printed page and is reproduced unaltered from the original; it is used solely in deriving the corresponding table of  $\{a\}, \{ab\}$  that follows.]

Here the symbols  $bc, ce$ , etc., denote differences;  $[ce]$  is the product of four differences; the arrangement in two lines is for convenience only.

We thus obtain the series of values of  $[a], [ab]$ , etc., which although only required as subsidiary to the determination of the corresponding values of  $\{a\}, \{ab\}$ , I nevertheless give in a table.

The signs are given as they were actually obtained, but as we are concerned only with the ratios of the functions, it is allowable to change all the signs in any column: and it appears that there are four columns in each of which the signs are or can be made all +; whereas in each of the remaining twelve columns the signs are or can be made six of them +, the other four −.

Passing to the values of  $\{a\}$ ,  $\{ab\}$ , etc., we have for example, from the  $ab$  column of the foregoing table,

$$\begin{aligned}\{c\} &= +\sqrt{c \cdot ac \cdot bc}, \\ \{d\} &= +\sqrt{d \cdot ad \cdot bd}, \\ &\vdots \\ \{ac\} &= -\sqrt{ac \cdot \frac{ac \cdot ae}{bc \cdot bf}}, \\ &\vdots\end{aligned}$$

where (since the radicals are all positive) the signs are correct: substituting for the quantities under the radical signs their full values, and squaring the rational parts in order to bring them also under the radical signs, this is

$$\begin{aligned}\{c\} &= +\sqrt{ab \cdot ad \cdot ae \cdot af \cdot bd \cdot be \cdot bf \cdot de \cdot df \cdot ef \cdot ac^2 \cdot bc^2}, \\ \{d\} &= +\sqrt{ab \cdot ac \cdot ae \cdot af \cdot bc \cdot be \cdot bf \cdot ce \cdot cf \cdot ef \cdot ad^2 \cdot bd^2}, \\ \{ac\} &= -\sqrt{ac \cdot af \cdot cf \cdot bd \cdot be \cdot de \cdot ac^2 \cdot ae^2 \cdot bc^2 \cdot bf^2},\end{aligned}$$

where all the expressions of this (the  $ab$ -column) have a common factor,

$$ac \cdot ad \cdot ae \cdot af \cdot bc \cdot bd \cdot be \cdot bf.$$

Omitting this factor, we find

$$\begin{aligned}\{c\} &= +\sqrt{ab \cdot ac \cdot bc \cdot de \cdot df \cdot ef}, \\ \{d\} &= +\sqrt{ab \cdot ad \cdot bd \cdot ce \cdot cf \cdot ef}, \\ \{ac\} &= -\sqrt{ad \cdot ae \cdot de \cdot bc \cdot bf \cdot cf};\end{aligned}$$

viz. recurring to the foregoing condensed notation, this is

$$\begin{aligned}\{c\} &= +\sqrt{de}, \\ \{d\} &= +\sqrt{ce}, \\ \{ac\} &= -\sqrt{bc},\end{aligned}$$

and, in fact, the terms in the several columns have only the ten values  $\sqrt{ab}$ ,  $\sqrt{ac}$ , etc. each with its proper sign. I repeat the meaning of the notation:  $ab$  stands in the first instance for the double triad  $abf \cdot cde$ , and then this denotes a product of differences  $ab \cdot af \cdot bf \cdot cd \cdot ce \cdot de$ . We have thus the following table in which I have in several cases changed the signs of entire columns.

TABLE II. OF THE FUNCTIONS  $\{a\}, \{ab\}$ , ETC., FOR THE SIXTEEN SPECIAL VALUES OF  $x, x'$ .

[The original printed Table II. is a  $16 \times 16$  array showing, for each of the sixteen special pairs  $x, x' = \infty\infty, ab, ac, ad, ae, af, bc, bd, be, bf, cd, ce, cf, de, df, ef$  (across the top), the corresponding values of the sixteen functions

$$\{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{f\}, \{ab\}, \{ac\}, \{ad\}, \{ae\}, \{bc\}, \{bd\}, \{be\}, \{cd\}, \{ce\}, \{de\}$$

(down the left side). Each non-zero entry is one of the ten radicals  $\pm\sqrt{ab}, \pm\sqrt{ac}, \dots, \pm\sqrt{de}$  with its proper sign; in many entries the sign of the entire column has been reversed (as compared with the values directly derived from Table I) so as to present the functions in their most convenient form. The full table occupies one printed page; it shows that the sixteen functions  $\{a\}, \{ab\}$ , take, over the sixteen special pairs  $x, x'$ , only the ten values  $\sqrt{ab}, \sqrt{ac}, \dots, \sqrt{de}$  each with a proper sign.]



Referring now to Göpel's memoir, *Crelle*, t. XXXV. (1847), pp. 277–312, we have the sixteen double  $\vartheta$ -functions

$$P, P_1, P_2, P_3; \ iQ, Q_1, iQ_2, Q_3; \ iR, iR_1, R_2, R_3; \ S, iS_1, iS_2, S_3,$$

where the six functions affected with the  $i (= \sqrt{-1})$  are odd functions, vanishing for the values  $u = 0, u' = 0$  of the arguments. It is convenient to take  $x = \infty, x' = \infty$  as the values of  $x, x'$  corresponding to these values  $u = 0, u' = 0$ : the expressions  $\{a\}$  will thus correspond to the six squares  $-Q^2, -Q_2^2, -R^2, -R_1^2, -S_1^2, -S_2^2$ , and the expressions  $\{ab\}$  to the remaining ten squares  $P^2, P_1^2, \dots, S_3^2$ ; and after some *tâtonnement*, I succeed in establishing the correspondence as follows:

$$\begin{aligned} & -S_2^2, -S_1^2, -R_1^2, -R^2, -Q_2^2, -Q^2, P_3^2, P_1^2, P^2, -S_3^2, P_2^2, P_1^2, S_1^2, Q_1^2, R_2^2, R_3^2, \\ & = \{a\}, \{b\}, \{c\}, \{d\}, \{e\}, \{f\}, \{ab\}, \{ac\}, \{ad\}, \{ae\}, \{bc\}, \{bd\}, \{be\}, \{cd\}, \{ce\}, \{de\}, \end{aligned}$$

viz. the sixteen squared double  $\vartheta$ -functions are proportional to the sixteen expressions  $\{a\}, \{ab\}$ , as hereby appearing.

We have, after Göpel (l.c. p. 283), a table showing how the ratios of the double  $\vartheta$ -functions are altered, when the arguments are increased by the quarter-periods

$$A, B, A+B, K, K+A, K+B,$$

that is, when  $u, u'$  are simultaneously changed into  $u+A, u'+A'$ , or into  $u+B, u'+B'$ , etc. If instead, we consider the squared functions, the table is very much simplified, inasmuch as in place of the coefficients  $\pm 1, \pm i$ , it will contain only the coefficients  $\pm 1$ : and we may complete the table by extending it to all the combinations  $0, A, B, A+B, K, K+A, K+B, K+A+B, L, L+A, L+B, L+A+B, K+L, K+L+A, K+L+B, K+L+A+B$  of the quarter-periods: we have thus a table included in the annexed Table III., viz. attending herein only to the capital letters  $P, Q, R, S$ , the sixteen columns of the table show how the ratios of the terms  $-S_2^2, -S_1^2$ , etc., of the first column are altered when the arguments are increased by the foregoing combinations of quarter-periods, as indicated by the headings  $0, A, B$ , etc., of the several columns.

TABLE III. OF THE SIXTEEN FORMS OF THE SQUARED DOUBLE  $\vartheta$ -FUNCTIONS.

[Table III. is a  $16 \times 16$  array whose left-hand column lists the sixteen squared functions

$$-S_2^2, -S_1^2, -R_1^2, -R^2, -Q_2^2, -Q^2, P_3^2, P_1^2, P^2, -S_3^2, P_2^2, P_1^2, S_1^2, Q_1^2, R_2^2, R_3^2,$$

and whose other fifteen columns are headed

$$A, B, A+B, K, K+A, K+B, K+A+B, L, L+A, L+B, L+A+B, K+L, K+L+A, K+L+B, K+L+A+B,$$

Each entry shows how the squared function in the row is transformed under the increment of the arguments by the quarter-period combination labelling the column, and is given as  $\pm$ (another squared function) followed by its value as one of the symbols  $a, b, c, \dots, ab, ac, \dots, de$  from the foregoing scheme. At the foot of each column is inserted the suffix ( $\infty, ab, cd$ , etc.) referring to the columns of Table II. The full  $16 \times 16$  table is reproduced unaltered from the original; the entries are used in the verification on the next page.]

But I have also in the table inserted the values to which  $-S_2^2, -S_1^2$ , etc., are respectively proportional, viz. the table runs  $-S_2^2 = a, -S_1^2 = b$ , etc., (read  $-S_2^2 = \{a\}, -S_1^2 = \{b\}$ , etc., the brackets  $\{ \}$  having been for greater brevity omitted throughout the table), and where it is of course to be understood that  $-S_2^2, -S_1^2$ , etc., are proportional only, not absolutely equal to  $\{a\}, \{b\}$ , etc. And I have also at the foot of the several columns inserted suffixes  $\infty\infty, ab, cd$ , etc., which refer to the columns of Table II.

Comparing the first with any other column of the table, for instance with the second column, the two columns respectively signify that

$$\begin{aligned} -S_2^2(u) &= \{a\}, & -S_2^2(u+A) &= -\{ae\}, \\ -S_1^2(u) &= \{b\}, & -S_1^2(u+A) &= -\{be\}, \\ &\vdots & &\vdots \\ Q_1^2(u) &= \{ab\}, & Q_1^2(u+A) &= -\{e\}, \end{aligned}$$

where, as before, the sign  $=$  means only that the terms are proportional;  $u$  is written for shortness instead of  $(u, u')$ , and so  $u+A$  for  $(u+A, u'+A')$ , etc.: the variables in the functions  $\{a\}, \{be\}$ , etc. are in each case  $x, x'$ . But if in the second column we write  $u-A$  for  $A$ , then the variables  $x, x'$  will be changed into new variables  $y, y'$ , or the meaning will be

$$\begin{aligned} x, x' & & y, y' \\ -S_2^2(u) &= \{a\}, & -S_2^2(u) &= -\{ae\}, \\ -S_1^2(u) &= \{b\}, & -S_1^2(u) &= -\{be\}, \\ &\vdots & &\vdots \\ Q_1^2(u) &= \{ab\}, & Q_1^2(u) &= -\{e\}, \end{aligned}$$

so that, omitting from the table the terms which contain the capital letters  $P, Q, R, S$ , except only the outside left-hand column  $-S_2^2, -S_1^2$ , etc., the table indicates that these functions  $-S_2^2, -S_1^2$ , etc., are proportional to the functions  $\{a\}, \{b\}$ , etc., of  $x, x'$  given in the first column; also to the functions  $-\{ae\}, -\{be\}$ , etc., of  $y, y'$  given in the second column; also to the functions  $-\{ae\}, -\{be\}$ , etc., of  $z, z'$  given in the third column; and so on, with a different pair of variables in each of the 16 columns.

Thus comparing any two columns, for instance the first and second, it appears that we can have simultaneously

$$\begin{aligned} x, x' & & y, y' \\ \{a\} &= -\{ae\}, \\ \{b\} &= -\{be\}, \\ &\vdots \\ \{ab\} &= -\{e\}, \end{aligned}$$

(fifteen equations, since the meaning is that the terms are only proportional, not absolutely equal), equivalent to two equations serving to determine  $x$  and  $x'$  in terms of  $y$  and  $y'$

or conversely  $y$  and  $y'$  in terms of  $x$  and  $x'$ . The functions in each column form in fact 16 sixes, such that any four belonging to the same six are linearly connected; and in any such linear relation between four functions in the left-hand column, substituting for these their values as functions in the right-hand column, we have the corresponding relations between four functions out of a set of six belonging to the right-hand column, or we have an identity  $0 = 0$ . I will presently verify this in a particular case.

If in any column we give to the variables the values  $x, x'$ , we obtain for the terms in the column the values which the terms of the first column assume on giving to  $x, x'$  the values shown at the foot of the column in question; thus, in the second column giving to the variables the values  $c, d$ , the column becomes

$$-\sqrt{be}, -\sqrt{ae}, 0, 0, -\sqrt{ab}, \sqrt{cd}, 0, \sqrt{ad}, \sqrt{ac}, 0, \sqrt{bd}, \sqrt{bc}, 0, 0, \sqrt{de}, \sqrt{ce}$$

which is, in fact, the  $cd$ -column of Table II.: this is of course as it should be, for the values in question are those of the functions  $-S_2^2, -S_1^2$ , etc., on writing therein

$$x, x' = c, d.$$

The formulae show that

$$\sqrt{ab}, \sqrt{ac}, \sqrt{ad}, \sqrt{ae}, \sqrt{bc}, \sqrt{bd}, \sqrt{be}, \sqrt{cd}, \sqrt{ce}, \sqrt{de},$$

are, in fact, proportional to

$$p_2^2, \varpi_2^2, \varpi_1^2, \varpi_3^2, \sigma_3^2, p_1^2, p_3^2,$$

( $k_1, k_2, \dots$  are Göpel's  $k', k'', \dots$ ). This gives rise to a remarkable theorem, for the ten squares are functions of only four quantities  $\alpha, \beta, \gamma, \delta$  (Göpel's  $t, u, v, w$ ). For greater clearness, I introduce single letters  $A, B, \dots, J$ , and write

$$\begin{aligned} A = abc \cdot def &= (\sqrt{def})^2 = p_2^2, &= \frac{1}{4}(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)^2, \\ B = abd \cdot cef &= (\sqrt{cef})^2 = p_2^2, &= \frac{1}{4}(\alpha\gamma + \beta\delta)^2, \\ C = abe \cdot cdf &= (\sqrt{cdf})^2 = k_3, &= \frac{1}{4}(\alpha\delta + \beta\gamma)^2, \\ D = abf \cdot cde &= (\sqrt{cde})^2 = k_1, &= \frac{1}{4}(\alpha^2 - \beta^2 - \gamma^2 + \delta^2)^2, \\ E = acd \cdot bef &= (\sqrt{bef})^2 = \varpi_3^2, &= \frac{1}{4}(\alpha\beta + \gamma\delta)^2, \\ F = ace \cdot bdf &= (\sqrt{bdf})^2 = \varpi_2^2, &= \frac{1}{4}(\alpha^2 + \beta^2 + \gamma^2 + \delta^2)^2, \\ G = acf \cdot bde &= (\sqrt{acf})^2 = \varpi_1^2, &= \frac{1}{4}(\alpha\delta - \beta\gamma)^2, \\ H = ade \cdot bcf &= (\sqrt{bcf})^2 = \varpi_1^2, &= \frac{1}{4}(\alpha\gamma - \beta\delta)^2, \\ I = adf \cdot bce &= (\sqrt{adf})^2 = \sigma_3^2, &= -\frac{1}{4}(\alpha\beta - \gamma\delta)^2, \\ J = aef \cdot bdc &= (\sqrt{aef})^2 = \sigma_3^2, &= \frac{1}{4}(\alpha^2 + \beta^2 - \gamma^2 - \delta^2)^2; \end{aligned}$$

viz. it has to be shown that  $A, B, \dots, J$ , considered as given functions of the six letters  $a, b, c, d, e, f$ , are really functions of four quantities  $\alpha, \beta, \gamma, \delta$ ; or, what is the same thing, that  $A, B, \dots, J$ , considered as functions of  $a, b, c, d, e, f$ , satisfy all those relations which they satisfy when considered as given functions of  $\alpha, \beta, \gamma, \delta$ .

Now considering them as given functions of  $\alpha, \beta, \gamma, \delta$ , they ought to satisfy six relations; and inasmuch as, so considered, they are, in fact, linear functions of

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4, \quad \alpha^2\beta^2 + \gamma^2\delta^2, \quad \alpha^2\gamma^2 + \beta^2\delta^2, \quad \alpha^2\delta^2 + \beta^2\gamma^2, \quad \alpha\beta\gamma\delta,$$

five of these relations will be linear: there is a sixth non-linear relation, expressible in a variety of different forms, one of them, as is easily verified, being

$$\sqrt{AJ} \pm \sqrt{CG} \pm \sqrt{DF} = 0.$$

Now considering  $A, B, \dots, J$  as given functions of  $a, b, c, d, e, f$ , there exist between them linear relations which may be obtained by the consideration of identities of the form

$$\overline{abcd} = 0, \quad \overline{abcdef} = 0,$$

where the left-hand side is used for shortness to denote the determinant

$$\begin{vmatrix} 1, & 1, & 1, & 1 \\ a, & b, & c, & d \\ a^2, & b^2, & c^2, & d^2 \end{vmatrix} = 0, \quad \begin{vmatrix} 1, & 1, & 1, & 1, & 1, & 1 \\ f, & a, & b, & c, & d, & e \\ f^2, & a^2, & b^2, & c^2, & d^2, & e^2 \end{vmatrix} = 0.$$

We thus obtain between them a system of fifteen linear relations, which present themselves in the form

- (1)  $A - J + E - B = 0,$
- (2)  $-A - I + F - C = 0,$
- (3)  $A - H + G - D = 0,$
- (4)  $-B - G + H + C = 0,$
- (5)  $B - F + I + D = 0,$
- (6)  $C - E + J - D = 0,$
- (7)  $-E - D - H + F = 0,$
- (8)  $E - C + G - I = 0,$
- (9)  $F - B - J - G = 0,$
- (10)  $H - A + J - I = 0,$
- (11)  $-J + D - G + I = 0,$
- (12)  $J + C - F + H = 0,$
- (13)  $I + B - E - H = 0,$
- (14)  $G + A + E - F = 0,$
- (15)  $D - A + B - C = 0,$

and these are all included in the equations (10), (4), (12), (15), (6), which serve to express  $G, B, E, F, I$  in terms of  $D, H, C, A, J$ , i.e.  $ac, ce, eb, bd, da$  in terms of  $G = ab, bc, cd, de, ea$ , if for the moment we write  $G = ac$ , etc. But the five linear relations in question are, it is at once seen, satisfied by  $A, B, \dots, J$  considered as given functions of  $\alpha, \beta, \gamma, \delta$ .

The equation  $\sqrt{AJ} \pm \sqrt{CG} \pm \sqrt{DF} = 0$ , substituting for  $A, B, \dots$ , their values in terms of  $a, b, c, d, e, f$ , becomes

$$\sqrt{abc \cdot def \cdot aef \cdot bcd} \pm \sqrt{abf \cdot cde \cdot ace \cdot bdf} \pm \sqrt{abe \cdot cdf \cdot acf \cdot bde} = 0,$$

which (omitting common factors) becomes  $\pm \sqrt{bc^2 \cdot ef^2} \pm \sqrt{bf^2 \cdot ce^2} \pm \sqrt{be^2 \cdot cf^2} = 0$ ; or, taking the proper signs, this is the identity  $bc \cdot ef + be \cdot cf - bf \cdot ce = 0$ .

It is to be noticed that

$$\begin{array}{lll} \delta^2 + \alpha^2 - \beta^2 - \gamma^2, & 2(\alpha\beta - \gamma\delta), & 2(\gamma\alpha + \beta\delta), \\ 2(\alpha\beta + \gamma\delta), & \delta^2 + \beta^2 - \gamma^2 - \alpha^2, & 2(\beta\gamma - \alpha\delta), \\ 2(\gamma\alpha - \beta\delta), & 2(\beta\gamma + \alpha\delta), & \delta^2 + \gamma^2 - \alpha^2 - \beta^2, \end{array}$$

each divided by  $\delta^2 + \alpha^2 + \beta^2 + \gamma^2$ , form a system of coefficients in the transformation between two sets of rectangular coördinates. We have therefore

$$\begin{array}{l} \sqrt{ab}, \sqrt{ad}, \sqrt{ce}, \\ \sqrt{be}, \sqrt{de}, \sqrt{ac}, \\ \sqrt{bf}, \sqrt{cd}, \sqrt{ae}, \end{array}$$

each divided by  $\sqrt{bd}$  and the several terms taken with proper signs, as a system of coefficients in the transformation between two sets of rectangular axes: a result which seems to be the same as that obtained by Hesse in the Memoir, "Transformations-Formeln für rechtwinklige Raum-Coordinaten"; *Crelle*, t. LXIII. (1864), pp. 247–251.

The composition of the last mentioned system of functions is better seen by writing them under the fuller form  $\sqrt{abf \cdot cde}$ , etc.; viz. omitting the radical signs, the terms are

$$\begin{array}{lll} abf \cdot cde, & adf \cdot bce, & abd \cdot cef, \\ bef \cdot acd, & def \cdot abc, & acf \cdot bde, \\ bcf \cdot ade, & cdf \cdot abe, & aef \cdot bcd, \end{array}$$

each divided by  $bdf \cdot ace$ ; or, in an easily understood algorithm, the terms are

$$\begin{array}{lll} a \cdot ce & bf \cdot d & df \cdot b \quad bd \cdot f \\ e \cdot ac & bf \cdot d & df \cdot b \quad bd \cdot f \\ c \cdot ae & bf \cdot d & df \cdot b \quad bd \cdot f \end{array}$$

each divided by  $bdf \cdot ace$ .

Reverting to the before-mentioned comparison of the first and second columns of Table III., four of the equations are

$$\begin{array}{ll} x, x' & y, y' \\ \{c\} = \{d\}, & \text{that is, } \sqrt{c}[c] = \sqrt{d}[d], \\ \{d\} = \{c\}, & \text{that is, } \sqrt{d}[d] = \sqrt{c}[c], \\ \{e\} = -\{ab\}, & \text{that is, } \sqrt{e}[e] = -\sqrt{ab}[ab], \\ \{f\} = -\{cd\}, & \text{that is, } \sqrt{f}[f] = -\sqrt{cd}[cd]; \end{array}$$

viz. the four terms on the left-hand side are not absolutely equal, but only proportional, to those on the right-hand side. Substituting for  $\sqrt{c}$ ,  $\sqrt{d}$ , etc., their values, and introducing on the right-hand side the factor

$$\sqrt{ac \cdot bc \cdot ce \cdot cf \cdot ad \cdot bd \cdot de \cdot df},$$

the equations become

$$\begin{array}{ll} xx' & yy' \\ [c] = ac \cdot bc \cdot ce \cdot ef [d], & \\ [d] = ad \cdot bd \cdot de \cdot df [c], & \\ [e] = -ce \cdot de [ab], & \\ [f] = -cf \cdot df [cd]. & \end{array}$$

The functions on the left-hand satisfy the identity

$$ef[c] - ef[d] + fcd[e] - cde[f] = 0,$$

or, as this may also be written,

$$def[c] - cef[d] + cdf[e] - cde[f] = 0.$$

Hence substituting the right-hand values, the whole equation divides by  $ce \cdot de \cdot cf \cdot df$ ; omitting this factor, it becomes

$$ef \cdot ac \cdot bc [d] - ef \cdot ad \cdot bd [c] - cd \{[ab] - [cd]\} = 0,$$

where the variables are  $y, y'$ : it is to be shown that this is in fact an identity, and (as it is thus immaterial what the variables are) I change them into  $x, x'$ .

We have  $ac \cdot bc [d] - ad \cdot bd [c]$

$$= (a - c)(b - c)(d - x)(d - x') - (a - d)(b - d)(c - x)(c - x')$$

suppose,

$$\begin{aligned} &= (c - d) \begin{vmatrix} 1, & x + x', & xx' \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix} \\ &= cd [xx'abcd]. \end{aligned}$$

We have moreover

$$\begin{aligned} [ab] - [cd] &= ef \frac{1}{(x - x')^2} \{abf' \cdot c'd'e' + a'b'f \cdot cde - cdf' \cdot a'b'e' - c'd'f \cdot abe\} \\ &= \frac{1}{(x - x')^2} (abc'd' - a'b'cd)(ef' - e'f), \end{aligned}$$

where for the moment  $a, b, a'$ , etc., are written to denote  $a - x, b - x, a - x'$ , etc.; we have then

$$e'f - ef' = (e - x')(f - x) - (e - x)(f - x') = -(e - f)(x - x') = -ef(x - x'),$$

and

$$\begin{aligned} abc'd' - a'b'cd &= (a - x)(b - x)(c - x')(d - x') - (a - x')(b - x')(c - x)(d - x) \\ &= (x - x') \begin{vmatrix} 1, & x + x', & xx' \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix} = -(x - x')[xx'abcd]. \end{aligned}$$

Hence  $[ab] - [cd] = ef[xx'abcd]$ , and the equation to be verified becomes

$$(ef \cdot cd - cd \cdot ef)[xx'abcd] = 0,$$

viz. this is, in fact, an identity.

*Cambridge, 14 March, 1877.*



## 664.

## ON THE 16-NODAL QUARTIC SURFACE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXIV. (1877), pp. 238–241.]

Prof. Borchardt in the Memoir “Ueber die Darstellung u. s. w.” Crelle, t. LXXXIII. (1877), pp. 234–243, shows that the coordinates  $x, y, z, w$  may be taken as proportional to four of the double  $\vartheta$ -functions, and that the equation of the surface is then Göpel’s relation of the fourth order between these four functions: and he remarks at the end of the memoir that it thus appears that the coordinates  $x, y, z, w$  of a point on the surface can be expressed as proportional to algebraic functions, involving square roots only, of two arbitrary parameters  $\xi, \xi'$ .

It is interesting to develop the theory from this point of view. Writing, as in my paper, “Further investigations on the double  $\vartheta$ -functions,” pp. 220–233, [663],

$$[a] = aa', \quad [b] = bb', \quad [c] = cc', \quad [d] = dd', \quad [e] = ee', \quad [f] = ff',$$

$$[ab] = \frac{1}{(\xi - \xi')^2} (\sqrt{abf \cdot c'd'e'} - \sqrt{a'b'f' \cdot cde})^2,$$

etc.,

where on the right-hand sides  $a, b, \dots, a', \dots$  denote  $a - \xi, b - \xi, \dots, a - \xi', \dots$  ( $\xi, \xi'$  being here written in place of the  $x, x'$  of my paper), then the sixteen functions

# Collected Mathematical Papers

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Arthur Cayley

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## 1 [665] A Memoir on the Double $\vartheta$ -Functions (conclusion)

[From the Philosophical Transactions of the Royal Society of London, vol. 169 (1878), pp. 663–730; continued from preceding pages.]

(Page 201.) We have, as the equation of a quadric surface enveloped by

$$\alpha(x - x') + \beta(y - y') + \gamma(z - z') + \delta(w - w') = 0,$$

which is the same form as before, the relations

$$ab' + ba' = 0, \quad cd' + dc' = 0, \quad \text{etc.},$$

or, more precisely,

$$\frac{\partial \Omega}{\partial x} \cdot X' + \frac{\partial \Omega}{\partial y} \cdot Y' + \frac{\partial \Omega}{\partial z} \cdot Z' + \frac{\partial \Omega}{\partial w} \cdot W' = 0.$$

It is to be remarked that there are 15 expressions such as  $A1B + B1A$ , and 45 expressions such as  $AC1BD + AD1BC$ ; and that each of these  $(15 + 45 =)60$  expressions is a quadric function of  $\partial_x, \partial_y, \partial_z, \partial_w$  alone.

In like manner  $\Omega, \Omega_1, \Omega_2, \Omega_3$  are quadric functions of  $(x, y, z, w)$ , and similarly  $\Omega', \Omega'_1, \Omega'_2, \Omega'_3$  of  $(x', y', z', w')$ ; and hence  $\Omega, \Omega_1, \Omega_2, \Omega_3$  satisfy the conditions

$$\frac{\partial \Omega}{\partial x} - 2x + \frac{\partial \Omega_1}{\partial y} \cdot 2 + \frac{\partial \Omega_2}{\partial z} \cdot 2 + \frac{\partial \Omega_3}{\partial w} \cdot 2 = 0,$$

and similarly for the conjugate functions.

*Expression of  $\Theta\Omega - \Omega\Theta'$ : second subheading.*

We assume  $\Theta\Omega - \Omega\Theta' = \frac{1}{2}M\Omega''$ , where  $M$  is a quadric function of  $\partial_x, \partial_y$ ; suppose

$$M \equiv \Theta(xx' + yy') + \Theta'(xx' + yy').$$

It is to be remarked that there are 15 expressions which can be presented in the form

$$\left(\Theta x' \frac{\partial}{\partial x} + \Theta y' \frac{\partial}{\partial y}\right)\sqrt{X} = \left(\Theta x' + \Theta y' \cdot \frac{xy'}{X}\right)\sqrt{X} + \left(\Theta y' \cdot \frac{xx'}{Y}\right)\sqrt{X};$$

it is clear that the term in question is at once expressible in the form indicated by means of these.

It is to be remarked that there are 15 expressions which form a quadric system of  $(x, y)$  alone; viz.,  $X, Y, \frac{xy'}{X}$  etc. are linear functions of  $(x, y)$  together with the symbols

$$\left(\frac{\partial \Theta}{\partial x} + \frac{\partial \Theta}{\partial y}\right)\sqrt{X} + \left(\frac{\partial \Theta}{\partial x'} + \frac{\partial \Theta}{\partial y'}\right)\sqrt{Y},$$

where the bracketed symbols satisfy the conditions

$$\xi\alpha + \eta\beta = 0, \quad \nu\alpha + \mu\beta = 0,$$

and the like for  $\xi', \eta', \nu', \mu'$ .

Putting for the moment

$$\lambda\alpha + \mu\beta = \nu\alpha, \quad \mu\beta + \nu\gamma = \mu\beta, \quad \nu\gamma + \lambda\delta = \nu\gamma, \quad p\alpha + q\beta = r\alpha + s\beta,$$

$\mu = ab, \nu = ac, \dots$ ; the values of  $\xi, \eta, \nu, \mu, \xi', \eta', \nu', \mu'$  are

$$\xi = 0, \quad \eta = 0, \quad \nu = 0, \quad \mu = 0,$$

respectively.

I found it convenient to assume

$$\Theta = -2(x'^2 + y'^2)E = bxx' - ay y' + p(yy' + xx') + q(yy' - xx'),$$

and that the system of seven is

$$\Theta = -2(x'^2 + y'^2)E - \alpha xy' + \beta yx' + (\gamma + \delta)xx' + (\gamma - \delta)yy'.$$

(Page 202.) Assuming now

$$\Theta = -2(x'^2 + y'^2)\varphi + (\alpha + \beta)(xy' - yx')$$

where  $\Theta$  has to be determined so that the first of the same conditions may be satisfied; then substituting this value of  $\Theta$ , we have

$$(\Theta y' - py' + qy' - ry' - sy')\sqrt{X} = (\Theta x' - px' + qx' - rx' - sx')\sqrt{Y};$$

that is,

$$-ahxy' - xbx'y\sqrt{X} = (-axx' + bxy' - byx')\sqrt{Y};$$

viz. in the inner subscription of  $\sqrt{X}$ , the function  $\Theta$  is to be diminished by

$$(abc + bca)\sqrt{abcdabcd} = (abcdabcd)\sqrt{abcdabcd},$$

or, what is the same thing,

$$(-hbxx' - haxx')\sqrt{X} = (-yhbxx' - hayxx')\sqrt{Y};$$

also,

$$\sqrt{Yx} = \frac{1}{2}a(abcdexy)\sqrt{xy},$$

and we thus reduce the equation to

$$-(\sqrt{abcdef} \cdot \sqrt{abcdefc})\sqrt{xy} \cdot (abcd \cdot \sqrt{abcdef}).$$

But, referring to the expression for  $\lambda\sqrt{ab}$ , we have, by a mere interchange of letters,

$$\frac{d}{dx}\sqrt{ab} = -(\sqrt{abcd} \cdot a_1 - \sqrt{abcd} \cdot a_2)\sqrt{ab};$$

and the formula thus becomes

$$-(\sqrt{abcdef} - \sqrt{abcdefc}) \cdot \frac{d}{dx}\sqrt{ab};$$

and, referring to the expression for  $\lambda\sqrt{ab}$ , we have, by a mere interchange of letters,

$$2\sqrt{ab} \cdot \frac{d}{dx}\sqrt{ab} = \frac{d}{dx} \cdot ab;$$

consequently

$$\Theta\alpha = -(\sqrt{abcd} - \sqrt{abcdc})\frac{1}{2}\frac{d}{dx} \cdot ab,$$

and the value of  $\Pi$  thus, suppose

$$\Pi = \frac{1}{2}(\sqrt{abcd} - \sqrt{abcdc})(a_1 - b_1) \cdot ab,$$

or so they may be written,

$$\Pi = \frac{1}{2} (\alpha b c - a b_1 c_1)(b c d - a_1 c_1 d_1) + a^2 - 2a x y y + a^2 c_1 \sqrt{X Y}.$$

(Page 203.) Hence

$$abc - a_1 b_1 c_1 = (a + b - a_1 + b_1 + c_1)(b + c - a_1 + b_1 + c_1) - (a + a_1)(b + b_1)^2 - 5c_1^2;$$

and similarly,

$$def - d_1 e_1 f_1 = (d + e - d_1 + e_1 + f_1)(e + f - d_1 + e_1 + f_1) - (d + d_1)(e + e_1)^2 - 5f_1^2.$$

The expression for  $a^2 c_1^2$  contains therefore the terms

$$-(a + b - a_1 + b_1 + c_1)(b + c - a_1 + b_1 + c_1) + (a + a_1)(b + b_1)^2 + 5c_1^2,$$

or that,  $A = abcc_1$ , where  $A$  is something like the coefficient of  $\beta\gamma$  in

$$(\alpha + \beta + \gamma + \delta - \epsilon - \eta)\sqrt{XY},$$

which, viz. in fact, the sum of Camy's form in  $\phi$ , equation (200), his  $\phi$  being

$$= 95'' - a.$$

It is to be observed that,  $\Pi$  is the difference of two squares of expressions

$$A\Pi'' B\Pi'',$$

the radical which multiplies the being

$$\frac{1}{a + b \cos T + c \cos T' \sqrt{ab}} \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'}.$$

the differential becomes

$$\frac{dT \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'}}{(\cos^2 T \cdot \frac{a+b}{1+\alpha \cos T'} + \beta \sin^2 T) \sqrt{a + b \cos T + c \cos T'}};$$

that is,

$$\frac{1}{\sqrt{P + Q \cos T + R \cos T' \sqrt{a + b \cos T + c \cos T'}}} (a + b \cos T + c \cos T')(1 + \alpha \cos T + \beta \cos T').$$

The denominator could, of course, be reduced to the form  $(\alpha 1, \cos T, \sin T)^2$ ; but the arrival form seems preferable, inasmuch as it puts in evidence the linear factors

$$\frac{1}{(a + b \cos T + c \cos T') \sqrt{a + b \cos T + c \cos T' \sqrt{XY}}} (\alpha + \beta \cos T + \gamma \cos T') dT;$$

and there seems to be no advantage in further reducing the integral.

(Page 204.) Assuming now

$$\Theta = -2x' \sqrt{a^2 + y^2} + (\alpha' + \beta') \sqrt{X} \cdot (\Theta x' - px' + qx') - rx'(\sqrt{X} - \sqrt{Y}),$$

where  $\Theta$  has to be determined so that the first of the same conditions may be satisfied. Then, substituting this value of  $\Theta$  in the equations

$$(\Theta y' - py' + qy' - ry') \sqrt{X} = (\Theta x' - px' + qx' - rx') \sqrt{Y},$$

that is,

$$\left(-axy' - xby'x'\right)\sqrt{X} = \left(-axx' + bxy' - byx'\right)\sqrt{Y},$$

viz. in the inner subscription of  $\sqrt{X}$ , the function  $\Theta$  is to be diminished by

$$(abc) \cdot \sqrt{abcdef \cdot \sqrt{XY}} - \sqrt{abcdef \cdot \sqrt{YX}} = -\frac{1}{2}\sqrt{X} - \frac{1}{2}\sqrt{Y};$$

or, what is the same thing,

$$(abc - a_1b_1c_1)\sqrt{X} = \sqrt{abcd \cdot X} - \sqrt{abcd \cdot Y},$$

that is,

$$abc - a_1b_1c_1 = \sqrt{abcd \cdot X/Y} - \sqrt{abcd \cdot Y/X}.$$

But, referring to the expression for  $\lambda\sqrt{ab}$ , we have, by a mere interchange of letters,

$$\frac{d}{dx} \sqrt{ab} = -(\sqrt{abcd} \cdot a_1 - \sqrt{abcd} \cdot a_2)\sqrt{ab},$$

and the formula thus becomes

$$-(\sqrt{abcdef} - \sqrt{abcdefc}) \cdot \frac{d}{dx} \sqrt{ab};$$

and, referring to the expression for  $\lambda\sqrt{ab}$ , we have, by a mere interchange of letters,

$$2\sqrt{ab} \cdot \frac{d}{dx} \sqrt{ab} = \frac{d}{dx}(ab);$$

consequently

$$\Theta a = -(\sqrt{abcd} - \sqrt{abcdc}) \cdot \frac{1}{2} \frac{d}{dx}(ab),$$

and the value of  $\Pi$  thus, suppose

$$\Pi = \frac{1}{2}(\sqrt{abcd} - \sqrt{abcdc})(a_1 - b_1)ab,$$

or so they may be written,

$$\Pi = \frac{1}{2}(abc - a_1b_1c_1)(bcd - a_1c_1d_1) + a^2 - 2axy + a^2c_1\sqrt{XY}.$$

(Page 205.) *Expression of  $\Theta\Omega - \Omega\Theta'$ : second subheading.*

Writing for shortness  $\Theta\Omega - \Omega\Theta' = R$ , so before, and as in other cases ... in general  $\Delta PQ = P\Delta Q + Q\Delta P$ . Hence starting from  $A = \Pi abc' - a'b'c$ , we have

$$\begin{aligned} \Delta A &= \Pi \Delta(abc') - \Pi \Delta(a'b'c) \\ &= 4c' \left( \Pi \frac{PU}{QV} - \Pi \frac{PV}{QU} \right) (\Omega - \Omega') c' \xi \xi' + 2b(b - b')(\xi - \xi')(\Omega - \Omega'), \end{aligned}$$

where

$$\Delta a = -a + (\xi - \xi')aa', \quad \xi - \xi' = 0, \quad \xi - \xi' = -1 \quad (\xi a - \xi' \xi),$$

so that

$$\Delta a = a - a + (\xi - \xi')aa' = \frac{1}{2}(a - a_1)(b + b_1)(\xi - \xi') = aa'(b - b_1).$$

We thus have

$$\Delta A = -\Pi abc'[(\xi - \xi')aa'c_1c'_1 + \Pi bb'c'c'_1c(\xi - \xi')] - \frac{1}{2}(a + a_1)(b + b_1)(c + c_1).$$

But the expression for  $\Pi$  contains the factors  $4c'$ ,  $a$ ,  $b$ ,  $c$ ; hence multiplying the whole by  $A$ , we have

$$\begin{aligned} \frac{4}{\Pi} \Delta A &= \Pi a^2 [\Pi (bc' - b'c)(bc - b'c') + 4\alpha b (\xi - \xi')] \\ &\quad - \frac{1}{2} \Pi b' (ab - a_1 b_1)(c - c_1) - \Pi ab (a + b)(\xi - \xi')(\sqrt{XY}) \\ &\quad + \frac{1}{2} \Pi (abcdc' - (\sqrt{XY})) - \omega_1 (\xi - \xi')(\xi - \xi'), \end{aligned}$$

where  $M$  has its foregoing value  $= [\Pi_1 - (\sqrt{XY})](\Pi' \cdot \Pi w + W(bw'))$ .

*First step of the reduction.*

Writing  $\Theta \xi' = U$ ,  $b_2 acdef = U_1$ , then  $X = aU$ ,  $Y = a_1 U_1$ , and consequently

$$X - U = a'U^2, \quad Y - U_1 = U'a_1 U_1^2;$$

the second expression in  $U$ ,  $U_1$  denoting differentiations in  $x$ ,  $y$  respectively; then

$$X' - X = -aX + 2\Theta x - \frac{1}{2}a' \left( \frac{\partial U}{\partial x} \right)^2 \cdot \sqrt{ab'/aac'},$$

and similarly

$$Y' - Y = -aY + 2\Theta y - \frac{1}{2}a' \left( \frac{\partial U}{\partial y} \right)^2 \cdot \sqrt{ab'/aac'},$$

where, as before,  $\Theta$  has been treated in the analytic form expressed in the formulae.

(Page 206.) Hence

$$\Theta \Omega = a \cdot \Omega - \frac{2}{a} \left( \Theta \frac{PV}{QU} - \Theta \frac{PU}{QV} \right) (\Omega - \Omega') 2\Theta (ba - b'a) - (aa_1)(b - b')(\xi - \xi');$$

where the first value,  $\Omega$ , substituting for  $X$ ,  $Y$ ,  $\Omega$ ,  $\Omega'$  their values, and arranging according to  $\partial x$ ,  $\partial y$ ,  $\Omega$ , is

$$\begin{aligned} &= \frac{2}{a} \left[ a'(\partial'_x P)(\partial'_x Q) - a'(\partial'_y Q)(\partial'_y P) - a' \partial'_x \partial'_y (\partial'_x \partial'_y) \right] + \dots \\ &\quad + 2b'(aa_1 - a_1 a)(xy' - x'y) + \dots \\ &\quad + (\partial x' - \partial' x')ab'_1 + a(b_1 b' - b b'_1)(\partial x' \cdot \partial'_x \partial'_y \cdot \partial'_y P \cdot \partial'_y Q) \cdot aa'. \end{aligned}$$

After collecting and simplifying, one is led to expressions of the form

$$\Theta \xi' = \frac{1}{2} \left( \frac{\partial \Pi}{\partial x} \cdot a' - \frac{\partial \Pi}{\partial y} \cdot a \right),$$

and these formulae are no advantage in further reducing the integral.

(Page 207.) *Third step of the reduction.*

First, for the coefficient of  $(\partial \omega')^2$ , multiplying that  $\Delta = a_1 \omega - \omega_1$ , we have

$$\begin{aligned} &2 \left( \frac{\partial U}{\partial x} \right)^2 a' - 2U(\alpha + a_1) - 2U'(\alpha' + a_1)(a + a_1)(\xi - \xi') - 2U'(a_1 b_1 - ab)(\xi - \xi') - 2U'(b'_1 a_1 - ba)(\xi - \xi') \\ &\quad - (\xi - \xi')(\xi - \xi')a = a(\partial x'^2 \cdot \partial y'^2 \cdot \partial x' \cdot \partial'_y \cdot a_1). \end{aligned}$$

Adding these, the right-hand side divides by  $(a_1 - a)$ ; that is, by  $2P$ ; and the resulting value is

$$= \frac{1}{2} a' \xi \xi' a b a' \sqrt{ab'abc'},$$

the term  $aaa'ac\xi\xi'$  is reduced to the foregoing formulae; by performing the additions, observing as before, that

$$\Theta = a'b'\Theta\xi'a' = a'(a_1b - 2a'b')(\xi - \xi') - 2aa_1b_1c_1\xi.$$

Adding these, the right-hand side divides by  $(a_1 - a)$ ; that is, by  $2P$ ; and the resulting value is

$$(\xi - \xi')(\Theta\xi')(a_1 - b_1)(c - c_1) - 2a + b_1 + b'(\xi - \xi') + b'(\xi - \xi')(\Theta\xi')(a_1 - b_1) = -2a + b_1 + b'(\xi - \xi'),$$

or that,  $A = abcc_1$ , where  $A$  is something like the coefficient of  $\beta\gamma$  in  $(\alpha + \beta + \gamma + \delta - \epsilon - \eta)\sqrt{XY}$ , in the manner above  $\xi \dots$  in fact, the sum of Camy's form in  $\phi$  equation (200), his  $\phi$  being  $= 95'' - a$ .

It is to be observed that,  $\Pi$  is the difference of two squares of expressions of the form

$$A\Pi^{1/2} = D\Pi^{1/2},$$

the radical which multiplies the being

$$\frac{1}{a + b \cos T + c \cos T' \sqrt{ab}} \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'};$$

the differential becomes

$$\frac{dT \cdot \sqrt{ab - a_1 \cos T - b_1 \cos T'}}{(\cos^2 T \cdot \frac{a+b}{1+\alpha \cos T'} + \beta \sin^2 T) \sqrt{a + b \cos T + c \cos T'}};$$

that is,

$$\frac{1}{\sqrt{P + Q \cos T + R \cos T' \sqrt{a + b \cos T + c \cos T'}}} (a + b \cos T + c \cos T') (1 + \alpha \cos T + \beta \cos T').$$

(Page 208.) The formula thus becomes

$$\begin{aligned} \frac{4}{\Pi} \Delta A = a a' \left[ M \left( \frac{\partial U}{\partial x} a' - \frac{\partial U}{\partial y} a \right) (\Omega \omega + a) 2b'\xi - \frac{\partial U}{\partial x} (ba - b'a)(\xi - \xi') \right] \\ - \frac{1}{2} \Pi b' (ab - a_1 b_1)(c - c_1) - \Pi ab (a + b)(\xi - \xi')(\sqrt{XY}) \\ + \frac{1}{2} \Pi (abcdc' - (\sqrt{XY})) - \omega_1 (\xi - \xi')(\xi - \xi'). \end{aligned}$$

The second formula thus, suppose

$$\begin{aligned} \Pi = -(a + b + c + d + e + f) - \frac{2}{3}(a + b + c + d + e + f)^2 - \frac{1}{3}(a + b + c + d + e + f)^3 \\ - \frac{1}{2} \Pi b' (abcdc' - (\sqrt{XY})), \end{aligned}$$

and the value of  $\Pi$  thus, suppose

$$\Pi = \frac{1}{2} (\sqrt{abcd} - \sqrt{abcdc'}) (a_1 - b_1) ab,$$

or so they may be written,

$$\Pi = \frac{1}{2} (abc - a_1 b_1 c_1)(bcd - a_1 c_1 d_1) + a^2 - 2axy + a^2 c_1 \sqrt{XY}.$$

The above two expressions involving  $M$  have on the right-hand side terms which, when collected with the second of those given in the formulae of §8, give the final expression  $\Delta A$ , viz.,

$$\Delta A = aa'M (\xi - \xi') \Pi (bc' - b'c)(bc - b'c') + \dots$$



which is obviously a sum of squares.

(Page 209.) *Completion of the reduction and final expression for  $\Delta A$ .*

But now substituting the values of  $\Pi$ ,  $\Pi'$ ,  $W$ , we find

$$\begin{aligned} \frac{2}{\Pi} \Delta A = aa' & \left[ (aa + ba + a'a + b'a)(\xi - \xi') a + (\xi - \xi') ab'_1 - (\xi ab'_1 a' b'_1) 2c_1 (\sqrt{XY}) \right. \\ & + (a'a'b'_1 - a'b'a'b_1)(\xi - \xi') a + (a'b'_1 - a'b a'_1 b'_1)(\sqrt{XY}) \\ & \left. - (\Pi a'b'_1 b_1 + aba'_1 b a'_1 c_1)(\Omega - \Omega') (bw + ac)(\xi - \xi') \right], \end{aligned}$$

viz., the coefficients belong to the partition which contains the factor  $aa'$ , viz., the expression for  $\frac{4}{\Pi} \Delta A$ ; the other portion contains the factor

$$(a' + a'a + ba_1 b'_1) \Pi \cdot (acb'cde' - \sqrt{XY} (bw' \cdot c_1) a'b_1 c'_1) ab a'b a'b',$$

where

$$X = b(dacdef), \text{ etc.}$$

We have then the complete result, viz. this is

$$\begin{aligned} \frac{2}{\Pi} \Delta A = aa' M & [(a + b) 2\Pi - a\Theta a + aba'_1 c_1 \xi (\Omega - \Omega') aaa' \Omega - a c' a' b' c a' b' c (\Omega - \Omega')] \\ & + \dots \end{aligned}$$

which is obviously a sum of squares.

(Page 210.) Also

$$-(a'a'U + U)(ba'U + U_1) = -\Theta b'a'(-aa + aa_1 - a'b' + bac) U_1 cb b' c' c_1 U_1 = -a + ba, + 2aca'a'_1;$$

adding these two quantities, the right-hand side divides by  $(a_1 - a)$ , viz., by  $\Theta$ , the resulting value is

$$= W = b'(aa + a'a' + aa'a'b'_1) + a'b_1(aa + ba'_1 baa') a'_1 b_1,$$

we then see that this radical coefficient of  $(ba')^2$  is

$$= 2\Theta b'(ba'a) 2c_1 W'b'.$$

To this is to be added the actual coefficient of  $2\Theta a$  in the next step

$$2\Theta b'(b'a'a) 2c_1 W'a',$$

and we see that, without difficulty

$$(\Theta b'(a'a) 2\Theta b' a'_1 b) W' c a' b' c a' c' a' b a a' a' c_1 c_1 W = 0.$$

Lastly, for the coefficient of  $(\partial \xi')^2$ , we have

$$\begin{aligned} & \left( 2 \frac{\partial U}{\partial x} a' - 2U(\alpha + a_1) \right)^2 - \left( 2 \frac{\partial U}{\partial y} a' - 2U'(\alpha' + a_1) \right)^2 \\ & = 4(\alpha - \alpha_1)(b + b_1)(c + c_1)(d + d_1)(e + e_1)(f + f_1)(b + b'_1) - 8acc'_1(ba_1 b'a + b_1 ac) (ba'a'b'b'b_1 + aca'_1 a) \\ & \quad + 8abcd(cc' - ba)(bc - b'c')(b'c - bc')(\sqrt{XY}); \end{aligned}$$

which is the required value.

(Page 211.) Adding it to the preceding expression, the sum is

$$\begin{aligned}
 &= (a' - a)(a + b)(a + c)(a + d)(\sqrt{XY})(aa'b'_1 + a'b a'_1 b'_1)(a + b)(c + aa'_1 + ba'_1 a + ac) a + a' a' c' c' (ab'_1 c'_1) \\
 &\quad + ] \cdots - 2b'(aa + aa'c) a'(aa + a'a'c'_1) c_1 a + 2c'(aaa + ac') a + 2b_1 a'_1 c'_1 (\sqrt{XY}) \\
 &\quad + \cdots + (2b' - baa' - a'b) 4a'_1 b_1 + 2c_1 (\Pi a' ba' + a'c) 4a' b_1 c_1.
 \end{aligned}$$

This is, in fact, divisible by  $(a_1 - a)$ ; that is, by  $\Pi$ ; for there exists between the symbols the relations

$$\begin{aligned}
 &ba'b' + a'c' + c a' a' c a' b' c' a' c' a' a', \\
 &b' c b a' a' c a' b a' a' c a' + a' b (c a' c' b' a') b' a', \\
 &c' a' b' c' a' (a a a' c'_1 c'_1 a' c'_1 a' c' a'_1) \\
 &\quad + (2b' - 2a' a a' c' a' c c) = 0,
 \end{aligned}$$

and we thus reduce the expression to

$$\frac{2}{\Pi} (ba' - b'a)(b + aa'a'b'b'b_1)(cb'b_1 b_1 + c'c'_1)(c'_1 - a'b'_1)(\xi - \xi')(c + c')$$

viz. effecting the division, the quotient is

$$= 2b' - 2b'c(b + a) - 4a'b'a'c'_1(c'_1 - cc'_1 a' c_1) - (aaa'c'_1)(\sqrt{XY});$$

and we thus reduce the coefficient of  $(\partial\xi)^2$  in like manner

$$\Pi'_a = (b' - a)(\sqrt{XY}) - (aa'c'_1)(\sqrt{XY});$$

viz., this is  $\Pi'_a$  multiplied by  $(b' - a)(ac' - a'c)(b'c - bc')(b - b') = -(b - b')(aa'c'_1)(\sqrt{XY})$ .

(Page 212.) or finally it is

$$\begin{aligned}
 \Pi'_a &= -2c'(a' - a)(2b + 2a' - a'c) + (a'b' - a'c)(b + b' - a) - 2(b + a' - a')(a + a' - a') \\
 &\quad + 2(aa' - a'b')(b - b' + 2a'c' + a'_1 c) 2b'c' (b + cb'a'_1)(b'_1 + a + 2bc'a'_1) \\
 &\quad + 2a'c' - 2b a' c a' c + b'a'(aaa'_1 + bb'c'_1 - a'b'_1) + b'a'a'c'c'_1 + aca'a'_1 + 2a'c'_1 \\
 &\quad + (c' - a'_1)(c'_1 + a'b)(\sqrt{XY}) - 2a(a'c)(ac' b_1)(c'_1 + b'c'_1 a'_1 b'_1) \\
 &\quad + 2bac'_1.
 \end{aligned}$$

It is to be observed that the investigation thus far has been entirely independent of the values of  $\Pi$ ,  $\Pi'$ ; those values are, in fact, such as to make the coefficients of  $\sqrt{XY}$  vanish; for the actual numerical values of these coefficients can then be found, and it may easily be confirmed with each other: the foregoing investigation of those values was therefore unnecessary.

(Page 213.) Hence the case of paper 665 ends with the establishment of the formula

$$A\Delta + A^2\Pi = 0;$$

or, more concisely,  $\Delta A = 0$ .

*Cambridge, 7th December, 1877.*

## 2 [666] Sur un exemple de réduction d'intégrales abéliennes aux fonctions elliptiques

[From the Comptes Rendus de l'Académie des Sciences de Paris, t. LXXXV (juillet-décembre, 1877), pp. 265–268; 373, 374; 426–429; 472–475.]

Je reprends l'investigation de M. Hermite sur lui rapport aux intégrales réductibles

$$\int \frac{(1, x) dx}{\sqrt{a \cdot 1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx}},$$

publiée au-dessus du titre : “Sur un exemple, etc.” (*Annales de la Société scientifique de Bruxelles*, 1876).

Nous avons les constantes  $a, b$  et les variables  $x, y, u, v$ ; et on posant

$$X = a \cdot 1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx,$$

$$Y = a \cdot 1 - y \cdot 1 + ay \cdot 1 + by \cdot 1 - aby,$$

(où  $\sqrt{a \cdot 1 - x \cdot 1 + bx} = X$ , etc.) et les fonctions de l'intégration, les fonctions elliptiques, par fonctions elliptiques, on a les équations différentielles

$$\frac{du}{\sqrt{X}} + \frac{dv}{\sqrt{Y}} = \frac{1}{\sqrt{a}}(du + dv),$$

$$\frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} = \frac{1}{\sqrt{a}\sqrt{b}}(du - dv).$$

Il y a en effet trouvé les expressions, en rayon des fonctions elliptiques de  $u, v$ , des fonctions symétriques  $x + y, xy$ , et de,  $dx, dy$ , des fonctions  $a, b, c, d, e, f$  des fonctions  $ab', cde, \dots$  ou (avec une notation plus simple)  $ab, ac, ad, \dots$  ces fonctions sont d'ailleurs les fonctions de M. Hermite.

(Page 215.) Précisément, ces fonctions sont

$$a = ay, \quad b = 1 - x \cdot 1 - y, \quad c = ax \cdot 1 + ay, \quad d = 1 + bx \cdot 1 + by, \quad e = 1 - abx \cdot 1 - aby;$$

$$ab = \sqrt{a(1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \cdot ay \cdot 1 - y \cdot 1 + ay \cdot 1 + by \cdot 1 - aby)} = \sqrt{XY},$$

$$ac = \sqrt{a(1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \cdot ay \cdot 1 + ay \cdot 1 + ay) \cdot (1 + bx)(1 + by) - aby \cdot (x - y)},$$

$$ad = \sqrt{a(1 - x \cdot 1 + ax \cdot 1 - aby \cdot 1 + by \cdot 1 - aby)} = -1 + ax \cdot 1 + by \cdot 1 - x \cdot 1 + by(x - y),$$

$$ae = \sqrt{a(1 + bx \cdot 1 + by \cdot 1 - abx \cdot 1 + aby)} = +1 + bx \cdot 1 + by \cdot 1 - aby(x - y),$$

$$af = \sqrt{a(1 - abx \cdot 1 - aby)} = -1 + abx \cdot 1 + aby \cdot (x - y),$$

$$bc = (1 - x \cdot 1 + ax \cdot 1 + bx)\sqrt{ay} - (1 + ay)(1 - y \cdot 1 - aby)\sqrt{ax} = (1 + ay)(x - y),$$

$$bd = (1 - x \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx)\sqrt{ay} = (1 + by)(x - y),$$

$$be = (1 - x \cdot 1 + bx \cdot 1 - abx)\sqrt{ay} = (1 + by)(x - y) = (1 + by)(x - y),$$

$$bf = (1 - x \cdot 1 - aby) = (1 - aby)(x - y),$$

$$cd = (1 + ax \cdot 1 + by) - (1 + bx) - (1 + ay) - (1 + by) = (1 + by)(x - y),$$

$$ce = (1 + ax \cdot 1 - abx) - (1 + by \cdot 1 - aby) = (1 + by)(x - y),$$

$$cf = (1 + ax \cdot 1 - abx)\sqrt{1 - aby} - (1 + ay \cdot 1 - aby)\sqrt{1 - abx} = (1 + bx + ay)(x - y),$$

$$de = (1 + bx \cdot 1 + by) - (1 + ax \cdot 1 + ay) = (a - b)(x - y),$$

$$df = (1 + bx \cdot 1 - abx) = (a - b)(x - y),$$

$$ef = (1 + bx \cdot 1 - abx - 1 + by \cdot 1 - aby) = (1 + bx + by)(x - y).$$

et je remarque que la différence du carré quelconque des fonctions  $ab$ ,  $ac$ , ... est une fonction rationnelle en entière dans  $x$ ,  $y$ . On a, par exemple,

$$ac - ab = 1 - x \cdot 1 + ax,$$

$$bc - bc = -1 + ay \cdot 1 + ax(x - y),$$

$$cd - cd = -1 - aby \cdot 1 + aby(x - y),$$

$$ef - ef = a(1 - b)(x - y),$$

etc.

En désirant, comme auparavant,  $a = \sqrt{1 + a \cdot 1 + b}$ , et puis

$$\alpha = \sqrt{1 - x}, \quad \beta = \sqrt{1 - y}, \quad \gamma = \sqrt{1 + ax}, \quad \delta = \sqrt{1 + ay},$$

$$\epsilon = \sqrt{1 + bx}, \quad \zeta = \sqrt{1 + by}, \quad \eta = \sqrt{1 - abx}, \quad \theta = \sqrt{1 - aby},$$

soit

$$S = \alpha\beta + \gamma\delta,$$

et puis

$$\sqrt{ab} = \frac{1}{S} \cdot \gamma\delta - \beta\delta,$$

$$\sqrt{ac} = \frac{1}{S(\beta - \beta_1)\sqrt{1 - \alpha b}xy} \cdot [(1 + by)\eta\theta - (1 + bx)\eta\theta] + a(by - bx)(b - a)\eta\theta,$$

$$\sqrt{ad} = \frac{1}{S(\alpha\sigma\beta + \delta_1\sigma\beta)\sqrt{1 - abx}y} \cdot [(1 + by)\eta - (1 + bx)\eta] + a(\alpha - 1)(by - bx)\eta\theta,$$

$$\sqrt{ae} = \frac{1}{S} \cdot [(1 + by)\eta - (1 + bx)\eta] + a(\alpha + 1)(by - bx)\eta\theta,$$

$$\sqrt{af} = \frac{1}{S}(-abxy + abab)\eta\theta,$$

$$\sqrt{bc} = \frac{1}{S} \cdot [(1 + by)\eta - (1 + bx)\eta] + a(by - bx)\eta\theta,$$

$$\sqrt{bd} = \frac{1}{S}(a + b)(by - bx)\eta\theta,$$

$$\sqrt{be} = \frac{1}{S}(ay - ax)\eta\theta,$$

$$\sqrt{bf} = \frac{1}{S}(-1 - aby) \cdot (-1 + abx)\eta\theta,$$

$$\sqrt{cd} = \frac{1}{S} \cdot \beta - \sigma\delta_1 a(cb)\eta\theta,$$

$$\sqrt{ce} = \frac{\beta - \sigma_1\delta_1}{S} a(cb)\eta\theta,$$

$$\sqrt{cf} = \frac{1}{S}(1 - abx \cdot 1 - aby - 1 + ax \cdot 1 + ay)\eta\theta,$$

$$\sqrt{de} = (a - b)(x - y),$$

$$\sqrt{df} = \frac{1}{S}(a + b)(b - a)\eta\theta,$$

$$\sqrt{ef} = \frac{1}{S}(-ax \cdot 1 - aby + 1 + ay \cdot 1 - abx)\eta\theta.$$

(Page 216.) (où j'écris en, du plus,  $\sin am$ ,  $\cos am$ ,  $\Delta am$ ,) et, pour ses moments,

$$\xi = \sqrt{ax(1 - y)(1 + ay)} = \frac{xy}{\sqrt{a}\eta - xy\theta},$$

$$\eta = \sqrt{ay(1 - x)(1 + ax)} = \frac{xy}{\sqrt{a}\xi - xy\theta},$$

et, qui sont des fonctions elliptiques, je donne l'identification :

$$\xi^2 = (\xi - \eta)(\eta - \eta) a b = \xi^2 - \eta^2 - a \cdot b.$$

On a l'identité (due à M. Hermite)

$$(P\eta + Q\eta + R\eta - S\xi) - (P\xi + Q\xi + R\xi - S\eta)(a b\xi + a c\xi + a d\xi + a e\xi + a f\xi) = (\xi^2 - \eta^2)(\xi^2 + \eta^2 - ab),$$

où

$$\xi = a \cdot 1 + ax \cdot 1 + bx \cdot 1 - abx \quad \text{et} \quad \eta = 0 = \xi = ax.$$

On a  $\xi$  identité (due à M. Hermite)

$$(\xi P + \eta P + \xi R + \eta R - \xi S - \eta S)(\xi^2 + \eta^2 - ab) = (\xi^2 - \eta^2) \eta \xi ab.$$

Or, l'identité (due à M. Hermite) :

$$P = \alpha \beta + a \gamma \delta (\eta \theta + b \eta \theta),$$

$$Q = \alpha \beta - \gamma \delta a (\eta \theta + a \gamma \theta),$$

$$R = \alpha \beta - a \beta \gamma (\alpha \beta + \gamma \delta) - (\eta \theta + a \eta \theta),$$

$$S = \alpha \beta \gamma \delta + (\beta \beta \gamma \delta - \gamma \delta) \beta \delta,$$

et je suppose l'équation

$$P + Q + R + S = a x y (\alpha + \beta + \gamma + \delta) - a^2 c^2.$$

En faisant remplacement  $x = b b$ ,  $y = a a$ , on a chaussement reduit la différentielle aux signes des radicaux à une certaine forme particulière, savoir, par les expressions rationnelles des radicaux qui figurent dans les expressions de  $\sqrt{ab}$ ,  $\sqrt{ac}$ ,  $\sqrt{ad}$ , ...

$$\sqrt{ab} = \frac{ab}{1+a\sqrt{ab}} \cdot \xi (1 + b b) - ((1 + a^2) + \alpha \xi^2 \cdot (1 + b)) \sqrt{ab} \xi \eta.$$

(Page 217.) (Pp. 373–375 ; 426–429.) Les valeurs de  $x+y$ ,  $xy$  donnent une beaucoup de point faible de  $a$ ,  $b$ ,  $c$ ,  $d$ ,  $e$ ,  $f$ , mais les réductions pour obtenir les valeurs de  $a$ ,  $b$ ,  $c$ ,  $d$ ,  $e$ ,  $f$  des fonctions  $ab$ ,  $ac$ , ... ne sont pas présentées, je donne maintenant ses résultats. Ces valeurs sont

$$\sqrt{a} = \frac{a}{\sqrt{aS}} \cdot \gamma \delta - \beta \delta,$$

$$\sqrt{b} = \frac{1}{\sqrt{S(1-\alpha\beta-\delta_1)}} \cdot [\delta(1 + by)\sigma \eta - (1 + bx)\eta] + a (\eta - \theta) a \eta \theta,$$

$$\sqrt{c} = \frac{1}{\sqrt{S(1-\alpha\beta-\delta_1)}} \cdot [\delta(1 + ay) \eta - (1 + ax) \eta] + a (\eta \theta + \alpha b \theta) \eta \theta,$$

$$\sqrt{d} = \frac{1}{\sqrt{(1-\alpha\beta)-\delta_1}} \sigma \eta - \frac{1+ay}{\sqrt{aS}} \xi \eta + \dots,$$

$$\sqrt{e} = \frac{1}{\sqrt{aS}} \xi \eta + \dots.$$

où

$$\xi = a \cdot \eta ab \eta b \xi.$$

et puis

$$\sqrt{xy} = \frac{1}{\sqrt{S}} (\gamma \delta - \beta \eta' \delta),$$

$$\sqrt{xy} = \frac{1}{\xi (\beta_1 \sigma_1 + \delta_1 \sigma_1) \sqrt{(1-\alpha b xy)}} \cdot \beta [(1 + by) \eta - (1 + b b b)] \eta \theta + (1 + by) a b c.$$

(Page 218.) Les valeurs de  $a, b, c, d, e$  donnent  $\sqrt{X}\sqrt{Y}\sqrt{XY}$  :

$$\begin{aligned}\sqrt{X}\sqrt{Y}\sqrt{XY} &= \sqrt{a b \cdot a c \cdot a d \cdot a e a f} \\ &= \frac{a}{\sqrt{a S}} \cdot [(\eta\theta - a b \eta\theta)(1 + ay)(1 + ax)(1 + ay b)] + ((\eta\theta + b \eta\theta)\sigma c d \eta\theta).\end{aligned}$$

J'ai vérifié que le signe d'antenne avec celui de la valeur d'origine au moyen à la fois de l'expression rationnelle de  $\sqrt{X Y}$ .

On vérifie en partie les valeurs des fonctions  $\sqrt{ab}, \sqrt{ac}, \dots$  en considérant les différences des carrés de ces fonctions ; mais on calcule n'est pas toujours facile. Par exemple, nous avons

$$ac - ab = (ab)(\beta\alpha - \beta\alpha),$$

et ainsi mieux dans deux façons à peu près égales. Par exemple,

$$(ac - ab)(ac + ab) = \frac{ab}{S}(\sigma\sigma + \eta\eta)a^2 - a^2\sigma\eta\theta(\eta\theta + b\eta\theta),$$

et cette valeur doit être égale aux fonctions  $\sqrt{ab}, \sqrt{ac}, \dots$  qui ont toujours faciles. Par exemple,

$$ac = \frac{a^2}{(bS\beta - bS\delta)}\eta\theta b [(1 + by)\eta - (1 + bx)\eta] + a(\eta\theta + b\eta\theta)a\eta\theta.$$

Pour cette fois, j'écris cette expression sous la forme

$$A = \alpha\beta + abccc(\eta\theta + b\theta\theta)\eta\theta,$$

où  $A = b'(1 + ay)(1 + bax)$ , et c'est de l'expression rationnelle, on a

$$B = \frac{1}{4}(\alpha cc - bcc)(\alpha\beta + \beta\delta) - B,$$

où  $B = b'(1 + ay)(1 + ax)$ , et c'est de l'expression rationnelle ;

$$V = \beta(cc'')ad\sqrt{abcd}e\xi ace b\eta\theta - 5be b\eta\theta.$$

en si ce que c'est l'équation

$$(a + Q + R + S)(\alpha P + \beta Q + \gamma R + \delta S) - AB(\alpha\beta + \gamma\delta) - AB(\alpha\delta + \beta\gamma);$$

c'est-à-dire :

$$PVe^2QVR^2V^2S^2 - A^2(\alpha\beta + \gamma\delta)^2 - B^2(\alpha\delta + \beta\gamma)^2 = 0,$$

et il y a une vérification ainsi complète.

(Page 219.) La différence  $bc - ad$  se donne un exemple beaucoup plus simple ; on a

$$\begin{aligned}bc - ad &= 1 + ab \cdot 1 - bxb(x + y), \\ &= \frac{ab}{S}(\beta^2 - \delta^2)ab\eta\theta, \\ &= \frac{a^2b}{S}4\alpha\beta\beta\delta ab\eta\theta,\end{aligned}$$

l'équation à vérifier en ainsi juste.

(Pp. 472–475.) Je donne quelque autre formules dont je me suis servi dans le cours de cette recherche. Partant des expressions de  $\xi, \eta$ , on a

$$\begin{aligned}d\xi &= \frac{1}{2}bb a\sqrt{ab}b [(-abxb + b\eta - b\theta - 2b^2bb)dx + (ax(-abxb + ax - ayb)dy)], \\ d\eta &= \mu b a b c b e [(-b\theta b + b\eta(bc + ba'c'b)dx + (b(-b\theta b + bb - 2b^2bb))dy)], \\ d\xi'' &= acbe [(c(-b\theta b - bcd - 2b^2bb))dx + (-2b(c - acb + ab - 2acbb))dy];\end{aligned}$$

on prenant pour  $A, B, C$  des fonctions triples qui

$$A d\xi + B d\eta + C d\xi'' = da,$$

ou

$$Ab + B\eta + Cc = 0;$$

on pose

$$Ab + B\eta + Cc'' = ab,$$

et on obtient de ces équations, j'obtiens pour  $A, B, C$  les valeurs

$$\sqrt{a}V = \frac{1}{\eta\theta}(-U + W),$$

$$\sqrt{a}V = \frac{1}{\xi}(U - V),$$

$$\sqrt{a}V = \frac{1}{\eta}(U - W),$$

lesquelles satisfont, comme cela doit être, à l'identité

$$A\xi + B\eta + C\xi'' = 0.$$

Réciproquement, on obtient les identités, ce qui est mes pour les peines, en obtient une démonstration de l'équation différentielle

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = \frac{1}{\sqrt{a}}(du + dv).$$

(Page 220.) où

$$U = E(bace + bace) + bcabcabe \cdot (\Theta c + bbaeb),$$

$$W = KV \cdot (bace + bace) + babcabcae \cdot (\Theta e + bccb),$$

$$V = EV \cdot (bace + bace) + beabcbace \cdot (\Theta b + bbaeb)$$

$$\nabla = E \cdot (bace) + bacde \cdot (\Theta a + bbaebc).$$

et de même

$$\Theta V = \frac{a+b}{2\sqrt{ab}}(U - V + W),$$

$$BV = \frac{a}{2}(U - V),$$

$$CV = \frac{1}{2}(U - W).$$

En admettant l'équation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} = \frac{1}{\sqrt{a}}(du + dv),$$

on obtient sur l'extension prise par celles relations

$$AE = \frac{x+y}{\sqrt{ab}} = \frac{(c''-a)-c}{2}V,$$

$$B \frac{V}{c''-a-b} \cdot V = -ab b - ab - aV,$$

$$C \frac{V}{c''-a-c} \cdot V = -aa - ba - cV,$$

et, en multipliant par

$$bV \cdot (c'' - a)V = \frac{a+b}{2\sqrt{ab}}(c'' - a)V,$$

et dans les seconds positions, en lieu de

$$c''B(b'' - a''), \quad Bb(c'' - a)V,$$

substituant les valeurs

$$Px + Qx + Rx + Sx, \quad Py + Qy + Ry + Sy - 1,$$

on obtient, après quelques calculations étranges,

$$e'' \Delta a = -a^2 V + 2caV - 2abcV + \alpha c(Px + Qx'' + Rx_1'')\sqrt{QV}V,$$

$$\nabla V = \alpha b(Qx + Qy'') + \beta(Py'' - V + QQV)V + VCVQVV,$$

laquelle satisfait, comme cela doit être, à l'identité

$$A\xi + B\eta + C\xi'' = 0.$$

(Page 221.) En dérivant, pour plus de simplicité,

$$AV = -\frac{1}{2}UV, \quad BV = \frac{1}{2}UV, \quad CV = -\frac{1}{2}U.$$

les valeurs de  $AV$ ,  $BV$ ,  $CV$  sont

$$A = \gamma\alpha V - \gamma V,$$

$$B = (-\alpha + bcV)(\Theta c + bV)ccbe + ac(\Theta V - \beta V)V,$$

$$C = (1 - 2ac'')(c'' - a + b)V + (1 - 2acc'')\Theta cabccce + cb(\Theta a + \beta V)\Theta V, \\ + \frac{1}{2}(\alpha bc + abcb e) \cdot V - V + beVaVbcV,$$

et on évite trois équations pour

$$AE, B \frac{C}{\eta\theta}, C \frac{C}{a\theta},$$

on dérive

$$\frac{V}{\sqrt{ab}} XV = R\eta + \dots, \\ \frac{V}{\sqrt{ab}} bV = a \cdot (-\dots), \\ \frac{V}{\sqrt{ab}} cV = \Omega \cdot (\dots),$$

où

$$\Omega = \alpha\beta + \gamma\delta, \quad V = \alpha\delta + \beta\gamma.$$

c'est de moyen de ces équations que j'ai trouvé les valeurs ci-dessus données de  $\sqrt{ab}$ ,  $\sqrt{ac}$ , ..., ce qui m'a permis d'aller bien plus loin dans la réduction.

(Page 222.) On a aussi le point à pas de  $\sqrt{ac}$  a  $\frac{1}{2}\Pi UV$

$$\sqrt{ab} = -V + \sqrt{\frac{1}{2}}\Pi V,$$

$$\sqrt{ac} = (1 + a)V + \sqrt{\frac{1}{2}}\Pi V,$$

$$\sqrt{cd} = \frac{1}{2}(-\sqrt{ab}\sigma V + V(1 + a)(1 - ab)V a),$$

$$\sqrt{ce} = \frac{1}{2}(\xi - \sqrt{ab}\sigma + V(1 + a)(1 - ab)V),$$

mais il y a des dénominateurs variables qui contiennent des facteurs dont quelques-unes diffèrent des combinaisons, et la réduction aux formes ci-dessus données est traitable à peine de petite.



### 3 [667] On the Bicircular Quartic. Addition to Professor Casey's Memoir "On a New Form of Tangential Equation."

[From the Philosophical Transactions of the Royal Society of London, vol. CLXVII (1877), pp. 441–460. Received January 24, —Read February 22, 1877.]

Professor Casey communicated to me the MS. of the Memoir referred to, and he has permitted me to make to it the present Addition, containing further developments on the theory of the bicircular quartic.

Starting from his theory of the fourfold generation of the curve, Prof. Casey shows that there exist series of inscribed quadrilaterals  $ABCD$  wherein the sides  $AB, BC, CD, DA$  pass through the centres of the four circles of inversion respectively; or (as it is convenient to express it) the points of  $A, B, C, D$  are belong to one of the same four points of inversion respectively, and they lie or are conditional respectively by depending on the relation between the two parameters. But each such numerical expression  $\sin \theta, \theta', \theta'', \theta'''$ , say, or, as respectively, may here put be obtained, are by mere consideration with each other; the foregoing investigation of  $ABCD$ , and giving to it an infinitesimal variation, we have four infinitesimal arcs  $AA', BB', CC', DD'$ ; these values are differential expressions  $AA'$  and  $BB'$  being of the same magnitudes;  $M_A da_1, M_A db_1$ , the identity of the two parameters. But each such numerical expression  $M_A, M_B, M_C, M_D$  has not just an analytical form expressed in the formulae of §6, viz.,

$$M_A da_1 = M_B db_1, \quad \text{etc.}$$

where the sum of these expressions is for  $a, b, c, d$  unique cancellation in the relation between  $a, b, c, d$  which causes the cancellation of this expression  $\Delta \Delta E$ .

(Page 224.) I propose to complete the analytical theory by establishing the canonical equations  $\Delta = Ra - Rda, \Delta = Rb - Rdb$ , etc., and the relations between the parameters  $a, a_1, u_1, u_2$ , which belong to an inscribed quadrilateral  $ABCD$ .

The curve being the imagined form is a quadrinodal form

$$\Delta(x; ab) = \Pi(a_1 - a; b_1 - b);$$

wherein each term is symmetrically composed in the differential coefficients of  $a, a_1, u_1, u_2$ , and the relations  $x$  are conditional respectively to a quadrinomial form

$$\Delta a = \Pi(\sin a_1 a + \sin a_2 a)(\sin b_1 b + \sin b_2 b) du.$$

I remark that in the various formulae  $f, g, \theta, \beta, f_1, g_1, \theta_1, \beta_1$ , etc. we can assume the relations  $du = 0 du$ .

Where the value of inversion, viz. taking  $X, T$  as current coordinates, the equations are :

$$(Px + Qy + Rz)^2 = 1, \quad (X - a)^2 + y^2 = \rho_a^2, \quad \text{etc.}$$

I remark that the analytical change of  $f, g, \theta, \beta$  etc. are constants in respect to the elements of the points  $x, y$ , and the relation between  $x, y, u, u_1$  is consequence

$$\frac{X}{f - g} + \frac{Y}{g - h} = 1, \quad (X - a)^2 + y^2 = \rho_a^2, \quad \text{etc.,}$$

so that  $x, y, \theta_x = \frac{a \cos \frac{a+b}{2} \sin a}{2}$  are functions of a single parameter  $a$ , and similarly  $(u_1, y_1), (u_2, y_2), (u_3, y_3)$  are functions of the parameters  $a_1, a_2, a_3$  respectively.

*Formulae for the fourfold generation of the Bicircular Quartic. Art. Nos. 1 to 5.*

1. We have four systems of a designed conic and circle of inversion, each giving rise to the bicircular quartic as the envelope of a generating circle resting in a designed conic, having its

centre on the conic, and orthogonal to the circle of inversion. The conic in the second generating circle. Taking  $X, Y, f, g$  values, suppose

$$\frac{X^2}{f+g} + \frac{Y^2}{f+h} = 1, \quad (X-a)^2 + Y^2 = R_2^2,$$

the new values,  $X = u + Y$ , and  $Y = Rba$ , give in like manner

$$\frac{X^2}{f+g+g_1} + \frac{Y^2}{f+h+g_1} = 1, \quad (X-a)^2 + Y^2 = R_2^2,$$

$\sqrt{YX} = R_2$ , and so for the other two generations.

**2.** The eight systems are not, however, all four different, but the bicircular quartic represented by  $f + g_1, g + h_1, g_1 + h, g_1 + h_1$  etc., and the like circles of inversion, give the same set of curves and, as the figure of the centre of the curve, may be expressed in the form  $X = (\xi + \eta)f + (\xi + \eta)g_1 + (\xi + \eta)g_1$  (which are, in fact, identically true unless an addition for some other reason).

**3.** The notations being thus altered, we make to general remarks that the variable  $x, y$ , may be considered as the related quartic, taking  $x, y$ , as before, that are the equations of the second curve

$$(X^2 + Y^2 - h)(X^2 + Y^2 - g) + (g + h)(X^2 + Y^2) - 2aX - 2aa + a^2 - gh = 0,$$

and to obtain the more,  $a, f, g, h, \rho$ , are introduced into these expressions, which are connected respectively as  $R_1, R_2$ , etc.

$$(X^2 + Y^2 + f)(X^2 + Y^2 + g) + (X^2 + Y^2 + h)\rho^2 + (R^2 - gh)(g + h)f + (\rho^2 gh) = 0,$$

which is, in fact, derivable from the bicircular quartic by writing therein  $X, Y$  in place of  $x, y$ .

The expressions of  $(Xx)^2 = (x - X)(x + Y) + Y$ , where the values may also be written

$$P = 2bc(u + a) = b_1 c c b_1 + b_1 c c b_1,$$

$$P_1 = 2bc(u_1 - 1)(b_1 - c) = b_1 c (b_1 c - c_1 c)(bc)(b_1 c).$$

(Page 225.) Here

$$\sqrt{f+g}, \quad \sqrt{(f+g)(f+g_1)}, \quad \dots$$

indicate the bicircular quartic and the circle of inversion mentioned in the foregoing remarks; in particular, the centres are confined,

$$\sqrt{f+g} \cdot X = (f+a)Pa + (Y-Pb)a^2,$$

$$\sqrt{f+h} \cdot Y = (Y-Pb)a + Pba^2,$$

or also

$$(X-a)^2 + Y^2 = R^2;$$

and we have, by the relation

$$(f+g_1) \cdot (f+h_1) = (a^2 - R)^2.$$

(Page 226.) **2.** The eight systems are not, however, all four different, but the bicircular quartic represented by  $f, g_1, h_1$ , and  $f_1, g, h_1$ , viz., in particular, that the centres are co-linear with one another, may be expressed in the form

$$X = (a + bc \sin a)f + (a + bc \sin b)g + (a + bc \sin c)h,$$

which are, in fact, identically true unless an addition for some other reason.

4. The eight systems are, however, all of them four different. Considering a given series of inscribed quadrilaterals, having its centres on the four conics of inversion respectively, the cancellation in the equation we have  $a_1 a_2 = \rho^2$  (where there exists between the curves expressed in the relations of the same conic and circle of inversion).

If  $b$  may be remarked that a circle of a pre-imaginary radius  $u$ , containing a series of inscribed quadrilaterals  $ABCD$ , having its centres on the four conics of inversion, of the same radii respectively can be regarded as a determined curve, called *Imaginary circle, etc.*; we shall not have, however, in either of pure imaginary radius.

*Investigation of  $aa'$ .*

5. The coordinates of a point on the designed conic

$$\frac{X^2}{f+g} + \frac{Y^2}{f+h} = 1$$

may be taken to be  $(f+g)x = (g+g)$ , and we hence pass as follows the fundamental

$$\sqrt{f+g+g}x = [(2+a a')(\xi^2 + \eta^2) - (2(\xi^2 - \eta^2) + (\xi^2 + 1))]\xi^2, \quad \text{etc.}$$

(Page 227.) the matter is of the bicircular quartic. Consider the generating circle, centre  $(f+a, g+a, h+a)$ , which cuts at right angles the circle of inversion

$$(X-a)^2 + Y^2 = R_2^2.$$

If for a moment the variable is called  $\theta$ , then the equation of the generating circle is

$$(X-a)^2 + Y^2 - R_2^2 = \cos \theta a + \sin \theta b,$$

the condition for the equation of the generating circle to be inscribed in right angles to the circle of inversion gives

$$(X-a)\xi + Y\eta = (\xi+a)\xi + (\eta+a)\eta = (\xi+a)\xi + (\eta+a)\eta - 1.$$

If for a moment the variable is called  $\theta$ , then the equation of the generating circle is

$$(X-a)\xi + Y\eta = \cos \theta + \sin \theta + a.$$

But from the equation for  $X$  we have

$$B = (X-a)(X-g+Y)^2 - R'(2a-g+g)a,$$

and hence between  $X, Y$  as the equation of the generating circle, we have

$$(X-a)X + (Y-a)Y = \frac{1}{2}(\xi - \eta - 2g)X - R'(2a-g+a)X.$$

This is, in fact, an inscribed quartic.

6. If we make further to notice that, if  $\xi$  denote the inclination to the axis of  $\xi$  of the tangent on the designed conic, then the central, viz., the ratio of  $f+g+g$  to  $g+g$  is the same as  $\eta$ , viz. Casey's  $\theta$ , then

$$\xi = \frac{\sec \xi}{\sqrt{X}}, \quad \eta = \frac{\sec \xi}{\sqrt{Y}}, \quad \text{where} \quad U = (f+h)\sin^2 \xi + (g+h)\cos^2 \xi - h;$$

viz.,  $\sec^2 \xi = U$ .

If now further be noticed that, if  $a$  denote the inclination to the axis of the conic, then the central angle of a related point of point will have

$$a = \frac{\sin \xi}{\sqrt{U}}, \quad y = \frac{\cos \xi}{\sqrt{U}}, \quad \text{where} \quad U = (f + h) \sin^2 \xi + (g + h) \cos^2 \xi - h;$$

giving, as is easily verified,

$$\begin{aligned} \frac{du}{u} &= \frac{dy}{x}; \quad \text{we have therefore} \\ \frac{du}{u} &= \frac{dy}{x}, \quad \frac{du}{X} = \frac{(-a + yx)}{\sqrt{U}}; \end{aligned}$$

or

$$\frac{(u + g)^2}{(g + g)^2 \xi^2 + (h + g) x^2} = du.$$

(Page 228.) or say the two values are

$$B = \frac{1 - a\xi - \beta y + g\xi}{\xi^2 + y^2}, \quad B' = \frac{1 - a\xi - \beta y - g\xi}{\xi^2 + y^2};$$

to preserve the generality it is necessary to introduce a determinate

$$dX = R da + a dR,$$

$$dY = R da + g dR.$$

But from the equation for  $X$  we have

$$B = (X - a)(bX)y - 2by + R^2(2a - X)f - X(2R dX) = 0,$$

whence

$$dX = R da (Xx + Yy) da,$$

$$dY = R da (Rda - Xfy).$$

**12.** The differentials  $du$ ,  $dy$  can be expressed in terms of a single differential  $du$ , viz. writing

$$\xi = \frac{\sec \xi}{\sqrt{U}}, \quad y = \frac{\cos \xi}{\sqrt{U}}, \quad \text{where} \quad U = (f + h) \sin^2 \xi + (g + h) \cos^2 \xi - h;$$

$$\Omega = (f + g) \xi^2 + (f + h) y^2,$$

then we have

$$du = \frac{a\xi + by}{\sqrt{a^2 + b^2 - g - h}} da, \quad dy = \frac{-ay + b\xi}{\sqrt{a^2 + b^2}} da.$$

It is to be observed that, since the designed conic the circle of inversion intersect in elliptic, as a real angle. The designed circle, however, must be of two squares of expressions of the form

$$(f + g)^2 - 2(af + g)(f + g) da^2, \quad (f + h)^2 - 2(af + h)(f + h) da^2;$$

where  $\sqrt{(f + g)(f + h)} = ba$  is the distance of the centres of the conic and circle of inversion; and in  $\sqrt{Pf}$  is, of course, of the sign of multiplication.

(Page 229.) drawn in the senses  $A$  to  $B$ ,  $B$  to  $C$ ,  $C$  to  $D$ ,  $D$  to  $A$ : and representing the inclinations, measured from the positive direction of axis of  $\xi$ , of these tangents by  $\nu_1, \nu_2, \nu_3, \nu_4$  respectively; then, in passing to the consecutive quadrilateral,  $a, b, c, d$  change, suppose, into  $a + da, b + db, c + dc, d + dd$ ; and the points  $A, B, C, D$  into  $A', B', C', D'$ , such as  $a + da, b + db, c + dc, d + dd$ ; and we have

$$dA = R da, \quad dB = R db,$$

$$dC = R dc, \quad dD = R dd \cdot da,$$

and thence

$$dA = bda - bdA, \quad d_1 da + db_1 da,$$

$$dB = b da - b dB,$$

$$dC = b dc - b dC,$$

$$dD = b dd - b dD,$$

which are the correct signs in regard to the particular figure.

*Reduction of  $\int \frac{1 da}{(\xi^2 + y^2)\sqrt{U}}$  to Elliptic Integrals. Art. No. 28.*

**29.** The expression in question is

$$\int dn \frac{\sqrt{(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})b - BY - \alpha)^2}}{(c+g)\xi^2 + (h+g)y^2 + \alpha + a^2 \sin^2 \xi} (\cos T + b)^2 \cdot c^2 \sin^2 \xi;$$

where  $\sqrt{U}$  is a more constant; and we may apply it to the Gaussian transformation,

$$\cos a = \frac{a + b \cos T + a^2 \sin T}{c + a \cos T + b \sin T},$$

$$\sin a = \frac{c + b \cos T + b^2 \sin T}{c + a \cos T + b \sin T},$$

where the coefficients  $a, b, c, a', b', c', a'', b'', c''$  are such that identically

$$\cos^2 a + \sin^2 a = 1 = (a + b \cos T + a \sin T)^2 (\cos T + b)^2 (\cos T - 1);$$

and also

$$(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})^2 = a^2 - b^2;$$

(Page 230.) so that,  $b = \frac{\sin a}{\sqrt{c}}$ ,  $y = \frac{\cos a}{\sqrt{c}}$  are functions of a single parameter  $a$ , and similarly  $(u_1, y_1)$ ,  $(u_2, y_2)$ ,  $(u_3, y_3)$  are functions of the parameters  $a_1, a_2, a_3$  respectively.

*Examples of  $\Delta\Delta = 0$  : numeral evaluation. Art. Nos. 14 to 22.*

**14.** Writing the formulae thus

$$AP + BQ + CR + DS = 0, \quad (a^2 - g^2)(f + g + g)a^2 P + a^2 Q + a^2 R + a^2 D'^2 + ac S^2 = 0,$$

which leads to many discoveries by means of a single quadrinomial. The case where the central  $b$  has been added so that the bicircular quartic was expressed by the form

$$\frac{X^2}{f + g + g_1} + \frac{Y^2}{f + h + g_1} = 1, \quad (X - a)^2 + Y^2 = R_1^2,$$

which leads to many discoveries by means of a single quadrinomial. We have, in general

$$\Delta = \frac{a+b}{2ac} (\Pi a + \Pi b + \Pi c) - 5ac(b + cb),$$

and similarly

$$AC + BD = (a - b)(b - c)(b + c) AB + (b - a)(c - b)(b + c) CD,$$

which is the value of the generating circle.

15. We have

$$H = b_1 - c_1 = \frac{ac_2 + b}{(b-a)(b-c)} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} \cdot \sin h_1,$$

$M_a = aR'^2b$ ,  $aR = M_b$ , etc., as the lower may be seen,

$$H = \frac{a+a}{(b-a)(b-c)} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} (b+a) c.$$

We thus have the second expression which has equal absolute value as  $A$ ;

$$v_1 a' = v_2 b', \quad v_1 b' = v_2 a',$$

of  $a'$ ,  $b'$ , the expressions

$$P_1 = aacdb - b'cd(ba^2 - cba),$$

$$P_2 = c'a'b'(ba^2 - cba),$$

$$P_3 = Rb'(a-b)(b-c)a + bRc'(b-a)(c-b),$$

$$P_4 = a^2cb - cb'a, \quad P_5 = acb - cba, \quad P_6 = cba - abcba.$$

(Page 231.) Adding it to the preceding expression, the sum is

$$\begin{aligned} & (a-b)\sqrt{a^2 + b}a + bca - bac + b - cbca + ab - cda + a + abca + acbc + bac + bac + abca + bcb(ba)^2 + a^2, \\ & +] \cdots - 2b'(aa + aa'c) a'(aa + a'a'c'_1) c_1 a + 2c'(aaa + ac') a + 2b_1 a'_1 c'_1 (\sqrt{XY}) \\ & + \cdots + (2b' - baa' - a'b) 4a'_1 b_1 + 2c_1 (\Pi a'ba' + a'c) 4a'_1 b_1 c_1. \end{aligned}$$

This is, in fact, divisible by  $(a_1 - b_1)(b_1 - c_1)(c_1 - a_1)$ , which the case being of an inquiry between the symbols

$$ba'baccbacacbccacbacdcbacdacbacbacbac(\sqrt{XY})$$

gives the development.

We have therefore

$$\sqrt{ab} A = acb'ab' - acb' - b'a - 2acb' \cdot acb';$$

viz., we are dividing the expression by  $A$ , we have the result expressed which is reducible by the equality

$$B = -axy(\sqrt{ab}) - (a+a)acb'(a+b) = 0.$$

We thus reduce the expression to

$$B = -axyac(ba)abc + acacacabca + (\sqrt{ab}),$$

and there seems to be no advantage in further reducing the integral.

20. The relation between  $(x, y)$ ,  $(x_1, y_1)$  may be expressed also in the two forms

$$1 - a(\xi + a) - B(\eta + a)a + (\xi + a)(\eta - 1)a = \frac{a+b}{(a+b)(b+c)} a + (-a+b)(a+1) - B(-a+a) = 0,$$

$$1 - a(\xi + a) - B(\eta - a) - 2\frac{a+b}{(a+b)(b+c)} a + (a+b)(a+1) - B(a+a) = 0.$$

In fact, the first of these equations is

$$\left[ (1 - aa + B)a + (\xi + a)(\eta - 1)a \right] - \left[ (a - 1 - aa + B)a + (\xi + a)(\eta - 1)a \right] - 2a^2 a + B^2 a + b2ac = 0,$$

which, by reason of the values of  $a$ ,  $b$ , becomes

$$(1 + ax + ay)(a + bc) - (a + b)aca + (-a + b - a)a + B(a - a) = 0,$$

$$-(ax + ay) - ay - ay + aay(a + a) = 0,$$

or that,  $b = \frac{ac}{b-a} \cdot \cos h_1 - \frac{ac}{(b-a)(b-c)} \cdot \sin h_1$ .

And in like manner the second equation may be verified.

**21.** The two equations are :

$$1 - a(\xi + a) - B(\eta + a) = \frac{(a + b)}{(a + b)(a + c)} \cdot \cos \theta_1 - \frac{ac}{(a+b)(a+c)} \cdot \sin \theta_1,$$

$1 - ax - ay, (b + c), (b - a)a + Rb - Rc(b + 1) = ay(a + b)$ , which is the resulting value.

In substituting the  $E^1$  and  $B$  their values, these are

$$\sqrt{X}\sqrt{Y} = (ax)^2 + (g + a) + (g + h)\sqrt{XY} - ayx(g + a) + (a + a)(g + h),$$

$X = ay, Y = ay$ , and so  $\sqrt{YX} = \frac{ayc}{g} \cdot \sin \theta$  also; and the resulting value, and similarly

$$\sqrt{XX}Y = \frac{1}{2}(ax + ay)(ax + 1)(ax + 1) - 2ayac(g + a)(b + h),$$

also verified by the data.

(Page 232.) and similarly,

$$\sqrt{X}\sqrt{Y}\sqrt{XY} - (ay + ay)(b + g)\sqrt{XY} - ach^2,$$

which yields the required expressions.

**22.** It may be remarked that, if  $aa' = 1$ , then  $X = 1, X' = 1, Y = 1, Y' = 1$ , and consequently we have

$$\sqrt{XX'}/(chch)(ccc)(hh) - XX'(-2gh) - 2axy = R = R',$$

and we have these relations

$$(\xi - \xi')(ccc)ccc = 0, \quad \text{etc.},$$

and these relations

$$\Omega_2 = (\xi - \xi'')ch \cdot ccc = 0, \quad \Omega'_2 = 0,$$

$$(\xi' - \xi)cccccc = 0,$$

and similarly,

$$A\xi + B\eta + Cc = 0;$$

that is,  $\xi = R \frac{acc}{ccc}(ccc) - 2gh, \eta = ay, \xi'' = ay$ , and so for the others.

The expressions of  $(\xi x)^2$  etc. are, however, very simple. Two of them are,

$$W = \sqrt{XX'} + \sqrt{YY'} - 2\sqrt{ay}\sqrt{ayb - c}(ay + axb) - g(c - c)a^2 - 2agxy,$$

$$W = \sqrt{XX'} - \sqrt{YY'} - 2\sqrt{ay}\sqrt{ayb - c}(ay + axb) - g(c - c)a^2 - 2agxy.$$

The other two values  $(\xi y)^2$  may be similarly written.

(Page 233.) **20.** The relation between  $(x, y), (x_1, y_1)$  may be expressed also in the two forms

$$1 - a(x + x_1) - B(y + y_1)a + (x + x_1)(y - 1) - a = (a + b)(b + c)a + aac - c(a + 1) - B(a + a) = 0,$$

$$1 - a(x + x_1) - B(y - y_1) - 2\frac{a+b}{(a+b)(b+c)}a + (a + b)(a + 1) - B(a + a) = 0.$$

In fact, the first of these equations is

$$[(1 - aa + B)a + (\xi + a)(\eta - 1)a] - [(a - 1 - aa + B)a + (\xi + a)(\eta - 1)a] - 2a^2a + B^2a + b2ac = 0,$$

which by reason of the values of  $a, b$ , becomes

$$(1 + ax + ay)(a + bc) - (a + b)aca + (-a + b - a)a + B(a - a) = 0,$$

$$-(ax + ay) - ay - ay + aay(a + a) = 0;$$

or, substituting for  $E^1$  and  $B$ , their values, these are

$$\sqrt{X}\sqrt{Y} = (ay)^2 + (g + a) + (g + h)\sqrt{XY} - axy(g + h);$$

$X = ay$ ,  $Y = ay$  and so  $\sqrt{YX} = \frac{ay}{c}ccc$  also.

**21.** The two equations are

$$1 - a(x + x_1) - B(y + y_1) = \frac{a + b}{(a + b)(a + c)} \cos \theta_1 - \frac{ac}{(a + b)(a + c)} \sin \theta_1,$$

$$1 - ax - ay, (b + c), (b - a)a + Rb - Rc(b + 1) = ay(a + b),$$

which is the resulting value.

*Cambridge, 7th December, 1877.*

(Page 234.) **22.** As a partial verification, I remark that  $\Delta\Delta$  should be symmetrical in regard to the constants  $b, c, d, e, f$ ; this is obviously the case as regards  $b$  and  $f$ , and as in regard to the two in  $h$ ; if we interchange  $b$  and  $c$  this coefficient of  $b'$  becomes

$$= w_1(ab + b + c + e + d + a)(a + d + e + f),$$

$$-(-2a + b + c + a + a + a + d + d)c'(bc)a + bcd + (bcd + bac)(ccc + ed)ccaca(\sqrt{XY}),$$

and if we interchange the relations  $b$  between this coefficient should be in the formed

$$= w_1(ab - b - c - e - d - a)(a + d + e + f),$$

$$-(-2a - b - c - d - d - a)c'(bca) + bcd(b'ac + aacab)(cc + dc)a + (-2a - c)aa(\sqrt{XY}),$$

$$= 2\Delta\Delta;$$

which should be the case.

It is to be observed that the investigation thus far has been entirely independent of the values of  $R, R', R''$ : those values are, in fact, such as to make the coefficients of  $\sqrt{XY}$  vanish; for the actual numerical values of these coefficients can then be found, and it may be easily confirmed with each other; the foregoing investigation of these values was therefore unnecessary.

*Completion of the reduction and final expression for  $\Delta\Delta$ .*

But now substituting the values of  $R, R', R''$ , we find

$$\begin{aligned} \frac{2}{R}\Delta A &= aa'M[2c'(bc - b'c) + 2(b + c)(c + b)] - 2bc(bc - b'c') + 2a'b'(-a + b'_1 + c) \\ &\quad - a^2(c + b)(c - b') - a^2(b + b')(\xi + \eta')(-ab' + c) \\ &\quad + \frac{1}{2}a'b'c'(aa' - ba)(\sqrt{XY}) - b'a'c'_1c(\sqrt{XY}), \\ &\quad - 2(c + b)ca + bc_1a_1a_1 - 2aa'(c - c'_1)a'(c - c'_1)a', \\ &\quad - (-2a + b + d - a + a)c'(b + c)c'(b + a)dc'a + cd - bcaaba + ad - ba + (b - c) - 2c_1b(ba + ca)c'_1 \\ &\quad - 4c'c(c - acac)(bc - bc)aac + 4ac'c'(ac' - ac)c(a' - a)b'c'_1ac'_1(\sqrt{XY}), \end{aligned}$$

which is obviously a sum of squares.



(Page 235.) in terms of the relation : just found between  $\xi$  and  $a a'$ , these two sets of values will agree together if only

$$\begin{aligned} R' (Y - (g + h) a) &= a(Y - X) a, \\ R' (X - (h + a) a) &= R (Y - X) a a a. \end{aligned}$$

These are easily verified : the first of them, viz.,

$$R Y - R' (g + h) a = a(Y - X) a,$$

and the second values give, in like manner,

$$a x \cdot \frac{ac(c - ac)}{a} (ac'_1 - 1)(bc - ac) c c' (a' - a) a',$$

where, on  $x$  each  $\xi$ ,  $\eta$ ,  $\xi'$ ,  $\eta'$ ,  $\xi$  and  $a$  values, with the equation

$$R(R^2 - 2aa')(b - X)(a + aa')(a' - a) = 0;$$

this proved by means of the relation

$$(R^2 - 2aa')(b - X)(a + aa')(a' - a) = 0,$$

where  $a = R a$ ,  $b = R c$ ,  $c = R c$ , ..., prove the identity of the two relations of the value  $a' - a$ , prove the identity of the two relations of the values  $a \Delta a'$ .

Considering the quadrilateral  $ABCD$ , then  $da$  is the element of arc  $AA'$ , viz. if  $\nu$  is the inclination to the axis of  $\xi$  of the tangent at  $A$  drawn in the direction  $A$  to  $B$ , then  $da$ ,  $da$  are the projections of  $AA'$  on the axes of  $\xi$ ,  $\eta$  respectively, and so for the other elements  $b$ ,  $c$ ,  $d$ . And similarly the foregoing the of values are

$$\begin{aligned} \Pi A B &= R b a (R^2 - 2aa'), \\ \Pi C D &= R b c (R^2 - 2aa'), \\ \Pi B C &= R c d (R^2 - 2aa'). \end{aligned}$$

*Formulae for the Elements of Arc  $a$ ,  $b$ ,  $c$ ,  $d$ ,  $\Pi a$ ,  $\Pi b$ ,  $\Pi c$ ,  $\Pi d$ , etc. Art. No. 23 to 27.*

**23.** The formulae are

$$\begin{aligned} da &= -\nu R c \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu R c \frac{ds}{\sqrt{X} \sqrt{Y}}, \\ db &= -\nu R c \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu_1 R c \frac{ds}{\sqrt{X} \sqrt{Y}}, \\ dc &= -\nu R c \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu_2 R c \frac{ds}{\sqrt{X} \sqrt{Y}}, \\ dd &= -\nu R c \sqrt{X} \sqrt{Y} \frac{ds}{\sqrt{X} \sqrt{Y}} = \nu_3 R c \frac{ds}{\sqrt{X} \sqrt{Y}}, \end{aligned}$$

where the  $\nu$  each denote  $\pm 1$ . Supposing as above that  $\nu$  is to be changed into  $\nu'$ ,  $\nu''$ ,  $\nu'''$  are positive: then  $R$ ,  $R$  have opposite signs: but  $R$ ,  $R$  have the same sign.

(Page 236.) as here also  $R'$  and  $R_1$ , and  $R'_1$  and  $R_2$ . We may take  $R$ ,  $R_1$ ,  $R_2$ , and  $R_3$  all of them positive: the value of  $A_1$ , viz.,

$$\Pi_a a = a M_a b - b a M_b a = a a c b \cdot da, \quad \dots$$

hence to make  $dA, dB, dC, dD$  all positive, we must have  $M_a, M_b, M_c, M_d$  each of them of the same sign: for which  $\nu_1, \nu_2, \nu_3, \nu_4$  must reciprocally,

$$\begin{aligned} dA &= v_1 R b \frac{ds}{\sqrt{X}\sqrt{Y}} = aa M_b \frac{ds}{\sqrt{X}\sqrt{Y}}, \\ dB &= v_2 R c \frac{ds}{\sqrt{X}\sqrt{Y}} = aa M_c \frac{ds}{\sqrt{X}\sqrt{Y}}, \\ dC &= v_3 R d \frac{ds}{\sqrt{X}\sqrt{Y}} = ac M_b \frac{ds}{\sqrt{X}\sqrt{Y}}, \\ dD &= v_4 R a \frac{ds}{\sqrt{X}\sqrt{Y}} = ad M_c \frac{ds}{\sqrt{X}\sqrt{Y}}; \end{aligned}$$

must hence follow  $dA, dB, dC, dD$  in either case in the form positive:  $a, b, c, d, \nu_1, \nu_2, \nu_3, \nu_4$  all to be 1 or 0, all to be either both positive or both negative respectively.

**24.** But from  $R = -2aa', b = 2aa', \dots$ , we have, in either case in the form positive

$$\begin{aligned} \frac{1}{(a' + y')\sqrt{X}\sqrt{Y}} &= \frac{1}{(a' + y')\sqrt{X}\sqrt{Y}} = \frac{1}{(a' + y')\sqrt{X}\sqrt{Y}} = P, \quad P_1, \quad P_2, \quad P_3, \\ dS &= dS_1 = 2aa'(b'a), \\ ds_1 &= 2acP_1, \\ dS_2 &= 2acP_2, \\ ds_3 &= 2acP_3, \end{aligned}$$

and consequently

$$\begin{aligned} dS &= \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2, \\ dS_1 &= \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2, \\ dS_2 &= \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2, \\ dS_3 &= \nu R (acP + acP_1) - \alpha c(a + acP_1)a + acP_2, \end{aligned}$$

which are the required formulae for the elements of arc.

**25.** The determination of the signs has been made by means of the particular figure: but it is easy to see that the points of terms could not for instance be all of them  $A, B, C$  at the interior corner of the curve, and at the exterior corner; or we could not have  $A, B, C, D$  all four positive,  $D$  at the exterior of the curve,  $A$  at the interior, and the like a similar examination would not be possible to eliminate  $dA, dB, dC, dD$ , and then obtains an equation such as

$$b'dA + 2a'bacP + a'cbacP_1 = 0;$$

(Page 237.) this would, by virtue of the relations between  $da, db, dc, dd$ , become

$$\frac{Yba}{(ax)^2 + Y a^2} dx + \frac{Xba}{(ax)^2 + Y a^2} dy + \frac{Xba}{(ax)^2 + Y a^2} dz + \frac{Yba}{(ax)^2 + Y a^2} dw = 0,$$

an equation between the variables in  $abc, \dots$  which cannot have a quadrinomial expression admitted by the variable form.

**26.** The differentials  $da, dy$  may be observed, the foregoing the of values  $\Pi dx$

$$= \frac{\xi dx}{(a+g)(b+h)} X + \frac{Y dy}{(a+g)(b+h)} Y,$$

where  $\sqrt{ab} = \sqrt{f + ga + bg}$ . It is also observed by the form  $\theta'$ , and

$$b'' = b + b'\xi\xi'\xi' - 2(b - h - \xi\xi - ga + a)a + aa'(c - c)\xi_1',$$

viz. in like manner be calculated in  $\theta$  as a function of  $a$  and  $a$ , an irrational function  $M\xi - \xi - X\sqrt{U}$ , and it would be the rest of such a function. Considering a diagonal conic and the corresponding circle of inversion, the corner of the curve, called  $\Pi$ , we have however, in elider:

$$\frac{d\Theta}{da} = \frac{f - a}{\Pi'\Pi a + fa + a},$$

and the formula thus becomes

$$2Vab\sqrt{ab}da = \frac{ab}{\Pi'}(\Pi a + faba)da,$$

consequently

$$\Theta a = -(\sqrt{ab} - \sqrt{abc})\frac{1}{2} \cdot a + ac'(\sqrt{abc})(ac'),$$

where  $k$  is the rather of the squares of the differentials; and these conditions and  $(a, b, c, d, e, f)$  are the are conditional respectively a determinate sign  $\nu$ .

**27.** The system of curves it

$$ds = aaacda + acdb + acbdc + acbda + acda,$$

$$ds_1 = aaacba + acbdc + acbbdc + acbda + acbda,$$

$$ds_2 = aaacdc + acbccda + acbda + acdb + acda,$$

$$ds_3 = aaacdc + acbdc + acbbdc + acbda + acbda,$$

where  $\nu_a = \pm 1$ ,  $\nu_b = \pm 1$ , ... for, etc., are conveniently used in writing it.

*Reference to Figure. Art. No. 28.*

**28.** I constructed a bicircular quartic consisting of an exterior and interior oval with the following numerical data:  $f, g, h, \rho, g_1, h_1, \rho_1, g_2, h_2, \rho_2, g_3, h_3, \rho_3, a, a_1, a_2, a_3, \dots$  etc. very convenient values, inasmuch as

(Page 238.) the exterior oval cuts (and crosses each, large; the annexed figure shows 0, 1, 2, 3, the centres of the circles of inversion, the interior oval, and a portion of the exterior oval, the centres 1, 2 outside the exterior oval; I add further the values

$$\sqrt{f + h_2}a = 95, \quad \sqrt{-(g + h_2)} = 244, \quad a_2 = 1019, \quad R_2 = -96,$$

$$\sqrt{f + h_1}a = 746, \quad \sqrt{g + h} = -241, \quad a = 873, \quad R_1 = 294,$$

$$\sqrt{f + h_1}a = 715, \quad \sqrt{g + h} = 245, \quad a = 873, \quad R_2 = -96,$$

$$\sqrt{f + h}a = 1934, \quad \sqrt{g + h} = -149, \quad a = 919, \quad R_3 = 23.$$

We thus see how there exists a series of quadrilaterals  $ABCD$ , where  $A, B$  are situate on the interior oval,  $C, D$  on the exterior oval. Considering the sides as

(Page 239.) drawn in the senses  $A$  to  $B$ ,  $B$  to  $C$ ,  $C$  to  $D$ ,  $D$  to  $A$ : and representing the inclinations, measured from the positive direction of axis of  $\xi$ , of these tangents by  $\nu_1, \nu_2, \nu_3, \nu_4$  respectively; then, in passing to the consecutive quadrilateral,  $a, b, c, d$  change, suppose, into  $a + da, b + db, c + dc, d + dd$ ; and the points  $A, B, C, D$  into  $A', B', C', D'$ , such as  $a + da, b + db, c + dc, d + dd$ ; and we have

$$dA = Rda, \quad dB = Rdb,$$

$$dC = R_2 dc, \quad dD = R_3 dd,$$

and thence

$$dA = bda - bdA, \quad d_1 da + db_1 da,$$

$$dB = bda - b dB,$$

$$dC = bdc - b dC,$$

$$dD = bdd - b dD,$$

which are the correct signs in regard to the particular figure.

*Reduction of  $\int \frac{1 da}{(\xi^2 + y^2)\sqrt{U}}$  to Elliptic Integrals. Art. No. 28.*

**29.** The expression in question is

$$\int dn \frac{\sqrt{(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})b - BY - \alpha)^2}}{(c+g)\xi^2 + (h+g)y^2 + \alpha + a^2 \sin^2 \xi} (\cos T + b)^2 \cdot c^2 \sin^2 \xi;$$

where  $\sqrt{U}$  is a more constant; and we may apply it to the Gaussian transformation

$$\cos a = \frac{a + b \cos T + a^2 \sin T}{c + a \cos T + b \sin T},$$

$$\sin a = \frac{c + b \cos T + b^2 \sin T}{c + a \cos T + b \sin T},$$

where the coefficients  $a, b, c, a', b', c', a'', b'', c''$  are such that identically

$$\cos^2 a + \sin^2 a = 1 = (a + b \cos T + a \sin T)^2 (\cos T + b)^2 (\cos T - 1);$$

and also

$$(\cos a + \sqrt{c-b})^2 + (\sin a + \sqrt{c-b})^2 = a^2 - b^2.$$

(Page 240.) Hence

$$(\cos \theta + \sqrt{c-b})^2 + (\sin \theta + \sqrt{c-b})^2 = a^2 - b^2,$$

also

$$\left[ \frac{f-a}{\cos \theta + \sqrt{c-b}} \right] = \frac{a-b}{f-h} (\cos \theta + \sqrt{c-b}) a^2;$$

$\Pi' = \sqrt{aa' - \sin^2 \theta (c-b)}$ , so that the integral becomes

$$\frac{-a}{\sqrt{(c+g)(c-b)c'}} dT,$$

which is, as it should be, a function of an elliptic integral.

*Cambridge, 7th December, 1877.*

## 4 [668] On Compound Combinations

[From the Proceedings of the Lit. Phil. Soc. Manchester, *t. XVI* (1877), pp. 113, 114; Memoirs, *ib.*, *SER. III.*, *t. VI.* (1879), pp. 99, 100.]

Prof. Clifford's paper, "On the Types of Compound Statement involving Four Classes," [volume of *Proceedings* quoted, pp. 88–110; *Mathematical Papers*, pp. 1–13], relates mathematically to a question of compound combinations; and it is worth while to consider its connexion with another question of compound combinations, the application of which is a very different one.

Starting with four symbols  $A, B, C, D$ , we have sixteen combinations of the form  $ABCD$ , ( $1 \cdot 4 \cdot 6 \cdot 4 \cdot 1 = 16$ ), as before). But in Prof. Clifford's question 1 means  $\Lambda WCD$ ,  $\Lambda$  means  $\Lambda HCD$ ; viz. each of the symbols means an aggregate of four assertions; and the 16 symbols are then all of them combinations of such assertions; and the question to be considered is to determine the number of types of the binary, ternary, ... combinations of the sixteen combinations; for, according as these are

$$\text{No. of types} = \frac{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}{1, 4, 6, 13, 19, 27, 37, 47, 51, 50, 47, 37, 27, 19, 6, 4}$$

together. In the first mentioned point of view the like question arises, in regard to the new belonging to the five different forms of compound combinations; and as it is worth while to consider its connexion with another question of compound combinations, the formal so the following development.

$$\text{No. of types} = \frac{1, 2, 3, 4, \dots, 24, 25, 26, 27}{1, 2, 5, 11, \dots, 11, 5, 2, 1}.$$

(Page 244.) as is very easily verified; but if the number of letters  $A, B, \dots$  be greater (say this = 6), or, instead of letters, writing the numbers 1, 2, 3, 4, 5, 6, 7, 8, then the question is that of the number of the types of combination of the 20 thirds 13, 14, ..., 78 taken 1, 2, ..., 27 together, a question presenting itself in geometry in regard to the bitangents of a quartic curve (see Salmon's *Higher Plane Curves*, Ed. 2 (1873), pp. 222, et seq.); the numbers, so far as they have been obtained, are

$$\text{No. of types} = \frac{1, 2, 3, 4, \dots, 24, 25, 26, 27}{1, 2, 5, 11, \dots, 11, 5, 2, 1}.$$

It might be interesting to complete the series, and, more generally, to determine the number of the types of combination of the  $\frac{1}{2}n(n-1)$  duads of  $n$  letters.

## 5 [669] On a Problem of Arrangements

[From the Proceedings of the Royal Society of Edinburgh, *t. IX.* (1878), pp. 338–342.]

It is a well-known problem to find for  $n$  letters the number of the arrangements in which no letter occupies its original place; and the solution of it is given by the following general theorem: viz., the number of the arrangements which satisfy  $ay$ ,  $r$  conditions is

$$\begin{aligned} & (1 - \Sigma(1) + \Sigma(2) - \dots (-1)^r \Sigma(r))(1 \cdot 2 \dots n), \\ & = 1 - \Sigma(1) + \Sigma(2) \dots \dots \pm (12 \dots r); \end{aligned}$$

where 1 denotes the whole number of arrangements; (1) the number of those which fail in regard to the first condition; (2) the number which fail in regard to the second condition; (12)

the number which fail in regard to the first and the second conditions; and so on, the second condition; the  $\Sigma$  refers to the combinations of the  $n$  conditions which can occur there; and lastly  $(1 \cdot 2 \cdots r)$  means  $1 \cdot 2 \cdot 3 \cdots r$ .

Then, in the special problem, the first condition is that the letter in the first place shall not be  $\alpha$ ; the second, the letter in the second place shall not be  $b$ ; and so on. Hence

$$\begin{aligned}\Sigma_0 &= \Pi n - \frac{n}{1} \Pi(n-1) + \frac{n \cdot n - 1}{1 \cdot 2} \Pi(n-2) + \dots \pm \frac{n \cdot n - 1 \cdot 1}{1 \cdot 2 \cdots n} \Pi 0, \\ &= 1 \cdot 2 \cdots n \left[ 1 - \frac{1}{1} + \frac{1}{1 \cdot 2} - \frac{1}{1 \cdot 2 \cdot 3} + \cdots \pm \frac{1}{1 \cdot 2 \cdots n} \right],\end{aligned}$$

giving for the several cases

$$n = 1, 2, 3, 4, 5, 6, 7, \dots,$$

$$N_0 = 1, 2, 9, 44, 265, 1854, \dots$$

I proceed to consider the following problem, suggested to me by Professor Tait, in connexion with his theory of knots: to find the number of the arrangements of  $n$  letters  $abc \dots k$ , in such manner as to fix the distances of the elements of all those non-pairs  $aa, bb, cc, \dots$  shall be the value  $k$ , where  $k$  is the value of inversion or expression. This contains the foregoing problem, as the present one is that in which  $k = 1$ .

(Page 246.) Numbering the conditions  $1, 2, 3, \dots, n$ , according to the places to which they relate, a single condition is called [1]; two conditions are called [2] or [1 1], according as the numbers are consecutive or non-consecutive; three conditions are called [3], [2 1], [1 1 1], according as the numbers are consecutive and one not consecutive, or all non-consecutive; and so on: the numbers which refer to the conditions being always written in their natural order, and it being understood that they follow each other cyclically, so that  $n = 6$ , the set 125 of conditions is [3], as consisting of 3 consecutive conditions; and similarly 1346 in [2 2].

Consider a single condition [1], say this is 1; the arrangements which fail in regard to this condition are those which contain  $aa$  in the first place or  $b$ ; whichever there are thus  $2(n-1)$  failing arrangements in any form whatever; and thus

$$2\Pi(n-1) = (n-1)\Pi(n-1).$$

Next for two conditions; these may be [2], say the conditions are 1 and 2; or else [1 1], say they are 1 and 3. In the former case, the arrangements which fail are those which contain  $ab$ , or  $ba$ , in the first and second places; there are thus, in the first place  $a$ , or  $b$ , and in the third place  $c$  or  $d$ ; the letters in those places are  $a, c, a, d, b, c, b, d$ , in the third place  $c$  or  $d$ ; the letters in those places are  $a, c, a, d, b, c, b, d$ ; and so on, the number of failing arrangements is thus  $= 2 \cdot \Pi(n-2)$ . And so, in general, when the conditions are  $[\alpha, \beta, \gamma, \dots]$ , the number of failing arrangements is

$$= (\alpha+1)(\beta+1)(\gamma+1) \dots \Pi(n-\alpha-\beta-\gamma-\dots).$$

But for  $[a]$ , that is, for the entire system of the  $n$  conditions, the number of failing arrangements is (not as by the rule it should be  $= n+1 \cdot bn$ )  $= 2$ . The number of failing arrangements in any form whatever; for the  $n$  conditions  $abc, abd, \dots$  in the conditions [2, 2], [1, 2], etc., would give  $\xi a$ .

Changing now the notation so that [1], [2], [1 1], [3], &c., shall denote the *number* of the conditions [1], [2], [1 1], [3], &c., respectively, it is easy to see the form of the general result. If, for greater clearness, we write  $n = 6$ , we have

$$1, \quad -\Sigma(1), \quad +\Sigma(12), \quad -\Sigma(123),$$

$$N_0 = 720 - \{[1] = 6\} 2 | 120 + \{[2] = 6\} 3 | 24 - \{[3] = 6\} 4 | 6$$

$$\begin{aligned}
 & +\{[1\ 1] = 9\} 2\ 2 + \{[2\ 1] = 12\} 3\ 2 \cdot 2, \\
 & +\Sigma(1234), \quad -\Sigma(12345), \quad +\Sigma(123456) \\
 & +\{[4] = 6\} 5\ 2 - \{[5] = 6\} 6\ 1 + \{[6] = 1\} 2 \\
 & +\{[3\ 1] = 6\} 4\ 2 + \{[3\ 2] = 6\} 4\ 3 \\
 & +\{[2\ 2] = 6\} 3 \cdot 3 + \{[1\ 1] = 6\} 3 \cdot 3.
 \end{aligned}$$

(Page 247.) or, reducing into numbers, this is

$$N_0 = 720 - 1296 + 1296 - 504 + 90 - 36 + 2 = 80.$$

The formula for the next succeeding case,  $n = 7$ , may be obtained in like manner

$$N_0 = 5040 - 10080 + 7080 - 3040 + 1560 - 4960 + 2 = 579.$$

Then for the preceding cases,  $n = 4, 5, 6$ , respectively we have

$$N_0 = 4, \quad 24 - 40 + 16 = 0, \quad 120 - 240 + 184 - 60 + 6 + 12 - 2 = 20,$$

$$N_0 = 1, 2, 0, 3, 80, 579, \dots$$

We have here  $r = 0, 4, 6, \dots$  which is,  $r = 0, 5, 7, \dots$  respectively\*.

The formula for general  $n$ , considering the second sums above-mentioned, becomes

$$N_0 = \Pi n - 720 + 1296 - 720 + (n - 1)(n - 2)\Pi(n - 3) - 36 + \dots = 0,$$

The formula for the preceding cases of an arrangement  $a, b, \dots$  may here be put in the form

$$N_0 = 24 - 40 + 16 = 0, \quad \dots$$

in respect of the result number of the symbols  $a, b, c, \dots$  are of  $n$  in cyclical order beginning with  $a, \dots *a$ , the letter in the last place not  $b$  or  $c, \dots$

For the actual variation of so  $r$  is  $= 0$ , and the foregoing formulae become

$$n = 4, 5, 6, 7, \dots$$

which the analysis a sum of squares.

(Page 248.) all the substitutions which serve every letter. Thus when  $n = 5$ , we obtain the 44 substitutions for these 4 letters  $b, c, d, e$ ; these are

(abcde, bdec, cebd, 24 symbols obtained by permuting in every way the four letters  $b, c, d, e$  at one time of  $n$

$$\begin{aligned}
 & (ab)(cd), (ac)(bd), (ad)(bc), \\
 & (abcd), (abdc), \\
 & (acbd), (acdb), \\
 & (adbc), (adcb).
 \end{aligned}$$

Here in the first column, performing on the symbol  $(abcd)$  the substitutions  $(abcd)$ , we obtain  $ab$ ;  $(cdcb)$ ,  $ab$ ; *i.e.* to perform a substitution which is again opening with  $a$ ; we must in the

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0. In explanation of the notation of substitutions, denote that  $(abcd)$  means that all the changes may be made in any way 1, 2, 3,  $\dots$ , but only at the same time; for instance,  $a$  is to be changed for  $b$ , this  $b$  is to be changed for  $c$ , this  $c$  is to be changed for  $d$ , this  $d$  is to be changed for  $a$ ,  $\dots$ . The set  $(1\ abcd)$  thus all of them denote the same substitution; the  $(a\ 1\ d)$  is to be changed for the cycle  $(12)$ .

manner end, the substitutions are nothing else than the substitutions which form the column system in the same way. The 13 substitutions are thus of these 4 different types, or any of the symbols which they belong, viz.,

$$\begin{aligned} &adcde, \quad bdcae, \quad cdeab, \quad ddeba, \\ &ddeab, \quad ebcad, \end{aligned}$$

are of 5 different types. The question is to determine for any value of  $n$ , the number of the different types, in this case there appear, in this case, but it is not at present yet to find it.

## 6 [670] [Note on Mr. Muir's Solution of a "Problem of Arrangement."]

[From the Proceedings of the Royal Society of Edinburgh, t. IX. (1878), pp. 388-391.]

The investigation may be carried further: writing for shortness  $w_1, u, v$ , &c. in place of  $\Psi(1)$ ,  $\Psi(2)$ , &c., the equations are

$$\begin{aligned} u_0 &= 1, \\ u_1 &= 0u_0, \\ u_2 &= 0u_1 + 1, \\ u_3 &= 0u_2 + 11u_1, \\ u_4 &= 0u_3 + 6u_2, \\ u_5 &= 0u_4 + 10u_3 + 12u_2 + 1. \end{aligned}$$

Hence assuming

$$u = u_0 + u_1x + u_2x^2 + u_3x^3 + \dots$$

we have

$$\begin{aligned} u &= \frac{1}{1-x^2} + u_0(2x + 6x^2 + 12x^3 + u_2x^4 + \dots) \\ &\quad + u_1(2x + 6x^2 + 10x^3 + 12x^4 + \dots) \\ &\quad + u_2(4x + 10x^2 + 18x^3 + 28x^4 + \dots) \\ &\quad + u_3(2x^2 + 6x^3 + 12x^4 + 20x^5 + \dots); \end{aligned}$$

so that, forming the equation

$$\begin{aligned} u' \frac{x^2}{(1-x)^2} &= u_0(x^2 + 2x^3 + 3x^4 + \dots) \\ &\quad + u_1(2x^2 + 6x^3 + 10x^4 + 12x^5 + \dots) \\ &\quad + u_2(4x^2 + 10x^3 + 18x^4 + 28x^5 + \dots) \\ &\quad + u_3(2x^3 + 6x^4 + 12x^5 + 20x^6 + \dots), \end{aligned}$$

(Page 250.) where  $u'$  denotes  $\frac{du}{dx}$ , we have

$$u - u \frac{x^2}{1-x^2} = \frac{1}{1-x^2} Q,$$

where

$$Q = u_0 \left( -\frac{x^2}{1-x} + u_1 2x + u_2 4x + u_3 2x^2 + \dots \right)$$



$$\begin{aligned}
 &= \frac{1}{1-x^2} \left[ 1 + \frac{2x}{1-x} + \frac{4x}{1-x} + \frac{2x^2}{1-x^2} \right] Q(1-ax) \\
 &= \frac{2x}{(1-x)^2} + Q \left[ 1 - \frac{2x}{1-x^2} \right] Q(1-ax).
 \end{aligned}$$

so that the equation is simplified: write

$$u = \frac{1}{1-x} Q - \left( -\frac{x^2}{1-x^2} + \frac{x^2}{1-x^2} \right) Q,$$

and the equation becomes

$$u' \frac{x^2}{(1-x)^2} Q = \frac{1}{1-x^2} Q + \frac{2x}{1-x^2} Q = \frac{1-x}{1+x},$$

then

$$\left( 1 - \frac{2}{x^2} \right) Q = Q \left( 1 - \frac{2}{x^2} \right),$$

or

$$-\frac{(1-x)x^2-x^2(1-x)^2}{x^2} = Q \left( 1 - \frac{2}{x^2} \right) = 1 - \frac{x}{1+x},$$

or finally

$$Q \left( 1 - \frac{x}{1+x} \right) = Q \frac{x^2}{(1+x)^2},$$

giving

$$Q = e^{-x} \left[ \int \frac{x^2}{(1+x)^2} e^x dx \right],$$

and thence

$$u = \frac{1-x}{x^2} + Q \left( \frac{1}{x^2} + \frac{1}{(1+x)^2} \right) \left[ \int \frac{x^2 e^x dx}{(1+x)^2} \right],$$

which is the value of the generating function,  $u = u_0 + u_1x + u_2x^2 + u_3x^3 + \dots$ .

## The Collected Mathematical Papers of Arthur Cayley

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Arthur Cayley

Cambridge, 1896

**667 (continued): ON THE BICIRCULAR QUARTIC.**

the generating circle. If we consider the two points on the generating circle, and write  $dS'$  for the element at the other point, then we have

$$dS' = \frac{2\rho}{\sqrt{n}} d\omega,$$

which is Casey's formula  $dS' - dS = 2\rho d\phi$  (273).

17. The foregoing forms of  $dX$ ,  $dY$  are those which give most directly the required value of  $dS$ ; but I had previously obtained them in a different form. Writing

$$\Lambda = \beta x - \alpha y + (f - g)xy,$$

then

$$x\Lambda = \beta x^2 - \alpha xy + (f - g)x^2y,$$

or since  $(f - g)x^2 = 1 - (g + \theta)r^2$ , this is

$$x\Lambda = \beta x^2 - \alpha xy + \{1 - (g + \theta)(x^2 + y^2)\}y = y\{1 - \alpha x - \beta y\} + (x^2 + y^2)\{\beta - (g + \theta)y\},$$

that is,  $= (x^2 + y^2)\{yR' + \beta - (g + \theta)y\} + y\sqrt{\Omega}$ ,

and similarly

$$\begin{aligned} x\Lambda - y\sqrt{\Omega} &= (x^2 + y^2)\{yR' + \beta - (g + \theta)y\}, \\ -y\Lambda - x\sqrt{\Omega} &= (x^2 + y^2)\{xR' + \alpha - (f + \theta)x\}. \end{aligned}$$

We have therefore

$$\begin{aligned} \frac{dX}{d\omega} &= \frac{R' d\omega}{(x^2 + y^2)\sqrt{\Omega}} (x\Lambda - y\sqrt{\Omega}), \\ \frac{dY}{d\omega} &= \frac{R' d\omega}{(x^2 + y^2)\sqrt{\Omega}} (y\Lambda + x\sqrt{\Omega}), \end{aligned}$$

and thence a value of  $dS$  which, compared with the former value, gives

$$\Omega + \Lambda^2 = (x^2 + y^2)h^2,$$

an equation which may be verified directly.

*Formulae for the Inscribed Quadrilateral. Art. Nos. 18 to 22.*

18. We consider on the curve four points,  $A, B, C, D$ , forming a quadrilateral,  $ABCD$ . The coordinates are taken to be  $(X, Y)$ ,  $(X_1, Y_1)$ ,  $(X_2, Y_2)$ ,  $(X_3, Y_3)$  respectively. It is assumed that  $(A, B)$ ,  $(B, C)$ ,  $(C, D)$ ,  $(D, A)$  belong to the generations 1, 2, 3, 0, and depend on the parameters  $(x_1, y_1)$ ,  $(x_2, y_2)$ ,  $(x_3, y_3)$ ,  $(x, y)$  respectively.

We write

$$\begin{aligned}\Omega &= (1 - \alpha x - \beta y)^2 - \gamma^2(x^2 + y^2), \\ \Omega_1 &= (1 - \alpha_1 x_1 - \beta_1 y_1)^2 - \gamma_1^2(x_1^2 + y_1^2), \\ \Omega_2 &= (1 - \alpha_2 x_2 - \beta_2 y_2)^2 - \gamma_2^2(x_2^2 + y_2^2), \\ \Omega_3 &= (1 - \alpha_3 x_3 - \beta_3 y_3)^2 - \gamma_3^2(x_3^2 + y_3^2),\end{aligned}$$

and then,  $\sqrt{\Omega}$  denoting as above a determinate value, positive or negative as the case may be, of the radical, and similarly  $\sqrt{\Omega_1}, \sqrt{\Omega_2}, \sqrt{\Omega_3}$  denoting determinate values of these radicals respectively, each radical having its own sign at pleasure, we further write

$$\begin{aligned} (x^2 + y^2) R' &= 1 - \alpha x - \beta y + \sqrt{\Omega}, & (x_1^2 + y_1^2) R'_1 &= 1 - \alpha_1 x_1 - \beta_1 y_1 + \sqrt{\Omega_1}, \\ (x_1^2 + y_1^2) R_1 &= 1 - \alpha_1 x_1 - \beta_1 y_1 - \sqrt{\Omega_1}, & (x_2^2 + y_2^2) R'_2 &= 1 - \alpha_2 x_2 - \beta_2 y_2 + \sqrt{\Omega_2}, \\ (x_2^2 + y_2^2) R_2 &= 1 - \alpha_2 x_2 - \beta_2 y_2 - \sqrt{\Omega_2}, & (x_3^2 + y_3^2) R'_3 &= 1 - \alpha_3 x_3 - \beta_3 y_3 + \sqrt{\Omega_3}, \\ (x_3^2 + y_3^2) R_3 &= 1 - \alpha_3 x_3 - \beta_3 y_3 - \sqrt{\Omega_3}, & (x^2 + y^2) R &= 1 - \alpha x - \beta y - \sqrt{\Omega}; \end{aligned}$$

and this being so, we must have

$$\begin{aligned} X &= \alpha + R'x = \alpha_1 + R_1x_1, & Y &= \beta + R'y = \beta_1 + R_1y_1, & R' &= \frac{1}{2}(X^2 + Y^2 - k), & R_1 &= \frac{1}{2}(X^2 + Y^2 - k_1), \\ X_1 &= \alpha_1 + R'_1x_1 = \alpha_2 + R_2x_2, & Y_1 &= \beta_1 + R'_1y_1 = \beta_2 + R_2y_2, & R'_1 &= \frac{1}{2}(X_1^2 + Y_1^2 - k_1), & R_2 &= \frac{1}{2}(X_1^2 + Y_1^2 - k_2), \\ X_2 &= \alpha_2 + R'_2x_2 = \alpha_3 + R_3x_3, & Y_2 &= \beta_2 + R'_2y_2 = \beta_3 + R_3y_3, & R'_2 &= \frac{1}{2}(X_2^2 + Y_2^2 - k_2), & R_3 &= \frac{1}{2}(X_2^2 + Y_2^2 - k_3), \\ X_3 &= \alpha_3 + R'_3x_3 = \alpha + Rx, & Y_3 &= \beta_3 + R'_3y_3 = \beta + Ry, & R'_3 &= \frac{1}{2}(X_3^2 + Y_3^2 - k_3), & R &= \frac{1}{2}(X_3^2 + Y_3^2 - k). \end{aligned}$$

and then from the values of  $X, Y, R', R$ , we have

$$\alpha - \alpha_1 + R'x - R_1x_1 = 0,$$

$$\beta - \beta_1 + R'y - R_1y_1 = 0,$$

giving

$$(\beta - \beta_1)(x - x_1) - (\alpha - \alpha_1)(y - y_1) + (\theta - \theta_1)(xy_1 - x_1y) = 0;$$

and similarly

$$(\beta_1 - \beta_2)(x_1 - x_2) - (\alpha_1 - \alpha_2)(y_1 - y_2) + (\theta_1 - \theta_2)(x_1y_2 - x_2y_1) = 0,$$

$$(\beta_2 - \beta_3)(x_2 - x_3) - (\alpha_2 - \alpha_3)(y_2 - y_3) + (\theta_2 - \theta_3)(x_2y_3 - x_3y_2) = 0,$$

$$(\beta_3 - \beta)(x_3 - x) - (\alpha_3 - \alpha)(y_3 - y) + (\theta_3 - \theta)(x_3y - xy_3) = 0,$$

which are the relations connecting the parameters  $(x, y), (x_1, y_1), (x_2, y_2), (x_3, y_3)$  of the quadrilateral.

19. We have thus apparently four equations for the determination of four quantities, or the number of quadrilaterals would be finite; but if from the first and second equations we eliminate  $(x_1, y_1)$ , and if from the third and fourth equations we eliminate  $(x_3, y_3)$ , we find in each case the same relation between  $(x, y), (x_2, y_2)$ , viz. this is found to be

$$\Omega\Omega_2 = (1 - \alpha x_2 - \beta y_2)^2 (1 - \alpha_2 x - \beta_2 y)^2;$$

and we have thus the singly infinite series of quadrilaterals. We have, of course, between  $(x_1, y_1), (x_3, y_3)$  the like relation,

$$\Omega_1\Omega_3 = (1 - \alpha_1 x_3 - \beta_1 y_3)^2 (1 - \alpha_3 x_1 - \beta_3 y_1)^2.$$

20. The relation between  $(x, y), (x_1, y_1)$  may be expressed also in the two forms

$$1 - \alpha(x + x_1) - \beta(y + y_1) + (f + \theta_1)xx_1 + (g + \theta_1)yy_1 + \frac{x^2 + y^2}{xy_1 - x_1y} \{(\alpha - \alpha_1)y_1 - (\beta - \beta_1)x_1\} = 0,$$

$$1 - \alpha_1(x + x_1) - \beta_1(y + y_1) + (f + \theta)xx_1 + (g + \theta)yy_1 + \frac{x_1^2 + y_1^2}{xy_1 - x_1y} \{(\alpha_1 - \alpha)y - (\beta_1 - \beta)x\} = 0.$$

In fact, the first of these equations is

$$\begin{aligned} &\{1 + (f + \theta_1)xx_1 + (g + \theta_1)yy_1\}(xy_1 - x_1y) - \{\alpha(x + x_1) + \beta(y + y_1)\}(xy_1 - x_1y) \\ &\quad + \{(\alpha - \alpha_1)y_1 - (\beta - \beta_1)x_1\}(x^2 + y^2) = 0, \end{aligned}$$

which, by virtue of the original form of relation, is

$$\begin{aligned} &\{1 + (f + \theta_1)xx_1 + (g + \theta_1)yy_1\}(xy_1 - x_1y) \\ &\quad - \{\alpha(x + x_1) + \beta(y + y_1)\}(xy_1 - x_1y) + \{(\alpha - \alpha_1)y_1 - (\beta - \beta_1)x_1\}(x^2 + y^2) = 0; \end{aligned}$$

or, in the first term, writing

$$f + \theta_1 = \frac{f - \theta}{f - \theta_1}(f + \theta'_1), \quad g + \theta_1 = \frac{g - \theta}{g - \theta_1}(g + \theta'_1),$$

and in the third term using the like expressions for  $\alpha - \alpha_1, \beta - \beta_1$ , this is verified.

In this equation the coefficients of  $\alpha$  and of  $\beta$  are separately  $= 0$ : in fact, the coefficient of  $\beta$  is

$$-y(xy_1 - x_1y) - x_1(x^2 + y^2) = -\{x(x_1^2 + y_1^2)\} + x_1(x^2 + y^2) = 0,$$

and similarly the coefficient of  $\alpha$  is  $= 0$ .

And in like manner the second equation may be verified.

21. The two equations are:

$$1 - \alpha x - \beta y - (x^2 + y^2)R' = \alpha x_1 + \beta y_1 - (f + \theta_1)xx_1 - (g + \theta_1)yy_1,$$

$$1 - \alpha_1 x_1 - \beta_1 y_1 - (x_1^2 + y_1^2)R_1 = \alpha_1 x + \beta_1 y - (f + \theta)xx_1 - (g + \theta)yy_1;$$

or, substituting for  $R'$  and  $R_1$  their values, these are

$$-\sqrt{\Omega} = -\alpha x_1 - \beta y_1 + (f + \theta_1)xx_1 + (g + \theta_1)yy_1, \quad \sqrt{\Omega_1} = -\alpha_1 x - \beta_1 y + (f + \theta)xx_1 + (g + \theta)yy_1;$$

and similarly

$$\begin{aligned} -\sqrt{\Omega_1} &= -\alpha_1 x_2 - \beta_1 y_2 + (f + \theta_2) x_1 x_2 + (g + \theta_2) y_1 y_2, & \sqrt{\Omega_2} &= -\alpha_2 x_1 - \beta_2 y_1 + (f + \theta_1) x_1 x_2 + (g + \theta_1) y_1 y_2, \\ -\sqrt{\Omega_2} &= -\alpha_2 x_3 - \beta_2 y_3 + (f + \theta_3) x_2 x_3 + (g + \theta_3) y_2 y_3, & \sqrt{\Omega_3} &= -\alpha_3 x_2 - \beta_3 y_2 + (f + \theta_2) x_2 x_3 + (g + \theta_2) y_2 y_3, \\ -\sqrt{\Omega_3} &= -\alpha_3 x_1 - \beta_3 y_1 + (f + \theta_1) x_3 x_1 + (g + \theta_1) y_3 y_1, & -\sqrt{\Omega} &= -\alpha x_3 - \beta y_3 + (f + \theta_3) x_3 x + (g + \theta_3) y_3 y. \end{aligned}$$

Differentiating the equation

$$(\beta - \beta_1)(x - x_1) - (\alpha - \alpha_1)(y - y_1) + (\theta - \theta_1)(xy_1 - x_1y) = 0,$$

we have

$$\begin{aligned} &[(\beta - \beta_1) + (\theta - \theta_1) y_1] dx - [(\alpha - \alpha_1) + (\theta - \theta_1) x_1] dy \\ &- [(\beta - \beta_1) + (\theta - \theta_1) y] dx_1 + [(\alpha - \alpha_1) + (\theta - \theta_1) x] dy_1 = 0; \end{aligned}$$

and writing herein

$$\begin{aligned} dx &= -\frac{(\beta + \theta y)}{\sqrt{\theta}} d\omega, & dy &= \frac{(\alpha + \theta x)}{\sqrt{\theta}} d\omega, \\ dx_1 &= -\frac{(\beta_1 + \theta_1 y_1)}{\sqrt{\theta_1}} d\omega_1, & dy_1 &= \frac{(\alpha_1 + \theta_1 x_1)}{\sqrt{\theta_1}} d\omega_1, \end{aligned}$$

we find

$$\begin{aligned} &-\frac{d\omega}{\sqrt{\theta}} \{(g + \theta)(\beta - \beta_1) y + (f + \theta)(\alpha - \alpha_1) x + (\theta - \theta_1)[(f + \theta) x x_1 + (g + \theta) y y_1]\} \\ &+ \frac{d\omega_1}{\sqrt{\theta_1}} \{(g + \theta_1)(\beta - \beta_1) y_1 + (f + \theta_1)(\alpha - \alpha_1) x_1 + (\theta - \theta_1)[(f + \theta_1) x x_1 + (g + \theta_1) y y_1]\} = 0; \end{aligned}$$

viz., dividing by  $\theta - \theta_1$ , this becomes

$$-\sqrt{\Omega_1} \frac{d\omega}{\sqrt{\theta}} - \sqrt{\Omega} \frac{d\omega_1}{\sqrt{\theta_1}} = 0, \quad \text{that is,} \quad \frac{d\omega}{\sqrt{\theta}\sqrt{\Omega}} + \frac{d\omega_1}{\sqrt{\theta_1}\sqrt{\Omega_1}} = 0;$$

or, completing the system, we have

$$\frac{d\omega}{\sqrt{\theta}\sqrt{\Omega}} = -\frac{d\omega_1}{\sqrt{\theta_1}\sqrt{\Omega_1}} = \frac{d\omega_2}{\sqrt{\theta_2}\sqrt{\Omega_2}} = -\frac{d\omega_3}{\sqrt{\theta_3}\sqrt{\Omega_3}},$$

which are the differential relations between the parameters  $\omega, \omega_1, \omega_2, \omega_3$ , or  $(x, y), (x_1, y_1), (x_2, y_2), (x_3, y_3)$ .

22. From the equations  $X = \alpha + R'x, Y = \beta + R'y$ , we found

$$\begin{aligned} dX &= \frac{R'd\omega}{(x^2 + y^2)\sqrt{\Omega}} \{Y - (g + \theta) y\}, \\ dY &= \frac{R'd\omega}{(x^2 + y^2)\sqrt{\Omega}} \{X - (f + \theta) x\}; \end{aligned}$$

the new values,  $X = \alpha_1 + R_1 x_1$  and  $Y = \beta_1 + R_1 y_1$ , give in like manner

$$\begin{aligned} dX &= \frac{R_1 d\omega_1}{(x_1^2 + y_1^2)\sqrt{\Omega_1}} \{X - (f + \theta_1) x_1\}, \\ dY &= \frac{R_1 d\omega_1}{(x_1^2 + y_1^2)\sqrt{\Omega_1}} \{Y - (g + \theta_1) y_1\}; \end{aligned}$$

in virtue of the relation just found between  $d\omega$  and  $d\omega_1$ , these two sets of values will agree together if only

$$\begin{aligned} R'\{Y - (g + \theta)y\} &= R_1\{Y - (g + \theta_1)y_1\}, \\ R'\{X - (f + \theta)x\} &= R_1\{X - (f + \theta_1)x_1\}. \end{aligned}$$

These are easily verified: the first is

$$R'Y - (g + \theta)(Y - \beta) = (R' - \theta + \theta_1)Y - (g + \theta_1)(Y - \beta_1),$$

viz. this is  $(g + \theta)\beta - (g + \theta_1)\beta_1 = 0$ , which is right; and similarly the second equation gives  $(f + \theta)\alpha - (f + \theta_1)\alpha_1 = 0$ , which is right.

From the first values of  $dX, dY$ , we have, as above,

$$dS = \frac{eR'S d\omega}{\sqrt{\Omega}\sqrt{\theta}},$$

and the second values give in like manner

$$dS = \frac{e_1R_1S_1 d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}},$$

where  $e, e_1$  is  $\pm 1$ . It will be observed that we have in effect, by means of the relation  $(\beta - \beta_1)(x - x_1) - (\alpha - \alpha_1)(y - y_1) + (\theta - \theta_1)(xy_1 - x_1y) = 0$ , proved the identity of the two values of  $dS$ .

Considering the quadrilateral  $ABCD$ , and giving it an infinitesimal variation, so as to change it into  $A'B'C'D'$ , then  $dS$  is the element of arc  $AA'$ ; and writing in like manner  $dS_1, dS_2, dS_3$  for the elements of arc  $BB', CC', DD'$ , we have, of course, a like pair of values for each of the elements  $dS_1, dS_2, dS_3$ .

*Formulæ for the elements of Arc  $dS, dS_1, dS_2, dS_3$ . Art. Nos. 23 to 27.*

23. The formulæ are

$$\begin{aligned} dS &= e'RS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}} = e_1R_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}}, \\ dS_1 &= e'_1R'_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}} = e_2R_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}}, \\ dS_2 &= e'_2R'_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}} = e_3R_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}}, \\ dS_3 &= e'_3R'_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}} = eRS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}}, \end{aligned}$$

where the  $e$ 's each denote  $\pm 1$ . Supposing as above that  $\gamma^2$  is negative, but that  $\gamma'_1, \gamma'_2, \gamma'_3$  are positive, then  $R, R'$  have opposite signs; but  $R'_1, R_1$  have the same sign,

as have also  $R'_2$  and  $R_2$ , and  $R'_3$  and  $R_3$ . We may take  $S, S_1, S_2$ , and  $S_3$  as each of them positive: the signs of

$$d\omega, d\omega_1, d\omega_2, d\omega_3 \text{ are } +, -, +, -, \text{ or } -, +, -, +;$$

hence to make  $dS, dS_1, dS_2, dS_3$  all positive,

$$\frac{e}{\sqrt{\theta}\sqrt{\Omega}}, \quad \frac{e'_1}{\sqrt{\Omega_1}\sqrt{\theta_1}}, \quad \frac{e_2}{\sqrt{\Omega_2}\sqrt{\theta_2}}, \quad \frac{e'_3}{\sqrt{\Omega_3}\sqrt{\theta_3}}$$

must have either the signs of

$$R', -R'_1, R'_2, -R'_3; -R, R_1, -R_2, R_3,$$

or else the reverse signs: hence in either case  $e' = -e_1 = e'_2 = -e_3$ ;  $e_1 = -e_2 = e'_3 = -e$ ; or the equations are

$$\begin{aligned} dS &= -eRS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}} = e_1R_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}}, \\ dS_1 &= -e_1R'_1S_1 \frac{d\omega_1}{\sqrt{\Omega_1}\sqrt{\theta_1}} = e_2R_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}}, \\ dS_2 &= -e_2R'_2S_2 \frac{d\omega_2}{\sqrt{\Omega_2}\sqrt{\theta_2}} = e_3R_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}}, \\ dS_3 &= -e_3R'_3S_3 \frac{d\omega_3}{\sqrt{\Omega_3}\sqrt{\theta_3}} = eRS \frac{d\omega}{\sqrt{\Omega}\sqrt{\theta}}. \end{aligned}$$

24. But we have  $R' - R = \frac{2\sqrt{\Omega}}{x^2+y^2}$ , &c.; and hence, putting for shortness

$$\frac{1}{(x^2+y^2)\sqrt{\theta}} = P, \quad \frac{1}{(x_1^2+y_1^2)\sqrt{\theta_1}} = P_1, \quad \frac{1}{(x_2^2+y_2^2)\sqrt{\theta_2}} = P_2, \quad \frac{1}{(x_3^2+y_3^2)\sqrt{\theta_3}} = P_3,$$

and consequently

$$\begin{aligned} dS + dS_3 &= +2ePS d\omega, \\ dS_1 - dS &= -2e_1P_1S_1 d\omega_1, \\ dS_2 - dS_1 &= -2e_2P_2S_2 d\omega_2, \\ dS_3 - dS_2 &= -2e_3P_3S_3 d\omega_3, \end{aligned}$$

$$\begin{aligned} dS &= +ePS d\omega + e_1P_1S_1 d\omega_1 + e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3, \\ dS_1 &= -ePS d\omega + e_1P_1S_1 d\omega_1 + e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3, \\ dS_2 &= ePS d\omega - e_1P_1S_1 d\omega_1 - e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3, \\ dS_3 &= ePS d\omega - e_1P_1S_1 d\omega_1 - e_2P_2S_2 d\omega_2 - e_3P_3S_3 d\omega_3, \end{aligned}$$

which are the required formulæ for the elements of arc.

25. The determination of the signs has been made by means of the particular figure; but it is easy to see that the pairs of terms could not for instance be  $dS - dS_3, dS_1 - dS, dS_2 - dS_1, dS_3 - dS$ , or any other pairs such that it would be possible to eliminate  $dS, dS_1, dS_2, dS_3$ , and thus obtain an equation such as

$$0 = ePS d\omega + e_1P_1S_1 d\omega_1 + e_2P_2S_2 d\omega_2 + e_3P_3S_3 d\omega_3;$$



this would, by virtue of the relations between  $d\omega, d\omega_1, d\omega_2, d\omega_3$ , become

$$\frac{eS\sqrt{\Omega}}{x^2 + y^2} - \frac{e_1S_1\sqrt{\Omega_1}}{x_1^2 + y_1^2} + \frac{e_2S_2\sqrt{\Omega_2}}{x_2^2 + y_2^2} - \frac{e_3S_3\sqrt{\Omega_3}}{x_3^2 + y_3^2} = 0,$$

an equation not deducible from the relations which connect  $\omega, \omega_1, \omega_2, \omega_3$ , and which therefore cannot be satisfied by the variable quadrilateral.

26. The differentials of the formulæ are, it will be observed, of the form  $P d\omega$

$$= \frac{S d\omega}{(x^2 + y^2)\sqrt{\theta}},$$

where  $\sqrt{\theta} = \sqrt{f + \theta \cdot g + \theta}$  is a mere constant, and

$$x = \frac{\cos \omega}{\sqrt{f + \theta}}, \quad y = \frac{\sin \omega}{\sqrt{g + \theta}},$$

viz. the form is

$$= \frac{\sqrt{(\cos \omega \sqrt{f + \theta} - \alpha)^2 + (\sin \omega \sqrt{g + \theta} - \beta)^2 - \gamma^2}}{\sqrt{\theta} \left( \frac{\cos^2 \omega}{f + \theta} + \frac{\sin^2 \omega}{g + \theta} \right)} d\omega,$$

which is, in fact, the same as Casey's form in  $\phi$ , equation (300), his  $\phi$  being  $= 90^\circ - \omega$ .

Writing as before  $v$  in place of his  $\theta$ , the differential expression becomes simply  $= S dv$ ; but  $S^2$  expressed as a function of  $v$  is an irrational function  $M + N\sqrt{V}$ , and  $S$  would be the root of such a function; so that, if the form originally obtained had been this form  $S dv$ , it would have been necessary to transform it into the first-mentioned form  $\frac{S d\omega}{(x^2 + y^2)\sqrt{\theta}}$ , in which  $S$  is expressed as a function of  $(x, y)$ , that is, of  $\omega$ .

27. The system of course is

$$\begin{aligned} dS &= eS dv + e_1S_1 dv_1 + e_2S_2 dv_2 + e_3S_3 dv_3, \\ dS_1 &= eS dv - e_1S_1 dv_1 + e_2S_2 dv_2 + e_3S_3 dv_3, \\ dS_2 &= eS dv - e_1S_1 dv_1 - e_2S_2 dv_2 + e_3S_3 dv_3, \\ dS_3 &= eS dv - e_1S_1 dv_1 - e_2S_2 dv_2 - e_3S_3 dv_3, \end{aligned}$$

where  $dv = \frac{d\omega}{(x^2 + y^2)\sqrt{\theta}}$ , &c.: and this is the most convenient way of writing it.

*Reference to Figure. Art. No. 28.*

28. I constructed a bicircular quartic consisting of an exterior and interior oval with the following numerical data:  $f + \theta_0 = 48$ ,  $f + \theta_1 = 56$ ,  $f + \theta_2 = 60$ ,  $f + \theta_3 = 80$ ;  $g + \theta_0 = -6$ ,  $g + \theta_1 = 2$ ,  $g + \theta_2 = 6$ ,  $g + \theta_3 = 26$ ,—not very convenient ones, inasmuch as

the exterior oval came out too large. The annexed figure shows 0, 1, 2, 3, the centres of the circles of inversion, the interior oval, and a portion of the exterior oval, also the origin and axes; it will be seen that the centres 0, 2 lie inside the interior oval, the centres 1, 3 outside the exterior oval: I add further the values

$$\begin{array}{llll}
 \sqrt{f + \theta_0} = 6 \cdot 93, & \sqrt{-(g + \theta_0)} = 2 \cdot 44, & \alpha_0 = 10 \cdot 19, & \beta_0 = - \cdot 98, \\
 \sqrt{f + \theta_1} = 7 \cdot 48, & \sqrt{g + \theta_1} = 1 \cdot 41, & \alpha_1 = 8 \cdot 73, & \beta_1 = +2 \cdot 94, \\
 \sqrt{f + \theta_2} = 7 \cdot 75, & \sqrt{g + \theta_2} = 2 \cdot 45, & \alpha_2 = 8 \cdot 15, & \beta_2 = + \cdot 98, \\
 \sqrt{f + \theta_3} = 8 \cdot 94, & \sqrt{g + \theta_3} = 5 \cdot 09, & \alpha_3 = 6 \cdot 10, & \beta_3 = 23.
 \end{array}$$

We thus see how there exists a series of quadrilaterals  $ABCD$ , where  $A, B$  are situate on the interior oval,  $C, D$  on the exterior oval. Considering the sides as

drawn in the senses  $A$  to  $B$ ,  $B$  to  $C$ ,  $C$  to  $D$ ,  $D$  to  $A$ : and representing the inclinations, measured from the positive infinity on the axis of  $x$  in the sense  $x$  to  $y$ , by  $v_1, v_2, v_3, v$  respectively: then, in passing to the consecutive quadrilateral  $A'B'C'D'$ , we have  $v_1$  and  $v_3$  decreasing,  $v_2$  and  $v$  increasing, that is,  $dv_1$  and  $dv_3$  negative,  $dv_2$  and  $dv$  positive; so that, reckoning the elements  $AA', BB', CC', DD'$ , that is,  $dS, dS_1, dS_2, dS_3$  as each of them positive, we have

$$\begin{aligned} dS_2 - dS_1 &= -2S_2 dv_2, \\ dS_3 - dS_2 &= -2S_3 dv_3, \\ dS - dS_3 &= +2S_1 dv_1, \\ dS_1 + dS &= +2S dv, \end{aligned}$$

and thence

$$\begin{aligned} dS &= S dv - S_1 dv_1 - S_2 dv_2 + S_3 dv_3, \\ dS_1 &= S dv + S_1 dv_1 + S_3 dv_3 - S_2 dv_2, \\ dS_2 &= S dv - S_1 dv_1 + S_2 dv_2 - S_3 dv_3, \\ dS_3 &= S dv - S_1 dv_1 - S_2 dv_2 - S_3 dv_3, \end{aligned}$$

which are the correct signs in regard to the particular figure.

$$\text{Reduction of } \int \frac{S d\omega}{(x^2 + y^2)\sqrt{\theta}} \text{ to Elliptic Integrals. Art. No. 29.}$$

29. The expression in question is

$$\int \frac{\sqrt{(\cos \omega \sqrt{f + \theta} - \alpha)^2 + (\sin \omega \sqrt{g + \theta} - \beta)^2 - \gamma^2}}{\sqrt{\theta} \left( \frac{\cos^2 \omega}{f + \theta} + \frac{\sin^2 \omega}{g + \theta} \right)} d\omega,$$

where  $\sqrt{\theta}$  is a mere constant; and we may apply it to the Gaussian transformation,

$$\begin{aligned} \cos \omega &= \frac{a + a' \cos T + a'' \sin T}{c + c' \cos T + c'' \sin T}, \\ \sin \omega &= \frac{b + b' \cos T + b'' \sin T}{c + c' \cos T + c'' \sin T}, \end{aligned}$$

where the coefficients  $a, b, c, a', b', c', a'', b'', c''$  are such that identically

$$\cos^2 \omega + \sin^2 \omega - 1 = \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\cos^2 T + \sin^2 T - 1);$$

and also

$$\sqrt{(\cos \omega \sqrt{f + \theta} - \alpha)^2 + (\sin \omega \sqrt{g + \theta} - \beta)^2 - \gamma^2} \quad (\text{see formula below}).$$

that is,

$$\begin{aligned} & \cos^2 \omega (f + \theta) + \sin^2 \omega (g + \theta) - 2\alpha\sqrt{f + \theta} \cos \omega - 2\beta\sqrt{g + \theta} \sin \omega + k \\ &= \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\theta_1 - \theta) \{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}. \end{aligned}$$

30. It is found that  $G_1, G_2, G_3$  are the roots of a cubic equation

$$(G + \theta - \theta_1)(G + \theta - \theta_2)(G + \theta - \theta_3) = 0,$$

which being so, we may assume  $G_1 = \theta_1 - \theta$ ,  $G_2 = \theta_2 - \theta$ ,  $G_3 = \theta_3 - \theta$ ; the second condition, in fact, then is

$$\begin{aligned} & (f + \theta) \cos^2 \omega + (g + \theta) \sin^2 \omega - 2\alpha\sqrt{f + \theta} \cos \omega - 2\beta\sqrt{g + \theta} \sin \omega + k \\ &= \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\theta_1 - \theta) \{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}; \end{aligned}$$

and this being so, we find without difficulty the values

$$\begin{aligned} a^2 &= \frac{g + \theta_1 \cdot f + \theta_2 \cdot f + \theta_3}{f - g \cdot \theta_1 - \theta_2 \cdot \theta_1 - \theta_3}, & b^2 &= \frac{f + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{g - f \cdot \theta_1 - \theta_2 \cdot \theta_1 - \theta_3}, & c^2 &= \frac{f + \theta_1 \cdot g + \theta_1}{\theta_2 - \theta_1 \cdot \theta_3 - \theta_1}, \\ a'^2 &= \frac{g + \theta_2 \cdot f + \theta_1 \cdot f + \theta_3}{f - g \cdot \theta_2 - \theta_1 \cdot \theta_2 - \theta_3}, & b'^2 &= \frac{f + \theta_2 \cdot g + \theta_1 \cdot g + \theta_3}{g - f \cdot \theta_2 - \theta_1 \cdot \theta_2 - \theta_3}, & c'^2 &= \frac{f + \theta_2 \cdot g + \theta_2}{\theta_1 - \theta_2 \cdot \theta_3 - \theta_2}, \\ a''^2 &= \frac{g + \theta_3 \cdot f + \theta_1 \cdot f + \theta_2}{f - g \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2}, & b''^2 &= \frac{f + \theta_3 \cdot g + \theta_1 \cdot g + \theta_2}{g - f \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2}, & c''^2 &= \frac{f + \theta_3 \cdot g + \theta_3}{\theta_1 - \theta_3 \cdot \theta_2 - \theta_3}. \end{aligned}$$

To make these positive, the order of ascending magnitude must, however, be not as heretofore  $\theta_0, \theta_1, \theta_2, \theta_3$ , but  $\theta_0, \theta_2, \theta_1, \theta_3$ , viz. we must have  $f + \theta_1, f + \theta_2, f + \theta_3, g + \theta_1, g + \theta_2, -(g + \theta_3), \theta_1 - \theta_2, \theta_1 - \theta_3, \theta_2 - \theta_3$  all positive.

31. The above are the values of the squares of the coefficients; we must have definite relations between the signs of the products  $aa', bb', ab$ , &c., viz. we may have

$$\begin{aligned} aa'' &= \frac{g + \theta_1}{f - g \cdot \theta_2 - \theta_3} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{\theta_3 - \theta_1 \cdot \theta_1 - \theta_2}}, & a'a'' &= \frac{g + \theta_2}{f - g \cdot \theta_3 - \theta_1} \sqrt{\frac{f + \theta_1 \cdot f + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3}}, \\ b'b'' &= \frac{f + \theta_1}{g - f \cdot \theta_2 - \theta_3} \sqrt{\frac{g + \theta_2 \cdot g + \theta_3}{\theta_3 - \theta_1 \cdot \theta_1 - \theta_2}}, & bb'' &= \frac{f + \theta_2}{g - f \cdot \theta_3 - \theta_1} \sqrt{\frac{g + \theta_1 \cdot g + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3}}, \\ b'c'' &= \frac{f + \theta_1}{g - f \cdot \theta_1 - \theta_2} \sqrt{\frac{g + \theta_2 \cdot g + \theta_3}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1}}, \\ b'c &= \frac{g + \theta_1}{g - f \cdot \theta_1 - \theta_2} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1}}. \end{aligned}$$

and further

$$b'c'' = \frac{f + \theta_2}{f - g \cdot \theta_2 - \theta_3 \cdot \theta_3 - \theta_1} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}}, \quad bc = \frac{f + \theta_1}{\theta_2 - \theta_3 \cdot \theta_1 - \theta_2} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}},$$

$$b''c'' = \frac{f + \theta_3}{f - g \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}}, \quad b''c = \frac{f + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3} \sqrt{\frac{g + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}},$$

and similarly

$$a'a'' = \frac{g + \theta_1}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{f - g}}, \quad ca' = \frac{g + \theta_1}{\theta_2 - \theta_3 \cdot \theta_3 - \theta_1} \sqrt{\frac{f + \theta_2 \cdot f + \theta_3}{f - g}},$$

$$c''a'' = \frac{g + \theta_2}{\theta_3 - \theta_1 \cdot \theta_1 - \theta_2} \sqrt{\frac{f + \theta_1 \cdot f + \theta_3}{f - g}}, \quad c'a'' = \frac{g + \theta_3}{\theta_1 - \theta_2 \cdot \theta_2 - \theta_3} \sqrt{\frac{f + \theta_1 \cdot f + \theta_2}{f - g}},$$

and also

$$b'c'' + b''c' = \pm \frac{2g + \theta_2 + \theta_3}{\theta_2 - \theta_3} \sqrt{\frac{g + \theta_1 \cdot f + \theta_2 \cdot f + \theta_3}{g - f \cdot \theta_1 - \theta_2 \cdot \theta_1 - \theta_3}},$$

$$= \frac{1}{f + \theta_1} \sqrt{\frac{f + \theta_1 \cdot g + \theta_2 \cdot g + \theta_3}{f - g}} \cdot \frac{2f + \theta_2 + \theta_3}{\theta_2 - \theta_3},$$

$$b''c + bc'' = \pm \frac{2g + \theta_3 + \theta_1}{\theta_3 - \theta_1} \sqrt{\frac{g + \theta_2 \cdot f + \theta_3 \cdot f + \theta_1}{g - f \cdot \theta_2 - \theta_1 \cdot \theta_2 - \theta_3}},$$

$$bc' + b'c = \pm \frac{2g + \theta_1 + \theta_2}{\theta_1 - \theta_2} \sqrt{\frac{g + \theta_3 \cdot f + \theta_1 \cdot f + \theta_2}{g - f \cdot \theta_3 - \theta_1 \cdot \theta_3 - \theta_2}}.$$

32. These values, in fact, satisfy the several relations which exist between the nine coefficients; viz. the original expressions of  $\cos \omega, \sin \omega$ , in terms of  $\cos T, \sin T$  give conversely expressions of  $\cos T, \sin T$  in terms of  $\cos \omega, \sin \omega$ , the two sets being

$$\cos T = \frac{-a + c \cos \omega + c'' a \sin \omega - c''}{a' \cos \omega + b' \sin \omega - c'}, \quad \sin T = \frac{a'' \cos \omega + b'' \sin \omega - c''}{a \cos \omega + b \sin \omega - c};$$

and we have then the relations

$$\cos^2 \omega + \sin^2 \omega - 1 = \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\cos^2 T + \sin^2 T - 1),$$

$$\cos^2 T + \sin^2 T - 1 = \frac{1}{(a \cos \omega + b \sin \omega - c)^2} (\cos^2 \omega + \sin^2 \omega - 1)',$$

$$(\theta + f) \cos^2 \omega + (\theta + g) \sin^2 \omega - 2\alpha \sqrt{\theta + f} \cos \omega - 2\beta \sqrt{\theta + g} \sin \omega + k$$

$$= \frac{1}{(c + c' \cos T + c'' \sin T)^2} (\theta_1 - \theta) \{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}$$

$$+ \frac{1}{(a \cos \omega + b \sin \omega - c)^2} \{(\theta + f) \cos^2 \omega + (\theta + g) \sin^2 \omega - 2\alpha \sqrt{\theta + f} \cos \omega - 2\beta \sqrt{\theta + g} \sin \omega + k\}.$$

giving the four sets each of six equations

$$\begin{aligned} a^2 + b^2 - c^2 &= -1, & a'a'' + b'b'' - c'c'' &= 0, \\ a'^2 + b'^2 - c'^2 &= +1, & a''a + b''b - c''c &= 0, \\ a''^2 + b''^2 - c''^2 &= +1, & aa' + bb' - cc' &= 0, \\ -a^2 + a'^2 + a''^2 &= +1, & -bc + b'c' + b''c'' &= 0, \\ -b^2 + b'^2 + b''^2 &= +1, & -ca + c'a' + c''a'' &= 0, \\ -c^2 + c'^2 + c''^2 &= -1, & -ab + a'b' + a''b'' &= 0, \end{aligned}$$

$$\begin{aligned} (\theta + f)a^2 + (\theta + g)b^2 - 2\alpha\sqrt{\theta + f}ac - 2\beta\sqrt{\theta + g}bc + kc^2 &= \theta_1 + \theta, \\ (\theta + f)a'^2 + (\theta + g)b'^2 - 2\alpha\sqrt{\theta + f}a'c' - 2\beta\sqrt{\theta + g}b'c' + kc'^2 &= -\theta_2 + \theta, \\ (\theta + f)a''^2 + (\theta + g)b''^2 - 2\alpha\sqrt{\theta + f}a''c'' - 2\beta\sqrt{\theta + g}b''c'' + kc''^2 &= -\theta_3 + \theta, \end{aligned}$$

$$\begin{aligned} (\theta + f)a'a'' + (\theta + g)b'b'' - \alpha\sqrt{\theta + f}(a'c'' + a''c') - \beta\sqrt{\theta + g}(b'c'' + b''c') + kc'c'' &= 0, \\ (\theta + f)a''a + (\theta + g)b''b - \alpha\sqrt{\theta + f}(a''c + ac'') - \beta\sqrt{\theta + g}(b''c + bc'') + kc''c &= 0, \\ (\theta + f)aa' + (\theta + g)bb' - \alpha\sqrt{\theta + f}(ac' + a'c) - \beta\sqrt{\theta + g}(bc' + b'c) + kcc' &= 0, \end{aligned}$$

$$\begin{aligned} (\theta_1 - \theta)a^2 - (\theta_2 - \theta)a'^2 - (\theta_3 - \theta)a''^2 &= \theta + f, \quad \text{or say} \quad (\theta_1 + f)a^2 - (\theta_2 + f)a'^2 - (\theta_3 + f)a''^2 = 0, \\ (\theta_1 - \theta)b^2 - (\theta_2 - \theta)b'^2 - (\theta_3 - \theta)b''^2 &= \theta + g, \quad (\theta_1 + g)b^2 - (\theta_2 + g)b'^2 - (\theta_3 + g)b''^2 = 0, \\ (\theta_1 - \theta)c^2 - (\theta_2 - \theta)c'^2 - (\theta_3 - \theta)c''^2 &= k, \quad \theta_1c^2 - \theta_2c'^2 - \theta_3c''^2 = k + \theta, \\ -(\theta_1 - \theta)bc + (\theta_2 - \theta)b'c' + (\theta_3 - \theta)b''c'' &= -\beta\sqrt{\theta + g}, \\ -(\theta_1 - \theta)ca + (\theta_2 - \theta)c'a' + (\theta_3 - \theta)c''a'' &= -\alpha\sqrt{\theta + f}, \\ -(\theta_1 - \theta)ab + (\theta_2 - \theta)a'b' + (\theta_3 - \theta)a''b'' &= 0; \end{aligned}$$

all which formulæ are in fact satisfied by the foregoing values of the expressions  $a^2, b^2, a'^2$ , &c.

33. We then have

$$d\omega = \frac{dT}{c + c' \cos T + c'' \sin T},$$

the radical which multiplies  $d\omega$  being

$$= \frac{1}{c + c' \cos T + c'' \sin T} \sqrt{(\theta_1 - \theta)\{(\theta_1 - \theta) - (\theta_2 - \theta) \cos^2 T - (\theta_3 - \theta) \sin^2 T\}};$$

the differential becomes

$$\frac{dT \sqrt{\theta_1 - \theta_2 \cos^2 T - \theta_3 \sin^2 T}}{\sqrt{\theta} \left( \frac{\cos^2 \omega}{f + \theta} + \frac{\sin^2 \omega}{g + \theta} \right) (c + c' \cos T + c'' \sin T)^3},$$

that is,

$$= \frac{dT \sqrt{\theta_1 - \theta_2 \cos^2 T - \theta_3 \sin^2 T}}{\frac{1}{f + \theta} (a + a' \cos T + a'' \sin T)^2 + \frac{1}{g + \theta} (b + b' \cos T + b'' \sin T)^2}.$$

The denominator could, of course, be reduced to the form  $(*)(1, \cos T, \sin T)^2$ ; but the actual form seems preferable, inasmuch as it puts in evidence the linear factors

$$\frac{1}{\sqrt{f + \theta}} (a + a' \cos T + a'' \sin T) \pm \frac{1}{\sqrt{g + \theta}} (b + b' \cos T + b'' \sin T),$$

and there seems to be no advantage in further reducing the integral.

## 668.

## ON COMPOUND COMBINATIONS.

[From the *Proceedings of the Lit. Phil. Soc. Manchester*, t. XVI. (1877), pp. 113, 114;  
*Memoirs*, ib., Ser. III., t. VI. (1879), pp. 99, 100.]

Prof. Clifford's paper, "On the Types of Compound Statement involving Four Classes," [volume of *Proceedings* quoted, pp. 88–101; *Mathematical Papers*, pp. 1–13], relates mathematically to a question of compound combinations; and it is worth while to consider its connexion with another question of compound combinations, the application of which is a very different one.

Starting with four symbols,  $A, B, C, D$ , we have sixteen combinations of the five types 1,  $A, AB, ABC, ABCD$ , ( $1 + 4 + 6 + 4 + 1 = 16$  as before). But in Prof. Clifford's question 1 means  $A'B'C'D'$ ,  $A$  means  $AB'C'D'$ , &c.; viz. each of the symbols means an aggregate of four assertions; and the 16 symbols are thus all of the same type. Considering them in this point of view, the question is as to the number of types of the binary, ternary, &c., combinations of the sixteen combinations; for, according as these are combined,

|                   |    |    |    |     |     |     |     |     |     |     |     |     |     |     |    |
|-------------------|----|----|----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|----|
| No. of types of = | 1, | 2, | 3, | 4,  | 5,  | 6,  | 7,  | 8,  | 9,  | 10, | 11, | 12, | 13, | 14, | 15 |
|                   | 1, | 4, | 6, | 19, | 27, | 47, | 55, | 78, | 55, | 47, | 27, | 19, | 6,  | 4,  | 1  |

together.

In the first mentioned point of view the like question arises, in regard to the sets belonging to the five different types separately or in combination with each other; for instance, taking only the six symbols of the type  $AB$ , these may be taken 1, 2, 3, 4, or 5 together, and we have in these cases respectively

$$\text{No. of types} = 1, 2, 3, 4, 5,$$

$$1, 2, 2, 2, 1.$$

as is very easily verified; but if the number of letters  $A, B, \dots$  be greater (say this = 8), or, instead of letters, writing the numbers 1, 2, 3, 4, 5, 6, 7, 8, then the question is that of the number of types of combination of the 28 duads 12, 13,  $\dots$ , 78, taken 1, 2, 3,  $\dots$ , 27 together, a question presenting itself in geometry in regard to the bitangents of a quartic curve (see Salmon's *Higher Plane Curves*, Ed. 2 (1873), pp. 222 *et seq.*): the numbers, so far as they have been obtained, are

No. of types = 1, 2, 3, 4,  $\dots$ , 24, 25, 26, 27

1, 2, 5, 11,  $\dots$ , 11, 5, 2, 1.

It might be interesting to complete the series, and, more generally, to determine the number of the types of combination of the  $\frac{1}{2}n(n-1)$  duads of  $n$  letters.



## 669.

## ON A PROBLEM OF ARRANGEMENTS.

[From the *Proceedings of the Royal Society of Edinburgh*, t. IX. (1878), pp. 338–342.]

It is a well-known problem to find for  $n$  letters the number of the arrangements in which no letter occupies its original place; and the solution of it is given by the following general theorem: viz., the number of the arrangements which satisfy any  $r$  conditions is

$$\begin{aligned} & (1-1)(1-2)\cdots(1-r), \\ & = 1 - \Sigma(1) + \Sigma(12) - \cdots + (12\cdots r), \end{aligned}$$

where 1 denotes the whole number of arrangements; (1) the number of them which fail in regard to the first condition; (2) the number which fail in regard to the second condition; (12) the number which fail in regard to the first condition, and also in regard to the second condition; and so on:  $\Sigma(1)$  means  $(1) + (2) + \cdots + (r)$ ;  $\Sigma(12)$  means  $(12) + (13) + \cdots + (2r) + \cdots + (r-1, r)$ ; and so on, up to  $(12\cdots r)$ , which denotes the number failing in regard to each of the  $r$  conditions.

Thus, in the special problem, the first condition is that the letter in the first place shall not be  $a$ ; the second condition is that the letter in the second place shall not be  $b$ ; and so on; taking  $r = n$ , we have the known result,

$$\begin{aligned} \text{No.} &= n^n - \frac{n}{1}(n-1)^n + \frac{n(n-1)}{1\cdot 2}(n-2)^n - \cdots \pm \frac{n!}{n!} \\ &= 1 \cdot 2 \cdot 3 \cdots n \left\{ 1 - 1 + \frac{1}{1\cdot 2} - \frac{1}{1\cdot 2\cdot 3} + \cdots \pm \frac{1}{1\cdot 2\cdots n} \right\}, \end{aligned}$$

giving for the several cases

$$n = 2, 3, 4, 5, 6, 7, \dots$$

$$\text{No.} = 1, 2, 9, 44, 265, 1854, \dots$$

I proceed to consider the following problem, suggested to me by Professor Tait, in connexion with his theory of knots: to find the number of the arrangements of  $n$  letters  $abc\dots jk$ , when the letter in the first place is not  $a$  or  $b$ , the letter in the second place not  $b$  or  $c$ ,  $\dots$ , the letter in the last place not  $k$  or  $a$ .

Numbering the conditions  $1, 2, 3, \dots, n$ , according to the places to which they relate, a single condition is called  $[1]$ ; two conditions are called  $[2]$  or  $[1, 1]$ , according as the numbers are consecutive or non-consecutive: three conditions are called  $[3]$ ,  $[2, 1]$ , or  $[1, 1, 1]$ , according as the numbers are all three consecutive, two consecutive and one not consecutive, or all non-consecutive; and so on: the numbers which refer to the conditions being always written in their natural order, and it being understood that they follow each other cyclically, so that  $1$  is consecutive to  $n$ . Thus,  $n = 6$ , the set  $125$  of conditions is  $[3]$ , as consisting of  $3$  consecutive conditions; and similarly  $1346$  is  $[2, 2]$ .

Consider a single condition  $[1]$ , say this is  $1$ ; the arrangements which fail in regard to this condition are those which contain in the first place  $a$  or  $b$ ; whichever it be, the other  $n - 1$  letters may be arranged in any form whatever; and there are thus  $2\Pi(n - 1)$  failing arrangements.

Next for two conditions; these may be  $[2]$ , say the conditions are  $1$  and  $2$ : or else  $[1, 1]$ , say they are  $1$  and  $3$ . In the former case, the arrangements which fail are those which contain in the first and second places  $ab, ac$ , or  $bc$ : and for each of these, the other  $n - 2$  letters may be arranged in any order whatever; there are thus  $3\Pi(n - 2)$  failing arrangements. In the latter case, the failing arrangements have in the first place  $a$  or  $b$ , and in the third place  $c$  or  $d$ , viz. the letters in these two places are  $a.c, a.d, b.c$ , or  $b.d$ , and in each case the other  $n - 2$  letters may be arranged in any order whatever: the number of failing arrangements is thus  $= 2 \cdot 2 \cdot \Pi(n - 2)$ .

And so, in general, when the conditions are  $[\alpha, \beta, \gamma, \dots]$ , the number of failing arrangements is

$$= (\alpha + 1)(\beta + 1)(\gamma + 1) \cdots \Pi(n - \alpha - \beta - \gamma \cdots).$$

But for  $[n]$ , that is, for the entire system of the  $n$  conditions, the number of failing arrangements is (not as by the rule it should be  $= n + 1$ , but)  $= 2$ , viz. the only arrangements which fail in regard to each of the  $n$  conditions are (as is at once seen),  $abc \dots jk$ , and  $bc \dots jka$ .

Changing now the notation so that  $[1], [2], [1, 1]$ , &c., shall denote the *number* of the conditions  $[1], [2], [1, 1]$ , &c., respectively, it is easy to see the form of the general result. If, for greater clearness, we write  $n = 6$ , we have

$$\begin{aligned} & 1 - \Sigma(1) + \Sigma(12) - \Sigma(123) \\ \text{No.} = & 720 - \{([1] = 6) 2\} 120 + \left\{ \begin{array}{l} ([2] = 6) 3 \\ + ([1, 1] = 9) 2.2 \end{array} \right\} 24 - \left\{ \begin{array}{l} ([3] = 6) 4 \\ + ([2, 1] = 12) 3.2 \\ + ([1, 1, 1] = 2) 2.2.2 \end{array} \right\} 6 \\ & + \Sigma(1234) - \Sigma(12345) + (123456) \\ & + \left\{ \begin{array}{l} ([4] = 6) 5 \\ + ([3, 1] = 6) 4.2 \\ + ([2, 2] = 3) 3.3 \end{array} \right\} 2 - \{([5] = 6) 6\} 1 + \{([6] = 1) 2\}; \end{aligned}$$

or, reducing into numbers, this is

$$\text{No.} = 720 - 1440 + 1296 - 672 + 210 - 36 + 2, = 80.$$

The formula for the next succeeding case,  $n = 7$ , gives

$$\text{No.} = 5040 - 10080 + 9240 - 5040 + 1764 - 392 + 49 - 2, = 579.$$

Those for the preceding cases,  $n = 3, 4, 5$ , respectively are

$$\begin{aligned} \text{No.} &= 6 - 12 + 9 - 2 &&= 1, \\ \text{No.} &= 24 - 48 + 40 - 16 + 2 &&= 2, \\ \text{No.} &= 120 - 240 + 210 - 100 + 25 - 2 &&= 13. \end{aligned}$$

We have in general  $[1] = n$ ,  $[2] = n$ ,  $[1, 1] = \frac{1}{2}n(n-3)$ ; and in the several columns of the formulæ the sums of the numbers thus represented are equal to the coefficients of  $(1+1)^n$ : thus, when  $n = 6$  as above, the sums are 6, 15, 20, 15, 6, 1. As regards the calculation of the numbers in question, any symbol  $[\alpha, \beta, \gamma]$  is a sum of symbols

$$[\alpha - \alpha' + \beta - \beta' + \gamma - \gamma' + \dots],$$

where  $\alpha' + \beta' + \gamma' + \dots$  is any partition of  $n - (\alpha + \beta + \gamma + \dots)$ ; read, of the series of numbers  $1, 2, 3, \dots, n$ , taken in cyclical order beginning with any number, retain  $\alpha$ , omit  $\alpha'$ , retain  $\beta$ , omit  $\beta'$ , retain  $\gamma$ , omit  $\gamma'$ , ... Thus in particular,  $n = 6$ ,  $[1, 1]$  is a sum of symbols  $[1 - 3 + 1 - 1]$  and  $[1 - 2 + 1 - 2]$ ; it is clear that any such symbol  $[\alpha - \alpha' + \beta - \beta' + \dots]$  is  $= n$  or a submultiple of  $n$  (in particular, if  $n$  be prime, the symbol is always  $= n$ ): and we thus in every case obtain the value of  $[\alpha, \beta, \gamma, \dots]$  by taking for the negative numbers the several partitions of  $n - (\alpha + \beta + \gamma + \dots)$ , and for each symbol

$$[\alpha - \alpha' + \beta - \beta' + \gamma - \gamma' + \dots],$$

writing its value,  $n$  or a given submultiple of  $n$ , as just mentioned. There would, I think, be no use in pursuing the matter further, by seeking to obtain an analytical expression for the symbols  $[\alpha, \beta, \gamma, \dots]$ .

For the actual formation of the required arrangements, it is of course easy, when all the arrangements are written down, to strike out those which do not satisfy the prescribed conditions, and so obtain the system in question. Or introducing the notion of substitutions<sup>1</sup>, and accordingly considering each arrangement as derived by a substitution from the primitive arrangement  $abcd \dots jk$ , we can write down the substitutions which give the system of arrangements in which no letter occupies its original place: viz. we must for this purpose partition the  $n$  letters into parts, no part less than 2, and then in each set taking one letter (say the first in alphabetical order) as fixed, permute in every possible way the other letters of the set; we thus obtain

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<sup>1</sup>In explanation of the notation of substitutions, observe that  $(abcde)$  means that  $a$  is to be changed into  $b$ ,  $b$  into  $c$ ,  $c$  into  $d$ ,  $d$  into  $e$ ,  $e$  into  $a$ ; and similarly  $(ab)(cde)$  means that  $a$  is to be changed into  $b$ ,  $b$  into  $a$ ,  $c$  into  $d$ ,  $d$  into  $e$ ,  $e$  into  $c$ .

all the substitutions which move every letter. Thus when  $n = 5$ , we obtain the 44 substitutions for the letters  $abcde$ , viz. these are

$(abcde)$ , &c., 24 symbols obtained by permuting in every way the four letters  $b, c, d, e$ ;

$(ab)(cde)$ , &c., 20 symbols corresponding to the 10 partitions  $ab, cde$ , and for each of them 2 arrangements such as  $cde, ced$ .

And then if we reject those symbols which contain in any ( ) two consecutive letters, we have the substitutions which give the arrangements wherein the letter in the first place is not  $a$  or  $b$ , that in the second place not  $b$  or  $c$ , and so on. In particular, when  $n = 5$ , rejecting the substitutions which contain in any ( ),  $ab, bc, cd, de$ , or  $ea$ , we have 13 substitutions, which may be thus arranged:

$$\begin{array}{lllll} (acbed), & (ac)(bed), & (acebd), & (adbec), & (aedbc), \\ (aebdc), & (bd)(aec), & & & \\ (acedb), & (ce)(adb), & & & \\ (aecbd), & (ad)(bce), & & & \\ (adceb), & (be)(acd), & & & \end{array}$$

Here in the first column, performing on the symbol  $(acbed)$  the substitution  $(abcde)$ , we obtain  $(bdcae)$ ,  $= (aebdc)$ , the second symbol; and so again and again operating with  $(abcde)$ , we obtain the remaining symbols of the column; these are for this reason said to be of the same type. In like manner, symbols of the second column are of the same type; but the symbols in the remaining three columns are each of them a type by itself; viz. operating with  $(abcde)$  upon  $(acebd)$ , we obtain  $(bdace)$ ,  $= (acebd)$ ; and the like as regards  $(adbec)$  and  $(aedbc)$  respectively. The 13 substitutions are thus of 5 different types, or say the arrangements to which they belong, viz.

$$\begin{array}{lllll} cebad, & ceabd, & cdeab, & deabc, & eabcd, \\ edacb, & edabc, & & & \\ caebd, & daebc, & & & \\ edbac, & debac, & & & \\ daecb, & deacb, & & & \end{array}$$

are of 5 different types. The question to determine for any value of  $n$ , the number of the different types, is, it would appear, a difficult one, and I do not at present enter upon it.

## 670.

[NOTE ON MR MUIR'S SOLUTION OF A "PROBLEM OF  
ARRANGEMENT."]

[From the *Proceedings of the Royal Society of Edinburgh*, t. IX. (1878), pp. 388–391.]

The investigation may be carried further: writing for shortness  $u_3, u_4$ , &c., in place of  $\Psi(3), \Psi(4)$ , &c., the equations are

$$\begin{aligned} v_3 &= 1, \\ v_4 &= 2u_3, \\ v_5 &= 2u_4 + 6u_3 + 1, \\ v_6 &= 4u_5 + 8u_4 + 12u_3, \\ v_7 &= 5u_6 + 10u_5 + 15u_4 + 18u_3 + 1. \end{aligned}$$

Hence assuming

$$u = u_3 + u_4x + u_5x^2 + u_6x^3 + u_7x^4 + \dots,$$

we have

$$\begin{aligned} v &= \frac{1}{1-x^2} + u_3(2x + 12x^2 + 15x^3 + \dots) \\ &\quad + u_4(3x^2 + 8x^3 + 15x^4 + 22x^5 + \dots) \\ &\quad + u_5(4x^3 + 10x^4 + 18x^5 + 26x^6 + \dots) \\ &\quad + u_6(5x^4 + 12x^5 + 21x^6 + 30x^7 + \dots); \end{aligned}$$

so that, forming the equation

$$\begin{aligned} u' - \frac{u}{(1-x)^2} &= u_3(x^2 + 2x^3 + 3x^4 + 4x^5 + \dots) \\ &\quad + u_4(2x^3 + 4x^4 + 6x^5 + 8x^6 + \dots) \\ &\quad + u_5(3x^4 + 6x^5 + 9x^6 + 12x^7 + \dots), \end{aligned}$$

where  $u'$  denotes  $\frac{du}{dx}$ , we have

$$u' - \frac{u}{(1-x)^2} = \frac{1}{1-x}u + \frac{2x}{(1-x)^2(1+x)}u + \frac{2x^3}{(1-x)^2(1+x)}u + \dots$$

or, what is the same thing,

$$u' = \left\{ \frac{1}{(1-x)^2} + \frac{2x}{(1-x)^2(1+x)} \right\} u - \frac{x^2}{(1-x)^2(1+x)} + \frac{1}{1-x^2}.$$

This equation may be simplified: write

$$u = \frac{1}{1-x^2}Q - \left( -\frac{1}{x} + \frac{1}{2} \right) Q,$$

then

$$u' = \left( -\frac{1}{x^2} - \frac{1}{2} \right) Q + \frac{1-x^2}{x^2} Q',$$

and the equation is

$$\left\{ -\frac{1-x^2}{x^2} + \frac{2}{x^2} \frac{1+x}{1+x} - \frac{2x^2}{x^2(1-x)} - \frac{2}{x^2} \frac{1}{x(1+x)} \right\} Q + \frac{1+x}{1-x^2} Q' = \frac{1-x}{1+x},$$

that is,

$$\left\{ -\frac{1}{x^2} + \frac{2}{x^3} \right\} Q + \frac{1+x^2}{x^2} Q' = \frac{1-x}{1+x},$$

or

$$\frac{-(1-x^2)}{x^4} Q + \frac{1-x^2}{x^2} Q' = \frac{1-x}{1+x},$$

or finally,

$$Q' \left( 1 - \frac{1}{x^2} \right) - \frac{1}{x^4} Q = \frac{1-x}{1+x},$$

giving

$$Q \left( 1 - \frac{1}{x^2} \right) - Q' \cdot \frac{1}{x^2} = \frac{1-x}{1+x},$$

and thence

$$Q = e^{(x-x^{-1})} \int e^{-(x-x^{-1})} \frac{x^2}{(x+1)^2} dx,$$

which is the value of the generating function

$$u = u_3 + u_4x + u_5x^2 + \&c.$$

But for the purpose of calculation it is best to integrate by a series the differential equation for  $Q$ : assuming

$$Q = q_3 + q_4x + q_5x^2 + \&c.,$$

we find

$$\begin{aligned} q_4 &= 4q_3 &= 1, \\ q_5 &= 5q_4 + q_3 &= -2, \\ q_6 &= 6q_5 + q_4 &= +3, \\ q_7 &= 7q_6 + q_5 &= -4, \\ q_8 &= 8q_7 + q_6 &= +5, \\ &\vdots \\ q_n &= nq_{n-1} + q_{n-2} + (-)^{n-1}(n-2). \end{aligned}$$

We have thus for  $q_3, q_4, q_5, \dots$  the values 1, 2, 14, 82, 593, 4820,  $\dots$ , and thence viz. writing

$$u = (1 - x^2)Q = (1 + x)(1 + 2x + 14x^2 + 82x^3 + 593x^4 + 4820x^5 + \dots),$$

|    |    |     |     |     |         |
|----|----|-----|-----|-----|---------|
| 1  | 2  | 14  | 82  | 593 | 4820... |
| -1 | -2 | -14 | -82 |     |         |
|    |    |     |     |     |         |

the values of  $u_3, u_4, \dots$  are 1, 2, 13, 80, 579, 4738,  $\dots$ , agreeing with the results found above.

In the more simple problem, where the arrangements of the  $n$  things are such that no one of them occupies its original place, if  $u_n$  be the number of arrangements, we have

$$\begin{aligned} u_2 &= 1 &= 1, \\ u_3 &= 2u_2 &= 2, \\ u_4 &= 3u_3 + (u_3 - u_2) &= 9, \\ u_5 &= 4u_4 + (u_4 - u_3) &= 44, \\ u_{n+1} &= nu_n + (u_n - u_{n-1}), \end{aligned}$$

and writing

$$u = u_2 + u_3x + u_4x^2 + \dots,$$

we find

$$1 + (2x + 3x^2 + \dots)u + (x^2 + x^3)u' = u(-1 + 2x + 3x^2 + \dots)u + (x^2 + x^3)u' = 1,$$

that is,

$$u(-1 + 2x + 3x^2) + (x^2 + x^3)u' = 1,$$

or, what is the same thing,

$$\frac{u}{x^2} + \left(\frac{3}{x} - 1\right)u' = \frac{1}{x^2},$$

whence

$$u = \frac{1}{x^2} e^{-\frac{1}{x}} \int e^{\frac{1}{x}} \frac{1}{1+x} dx.$$

But the calculation is most easily performed by means of the foregoing equation of differences, itself obtained from the differential equation written in the foregoing form,

$$u(-1 + 2x + 3x^2) + (x^2 + x^3)u' = 1.$$

## 671.

## ON A SIBI-RECIPROCAL SURFACE.

[From the *Berlin. Akad. Monatsber.*, (1878), pp. 309–313.]

The question of the generation of a sibi-reciprocal surface—that is, a surface the reciprocal of which is of the same order and has the same singularities as the original surface—was considered by me in the year 1868, see *Proc. London Math. Soc.* t. II. pp. 61–63, [part of 387], where it is remarked that if a surface be considered as the envelope of a quadric surface varying according to given conditions, then the reciprocal surface is given as the envelope of a quadric surface varying according to the reciprocal conditions; whence, if the conditions be sibi-reciprocal, it follows that the surface is a sibi-reciprocal surface. And I gave as instances the surface which is the envelope of a quadric surface touching each of 8 given lines; and also the surface called the “tetrahedroid,” which is a homographic transformation of Fresnel’s Wave Surface and a particular case of the quartic surface with 16 nodes.

The interesting surface of the order 8, recently considered by Herr Kummer, *Berl. Monatsber.*, Jan. 1878, pp. 25–36, is included under the theory. In fact, if we consider a line  $L$ , whereof the six coordinates

$$a, b, c, f, g, h,$$

satisfy each of the three linear relations

$$\begin{aligned} f_1a + g_1b + h_1c + a_1f + b_1g + c_1h &= 0, \\ f_2a + g_2b + h_2c + a_2f + b_2g + c_2h &= 0, \\ f_3a + g_3b + h_3c + a_3f + b_3g + c_3h &= 0, \end{aligned}$$

the locus of this line is a quadric surface the equation of which is

$$\begin{aligned} T &= (agh)x^2 + (bhf)y^2 + (cfg)z^2 + (abc)w^2 \\ &+ [(abg) - (cah)]xw + [(bfg) + (chf)]yz \\ &+ [(bch) - (abf)]yw + [(cgh) - (afg)]zx \\ &+ [(caf) - (bcg)]zw + [(ahf) + (bgh)]xy = 0. \end{aligned}$$



where  $(agh)$  is used to denote the determinant  $\begin{vmatrix} a_1 & g_1 & h_1 \\ a_2 & g_2 & h_2 \\ a_3 & g_3 & h_3 \end{vmatrix}$ , and so for the other symbols.

Considering the reciprocal of the line  $L$  in regard to the quadric surface  $X^2 + Y^2 + Z^2 + W^2 = 0$ , the six coordinates of the reciprocal line are

$$f, g, h, a, b, c,$$

and it is hence at once seen that the locus of the reciprocal line is the quadric surface  $T' = 0$  obtained from the equation by interchanging therein the symbolical quantities  $a, b, c$  and  $f, g, h$ : viz. writing also  $(\xi, \eta, \zeta, \omega)$  in place of  $(x, y, z, w)$ , the new equation is

$$\begin{aligned} T' = & (fbc)\xi^2 + (gca)\eta^2 + (hab)\zeta^2 + (fgh)\omega^2 \\ & + [(fgh) - (hfc)]\zeta\omega + [(fab) + (hca)]\eta\zeta \\ & + [(ghc) - (fga)]\eta\omega + [(gbc) + (fab)]\xi\omega \\ & + [(hbc) - (ghb)]\xi\zeta + [(hca) + (gbc)]\xi\eta = 0; \end{aligned}$$

or, what is the same thing, this equation  $T' = 0$  is the equation of the original quadric surface (the locus of  $L$ ) expressed in terms of the plane-coordinates  $\xi, \eta, \zeta, \omega$ .

Now considering each of the quantities  $a_1, b_1, c_1, f_1, g_1, h_1, a_2, b_2$ , etc.,  $a_3, b_3$ , etc., as a given linear function of a variable parameter  $\lambda$ , say  $a_1 = a'_1 + a''_1\lambda$ ,  $b_1 = b'_1 + b''_1\lambda$ , etc., the equation  $T = 0$  takes the form

$$A\lambda^3 + 3B\lambda^2 + 3C\lambda + D = 0,$$

where  $A, B, C, D$  are given quadric functions of the coordinates  $x, y, z, w$ ; and the envelope of the quadric surface  $T = 0$  is Herr Kummer's surface of the eighth order

$$(AD - BC)^2 - 4(AC - B^2)(BD - C^2) = 0.$$

In like manner the equation  $T' = 0$  takes the form

$$A'\lambda^3 + 3B'\lambda^2 + 3C'\lambda + D' = 0,$$

where  $A', B', C', D'$  are given functions of the coordinates  $\xi, \eta, \zeta, \omega$ ; and we have

$$(A'D' - B'C')^2 - 4(A'C' - B'^2)(B'D' - C'^2) = 0,$$

as the equation of the reciprocal surface; or (what is the same thing) as that of the original surface, regarding  $\xi, \eta, \zeta, \omega$  as plane-coordinates.

In regard to the foregoing equation  $T = 0$ , it is to be noticed that, if  $a_1, b_1, c_1, f_1, g_1, h_1, a_2, b_2$ , etc.,  $a_3, b_3$ , etc., instead of being arbitrary coefficients, were the coordinates of three given lines  $L_1, L_2, L_3$  respectively; that is, if we had

$$\begin{aligned} a_1f_1 + b_1g_1 + c_1h_1 &= 0, \\ a_2f_2 + b_2g_2 + c_2h_2 &= 0, \\ a_3f_3 + b_3g_3 + c_3h_3 &= 0, \end{aligned}$$

then the three linear relations satisfied by  $(a, b, c, f, g, h)$  would express that the line  $L$  was a line meeting each of the three given lines  $L_1, L_2, L_3$ : the locus is therefore the quadric surface which passes through these three lines; and I have in my paper “On the six coordinates of a Line,” *Camb. Phil. Trans.*, t. XI. (1869), pp. 290–323, [435], found the equation to be the foregoing equation  $T = 0$ . But it is easy to see that the same equation subsists in the case where the three equations  $a_1f_1 + b_1g_1 + c_1h_1 = 0$ , etc., are not satisfied. For the several coefficients being perfectly general, any one of the three linear relations may be replaced by a linear combination of these equations; that is, in place of  $a_1, b_1, c_1, f_1, g_1, h_1$ , we may write  $a'_1, b'_1, c'_1, f'_1, g'_1, h'_1$ , where  $a'_1 = \theta_1a_1 + \theta_2a_2 + \theta_3a_3$ ,  $b'_1 = \theta_1b_1 + \theta_2b_2 + \theta_3b_3$ , etc.; and these factors  $\theta_1, \theta_2, \theta_3$  may be conceived to be such that the condition in question  $a'_1f'_1 + b'_1g'_1 + c'_1h'_1 = 0$  is satisfied. Similarly the second set of coefficients may be replaced by  $a'_2, b'_2, c'_2, f'_2, g'_2, h'_2$ , where  $a'_2 = \phi_1a_1 + \phi_2a_2 + \phi_3a_3$ , etc., and the condition  $a'_2f'_2 + b'_2g'_2 + c'_2h'_2 = 0$  is satisfied: and the third set by  $a'_3, b'_3, c'_3, f'_3, g'_3, h'_3$ , where  $a'_3 = \psi_1a_1 + \psi_2a_2 + \psi_3a_3$ , etc., and the condition  $a'_3f'_3 + b'_3g'_3 + c'_3h'_3 = 0$  is satisfied. We have therefore an equation  $T' = (a'g'h')x^2 + \text{etc.}$ , which only differs from the equation  $T = 0$  by having therein the accented letters in place of the unaccented ones: and, substituting for the accented letters their values, the whole divides by the determinant  $(\theta\phi\psi)$ , and throwing this out we obtain the required equation  $T = 0$ .

But it is easier to obtain the equation  $T = 0$  directly. We have

$$\begin{aligned} hy - gz &+ aw = 0, \\ -hx &+ fz + bw = 0, \\ gx - fy &+ cw = 0, \\ -ax - by - cz &= 0; \end{aligned}$$

viz. in virtue of the equation  $af + bg + ch = 0$  which connects the six coordinates, these four equations are equivalent to two independent equations which are the equations of the line  $(a, b, c, f, g, h)$ : or, what is the same thing, any three of these equations imply the fourth equation and also the relation  $af + bg + ch = 0$ .

We might, from the three linear relations and any three of the last-mentioned four equations, eliminate  $a, b, c, f, g, h$  and so obtain the required equation  $T = 0$ ; but it is better, introducing the arbitrary coefficients  $\alpha, \beta, \gamma, \delta$ , to employ all the four equations. The result of the elimination is thus given in the form

$$\begin{vmatrix} \alpha, & w, & -z, & y & & & \\ \beta, & & w, & z, & -x & & \\ \gamma, & & & w, & -y, & x & \\ \delta, & x, & y, & z, & & & \\ & f_1, & g_1, & h_1, & a_1, & b_1, & c_1 \\ & f_2, & g_2, & h_2, & a_2, & b_2, & c_2 \\ & f_3, & g_3, & h_3, & a_3, & b_3, & c_3 \end{vmatrix} = 0,$$

viz. the left-hand side here contains the factor  $-(\alpha x + \beta y + \gamma z + \delta w)$ ; throwing this out, we obtain the required quadric equation  $T = 0$ . If for the calculation of  $T$  we compare the terms containing  $\delta$ , we have

$$Tw = \begin{vmatrix} w, & -z, & y & & & \\ & w, & z, & -x & & \\ & & w, & -y, & x & \\ f_1, & g_1, & h_1, & a_1, & b_1, & c_1 \\ f_2, & g_2, & h_2, & a_2, & b_2, & c_2 \\ f_3, & g_3, & h_3, & a_3, & b_3, & c_3 \end{vmatrix},$$

where observe that, writing  $w = 0$ , the right-hand side vanishes as containing the factor

$$\begin{vmatrix} & -z, & y \\ z, & & -x \\ -y, & x, & \end{vmatrix}.$$

Hence the right-hand side divides by  $w$ ; and one of its terms being evidently  $w^2(abc)$ ,  $T$  contains as it should do the term  $(abc)w^2$ : the remaining terms can be found without any difficulty, and the foregoing expression for  $T$  is thus verified.

## 672.

## ON THE GAME OF MOUSETRAP.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 8–10.]

In the note “A Problem in Permutations,” *Quarterly Mathematical Journal*, t. I. (1857), p. 79, [161], I have spoken of the problem of permutations presented by this game.

A set of cards—ace, two, three, &c., say up to thirteen—are arranged (in any order) in a circle with their faces upwards; you begin at any card, and count one, two, three, &c., and if upon counting, suppose the number five, you arrive at the card five, the card is thrown out; and beginning again with the next card, you count one, two, three, &c., throwing out (if the case happen) a new card as before, and so on until you have counted up to thirteen, without coming to a card which has to be thrown out. The original question proposed was: for any given number of cards to find the arrangement (if any) which would throw out all the cards in a given order; but (instead of this) we may consider all the different arrangements of the cards, and inquire how many of these there are in which all or any given smaller number of the cards will be thrown out; and (in the several cases) in what orders the cards are thrown out. Thus to take the simple case of four cards, the different arrangements, with the cards thrown out in each, are

|            |             |
|------------|-------------|
| 1, 2, 3, 4 | 1,          |
| 1, 2, 4, 3 | 1,          |
| 1, 3, 2, 4 | 1, 2, 3, 4, |
| 1, 3, 4, 2 | 1,          |
| 1, 4, 2, 3 | 1,          |
| 1, 4, 3, 2 | 1,          |
| 2, 1, 3, 4 | 3, 4,       |
| 2, 1, 4, 3 | — — —       |
| 2, 3, 4, 1 | — — —       |
| 2, 3, 1, 4 | 4,          |
| 2, 4, 1, 3 | — — —       |
| 2, 4, 3, 1 | — — —       |
| 3, 1, 2, 4 | 4,          |
| 3, 1, 4, 2 | 3, 2,       |
| 3, 2, 1, 4 | — — —       |
| 3, 2, 4, 1 | 4, 2, 1, 3, |
| 3, 4, 1, 2 | — — —       |
| 3, 4, 2, 1 | — — —       |
| 4, 1, 2, 3 | — — —       |
| 4, 1, 3, 2 | 3,          |
| 4, 2, 1, 3 | 2, 1, 3, 4, |
| 4, 2, 3, 1 | 3, 1, 2, 4, |
| 4, 3, 1, 2 | — — —       |
| 4, 3, 2, 1 | — — —       |

Classifying these so as to show in how many arrangements a given card or permutation of cards is thrown out, we have the table

| No.   | Thrown out  |
|-------|-------------|
| 9     | none        |
| 4     | 1           |
| 1     | 3           |
| 2     | 4           |
| 1     | 3, 2        |
| 1     | 2, 3        |
| 1     | 3, 4        |
| 1     | 1, 3, 4, 2  |
| 1     | 1, 2, 3, 4  |
| 1     | 4, 2, 1, 3  |
| 1     | 2, 1, 3, 4  |
| 1     | 3, 1, 2, 4, |
| <hr/> |             |
| 24    |             |

viz. there are nine arrangements in which no card is thrown out, four arrangements in which only the card 1 is thrown out, one arrangement in which only the card 3 is thrown out, and so on.

It will be observed that there are five arrangements in which all the cards are thrown out, each throwing them out in a different order; there are thus only five orders in which all the cards are thrown out.

The general question is of course to form a like table for the numbers 5, 6, ..., or any greater number of cards.

## 673.

## NOTE ON THE THEORY OF CORRESPONDENCE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 32, 33.]

If the point  $P$  on a given curve  $U$  of the order  $m$ , and the point  $Q$  on a given curve  $V$  of the order  $m'$ , have a  $(1, 1)$  correspondence, this implying that the two curves have the same deficiency; then if  $PQ$  intersects the consecutive line  $P'Q'$  in a point  $R$ , the locus of  $R$  is a curve  $W$  of the class  $m' + m$ , and the point  $R$  on this curve has, in general (but not universally), a  $(1, 1)$  correspondence with the point  $P$  on  $U$ , or with the point  $Q$  on  $V$ . For, considering the correspondence of the points  $P$  and  $R$ , to a given position of  $P$  there corresponds, it is clear, a single position of  $R$ ; on the other hand, starting from  $R$ , the tangent at this point to the curve  $W$  meets the curve  $U$  in  $m$  points and the curve  $V$  in  $m'$  points, but it is in general only one of the  $m$  points and only one of the  $m'$  points which are corresponding points on the curves  $U$  and  $V$ ; that is, it is only one of the  $m$  points which is a point  $P$ ; and the correspondence of  $(P, R)$  is thus a  $(1, 1)$  correspondence.

But the curves  $U, V$  may be such that the correspondence of  $(P, R)$  is not a  $(1, 1)$  but a  $(k, 1)$  correspondence; viz., that to a given position of  $P$  there corresponds a single position of  $R$ , but to a given position of  $R$ ,  $k$  positions of  $P$ . To show that this is so, imagine through  $P$  a line  $\Pi$  having therewith a  $(k, 1)$  correspondence;  $P$  being, as above, a point on the curve  $U$ , the line in question envelopes a curve  $W$ ; and the correspondence is such that, for any given position of  $P$  on the curve  $U$ , we have through it a single position of the line: but, for a given tangent of the curve  $W$ , we have upon it  $k$  positions of the point  $P$ , viz.  $k$  of the intersections of the line with the curve  $U$  are points corresponding to the line; this, of course, implies that the curve  $U$  is not any curve whatever of the order  $m$ , but a curve of a peculiar nature.

Imagine now that we have on the line  $\Pi$  a point  $Q$ , having with  $P$  a  $(1, 1)$  correspondence of a given nature: to fix the ideas, suppose  $P, Q$  are harmonics in regard to a given conic: since on each of the lines  $\Pi$  there are  $k$  positions of  $P$ , there are also on the line  $k$  positions of  $Q$ , and the locus of these  $k$  points  $Q$  is a curve  $V$ , say of the order  $m'$ .

The point  $P$  on the curve  $U$  and the point  $Q$  on the curve  $V$  have a  $(1, 1)$  correspondence. For, consider  $P$  as given: there is a single position of the line  $\Pi$  intersecting  $V$  in  $m'$  points, but obviously only one of these is the point  $Q$ . And consider  $Q$  as given: then through  $Q$  we have say  $\mu$  tangents of the curve  $W$ ; each of these tangents intersects the curve  $U$  in  $m$  points,  $k$  of which are points  $P$ , but for a tangent taken at random no one of these is the correspondent of  $Q$ ; it is, in general, only one of the  $\mu$  tangents which has upon it  $k$  points  $P$ , one of them being the point corresponding to  $Q$ ; that is, to a given position of  $Q$  there corresponds a single position of  $P$ ; and the correspondence of the points  $(P, Q)$  is thus a  $(1, 1)$  correspondence.

We have thus the point  $P$  on the curve  $U$  and the point  $Q$  on the curve  $V$ , which points have with each other a  $(1, 1)$  correspondence; and the line  $\Pi$  is the line joining these points; this intersects the consecutive line  $PQ$  in a point  $R$ ; and the locus of  $R$  is the curve  $W$ . To a given position of  $P$  there corresponds a single line  $\Pi$ , and therefore a single position of  $R$ ; but to a given position of  $R$  there correspond  $k$  positions of  $P$ , viz. drawing at  $R$  the tangent to the curve  $W$ , this is a line  $\Pi$  having upon it  $k$  points  $P$ , or the correspondence of  $(P, Q)$  is, as stated, a  $(k, 1)$  correspondence.

The foregoing considerations were suggested to me by the theory of parallel curves. Take a curve parallel to a given curve, for example, the ellipse; this is a curve of the order 8, such that every normal thereto is a normal at two distinct points; and the curve has as its evolute the evolute of the ellipse, or, more accurately, the evolute of the ellipse taken *twice*; but, attending only to the evolute taken once, each tangent of the evolute is a normal of the parallel curve at two distinct points thereof, and the points of the parallel curve have with those of the evolute not a  $(1, 1)$  but a  $(2, 1)$  correspondence.



## 674.

## NOTE ON THE CONSTRUCTION OF CARTESIANS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), p. 34.]

If  $\rho = a + b \cos \theta$ , and  $r = \frac{1}{2}\{\rho \pm \sqrt{(\rho^2 - c^2)}\}$ , then obviously  $r^2 - r\rho + \frac{1}{4}c^2 = 0$ , that is,

$$r^2 - r(a + b \cos \theta) + \frac{1}{4}c^2 = 0,$$

which is the equation of a Cartesian. Here  $\rho = a + b \cos \theta$  is the equation of a limaçon or nodal Cartesian, having the origin for the node; and for any given value of  $\theta$ , deducing from the radius vector of the limaçon the new radius vector  $r$  by the above formula  $r = \frac{1}{2}\{\rho \pm \sqrt{(\rho^2 - c^2)}\}$ , we obtain a Cartesian, or by giving different values to  $c$ , a series of Cartesians having the origin for a common focus. The construction is a very convenient one.

## 675.

## ON THE FLEFLECNODAL PLANES OF A SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 49–51.]

If at a node (or double point) of a plane curve there is on one of the branches an inflexion, (that is, if the tangent has a 3-pointic intersection with the branch), the node is said to be a flecnod; and if there is on each of the branches an inflexion, then the node is said to be a fleflecnode. The tangent plane of a surface intersects the surface in a plane curve having at the point of contact a node; if this is a flecnod or a fleflecnode, the tangent plane is said to be a flecnodal or a fleflecnodal plane accordingly. For a quadric surface each tangent plane is fleflecnodal: this is obvious geometrically (since the section is a pair of lines), and it will presently appear that the analytical condition for such a plane is satisfied. In fact, if the origin be taken at a point of a surface, so that  $z = 0$  shall be the equation of the tangent plane, then in the neighbourhood of the point we have

$$z = (x, y)^2 + (x, y)^3 + \&c.;$$

and the condition for a fleflecnodal plane is that the term  $(x, y)^2$  shall be a factor of the succeeding term  $(x, y)^3$ . Now for a quadric surface the equation is

$$z = \frac{1}{2}\{ax^2 + 2hxy + by^2 + 2(fy + gx)z + cz^2\}$$

that is,

$$z(1 - fy - gx - \frac{1}{2}cz) = \frac{1}{2}(ax^2 + 2hxy + by^2),$$

or developing as far as the third order in  $(x, y)$ , we have

$$z = \frac{1}{2}(ax^2 + 2hxy + by^2)(1 + fy + gx),$$

so that the condition in question is satisfied.

In what follows, I take for greater simplicity  $h = 0$ , (viz.  $x = 0, y = 0$  are here the tangents to the two curves of curvature at the point in question), and to avoid fractions write  $2f, 2g$  in place of  $f, g$  respectively; the developed equation of the quadric surface is thus

$$z = \frac{1}{2}(ax^2 + by^2) + (ax^2 + by^2)(gx + fy).$$

I consider the parallel surface, obtained by measuring off on the normal a constant length  $k$ . If, as usual,  $p, q$  denote  $\frac{dz}{dx}$  and  $\frac{dz}{dy}$  respectively, then, in general,  $(X, Y, Z)$  being the coordinates of the point on the parallel surface,

$$Z = z + \frac{k}{\sqrt{1 + p^2 + q^2}},$$

$$X = x - \frac{kp}{\sqrt{1 + p^2 + q^2}},$$

$$Y = y - \frac{kq}{\sqrt{1 + p^2 + q^2}}.$$

But in the present case

$$p = ax + 3agx^2 + 2afxy + bgy^2,$$

whence

$$q = by + afx^2 + 2bgxy + 3bfy^2,$$

$$X = x - k\{ax + 3agx^2 + 2afxy + bgy^2\},$$

$$Y = y - k\{by + afx^2 + 2bgxy + 3bfy^2\};$$

or, putting for convenience,

$$X = (1 - ka)\xi, \quad Y = (1 - kb)\eta,$$

then, for a first approximation  $x = \xi, y = \eta$ ; whence, writing

$$P = 3ag\xi^2 + 2af\xi\eta + bg\eta^2, \quad Q = af\xi^2 + 2bg\xi\eta + 3bf\eta^2,$$

we find

$$x = \xi + \frac{kP}{1 - ka}, \quad y = \eta + \frac{kQ}{1 - kb},$$

and thence

$$p = a\left(\xi + \frac{kP}{1 - ka}\right) + P = a\xi + \frac{P}{1 - ka},$$

$$q = b\eta + \frac{Q}{1 - kb}.$$

Hence

$$Z = k + \frac{1}{2}\{a(1 - ka)\xi^2 + b(1 - kb)\eta^2\} + (a\xi^2 + b\eta^2)(g\xi + f\eta).$$

where, changing the origin to the point  $x = 0, y = 0, z = k$  on the parallel surface, the coordinates of the consecutive point are  $Z - k, X = (1 - ka)\xi$ , and  $Y = (1 - kb)\eta$ .

We cannot, by any determination of the value of  $k$ , make the plane  $Z - k = 0$  a flefnodal plane of the parallel surface; but if

$$\frac{a\xi^2 + b\eta^2}{a^2\xi^2 + b^2\eta^2} = k,$$

then

$$1 - ka = \frac{b\eta^2(a - b)}{a^2\xi^2 + b^2\eta^2}, \quad 1 - kb = \frac{a\xi^2(b - a)}{a^2\xi^2 + b^2\eta^2},$$

and the equation becomes

$$Z - k = \frac{1}{2} \frac{ab(b - a)}{a^2\xi^2 + b^2\eta^2} (\xi^2 - \eta^2)(a^2\xi^2 + b^2\eta^2) (g\xi + f\eta),$$

viz. the term of the second has here a factor  $g\xi + f\eta$  which divides the term of the third order, and the equation  $Z - k = 0$  is a flecnodal plane of the parallel surface.

## 676.

## NOTE ON A THEOREM IN DETERMINANTS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 55–57.]

It is well known that if 12, &c., denote the determinants formed with the matrix

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \end{vmatrix},$$

then, identically,

$$12.34 + 13.42 + 14.23 = 0.$$

The proper proof of the theorem is obtained by remarking that we have

$$0 = \begin{vmatrix} \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \\ \alpha, & \beta, & \gamma, & \delta \\ \alpha', & \beta', & \gamma', & \delta' \end{vmatrix},$$

as at once appears by subtracting the first and second lines from the third and fourth lines respectively; and, this being so, the development of the determinant gives the theorem. The theorem might, it is clear, have been obtained in four different forms according as in the determinant the missing terms were taken to be as above  $(\delta, \delta')$ , or to be  $(\alpha, \alpha')$ ,  $(\beta, \beta')$ , or  $(\gamma, \gamma')$ ; but the four results are equivalent to each other.

There is obviously a like theorem for the sums of products of determinants formed with the matrix

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \end{vmatrix},$$

viz. the theorem is obtained by development of the determinant in an identical equation, such as

$$0 = \begin{vmatrix} \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \\ \alpha, & \beta, & \gamma, & \delta, & \epsilon, & \zeta \\ \alpha', & \beta', & \gamma', & \delta', & \epsilon', & \zeta' \\ \alpha'', & \beta'', & \gamma'', & \delta'', & \epsilon'', & \zeta'' \end{vmatrix},$$

but we thus obtain 15 results which are not all equivalent.

If, for shortness, we write

$$\begin{aligned} A &= 123 \cdot 456, \\ -B &= 124 \cdot 356, \\ -C &= 125 \cdot 346, \\ D &= 126 \cdot 345, \\ -E &= 134 \cdot 256, \\ -F &= 135 \cdot 246, \\ G &= 136 \cdot 245, \\ -H &= 145 \cdot 236, \\ I &= 146 \cdot 235, \\ J &= 156 \cdot 234, \end{aligned}$$

then the fifteen results are

$$\begin{aligned} A + B - C - D &= 0, \\ A + B - E - J &= 0, \\ A - C + F - I &= 0, \\ A - D + G - H &= 0, \\ A - E + F + G &= 0, \\ A - H - I - J &= 0, \\ B - C - G + H &= 0, \\ B - D - F + I &= 0, \\ B - E + H + I &= 0, \\ B - F - G - J &= 0, \\ C + D - E - J &= 0, \\ C - E + G + I &= 0, \\ C - F - H - J &= 0, \\ D - E + F + H &= 0, \\ D - G - I - J &= 0, \end{aligned}$$

which are all satisfied if only

$$\begin{aligned} A &= \dots + H + I + J, \\ B &= F + G + \dots + J, \\ C &= F + \dots + H + J, \\ D &= \dots + G + \dots + J, \\ E &= F + G + H + I + J; \end{aligned}$$

and we thus have these five relations between the ten products of determinants  $A, B, C, D, E, F, G, H, I, J$ .

## 677.

## [ADDITION TO MR GLAISHER'S PAPER "PROOF OF STIRLING'S THEOREM."]

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 63, 64.]

It is easy to extend Mr Glaisher's investigation so as to obtain from it the more approximate value

$$\Pi n = \sqrt{(2\pi)} n^{n+\frac{1}{2}} e^{-n-\frac{1}{12n}}.$$

We, in fact, have

$$\psi x = e^{ax+bx^2+cx^3+\dots},$$

where  $a, b, \dots$  are given functions of  $n$ , viz.

$$a = \frac{1}{2} \left\{ \frac{1}{3^2} + \frac{1}{5^2} + \dots + \frac{1}{(2n+1)^2} \right\},$$

$$b = \frac{1}{3} \left\{ \frac{1}{3^3} + \frac{1}{5^3} + \dots + \frac{1}{(2n+1)^3} \right\},$$

&c.

And hence writing  $x = 1$ , we have

$$\psi(1) = \frac{1}{2} \sum_0^n \frac{1}{(2n+2)^{n+1}} (2n+2)^{n+1} = e^{a+b+c+\dots},$$

that is,

$$\begin{aligned} \Pi n &= \left( \frac{2n+2}{e} \right)^{n+1} e^{-(n+1)+(a+b+\dots)} \\ &= (n+1)^{n+1} e^{-n-1+(a+b+\dots)} \\ &= n^{n+1} \left( 1 + \frac{1}{n} \right)^{n+1} e^{-n-1+(a+b+\dots)}. \end{aligned}$$



Hence for  $(1 + \frac{1}{n})^{n+1}$  writing  $e^{(n+1)\log(1+\frac{1}{n})}$ , the whole exponent of  $e$  is

$$\begin{aligned} & (n+1)\log\left(1 + \frac{1}{n}\right) - n - 1 + (a + b + \dots) \\ &= (n+1)\left\{\frac{1}{n} - \frac{1}{2n^2} + \frac{1}{3n^3} - \dots\right\} - n - 1 + (a + b + \dots) \\ &= -n + 1 + \frac{1}{2} - \frac{1}{3n} + \frac{1}{4n^2} - \dots - n - 1 + (a + b + \dots) \\ &= -\frac{1}{2}(a + b + \dots). \end{aligned}$$

We have

$$\frac{1}{1} \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{5} + \frac{1}{3} \cdot \frac{1}{7} + \text{const.} + \frac{1}{(2n+1)} + \text{terms in } \frac{1}{n}, \frac{1}{n^2}, \&c.$$

(the constant is in fact  $= \frac{1}{2}\pi$ , but the value is not required), hence  $a = \text{const.} - \frac{1}{2n} + \text{terms in } \frac{1}{n^2}, \frac{1}{n^3}, \&c.$ ; as regards  $b, c, \&c.$ , there are no terms in  $\frac{1}{n}$ , but we have  $b = \text{const.} + \text{terms in } \frac{1}{n^2}, \&c.$ ,  $c = \text{const.} + \text{terms in } \frac{1}{n^2}, \&c.$  Hence the whole exponent of  $e$  is

$$= -n + C^* + \frac{1}{2} - \frac{1}{12n} + \text{terms in } \frac{1}{n^2}, \&c.$$

As in Mr Glaisher's investigation, it is shown that  $e^{C^* + \frac{1}{2}} = \sqrt{(2\pi)}$ , and hence neglecting the terms in  $\frac{1}{n^2}, \&c.$ , the final result is

$$\Pi n = \sqrt{(2\pi)} n^{n+\frac{1}{2}} e^{-n-\frac{1}{12n}}.$$

## 678.

## ON A SYSTEM OF QUADRIC SURFACES.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 124, 125.]

The following theorem was communicated to me by Dr Klein; “given in regard to a quadric surface two sibi-reciprocal line-pairs, the two tractors (or lines meeting each of the four lines) form a sibi-reciprocal line-pair.” This may be presented under a more general form as a theorem relating to the tractors of any two line-pairs. In fact, if a given line-pair is taken to be sibi-reciprocal in regard to a quadric surface, we thereby establish only a four-fold relation between the coefficients of the surface, and the surface will still depend on five arbitrary parameters. Whence if two given line-pairs are taken to be each of them sibi-reciprocal in regard to one and the same quadric surface, we thereby establish only an eight-fold relation and the surface will still depend upon one arbitrary parameter. The theorem thus is: given any two line-pairs, then each of these, and also the pair of tractors, are sibi-reciprocal in regard to a singly infinite system of quadric surfaces.

The question arises, what is this system of quadric surfaces? It is, in fact, the system of surfaces having in common a skew quadrilateral constructed as follows: starting from the two given line-pairs, construct the two tractors, each of them intersected by the given line-pairs in two point-pairs; and on each tractor construct the double or sibi-reciprocal points of the involution thus determined; these double points are the vertices (those on the same tractor being opposite vertices) of the skew quadrilateral; which is consequently at once obtained by joining the two double points on the one tractor with the two double points on the other tractor. The construction is an immediate consequence of the following theorem: consider a skew quadrilateral, and drawing its two diagonals, take a pair of lines cutting each diagonal harmonically; these will be sibi-reciprocal in regard to any quadric surface through the skew quadrilateral.

The condition of passing through a skew quadrilateral is that of passing through a certain system of eight points; in fact, the eight points may be taken to be the four vertices and any four points on the four sides respectively. But observe that the system of the quadric surfaces through any eight points has the characteristics  $(1, 2, 3)$ ; viz. there are in the system 1 surface passing through a given point, 2 touching a given line, 3 touching a given plane; the system of surfaces through the same skew quadrilateral has the characteristics  $(1, 2, 1)$ .

## 679.

## ON THE REGULAR SOLIDS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 127–131.]

In a regular solid, or say in the spherical figure obtained by projecting such solid, by lines from the centre, on the surface of a concentric sphere, we naturally consider 1° the summits, 2° the centres of the faces, 3° the mid-points of the sides. But, imagining the five regular figures drawn in proper relation to each other on the same spherical surface, the only points which have thus to be considered are 12 points  $A$ , 20 points  $B$ , 30 points  $\Theta$ , and 60 points  $\Phi$ . These may be, in the first instance, described by reference to the dodecahedron; viz. the points  $A$  are the centres of the faces, the points  $B$  are the summits, the points  $\Theta$  are the mid-points of the sides, and the points  $\Phi$  are the mid-points of the diagonals of the faces (viz. there are thus 5 points  $\Phi$  in each face of the dodecahedron, or in all 60 points  $\Phi$ ). But reciprocally we may describe them in reference to the icosahedron; viz. the points  $A$  are the summits, the points  $B$  the centres of the faces, the points  $\Theta$  the mid-points of the sides, (viz. each point  $\Theta$  is the common mid-point of a side of the dodecahedron and a side of the icosahedron, which sides there intersect at right angles), and the points  $\Phi$  are points lying by 3's on the faces of the icosahedron, each point of the face being given as the intersection of a perpendicular  $A\Theta$  of the face by a line  $BB$ , joining the centres of two adjacent faces and intersecting  $A\Theta$  at right angles.

The points  $A$  lie opposite to each other in pairs in such wise that, taking any two opposite points as poles, the relative situation is as follows:

| $A$ | Longitudes                                             |
|-----|--------------------------------------------------------|
| 1   | —                                                      |
| 5   | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$   |
| 5   | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$ |
| 1   | —                                                      |

where the points  $A$  in the same horizontal line form a zone of points equidistant from the point taken as the North Pole. And the points  $B$  lie also opposite to

each other in such wise that, taking two opposite points as poles, the relative situation is as follows:

| $B$ | Longitudes                                        |
|-----|---------------------------------------------------|
| 1   | —                                                 |
| 3   | $0^\circ, 120^\circ, 240^\circ$                   |
| 6   | $(0^\circ, 120^\circ, 240^\circ) + 22^\circ 14'$  |
| 6   | $(60^\circ, 180^\circ, 300^\circ) + 22^\circ 14'$ |
| 3   | $60^\circ, 180^\circ, 300^\circ$                  |
| 1   | —                                                 |

where the points  $B$  in the same horizontal line form a zone of points equidistant from the point taken as the North Pole. Neglecting the  $3 + 3$   $B$  points which lie adjacent to the poles, the remaining 14 points  $B$  may be arranged as follows ( $\beta = 22^\circ 14'$  as above):

| $B$ | Longitudes                                                                                                       |
|-----|------------------------------------------------------------------------------------------------------------------|
| 1   | —                                                                                                                |
| 6   | $120^\circ + \beta, 240^\circ + \beta; 120^\circ - \beta, 240^\circ - \beta$                                     |
| 6   | $60^\circ + \beta, 180^\circ + \beta, 300^\circ + \beta; 60^\circ - \beta, 180^\circ - \beta, 300^\circ - \beta$ |
| 1   | —                                                                                                                |

And taking the two poles separately with each system of the remaining poles, we have 2 systems each of 8 points  $B$ , which are, in fact, the summits of a cube (hexahedron); each point  $B$  taken as North Pole thus belongs to two cubes; but inasmuch as the cube has 8 summits, the number of the cubes thus obtained is  $20 \times 2 \div 8, = 5$ ; viz. the 20 points  $B$  form the summits of 5 cubes, each point of course belonging to 2 cubes.

It is to be added that, considering the 5 points  $B$  which form a face of the dodecahedron, any diagonal  $BB$  of this dodecahedron is a side of a cube. We have thus  $12 \times 5, = 60$ , the number of the sides of the 5 cubes.

It is at once seen that the centres of the faces of a cube are points  $\Theta$ , and that the mid-points of the sides of the cube are points  $\Phi$ .

To each cube there corresponds of course an octahedron, the summits being points  $\Theta$ , the centres of the faces points  $B$ , and the mid-points of the sides points  $\Phi$ ; thus, for the five octahedra the summits are the  $5 \times 6, = 30$ , points  $\Theta$ ; the centres of the faces are  $5 \times 8, = 40$ , points  $B$  (each point  $B$  being thus a centre of face for two octahedra), and the mid-points of the sides being the  $5 \times 12, = 60$ , points  $\Phi$ .

Finally, considering the 8 points  $B$  which belong to a cube, we can, in four different ways, select thereout 4 points  $B$  which are the summits of a tetrahedron:

the remaining 4 points  $B$  are then the centres of the faces, and the mid-points of the sides are points  $\Theta$ : there are thus  $5 \times 4, = 20$ , tetrahedra having  $20 \times 4$  summits which are the 20 points  $B$  each 4 times;  $20 \times 4$  centres of faces which are the 20 points  $B$  each 4 times; and  $20 \times 6$  mid-points of sides which are the 30 points  $\Theta$  each 4 times.

It thus appears that, as mentioned above, the five regular figures depend only on the points  $A, B, \Theta$ , and  $\Phi$ .

We might take as poles two opposite points  $A, B, \Theta$ , or  $\Phi$ ; and in each case determine in reference to these the positions of the other points; but for brevity I consider only the case in which we take as poles two opposite points  $A$ . We have the following table:

**Poles two opposite points  $A$ .**

|              | N. P. D.        | Longitudes                                                              |
|--------------|-----------------|-------------------------------------------------------------------------|
| $A_1$        | $0^\circ$       | $0^\circ$                                                               |
| $5A_1$       | $63^\circ 26'$  | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $5A_2$       | $116^\circ 34'$ | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $A_2$        | $180^\circ$     |                                                                         |
| $5B_1$       | $37^\circ 22'$  | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $5B_2$       | $79^\circ 12'$  | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $5B_3$       | $100^\circ 48'$ | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $5B_4$       | $142^\circ 38'$ | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $5\Theta_1$  | $31^\circ 43'$  | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $5\Theta_2$  | $58^\circ 17'$  | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $10\Theta_3$ | $90^\circ$      | $(0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ) + 18^\circ$       |
| $5\Theta_4$  | $121^\circ 43'$ | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $5\Theta_5$  | $148^\circ 17'$ | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $5\Phi_1$    | $13^\circ 16'$  | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $10\Phi_2$   | $52^\circ 52'$  | $(0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ) + 9^\circ 44'$    |
| $10\Phi_3$   | $68^\circ 10'$  | $(0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ) + 13^\circ 35'$   |
| $10\Phi_4$   | $76^\circ 42'$  | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $10\Phi_5$   | $103^\circ 18'$ | $0^\circ, 72^\circ, 144^\circ, 216^\circ, 288^\circ$                    |
| $5\Phi_6$    | $111^\circ 50'$ | $36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ$                  |
| $5\Phi_7$    | $127^\circ 8'$  | $(36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ) + 13^\circ 35'$ |
| $5\Phi_8$    | $166^\circ 44'$ | $(36^\circ, 108^\circ, 180^\circ, 252^\circ, 324^\circ) + 9^\circ 44'$  |
|              |                 | $108^\circ.$                                                            |

I add for greater completeness the following results, some of which were used in the calculation of the foregoing table. Considering successively (1) the tetrahedral triangle, summits 3 points  $B$ , centre a point  $B$ ; (2) the hexahedral square, summits 4 points  $B$ , centre a point  $\Theta$ ; (3) the octahedral triangle, summits 3 points  $\Theta$ , centre a point  $B$ ; (4) the icosahedral triangle, summits 3 points  $A$ , centre a point  $B$ ; (5) the dodecahedral pentagon, summits 5 points  $A$ , centre a point  $B$ ; and (6), what may be called the small pentagon, summits 5 points  $\Phi$  lying within a dodecahedral pentagon, and having therewith the common centre  $B$ ; we may in each case write  $s$  the side,  $r$  the radius or distance of the centre from a summit,  $p$  the perpendicular or distance of the centre from a side. And the values then are

|                   | $s$      | $r$     | $p$     |
|-------------------|----------|---------|---------|
| Tet. $\triangle$  | 109° 30' | 70° 30' | 54° 45' |
| Hex. square       | 90       | 54 45   | 45      |
| Oct. $\triangle$  | 63 26    | 37 22   | 35 15   |
| Icos. $\triangle$ | 41 50    | 37 22   | 20 55   |
| Dod. pentagon     | 15 30    | 13 16   | 31 43   |
| Small pentagon    |          |         | 10 48   |

## 680.

## ON THE HESSIAN OF A QUARTIC SURFACE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 141–144.]

The surface considered is

$$U = k^2 w^2 \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) - (x^2 + y^2 + z^2)^2 = 0,$$

or say

$$U = k^2 w^2 P - Q^2 = 0,$$

viz. this may be considered as the central inverse of the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1.$$

The values of the second derived functions, or terms of the Hessian determinant

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix},$$

are

$$\begin{array}{cccc} \frac{k^2}{a^2} w^2 - 2Q - 4x^2, & -4xy, & -4xz, & \frac{2k^2}{a^2} wx \\ -4xy, & \frac{k^2}{b^2} w^2 - 2Q - 4y^2, & -4yz, & \frac{2k^2}{b^2} wy \\ -4xz, & -4yz, & \frac{k^2}{c^2} w^2 - 2Q - 4z^2, & \frac{2k^2}{c^2} wz \\ \frac{2k^2}{a^2} wx, & \frac{2k^2}{b^2} wy, & \frac{2k^2}{c^2} wz, & 2k^2 P \end{array}$$

and we thence have

$$bc - f^2 = \frac{k^4}{b^2c^2}w^4(3Q + 4y^2 + 4z^2) - \frac{2k^2w^2}{b^2c^2}(3Q + 4y^2 + 4z^2)Q + 4Q^2 - 24Q^3,$$

whence, forming the analogous quantities  $-ca - g^2$ , &c., it is easy to obtain

$$\begin{aligned} & abc - af^2 - bg^2 - ch^2 + 2fgh \\ &= -\frac{k^6w^6}{a^2b^2c^2} \left\{ 4Q^3 + 2Q^2 \left( \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) + 4 \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) P \right\} \\ & \quad + \frac{2k^4w^4}{a^2b^2c^2} \left\{ 12Q \left( \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) + 4 \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) \right\} \\ & \quad - 24Q^3, \end{aligned}$$

which is to be multiplied by  $d, = k^2P$ . And

$$\begin{aligned} & -[l^2(bc - f^2) + m^2(ca - g^2) + n^2(ab - h^2) \\ & \quad + 2mn(gh - af) + 2nl(hf - bg) + 2lm(fg - ch)] \\ &= -\frac{4k^6w^6}{a^2b^2c^2} \left[ Q \left\{ \frac{x^2}{a^2} \left( \frac{1}{b^2} + \frac{1}{c^2} \right) + \frac{y^2}{b^2} \left( \frac{1}{c^2} + \frac{1}{a^2} \right) + \frac{z^2}{c^2} \left( \frac{1}{a^2} + \frac{1}{b^2} \right) \right\} \right. \\ & \quad \left. + 4 \left( \frac{x^4}{a^2b^2c^2} \left( \frac{1}{b^2} + \frac{1}{c^2} \right) + \dots \right) - 4 \left( \frac{x^2y^2}{a^2b^2} \left( \frac{1}{a^2} + \frac{1}{b^2} \right) + \dots \right) \right] \\ & \quad + 4k^4w^4 \left[ 3Q \left( \frac{x^4}{a^4} + \frac{y^4}{b^4} + \frac{z^4}{c^4} \right) - 4P \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) \right. \\ & \quad \left. + 12 \left( \frac{x^4}{a^4}Q^{-1} + \dots \right) - 4P \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) Q + 8P^2Q \right. \\ & \quad \left. + 24P^2Q \right] \end{aligned}$$

which is

$$\begin{aligned} & 4k^6w^6 \Delta P \\ & + 8w^4 \left[ -48Q + \left( \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) Q + 32P^2Q \right]. \end{aligned}$$



Hence, uniting the two parts, we have

$$\begin{aligned}
 H &= 8w^4 \left( -\frac{8}{a^2b^2c^2} P \right) \\
 &+ 2k^4w^4 \left\{ -6 \left( \frac{x^4}{a^4b^2c^2} + \frac{y^4}{a^2b^4c^2} + \frac{z^4}{a^2b^2c^4} \right) PQ \right. \\
 &\quad \left. - \frac{24Q^3}{a^2b^2c^2} + 12 \left( \frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) P \right\} \\
 &+ 2k^4w^4 \left\{ 12 \left( \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2} \right) PQ \right. \\
 &\quad \left. - 48 \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right) Q^2 \right. \\
 &\quad \left. + 24P^2Q \right\} \\
 &+ k^2w^2(-24PQ^3).
 \end{aligned}$$

Writing herein  $Q^2 = k^2w^2P - U$ , and transposing all terms which contain  $U$ , we have

$$\begin{aligned}
 H &= k^6w^6P \left\{ -\frac{8}{a^2b^2c^2} P + 2 \left( \frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q - 4 \left( \frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) \right\} \\
 &- 48k^4w^4P^2 + 12k^4w^4 \left[ -\frac{2Q^2}{a^2b^2c^2} + 2 \left( \frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q - 8k^2w^2 \left( \frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) P - 24k^2w^2Q \right]
 \end{aligned}$$

where, in the term in  $\{ \}$ , the last four lines are

$$\begin{aligned}
 &= -16 \left( \frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q \\
 &\quad - 32 \left( \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right).
 \end{aligned}$$

Hence, writing for shortness

$$\Theta = \dots - \frac{32}{a^2b^2c^2}w^2 + 2k^2w^2 \left( \frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2} \right) Q - 4k^2w^2 \left( \frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4} \right) - 32PQ.$$

we have

$$H = 12k^6w^6P \cdot \Theta = 16k^4w^4 \left\{ -\frac{32}{a^2b^2c^2}w^2 + 2\left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2}\right)Q - 4\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right) \right\}.$$

Hence, recollecting that  $U = k^2w^2P - Q^2$ , the Hessian curve of the order 32 breaks up into

$$U = 0, w^4 = 0, \text{ that is, } Q^2 = 0, w^4 = 0, \text{ or the nodal conic,}$$

$$w = 0, Q = 0, \text{ 8 times (order 16),}$$

$$U = 0, P = 0, \text{ that is, } Q^2 = 0, P = 0, \text{ or the quadriquadric,}$$

$$P = 0, Q = 0, \text{ 2 times (order 8),}$$

and into a curve (order 8) which is

$$-\frac{32}{a^2b^2c^2}w^2 + 2k^2w^2\left(\frac{1}{a^2b^2} + \frac{1}{b^2c^2} + \frac{1}{c^2a^2}\right)Q - 4k^2w^2\left(\frac{x^2}{a^4} + \frac{y^2}{b^4} + \frac{z^2}{c^4}\right) = 0,$$

viz. this, the intersection of the surface with a quadric surface, is the proper Hessian curve.

## 681.

## ON THE DERIVATIVES OF THREE BINARY QUANTICS.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 157–168.]

For a reason which will appear, instead of the ordinary factorial notation, I write  $\{\alpha 012\}$  to denote the factorial  $\alpha \cdot \alpha + 1 \cdot \alpha + 2$ , and so in other cases; and I consider the series of equations

$$\begin{aligned} (1) &= X, \\ (2) &= (\{\alpha 0\}, \{\beta 0\})(Y, Y'), \\ (3) &= (\{\alpha 01\}, 2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(Z, Z', Z''), \\ (4) &= (\{\alpha 012\}, 3\{\alpha 12\}\{\beta 2\}, 3\{\alpha 2\}\{\beta 12\}, \{\beta 012\})(W, W', W'', W'''), \\ &\quad \&c. \end{aligned}$$

where

$$\begin{aligned} X &= Y + Y', \\ Y &= Z + Z', \quad Y' = Z' + Z'', \\ Z &= W + W', \quad Z' = W' + W'', \quad Z'' = W'' + W''', \\ &\quad \&c. \end{aligned}$$

We have thus a series of linear equations serving to determine  $X$ ;  $Y, Y'$ ;  $Z, Z', Z''$ ;  $W, W', W'', W'''$ ; &c. We require in particular the values of  $X$ ;  $Y, Y'$ ;  $Z, Z''$ ;  $W, W'''$ ; &c., and I write down the results as follow:

$$\begin{aligned} X &= (1), \\ \{\alpha + \beta\}Y &= \{\alpha 0\}(1), \quad +1, \\ \{\alpha + \beta\}Y' &= \{\alpha 0\}(1), \quad -1; \end{aligned}$$

$$\begin{aligned}
 &\{\alpha + \beta 2\}(1), \quad \{\alpha + \beta 1\}(2), \quad \{\alpha + \beta 01\}(3), \\
 &\{\alpha + \beta 012\}Z = \{\beta 01\}, \quad +2\{\beta 1\}, \quad +1, \\
 &\{\alpha + \beta 012\}Z'' = \{\alpha 01\}, \quad -2\{\alpha 1\}, \quad +1; \\
 &\{\alpha + \beta 34\}(1), \quad \{\alpha + \beta 4\}(2), \quad \{\alpha + \beta 03\}(3), \quad \{\alpha + \beta 01\}(4); \\
 &\{\alpha + \beta 01 \dots 4\}W = \{\beta 012\}, \quad +3\{\beta 12\}, \quad +3\{\beta 2\}, \quad +1, \\
 &\{\alpha + \beta 01 \dots 4\}W''' = \{\alpha 012\}, \quad -3\{\alpha 12\}, \quad +3\{\alpha 2\}, \quad -1; \\
 &\{\alpha + \beta 456\}(1), \quad \{\alpha + \beta 156\}(2), \quad \{\alpha + \beta 036\}(3), \quad \{\alpha + \beta 015\}(4), \quad \{\alpha + \beta 012\}(5); \\
 &\{\alpha + \beta 01 \dots 6\}U = \{\beta 0123\}, \quad +4\{\beta 123\}, \quad +6\{\beta 23\}, \quad +4\{\beta 3\}, \quad +1, \\
 &\{\alpha + \beta 01 \dots 6\}U'''' = \{\alpha 0123\}, \quad -4\{\alpha 123\}, \quad +6\{\alpha 23\}, \quad -4\{\alpha 3\}, \quad +1;
 \end{aligned}$$

&c.

read

$$\alpha + \beta \cdot Y = \beta(1) + (2),$$

$$\alpha + \beta \cdot Y' = \alpha(1) - (2),$$

$$\alpha + \beta \cdot \alpha + \beta + 1 \cdot \alpha + \beta + 2 \cdot Z = \beta \cdot \beta + 1 \cdot \alpha + \beta + 2 \cdot (1) + 2 \cdot \beta + 1 \cdot \alpha + \beta + 1 \cdot (2) + \alpha + \beta \cdot (3),$$

$$\alpha + \beta \cdot \alpha + \beta + 1 \cdot \alpha + \beta + 2 \cdot Z'' = \alpha \cdot \alpha + 1 \cdot \alpha + \beta + 2 \cdot (1) + 2 \cdot \alpha + 1 \cdot \alpha + \beta + 1 \cdot (2) + \alpha + \beta \cdot (3),$$

&c.,

the law being obvious, except as regards the numbers which in the top lines occur in connexion with  $\alpha + \beta$  in the  $\{ \}$  symbols. As regards these, we form them by successive subtractions as shown by the diagrams

|         |           |             |
|---------|-----------|-------------|
| 3 4     | 4 5 6     | 5 6 7 8     |
| 2 1 4   | 3 1 5 6   | 4 1 6 7 8   |
| 1 1 0 3 | 1 2 0 3 6 | 1 3 0 3 7 8 |
|         | 2 1 0 1 5 | 2 2 0 1 5 8 |
|         | 3 0 1 2   | 3 1 0 1 2 7 |
|         |           | 4 0 1 2 3   |

and the statement of the result is now complete.

In part verification, starting from the  $Y$ -formulas (which are obtained at once), assume

$$\{\alpha + \beta 2\}(1), \quad \{\alpha + \beta 1\}(2), \quad \{\alpha + \beta 0\}(3),$$

we must have

$$\{\alpha + \beta 012\}Z = \lambda, \quad \lambda', \quad \lambda'',$$

$$\{\alpha + \beta 012\}Z' = \mu, \quad \mu', \quad \mu'',$$

$$\{\alpha + \beta 012\}Z'' = \nu, \quad \nu', \quad \nu'',$$

$$\{\alpha + \beta 012\}Y = \{\alpha + \beta 012\}Z' = \{\alpha + \beta 12\}(\{\beta 0\}, +1),$$

that is,

$$\{\alpha + \beta 012\}Z' + Z'' = \{\alpha + \beta 12\}Y', \quad = \{\alpha + \beta 12\}(\{\alpha 0\}, -1),$$

$$\{\alpha + \beta 2\}X + X' = \{\alpha + \beta 12\}\{\beta 0\},$$

$$\{\alpha + \beta 2\}X' + X'' = \{\alpha + \beta 12\}\{\alpha 0\},$$

and further

$$\{\alpha + \beta 2\}(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(X, X', X'') = 0,$$

or, what is the same thing,

$$X + X' = \{\alpha + \beta 1\}\{\beta 0\},$$

$$X' + X'' = \{\alpha + \beta 1\}\{\alpha 0\},$$

$$(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(X, X', X'') = 0.$$

And in like manner we have

$$\mu + \mu' = \{\alpha + \beta 2\} \cdot 1,$$

$$\mu' + \mu'' = \{\alpha + \beta 2\} \cdot 1,$$

$$(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(\mu, \mu', \mu'') = 0;$$

and

$$\nu + \nu' = 0, \quad \nu' + \nu'' = 0,$$

$$(\{\alpha 01\}, -2\{\alpha 1\}\{\beta 1\}, \{\beta 01\})(\nu, \nu', \nu'') = 0.$$

We hence find without difficulty

$$\lambda, \mu, \nu = \beta \cdot \beta + 1 \cdot \alpha + \beta + 2 \cdot \beta + 1, \quad +1, = \{\beta 01\}, \quad 2\{\beta 1\}, \quad 1,$$

$$\lambda', \mu', \nu' = \alpha \cdot \beta, \quad \alpha - \beta, \quad -1, = \{\alpha 0\}\{\beta 0\}, \quad \alpha - \beta, \quad -1,$$

$$\lambda'', \mu'', \nu'' = \alpha \cdot \alpha + 1, \quad -2 \cdot \alpha + 1, \quad 1, = \{\alpha 01\}, \quad -2\{\alpha 1\}, \quad 1;$$

viz. for verification of the  $\lambda$ -equations we have

$$\beta \cdot \beta + 1 \cdot \alpha + \beta, \quad \text{that is, } \alpha + \beta \cdot \beta = \{\alpha + \beta 1\}\{\beta 0\},$$

$$\alpha \cdot \beta + \alpha \cdot \alpha + 1, \quad \text{that is, } \alpha + 1 + \beta \cdot \alpha = \{\alpha + \beta 1\}\{\alpha 0\},$$

and

$$(\alpha \cdot \alpha + 1, \quad -2 \cdot \alpha + 1 \cdot \beta + 1, \quad \beta \cdot \beta + 1)(\beta \cdot \beta + 1, \quad \alpha \cdot \beta, \quad \alpha \cdot \alpha + 1) = 0,$$

that is,

$$\alpha \cdot \alpha + 1 \cdot \beta \cdot \beta + 1 \cdot -2 \cdot \alpha + 1 \cdot \beta + 1 \cdot \alpha \cdot \beta + \beta \cdot \beta + 1 \cdot \alpha \cdot \alpha + 1 = 0;$$

and similarly the  $\mu$ - and  $\nu$ -equations may be verified.

We have thus for the  $Z$ 's the equations

$$\{\alpha + \beta 2\}(1), \quad \{\alpha + \beta 1\}(2), \quad \{\alpha + \beta 0\}(3),$$

$$\{\alpha + \beta 012\}Z = \{\beta 01\}, \quad 2\{\beta 1\}, \quad +1,$$

$$\{\alpha + \beta 012\}Z' = \{\alpha 0\}\{\beta 0\}, \quad \alpha - \beta, \quad -1,$$

$$\{\alpha + \beta 012\}Z'' = \{\alpha 01\}, \quad -2\{\alpha 1\}, \quad +1,$$

which include the foregoing expressions for  $Z$  and  $Z''$ .

We may then take the expressions for the  $W$ 's to be

$$\{\alpha + \beta 34\}(1), \quad \{\alpha + \beta 14\}(2), \quad \{\alpha + \beta 03\}(3), \quad \{\alpha + \beta 01\}(4),$$

$$\{\alpha + \beta 0123\}W = \lambda,$$

$$\{\alpha + \beta 0123\}W' = \lambda',$$

$$\{\alpha + \beta 0123\}W'' = \lambda'',$$

$$\{\alpha + \beta 0123\}W''' = \lambda''',$$

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Arthur Cayley

Cambridge, 1896

**681 (continued):** ON THE DERIVATIVES OF THREE BINARY QUANTICS.

and we obtain in like manner the equations

$$\begin{aligned}
\lambda + \lambda' &= \{\alpha + \beta 234\}\{\beta 01\}, \\
\lambda' + \lambda'' &= \{, \}\{\alpha 0\}\{\beta 0\}, \\
\lambda'' + \lambda''' &= \{, \}\{\alpha 01\}, \\
(\{\alpha 012\}, -3\{\alpha 12\}\{\beta 2\}, +3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \lambda, \lambda', \lambda'', \lambda''') &= 0; \\
\mu + \mu' &= \{\alpha + \beta 134\} \cdot 2\{\beta 1\}, \\
\mu' + \mu'' &= \{, \} \cdot \alpha - \beta, \\
\mu'' + \mu''' &= \{, \} \cdot -2\{\alpha 1\}, \\
(\{\alpha 012\}, -3\{\alpha 12\}\{\beta 2\}, +3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \mu, \mu', \mu'', \mu''') &= 0; \\
\nu + \nu' &= \{\alpha + \beta 034\} \cdot 1, \\
\nu' + \nu'' &= \{, \} \cdot -1, \\
\nu'' + \nu''' &= \{, \} \cdot 1, \\
(\{\alpha 012\}, -3\{\alpha 12\}\{\beta 2\}, 3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \nu, \nu', \nu'', \nu''') &= 0; \\
\rho + \rho' &= 0, \quad \rho' + \rho'' = 0, \quad \rho'' + \rho''' = 0, \\
(\{\alpha 012\}, 3\{\alpha 12\}\{\beta 2\}, 3\{\alpha 2\}\{\beta 12\}, -\{\beta 012\} \$ \rho, \rho', \rho'', \rho''') &= \{\alpha + \beta 01234\}.
\end{aligned}$$

These give for the  $\psi'''$  square the values

$$\begin{aligned}
&\{\beta 012\}, \quad 3\{\beta 12\}, \quad 3\{\beta 2\}, \quad +\dots, \quad 1, \\
&\{\alpha 0\}\{\beta 01\}, \quad -2\alpha\beta \cdot \{\beta 1\}, \quad -\alpha^2\beta^2, \quad 1, \\
&\{\alpha 01\}\{\beta 0\}, \quad +\alpha^2\beta \cdot \{\alpha 1\}, \quad 2\alpha\beta^2, \quad 1, \\
&-\{\alpha 012\}, \quad 3\{\alpha 12\}, \quad 3\{\alpha 2\}, \quad 1,
\end{aligned}$$

and so on; the law however of the terms in the intermediate lines is not by any means obvious.

Consider now the binary quantics  $P, Q, R$ , of the forms  $(* \$ x, y)^p, (* \$ x, y)^q, (* \$ x, y)^r$ ; we have for any, for instance for the fourth, order, the derivates

$$P(Q, R)^4, (P, (Q, R)^3)^1, (P, (Q, R)^2)^2, (P, (Q, R)^1)^3, (P, QR)^4;$$

and it is required to express

$$Q(P, R)^4 \text{ and } R(P, Q)^4,$$

each of them as a linear function of these.

I recall that we have  $(P, Q)^0 = PQ$ , so that the first and the last terms of the series might have been written  $(P, (Q, R)^4)^0$  and  $(P, (Q, R)^0)^4$  respectively; and, further, that  $(P, Q)^2$  denotes  $d_x P \cdot d_y Q - d_y P \cdot d_x Q$ ;  $(P, Q)^2$  denotes

$$d_x^2 P \cdot d_y^2 Q - 2d_x d_y P \cdot d_x d_y Q + d_y^2 P \cdot d_x^2 Q;$$

and so on.

I write  $(a, b, c, d, e)$  for the fourth derived functions of any quantic  $U, = (*\$x, y)^m$ ; we have, in a notation which will be at once understood,

$$\begin{aligned} U &= (a, b, c, d, e \$x, y)^4 \div [m]^4, \\ (d_x, d_y) U &= ((a, b, c, d), (b, c, d, e) \$x, y)^3 \div [m-1]^3, \\ (d_x, d_y)^2 U &= ((a, b, c), (b, c, d), (c, d, e) \$x, y)^2 \div [m-2]^2, \\ (d_x, d_y)^3 U &= ((a, b), (b, c), (c, d), (d, e) \$x, y) \div [m-3], \\ (d_x, d_y)^4 U &= (a, b, c, d, e); \end{aligned}$$

and then, taking

$$(a_1, b_1, c_1, d_1, e_1), (a_2, b_2, c_2, d_2, e_2), (a_3, b_3, c_3, d_3, e_3),$$

to belong to  $P, Q, R$ , respectively, we must, instead of  $m$ , write  $p, q, r$  for the three functions respectively.

If we attend only to the highest terms in  $x$ , we have

$$\begin{aligned} U &= ax^m + [m-1], \\ (d_x, d_y) U &= (a, b) x^{m-1} + [m-2], \\ (d_x, d_y)^2 U &= (a, b, c) x^{m-2} + [m-3], \\ (d_x, d_y)^3 U &= (a, b, c, d) x^{m-3} + [m-4], \\ (d_x, d_y)^4 U &= (a, b, c, d, e). \end{aligned}$$

Consider now  $P(Q, R)^4, (P, (Q, R)^3)^1$ , etc.; in each case attending only to the term in  $a_1$ , and in this term to the highest term in  $x$ , we have

- (1)  $[p]^0 P(Q, R)^4 = a_1 \{a_3 e_3 - 4b_3 d_3 + 6c_2 c_3 - 4d_2 b_3 + e_2 a_3\} \quad (X),$
- (2)  $[p-1]^1 [q-3]^3 [r-3]^3 (P, (Q, R)^3)^1 = -[q-3]^3 \cdot b_1 \{d_2 \cdot 3c_2 c_3 - 3d_2 b_3 - e_2 a_3\} \quad (-Y),$   
 $+ [r-3]^3 \cdot a_1 a_2 e_3 - 3b_2 d_3 + 3c_2 c_3 - d_2 b_3 \quad (Y),$
- (3)  $[p-2]^2 [q-2]^2 [r-2]^2 (P, (Q, R)^2)^2 = [q-2]^2 c_1 c_2 a_3 - 2d_2 b_3 + e_2 a_3 \quad (Z''),$   
 $+ 2[q-2]^2 [r-2]^2 \cdot b_1 d_2 - 2c_2 c_3 + d_2 b_3 \quad (-Z'),$   
 $+ [r-2]^2 c_1 e_3 - 2b_2 d_3 + c_2 c_3 \quad (Z),$
- (4)  $[p-3]^3 [q-1]^1 [r-1]^1 (P, (Q, R)^1)^3 = [q-1]^1 d_1 b_3 - c_2 a_3 \quad (-W''),$   
 $+ 3[q-1]^1 [r-1]^1 c_1 c_3 - d_2 b_3 \quad (W''),$   
 $+ 3[q-1]^1 [r-1]^1 b_1 d_3 - c_2 c_3 \quad (-W'),$   
 $+ [r-1]^1 a_1 e_3 - b_2 d_3 \quad (W),$

$$\begin{aligned}
 (5) \quad & [p-4]^4 [q]^0 [r]^0 (P, QR)^4 = e_1 a_3 a_3 (U''''), \\
 & + 4[q]^0 [r]^0 d_1 b_3 (-U'''), \\
 & + 6[q]^0 [r]^0 c_1 c_3 (U''), \\
 & + 4[q]^0 [r]^0 b_1 d_3 (-U'), \\
 & + [q]^0 [r]^0 a_1 e_3 (U).
 \end{aligned}$$

Thus, for the second of these equations,

$$(P, (Q, R)^3)^1 = d_x P \cdot d_y (Q, R)^3 - \&c.$$

the term in  $a_1$  is  $d_y(Q, R)^3 = (d_x Q, R)^3 + (Q, d_y R)^3$ , the whole being divided by  $[p-1]^1$ ; where attending only to the highest terms in  $x$ , the two terms are respectively

$$-(b_2 d_3 - 3c_2 c_3 + 3d_2 b_3 - e_2 a_3) \div [r-3]^3,$$

and

$$(a_2 e_3 - 3b_2 d_3 + 3c_2 c_3 - d_2 b_3) \div [q-3]^3,$$

which are each divided by  $[p-1]^1$  as above; whence, multiplying by

$$[p-1]^1 [q-1]^1 [r-1]^1,$$

we have the formula in question; and so for the other cases.

Writing now (1), (2), (3), (4), (5) for the left-hand sides of the five equations respectively; and

$$-Y', Y; \quad Z'', -Z', Z; \quad -W''', W'', -W', W; \quad U'''', -U''', U'', -U', U;$$

for the literal parts on the right-hand sides of the same equations respectively, then we have

$$X = Y + Y',$$

$$Y = Z + Z', \quad Y' = Z' + Z'',$$

$$\&c.,$$

and the equations become

$$(1) = X,$$

$$(2) = [r-1]^1 Y - [q-3]^3 Y',$$

$$(3) = [r-2]^2 Z - 2[r-2]^2 [q-2]^2 Z' + [q-2]^2 Z'',$$

$$(4) = [r-1]^1 W'' - 3[r-1]^1 [q-1]^1 W' + 3[r-1]^1 [q-1]^1 W - [q-1]^1 W''',$$

$$(5) = [r]^0 U - 4[r]^0 [q]^0 U' + 6[r]^0 [q]^0 U'' - 4[r]^0 [q]^0 U''' - [q]^0 U'''' ,$$

which are, in fact, the equations considered at the beginning of the present paper, putting therein  $\alpha = r-3$  and  $\beta = q-3$ , they consequently give

$$\{q+r-6, 456\}(1), \{q+r-6, 156\}(2), \{q+r-6, 036\}(3), +\{q+r-6, 015\}(4), \{q+r-6, 012\}(5),$$

$$\{q+r-6, 01\dots 6\}Q = \{q-3, 0123\} + 4\{q-3, 123\} + 6\{q-3, 23\} + 4\{q-3, 3\} + 1,$$

$$U'''' = \{r-3, 0123\} - 4\{r-3, 123\} + 6\{r-3, 23\} - 4\{r-3, 3\} + 1.$$



Also, attending as before only to the terms in  $a_1$ , and therein to the highest power of  $x$ , we have

$$Q(P, R)^4 = a_2 a_1 \div [q]^0,$$

that is,

$$[q]^2 Q(P, R)^4 = U, \quad [r]^0 R(P, Q)^4 = U'''';$$

and, observing that  $\{q + r - 6, 01 \dots 6\}$  is  $= [q + r]^6$ , and that  $\{q + r - 6, 456\}$ , etc., may be written  $\{q + r, \bar{2}\bar{1}\bar{0}\}$ , etc., where the superscript bars are the signs  $-$ , the formulae become

$$\begin{aligned} &\{q + r, \bar{2}\bar{1}\bar{0}\}(1), \{q + r, \bar{5}\bar{1}\bar{0}\}(2), \{q + r, \bar{6}\bar{3}\bar{0}\}(3), \{q + r, \bar{6}\bar{5}\bar{1}\}(4), \{q + r, \bar{6}\bar{5}\bar{4}\}(5), \\ &[q + r]^6 [q]^0 Q(P, R)^4 = [q]^0 + 6[q]^0 + 4[q]^0 + [q]^0 + 1, \\ &[q + r]^6 [r]^0 R(P, Q)^4 = [r]^0 - 4[r]^0 + 6[r]^0 - 4[r]^0 + 1. \end{aligned}$$

Written at full length, the first of these equations (which, as being the fourth in a series, I mark 4th equation) is

$$\begin{aligned} [q + r]^6 [q]^0 Q(P, R)^4 &= 1 \cdot q + r \cdot q + r - 1 \cdot q + r - 2 \cdot [p]^0 [q]^0 \cdot P(Q, R)^4 \quad (4\text{th equation}) \\ &+ 4 \cdot q + r \cdot q + r - 1 \cdot q + r - 5 \cdot [p - 1]^1 [q]^0 [q - 3]^3 [r - 1]^1 \cdot (P, (Q, R)^3)^1 \\ &+ 6 \cdot q + r \cdot q + r - 5 \cdot q + r - 6 \cdot [p - 2]^2 [q]^0 [q - 2]^2 [r - 2]^2 \cdot (P, (Q, R)^2)^2 \\ &+ 4 \cdot q + r - 1 \cdot q + r - 5 \cdot q + r - 6 \cdot [p - 3]^3 [q]^0 [q - 1]^1 [r - 1]^1 \cdot (P, (Q, R)^1)^3 \\ &+ 1 \cdot q + r - 1 \cdot q + r - 5 \cdot q + r - 6 \cdot [q]^0 \cdot P(Q, R)^4 [r]^0; \end{aligned}$$

and the other is, in fact, the same equation with  $q, Q, r, R$  interchanged with  $r, R, q, Q$ ; the alternate  $+$  and  $-$  signs arise evidently from the terms

$$(R, Q)^4 = (Q, R)^4; (R, Q)^3 = -(Q, R)^3; \text{ etc.},$$

which present themselves on the right-hand side.

It will be observed that the identity has been derived from the comparison of the terms in  $a_1$ , which are the highest terms in  $x$ , the other terms not having been written down or considered; but it is easy to see that an identity of the form in question exists, and, this being admitted, the process is a legitimate one.

The preceding equations of the series are

$$\begin{aligned} [q + r]^2 [q]^0 Q(P, R)^2 &= 1 \cdot [p]^0 [q]^0 \cdot P(Q, R)^2 \quad (1\text{st equation}) \\ &+ 1 \cdot [q]^0 [r]^0 \cdot (P, QR)^2; \\ [q + r]^3 [q]^0 Q(P, R)^3 &= 1 \cdot q + r \cdot [p]^0 [q]^0 \cdot P(Q, R)^3 \quad (2\text{nd equation}) \\ &+ 2 \cdot q + r - 1 \cdot [p - 1]^1 [q]^0 [q - 1]^1 [r - 1]^1 \cdot (P, (Q, R)^1)^1 \\ &+ 1 \cdot q + r - 2 \cdot [q]^0 [r]^0 \cdot (P, QR)^3; \\ [q + r]^4 [q]^0 Q(P, R)^4 &= 1 \cdot q + r \cdot q + r - 1 \cdot [p]^0 [q]^0 \cdot P(Q, R)^4 \quad (3\text{rd equation}) \\ &+ 3 \cdot q + r \cdot q + r - 3 \cdot [p - 1]^1 [q]^0 [q - 2]^2 [r - 2]^2 \cdot (P, (Q, R)^2)^1 \\ &+ 2 \cdot q + r - 1 \cdot q + r - 4 \cdot [p - 2]^2 [q]^0 [q - 1]^1 [r - 1]^1 \cdot (P, QR)^2 \\ &+ 1 \cdot q + r - 3 \cdot q + r - 4 \cdot [q]^0 [r]^0 \cdot (P, QR)^4. \end{aligned}$$

From these four equations the law is evident, except as to the numbers subtracted from  $q + r$ . These are obtained, as explained above, in regard to the numbers added to  $\alpha + \beta$  in the  $\{ \}$  symbols; transforming the diagrams so as to be directly applicable to the case now in question, they become

$$\begin{array}{cccccc}
 01 & 01 & 012 & 012 & 0123 & 0123 \\
 1\ 1 & 2\ 03 & 3\ 015 & 4\ 0127 & & \\
 1\ 2 & 1\ 1\ 14 & 21\ 036 & 31\ 0158 & & \\
 & 2\ 34\ 12\ 156 & & 22\ 0378 & & \\
 & & 3\ 456\ 13 & & 1678 & \\
 & & & 4\ 5678 & & 
 \end{array}$$

showing how the numbers are obtained for the equations 2, 3, 4, 5 respectively.

The first equation is

$$(q^2 + qr) Q(P, R) = pq P(Q, R) + qr [Q(P, R) + (P, Q)],$$

viz. this is

$$(q^2 + qr) Q(P, R) = pq P(Q, R) - qr Q(R, P) + qr R(P, Q)$$

or, dividing by  $q$ , this is

$$(q^2 + qr) Q(R, P) = p P(Q, R) + q Q(R, P) + r R(P, Q),$$

which is a well-known identity.

We may verify any of the equations, though the process is rather laborious, for the particular values

$$P = Q = R = x^{\frac{1}{2}(p+\alpha)} y^{\frac{1}{2}(p-\alpha)}, \quad x^{\frac{1}{2}(q+\beta)} y^{\frac{1}{2}(q-\beta)}, \quad x^{\frac{1}{2}(r+\gamma)} y^{\frac{1}{2}(r-\gamma)}.$$

Thus, taking the second equation, we have, omitting common factors,

$$\begin{aligned}
 (Q, R)^2 &= q + \beta \cdot q + \beta - 2 \cdot r - \gamma \cdot r - \gamma - 2 \\
 &\quad - 2 \cdot q + \beta \cdot q - \beta \cdot r + \gamma \cdot r - \gamma \\
 &\quad + q - \beta \cdot q - \beta - 2 \cdot r + \gamma \cdot r + \gamma - 2 \\
 &= \beta^2(r^2 - r) + \gamma^2(q^2 - q) - 2\beta\gamma(q - 1)(r - 1) - qr(q + r - 2),
 \end{aligned}$$

$$\begin{aligned}
 (P, (Q, R)^1)^1 &= (q + \beta \cdot r - \gamma - q - \beta \cdot r + \gamma)(p + \alpha \cdot q + r - \beta - \gamma - 2 - p - \alpha \cdot q + r + \beta + \gamma - 2) \\
 &= (\beta r - q\gamma)(\alpha \cdot q + r - 2 - p \cdot \beta + \gamma) \\
 &= \alpha\beta r(r + q - 2) - \alpha\gamma q(q + r - 2) - pr\beta^2 + p(q - r)\beta\gamma + pq\gamma^2,
 \end{aligned}$$

and from the first of these the expressions of  $Q(P, R)^2$  and  $(P, QR)^2$  are at once obtained. The identity to be verified then becomes

$$\begin{aligned}
 &[q + r]^2 [q]^2 \{ \alpha^2(r^2 - r) + \gamma^2(p^2 - p) - 2\alpha\gamma(p - 1)(r - 1) - pr(p + r - 2) \} \\
 &= (q + r)[q]^2 [p]^2 \{ \beta^2(r^2 - r) + \gamma^2(q^2 - q) - 2\beta\gamma(q - 1)(r - 1) - qr(q + r - 2) \} \\
 &\quad + 2(q + r - 1)[q]^2 (p - 1)(r - 1) \{ \alpha\beta r(q + r - 2) - \alpha\gamma q(q + r - 2) \\
 &\quad \quad - pr\beta^2 + p(q - r)\beta\gamma + pq\gamma^2 \} \\
 &\quad + (q + r - 2)[q]^2 [r]^2 \{ \alpha^2(q + r)(q + r - 1) + (\beta + \gamma)^2(p^2 - p) \\
 &\quad \quad - 2\alpha(\beta + \gamma)(p - 1)(q + r - 1) - p(q + r)(p + q + r - 2) \},
 \end{aligned}$$

which is easily verified, term by term; for instance, the terms with  $\alpha$ ,  $\beta$ , or  $\gamma$ , give

$$\begin{aligned} [q+r]^2[q]^2 pr(p+r-2) &= (q+r)[q]^2[p]^2 qr(q+r-2) \\ &\quad + (q+r-2)[q]^2[r]^2 p(q+r)(p+q+r-2), \end{aligned}$$

which, omitting the factor  $(q+r)(q+r-2)[q]^2 pr$ , is

$$(q+r-1)(p+r-2) = (p-1)q + (r-1)(p+q+r+2);$$

viz. the right-hand side is

$$(p-1)q + (r-1)q + (r-1)(p+r-2) = (q+r-1)(p+r-2),$$

as it should be.

The equations are useful for the demonstration of a subsidiary theorem employed in Gordan's demonstration of the finite number of the covariants of any binary form  $U$ . Suppose that a system of covariants (including the quantic itself) is

$$P, Q, R, S, \dots;$$

this may be the complete system of covariants; and if it is so, then,  $T$  and  $V$  being any functions of the form  $P^\alpha Q^\beta R^\gamma \dots$ , every derivative  $(T, V)^s$  must be a term or sum of terms of the like form  $P^\alpha Q^\beta R^\gamma \dots$ ; the subsidiary theorem is that in order to prove that the case is so, it is sufficient to prove that every derivative  $(P, Q)^s$ , where  $P$  and  $Q$  are any two terms of the proposed system, is a term or sum of terms of the form in question  $P^\alpha Q^\beta R^\gamma \dots$ .

In fact, supposing it shown that every derivative  $(T, V)^s$  up to a given value  $\theta$  of  $s$  is of the form  $P^\alpha Q^\beta R^\gamma \dots$ , we can by successive application of the equation for  $Q(P, R)^{\theta+1}$ , regarded as an equation for the reduction of the last term on the right-hand side  $(P, QR)^{\theta+1}$ , bring first  $(P, QR)^{\theta+1}$ , and then  $(P, QRS)^{\theta+1}, \dots$ , and so ultimately any function  $(P, V)^{\theta+1}$ , and then again any functions  $(PQ, V)^{\theta+1}, (PQR, V)^{\theta+1}, \dots$ , and so ultimately any function  $(T, V)^{\theta+1}$ , into the required form  $P^\alpha Q^\beta R^\gamma \dots$ : or the theorem, being true for  $\theta$ , will be true for  $\theta + 1$ ; whence it is true generally.

## 682.

## FORMULÆ RELATING TO THE RIGHT LINE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 169–171.]

1. Let  $\lambda, \mu, \nu$  be the direction-angles of a line;  $\alpha, \beta, \gamma$  the coordinates of a point on the line; and write

$$\begin{aligned} a &= \cos \lambda, & f &= \beta \cos \nu - \gamma \cos \mu, \\ b &= \cos \mu, & g &= \gamma \cos \lambda - \alpha \cos \nu, \\ c &= \cos \nu, & h &= \alpha \cos \mu - \beta \cos \lambda, \end{aligned}$$

whence

$$a^2 + b^2 + c^2 = 1,$$

$$af + bg + ch = 0,$$

or the six quantities  $(a, b, c, f, g, h)$ , termed the coordinates of the line, depend upon four arbitrary parameters.

2. It is at once shown that the condition for the intersection of any two lines  $(a, b, c, f, g, h)$ ,  $(a', b', c', f', g', h')$ , is  $af' + bg' + ch' + a'f + b'g + c'h = 0$ .

3. Given two lines  $(a, b, c, f, g, h)$ ,  $(a', b', c', f', g', h')$ , it is required to find their shortest distance, and the coordinates of their line of shortest distance.

Let

$$Ax + By + Cz + D = 0,$$

$$Ax + By + Cz + D' = 0,$$

be parallel planes containing the two lines respectively; then the first plane contains the point  $\alpha + r \cos \lambda$ ,  $\beta + r \cos \mu$ ,  $\gamma + r \cos \nu$ , and the second contains the point  $\alpha' + r' \cos \lambda'$ ,  $\beta' + r' \cos \mu'$ ,  $\gamma' + r' \cos \nu'$ ; that is, we have

$$A\alpha + B\beta + C\gamma + D = 0,$$

$$A\alpha' + B\beta' + C\gamma' + D' = 0,$$

$$A \cos \lambda + B \cos \mu + C \cos \nu = 0,$$

$$A \cos \lambda' + B \cos \mu' + C \cos \nu' = 0,$$

which last equations may be written

$$\begin{aligned} Aa + Bb + Cc &= 0, \\ Aa' + Bb' + Cc' &= 0, \end{aligned}$$

giving

$$A : B : C = bc' - b'c : ca' - c'a : ab' - a'b,$$

or, if we write

$$\theta = aa' + bb' + cc',$$

and assume, as is convenient,

$$A^2 + B^2 + C^2 = 1,$$

then

$$\sqrt{1 - \theta^2} A, B, C = bc' - b'c, ca' - c'a, ab' - a'b,$$

where  $\theta = \text{cosine-inclination} = aa' + bb' + cc'$ .

Hence, shortest distance  $= D' - D$

$$\begin{aligned} &= A(\alpha - \alpha') + B(\beta - \beta') + C(\gamma - \gamma') \\ &= \frac{1}{\sqrt{1 - \theta^2}} \{ (bc' - b'c)(\alpha - \alpha') + (ca' - c'a)(\beta - \beta') + (ab' - a'b)(\gamma - \gamma') \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} \{ a'(c\beta - b\gamma) + b'(a\gamma - c\alpha) + c'(b\alpha - a\beta) \\ &\quad + a(c'\beta' - b'\gamma') + b(a'\gamma' - c'\alpha') + c(b'\alpha' - a'\beta') \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} (af' + bg' + ch' + a'f + b'g + c'h), = \mathfrak{D} \text{ suppose.} \end{aligned}$$

The six coordinates of the line of shortest distance are  $A, B, C, F, G, H$ , where  $A, B, C$  denote as before, and  $F, G, H$  are to be determined.

Since the line meets each of the given lines, we have

$$\begin{aligned} Af + Bg + Ch + Fa + Gb + Hc &= 0, \\ Af' + Bg' + Ch' + Fa' + Gb' + Hc' &= 0, \\ FA + GB + HC &= 0, \end{aligned}$$

which equations give  $F, G, H$ . Multiplying the first equation by  $-b'C + c'B$ , the second by  $Bc - Cb$ , and the third by  $-bc' + b'c$ , we find

$$(b'C - c'B)(Af + Bg + Ch) + (Bc - Cb)(Af' + Bg' + Ch') + F \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ A, & B, & C \end{vmatrix} = 0.$$

Here

$$\begin{aligned} b'C - c'B &= \frac{1}{\sqrt{1 - \theta^2}} \{ b'(ab' - a'b) - c'(ca' - c'a) \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} \{ a(a'^2 + b'^2 + c'^2) - a'(aa' + bb' + cc') \} \\ &= \frac{1}{\sqrt{1 - \theta^2}} (a - a'\theta), \end{aligned}$$

and similarly

$$cB - bC = \frac{1}{\sqrt{1-\theta^2}}(a' - a\theta).$$

Also, putting for shortness

$$\Omega = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ f, & g, & h \end{vmatrix}, \quad \Omega' = \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ f', & g', & h' \end{vmatrix},$$

we have

$$Af + Bg + Ch = \frac{1}{\sqrt{1-\theta^2}} \Omega, \quad Af' + Bg' + Ch' = \frac{1}{\sqrt{1-\theta^2}} \Omega',$$

and finally, the determinant which multiplies  $F$  is

$$\frac{1}{\sqrt{1-\theta^2}} \{(bc' - b'c)^2 + (ca' - c'a)^2 + (ab' - a'b)^2\} = \frac{1}{\sqrt{1-\theta^2}}(1 - \theta^2), = \sqrt{1 - \theta^2}.$$

We have thus the value of  $F$ ; forming in the same way those of  $G$  and  $H$ , we find

$$\begin{aligned} F &= \frac{-1}{(1 - \theta^2)^{\frac{3}{2}}} \{(a - a'\theta) \Omega + (a' - a\theta) \Omega'\}, \\ G &= \frac{-1}{(1 - \theta^2)^{\frac{3}{2}}} \{(b - b'\theta) \Omega + (b' - b\theta) \Omega'\}, \\ H &= \frac{-1}{(1 - \theta^2)^{\frac{3}{2}}} \{(c - c'\theta) \Omega + (c' - c\theta) \Omega'\}, \end{aligned}$$

which, with the foregoing equations for  $A, B, C$ , give the six coordinates of the line of shortest distance.

## 683.

ON THE FUNCTION  $\arcsin(x + iy)$ .

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 171–174.]

The determination of the function in question, the arc to a given imaginary sine, is considered in Cauchy's *Exercices d'Analyse, &c.*, t. iii. (1844), p. 382; but it appears, by two hydrodynamical papers by Mr Ferrers and Mr Lamb, *Quarterly Mathematical Journal*, t. XIII. (1874), p. 115, and t. XIV. (1875), p. 40, that the question is connected with the theory of confocal conics.

Taking  $c = \sqrt{a^2 - b^2}$  a positive real quantity which may ultimately be put  $= 1$ , the question is to find the real quantities  $\xi$ ,  $\eta$ , such that

$$\xi + i\eta = \arcsin \frac{1}{c}(x + iy),$$

or say

$$x + iy = c \sin(\xi + i\eta),$$

so that

$$x = c \sin \xi \cos i\eta, \quad iy = c \cos \xi \sin i\eta.$$

It is convenient to remark that if a value of  $\xi + i\eta$  be  $\xi' + i\eta'$ , then the general value is  $2m\pi + \xi' + i\eta'$  or  $(2m + 1)\pi - (\xi' + i\eta')$ ; hence,  $\eta$  may be made positive or negative at pleasure;  $\cos i\eta$  is in each case positive, but  $-\sin i\eta$  has the same sign as  $\eta$ ; hence  $\cos \xi$  has the same sign as  $x$ , but  $\sin \xi$  has the same sign as  $y$  or the reverse sign, according as  $\eta$  is positive or negative; for any given values of  $x$  and  $y$ , we obtain, as will appear, determinate positive values of  $\sin^2 \xi$  and  $\cos^2 \xi$ ; and the square roots of these must therefore be taken so as to give to  $\sin \xi$ ,  $\cos \xi$  their proper signs respectively.

Suppose that  $\lambda, \mu$  are the elliptic coordinates of the point  $(x, y)$ ; viz. that we have

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1,$$

$$\frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} = 1,$$

where  $a^2 + \lambda, b^2 + \lambda$ , and  $a^2 + \mu$  are positive, but  $b^2 + \mu$  is negative. Calling  $\rho, \sigma$  the distances of the point  $x, y$  from the points  $(c, 0)$  and  $(-c, 0)$ , that is, assuming

$$\rho = \sqrt{(x - c)^2 + y^2},$$

$$\sigma = \sqrt{(x + c)^2 + y^2},$$

then we have

$$\begin{aligned} \sqrt{a^2 + \lambda} &= \frac{1}{2}(\sigma + \rho), & \text{whence also } \sqrt{b^2 + \lambda} &= \frac{1}{2}\sqrt{(\sigma + \rho)^2 - 4c^2}, \\ \sqrt{a^2 + \mu} &= \frac{1}{2}(\sigma - \rho), & \sqrt{b^2 + \mu} &= \frac{1}{2}\sqrt{(\sigma - \rho)^2 - 4c^2}, \end{aligned}$$

which equations determine  $\lambda, \mu$  as functions of  $x, y$ .

Now we have

$$\begin{aligned} \rho\sigma &= \sqrt{(x^2 + y^2 - c^2)^2 - 4c^2y^2}, \\ \rho^2 + \sigma^2 &= 2(x^2 + y^2 + c^2); \end{aligned}$$

substituting herein for  $x, y$  their values

$$c \sin \xi \cos i\eta, \quad ci \cos \xi \sin i\eta,$$

we find

$$\begin{aligned} x^2 - y^2 - c^2 &= c^2(\sin^2 \xi \cos^2 i\eta + \cos^2 \xi \sin^2 i\eta - \sin^2 \xi - \cos^2 \xi) \\ &= -c^2(\sin^2 \xi \sin^2 i\eta + \cos^2 \xi \cos^2 i\eta), \end{aligned}$$

whence

$$\begin{aligned} (x^2 - y^2 - c^2)^2 &= c^4(\cos^2 \xi \cos^2 i\eta + \sin^2 \xi \sin^2 i\eta)^2, \\ 4x^2y^2 &= 4c^4 \sin^2 \xi \cos^2 \xi \sin^2 i\eta \cos^2 i\eta \\ &= c^4(\cos^2 \xi \cos^2 i\eta - \sin^2 \xi \sin^2 i\eta)^2. \end{aligned}$$

Hence  $\rho\sigma = 2c^2(\cos^2 \xi \cos^2 i\eta - \sin^2 \xi \sin^2 i\eta)$ , and

$$\begin{aligned} \rho^2 + \sigma^2 &= 2c^2(\sin^2 \xi \cos^2 i\eta + \cos^2 \xi \sin^2 i\eta + 1) \\ (\rho + \sigma)^2 &= 2c^2(\cos^2 i\eta + \sin^2 i\eta + 1) = 4c^2 \cos^2 i\eta, \\ (\rho - \sigma)^2 &= 2c^2(\sin^2 \xi - \cos^2 \xi + 1) = 4c^2 \sin^2 \xi. \end{aligned}$$

Consequently

$$\begin{aligned} a^2 + \lambda &= c^2 \cos^2 i\eta, & b^2 + \lambda &= -c^2 \sin^2 i\eta, \\ a^2 + \mu &= c^2 \sin^2 \xi, & b^2 + \mu &= -c^2 \cos^2 \xi, \end{aligned}$$

values which verify as they should do the equations

$$\frac{x^2}{a^2 + \lambda} + \frac{y^2}{b^2 + \lambda} = 1,$$

$$\frac{x^2}{a^2 + \mu} + \frac{y^2}{b^2 + \mu} = 1.$$



viz. these become

$$\frac{x^2}{c^2 \cos^2 i\eta} + \frac{y^2}{-c^2 \sin^2 i\eta} = \sin^2 \xi + \cos^2 \xi = 1,$$

$$\frac{x^2}{c^2 \sin^2 \xi} + \frac{y^2}{-c^2 \cos^2 \xi} = \cos^2 i\eta - \sin^2 i\eta = 1.$$

The same equations, or as we may also write them,

$$\lambda = -a^2 \sin^2 i\eta - b^2 \cos^2 i\eta,$$

$$\mu = -a^2 \cos^2 \xi - b^2 \sin^2 \xi,$$

determine  $\eta$  as a function of  $\lambda$ , and  $\xi$  as a function of  $\mu$ ;  $\lambda, \mu$  being by what precedes, given functions of  $x, y$ .

Or more simply, starting from the last-mentioned values of  $\lambda, \mu$ , and substituting these in the expressions

$$x^2 = \frac{a^2 + \lambda}{a^2 - b^2} \cdot \frac{a^2 + \mu}{1}, \quad y^2 = \frac{b^2 + \lambda}{b^2 - a^2} \cdot \frac{b^2 + \mu}{1},$$

we find

$$x^2 = c^2 \sin^2 \xi \cos^2 i\eta, \quad y^2 = -c^2 \cos^2 \xi \sin^2 i\eta,$$

or say

$$x = c \sin \xi \cos i\eta, \quad iy = c \cos \xi \sin i\eta,$$

whence

$$x + iy = c \sin(\xi + i\eta),$$

the original relation between  $x, y$  and  $\xi, \eta$ .

684.

ON A RELATION BETWEEN CERTAIN PRODUCTS OF  
DIFFERENCES.[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 174, 175.]

Consider the function

$$3 \left\{ \begin{array}{l} abc \cdot de \\ +bcd \cdot ea \\ +cde \cdot ab \\ +dea \cdot bc \\ +eab \cdot cd \end{array} \right\} - \left\{ \begin{array}{l} abd \cdot ce \\ +bce \cdot da \\ +cda \cdot eb \\ +deb \cdot ac \\ +eac \cdot bd \end{array} \right\}$$

where

$$abc = (a-b)(b-c)(c-a),$$

$$ab = (a-b)(b-a) = -(a-b)^2,$$

therefore

$$abc = bca = cab = -bac, \&c.,$$

$$ab = ba.$$

It is to be shown that the function vanishes if  $e = d$ . Writing  $e = d$ , the value is

$$\begin{aligned} & 3(bcd \cdot da + dab \cdot cd) - abd \cdot cd \\ & \quad - bcd \cdot da \\ & \quad - cda \cdot db \\ & \quad - dac \cdot bd, \end{aligned}$$

viz. this is

$$\begin{aligned}
 & -3bcd \cdot ad - abd \cdot cd \\
 & +3abd \cdot cd - bcd \cdot ad \\
 & \quad - 2acd \cdot bd \\
 & = 2bcd \cdot ad - 2acd \cdot bd + 2abd \cdot cd \\
 & = 2(bcd \cdot ad + cad \cdot bd + abd \cdot cd),
 \end{aligned}$$

which is easily seen to vanish; the value is

$$\begin{aligned}
 & (b-c)(c-d)(d-b)(a-d)^2 - (b-c)(a-d)^2(b-d)(c-d) \\
 & + (c-a)(a-d)(d-c)(b-d)^2 - (c-a)(a-d)(b-d)^2(c-d) \\
 & + (a-b)(b-d)(d-a)(c-d)^2 - (a-b)(a-d)(b-d)(c-d)^2.
 \end{aligned}$$

viz. omitting the factor  $(a-d)(b-d)(c-d)$ , this is

$$-(b-c)(a-d) - (c-a)(b-d) - (a-b)(c-d),$$

which vanishes. Hence the function also vanishes if  $e = a$ , or  $a = b$  or  $b = c$ , or  $c = d$ ; and it is thus a mere numerical multiple of  $(a-b)(b-c)(c-d)(d-e)(e-a)$ , or say it is  $= M \cdot abcde$ .

To find  $M$  write  $e = c$ , the equation becomes

$$\begin{aligned}
 3abc \cdot dc - cda \cdot cb &= M \cdot abcde, = M \cdot abc \cdot dc, \\
 & +3bcd \cdot ca - ac \\
 & +3dea \cdot bc \\
 & +3cab \cdot cd,
 \end{aligned}$$

viz. this is  $6abc \cdot dc + 4dbc \cdot ac + 4cadc \cdot bc = M \cdot abc \cdot dc$ ,

giving  $M = 10$ . In fact, we then have

$$4abc \cdot dc + 4dbc \cdot ac - 4adc \cdot bc = 0,$$

that is,

$$-abc \cdot dc - bdc \cdot ac - dac \cdot bc = 0,$$

which is right. And we have thus the identity

$$3 \left\{ \begin{array}{l} abc \cdot de \\ +bcd \cdot ea \\ +cde \cdot ab \\ +dea \cdot bc \\ +eab \cdot cd \end{array} \right\} - \left\{ \begin{array}{l} abd \cdot ce \\ +bce \cdot da \\ +cda \cdot eb \\ +deb \cdot ac \\ +eac \cdot bd \end{array} \right\} = 10 \cdot abcde,$$

or say

$$3[abcde] - [acebd] = 10\{abcde\}.$$

## 685.

## ON MR COTTERILL'S GONIOMETRICAL PROBLEM.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 196–198.]

The very remarkable formulæ contained in Mr Cotterill's paper, "A goniometrical problem, to be solved analytically in one move, or more simply synthetically in two moves," *Quarterly Mathematical Journal*, t. vii. (1866), pp. 259–272, are presented in a form which, to say the least, is not as easily intelligible as might be; and they have not, I think, attracted the attention which they well deserve.

Using his notation, except that I write for angles small roman letters, in order to be able to have the corresponding italic small letters and capitals for the sines and cosines respectively of the same angles, we consider nine angles

$$\begin{array}{ccc} a, & b, & c, \\ d, & e, & f, \\ x, & y, & z, \end{array}$$

which are such that the sum of three angles in the same line, or in the same column, is an odd multiple of  $\pi$ . Of course, any four angles such as  $a, b, d, e$  are

arbitrary, and each of the remaining angles is then determinate save as to an even multiple of  $\pi$ . And it may be remarked that these angles  $a, b, d, e$  may represent the inclinations of any four lines to a fifth line, and that the remaining angles are then at once obtained, as in the figure. The small roman letters are here used to denote as well angles as points, being so placed as to show what the angles are which they respectively denote; the points  $*$ ,  $*$  are constructed as the intersections of the lines  $ac, bc$  by the circle circumscribed about  $axy$ , and the angle  $z$  is the angle which the points  $*$ ,  $*$  subtend at  $x$  or  $y$ . It will be observed that the sum of the three angles in a line or column is in each case  $= \pi$ .

But this in passing: the analytical theorem is, first, we can form with the sines and cosines of the angles in any two lines or columns a function  $S$  presenting itself under two distinct forms, which are in fact equal in value, or say  $S$  is a symmetrical function of the two lines or columns, viz. for the first and second lines this is

$$\begin{aligned} S\left(\begin{matrix} a, & b, & c \\ d, & e, & f \end{matrix}\right) &= d^2 Abc + e^2 Bca + f^2 Cab \\ &= a^2 Def + b^2 Efd + c^2 Fde, \end{aligned}$$

where, as already mentioned,  $a, A$  denote  $\sin a, \cos a$ , and so for the other letters.

Secondly, if to the  $S$  of any two lines or columns we add twice the product of the six sines, we obtain a sum  $M$  which has the same value from whichever two lines or columns we obtain it; or, say  $M$  is a symmetrical function of the matrix of the nine angles. Thus

$$M = S\left(\begin{matrix} a, & b, & c \\ d, & e, & f \end{matrix}\right) + 2abcdef,$$

which is one of a system of six forms each of which (on account of the two forms of the  $S$  contained in it) may be regarded as a double form, and the twelve values are all of them equal. There are, moreover, 15 other forms of  $M$ , viz. 3 line-forms, such as

$$bcdx + caey + abfz \quad (\text{belongs to line } a, b, c),$$

3 column-forms, such as

$$dxbc + xae f + adyz \quad (\text{belongs to column } a, d, x),$$

and 9 term-forms, such as

$$e^2 z^2 + f^2 y^2 + 2efyzA \quad (\text{belongs to term } a),$$

and the  $12 + 15 = 27$  values are all equal.

The several identities can of course be verified by means of the relations between the nine angles, or rather the derived sine- and cosine-relations

$$C = ab - AB,$$

$$c = aB + bA, \&c.$$

Thus, as regards the two forms of  $S\left(\begin{smallmatrix} a, & b, & c \\ d, & e, & f \end{smallmatrix}\right)$ , the identity to be verified may be written

$$c(d^2Ab + e^2Ba - cFde) = f(a^2De + b^2Ed - fCab).$$

Proceeding to reduce the factor  $a^2De + b^2Ed - fCab$ , if we first write herein  $f = eD + dE$ , it becomes

$$a^2De + b^2Ed - (eD + dE)Cab,$$

which is

$$= aDe(a - bC) + bEd(b - aC),$$

and then writing  $C = ab - AB$ , we have  $a - bC = a(1 - b^2) + bAB = B(aB + bA) = Bc$ ; and, similarly,  $b - aC = Ac$ ; whence the term is  $= c(aeBD + bdAE)$ ; or, in the equation to be verified, the right-hand side is  $= cf(aeBD + bdAE)$ , and by a similar reduction, the left-hand side is found to have the same value.

The paper contains various other interesting results.

## 686.

## ON A FUNCTIONAL EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 315–325; *Proceedings of the London Mathematical Society*, vol. ix. (1878), p. 29.]

I was led by a hydrodynamical problem to consider a certain functional equation viz. writing for shortness  $x_1 = \frac{ax+b}{cx+d}$ , this is

$$\phi x - \phi x_1 = (x - x_1) \frac{Ax + B}{Cx + D}.$$

I find by a direct process, which I will afterwards explain, the solution

$$\phi x = \frac{A}{C}x + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(dC - cD)} \int_0^\infty \frac{\sin \xi t \sin \eta t dt}{\sin \zeta t \sinh \pi t},$$

where  $\zeta$  is a constant, but  $\xi, \eta$  are complicated logarithmic functions of  $x$  ( $\xi, \eta, \zeta$  depend also on the quantities  $a, b, c, d, C, D$ );  $\sinh \pi t$  denotes as usual the hyperbolic sine,  $\frac{1}{2}(e^{\pi t} - e^{-\pi t})$ .

The values of  $\xi, \eta, \zeta$  are given by the formulæ

$$\lambda + \frac{1}{\lambda} = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

$$\mathfrak{a} = ax + b, \quad \mathfrak{b} = -dx + b,$$

$$\mathfrak{c} = cx + d, \quad \mathfrak{d} = cx - a,$$

$$W = C\mathfrak{a} + D\mathfrak{c},$$

$$Z = C\mathfrak{b} + D\mathfrak{d},$$

$$R = \lambda\mathfrak{c} + \lambda\mathfrak{d},$$

$$S = -\mathfrak{c} - \mathfrak{d},$$

$$R' = W + \frac{1}{\lambda}Z,$$

$$S' = -W - \lambda Z,$$

which determine  $\lambda, R, S, R', S'$  and then

$$\xi = \frac{1}{2} \log \lambda, \quad \eta = \frac{1}{2} \log \frac{RR'}{SS'}, \quad \zeta = \frac{1}{2} \log \frac{RR'}{SS'}, \quad \zeta = \frac{1}{2} \log \lambda.$$

There is some difficulty as to the denominator factor  $\sin \zeta t$ , which becomes  $= 0$  for the series of values  $t = \frac{n\pi}{\zeta}$ , but this is a point which I do not enter into.

I will in the first instance verify the result. Writing  $x_1$  in place of  $x$ , and taking  $\xi_1, \eta_1$  to denote the corresponding values of  $\xi, \eta$ , it will be shown that

$$\xi_1 = \xi, \quad \eta_1 = \eta + 2\zeta, \quad \text{see post, (1).}$$

Hence in the difference  $\phi x - \phi x_1$  we have the integral

$$\int \frac{\sin \xi t \{ \sin \eta t - \sin(\eta + 2\zeta)t \} dt}{\sin \zeta t \sinh \pi t}$$

(where and in all that follows the limits are  $0, \infty$  as before); here, since

$$\sin \eta t - \sin(\eta + 2\zeta)t = -2 \sin \zeta t \cos(\eta + \zeta)t,$$

the factor  $\sin \zeta t$  divides out, and the numerator is

$$= -2 \sin \xi t \cos(\eta + \zeta)t,$$

which is

$$= \sin(\eta + \zeta - \xi)t - \sin(\eta + \zeta + \xi)t.$$

Hence the integral in question is

$$- \int \frac{\sin(\eta + \zeta - \xi)t dt}{\sinh \pi t} + \int \frac{\sin(\eta + \zeta + \xi)t dt}{\sinh \pi t}.$$

Now we have in general

$$\frac{1}{1 + \exp .a} - \frac{1}{2} = - \int \frac{\sin at dt}{\sinh \pi t},$$

(this is, in fact, Poisson's formula

$$\frac{1}{1 + e^a} - \frac{1}{2} = -2 \int \frac{\sin 2n \log \frac{1}{k} \log k \cdot t dt}{e^{2\pi t} - 1}$$

in the second Memoir on the distribution of Electricity, &c., *Mém. de l'Inst.*, 1811, p. 223); and hence the value is

$$\frac{1}{1 + \exp .(\eta + \zeta - \xi)} - \frac{1}{1 + \exp .(\eta + \zeta + \xi)},$$

or since

$$\eta + \zeta = \frac{1}{2} \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'}, \quad \xi = \frac{1}{2} \log \lambda,$$

we have

$$\eta + \zeta - \xi = \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'} = \log \lambda \frac{R}{S},$$

$$\eta + \zeta + \xi = \log \lambda + \frac{1}{2} \log \frac{R'^2}{S'^2} = \log \lambda \frac{R'}{S'},$$



and the value is thus

$$\frac{1}{1 + \lambda \frac{R}{S}} - \frac{1}{1 + \lambda \frac{R'}{S'}} = \frac{(RS' - R'S)\lambda}{(\lambda R + S)(\lambda R' + S')}.$$

Hence, from the assumed value of  $\phi x$ , we obtain

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(dC - cD)} \frac{(RS' - R'S)\lambda}{(\lambda R + S)(\lambda R' + S')}.$$

We have

$$RS' - R'S = \frac{(\lambda - 1)(\lambda^2 - 1)}{\lambda}(dC - cD)\{cx^2 + (d - a)x - b\},$$

$$R\lambda + S = (\lambda^2 - 1)(cx + d), \quad \text{see post, (2),}$$

or since

$$R'\lambda + S' = (\lambda - 1)(a + d)(Cx + D),$$

this is

$$\phi x - \phi x_1 = (x - x_1) \frac{A}{C} + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(dC - cD)} \frac{cx^2 + (d - a)x - b}{(a + d)\lambda} \lambda'.$$

But from the value of  $\lambda$ ,

$$\frac{\lambda}{\lambda^2 - 1} = \frac{ad - bc}{(a + d)\sqrt{(a - d)^2 + 4bc}},$$

and the equation thus is

$$\phi x - \phi x_1 = (x - x_1) \left\{ \frac{A}{C} - \frac{AD - BC}{C(Cx + D)} \right\} = (x - x_1) \frac{Ax + B}{Cx + D},$$

as it should be.

(1) For the foregoing values of  $\xi_1, \eta_1$ , we require  $R_1, S_1, R'_1, S'_1$ , the values which  $R, S, R', S'$  assume on writing therein  $x_1$  for  $x$ . We have

$$R_1 = \lambda(cx_1 + d) + (cx_1 - a),$$

$$S_1 = -(cx_1 + d) - \lambda(cx_1 - a);$$

substituting for  $x_1$  its value, we find

$$R_1(cx + d) = (a + d)\lambda(cx + d) - (ad - bc)(\lambda + 1),$$

or writing herein

$$ad - bc = \frac{(a + d)^2 x}{\lambda + 1},$$

this is

$$R_1(cx + d) = \frac{a + d}{\lambda + 1} R,$$

and similarly

$$S_1(cx + d) = \frac{a + d}{\lambda + 1} S.$$

We have in like manner

$$R'_1 = W_1 + \frac{1}{\lambda}Z_1, \text{ where } W_1 = C(ax_1 + b) + D(cx_1 + d),$$

$$S'_1 = -W_1 - \lambda Z_1, \text{ where } Z_1 = C(-dx_1 + b) + D(cx_1 - a).$$

Substituting for  $x_1$  its value, we find

$$W_1(cx + d) = C[(a + d)(ax + b) - (ad - bc)x] + D[(a + d)(cx + d) - (ad - bc)],$$

$$Z_1(cx + d) = C[-(ad - bc)x] + D[-(ad - bc)],$$

hence, substituting for  $ad - bc$  as before,

$$W_1(cx + d) = \frac{a + d}{(\lambda + 1)^2} \{(\lambda + 1)^2 W - (a + d)\lambda(Cx + D)\},$$

$$Z_1(cx + d) = \frac{a + d}{(\lambda + 1)^2} \{-(a + d)\lambda(Cx + D)\},$$

whence without difficulty

$$\frac{R'_1}{R_1} = \frac{R'}{R}; \quad \frac{S'_1}{S_1} = \frac{S'}{S} \cdot \frac{1}{\lambda},$$

consequently

$$\frac{R_1 R'_1}{S_1 S'_1} = \frac{RR'}{SS'} \cdot \lambda, \quad \eta_1 = \log \lambda + \eta = \eta + 2\zeta, \quad \xi_1 = \xi,$$

which are the formulæ in question.

(2) For the value of  $RS' - R'S$ , we have

$$\begin{aligned} RS' - R'S &= (\lambda c + d)(-W - \lambda Z) - (-\lambda d - c)(W + \frac{1}{\lambda}Z) \\ &= (-\lambda^2 + 1)cZ + (\lambda + 1)\{(d - c)W - dR\} \div \frac{1}{\lambda} \\ &= -(\lambda - 1) \left\{ \left( 1 + \lambda + \frac{1}{\lambda} \right) cZ + (c - d)W + dZ \right\}; \end{aligned}$$

or substituting for  $\lambda + \frac{1}{\lambda}$ ,  $W$  and  $Z$  their values, this is

$$\begin{aligned} &= \frac{-(\lambda - 1)}{ad - bc} \{(a^2 + d^2 + ad + bc)c(bC + dD) \\ &\quad + (ad - bc)[(c - d)(aC + cD) + d(bC + dD)]\}. \end{aligned}$$

In the term in  $\{ \}$ , the coefficient of  $C$  is

$$\begin{aligned} &[(a^2 + d^2 + ad + bc)b + (ad - bc)a]c - d(a - b)(ad - bc) \\ &= (a + d)(db - bd)c - (a + d)dx(ad - bc), \end{aligned}$$

and similarly the coefficient of  $D$  is

$$\begin{aligned} & [(a^2 + d^2 + ad + bc)d + (ad - bc)c]c - d(c - d)(ad - bc) \\ &= (a + d)(ad - bc)c - (a + d)d(ad - bc). \end{aligned}$$

Hence the whole term in  $\{ \}$  is

$$= (a + d)\{[(db - bd)c - d(ad - bc)x]C + [(ad - cb)c - d(ad - bc)]D\},$$

which is readily reduced to

$$(a + d)(ad - bc)(-dC + cD);$$

also

$$ad - bc = (a + d)\{cx^2 + (d - a)x - b\};$$

so that we have

$$RS' - R'S = \frac{(\lambda - 1)(\lambda^2 - 1)}{\lambda}(dC - cD)\{cx^2 + (d - a)x - b\},$$

which is the required value of  $RS' - R'S$ ; and there is no difficulty in obtaining the other two formulæ,

$$R\lambda + S = (\lambda^2 - 1)(cx + d),$$

$$R'\lambda + S' = (\lambda - 1)(a + d)(Cx + D);$$

the verification is thus completed.

To show how the formula was directly obtained, we have

$$\begin{aligned} \frac{Ax + B}{Cx + D} &= \frac{A}{C} - \frac{AD - BC}{C} \cdot \frac{1}{Cx + D} \\ &= \frac{A}{C} + \beta x, \text{ suppose;} \end{aligned}$$

the equation then is

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) + (x - x_1)\beta x.$$

Hence, if  $x_1, x_2, x_3, \dots$  denote the successive functions  $\psi x, \psi^2 x, \psi^3 x$ , &c., we have

$$\phi x - \phi x_1 = \frac{A}{C}(x - x_1) + (x - x_1)\beta x,$$

$$\phi x_1 - \phi x_2 = \frac{A}{C}(x_1 - x_2) + (x_1 - x_2)\beta x_1,$$

whence adding, and neglecting  $\phi x_\infty$  and  $x_\infty$ , we have

$$\phi x = \frac{A}{C}x + [(x - x_1)\beta x + (x_1 - x_2)\beta x_1 + (x_2 - x_3)\beta x_2 + \dots],$$

where the term in  $[ ]$ , regarding therein  $x_1, x_2, x_3, \dots$  as given functions of  $x$ , is itself a given function of  $x$ ; and it only remains to sum the series.

Starting from

$$x_1 = \psi x = \frac{ax + b}{cx + d},$$

and writing

$$\lambda + \frac{1}{\lambda} = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

then the  $n$ th function is given by the formula

$$\begin{aligned} x_n = \psi^n x &= \frac{(\lambda^{n+1} - 1)(ax + b) + (\lambda^n - \lambda)(-dx + b)}{(\lambda^{n+1} - 1)(cx + d) + (\lambda^n - \lambda)(cx - a)} \\ &= \frac{(\lambda^{n+1} - 1)a + (\lambda^n - \lambda)b}{(\lambda^{n+1} - 1)c + (\lambda^n - \lambda)d} \quad (\text{numerator}) \div (\text{denominator}) \\ &= \frac{\lambda^n P + Q}{\lambda^n R + S}, \end{aligned}$$

if  $P = \lambda a + b$ ,  $Q = -a - \lambda b$ , and  $R = \lambda c + d$ ,  $S = -c - \lambda d$ , as before.

I stop to remark that  $\lambda$  being real, then if  $\lambda > 1$  we have  $\lambda^n$  very large for  $n$  very large, and  $x_n = \frac{P}{R}$  which is independent of  $n$ ; the value in question is

$$x_n = \frac{\lambda(ax + b) + (-dx + b)}{\lambda(cx + d) + (cx - a)},$$

which, observing that the equation in  $\lambda$  may be written

$$\frac{\lambda a - d}{c(\lambda + 1)} = \frac{b(\lambda + 1)}{\lambda d - a},$$

is, in fact, independent of  $x$ , and  $= \frac{\lambda a - d}{c(\lambda + 1)}$  or  $\frac{b(\lambda + 1)}{\lambda d - a}$ : we have  $x_{n-1} = x_n$ , or calling each of these two equal values  $x$ , we have

$$x = \frac{ax + b}{cx + d},$$

which is the same equation as is obtainable by the elimination of  $\lambda$  from the equations

$$x = \frac{\lambda a - d}{c(\lambda + 1)} = \frac{b(\lambda + 1)}{\lambda d - a}.$$

The same result is obtained by taking  $\lambda < 1$  and consequently  $x_n = \frac{Q}{S}$ .

We find

$$x_{n-1} - x_n = \frac{\lambda^{n-1}P + Q}{\lambda^{n-1}R + S} - \frac{\lambda^n P + Q}{\lambda^n R + S},$$

where

$$PS - QR = -(\lambda^2 - 1)(ad - bc) = -(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\},$$

and therefore

$$x_{n-1} - x_n = \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^n}{\lambda(\lambda^{n-1}R + S)(\lambda^n R + S)}.$$

Also

$$\beta x_n = -\frac{AD - BC}{C} \cdot \frac{1}{Cx_n + D},$$

where

$$Cx_n + D = \frac{C(\lambda^n P + Q) + D(\lambda^n R + S)}{\lambda^n R + S} = \frac{R'\lambda^n + S'}{\lambda^n R + S},$$

where

$$\begin{aligned} R' &= CP + DR, = C\left(-b + \frac{1}{\lambda}\right) + D\left(c + \frac{1}{\lambda}\right), \\ S' &= CQ + DS, = C(-b - \lambda) + D(-c - d\lambda); \end{aligned}$$

viz.

$$R' = W + \frac{1}{\lambda}Z, \quad S' = -W - \lambda Z,$$

where  $Z$  and  $W$  denote  $aC + cD$  and  $bC + dD$  as before.

We hence obtain

$$\begin{aligned} (x_{n-1} - x_n)\beta x_n &= \frac{-(AD - BC)}{C} \cdot \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^n}{\lambda(R\lambda^n + S)(R'\lambda^n + S')} \\ &= \frac{-(AD - BC)}{C} \cdot \frac{(\lambda - 1)(\lambda^2 - 1)(a + d)\{cx^2 + (d - a)x - b\}\lambda^n}{\lambda(dC - cD)(R\lambda^n + S)(R'\lambda^n + S')}, \end{aligned}$$

or, substituting for  $RS' - R'S$  its value in the denominator, this is

$$\begin{aligned} (x_{n-1} - x_n)\beta x_n &= \frac{AD - BC}{C} \cdot \frac{-(ad - bc)(\lambda^n - 1)(RS' - R'S)\lambda^n}{(a + d)\lambda(cD - dC)(R\lambda^n + S)(R'\lambda^n + S')} \\ &= \frac{\sqrt{(a - d)^2 + 4bc}(AD - BC)(RS' - R'S)\lambda^n}{C(cD - dC)(R\lambda^n + S)(R'\lambda^n + S')}, \end{aligned}$$

and thence

$$\phi x = \frac{A}{C}x + \Sigma \frac{\sqrt{(a - d)^2 + 4bc}(AD - BC)(RS' - R'S)\lambda^n}{C(cD - dC)(R\lambda^n + S)(R'\lambda^n + S')},$$

the summation extending from 1 to  $\infty$ .

Now the before-mentioned integral formula gives

$$\begin{aligned} \frac{1}{1 + k\lambda^n} - \frac{1}{2} &= - \int \frac{\sin(n \log \lambda + \log k)t \, dt}{\sinh \pi t}, \\ \frac{1}{1 + k'\lambda^n} - \frac{1}{2} &= - \int \frac{\sin(n \log \lambda + \log k')t \, dt}{\sinh \pi t}. \end{aligned}$$

Taking the difference, and then writing  $k = \frac{R}{S}$ ,  $k' = \frac{R'}{S'}$ , we have under the integral sign

$$\sin \left( n \log \lambda + \log \frac{R}{S} \right) t - \sin \left( n \log \lambda + \log \frac{R'}{S'} \right) t,$$

which is

$$= 2 \sin \frac{1}{2} \left( \log \frac{RR'}{SS'} \right) t \cos \left( n \log \lambda + \frac{1}{2} \log \frac{RR'}{SS'} \right) t,$$

which attending to the before-mentioned values of  $\xi$ ,  $\eta$ ,  $\zeta$  is

$$= 2 \sin \zeta t \cos(2n\zeta - \xi + \eta)t,$$

and the formula thus is

$$\frac{S}{R\lambda^n + S} - \frac{S'}{R'\lambda^n + S'} = \frac{(RS' - R'S)\lambda^n}{(R\lambda^n + S)(R'\lambda^n + S')} = -2 \int \frac{\sin \zeta t \cos(2n\zeta - \xi + \eta)t dt}{\sinh \pi t}.$$

We have here

$$\cos(2n\zeta - \xi + \eta)t = \cos 2n\zeta t \cos(\eta - \xi)t - \sin 2n\zeta t \sin(\eta - \xi)t,$$

whence summing from 1 to  $\infty$  by means of the formulæ

$$\cos 2\zeta t + \cos 4\zeta t + \dots = -\frac{1}{2},$$

$$\sin 2\zeta t + \sin 4\zeta t + \dots = \frac{1}{2} \cot \zeta t,$$

(which series however are not convergent), the numerator under the integral sign becomes

$$- \sin \zeta t \{ -\cos(\eta - \xi)t - \cot \zeta t \sin(\eta - \xi)t \},$$

which is

$$= \frac{\sin \xi t \sin \eta t}{\sin \zeta t},$$

and the formula thus is

$$\Sigma \frac{(RS' - R'S)\lambda^n}{(R\lambda^n + S)(R'\lambda^n + S')} = - \int \frac{\sin \xi t \sin \eta t}{\sin \zeta t \sinh \pi t} dt,$$

and we therefore find

$$\phi x = \frac{A}{C}x + \frac{\sqrt{(a-d)^2 + 4bc}(AD - BC)}{C(cD - dC)} \int \frac{\sin \xi t \sin \eta t dt}{\sin \zeta t \sinh \pi t},$$

which is the result in question.

The solution is a particular one; calling it for a moment  $(\phi x)$ , then, if the general solution be  $\Phi x = \phi x + \varpi x$ , it at once appears that we must have  $\varpi x = \varpi x_1$ ; and as it has been shown that  $\frac{RR'}{SS'}$  is a function of  $x$  which remains unaltered by the change of  $x$  into  $x_1$ , this is satisfied by assuming  $\varpi x = f\left(\frac{RR'}{SS'}\right)$ , an arbitrary function of  $\frac{RR'}{SS'}$ . Hence we may to the foregoing expression of  $\phi x$  add this term  $f\left(\frac{RR'}{SS'}\right)$ .

*Postscript.* The new formula

$$x_n = \frac{(\lambda^{n+1} - 1)(ax + b) + (\lambda^n - \lambda)(-dx + b)}{(\lambda^{n+1} - 1)(cx + d) + (\lambda^n - \lambda)(cx - a)},$$

where

$$\lambda + \frac{1}{\lambda} = \frac{a^2 + d^2 + 2bc}{ad - bc},$$

for the  $n$ th repetition of  $\psi x$ ,  $= \frac{ax+b}{cx+d}$ , is a very interesting one. It is to be remembered that, when  $n$  is even the numerator and denominator each divide by  $\lambda-1$ , but when  $n$  is odd they each divide by  $\lambda^2-1$ ; after such division, then further dividing by a power of  $\lambda$ , they each consist of terms of the form  $\lambda^k + \frac{1}{\lambda^k}$ , that is, they are each of them a rational function of  $\lambda + \frac{1}{\lambda}$ . Substituting and multiplying by the proper power of  $ad-bc$ , the numerator and denominator become each of them a rational and integral function of  $a, b, c, d$  of the order  $n+1$  when  $n$  is even, but of the order  $n$  when  $n$  is odd; in the former case, however, the numerator and denominator each divide by  $a+d$ , so that ultimately, whether  $n$  be even or odd, the order is  $n$  as it should be.

For example, when  $n=2$ , the value is

$$\frac{(\lambda^3-1)a + (\lambda^2-\lambda)b}{(\lambda^3-1)c + (\lambda^2-\lambda)d} = \frac{(\lambda^2+\lambda+1)a + \lambda b}{(\lambda^2+\lambda+1)c + \lambda d} = \frac{(\lambda + \frac{1}{\lambda} + 1)a + b}{(\lambda + \frac{1}{\lambda} + 1)c + d},$$

or, as this may be written,

$$= \frac{(\lambda + \frac{1}{\lambda} + 2)a - a + b}{(\lambda + \frac{1}{\lambda} + 2)c - c + d},$$

where, observing that

$$\lambda + \frac{1}{\lambda} + 2 = \frac{(a+d)^2}{ad-bc}, \quad -a+b = -(a+d) \cdot \frac{ad-bc}{(a+d)^2},$$

the numerator and denominator each divide by  $a+d$ , and the final value is

$$= \frac{(a+d)(ax+b) - (ad-bc)x}{(a+d)(cx+d) - (ad-bc)} = \frac{(a^2+bc)x + b(a+d)}{c(a+d)x + bc + d^2},$$

which is the proper value of  $\psi^2 x$ . But, when  $n=3$ , the value is

$$\frac{(\lambda^4-1)a + (\lambda^3-\lambda)b}{(\lambda^4-1)c + (\lambda^3-\lambda)d} = \frac{(\lambda^2+1)a + \lambda b}{(\lambda^2+1)c + \lambda d}$$

and this is

$$= \frac{(a^2+d^2+2bc)(ax+b) + (ad-bc)(-dx+b)}{(a^2+d^2+2bc)(cx+d) + (ad-bc)(cx-a)}$$

or finally

$$= \frac{(a^3+2abc+bcd)x + b(a^2+ad+bc+d^2)}{c(a^2+ad+bc+d^2)x + (abc+2bcd+d^3)},$$

which is the proper value of  $\psi^3 x$ .

## 687.

NOTE ON THE FUNCTION  $\psi x = a^2(c - x) \div \{c(c - x) - b^2\}$ .[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 338–340.]

Starting from the general form

$$\psi x = \frac{\alpha x + \beta}{\gamma x + \delta},$$

we have

$$\psi^n x = \frac{(\lambda^{n+1} - 1)(\alpha x + \beta) + (\lambda^n - \lambda)(-\delta x + \beta)}{(\lambda^{n+1} - 1)(\gamma x + \delta) + (\lambda^n - \lambda)(\gamma x - \alpha)},$$

where

$$\lambda + \frac{1}{\lambda} = \frac{\alpha^2 + \delta^2 + 2\beta\gamma}{\alpha\delta - \beta\gamma}.$$

For the function in question

$$\psi x = \frac{a^2(c - x)}{c(c - x) - b^2},$$

(a form which presents itself in the problem of the distribution of electricity upon two spheres), the values of  $\alpha, \beta, \gamma, \delta$  are

$$\alpha = -a^2, \quad \beta = a^2c, \quad \gamma = -c, \quad \delta = c^2 - b^2;$$

the equation for  $\lambda$  therefore is

$$\lambda + \frac{1}{\lambda} = \frac{a^4 + (c^2 - b^2)^2 - 2a^2c^2}{a^2b^2},$$

or, what is the same thing,

$$\frac{(\lambda + 1)^2}{\lambda} = \frac{(a^2 + b^2 - c^2)^2}{a^2b^2}.$$



Suppose that  $a, b, c$  are the sides of a triangle the angles whereof are  $A, B, C$ ; then  $c^2 = a^2 + b^2 - 2ab \cos C$ , or we have

$$\frac{(\lambda + 1)^2}{\lambda} = 4 \cos^2 C;$$

or, writing this under the form

$$\sqrt{\lambda} + \frac{1}{\sqrt{\lambda}} = 2 \cos C,$$

the value of  $\lambda$  is at once seen to be  $= e^{2iC}$ ; and it is interesting to obtain the expression of the  $n$ th function in terms of the sides and angles of the triangle.

The numerator and the denominator are

$$\lambda^n P + Q,$$

$$\lambda^n R + S,$$

where

$$P = \lambda(\alpha x + \beta) + (-\delta x + \beta), \quad R = \lambda(\gamma x + \delta) + \gamma x - \alpha,$$

$$Q = -(\alpha x + \beta) - \lambda(-\delta x + \beta), \quad S = -(\gamma x + \delta) - \lambda(\gamma x - \alpha).$$

Hence, writing the numerator and the denominator in the forms

$$\lambda^{\frac{1}{2}n} P + \lambda^{-\frac{1}{2}n} Q,$$

these are

$$(P + Q) \cos nC + (P - Q)i \sin nC,$$

viz. they are

$$(R + S) \cos nC + (R - S)i \sin nC;$$

$$(\lambda - 1)(\alpha + \delta)x \cos nC + (\lambda + 1)\{(\alpha - \delta)x + 2\beta\}i \sin nC,$$

$$-(\lambda - 1)(\alpha + \delta) \cdot \cos nC + (\lambda + 1)\{2\gamma x - (\alpha - \delta)\}i \sin nC,$$

or, observing that  $\frac{\lambda - 1}{\lambda + 1} = i \tan C$  and removing the common factor  $i(\lambda + 1)$ , they may be written

$$\tan C(\alpha + \delta)x \cos nC + \{(\alpha - \delta)x + 2\beta\} \sin nC,$$

$$\tan C(\alpha + \delta) \cdot \cos nC + \{2\gamma x - (\alpha - \delta)\} \sin nC.$$

Substituting for  $\alpha, \beta, \gamma, \delta$  their values, these are

$$\tan C\{(c^2 - a^2 - b^2)x \cos nC\} + \{(b^2 - a^2 - c^2)x + 2a^2c\} \sin nC,$$

$$-a - \tan C\{(c^2 - a^2 - b^2) \cdot \cos nC\} + \{-2cx - (b^2 - a^2 - c^2)\} \sin nC,$$

$$= \tan C\{-ab \cos C \cdot x \cos nC\} + \{-ac \cos B \cdot x + a^2c\} \sin nC,$$

$$\tan C\{-ab \cos C \cdot x \cos nC\} + \{-cx + ac \cos B\} \sin nC,$$

$$= x\{-ab \sin C \cos nC - ac \cos B \sin nC\} + a^2c \sin nC,$$

$$-cx \sin nC + \{ac \cos B \sin nC - ab \sin C \cos nC\};$$

or, writing herein  $b \sin C = c \sin B$ , these are

$$\begin{aligned} & -acx\{\sin B \cos nC + \cos B \sin nC\} + a^2c \sin nC, \\ & -cx \sin nC + ac\{\cos B \sin nC - \sin B \cos nC\}, \end{aligned}$$

whence finally

$$\psi^n x = \frac{a^2 \sin nC - ax \sin(nC + B)}{a \sin(nC - B) - x \sin nC}.$$

As a verification, writing  $n = 1$ , we have

$$\psi x = \frac{a^2 \sin C - ax \sin A}{a \sin(C - B) - x \sin C} = \frac{a^2 c - acx \frac{\sin A}{\sin C}}{a \frac{\sin(C - B)}{\sin C} \cdot c - cx},$$

or observing that

$$ac \frac{\sin(C - B)}{\sin C} = c^2 - b^2,$$

(for this is  $\sin A = \sin(C - B) \sin^2 C - \sin^2 B$ ), we have

$$\psi x = \frac{a^2(c - x)}{c^2 - b^2 - cx},$$

as it should be. If in the formula for  $\psi^n x$  we write  $x = 0$ , we have a formula given in the Senate-House Problems, January 14, 1878: it was thus that I was led to investigate the general expression.

## 688.

## GEOMETRICAL CONSIDERATIONS ON A SOLAR ECLIPSE.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XV. (1878), pp. 340–347.]

I consider, from a geometrical point of view, the phenomena of a solar eclipse over the earth generally; attending at present only to the penumbral cone, the vertex of which I denote by  $V$ . It is convenient to regard the earth as fixed, and the sun and moon as moving each of them with its proper motion, and also with the diurnal motion. The penumbral cone meets the earth's surface in a curve which may be called the penumbral curve; viz. when the cone is not completely traversed by the earth's surface, (that is, when only some of the generating lines of the cone meet the earth's surface), the penumbral curve is a single (convex or hour-glass-shaped) oval; separated, as afterwards mentioned, into two parts, one of them lying away from the sun, and having no astronomical significance; but when the cone is completely traversed by the earth's surface, then the penumbral curve consists of two separate (convex) ovals; one of them lying away from the sun and having no astronomical significance, the other lying towards the sun. The intermediate case is when the cone just traverses the earth's surface, or is touched internally by the earth's surface; the penumbral curve is then a figure of eight, one portion of which lies away from the sun, and has no astronomical significance: there is another limiting case when the cone is touched externally by the earth's surface, the penumbral curve being then a mere point.

It is necessary to consider on the earth's surface a curve which may for shortness be termed the horizon; viz. this is the curve of contact of the cone, vertex  $V$ , circumscribed about the earth; it is a small circle nearly coincident with the great circle, which is the intersection by a plane through the centre of the earth at right angles to the line from this point to the centre of the sun.

Regarding  $V$  as a point in the heavens, capable of being viewed notwithstanding the interposition of the moon; the horizon, as above defined, is the curve separating

the portions of the earth's surface for which  $V$  is visible and invisible respectively. The horizon does or does not meet the penumbral curve, according as this last consists of a single oval or of two distinct ovals; viz. in the latter case the horizon lies between the two ovals, in the former case the horizon traverses the area of the oval (separating this area into two parts), thus meeting the oval, or penumbral curve, in two points, or say these points separate the oval into two parts; from any point of the one part  $V$  is visible, from any point of the other part  $V$  is invisible; and from each of the two points themselves  $V$  is visible as a point on the horizon in the ordinary sense of the word; that is, there is an exterior contact of the sun and moon visible on the horizon. It is to be observed that, in the limiting cases where the penumbral curve is a mere point and a figure of eight respectively, the horizon passes through the mere point and through the node of the figure of eight respectively.

The two points of intersection of the penumbral curve with the horizon may for shortness be termed *critic points*. The lines which present themselves in a diagram of a solar eclipse, (see *Nautical Almanac*;) are the “northern and south lines of simple contact,” say for shortness the “limits”; viz. these are the envelope or, geometrically, a portion of the envelope of the penumbral curve; and the lines of “eclipse begins or ends at sunrise or sunset,” say for shortness the *critic lines*; viz. these are the locus of the *critic points*.

The point  $V$  considered as a point in the heavens is a point occupying a position intermediate between those of the centres of the sun and moon; hence referring it to the surface of the earth by means of a line drawn from the centre, its position on the earth's surface is nearly coincident with that point to which the sun is then vertical; and its motion on the earth's surface is from east to west approximately along the parallel of latitude = sun's declination, and with a velocity of approximately  $15^\circ$  per hour. For any given position of  $V$  on the earth's surface, describing with a given angular radius nearly  $= 90^\circ$  a small circle (nearly a great circle), this is the horizon; as  $V$  moves upon the surface of the earth, the horizon envelopes a curve which is very nearly a parallel, angular radius = sun's declination (there are two such curves in the northern and southern hemispheres respectively, but I attend only to one of them in the proper hemisphere, as will be explained), say this is the horizon-envelope; the horizon in each of its successive positions is thus a curvilinear tangent (nearly a great circle) to this horizon-envelope. If for a given position of  $V$ , and also for the consecutive position we consider the corresponding horizons, these intersect in a point  $K$  on the horizon-envelope, and the horizon for  $V$  is the circle centre  $V$  and angular radius  $VK$ ;  $K$  is a point which is very nearly upon, and which may be taken to be upon, the meridian through  $V$ ; the horizon may be regarded as a tangent which sweeps round the horizon-envelope; to each position thereof there corresponds a position of  $V$ , and consequently also a penumbral curve; and (when this is a single oval) the horizon meets it in two points, which are the *critic points*. It is to be added that, if for a given position of the horizon we consider as well  $K$  as the opposite point  $K_1$ , (viz.,  $K_1$  lies on the great circle  $KV$ ), then the points  $K$  and  $K_1$  divide the horizon into two portions; for any point on one of these portions

$V$  (considered as a point in the heavens) is rising, for a point on the other of them it is setting; and for the points  $K$  and  $K_1$  respectively it is moving horizontally; that is, first rising and then setting, or vice versa.

A solar eclipse is of one of two classes; viz. either the penumbral cone completely traverses the earth, so that towards the middle of the eclipse the penumbral curve consists of two separate ovals: or the penumbral cone does not completely traverse the earth, so that throughout the eclipse the penumbral curve consists of a single oval only. In the former case, we have to consider the commencement, during which the penumbral curve passes from a mere point to a figure of eight: the middle, during which it passes from a figure of eight through two ovals to a figure of eight: and the termination, during which it passes from a figure of eight to a mere point. In the latter case, we consider the whole eclipse during which the penumbral curve passes from a mere point through a single oval to a mere point.

In an eclipse of the first class: for the commencement, the penumbral curve is at first a mere point (point of first contact); it then becomes a convex oval, each oval in the first instance inclosing the preceding ones, so that there is not any intersection of two consecutive ovals. We come at last to an oval which is touched north by its consecutive oval, and to an oval which is touched south by its consecutive oval (I presume that the contacts north and south do not take place on the same oval, but I am not sure); and after this, the ovals assume the hour-glass form, each oval intersecting the consecutive oval in two points north and two points south; the ovals thus beginning to form an envelope or limit. There are on each of the ovals two critic points, and we have thus a critic curve commencing at the mere point (point of first contact) and extending in each direction from this point. The point, where an oval is touched by the consecutive oval, is not so far as appears a critic point; that is, the critic curve does not at this point unite itself with the envelope or limit. But the critic curve comes subsequently to unite itself each way with the limit; and, since clearly it cannot intersect the limit, it will at each of these points touch the limit; that is, we have a critic curve extending each way from the point of first contact until it touches the northern limit and until it touches the southern limit. Observe that the penumbral curve, as being at first a mere point or an indefinitely small oval, does not at first contain within itself the point  $K$  or  $K_1$ ; it can only come to do this by passing through a position where the curve passes through  $K$  or  $K_1$ ; viz.  $K$  or  $K_1$  would then be a critic point; and I assume for the present that this does not take place. The critic curve at the point of first contact is a curve "eclipse begins at sunrise," and as not coming to pass through a point  $K$  or  $K_1$ , it cannot alter its character; that is, the critic curve, as extending each way from the point of first contact until it comes to touch the northern and southern limits respectively, is a curve "eclipse begins at sunrise"; at the terminal points in question, there is a mere contact of the sun and moon, so that they are points, where the eclipse begins and simultaneously ends at sunrise. Continuing the series of ovals until we arrive at the figure of eight, there are on each of them two critic points, which ultimately unite in the node of the figure of eight; these constitute a critic curve, extending each way from the node of the figure of eight to

the contacts with the northern and southern limits respectively. There is, as before, no passage through a point  $K$  or  $K_1$ , the curve in question thus retains throughout the same character; and by consideration of the two terminal points it at once appears that it is a curve “eclipse ends at sunrise.” The above-mentioned critic curves form together an oval touching the northern and southern limits respectively; say this is the sunrise oval.

The termination of the eclipse is similar to this, only the events happen in the reverse order; we have a critic line starting from the node of the figure of eight and extending each way until it comes to touch the northern and southern limits respectively, viz. this is the line “eclipse begins at sunset”; and then, extending each way from the points of contact to reunite itself at the point of last contact, this being the line “eclipse ends at sunset,” and the two portions together form an oval touching the northern and southern limits respectively; say this is the sunset oval. It is to be noticed that certain portions of the two limits are generated as the envelope of the penumbral curve during the commencement and during the termination of the eclipse.

For the middle of the eclipse; the penumbral curve, in the first instance a figure of eight, breaks up into two ovals, but only one of these is attended to; and ultimately the oval unites itself with another oval so as to give rise to a new figure of eight. There is thus throughout the middle of the eclipse a single oval: this has, north and south, an envelope which joins itself on to the portions enveloped during the latter part of the commencement and the former part of the termination of the eclipse, and constitutes therewith the northern and southern limits respectively, viz. each of these is considered as extending from a point of contact with the sunrise oval to a point of contact with the sunset oval.

The line  $K_1VK$ , or say the meridian line through  $V$ , travels westwardly, while the penumbral curve travels eastwardly; the two come to touch each other, and there are then two intersections which ultimately come to the northern and southern limits respectively: the locus of these is a line of “eclipse commences at midday”; as the motion continues, the points of intersection move away from the two limits respectively and ultimately unite at the point where the line  $KVK_1$  again touches the penumbral curve; the locus is the line of “eclipse terminates at midday,” the two lines together forming an oval which touches the northern and southern limits respectively and which may be termed the midday oval. In all that precedes, no distinction has been made between the two portions of the horizon-envelope, or the points  $K$  and  $K_1$ , and either curve and point indifferently may be alone attended to.

Considering now an eclipse of the second kind, the penumbral curve is at first a mere point (the point of first contact) and it then becomes an oval, the successive ovals not at first intersecting each other, but each oval inclosing within itself the preceding ones. Any oval is met by the corresponding horizon in two points  $P$  and  $P'$ , at first coinciding with each other at the point of first contact, and then separating from each other, one of them, say  $P$ , moving down towards and ultimately arriving at one of the horizon-envelopes, say to fix the ideas the southern one (which curve

is henceforth selected as being, and is called, the horizon-envelope, and the points on this curve are taken to be the points  $K$ ), viz.  $P$  is then a point on the penumbral curve, I call it  $K_1$ . The successive ovals will in the meantime have begun to intersect each other so as to give rise to a northern limit; this will touch the critic line (locus of  $P, P'$ ), and we have a portion of the critic line extending from the point of first contact, in one direction to the point of contact with the northern limit, and in the other direction to the point  $K_1$  on the horizon-envelope: this is the line “eclipse begins at sunrise.” As the horizon continues to sweep on, the other point  $P'$ , which has not yet reached the horizon-envelope, will gradually approach and ultimately arrive at the horizon-envelope, say at the point  $K_2$ ; we have thus a second portion of the critic line extending from the contact with the northern envelope to the point  $K_2$ ; this is the line “eclipse ends at sunrise.” The horizon continuing to sweep on, the point  $P$  beginning with the position  $K_1$ , which is now on the other side of the point of contact of the horizon with the horizon-envelope, will trace out a portion of the critic curve extending from  $K_1$  to a second point of contact with the northern limit; this will be the line of “eclipse begins at sunset.” And, finally, the point  $P$  from the last-mentioned point of contact, and the point  $P'$  from its position  $K_2$ , which is now on the other side of the point of contact of the horizon with the horizon-envelope, (that is,  $P, P'$  have now each passed through the point of contact of the horizon with the horizon-envelope, and are both of them on the same side thereof, viz. the side opposite to their original side), will come to unite at the point of the last contact; we have thus a fourth portion of the critic curve extending from  $K_2$  to the second point of contact with the northern limit, viz. this is the line “eclipse ends at sunset.” The description will be more intelligible by means of the figure, in which  $1, 1', 2, 2', \dots, 8, 8'$  represent successive corresponding positions of the points  $P, P'$ , the successive positions of the horizon being given by the right lines  $11', 22', \&c.$ , all of them tangents to the dotted circle or horizon-envelope.

The entire critic line is thus a figure of eight, twice touching the horizon-envelope

and also twice touching the limit. If we consider, as before, the intersections of  $KV$  with the corresponding penumbral curve, this will be a curve extending from  $K_1$  so as to touch the limit, and thence onward to  $K_2$ , the portion from  $K_1$  to the contact with the limit being the line “eclipse begins at transit,” and the portion from the limit to  $K_2$  the line “eclipse ends at transit.” I say “transit” instead of midday, since for a circumpolar place the phenomenon may happen at one or the other transit of the sun over the meridian. It is to be remarked, that the node of the figure of eight is a point, such that the eclipse there begins at sunrise and ends at sunset; this point does not appear to be an important one in the geometrical theory.

The two loops of the critic line may be of very unequal magnitudes, and in particular one of them may actually vanish; viz. the points  $K_1$  and  $K_2$  then coincide together, and the critic curve is a closed cuspidal curve touching the horizon-envelope at the cusp; moreover, instead of two contacts with the limit there is one proper contact, and an improper contact at the cusp, that is, the limit simply passes through the cusp. And through this special separating case, we pass to the case where, instead of the figure of eight, we have a single oval, not touching the horizon-envelope (viz. the points  $K_1, K_2$  have become imaginary), but still touching the limit twice: this is a distinct type for an eclipse of the second class.

And, similarly, in an eclipse of the first class, where the points  $K_1, K_2$  do not in general exist (viz. geometrically they are imaginary), these points may present themselves in the first instance as two coincident points, viz. instead of the sunrise oval or the sunset oval (as the case may be), we have then a cuspidal curve; or they may be two real points, viz. instead of the same oval, we have then a figure of eight touching the horizon-envelope twice, and also touching each of the two limits. These are thus the several cases.

When the Earth traverses the penumbral cone, the critic curve is

1. A pair of ovals;
2. An oval and a cuspidate oval;
3. An oval and a figure of eight.

And when the Earth does not traverse the penumbral cone, the critic curve is

4. A figure of eight;
5. A cuspidate oval;
6. An oval.

To which may be added the transition case which separates 1 and 4, viz. here the Earth just has an internal contact with the penumbral cone, and the critic curve is

7. Two ovals touching each other.

But of course 2, 5, and 7 are so special that they may be disregarded altogether; and 3 and 6 are of rare occurrence. I have not sufficiently examined the conditions for the occurrence of these forms 3 and 6; my attention was called to them, and indeed to the whole theory, by a question proposed by Prof. Adams in the Cambridge Smith's Prize Examination for 1869.



## 689.

ON THE GEOMETRICAL REPRESENTATION OF IMAGINARY  
VARIABLES BY A REAL CORRESPONDENCE OF TWO PLANES.

[From the *Proceedings of the London Mathematical Society*, vol. ix. (1878), pp. 31–39. Read December 13, 1877.]

In my recently published paper, “Geometrical Illustration of a Theorem relating to an Irrational Function of an Imaginary Variable,” *Proceedings of the London Mathematical Society*, t. viii. (1877), pp. 212–214, [627], I remark as follows:—“If we have  $v$  a function of  $u$  determined by an equation  $f(u, v) = 0$ , then to any given imaginary value  $x + iy$  of  $u$  there belong two or more values, in general imaginary, of  $v$ ; and for the complete understanding of the relation between the two imaginary variables we require to know the series of values  $x' + iy'$  which correspond to a given series of values  $x + iy$  of  $v, u$  respectively. We must, for this purpose, take  $x, y$  as the coordinates of a point  $P$  in a plane  $\Pi$ , and  $x', y'$  as the coordinates of a corresponding point  $P'$  in another plane  $\Pi'$ ; and I then proceed to consider the particular case where the equation between  $u, v$  is  $u^2 + v^2 = a^2$ , that is, where

$$(x + iy)^2 + (x' + iy')^2 = a^2.$$

The general case is that of an equation  $(*)(u, 1)^m(v, 1)^n = 0$ , where to each given value, real or imaginary, of  $u$ , there correspond  $n$  real or imaginary values of  $v$ ; and to each given value, real or imaginary, of  $v$ , there correspond  $m$  real or imaginary values of  $u$ . And then, writing  $u = x + iy$  and  $v = x' + iy'$ , and regarding  $(x, y), (x', y')$  as the coordinates of the points  $P, P'$  in the two planes  $\Pi, \Pi'$  respectively, we have a real  $(m, n)$  correspondence between the two planes; viz. to each real point  $P$  in the first plane there correspond  $n$  real points  $P'$  in the second plane, and to each real point  $P'$  in the second plane there correspond  $m$  real points  $P$  in the first plane. But such real correspondence of two planes does not of necessity arise from an equation between the two imaginary variables  $u, v$ ; and the

question of the real correspondence of two planes may be considered in itself, without any reference to such origin.

I was under the impression that the theory was a known one; but I have not found it anywhere set out in detail. It is to be noticed that, although intimately connected with, it is quite distinct from (and seems to me to go beyond) that of a Riemann's surface. Riemann represents the value  $u, = x + iy$ , by a point  $P$  whose coordinates are  $x, y$ ; but he considers  $u', = x' + iy'$ , as a given imaginary value attached to the point  $P$ , without representing this value by a point  $P'$ , coordinates  $x', y'$ .

I proceed to consider the general theory of the real  $(m, n)$  correspondence. Points in the first plane are denoted by the unaccented letters  $P, Q, \dots$ ; and the corresponding points in the second plane are in general denoted by the same letters accented; but there are, as will be explained, special points  $V, W$ , where the points are interchanged; viz. to the points  $V$  or  $W$  in the first plane correspond  $W$  or  $V$  in the second plane.

1. To a point  $P$  there correspond in general  $n$  distinct points  $P'$ ; and as  $P$  varies continuously, each of the points  $P'$  also varies continuously.

2. There are certain points  $V$  called branch-points (*Verzweigungspunkte*), such that to each point  $V$  there correspond two united points, represented by  $(W)$ , and  $n - 2$  other distinct points  $W$ . The points  $(W)$  are called cross-points, and the number of them is of course equal to that of the branch-points  $V$ .

It is throughout assumed that a point denoted by a letter other than  $V$  is not a point  $V$ .

3. If the point  $P$ , moving continuously, describe a closed curve so as to return to its original position, then, if this curve includes within it no point  $V$  (or all the points  $V$ )<sup>1</sup>, each of the corresponding points  $P'$  will describe continuously a closed curve returning into its original position. Supposing that the curve described by  $P$  is an oval (non-autotomic closed curve), and taking this to be in the first instance an indefinitely small oval, then the curves described by the points  $P'$  will in the first instance be each of them an indefinitely small oval; but it is worth while to notice how, as the oval described by  $P$  increases, any one of the ovals described by a point  $P'$  may become autotomic; viz. if the oval described by  $P$  passes through two points  $Q, Q$  of the  $m$  points  $Q$  which correspond in the first plane to the same  $Q'$  in the second plane, then  $Q'$  will be a node in the closed curve described by that point  $P'$  which in the course of its motion comes to pass through  $Q'$ . This curve is in general an inloop curve composed of two loops, one wholly within the other (united at the point  $Q'$ ), and such that they each include one and the same point  $V$  (viz.  $V$  is included within the inner loop): as to this, see post, Nos. 9 and 10. It will be observed that this node  $Q'$  is not a point  $(W)$  nor any other special point of the second plane.

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<sup>1</sup>The two cases of the closed curve including no point  $V$ , and including all the points  $V$ , are really identical, as the discontinuity at infinity may be disregarded. It is to be observed that, this being so, it follows that the number of the points  $V$  must be even.

4. Consider, as before,  $P$  as describing a closed curve which does not include within it any point  $V$ , and the corresponding points  $P'$  as describing each of them a closed curve. As the curve described by  $P$  approaches a point  $V$ , the curves described by two of the points  $P'$  will approach the corresponding point ( $W$ ); and when the curve described by  $P$  passes through  $V$ , the curves described by the two points  $P'$  will unite together at this point ( $W$ ) as a node; viz. they will form a figure of eight<sup>2</sup>, the crossing being at the cross-point ( $W$ ), which corresponds to the branch-point  $V$ . And, corresponding to the closed curve described by  $P$ , we have this figure of eight (replacing two of the original  $n$  closed curves), and  $n - 2$  closed curves described by the other points  $P'$ .

5. Supposing, next, that the closed curve described by  $P$  (instead of passing through the point  $V$ ) includes within it the point  $V$ , then the figure of eight transforms itself into a twice-indented oval. There are on this curve two of the points  $P'$  which correspond to the given point  $P$ ; and as  $P$ , moving continuously in its closed curve, returns to its original position, the first of these points  $P'$ , moving continuously along a portion of the curve, comes to coincide with the original position of the second point  $P'$ ; while the second point  $P'$ , moving continuously along the remaining portion of the curve, comes to coincide with the original position of the first point  $P'$ ; viz. the two portions of the curve are described by the two points  $P'$  respectively. The curve may thus be regarded as a bifid curve, belonging to these two points  $P'$ . And, corresponding to the closed curve described by  $P$ , we have this bifid curve belonging to the two points  $P'$ , and  $n - 2$  single closed curves belonging to the other  $n - 2$  points  $P'$  respectively.

6. If the closed curve described by  $P$  (including within it a point  $V$ ) comes to pass through a second point  $V$ , the effect will be a new node at the corresponding point ( $W$ ); viz. at this point ( $W$ ) either the bifid curve unites itself with one of the single curves, or two of the single curves unite together, or the bifid curve there cuts itself. And, if the curve described by  $P$  comes to include within it this second point  $V$ , then in the three cases respectively: the bifid curve takes to itself the single curve, so that the system then is a trifold curve and  $n - 3$  single curves; or the two single curves give rise to a bifid curve, so that the system is two bifid curves and  $n - 4$  single curves; or, lastly, the bifid curve breaks up into two single curves, so that the system resumes its original form of  $n$  single curves.

7. We thus see how the closed curve described by  $P$ , including within it certain of the points  $V$ , may be such as to have corresponding to it an  $\alpha$ -fid curve, a  $\beta$ -fid curve, &c., ( $\alpha + \beta + \dots = n$ ); viz. an  $\alpha$ -fid curve contains upon it  $\alpha$  of the points  $P'$  which correspond to the original position of  $P$ ; and then, as  $P$  describes

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<sup>2</sup>The name *figure of eight* refers to the case where the two curves which come to unite at ( $W$ ) are proper ovals (non-autotomic closed curves). They might have one or both of them a node or nodes, as explained in No. 3; and the term would then be inappropriate. And so, lower down, the name twice-indented oval is used to express the form into which a proper figure of eight is changed by the disappearance of the node.

continuously its closed curve, returning to its original position, each of these points  $P'$  describes a portion of the  $\alpha$ -fid curve, passing from its original position to the original position of a point  $P'$  next to it upon the  $\alpha$ -fid curve; and the like as to a  $\beta$ -fid curve, &c. The numbers  $\alpha, \beta, \dots$  are not of necessity unequal, and we may have sets of equal numbers in any manner. It is hardly necessary to remark that, if the curve described by  $P$  passes through any point or points  $V$ , then two of the curves described by the points  $P'$  will unite together, or it may be that one of these will cut itself at the corresponding point or points ( $W$ ); and further that, as in No. 3, if the curve described by  $P$  passes through two or more of the points  $Q$  which correspond to the same point  $Q'$ , then any such point  $Q'$  will present itself as a node upon the curve belonging to some point, or set of points,  $P'$ . But the order of succession in which the original  $n$  single curves unite themselves together into multifid curves, or again break up into single curves, cannot, it would appear, be explained in any general manner, and would in each case depend on the nature of the particular correspondence.

8. We may consider the case where the closed curve described by  $P$  cuts itself. The curve may here be considered as made up of two or more ovals, or, to use a more appropriate term, say loops, each such loop being a curve not cutting itself; and the case is thus reducible to that before considered, where the curve does not cut itself. Thus, to fix the ideas, let the curve be a figure of eight, the initial position of  $P$  being at the crossing, and let neither of the loops contain within it a point  $V$ . Then, as  $P$  passes continuously along one of the loops, returning to its original position, each of the corresponding points  $P'$  describes a closed curve, which will be in the nature of a loop, viz. the initial and final directions of the motion of  $P$  not being continuous with each other, the initial and final directions of the motions of each point  $P'$  will not be continuous with each other, or there will be at the point  $P'$  an abrupt change in the direction of the curve. Similarly, as  $P$  describes the other loop of the figure of eight, each of the points  $P'$  will describe another loop; and the two loops belonging to the same point  $P'$  will unite together so as to form a figure of eight; viz. to the figure of eight described by  $P$  there will correspond figures of eight described by the  $n$  points  $P'$  respectively.

9. But consider next the case where the two loops of the curve described by  $P$  include each of them one and the same point  $V$ . This implies that one of the two loops lies inside the other, or that the curve is what has been called an inloop curve. As  $P$ , which is in the first instance taken to be at the node, passes continuously along one of the loops and returns to its original position, there are two of the points  $P'$  such that the first of these passes from its original position to the original position of the second, and the second of them passes from its original position to the original position of the first of them. We have thus two arcs between these two points  $P'$ ; but inasmuch as the initial and the final directions of motion of the point  $P$  are not continuous with each other, these two arcs are not continuous in direction at the two points  $P'$ , but at each of these points  $P'$  the two arcs meet at an angle. As  $P$  describes the other loop, we have in like manner two arcs between the same two points  $P'$ , these arcs at each of the points  $P'$  meeting at an

angle; but they join on to the first-mentioned two arcs in such manner as to form two ovals intersecting each other in the two points  $P'$ . Corresponding to the inloop curve described by  $P$ , we have this pair of intersecting ovals described by two of the points  $P'$ , and  $n - 2$  other curves described by the other points  $P'$ , and being each of them (I assume) an inloop curve.

10. If we attend only to one of the two intersecting ovals, we have in the first plane an inloop curve, and corresponding thereto in the second plane an oval passing through two of the points  $P'$  which correspond to the node  $P$  of the inloop curve. Interchanging the two planes, and writing  $Q$  instead of  $P$ , we have in the first plane an oval passing through two of the points  $Q$  which correspond to a point  $Q'$ ; and corresponding to this oval we have in the second plane an inloop curve having this point  $Q'$  for its node, viz. these are the corresponding figures mentioned in No. 3.

11. Consider a given point  $Q$ ; and let the corresponding points  $Q'$  be called (selecting the suffixes at pleasure)  $Q'_1, Q'_2, \dots, Q'_n$ . Taking then a point  $O$  indefinitely near to  $Q$ , the corresponding points  $O'$  will be indefinitely near to  $Q'_1, Q'_2, \dots, Q'_n$  respectively, and they will be called  $O'_1, O'_2, \dots, O'_n$  accordingly. It is to be observed that by the indefinitely near point  $O$  is meant a point such that the distance from  $O$  to  $Q$  is indefinitely small in comparison with the distance of either of these points from any point  $V$ ; so that we cannot have from  $Q$  to  $O$  two indefinitely short paths including between them a point  $V$ ; or say so that the indefinitely short path from  $Q$  to  $O$  is determinate.

Proceeding in this manner from  $Q$  to  $O$ , and so through a succession of indefinitely near points to a distant point  $S$ , we seem to determine the suffixes of the corresponding points  $S'$ ; but, by what precedes, it appears that such determination for a point  $S$  is dependent on the path from  $Q$  to  $S$ ; and consequently that we do not thus obtain a proper determination of the suffixes of the points  $S'$ . In fact, if we were to pass from  $Q$  by a path including one or more of the points  $V$  back to  $Q$ , we should obtain for the several points  $Q'$  respectively suffixes which are in general different from the suffixes originally given to these points respectively.

12. The difficulty is got over as follows:—Considering as before the given point  $Q$ , and calling the corresponding points  $Q'_1, Q'_2, \dots, Q'_n$  at pleasure, we pass from  $Q$  to the indefinitely near point  $O$ , and thence, by so many paths chosen at pleasure, to the several branch-points  $V$ ; these paths from  $O$  to the several points  $V$  are called barriers. To fix the ideas, we may consider these as non-autotomic non-intersecting lines drawn from  $O$  to the several points  $V$ . Consider the barrier from  $O$  to one of these points  $V$ ; as  $P$  passes along this barrier from  $O$  to  $V$ , two of the corresponding points  $P'$  will pass from two of the corresponding points  $O'$  to the corresponding cross-point ( $W$ ); the paths of these two points are called the counter-barrier corresponding to the barrier in question; and we have thus in the second plane a system of counter-barriers, each drawn from two points  $O'$  to meet in a point ( $W$ ). By what precedes, the points  $O'$  have each of them a determinate suffix; a counter-barrier is thus drawn from two points with given suffixes, suppose  $O'_1$  and  $O'_2$ , to a

point ( $W$ ), and this may be distinguished accordingly as a counter-barrier 12; and in like manner the cross-point ( $W$ ) through which it passes will be called a cross-point ( $W_{12}$ ); and the barrier corresponding hereto, and the branch-point  $V$  at which it terminates, will in like manner be called a barrier 12, and a branch-point  $V_{12}$ . Each barrier and branch-point will thus have a pair of suffixes; and the corresponding counter-barrier and cross-point will have the same pair of suffixes. It is to be observed that two or more of these corresponding figures may very well have the same pair of suffixes; but that such corresponding figures must be distinguished from each other; thus, if there are two branch-points  $V_{12}$ , these may be distinguished as the branch-points  $\alpha V_{12}$  and  $\beta V_{12}$ , and the barriers, counter-barriers, and cross-points by means of these same letters  $\alpha$  and  $\beta$ , (or otherwise), as may be convenient. It would seem that not only the number of the points  $V$  must be even, but the number of each set of points  $V_{12}$  must also be even (see post, No. 15).

13. It is also to be noticed that the determination of the suffixes of the several points  $V$ , &c., depends first upon the arbitrary choice of the suffixes of the points  $Q'$ , and next on the choice of the system of barriers; but that, these being assumed, the suffixes of the several points  $V$ , &c., are completely determinate.

14. Taking now any point  $S$  whatever, and supposing that  $P$  moves from  $Q$  continuously to  $S$  by a path which does not meet a barrier, the points  $P'$  will move from the several points  $Q'$  to the several points  $S'$  by paths not meeting the counter-barriers; viz. to each point  $S'$  there will be a path from some point  $Q'$ ; and giving to such point  $S'$  the suffix of the point  $Q'$ , the suffixes of the several points  $S'$  which correspond to any point whatever,  $S$ , will be completely determined. The determination depends of course on the assumptions referred to No. 13, but not in anywise on the position of the point  $S$ .

It will be noticed that, as all the points  $V$  are connected together by the barriers, the only closed paths from a point to itself are paths not including any, or including all, of the points  $V$ ; and that between such paths there is no real distinction.

15. Consider a point  $P$  moving continuously in any manner. The several corresponding points  $P'_1, P'_2, \dots, P'_n$  will each of them move continuously, but the suffixes interchange; viz. when  $P$  arrives at and then passes over a barrier  $\alpha\beta$ , the corresponding points  $P'_\alpha$  and  $P'_\beta$  will each arrive at the corresponding counter-barrier  $\alpha\beta$ , and, on passing over this,  $P'_\alpha$  will be changed into  $P'_\beta$  and  $P'_\beta$  into  $P'_\alpha$ , the other points  $P'$  remaining unchanged; and the like in other cases. This in fact includes the whole or the greater part of the foregoing theory. Thus, if  $P$  describe a closed curve not cutting any barrier, there will be no change of suffix; and when  $P$  returns to its original position each of the corresponding points  $P'_1, P'_2, \dots, P'_n$  will describe a closed curve, returning to its original position. But suppose that  $P$  describes a closed curve, cutting once only a barrier 12; suppose that the path is from  $P$  to  $Q$ , and then crossing the barrier to  $R$ , and thence again to  $P$ ;  $P'_1$  passes to  $Q'_1$ , and then crossing the counter-barrier it passes from  $P'_1$  to  $P'_2$ ; while at the same time  $P'_2$  passes to  $Q'_2$ , and then crossing the counter-barrier it passes from  $R'_2$  to  $P'_1$ ;

viz. we have  $P'_1, P'_2$  describing the two portions of a bifid curve. If there were only a single branch-point  $V_{12}$ , and therefore only a single barrier  $OV_{12}$ , then we might have through  $P$  a closed curve cutting  $OV_{12}$  once only, and including within it the point  $O$ , but not including within it the point  $V_{12}$ ; and here there ought not to be a bifid curve, but the points  $P'_1, P'_2$  ought to describe each of them a single curve. But suppose there are two points  $V_{12}$ , and consequently two barriers  $OV_{12}$  (meeting in  $O$ ); then the closed curve, meeting once only a barrier 12, (viz. it meets only one such barrier, and that once only), must include within one and only one of the two points  $V_{12}$ ; and in this case there ought to be a bifid curve. It is by such reasoning as this that I infer the foregoing theorem (No. 12), that the number of each set of points  $V_{12}$  is even.

16. We may consider how the suffixes are affected when, instead of the original system of barriers, we have a new system of barriers. I suppose that we have in the two cases respectively the same point  $Q$ , and the same suffixes for the points  $Q'_1, Q'_2, \dots, Q'_n$  which correspond thereto. In the first case, passing from  $Q$  to an indefinitely near point  $O$ , say the red  $O$ , we draw from this point to the several points  $V$  a set of barriers, say the red barriers; while in the second case, passing from  $Q$  to an indefinitely near point  $O$ , say the blue  $O$ , we draw from this point to the several points  $V$  a set of barriers, say the blue barriers; and we then proceed as before, viz. in the first case, drawing from  $Q$  to the point  $S$  a curve which does not meet any of the red barriers, we determine accordingly the suffixes (say the red suffixes) of the several corresponding points  $S'$ ; and in the second case, drawing in like manner from  $Q$  to  $S$  a curve which does not meet any of the blue barriers, we determine accordingly the suffixes (say the blue suffixes) of the same points  $S'$ . Now the curve drawn from  $Q$  to  $S$  so as not to cut any of the red barriers, and which is used for the determination of the red suffixes of the several points  $S'$ , will in general cut certain of the blue barriers; and, by examining the suffixes of the blue barriers which are thus cut, we determine the blue suffixes of the same points  $S'$ ; the result of course depending only on the situation of  $S$  in one or other of the regions formed by the red barriers and the blue barriers conjointly. In particular, the point  $S$  may be so situate that we can from  $Q$  to  $S$  draw a curve not meeting any red barrier or any blue barrier; and in this case the red suffixes and the blue suffixes are identical.

17. We may imagine the first plane as consisting of  $n$  superimposed planes or sheets, say the sheets  $1, 2, \dots, n$ . Each barrier 12 is considered as a line drawn in the two sheets 1 and 2; and so on in other cases. The point  $P$  is considered as a set of superimposed points  $P_1, P_2, \dots, P_n$  moving in the several sheets respectively; under the convention that  $P_1$ , moving in the sheet 1, and coming to cross a barrier 12, passes into the sheet 2 and becomes  $P_2$ ; and the like in other cases. And this being so, we say that to a point  $P$ , considered as a point  $P_\alpha$  in the sheet  $\alpha$ , there corresponds in the second plane one and only one point  $P'_\alpha$ ; and that  $P$  moving continuously in any manner (subject to the change of sheet as just explained), each of the  $n$  corresponding points  $P'$  will also move continuously, and so that each such point  $P'_\alpha$  will return

to its original position, upon the corresponding point  $P_\alpha$  returning to its original position and sheet. This is, in fact, Riemann's theory, only instead of the points  $P'$  we must speak of the values  $x' + iy'$  of the irrational function of  $x + iy$ .

18. Everything is of course symmetrical as regards the two planes; we have therefore, in the second plane, a system of points  $V$  and of barriers, and in the first plane a system of points  $(W)$  and of counter-barriers. To a given point  $P'$  in the second plane there correspond  $m$  points  $P$  in the first plane; and we can (the determination depending on the system of barriers in the second plane) assign to the  $m$  points suffixes, thereby distinguishing them as the corresponding points  $P_1, P_2, \dots, P_m$ . And we may imagine the second plane as consisting of  $m$  superimposed planes or sheets, say the sheets  $1, 2, 3, \dots, m$ ; the general theorem then is that to a point  $P$  or  $P'$  in either plane, considered as a point  $P_\alpha$  or  $P'_{\alpha'}$  in the sheet  $\alpha$  or  $\alpha'$ , there corresponds in the other plane one and only one point  $P'_{\alpha'}$  or  $P_\alpha$ ; and that the first-mentioned point in either plane moving continuously in any manner (subject to the proper change of sheet), the corresponding point in the other plane will also move continuously, and will return to its original position and sheet, upon the first-mentioned point returning to its original position and sheet.

19. In all that precedes it has been assumed that, to a branch-point  $V$ , there correspond two united points represented by  $(W)$  and  $n-2$  distinct points  $W$ ; the cases of a point  $(W)$  composed of three or more united points, or of the points  $W$  uniting themselves in sets in any other manner, would give rise to further specialities.



## 690.

## ON THE THEORY OF GROUPS.

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I recapitulate the general theory so far as is necessary in order to render intelligible the quasi-geometrical representation of it which will be given.

Let  $\alpha, \beta, \dots$  be functional symbols each operating upon one and the same number of letters, and producing as its result the same number of functions of these letters. For instance,  $\alpha(x, y, z) = (X, Y, Z)$ , where the capitals denote each of them a given function of  $(x, y, z)$ .

Such symbols are susceptible of repetition and combination

$$\alpha^2(x, y, z) = \alpha(X, Y, Z),$$

or

$$\beta\alpha(x, y, z) = \beta(X, Y, Z),$$

in each case equal to three given functions of  $(x, y, z)$ ; and similarly for  $\alpha^3, \alpha^2\beta$ , etc.

The symbols are not in general commutative,  $\alpha\beta$  not  $= \beta\alpha$ ; but they are associative,  $\alpha\beta \cdot \gamma = \alpha \cdot \beta\gamma$ , each  $= \alpha\beta\gamma$ , which has thus a determinate meaning.

Unity as a functional symbol denotes that the letters are unaltered,

$$1(x, y, z) = (x, y, z);$$

whence

$$1\alpha = \alpha 1 = \alpha.$$

The functional symbols may be substitutions;  $\alpha(x, y, z) = (y, z, x)$ , the same letters in a different order. Substitutions can be represented by the notation  $\begin{pmatrix} x & y & z \\ y & z & x \end{pmatrix}$ , the substitution which changes  $xyz$  into  $yzx$ , or, as products of cyclical substitutions,  $\begin{pmatrix} x & y & z & u & v & w \\ y & z & x & w & v & u \end{pmatrix} = (xyz)(uw)$ , the product of the cyclical substitutions  $x$  into  $y$ ,  $y$  into  $z$ ,  $z$  into  $x$ , and  $u$  into  $w$ ,  $w$  into  $u$ , the letter  $v$  being unaltered.

A set of symbols  $\alpha, \beta, \gamma, \dots$ , such that the product  $\alpha\beta$  of each two of them (in each order,  $\alpha\beta$  and  $\beta\alpha$ ) is a symbol of the set, is a group. It is easily seen that 1 is a symbol of every group, and we may therefore give the definition in the form that a set of symbols  $1, \alpha, \beta, \gamma, \dots$  satisfying the foregoing condition is a group. When the number of symbols (or terms) is  $= n$ , then the group is of the order  $n$ ; and each symbol  $\alpha$  is such that  $\alpha^n = 1$ , so that a group of the order  $n$  is in fact a group of symbolical  $n$ th roots of unity.

A group is defined by means of the laws of combinations of its symbols. For the statement of these we may either (by the introduction of powers and products) diminish as much as may be the number of distinct functional symbols; or else, using distinct letters for the several terms of the group, employ a square diagram, as presently mentioned.

Thus, in the first mode, a group is  $1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2$ , ( $\alpha^2 = 1, \beta^3 = 1, \alpha\beta^2 = \beta\alpha$ ), where observe that these conditions imply also  $\alpha\beta^2 = \beta\alpha$ .

Or in the second mode, calling the symbols  $(1, \alpha, \beta, \alpha\beta, \beta^2, \alpha\beta^2)$  of the same group  $(1, \alpha, \beta, \gamma, \delta, \epsilon)$ , or, if we please,  $(a, b, c, d, e, f)$ , the laws of combination are given by one or other of the square diagrams

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$      | $\epsilon$ |
|------------|------------|------------|------------|------------|---------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$      | $\epsilon$ |
| $\alpha$   | $\alpha$   | 1          | $\gamma$   | $\beta$    | $\epsilon$    | $\delta$   |
| $\beta$    | $\beta$    | $\epsilon$ | $\delta$   | $\alpha$   | 1             | $\gamma$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha\beta$ |            |
| $\delta$   | $\delta$   | $\gamma$   | 1          | $\epsilon$ | $\beta$       | $\alpha$   |
| $\epsilon$ | $\epsilon$ | $\beta$    | $\alpha$   | $\delta$   | $\gamma$      | 1          |

| $f$ | $a$ | $b$ | $c$ | $d$ | $e$ | $f$ |
|-----|-----|-----|-----|-----|-----|-----|
| $f$ | $b$ | $a$ | $d$ | $c$ | $f$ | $e$ |
| $f$ | $c$ | $e$ | $b$ | $a$ | $d$ |     |
| $f$ | $d$ | $e$ | $a$ | $b$ | $c$ |     |
| $f$ | $e$ | $d$ | $a$ | $c$ | $b$ |     |
| $f$ | $c$ | $b$ | $e$ | $d$ | $a$ |     |

where, taking for greater symmetry the second form of the square, observe that the square is such that no letter occurs twice in the same line, or in the same column (or what is the same thing, each of the lines and of the columns contains all the letters). But this is not sufficient in order that the square may represent a group; the square must be such that the substitutions by means of which its several lines are derived from any line thereof are (in a different order) the same substitutions by which the lines are derived from a particular line, or say from the top line. These, in fact, are

$$\begin{aligned}
 &ab \cdot cd \cdot ef, \\
 &ace \cdot bfd, \\
 &ad \cdot be \cdot cf, \\
 &aec \cdot bdf, \\
 &af \cdot be \cdot de,
 \end{aligned}$$

where, for shortness,  $ab$ ,  $ace$ , &c., are written instead of  $(ab)$ ,  $(ace)$ , &c., to denote the cyclical substitutions  $a$  into  $b$ ,  $b$  into  $a$ ; and  $a$  into  $c$ ,  $c$  into  $e$ ,  $e$  into  $a$ , &c.; and it is at once seen that by the same substitutions the lines may be derived from any other line.

It will be noticed that in the foregoing substitution-group each substitution is regular, that is, composed of cyclical substitutions each of the same number of letters; and it is easy to see that this property is a general one; each substitution of the substitution-group must be regular.

By what precedes, the group of any order composed of the functional symbols is replaced by a substitution-group upon a set of letters the number of which is equal to the order of the group, and wherein all the substitutions are regular.

The general theory being thus explained, I endeavour to form a substitution-group with the twelve letters  $abcdefghijkl$ ; and I assume that there is one substitution, such as  $abc \cdot def \cdot ghi \cdot jkl$ , and another substitution, such as  $agj \cdot bfi \cdot cek \cdot dhl$ . Observe that, if the twelve letters are to be thus arranged in two different ways as a set of four triads, without repetition of any duad, all the ways in which this can be done are essentially similar, and there is no loss of generality in taking the two sets of triads to be those just written down. But the substitution to be formed with either set of triads will be different according as any triad thereof, for instance  $agj$ , is written in this form or in the reversed form  $ajg$ . There are thus in all sixteen substitutions which can be formed with the first set of triads, and sixteen substitutions which can be formed with the second set of triads; and the relation of a triad of the first set to a triad of the second set is by no means independent of the selection of the triads out of the two sets respectively. To show this, take the two substitutions quite at random; suppose they are those written down above, say

$$\alpha = abc \cdot def \cdot ghi \cdot jkl, \quad \beta = agj \cdot bfi \cdot cek \cdot dhl$$

and perform these in succession on the primitive arrangement  $\Omega = abcdefghijkl$ . The operation stands thus:

$$\beta\alpha\Omega = fegkihbjcda,$$

$$\alpha\Omega = bcaefdhigklj,$$

whence

$$\Omega = abcdefghijkl,$$

$$\beta\alpha = abeijcgl \cdot dk,$$

is not a regular substitution; and, by what precedes,  $\alpha, \beta$  cannot belong to a group.

But take the substitutions to be

$$\alpha \text{ (as before)} = abc \cdot def \cdot ghi \cdot jkl, \quad \beta = ajg \cdot bif \cdot cek \cdot dhl,$$

then we have

$$\beta\alpha\Omega = iejkbhlfacdg,$$

$$\alpha\Omega = bcaefdhigklj,$$

$$\Omega = abcdefghijkl,$$

whence

$$\beta\alpha = ai \cdot bc \cdot cj \cdot dk \cdot fh \cdot gl,$$

a regular substitution; and, for anything that appears to the contrary,  $\alpha, \beta$  may belong to a group. It is convenient to mention at once that these two substitutions do, in fact, give rise to a group; viz. the square diagram is

|          |          |          |          |          |          |          |          |          |          |          |          |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| <i>a</i> | <i>b</i> | <i>c</i> | <i>d</i> | <i>e</i> | <i>f</i> | <i>g</i> | <i>h</i> | <i>i</i> | <i>j</i> | <i>k</i> | <i>l</i> |
| <i>b</i> | <i>c</i> | <i>a</i> | <i>f</i> | <i>d</i> | <i>e</i> | <i>h</i> | <i>i</i> | <i>g</i> | <i>k</i> | <i>l</i> | <i>j</i> |
| <i>c</i> | <i>a</i> | <i>b</i> | <i>e</i> | <i>f</i> | <i>d</i> | <i>i</i> | <i>g</i> | <i>h</i> | <i>l</i> | <i>j</i> | <i>k</i> |
| <i>d</i> | <i>l</i> | <i>h</i> | <i>g</i> | <i>j</i> | <i>c</i> | <i>e</i> | <i>k</i> | <i>f</i> | <i>i</i> | <i>b</i> | <i>a</i> |
| <i>e</i> | <i>j</i> | <i>i</i> | <i>b</i> | <i>h</i> | <i>k</i> | <i>a</i> | <i>l</i> | <i>d</i> | <i>g</i> | <i>c</i> | <i>f</i> |
| <i>f</i> | <i>k</i> | <i>g</i> | <i>c</i> | <i>i</i> | <i>l</i> | <i>b</i> | <i>j</i> | <i>e</i> | <i>h</i> | <i>a</i> | <i>d</i> |
| <i>g</i> | <i>f</i> | <i>k</i> | <i>l</i> | <i>e</i> | <i>i</i> | <i>j</i> | <i>a</i> | <i>c</i> | <i>b</i> | <i>d</i> | <i>h</i> |
| <i>h</i> | <i>d</i> | <i>l</i> | <i>j</i> | <i>a</i> | <i>g</i> | <i>k</i> | <i>c</i> | <i>b</i> | <i>f</i> | <i>i</i> | <i>e</i> |
| <i>i</i> | <i>e</i> | <i>j</i> | <i>k</i> | <i>b</i> | <i>h</i> | <i>l</i> | <i>f</i> | <i>a</i> | <i>c</i> | <i>d</i> | <i>g</i> |
| <i>j</i> | <i>i</i> | <i>e</i> | <i>h</i> | <i>k</i> | <i>b</i> | <i>a</i> | <i>d</i> | <i>l</i> | <i>f</i> | <i>g</i> | <i>c</i> |
| <i>k</i> | <i>g</i> | <i>f</i> | <i>i</i> | <i>l</i> | <i>c</i> | <i>b</i> | <i>j</i> | <i>d</i> | <i>h</i> | <i>a</i> | <i>e</i> |
| <i>l</i> | <i>h</i> | <i>d</i> | <i>g</i> | <i>j</i> | <i>a</i> | <i>f</i> | <i>k</i> | <i>e</i> | <i>i</i> | <i>b</i> | <i>c</i> |

and the substitutions, obtained therefrom by writing successively each line over the top line, are

$$\begin{aligned}
 1 &= 1, \\
 abc \cdot def \cdot ghi \cdot jkl &= \alpha, \\
 acb \cdot dfe \cdot gih \cdot jlk &= \alpha^2, \\
 ad \cdot bl \cdot ch \cdot eg \cdot fj \cdot ik &= \beta\alpha^2, \\
 aeh \cdot bjd \cdot cli \cdot fkg &= \beta\alpha^2, \\
 afl \cdot bkh \cdot cgd \cdot eij &= \beta^2\alpha, \\
 agj \cdot bfi \cdot cek \cdot dlh &= \beta\alpha, \\
 ahe \cdot bdj \cdot cli \cdot fgk &= \beta\alpha^2\beta\alpha^2, \\
 ai \cdot bc \cdot cj \cdot dk \cdot fh \cdot gl &= \beta\alpha, \\
 ajg \cdot bif \cdot cek \cdot dhl &= \beta, \\
 ak \cdot bg \cdot cf \cdot di \cdot el \cdot hj &= \beta\alpha\beta, \\
 alf \cdot bkh \cdot cdg \cdot eji &= \beta^2\alpha^2\beta\alpha.
 \end{aligned}$$

To explain the theory, I introduce the notion of a hemipolyhedron, or say a hemihedron, viz. this is a figure obtained from a polyhedron by the removal of certain faces. In a polyhedron each edge occurs twice (more properly it occurs in the two forms  $ab$  and  $ba$ ), as belonging to two faces; but in a hemihedron one of these faces must always be removed, so that the edge may occur once only; and again (what is apparently, although not really, a different thing), we may remove two intersecting faces, leaving their edge of intersection; this edge is, in fact, then considered as a bilateral face  $ab = ab \cdot ba$ , just as  $abc$  is a trilateral face  $abc = ab \cdot bc \cdot ca$ . Thus, if in a prism we remove the lateral faces, leaving the lateral edges, and leaving also the terminal faces, we have a hemihedron: thus, the prism being trilateral, say the faces of the hemihedron are  $abc, def, ad, be, cf$ , where  $ad, be, cf$  are the edges regarded as bilateral faces. And, for the present purpose,  $abc$  denotes the cyclical substitution  $a$  into  $b$ ,  $b$  into  $c$ ,  $c$  into  $a$ ; and  $ad$  denotes in like manner the cyclical substitution (or interchange)  $a$  into  $d$ ,  $d$  into  $a$ .

But the hemihedron about to be considered has no bilateral faces; it is, in fact, the figure composed of the 8 triangular faces of the octo-hexahedron or figure obtained by truncating the summits of a hexahedron (or of an octahedron) so as to obtain a polyhedron of 8 triangular faces and 6 square faces, representing the faces of the octahedron and the hexahedron respectively. The faces of the octo-hexahedron may be taken to be

$$\begin{aligned} &abc, def, ghi, jkl, \\ &agj, bfi, cek, dhl, \\ &cbfe, fihd, hgjl, jack, agib, klde, \end{aligned}$$

(where I observe in passing that the symbols are written in such manner that each edge  $ab$  occurs under the two opposite forms  $ab$  in  $abc$  and  $ba$  in  $agib$ ). And then, omitting the square faces, represented by the third line, we have the hemihedron, wherein as before  $abc$  denotes the cyclical substitution  $a$  into  $b$ ,  $b$  into  $c$ ,  $c$  into  $a$ ; and so for the other faces.

I represent this by a diagram, the lines of which were red and black, and they will be thus spoken of, but the black lines are in the woodcut continuous lines, and the red lines broken lines: each face indicates a cyclical substitution, as shown by the arrows. The figure should be in the first instance drawn with the arrows, but without the letters, and these may then be affixed to the several points in a perfectly arbitrary manner; but I have in fact affixed them in such wise that the group given

by the diagram, as presently appearing, may (instead of being any other equivalent group) be that group which contains the before-mentioned substitution

$$\alpha = abc \cdot def \cdot ghi \cdot jkl, \quad \text{and} \quad \beta = ajg \cdot bif \cdot cek \cdot dhl.$$

Observe that in the diagram, considering the lines to be drawn as shown by the arrows, there is from any given point whatever only one black line, and only one red line. Let  $B$  denote motion along a black line,  $R$  motion along a red line (always from a point to the next point); then  $B^2$  will denote motion along two black lines successively,  $RB$  (any such symbol being read always from right to left) will denote motion first along a red line, and then along a black line, and so in other cases; a symbol or "route"  $\dots R^\rho B^\beta R^{\rho'}$  has thus a perfectly definite signification, determining the path when the initial point is given.

The diagram has the property that every route, leading from any one letter to itself, leads also from every other letter to itself; or say a route leading from  $a$  to  $a$ , leads also from  $b$  to  $b$ , from  $c$  to  $c$ ,  $\dots$ , from  $l$  to  $l$ ; and we can thus in the diagram speak absolutely (that is, without restriction as to the initial point) of a route as leading from a point to itself, or say as being equal to unity; it is in virtue of this property that the diagram gives a group.

For, assuming the property, it at once follows (1) that two routes, each leading say from the point  $a$  to the same point  $f$ , lead also from any other point  $b$  to one and the same point  $g$ . Such routes are said to be equivalent, or equal to each other; and the number of distinct routes (including the route unity) is thus equal to the numbers of the letters, viz. we have only the routes from  $a$  to  $a$ , to  $b$ ,  $\dots$ , to  $l$ , respectively; (2) a route, leading from a point  $a$  to a point  $f$ , leads from any other point  $b$  to a different point  $g$ ; and (3) two routes, leading from the same point  $a$  to different points  $b$  and  $c$ , lead also from any other point  $f$  to different points  $k$  and  $l$ . Hence a given route leads from the several points  $abc \dots l$  successively to the same series of points taken in a different order, or we thus obtain a new arrangement of the points; and dealing in this manner successively with the routes from  $a$  to  $a$ , to  $b$ ,  $\dots$ , to  $l$ , we obtain so many distinct arrangements, beginning with the letters  $a, b, c, \dots, l$  respectively, such that in no two of them does the same letter occupy the same place; we thus obtain a square of 12 such as that already written down, and which is, in fact, the same square, the several routes of course corresponding to the substitutions of the square. The hemihedron thus gives the foregoing group of 12.

Observe that the diagram is composed of the four black triangles representing the substitution  $abc \cdot def \cdot ghi \cdot jkl$ , and of the four red triangles representing the substitution  $ajg \cdot bif \cdot cek \cdot dhl$ ; viz. these are independent substitutions which by their powers and products serve to express all the substitutions of the group; that they are sufficient appears by the diagram itself, in that every point thereof is (by black and red lines) connected with every other point thereof. The group might have contained three or more independent substitutions, and the diagram would then have contained the like number of differently coloured sets of lines. The essential characters are that the lines of any given colour shall form polygons of the same number of sides (but for different

colours the polygons may have different numbers of sides; in particular, for any given colour or colours, the polygons may be bilaterals, represented each by a line with a double arrow pointing opposite ways); that there shall be from each point only one line of the same colour; that every point shall be connected with every other point; and finally, that every route leading from one point to itself shall lead also from every other point to itself. When these conditions are satisfied the foregoing investigation in fact shows that the diagram, or say the hemihedron, gives rise to a group.

It may be remarked that we can, if we please, introduce into the diagram a set of lines of a new colour to represent any dependent substitution of the group; thus, in the example considered, a substitution is  $ae h \cdot b j d \cdot c i l \cdot f k g$ , and if we draw these triangles in green (the arrows being from  $a$  to  $e$ ,  $e$  to  $h$ ,  $h$  to  $a$ , &c.), then there will be from each point one black line, one red line, and one green line; any route  $\dots G^\gamma B^\beta R^\rho$  will thus be perfectly definite, and will have the same properties as a route composed of black and red lines only; and the theory thus subsists without alteration.

I remark, in conclusion, that the group of 12 considered above is, in fact, the group of 12 positive substitutions upon 4 letters  $abcd$ ; viz. the substitutions are 1,  $abc$ ,  $acb$ ,  $abd$ ,  $adb$ ,  $acd$ ,  $adc$ ,  $bcd$ ,  $bdc$ ,  $ab \cdot cd$ ,  $ac \cdot bd$ ,  $ad \cdot bc$ ; the groups each contain unity, three substitutions of the order (or index) 2, and 8 substitutions of the order (or index) 3, and their identity can be easily verified.

# 691.

## NOTE ON MR MONRO'S PAPER "ON FLEXURE OF SPACES."

[From the Proceedings of the London Mathematical Society, vol. ix. (1878), pp. 171, 172.  
Read June 13, 1878.]

CONSIDER an element of surface, surrounding a point  $P$ ; the flexure of the element may be interfered with by the continuity round  $P$ , and it is on this account proper to regard the element as cut or slit along a radius drawn from  $P$  to the periphery of the element. This being understood, we have the well-known theorem that, considering in the neighbourhood of the origin elements of the surfaces

$$z = \frac{1}{2}(ax^2 + 2hxy + by^2), \quad \text{and} \quad z' = \frac{1}{2}(a'x'^2 + 2h'x'y' + b'y'^2),$$

these will be applicable the one on the other, provided only  $ab - h^2 = a'b' - h'^2$ . But in connexion with Mr Monro's paper it is worth while to give the proof in detail.

It is to be shown that  $z, z'$  denoting the above-mentioned functions of  $(x, y)$  and  $(x', y')$  respectively, it is possible to find (for small values)  $x', y'$  functions of  $x, y$  such that identically

$$dx'^2 + dy'^2 + dz'^2 = dx^2 + dy^2 + dz^2.$$

The solution is taken to be  $x' = x + \xi, y' = y + \eta$ , where  $\xi, \eta$  denote cubic functions of  $x, y$ . We have then, attending only to the terms of an order not exceeding 3 in  $x, y$ ,

$$\begin{aligned} dx^2 + dy^2 + 2(dx d\xi + dy d\eta) + \{(a'x + h'y)dx + (h'x + b'y)dy\}^2 \\ = dx^2 + dy^2 + \{(ax + hy)dx + (hx + by)dy\}^2, \end{aligned}$$

so that the terms  $dx^2 + dy^2$  disappear; and then writing

$$d\xi = \frac{d\xi}{dx} dx + \frac{d\xi}{dy} dy, \quad d\eta = \frac{d\eta}{dx} dx + \frac{d\eta}{dy} dy,$$

the equation will be satisfied identically as regards  $dx, dy$  if only

$$\begin{aligned} 2 \frac{d\xi}{dx} &= (ax + hy)^2 - (a'x + h'y)^2, \\ \frac{d\xi}{dy} + \frac{d\eta}{dx} &= (ax + hy)(hx + by) - (a'x + h'y)(h'x + b'y), \\ 2 \frac{d\eta}{dy} &= (hx + by)^2 - (h'x + b'y)^2. \end{aligned}$$

Calling the terms on the right-hand side  $2\mathfrak{A}, \mathfrak{B}, 2\mathfrak{C}$  respectively, we have

$$\frac{d^2\mathfrak{A}}{dy^2} - \frac{d^2\mathfrak{B}}{dx dy} + \frac{d^2\mathfrak{C}}{dx^2} = 0,$$

that is,

$$(h^2 - h'^2) + (h^2 - h'^2) - \{(ab - h^2) - (a'b' - h'^2)\} = 0,$$

or, what is the same thing,

$$a'b' - h'^2 = ab - h^2,$$

a relation which must exist between the constants  $(a, b, h)$  and  $(a', b', h')$ .

It is easy to find the actual values of  $\xi, \eta$ ; viz. these are

$$\begin{aligned} \xi &= \frac{1}{4}(a^2 - a'^2)x^3 + \frac{1}{2}(ah - a'h')x^2y + \frac{1}{4}(h^2 - h'^2)xy^2 + \frac{1}{6}(bh - b'h')y^3, \\ \eta &= \frac{1}{6}(ah - a'h')x^3 + \frac{1}{4}(h^2 - h'^2)x^2y + \frac{1}{2}(bh - b'h')xy^2 + \frac{1}{4}(b^2 - b'^2)y^3, \end{aligned}$$

or, what is the same thing, we have

$$\xi = \frac{1}{4} \frac{d\Omega}{dx}, \quad \eta = \frac{1}{4} \frac{d\Omega}{dy},$$

where

$$\begin{aligned} \Omega &= (a^2 - a'^2)x^4 + 4(ah - a'h')x^3y + 6(h^2 - h'^2)x^2y^2 + 4(bh - b'h')xy^3 + (b^2 - b'^2)y^4 \\ &= (ax^2 + 2hxy + by^2)^2 - (a'x^2 + 2h'xy + b'y^2)^2 = 4(z^2 - z'^2), \end{aligned}$$

in virtue of the relation  $ab - h^2 = a'b' - h'^2$ . The resulting values  $x' = x + \xi, y' = y + \eta$  are obviously the first terms of two series which, if continued, would contain higher powers of  $(x, y)$ .



# 692.

## ADDITION TO THE MEMOIR ON THE TRANSFORMATION OF ELLIPTIC FUNCTIONS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXIX. Part II. (1878), pp. 419–424. Received February 6,—Read March 7, 1878.]

I HAVE recently succeeded in completing a theory considered in my “Memoir on the Transformation of Elliptic Functions,” *Phil. Trans.*, vol. CLXIV. (1874), pp. 397–456, [578],—that of the septic transformation,  $n = 7$ . We have here

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left( \frac{\alpha - \beta x + \gamma x^2 - \delta x^3}{\alpha + \beta x + \gamma x^2 + \delta x^3} \right)^2,$$

a solution of

$$\frac{M dy}{\sqrt{1-y^2} \cdot 1-v^2 y^2} = \frac{dx}{\sqrt{1-x^2} \cdot 1-u^2 x^2},$$

where  $\frac{1}{M} = 1 + \frac{2\beta}{\alpha}$ ; and the ratios  $\alpha : \beta : \gamma : \delta$ , and the  $uv$ -modular equation are determined by the equations

$$\begin{aligned} u^2(2\alpha\gamma + 2\alpha\beta + \beta^2) &= v^2(\gamma^2 + 2\gamma\delta + 2\beta\delta), \\ \gamma^2 + 2\beta\gamma + 2\alpha\delta + 2\beta\delta &= \frac{1}{M^2} u^2 v^2 (2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2), \\ \delta^2 + 2\gamma\delta &= \frac{1}{M^2} v^2 u^4 (\alpha^2 + 2\alpha\beta); \end{aligned}$$

or, what is the same thing, writing  $\alpha = 1$ , the first equation may be replaced by

$$\delta = \frac{v^2}{u},$$

and then,  $\alpha, \delta$  having these values, the last three equations determine  $\beta, \gamma$  and the modular equation. If instead of  $\beta$  we introduce  $M$ , by means of the relation

$\frac{1}{M} = 1 + 2\beta$ , that is,  $2\beta = \frac{1}{M} - 1$ , then the last equation gives  $2\gamma = uv^3 \left( \frac{1}{M} - \frac{u^4}{v^2} \right)$ ; and  $\alpha, \beta, \gamma, \delta$  having these values, we have the residual two equations

$$\begin{aligned} u^2(2\alpha\gamma + 2\alpha\beta + \beta^2) &= v^2(\gamma^2 + 2\gamma\delta + 2\beta\delta), \\ \gamma^2 + 2\beta\gamma + 2\alpha\delta + \beta\delta &= u^2 v^2 (2\alpha\gamma + 2\beta\gamma + 2\alpha\delta + \beta^2), \end{aligned}$$

viz. each of these is a quadric equation in  $\frac{1}{M}$ ; hence eliminating  $\frac{1}{M}$ , we have the modular equation; and also (linearly) the value of  $\frac{1}{M}$ , and thence the values of  $\alpha, \beta, \gamma, \delta$  in terms of  $u, v$ .

Before going further it is proper to remark that, writing as above  $\alpha = 1$ , then if  $\delta = \beta\gamma$ , we have

$$\begin{aligned} 1 - \beta x + \gamma x^2 - \delta x^3 &= (1 - \beta x)(1 + \gamma x^2), \\ 1 + \beta x + \gamma x^2 + \delta x^3 &= (1 + \beta x)(1 + \gamma x^2), \end{aligned}$$

and the equation of transformation becomes

$$\frac{1-y}{1+y} = \frac{1-x}{1+x} \left( \frac{1-\beta x}{1+\beta x} \right)^2,$$

viz. this belongs to the cubic transformation. The value of  $\beta$  in the cubic transformation was taken to be  $\beta = \frac{u^2}{v}$ , but for the present purpose it is necessary to pay attention to an omitted double sign, and write  $\beta = \pm \frac{u^2}{v}$ ; this being so,  $\delta = \beta\gamma$ , and giving to  $\gamma$  the value  $\mp u^4$ ,  $\delta$  will have its foregoing value  $-\frac{u^2}{v}$ . And from the theory of the cubic equation, according as  $\beta = \frac{u^2}{v}$  or  $-\frac{u^2}{v}$ , the modular equation must be

$$u^4 - v^4 + 2uv(1 - u^2 v^2) = 0, \quad \text{or} \quad u^4 - v^4 - 2uv(1 - u^2 v^2) = 0.$$

We thus see *à priori*, and it is easy to verify that the equations of the septic transformation are satisfied by the values

$$\alpha = 1, \beta = \frac{u^2}{v}, \gamma = -u^4, \delta = -\frac{u^2}{v}, \text{ and } u^4 - v^4 + 2uv(1 - u^2v^2) = 0;$$

$$\alpha = 1, \beta = -\frac{u^2}{v}, \gamma = -u^4, \delta = -\frac{u^2}{v}, \text{ and } u^4 - v^4 - 2uv(1 - u^2v^2) = 0;$$

and it hence follows that in obtaining the modular equation for the septic transformation, we shall meet with the factors  $u^4 - v^4 \pm 2uv(1 - u^2v^2)$ . Writing for shortness  $uv = \theta$ , these factors are  $u^4 - v^4 \pm 2\theta(1 - \theta^2)$ ; the factor for the proper modular equation is  $u^8 + v^8 - \Theta$ , where

$$\Theta = 8\theta - 28\theta^2 + 56\theta^3 - 70\theta^4 + 56\theta^5 - 28\theta^6 + 8\theta^7,$$

viz. the equation  $(1 - u^8)(1 - v^8) - (1 - uv)^8 = u^8 + v^8 - \Theta = 0$ ; and the modular equation, as obtained by the elimination from the two quadric equations, presents itself in the form

$$(u^4 - v^4 + 2\theta)^2(u^4 - v^4 + 2\theta)(u^4 + v^4 - \Theta) = 0.$$

Proceeding to the investigation, we substitute the values

$$\alpha = 1, \beta = \frac{1}{2} \left( \frac{1}{M} - 1 \right), \gamma = \frac{1}{2} uv^3 \left( \frac{1}{M} - \frac{u^4}{v^2} \right), \delta = \frac{v^2}{u},$$

in the residual two equations, which thus become

$$\frac{1}{M^2}(1 - v^4) + \frac{2}{M}(1 - uv)^4(1 + uv) + \{(1 - v^8) - 4(1 - uv)(1 + uv)^3\} = 0,$$

$$\frac{1}{M^2}\{-u^4v^2(1 - uv)^4(1 + uv)\} + \frac{2}{M}\{u^2v^2(1 - v^4) + \frac{u^2}{v}(1 + u^2v^2)(u^4 - v^4)\} + \{u^8 + 6\frac{u^4}{v^2}(1 - u^2v^2) - \frac{u^4}{v^2}\}(1 - v^4) = 0,$$

the first of which is given p. 432 of the “Memoir,” [Coll. Math. Papers, vol. ix., p. 150]. Calling them

$$(a, b, c \S \frac{1}{M}, 1)^2 = 0, \quad (a', b', c' \S \frac{1}{M}, 1)^2 = 0,$$

we have

$$\frac{-1}{bc' - b'c} : \frac{2}{ca' - c'a} : \frac{1}{ab' - a'b},$$

and the result of the elimination therefore is

$$(ca' - c'a)^2 - 4(bc' - b'c)(ab' - a'b) = 0.$$

Write as before  $uv = \theta$ . In forming the expressions  $ca' - c'a$ , &c., to avoid fractions we must in the first instance introduce the factor  $v^2$ : thus

$$v^2(ca' - c'a) = v^2\{v^2(1 - u^8) - 4(1 - \theta)(1 + \theta)^3\} \cdot [-u^4v^2(1 - \theta)^4(1 + \theta)]$$

$$- \{u^8 + 6u^4\theta^2(1 - \theta^2) - v^4\theta^2\}\{1 - v^4\},$$

$$= -u^4v^4(1 + \theta)(1 - \theta)^4\{v^2(-3 + 4\theta) + u^4(-4\theta + 3\theta^2)\}$$

$$- \{u^8 + 6u^4(\theta - \theta^3) - v^4\theta^2\}(1 - v^4)$$

but instead of  $\theta^2v^2$  writing  $v^6$ , the expression on the right-hand side becomes divisible by  $v^4$ ; and we find

$$\frac{1}{v^4}(ca' - c'a) = -(1 + \theta)(1 - \theta)^4\{v^4(-3 + 4\theta) + u^4(-4\theta + 3\theta^2)\}$$

$$- \{u^8 + 6u^4(\theta - \theta^3) - v^4\}(1 - v^4),$$

and thence

$$\begin{aligned} \frac{1}{v^4}(ca' - c'a) &= u^8 + u^4(6\theta - 10\theta^2 + 11\theta^3 - 6\theta^4 - 8\theta^5 + 10\theta^6 - 4\theta^7) \\ &\quad + v^4(-4 + 10\theta - 8\theta^2 - 6\theta^3 + 11\theta^4 - 10\theta^5 + 6\theta^6) + v^8. \end{aligned}$$

Similarly we have

$$\begin{aligned} \frac{1}{v^4}(bc' - b'c) &= u^8(5 - 5\theta + 4\theta^2 - 5\theta^3 + 2\theta^4) + u^4(9\theta - 16\theta^2 + \theta^3 + 10\theta^4 + \theta^5 - 16\theta^6 + 9\theta^7) \\ &\quad + v^4(2 - 5\theta + 4\theta^2 - 5\theta^3 + 5\theta^4), \\ \frac{1}{v^4}(ab' - a'b) &= u^8(\theta + \theta^2 - \theta^4) + u^4(2 - 5\theta + 4\theta^2 + 3\theta^3 - 10\theta^4 + 3\theta^5 + 4\theta^6 - 5\theta^7 + 2\theta^8) \\ &\quad + v^4(-1 + \theta + \theta^2); \end{aligned}$$

say these values are

$$u^8 + pu^4 + qv^4 + v^8, \quad \lambda u^8 + \mu u^4 + \nu v^4, \quad \rho u^8 + \sigma u^4 + \tau v^4.$$

The required equation is thus

$$(u^8 + pu^4 + qv^4 + v^8)^2 - 4(\lambda u^8 + \mu u^4 + \nu v^4)(\rho u^8 + \sigma u^4 + \tau v^4),$$

viz. the function is

$$\begin{aligned} &u^{16} \\ &+ u^{12}(2p - 4\lambda\rho) \\ &+ u^8(2q\theta^4 + p^2 - 4\lambda\sigma\theta^4 - 4\mu\rho) \\ &+ (2p\theta^4 + 2pq\theta^4 - 4\lambda\tau\theta^8 - 4\mu\sigma\theta^4 - 4\nu\rho\theta^4) \\ &+ u^4(2p\theta^4 + q^2\theta^4 - 4\mu\tau\theta^8 - 4\nu\sigma\theta^4) \\ &+ v^{12}(2q - 4\nu\tau) \\ &+ v^{16}, \end{aligned}$$

or say it is

$$= (1, b, c, d, e, f \text{ } \$ u^{16}, u^{12}, u^8, 1, v^8, v^{12}, v^{16}).$$

Supposing that this has a factor  $u^8 + v^8 - \Theta$ , the form is

$$(u^{16} + Bu^8 + Cu^4v^4 + Dv^8 + v^{16})(u^8 + v^8 - \Theta)$$

and comparing coefficients we have

$$\begin{aligned} B - \Theta &= b, \\ C - \Theta B + \Theta^2 &= c, \\ D\Theta^2 - \Theta C + B\Theta^3 &= d, \\ \Theta^2 - \Theta D + C &= e, \\ -\Theta + D &= f, \end{aligned}$$

where  $\Theta$  has the before-mentioned value

$$\Theta = (8, -28, +56, -70, +56, -28, +8 \text{ } \$ \theta, \theta^2, \theta^3, \theta^4, \theta^5, \theta^6, \theta^7).$$

From the first, second, and fifth equations,  $B = b + \Theta$ ,  $D = f + \Theta$ ,  $C = c + \Theta b + \Theta^2$ ; and

the third and fourth equations should then be verified identically. Writing down the coefficients of the different powers of  $\theta$ , we find

$$\begin{aligned} 2p &= 0 + 12 \quad 0 - 20 + 22 - 12 - 16 + 20 - 8 \quad (\theta, \dots, \theta^7) \\ -4\lambda\rho &= -20 + 20 - 36 + 60 - 44 + 36 - 28 + 8 \\ b &= 0 - 8 + 20 - 56 + 82 - 56 + 20 - 8 \\ e &= 0 + 8 - 28 + 56 - 70 + 56 - 28 + 8 \\ B &= 0 - 8 \quad + 12 \quad 0 - 8 \end{aligned}$$

that is,

$$B = -8\theta + 12\theta^2 - 8\theta^3;$$

and in precisely the same way the fifth equation gives the corresponding value of  $D$ .

We find similarly  $C$  from the second equation: writing down first the coefficients of  $p^2$ ,  $2q\theta^4$ ,  $-4\lambda\sigma\theta^4$ , and  $4\mu\rho$ , the sum of these gives the coefficients of  $c$ ; and then writing underneath these the coefficients of  $B\Theta$  and of  $-\Theta^2$ , the final sum gives the coefficients of  $C$ : the coefficients of each line belong to  $(\theta^0, \theta^1, \dots, \theta^{12})$ .

$$\begin{array}{r} 36 \quad 0 - 120 + 132 + 28 - 316 + 361 - 20 - 340 + 396 - 144 - 112 + 164 - 80 + 16 \\ \quad \quad \quad - 8 + 20 - 16 - 12 + 22 - 20 \quad 0 + 12 \\ -40 + 140 - 212 + 140 + 80 - 188 + 168 - 92 - 64 + 176 - 164 + 80 - 16 \\ \quad \quad \quad - 36 + 64 - 40 + 60 - 72 + 28 \quad 0 + 68 - 100 + 36 \\ 0 + 64 - 208 + 352 - 272 - 160 + 463 - 160 - 272 + 352 - 208 + 64 \\ \quad \quad \quad - 64 + 224 - 352 + 224 + 160 - 392 + 160 + 224 - 352 + 224 - 64 \\ \hline \quad \quad \quad 0 \quad 0 \quad 0 \quad 0 + 16 \quad 0 - 48 \quad 0 + 70 \quad 0 - 48 \quad 0 + 16 \quad 0, \end{array}$$

that is,

$$C = 16\theta^4 - 48\theta^5 + 70\theta^6 - 48\theta^7 + 16\theta^8;$$

and in precisely the same way this value of  $C$  would be found from the fourth equation. There remains to be verified only the fourth equation  $(D + B)\Theta^2 - \Theta C = d$ , that is,

$$2f\Theta^2(-8\theta + 12\theta^2 - 8\theta^3) - \Theta C = (2p - 4\lambda\tau)\theta^8 + (2pq - 4\mu\sigma - 4\nu\rho)\theta^4,$$

and this can be effected without difficulty.

The factor of the modular equation thus is

$$u^{16} + v^{16} + (-8\theta + 12\theta^2 - 8\theta^3)(u^8 + v^8) + 16\theta^4 - 48\theta^5 + 70\theta^6 - 48\theta^7 + 16\theta^8$$

viz. this is

$$\begin{aligned} &(u^8 + v^8)^2 + (-4\theta + 6\theta^2 - 4\theta^3) \cdot 2(u^8 + v^8) + 16\theta^2 - 48\theta^3 + 68\theta^4 - 48\theta^5 + 16\theta^6 \\ &= \{u^8 + v^8 - 4\theta + 6\theta^2 - 4\theta^3\}^2 \\ &= \{(u^4 - v^4)^2 - 4\theta(1 - \theta^2)\}^2, \end{aligned}$$

that is,

$$\{u^4 - v^4 - 2\theta(1 - \theta^2)\}^2 \{u^4 - v^4 + 2\theta(1 - \theta^2)\}^2;$$

or the modular equation is

$$\{u^4 - v^4 - 2\theta(1 - \theta^2)\}^2 \{u^4 - v^4 + 2\theta(1 - \theta^2)\}^2 (u^8 + v^8 - \Theta) = 0;$$

viz. the first and second factors belong to the cubic transformation; and we have for the proper modular equation in the septic transformation  $u^8 + v^8 - \Theta = 0$ , or what is the same thing  $(1 - u^8)(1 - v^8) - (1 - \theta)^8 = 0$ , that is,  $(1 - u^8)(1 - v^8) - (1 - uv)^8 = 0$ , the known result; or, as it may also be written,

$$\{(1 - u^2)(1 - v^2)\}^4 = (1 - uv)^8.$$

The value of  $M$  is given by the foregoing relations

$$\frac{-1}{\lambda u^8 + \mu u^4 + \nu v^4} : \frac{2}{u^8 + pu^4 + qv^4 + v^8} : \frac{1}{\rho u^8 + \sigma u^4 + \tau v^4};$$

but these can be, by virtue of the proper modular equation  $u^8 + v^8 - \Theta = 0$ , reduced into the form viz. the equality of these two sets of ratios depends upon the following identities,

$$\begin{aligned} & (-\lambda u^8 + \nu v^8)(u^8 + pu^4 + qv^4 + v^8) + (\theta - 2\theta^2 + 2\theta^3 - \theta^4)(\rho u^8 + \sigma u^4 + \tau v^4) \\ & = \{-\tfrac{1}{2}u^8 + (1 - \theta)(-4 - \theta + 5\theta^2 - \theta^3 - 4\theta^4)v^4 + v^8\}(u^8 - \Theta + v^8), \\ & (\theta - u^4)(\rho u^8 + \sigma u^4 + \tau v^4) - (\theta - v^4)(\lambda u^8 + \mu u^4 + \nu v^4) \\ & = \{(2\theta + 5\theta^2 + 3\theta^3 - 2\theta^4 - 2\theta^5)u^4 + (2 + 2\theta - 3\theta^2 - 5\theta^3 - 2\theta^4)v^4\}(u^8 - \Theta + v^8), \\ & - 2(\theta - 2\theta^2 + 2\theta^3 - \theta^4)(\lambda u^8 + \mu u^4 + \nu v^4) + (u^8 - \theta)(u^8 + pu^4 + qv^4 + v^8) \\ & = \{u^8 + \theta(1 - \theta)(3 + 5\theta + 3\theta^2)u^4 - \theta^4v^4\}(u^8 - \Theta + v^8), \end{aligned}$$

which can be verified without difficulty: from the last-mentioned system of values, replacing  $\theta$  by its value  $uv$ , we then have

$$\frac{-1}{\frac{1}{2}u(v - u^7)} : \frac{2}{1 - uv(1 - uv)(1 - uv + u^2v^2)} : \frac{1}{-v(u - v^7)},$$

which agree with the values given p. 432 of the “Memoir”; and the analytical theory is thus completed.

### 693.

#### A TENTH MEMOIR ON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CLXIX., Part II. (1878), pp. 603–661. Received June 12,—Read June 20, 1878.]

The present Memoir, which relates to the binary quintic  $(x, y)^5$ , has been in hand for a considerable time: the chief subject-matter was intended to be the theory of a canonical form which was discovered by myself and is briefly noticed in Salmon’s *Higher Algebra*, 3rd Ed. (1876), pp. 217, 218; writing  $a, b, c, d, e, f, g, \dots, u, v, w$  to denote the 23 covariants of the quintic, then  $a, b, c, d, f$  are connected by the relation

$$f^2 = -a^3d + a^2bc - 4c^3;$$

and the form contains these covariants thus connected together, and also  $e$ ; it, in fact, is

$$(1, 0, c, f, a^2b - 3c^2, a^2e - 2cf, x, y)^5.$$

But the whole plan of the Memoir was changed by Sylvester’s discovery of what I term the Numerical Generating Function (N.G.F.) of the covariants of the quintic, and my own subsequent establishment of the Real Generating Function (R.G.F.) of the same covariants. The effect of this was to enable me to establish for any given degree in the coefficients and order in the variables, or as it is convenient to express it, for any given deg-order whatever, a selected system of powers and products of the covariants, say a system of “segregates”: these are asyzygetic, that is, not connected together by any linear equation with numerical coefficients; and they are also such that every other combination of covariants of the same deg-order, say every “congregate” of the same deg-order, can be expressed (and that, obviously, in one way only) as a linear function, with numerical coefficients, of the segregates of that deg-order. The number of congregates of a given deg-order is precisely equal to the number of the independent syzygies of the same deg-order, so that these syzygies give in effect the congregates in terms of the segregates: and the proper form in which to exhibit the

syzygies is thus to make each of them give a single congregate in terms of the segregates: viz. the left-hand side can always be taken to be a monomial congregate  $a^\alpha b^\beta \dots$ , or, to avoid fractions, a numerical multiple of such form; and the right-hand side will then be a linear function, with numerical coefficients, of the segregates of the same deg-order. Supposing such a system of syzygies obtained for a given deg-order, any covariant function (rational and integral function of covariants) is at once expressible as a linear function

of the segregates of that deg-order: it is, in fact, only necessary to substitute therein for every monomial congregate its value as a linear function of the segregates. Using the word covariant in its most general sense, the conclusion thus is that every covariant can be expressed, and that in one way only, as a linear function of segregates, or say in the segregate form.

Reverting to the theory of the canonical form, and attending to the relation

$$f^2 = -a^3d + a^2bc - 4c^3,$$

it thereby appears that every covariant multiplied by a power of the quintic itself  $a$ , can be expressed, and that in one way only, as a rational and integral function of the covariants  $a, b, c, d, e, f$ , linear as regards  $f$ : say every covariant multiplied by a power of  $a$  can be expressed, and that in one way only, in the “standard” form as an illustration, take

$$a^2h = 6acd + 4bc^2 + ef.$$

Conversely, an expression of the standard form, that is, a rational and integral function of  $a, b, c, d, e, f$ , linear as regards  $f$ , not explicitly divisible by  $a$ , may very well be really divisible by a power of  $a$  (the expression of the quotient of course containing one or more of the higher covariants  $g, h$ , &c.), and we say that in this case the expression is divisible, and has for its divided form the quotient expressed as a rational and integral function of covariants. Observe that in general the divided form is not perfectly definite, only becoming so when expressed in the before-mentioned segregate form, and that this further reduction ought to be made. There is occasion, however, to consider these divided forms, whether or not thus further reduced; and moreover it sometimes happens that the non-segregate form presents itself, or can be expressed, with integer numerical coefficients, while the coefficients of the corresponding segregate form are fractional.

The canonical form is peculiarly convenient for obtaining the expressions of the several derivatives (Gordan’s *Uebereinanderschiebungen*)  $(a, b)^1$ ,  $(a, b)^2$ , &c., (or as I propose to write them  $ab1$ ,  $ab2$ , &c.), which can be formed with two covariants, the same or different, as rational and integral functions of the several covariants. It will be recollected that in Gordan’s theory these derivatives are used in order to establish the system of the 23 covariants: but it seems preferable to have the system of covariants, and by means of them to obtain the theory of the derivatives.

I mention at the end of the Memoir two expressions (one or both of them due to Sylvester) for the N.G.F. of a binary sextic.

The several points above adverted to are considered in the Memoir; the paragraphs are numbered consecutively with those of the former Memoirs upon Quantics.

*The Numerical and Real Generating Functions.* Art. Nos. 366 to 374,  
and Table No. 96.

366. I have, in my Ninth Memoir (1871) [462], given what may be called the Numerical Generating Function (N.G.F.) of the covariants of a quartic; this was

$$A(x) = \frac{1 - a^6x^{12}}{1 - ax^4 \cdot 1 - a^2x^4 \cdot 1 - a^2 \cdot 1 - a^3 \cdot 1 - a^3x^6},$$

the meaning being that the number of asyzygetic covariants  $a^\theta x^\mu$ , of the degree  $\theta$  in the coefficients and order  $\mu$  in the variables, or say of the deg-order  $\theta.\mu$ , is equal to the coefficient of  $a^\theta x^\mu$  in the development of this function. And I remarked that the formula indicated that the covariants were made up of  $(ax^4, a^2x^4, a^2, a^3, a^3x^6)$ , the quartic itself, the Hessian, the quadrinvariant, the cubinvariant, and the cubi-covariant, these being connected by a syzygy  $a^6x^{12}$  of the degree 6 and order 12. Calling these covariants  $a, b, c, d, e$ , so that these italic small letters stand for covariants,

| deg-order. |       |
|------------|-------|
| 1. 4       | $a$ , |
| 2. 0       | $b$ , |
| 2. 4       | $c$ , |
| 3. 0       | $d$ , |
| 3. 6       | $e$ , |

then it is natural to consider what may be called the Real Generating Function (R.G.F.): this is

$$\frac{1 - e^2}{1 - a \cdot 1 - b \cdot 1 - c \cdot 1 - d \cdot 1 - e},$$

the development of this contains, as it is easy to see, only terms of the form  $a^\alpha b^\beta c^\gamma d^\delta$  and  $a^\alpha b^\beta c^\gamma d^\delta e$ , each with the coefficient +1, so that the number of terms of a given deg-order  $\theta.\mu$  is equal to the coefficient of  $a^\theta x^\mu$  in the first-mentioned function: and these terms of a given deg-order represent the asyzygetic covariants of that deg-order: any other covariant of the same deg-order is expressible as a linear function of them.

For instance, deg-order 6.12, the terms of the R.G.F. are  $a^3d$ ,  $a^2bc$ ,  $c^3$ ; there is one more term  $e^2$  of the same deg-order; hence  $e^2$  must be a linear function of these and in fact

$$e^2 = -a^3d + a^2bc - 4c^3,$$

viz. this is the equation

$$\Phi^2 = -U^3J + U^2IH - 4H^3.$$

367. Sylvester obtained an expression for the N.G.F. of the quintic: this is

$$\begin{aligned}
 & a^0 \cdot 1 \\
 & + a^3 \cdot x^3 + x^5 + x^9 \\
 & + a^4 \cdot x^4 + x^6 \\
 & + a^5 \cdot x + x^3 + x^7 \\
 & + a^6 \cdot x^2 + x^4 \\
 & + a^7 \cdot x + x^3 - x^5 - x^7 \\
 & + a^8 \cdot x^2 - x^4 \\
 & + a^9 \cdot x + a^3 \cdot x^3 \\
 & + a^{10} \cdot x^2 + x^4 - x^6 \\
 & + a^{11} \cdot x - x^3 - x^5 \\
 & + a^{12} \cdot x^0 - x^2 - x^6 - x^{10} \\
 & + a^{13} \cdot x - x^3 - x^7 \\
 & + a^{14} \cdot x^2 - x^4 - x^6 \\
 & + a^{15} \cdot -x^3 - x^5 \\
 & + a^{16} \cdot -x^2 - x^4 - x^8 \\
 & + a^{17} \cdot -x - x^3 \\
 & + a^{18} \cdot -x^4 - x^6 - x^{10} \\
 & + a^{19} \cdot x^5 - x^7 \\
 & + a^{23} \cdot -x^3 - x^5 - x^9
 \end{aligned}$$

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$$1 - ax^5 \cdot 1 - a^2x^2 \cdot 1 - a^2x^6 \cdot 1 - a^3x^3 \cdot 1 - a^4 \cdot 1 - a^8$$

viz. expanding this function in ascending powers of  $a, x$ , then, if a term is  $Na^\theta x^\mu$ , this means that there are precisely  $N$  aszygetic covariants of the deg-order  $\theta.\mu$ .

368. It is known that the number of the irreducible covariants of the binary quintic is = 23; representing



these by the letters  $a, b, c, d, e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w$  ( $a$  the quintic itself), the deg-orders of these, and the references\* to the tables which give them are

[\* See also the paper, 143, in the second volume of this collection.]

| Deg-order. |             | Tab. | Mem.                   |
|------------|-------------|------|------------------------|
| 1.5        | $a$         | 13   | 2                      |
| 2.2        | $b$         | 14   |                        |
| „.6        | $c$         | 15   |                        |
| 3.3        | $d$         | 16   |                        |
| „.5        | $e$         | 17   |                        |
| „.9        | $f$         | 18   |                        |
| 4.0        | $g$         | 19   |                        |
| „.4        | $h$         | 20   |                        |
| „.6        | $i$         | 21   |                        |
| 5.1        | $j$         | 22   |                        |
| „.3        | $k$         | 23   |                        |
| „.7        | $l$         | 24   |                        |
| 6.2        | $m$         | 83   | 8                      |
| „.4        | $n$         | 84   |                        |
| 7.1        | $o$         | 90*  | 9                      |
| „.5        | $p$         | 91   | 2                      |
| 8.0        | $q$         | 25   | 2 [See also paper 143] |
| „.2        | $r$         | 92   | 9                      |
| 9.3        | $s^\dagger$ | 94   | 9                      |
| 11.1       | $t$         | 29   | 3                      |
| 12.0       | $u$         | 95   | 9                      |
| 13.1       | $v$         | 29a  | 5.                     |
| 18.0       | $w$         |      |                        |

Starting from the foregoing expression of the N.G.F. of the quintic, we can, instead of each term  $a^\theta x^\mu$ , introduce a covariant or product of covariants of the proper deg-order  $\theta, \mu$ : the mode of doing this depends of course on the different admissible partitions of  $\theta, \mu$ , and it is for some of the terms very indeterminate: for instance,  $a^4 x^6$  is  $ai$ ,  $hj$ , or  $ce$ . I found it possible to perform the whole process so as to satisfy a condition which will be presently referred to; and I found

[\* See vol. VII. of this collection, p. 348.]

† See end of Memoir. The  $S$  of Table 93 has the value  $-96(D, M) + 16BO - 7GA'$ , but it is better to use the simple value  $-(D, M)$ ; and the  $S$  of the present Memoir has this value, say  $S = -(d, m)$ .

*R.G.F. of quintic:*

|                            |             |
|----------------------------|-------------|
| 1.                         | Deg-orders. |
|                            | 0.0–10.10   |
| $+d \cdot 1 - ag^2$        | 3.3–12.8    |
|                            | 3.5–7.9     |
| $+e \cdot 1 - b^2$         | 3.9–5.11    |
|                            | 4.4–13.9    |
| $+f \cdot 1 - b$           | 4.6–12.10   |
| $+h \cdot 1 - ag^2$        | 5.1–14.6    |
|                            | 5.3–9.7     |
| $+i \cdot 1 - b^2g$        | 5.7–11.9    |
| $+j \cdot 1 - ag^2$        | 6.2–15.7    |
|                            | 6.4–14.8    |
| $+k \cdot 1 - b^2$         | 7.1–13.7    |
|                            | 7.5–15.9    |
| $+l \cdot 1 - bg$          | 8.2–16.6    |
|                            | 8.4–17.9    |
| $+m \cdot 1 - ag^2$        | 9.3–16.10   |
| $+n \cdot 1 - b^2g$        | 9.5–18.10   |
| $+o \cdot 1 - b^2$         | 10.2–19.7   |
|                            | 10.4–18.8   |
| $+p \cdot 1 - b^2g$        | 11.1–17.7   |
| $+r \cdot 1 - \frac{1}{6}$ | 11.3–20.8   |
| $+dj \cdot 1 - ag^2$       | 12.2–18.4   |

where observe that each negative term of the numerator is equal to a positive term multiplied by a power or product of terms  $a, b, g$ , contained in the denominator: this is the condition above referred to. The expansion thus consists only of terms each with the coefficient  $+1$ . For instance, a part of the function is

$$\frac{s(1 - abg)}{1 - a \cdot 1 - b \cdot 1 - c \cdot 1 - g \cdot 1 - q \cdot 1 - u} = \frac{s}{1 - c \cdot 1 - q \cdot 1 - u} \cdot \frac{1 - abg}{1 - a \cdot 1 - b \cdot 1 - g},$$

where the first factor is the entire series of terms  $sc^\alpha q^\beta u^\gamma$ , and the second factor is the series of terms  $a^\alpha b^\beta g^\gamma$  omitting only those terms which are divisible by  $abg$ : and in the product of the two factors the terms are all distinct, so that the coefficients are still each  $= 1$ . The same thing is true for every other pair of numerator terms and since the terms arising from each such pair are distinct from each other, in the expansion of the entire function the coefficients are each  $= +1$ . Hence (as in the case of the quartic) for any given deg-order, the terms in the expansion of the R.G.F. may be taken for the asyzygetic covariants of that deg-order; and if there are any other terms of the same deg-order, each of these must be a linear function, with numerical coefficients, of these asyzygetic covariants: thus deg-order 6.14, the expansion contains only the terms  $a^2h, acd, bc^2$ ; there is besides a term of the same deg-order,  $ef$ , which is not a term of the expansion, and hence  $ef$  must be a linear function of  $a^2h, acd, bc^2$ ; we in fact have  $ef = a^2h - 6acd - 4bc^2$ .

The terms in the expansion of the R.G.F. may be called “segregates,” and the terms not in the expansion “congregates”; the theorem thus is: every congregate is a linear function, with determinate numerical coefficients, of the segregates of the same deg-order.

369. I stop to remark that the numerator of the R.G.F. may be written in the more compendious form

$$\begin{aligned} & (1 - b^2)(1 - v) + (1 - b^2)(o + t) + (1 - b)(e + k) + (1 - b)f \\ & + (1 - ag^2)(d + h + j + m + dj + hj + j^2 + jm) \\ & + (1 - bg)(l + jo + js) \\ & + (1 - b^2g)(n + p + i + jk) \\ & + (1 - abg)s \\ & + (1 - g)jt \\ & + (1 - a)w; \end{aligned}$$

but the first-mentioned form is, I think, the more convenient one.

370. It is to be noticed that the positive terms of the numerator are unity, the seventeen covariants  $d, e, f, h, i, j, k, l, m, n, o, p, r, s, t, v, w$ , and the products of  $j$  by  $(d, h, j, k, m, o, s, t)$ , where  $j^2$  is reckoned as a product; in all, 26 terms. Disregarding the negative terms of the numerator the expansion would consist of these 26 terms, each multiplied by every combination whatever  $a^\alpha b^\beta c^\gamma g^\delta q^\epsilon u^\zeta$  of the denominator terms  $a, b, c, g, q, u$  (which for this reason might be called “reiterative”): the effect of the negative terms of the numerator is to remove from the expansion certain of the terms in question, thereby diminishing the number of the segregates: thus as regards the terms belonging to unity, any one of these which contains the factor  $b^2$  is not a segregate but a congregate: and so as regards the terms belonging to  $d$ , any one of these which contains the factor  $ag^2$  is a congregate: and the like in other cases.

For a given deg-order we have a certain number of segregates and a certain number of congregates: and the number of independent syzygies of that deg-order is precisely equal to the number of congregates: viz. each such syzygy may be regarded as giving a congregate in terms of the segregates: we have on the left-hand side a congregate, or, to avoid fractions, a numerical multiple of the congregate, and on the right-hand side a linear function, with numerical coefficients, of the segregates.

371. The syzygy is irreducible or reducible; and in the latter case it is, or is not, simply divisible: viz. if the congregate on the left-hand side contains any congregate factor (the other factor being literal), then the syzygy is reducible: it is, in fact, obtainable from the syzygy (of a lower deg-order) which gives the value of such congregate factor. But there are here two cases; multiplying the lower syzygy by the proper factor, the right-hand side may still contain segregates only, and then no further step is required: the original syzygy is nothing else than this lower syzygy, each side multiplied by the factor in question, and it is accordingly said to be simply divisible (S.D.). But contrariwise, the right-hand side, as multiplied, may contain congregates which have to be replaced by their values in terms of the segregates of the same deg-order: the resulting expression is then no longer explicitly divisible by the introduced factor: and the

original syzygy, although arising as above from a lower syzygy, is not this lower syzygy each side multiplied by a factor: viz. it is in this case not simply divisible.

For example (see the subsequent Table No. 96, under the indicated deg-orders) (6.6), from the syzygy

$$9d^2 = aj - b^2 + 2bh - cg,$$

we deduce (7.11) the syzygy

$$9da^2 = a^2j - ab^2 + 2abh - acg,$$

which (all the terms on the right-hand being segregates) requires no further reduction it is a reducible and simply divisible syzygy. But we have (6.8) a syzygy giving  $de$ , and also (6.10) a syzygy giving  $e^2$ ; multiplying the former of these by  $e$  or the latter of them by  $d$ , we obtain values of  $de^2$ , but in each case the right-hand sides contain terms which are not segregates, and have thus to be further reduced; the final formula (9.13) is

$$3a^2de^2 = -3a^2d^2f + 4a^2bj + 8abh^2 - 4abcg + 12b^2cd,$$

which is not divisible by any factor: the syzygy is thus reducible, but not simply divisible.

A syzygy, which is not in the sense explained reducible, is said to be irreducible.

372. The number of irreducible syzygies is obviously finite: it has, however, the large value 179 as appears from the annexed diagram, showing the congregates determined by these several syzygies, and the deg-orders of the syzygies:

*[Figure: Annexed diagram showing the congregates determined by the irreducible syzygies, with deg-orders and tabulated entries; spans p. 347 in the printed text.]*

Observe as regards the foregoing diagram, that  $dj^2$  is irreducible (since neither  $dj$  nor  $j^2$  is segregate), and similarly  $j^2h$ ,  $j^4$ , &c., are irreducible: we have thus the last or  $j^2$  column of the diagram.

The simply divisible syzygies are infinite in number, as are also the reducible syzygies not simply divisible. There is obviously no use in writing down a simply divisible syzygy; but as regards the reducible syzygies not simply divisible, these require a calculation, and it is proper to give them as far as they have been obtained.

373. The following Table, No. 96, replaces Tables 88 and 89 of my Ninth Memoir. The arrangement is according to deg-orders, and the table is complete up to the deg-order 8.40: it shows for each deg-order the segregate covariants, and also the congregate covariants (if any), and the syzygies which are the expressions of these in terms of the segregates. When there are only segregates these are given in the same horizontal line with the deg-order; for instance, 5.9  $ab^2$ ,  $ah$ ,  $cd$ , shows that for the deg-order 5.9 the only covariants are the segregates  $ab^2$ ,  $ah$ ,  $cd$ ; but when there are also congregates, the segregates are arranged in the same horizontal line with the deg-order, and the congregates, each in its own horizontal line together with its expression as a linear function of the segregates: thus

$$5.11 \quad * \quad ai, ce \quad bf \left| \begin{array}{cc} ai & ce \\ -1 & +1 \end{array} \right.$$

the segregates are  $ai$ ,  $ce$ , and there is a congregate  $bf$  which is a linear function of these,  $= -ai + ce$ . The table gives the irreducible syzygies and also the reducible syzygies which are not simply divisible, but the simply divisible syzygies are indicated each by a reference to the divided syzygy which occurs previously in the table.

374. Any syzygy might of course be directly verified by substituting for the several covariants contained therein their expressions in terms of the coefficients and facients of the quintic. But it is to be remarked that among the syzygies, or easily deducible from them, we have (6.18) the before-mentioned equation  $f^2 = -a^3d + a^2bc - 4c^3$ , and also a set of 17 syzygies, the left-hand sides of which are the covariants  $g, h, \dots, u, v, w$ , each multiplied by  $a$  or  $a^2$ , and which lead ultimately to the standard expressions of these covariants respectively, viz. each covariant multiplied by a proper power of  $a$  can be expressed as a rational and integral function of  $a, b, c, d, e, f$ , linear as regards  $f$ . Supposing them thus expressed, a far more simple verification of any syzygy would consist in substituting therein for the several covariants their expressions in the standard form, reducing if necessary by the equation  $f^2 = -a^3d + a^2bc - 4c^3$ : but of course, as to the syzygies used for obtaining the standard forms, this is only a verification if the standard forms have been otherwise obtained, or are assumed to be correct.

The 17 syzygies above referred to are

|          |          |                                                                        |
|----------|----------|------------------------------------------------------------------------|
| Deg-ord. |          |                                                                        |
| 6.10     | $a^2g =$ | $12abd + 4b^2c + e^2,$                                                 |
| 6.14     | $a^2h =$ | $6acd + 4bc^2 + ef,$                                                   |
| 5.11     | $ai =$   | $-bf + ce,$                                                            |
| 6.6      | $aj =$   | $b^2 - 2bh + cg + 9d^2,$                                               |
| 6.8      | $ak =$   | $2bi - 3de,$                                                           |
| 6.12     | $al =$   | $kci - 4df,$                                                           |
| 7.7      | $am =$   | $b^2d - cj + 3dh,$                                                     |
| 7.9      | $an =$   | $b^2e - 6bl - 2ck - fg,$                                               |
| 8.6      | $ao =$   | $2b^2 - ej,$                                                           |
| 8.10     | $ap =$   | $-2cn - fj,$                                                           |
| 9.5      | $aq =$   | $2bg - 4d^2f + 12dj^2 + hj,$                                           |
| 9.7      | $ar =$   | $b^2k - bp + co - hi,$                                                 |
| 10.8     | $as =$   | $3bdk + 3djf + 2im,$                                                   |
| 12.6     | $at =$   | $vj^2 + jp - 2mn,$                                                     |
| 13.5     | $18au =$ | $2agq + b^2j + 6bmj - 9dt - ghj + no,$                                 |
| 14.6     | $3au =$  | $-6b^2d - 8b^2f + 2b^2g - 2b^2q + 12b^2h^2 - 3et,$                     |
| 19.5     | $18aw =$ | $3b^2g + b^2qo - 4bfo - bgmo + 18bm^2 + 3dgjo - 18djt - 3ght - 6m^2o,$ |

the last four of these being, however, beyond the limits of the table: the expressions of  $g, h, i$  are here in the standard form: the standard forms of the other covariants  $j, k, \dots, u, v, w$ , will be given further on.

*Table No. 96 (Segregates, Congregates, and Syzygies).*

| Deg-ord. | Congs. | Segregates. |
|----------|--------|-------------|
| 1 . 1    |        | 1           |
| 3        |        |             |
| 5        |        | $a$         |
| 2 . 2    |        | $b$         |
| 4        |        |             |
| 6        |        | $c$         |
| 8        |        |             |
| 10       |        | $a^2$       |
| 3 . 1    |        |             |
| 3        |        | $d$         |
| 5        |        | $e$         |
| 7        |        | $ab$        |
| 9        |        | $f$         |
| 11       |        | $ac$        |
| 13       |        |             |
| 15       |        | $a^3$       |

*Table No. 96 (continued).*

| Deg-ord. | Congs.         | Segregates.                                  |
|----------|----------------|----------------------------------------------|
| 4.0      |                | $g$                                          |
| 2        |                | $b^2, h$                                     |
| 4        |                |                                              |
| 6        |                | $ad, bc$                                     |
| 8        |                | $ae$                                         |
| 10       |                |                                              |
| 12       |                | $b, a^2b$                                    |
| 14       |                |                                              |
| 16       |                | $af$                                         |
| 18       |                |                                              |
| 20       |                | $a^2e$<br>$a^4$                              |
| 5.1      |                | $j$                                          |
| 3        |                | $k$                                          |
| 5        |                | $ag, bd$                                     |
| 7        |                |                                              |
| 9        |                | $be, l, ab^2, ah, cd$                        |
| 11       | $*$<br>$bf$    | $ai, ce$<br>$-1 +1$                          |
| 13       |                | $a^2g, abc$                                  |
| 15       |                |                                              |
| 17       |                | $a^2e, cf$                                   |
| 19       |                |                                              |
| 21       |                | $a^3b, ac^2$                                 |
| 23       |                |                                              |
| 25       |                | $a^3f$                                       |
| 6.0      |                | $a^2e$                                       |
| 2        |                | $m, bg$                                      |
| 4        |                | $n$                                          |
| 6        | $*$<br>$d^2.9$ | $aj, b^2, bh, cg$<br>$+1 -1 +2 -1$           |
| 8        | $*$<br>$de.3$  | $bi, ak$<br>$+1 +2$                          |
| 10       |                | $c^2g, abd, b^2c, eh$<br>$+1 -12 -4 \quad .$ |
| 12       | $*$<br>$df.3$  | $cdc, al, ci$<br>$. -1 +2$                   |
| 14       | $*$            | $a^2b^2, a^2h, acd, bc^2$<br>$+1 -6 -4$      |
| 16       | $*$<br>$abf$   | $a^2b, ace$                                  |

Table No. 96 (continued).

| Deg-ord. | Congs. | Segregates.                         |
|----------|--------|-------------------------------------|
| 6 . 18   | *      | $a^3d, a^2bc, c^3$                  |
|          | $f^2$  | $-1 +1 -4$                          |
| 20       |        | $a^3e, acf$                         |
| 22       |        | $a^4b, a^2c^2$                      |
| 24       |        |                                     |
| 26       |        | $a^3f$                              |
| 28       |        |                                     |
| 30       |        | $a^5c$                              |
| 7 . 1    |        | $o$                                 |
| 3        |        | $bj, dg$                            |
| 5        | *      | $p, hk, eg$                         |
|          | $dk.3$ | $0 +1 +2 +1$                        |
| 7        |        | $abg, am, b^2d, ej$                 |
| 9        | *      | $an, bi, ck, fg$                    |
|          | $di.3$ | $+1 +6 +2 +1$                       |
|          | $eh$   | $+1 +1$                             |
| 11       | *      | $a^2j, ab^2, abh, acg, bcd$         |
|          | $ad^2$ | $+1 +1 +4 +2 +1$ S.D. 6.6, $d^2$    |
|          |        | $-1 +6 -6$                          |
| 13       | *      | $c^2k, abi, bce, el$                |
|          | $cde$  | S.D. 6.8, $de$                      |
|          | $fh.3$ | $-1 -2 +3 -6$ S.D. 5.11, $bf$       |
| 15       | *      | $a^2g, a^2bd, ab^2c, ach, c^2d$     |
|          | $ae^2$ | S.D. 6.10, $e^2$                    |
|          | $fi$   | $+1 -1 +1 -6$                       |
| 17       | *      | $a^2f, a^2be, a^2d, c^2e$           |
|          | $adf$  | S.D. 6.12, $df$                     |
|          | $bef$  | S.D. 5.11, $bf$                     |
| 19       |        | $a^3b^2, a^3h, a^2cd, abc^2$        |
|          |        | S.D. 6.14, $ef$                     |
| 21       | *      | $a^3f, a^3ce, c^3f$                 |
|          |        | $a^4d, a^3bc, ac^3$ S.D. 5.11, $bf$ |
| 23       | *      | $a^2bf$ S.D. 6.18, $f^2$            |

Table No. 96 (continued).

| Deg-ord. | Congs.     | Segregates.                                    |
|----------|------------|------------------------------------------------|
| 7.25     |            | $a^3e, a^2cf$                                  |
| 27       |            | $a^4b, a^2c^2$                                 |
| 29       |            | $a^3f$                                         |
| 31       |            |                                                |
| 33       |            | $a^5c$                                         |
| 35       |            | $r$                                            |
| 8.0      |            | $q$                                            |
| 2        |            | $b^2g, bm, dj, gh$                             |
| 4        |            |                                                |
| 6        | *          | $ao, bn, gi$                                   |
|          | $dk.3$     | +1 -3 -1                                       |
|          | $ej$       | +1 -2                                          |
| 8        | *          | $a^2i, ab^2, b^3, b^2h, bcg, cm$               |
|          | $bd^2$     | -4 +3 +4 -6 S.D. 6.6, $d^2$                    |
|          | $ek$       | +4 -3 -4 +8 +1 +12 S.D. 6.8, $de$              |
|          | $h^2.3$    |                                                |
| 10       |            | $abk, aeg, ap, b^2l, cn$                       |
|          |            | +2 . +3 +1 +3                                  |
|          | $bhe$      | -1 . . -2                                      |
|          | $dl.3$     |                                                |
|          | $fj$       | +1 . . +2 -3                                   |
|          | $hi.3$     |                                                |
| 12       |            | $a^2bg, abn, ab^2d, acj, b^2c, bch, c^2g$      |
|          | $adh$      | S.D. 7.7, $dh$                                 |
|          | $be^2$     | S.D. 6.10, $e^2$                               |
|          | $cd^2$     | S.D. 6.6, $d^2$                                |
|          | $d$        | -1 -2 . +1 -2 +2                               |
|          |            | -1 +4 -6 +2                                    |
|          | $fk$       | +1 . . +1 -2 +1                                |
| 14       |            | $a^2n, abl, ack, afg, bci$                     |
|          | $adi$      | S.D. 7.9, $bi$                                 |
|          | $aei$      | S.D. 7.9, $dh$                                 |
|          | $bdf$      | S.D. 7.9, $eh$                                 |
|          | $cde$      | S.D. 6.12, $df$                                |
| 16       |            | S.D. 6.8, $de$                                 |
|          | $a^2d^2$   |                                                |
|          | $aei, bef$ | $a^2j, a^2b^2, a^2bh, a^2cg, abcd, b^2c, c^2h$ |
|          | $fl.3$     | S.D. 6.6, $d^2$                                |
|          |            | S.D. 7.11, $ei$                                |
|          |            | S.D. 6.14, $ef$                                |
|          |            | S.D. 6.10, $e^2$                               |
|          |            | +1 -1 +2 . -6 +6                               |

Table No. 96 (continued).



| Deg-ord. | Congs.   | Segregates.                               |
|----------|----------|-------------------------------------------|
| 8 . 18   | *        | $a^2k, a^2bi, abce, acl, c^2l$            |
|          | $abde$   | S.D. 6.8, $de$                            |
|          | $abf$    | S.D. 5.11, $bf$                           |
|          | $afh$    | S.D. 7.13, $fh$                           |
|          | $cdf$    | S.D. 6.12, $df$                           |
| 20       | *        | $a^2g, a^2bd, a^2b^2c, a^2ch, ac^2d, bc$  |
|          | $aby$    | S.D. 6.10, $e^2$                          |
|          | $afh$    | S.D. 7.15, $fi$                           |
|          | $cdf$    | S.D. 6.18, $f^2$                          |
|          |          | S.D. 6.14, $ef$                           |
|          | *        | S.D. 6.12, $df$                           |
|          | $afi$    | S.D. 6.14, $ef$                           |
|          | $bf^2$   | S.D. 6.18, $f^2$                          |
|          | $cef$    | S.D. 5.11, $bf$                           |
| 22       | *        | $a^2be, a^2f, a^2ci, abcf, ac^2e$         |
|          |          | S.D. 6.18, $f^2$                          |
|          | $a^2df$  |                                           |
| 24       | *        | $a^3b, a^3b^2, a^3cd, a^2bc^2, c^4$       |
| 26       |          |                                           |
|          | $a^3kf$  |                                           |
| 28       | *        | $a^4d, a^4bc, a^3c^2$                     |
| 30       |          | $a^4e, a^3cf$                             |
| 32       |          |                                           |
| 34       |          | $a^5f$                                    |
| 36       |          | $bo, gk, s$                               |
| 38       |          |                                           |
| 40       | *        | $af, aq, b^2j, bdg, hj$                   |
| 9 . 1    | $dm.12$  | $-1 -2 +1 +1$                             |
| 3        | *        | $ar, beg, bp, co, gl$                     |
| 5        | $b^2k.3$ | $+1 \quad . \quad -5 -1 \quad . \quad +1$ |
| 7        |          | $-1$                                      |
| 9        | $dn.3$   | $. \quad . \quad -1 -1$                   |
|          | $em.3$   |                                           |
|          | $hk.3$   | $+2 \quad . \quad +2 +1 -1$               |
|          | $j^2$    | $+2 \quad . \quad +2 +4 -1$               |
|          | *        | $. \quad +1 +1$                           |
|          | $bdh.3$  | $a^2b^2g, abm, adj, agh, b^2d, bcj, cdg$  |
|          | $d^3.27$ |                                           |
|          | $en$     | $+2 +3 \quad . \quad +1 +2 -3$            |
|          | $ik$     | $-1 \quad . \quad . \quad . \quad -3 +3$  |

Table No. 96 (continued).

| Deg-ord. | Congs.      | Segregates.                                                  |
|----------|-------------|--------------------------------------------------------------|
| 9 . 11   | *           | $a^2o, abn, agi, b^2l, bek, ceg, cp$                         |
|          | $adk$       | S.D. 8.6, $dk$                                               |
|          | $ag$        | S.D. 8.6, $ej$                                               |
|          | $b^2e$      | S.D. 7.9, $bi$                                               |
|          | $bf^2$      | . +1 -1 +6 +2 +1 S.D. 5.11, $bf$                             |
|          | $bdi$       | . . -1 +4 +2 +1                                              |
|          | $beh$       | S.D. 6.6, $d^2$                                              |
|          | $eg$        | +1 -3 -1 +2 +2 S.D. 8.8, $ek$                                |
|          | $d^2e.9$    | +1 +3 +1 . -3 . -3 S.D. 8.8, $h^2$                           |
|          | $fm.3$      | +1 -3 -1 +2 +5 . -6 S.D. 7.11, $ei$                          |
| 13       |             | $a^2bj, a^2dg, ab^3, ab^2h, abcg, acm, b^2cd, c^3j$          |
|          | $hi.3$      | S.D. 7.7, $dh$                                               |
|          | $abd^2$     |                                                              |
|          | $aei$       | -8 +4 -4 +3 +4 . -12                                         |
|          | $ah^2$      | -2 +1 +2 -3 +1 -2                                            |
|          | $bei$       |                                                              |
|          | $cdh$       |                                                              |
|          | $d^2c.3$    |                                                              |
|          |             | +2                                                           |
|          |             | -1 . . +1 -2 +1 -3 -3 +3                                     |
| 10 . 0   |             | $bg^2, hg, g^2, z, i$                                        |
| 2        | *           | $br, gn, jk$                                                 |
| 4        | $do.6$      | +2 . -1                                                      |
| 6        | *           | $aa_j, b^2g, b^2m, bdj, bgh, cg^2, cq$                       |
|          | $dfg$       | S.D. 6.6, $d^2$                                              |
|          | $eo$        | 1 +6 +1 +2 +12 -12 -2                                        |
|          | $hm.6$      | -1 -4 -1 +4 +12 -2 +1 -3                                     |
|          | $bdk$       | -12 +2 -1                                                    |
|          | $bej$       | +1                                                           |
|          | $deg$       | $abo, ag^2, as, b^2n, bgi, cr$                               |
| 8        | *           | S.D. 8.6, $dk$                                               |
|          | $dp$        | S.D. 8.6, $ej$                                               |
|          | $hn.3$      | S.D. 6.8, $de$                                               |
|          | $hi.3$      | -5 . . +3 +15 +5 -6                                          |
|          | $jl.3$      | -6 +1 . . +18 +15 -12                                        |
|          | $adm$       | +13 +1                                                       |
| 10       |             | +1 . . . +3                                                  |
|          | $b^2r$      | -5 . +3 +15 +5 -12                                           |
|          | $b^2d^2.72$ | $ag^2, a^2q, ab^2j, cb^2dg, abj, b^2h, b^2cg, bcm, cdj, cgh$ |
|          | $bek.2$     | $dm$ S.D. 9.5.                                               |
|          | $bh^2.6$    |                                                              |
|          | $d^2h.27$   |                                                              |
|          | $eg$        | +1 +12 . -9 -3 +4 +3 +12 . +32 -12 -4                        |
|          | $g^2.2$     | +1 +2 . . . . -8 -32 +12 +4                                  |
|          |             | -1 -2 . +9 -3 +4 +3 . +4 +32 -12 -4                          |
|          | $in.4$      | . . . -3 . . . -2 -8 +12 -4                                  |
|          |             | +1 -2 . . . . . . . .                                        |
|          | $jl.4$      | . . . +3 . . . -8 +12 +1                                     |
|          |             | +3                                                           |
|          |             | S.D. 6.10, $e^2$                                             |
|          |             | . . . . . -1 . -2 -12 +6 +2                                  |
|          |             | +1 +2 -3 +7 . . . +4 +24 -24 -4                              |
|          |             | -1 -2 +3 -5 . . . . . . .                                    |
|          |             | 1 +2 . . . . . . . +12                                       |
|          |             | . . . -1 -1 . . . +16 +12                                    |

*Table No. 96 (concluded).*

| Deg-ord. | Congs.       | Segregates.                                                                           |
|----------|--------------|---------------------------------------------------------------------------------------|
| 11 . 1   |              | $go, t$                                                                               |
| 3        |              | $bgj, df, dq, jm$                                                                     |
| 5        | *            | $b^2o, bgk, bs, cg^2, cq, gp$                                                         |
|          | <i>dr.18</i> | -2 +5 -6     .     -3 +3                                                              |
|          | <i>ho.3</i>  | -2 +11 -24     .     -3 +6                                                            |
| 12 . 2   |              | $jn$                                                                                  |
| 4        |              | $km.6$ -1 +3 -1 +26     .     .     +1 -2                                             |
| 13 . 1   |              | .    -3 +3                                                                            |
| 3        | *            | $gr, jo$                                                                              |
|          | <i>ko</i>    |                                                                                       |
| 14 . 0   |              | $bv, b^2q, bgm, bf, dgj, g^2h, hg$                                                    |
| 2        | *            | .    -2 -2 -4 +3                                                                      |
| 4        | <i>mλ.12</i> | .    +2 +1 +4 -3     .     -3                                                         |
|          | <i>jr.2</i>  | $fj, jq, b^2$                                                                         |
|          | <i>nu.2</i>  | $bgo, bt, g^2k, gs, kg$                                                               |
|          | *            |                                                                                       |
|          | <i>dgo</i>   | .    -2 +1 -1 .                                                                       |
|          | <i>dt.18</i> | .    -4       +1 -1                                                                   |
|          | <i>mr.12</i> |                                                                                       |
|          |              | $b^2, bgq, bu, fm, gj^2, mq, o^2$                                                     |
|          |              | $bgr, bjo, ghi, gjk, js, nq$                                                          |
|          |              | +1 +2     .     .     .     -1 +6 +3     .     .     .     .     S.D. 10.4, <i>do</i> |
|          |              | +1 +2     .     .     .     -1     .     +3                                           |

*Theory of the Canonical Form.* Art. Nos. 375 to 381, and Tables Nos. 97 and 98.

375. As the small italic letters have been used to represent the covariants, different letters are required for the coefficients of the quintic: using also new letters for the facients, I take the quintic to be  $(a, b, c, d, e, f \, \S \, \xi, \eta)^5$ . Considering a linear transformation of  $\frac{1}{a}(a, b, c, d, e, f \, \S \, \xi, \eta)^5$ , viz.

$$\frac{1}{a}(a, b, c, d, e, f \S \xi - b\eta, a\eta)^5,$$

# A Tenth Memoir on Quantics

(Continuation from earlier pages, Paper No. 693)

*Philosophical Transactions of the Royal Society of London*, **169** (1878), pp. 603–662

[Vol. X, printed page 356; PDF page 376.]

[The preceding paragraphs (printed pages 354–355) construct from the original quintic  $(a, b, c, d, e, f \mid \xi, \eta)^5$  the canonical reduced form and explain the genesis of the capital covariants  $A, B, C, \dots, V, W$ .]

This is

$$\xi^5 + 5\xi^4\eta + 10\xi^3\eta^2 + 10\xi^2\eta^3 + 5\xi\eta^4 + \eta^5,$$

which is

$$\begin{aligned} &= (1 - b + b^2 - b^3 + b^4 - b^5) \\ &\quad - 2b + 3b^2 + 5b^4 \\ &+ b(1 - 3b - 4b^2) \\ &+ ac(1 - 10b^2) \\ &+ a^2d(1 + 6b^2) \\ &+ a^3e(-4b + 10b^2) \\ &+ af(-5b) \\ &\quad 1 \\ &\quad 1). \end{aligned}$$

$$= (1 \quad ae + 1 \quad a^2d + 1 \quad a^3e + 1 \quad a^4f + 1 \mid \xi, \eta)^5$$

with the subsidiary numerical schedule

$$\begin{array}{cccc} b^2 - 1 & abc - 3 & a^2bd - 4 & a^3be - 5 \\ b^3 + 2 & ab^2c + 6 & a^2bcd + 10 & a^2b^2c \text{ (etc.)} \\ b^4 - 3 & ab^3c - 10 & & \\ b^5 + 4 & & & \end{array}$$

The values of  $a, b, c, d, e, f$ , considered for a moment as denoting the leading coefficients of the several covariants ultimately represented by these letters respectively, are

$$\begin{array}{cccccc} a & b & c & d & e & f \\ a + 1 & ae + 1 & ac + 1 & ace + 1 & a^2f + 1 + a^2c^2d & 1 \\ & & & & abe + 5 & \\ bd - 4 & b^2 - 1 & ad^2 - 1 & & ac^2d + 2 & abc - 3 \\ c^2 + 3 & & & & ab^2d + 8 & b^3 - 2 \\ & & & -b^2e \cdot 1 & bcd + 2 & bc^2 - 10 \\ & & & bcd + 2 & -c^3 \cdot 1 & \end{array}$$

satisfying, as they should do, the relation

$$f^2 = -a^2d + a^2bc - 4c^3.$$

Hence forming the values of  $a^4b \cdot 5c^2$  and  $a^3e \cdot 2cf$ , it appears that the value of the last-mentioned quintic function is

$$(1, 0, c, f, a^2b - 3c^2, a^3e - 2cf \mid \xi, \eta)^5.$$

Writing herein  $x, y$  in place of  $\xi, \eta$ , and now using  $a, b, c, d, e, f$  to denote, not the leading coefficients but the covariants themselves ( $a$  denoting the original quintic, with  $\xi, \eta$  as facients), we have the form

$$\mathfrak{A} = (1, 0, c, f, a^2b - 3c^2, a^3e - 2cf \mid x, y)^5,$$

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a new quintic, which is the canonical form in question: the covariants hereof (reckoning the quintic itself as a covariant) will be written  $A, B, C, \dots, V, W$ , and will be spoken of as *capital covariants*.

**376.** The fundamental property is: Every capital covariant, say  $I$ , has for its leading coefficient the corresponding covariant  $i$  multiplied by a power of  $a$ : and this follows as an immediate consequence of the foregoing genesis of  $A$ . The covariant  $i$  of the form

$$\frac{1}{n}(a, b, c, d, e, f \mid \xi, \eta)^n$$

has a leading coefficient

$$= \frac{1}{n}(a^2cf - a^2be + \&c.),$$

which, when  $a, b, c, d, e, f, \dots, i$  denote leading coefficients, is  $i$  multiplied by a power of  $a$ : and upon substituting for the quintic the linear transformation thereof

$$(1, 0, c, f, a^2b - 3c^2, a^3e - 2cf \mid \xi, \eta)^5,$$

(observing that, in the transformation  $\xi, \eta$  into  $\xi + h\eta, a\eta$ , the determinant of substitution is  $= a$ ), the value is still  $= i$  multiplied by a power of  $a$ ; or using the relation  $\mathfrak{a} = a$ , say the value is  $= i$  multiplied by a power of  $a$ . Now the covariant  $i$  is the same function of the covariants  $a, b, c, d, e, \dots$  that the leading coefficient  $i$  is of the leading coefficients  $a, b, c, d, e, f$ ; hence, the italic letters now denoting covariants, the leading coefficient still is  $= i$  multiplied by a power of  $a$ : which is the above-mentioned theorem.

**377.** To show how the transformation is carried out, consider, for example, the covariant  $B$ . This is obtained from the corresponding covariant of  $(a, b, c, d, e, f \mid \xi, \eta)^5$ , that is,

$$(ae \cdot 1 \quad af \cdot 1 \quad bf \cdot 1 \mid \xi, \eta)^2,$$

$$\begin{array}{ccc} bd - 4 & be - 3 & ce - 4 \\ & cd + 1 & \\ c^2 + 1 & & d^2 + 3 \end{array}$$

by changing the variables, and for the coefficients

$$a, b, c, d, e, f$$

writing

$$1, 0, c, f, a^2b - 3c^2, a^3e - 2cf;$$

thus the coefficients are

| First.           | Second.          | Third.                     |
|------------------|------------------|----------------------------|
| $1(a^2b - 3c^2)$ | $-1(a^3e - 2cf)$ | $-4c(a^2b - 3c^2)$         |
|                  | $+2cf$           |                            |
| $+3c^2$          | $= a^3e$         | $+3f^2$                    |
| $= a^2b$         |                  | $= -4a^2bc + 12c^4$        |
|                  |                  | $+3(-a^2d + a^2bc - 4c^3)$ |
|                  |                  | $= a^2(-3ad - bc);$        |

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and we have thus the expression of  $B$  (see the Table No. 97); and similarly for the other capital covariants  $C, D, \dots, V, W$ : in every case the coefficients are obtained in the standard form, that is, as rational and integral functions of  $a, b, c, d, e, f$ , linear as regards  $f$ .

**378.** It will be observed that there is in each case a certain power of  $a$  which explicitly divides all the coefficients and is consequently written as an exterior factor: disregarding these exterior factors, the leading coefficients for  $B, C, D, E, F$  are  $b, c, ad, e, f$  respectively; that for  $G$  is  $12abd + 4b^2c + e^2$ , which must be  $=g$  multiplied by a power of  $a$ , and (in Table 97) is given as  $=a^2g$ ; similarly, that for  $H$  is  $6acd + \&c.$ , which must be  $=h$  multiplied by a power of  $a$ , and is given as  $=a^4h$ : and so in the other cases. The index of  $a$  is at once obtained by means of the deg-order, which is in each case inserted at the foot of the coefficient.

For  $A, B, C, E, F$  there is no power of  $a$  as an interior factor: and for the invariants  $G, Q, \dots$  we may imagine the interior factor thrown together with the exterior factor, ( $G = a^2g, \&c.$ ): whence disregarding the exterior factors, we may say that for  $A, B, C, E, F, G, Q, \dots$  the standard forms are also “divided” forms.

But take any other covariant, for instance,  $D$ : the leading coefficient is  $ad$ , having the interior factor  $a$ ; and this being so it is found that all the following coefficients will divide by  $a$  (the quotients being of course expressible only in terms of the covariants subsequent to  $f$ ): thus the second coefficient of  $D$  is  $-bf + ce$ , and (5.11) we have  $-bf + ce = ai$ , or the coefficient divided by  $a$  is  $i$ ; and so for the other coefficients of  $D$ ; or throwing out the factor  $a$ , we obtain for  $D$  an expression of the form  $(d, i, \dots | x, y)^4$ , see the Table 98: this is the “divided” form of  $D$ : and we have similarly a divided form for every other capital covariant. All that has been required is that each coefficient of the divided form shall be expressed as a rational and integral function of the covariants  $a, b, c, \dots, v, w$ : and the form is not hereby made definite: to render it so, the coefficient must be expressed in the *segregate* form. But there is frequently the disadvantage that we thus introduce fractions; for instance, the last coefficient of  $D$  is  $ci + df$ , where to get rid of the congregate term  $df$  we have (6.12),  $3df = -al + 2ct$ , and the segregate form of the coefficient is

$$= \frac{1}{3}al + \frac{1}{3}ct.$$

**379.** We have in regard to the canonical form, a differential operator which is analogous to the two differential operators  $xd_y \{xd_y\}, yd_x \{yd_x\}$  considered in the Introductory Memoir (1854), [139]. Let  $\delta$  denote a differentiation in regard to the constants under the conditions

$$\begin{aligned}\delta a &= 0, \\ \delta b &= e, \\ \delta c &= 3f, \\ \delta d &= \frac{1}{a}(-bf + ce), \quad (= i), \\ \delta e &= -6ad - 10bc, \\ \delta f &= -2a^2b - 18c^2,\end{aligned}$$

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which (as is at once verified) are consistent with the fundamental relation

$$f^2 = -a^2d + a^2bc - 4c^3;$$

then it is easy to verify that

$$\left(x \frac{d}{dy} - 4cy \frac{d}{dx} - \delta\right) A = 0;$$

and this being so, any other covariant whatever, expressed in the like standard form, is reduced to zero by the operator

$$x \frac{d}{dy} - 4cy \frac{d}{dx} - \delta;$$

and we have thus the means of calculating the covariant when the leading coefficient is known.

Thus, considering the covariant  $B$ , the expression of which has just been obtained,  $= (B_0, B_1, B_2 \mid x, y)^2$ , suppose: the equation to be satisfied is

$$\begin{aligned} & x(B_1x + 2B_2y - 4cy(2B_0x + B_1y) \\ & - x^2\delta B_0 - xy\delta B_1 - y^2\delta B_2) = 0, \end{aligned}$$

viz. we have

$$\begin{aligned} B_1 - \delta B_0 &= 0, \\ 2B_2 - 8cB_0 - \delta B_1 &= 0, \\ -4cB_1 - \delta B_2 &= 0; \end{aligned}$$

which (omitting, as we may do, the outside factor  $a^2$ ) are satisfied by the foregoing values  $B_0, B_1, B_2, = b, e, -3ad - bc$ . And if we assume only  $B_0 = b$ , then the first equation gives at once the value  $B_1 = e$ , the second equation then gives  $B_2 = -3ad - bc$ ; and the third equation is satisfied identically, viz. the equation is

$$-4ce + 8(3ad + bc) = 0,$$

that is,

$$-4ce = -4ce = 0,$$

$$\begin{aligned} & + c\delta b + c \cdot e \\ & + b\delta e + b \cdot 3f \\ & + 3a\delta d + 3(-bf + ce) \end{aligned}$$

which is right.

Of course every invariant must be reduced to zero by the operation  $\delta$ : thus we have, see the Table No. 97,

$$a^2g = 12abd + 4b^2c + 1e^2,$$

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and thence

$$\begin{aligned} a^2\delta g &= (12ad + 8bc)\delta b &= (12ad + 8bc)e \\ &+ 4b^2 \cdot \delta c &+ 4b^2 \cdot 3f \\ &+ 12ab \cdot \delta d &+ 12b(-bf + ce) \\ &+ 2e \cdot \delta e &+ 2e(-6ad - 10bc), \end{aligned}$$

with row-totals (in the columns  $ade, b^2f, bce$ ):

|  | <i>ade</i> | <i>b<sup>2</sup>f</i> | <i>bce</i> | Sum |
|--|------------|-----------------------|------------|-----|
|  | +12        |                       | +8         |     |
|  |            | +12                   |            |     |
|  | -12        |                       | +12        |     |
|  | -12        |                       | -20        |     |
|  | 0          | 0                     | 0          |     |

which is = 0, as it should be.

**380.** As already remarked, the leading coefficients of  $H, I, J$ , &c., are each of them equal to a power of  $a$  multiplied by the corresponding covariant  $h, i, j, \dots$ ; hence, supposing these leading coefficients, or, what is the same thing, the standard expressions of the covariants  $h, i, j, \dots, v, w$  to be known, we can calculate the values of  $\delta h, \delta i, \delta j, \dots, \delta v, \delta w$  ( $= 0$ , since  $w$  is an invariant): and the operation  $\delta$ , instead of being applicable only to the forms containing  $a, b, c, d, e, f$ , becomes applicable to forms containing any of the covariants. The values of  $\delta a, \delta b, \dots, \delta v, \delta w$  can, it is clear, be expressed in terms of segregates; and this is obviously the proper form: but for  $\delta r, \delta t$ , and  $\delta v$ , for which the segregate forms are fractional, I have given also forms with integer coefficients. The entire series is



**Deg-order.**

$$2.8 \quad \delta a = 0,$$

$$3.5 \quad \delta b = e,$$

$$3.9 \quad \delta c = 3f,$$

$$4.6$$

$$4.8 \quad \delta d = i,$$

$$4.12$$

$$5.3 \quad \delta e = -6ad - 10bc,$$

$$5.7$$

$$5.9 \quad \delta f = -2a^2b - 18c^2,$$

$$6.4$$

$$6.6 \quad \delta g = 0,$$

$$6.10$$

$$7.5 \quad \delta h = 2je - 4i,$$

$$7.7$$

$$8.4 \quad \delta i = -2ab^2 + 2ah - 18cd,$$

$$8.8$$

$$9.3 \quad \delta j = -n,$$

$$9.5$$

$$10.6 \quad \delta k = 2aj + 6b^2j - 9bh + 3c^2,$$

$$12.4$$

$$13.3 \quad \delta l = -3abd - 7b^2c + 7ch,$$

$$14.4$$

$$19.3 \quad \delta m = -bk - p,$$

$$\delta n = 4cj,$$

$$\delta o = \dots$$

$$\delta p = b^2g + 6bm - 6dj - gh,$$

$$8ahj - 5adg - 10b^4 + 15b^2c^2 - 5bcg + 10cm,$$

$$\delta q = 0,$$

$$\delta r = \frac{1}{3}(aq + 6b^2j - 5bdg - jh), \quad = 2b^2j - 2bdg - 6dm,$$

$$\delta s = 2agj + 2f^2g + 3b^2m + 21bdj - 4bgh + 2c^2f^2 - 3cq,$$

$$\delta t = \frac{1}{3}(bgm + \frac{4}{3}bf^2 - 3dgj - kq), \quad = -b^2q + hq + 6m^2,$$

$$\delta u = 0,$$

$$\delta v = \frac{1}{3}(-5bgr - 10bjo + 5gjk - 12js - 9nq), \quad = -6dt - 6mr + nq,$$

$$\delta w = 0.$$

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It is obvious that for every covariant whatever written in the denumerate form  $(I_0, I_1, \dots | x, y)^n$ , the second coefficient is equal to the first coefficient operated upon by  $\delta$ ; so that the foregoing formulae give, in fact, the second coefficients of the several covariants.

**381.** It is worth noticing how very much the formulae of Table No. 97 simplify themselves, if one of the covariants  $b, c, d, e$  vanishes, in particular, if  $b$  vanishes. Suppose  $b = 0$ ; writing also (although this makes but little difference)  $a = 1$ , we have

$$\begin{aligned} a &= 1, \\ b &= 0, \\ c &= c, \\ d &= d, \\ e &= e, \\ f^2 &= -d - 4c^3, \\ g &= e^2, \\ h &= 6cd + ef, \\ i &= ce, \\ j &= 9d^2 + c^4, \\ k &= 3de, \\ l &= -3df + 2c^2e, \\ m &= 9cd^2 + 3def - c^4e, \\ n &= -6cde - e^2f, \\ o &= 9d^2e + c^2e^3, \\ p &= +9d^2f + 12c^3de - c^2ef, \\ q &= 54cd^3 - 27d^2ef + 18c^3de^2 + c^6f, \\ r &= 9cd^2e + 3de^3f - c^5e^2, \\ s &= -27d^3f + 54c^2d^2e + 9cde^3f - 2c^5e^3, \\ t &= -81d^4f - 6d^2f^3 + 216c^3d^2e + 54cd^3f - 24c^4de^2 - c^7ef, \\ u &= -27d^4 - 18cd^2e^2 - 4d^2e^2f + c^6de^4, \\ v &= 81d^4ef - 6d^4f^3 + 216c^3d^2e^2 + 54cd^3ef - 24c^4de^4 - 7c^7e^2f, \\ w &\text{ (not calculated).} \end{aligned}$$

These values are very convenient for the verification of syzygies, &c. Take, for instance, the before-mentioned relation  $\delta v = -6dt - 6mr + nq$ , that is, if  $V = (V_0, V_1 | x, y)$ , then  $V_1 = -6dt - 6mr + nq$ : calculating the three products on the right-hand side, observing

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that  $f^2$  when it occurs is to be replaced by its value  $-d - 4c^3$ , and taking their sum, the figures are as follows:

|               | $-6dt$ | $-6mr$ | $+nq$ | Sum   |
|---------------|--------|--------|-------|-------|
| $d^5 f$       | +486   |        |       | +486  |
|               | +36    |        |       |       |
| $d^3 f^3$     |        | +54    | -27   | +63   |
|               | -1296  |        |       |       |
| $c^3 d^3 e^2$ | -324   | -486   | +324  | -1458 |
| $cd^4 e^2 f$  |        | -324   | +216  | -432  |
|               |        |        | +1    | +1    |
|               |        |        | -216  |       |
| $c^4 d^2 e^4$ | +144   | +324   | -24   | +252  |
|               | +6     | +36    |       |       |
| $c^6 de^4 f$  |        |        | +4    | +18   |
|               |        | -6     |       |       |
|               |        |        | -2    |       |

where the last column is, in fact, what  $V_1$  becomes on writing therein  $a = 1, b = 0$ . The verification would not of course apply to terms which contain  $b$ ; thus, (13.3), a derived syzygy is  $\delta l = 6t + mo$ ; and the foregoing values give, as they should do,  $\delta l = mo$ : we might for the verification of most of the terms in  $b$  use values  $a, b, c, d, e, f^2 = -d$ : 1,  $b, 0, d, e$ , the only failure would be for terms containing  $bc$ .

**Table No. 97.**

*Covariants of A, in the af- or standard forms; W is not given.*

[The Table presents the capital covariants  $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V$  each expressed as a polynomial in  $x, y$  whose coefficients are listed in vertical columns of monomials in the lowercase letters  $a, b, c, d, e, f$ , with numerical multipliers and deg-order labels at the foot of each column. The structure is reproduced below.]

**Covariant A:** the original quintic. Coefficients of  $(x, y)^5$ :

$$A = (1 ; c + 10 ; f + 10, c^2 - 15 ; a^2 b + 5, cf - 2 ; a^3 e + 1 \mid x, y)^5$$

with deg-orders 0.0, 1.3, 2.6, 3.9, 4.12, 5.15.

**Covariant B**  $= a^2$  of order 2 in  $(x, y)$ :

$$B = a^2 (b + 1 ; e + 1 ; ad - 3, bc - 1 \mid x, y)^2$$

with deg-orders 2.2, 3.5, 4.8.

**Covariant C:** order 6,

$$C = (c + 1 ; f + 1 ; a^2 b + 3, c^2 - 15 ; a^2 b + 1, cf - 10 ; a^2 d + 6, e^2 + 15 ; \\ af^2 \cdot 1, a^2 ce + 3, a^2 cf + 1, a^2 cf + 1 ; a^3 b^2 - 1, a^2 cd + 2, a^2 b^2 c + 4, a^2 ef + 1, a^3 f^2 - 1 \mid x, y)^6$$

deg-orders 2.6, 3.9, 4.12, 5.15, 6.18, 7.21, 8.24.

[Vol. X, printed page 363; PDF page 383. Table No. 97 (continued).]

**Covariant D**  $= a \cdot$  order 4:

$$D = a (ad + 1 ; bf - 1, ce + 1 ; a^2 b^2 - 1, acd + 3, a^2 bc^2 + 4, ef + 1 ; \dots \mid x, y)^4$$

deg-orders 4.8, 5.11, 6.14, 7.17.

**Covariant**  $E = a^2 \cdot$  order 5:

$$E = a^2(e + 1 ; ad \cdot 6, a^2bc \cdot 10 ; bf - 12, ce + 2 ; a^2b^2 \cdot 8, acd \cdot 36, a^2bc^2 + 12, a^2ef \cdot 2 ; \dots | x, y)^5$$

further columns:  $a^2be \cdot 5$ ,  $adf - 1$ ,  $a^2bcf \cdot 4$ ,  $a^2c^2e + 2$ ;  $a^3bd \cdot 6$ ,  $a^2b^2c^2 \cdot 2$ ,  $a^2e^2 \cdot 1$ ,  $ac^3d + 18$ ,  $a^2b^2c + 6$ ,  $a^2cef + 2$ .  
deg-orders 3.5, 4.8, 5.11, 6.14, 7.17, 8.20.

**Covariant**  $F$ : order 4 (very long — shown across columns):

$$F = (f + 1 ; a^2b + 2, a^3c - 18 ; a^2e + 1, a^3c - 36, ac^2 + 168 ; a^2d + 34, a^2bc \cdot 42, a^3cf - 126 ; \dots | x, y)^4$$

further columns:  $a^2bf \cdot 40$ ,  $a^2ce + 5$ ,  $a^2bc^2 + 134$ ,  $a^2ef \cdot 5$ ,  $a^2c^3 \cdot 252$ ;  $a^4b^2 \cdot 16$ ,  $a^2cd + 6$ ,  $a^2bcf + 8$ ,  $a^2c^2e + 65$ ,  $a^2c^2f \cdot 84$ ;  $\dots$ ;  $a^2b^2d \cdot 12$ ,  $a^2bcd + 6$ ,  $\dots$ ,  $a^2c^4 \cdot 2$ .  
Deg-orders 3.9, 4.12, 5.15, 6.18, 7.21, 8.24, 9.27, 10.30, 11.33, 12.36.

**Covariant**  $G = a^2g$  (invariant): the single coefficient

$$G = a^2(12abd + 4b^2c + e^2) \quad (\text{deg-order } 6.10).$$

[Vol. X, printed page 364; PDF page 384. Table No. 97 (continued).]

**Covariant**  $H = a^4h$ : order 4:

$$\begin{aligned} H = a^4 & (acd + 3, a^2bc^2 + 4, a^2ef + 1; a^2be + 2, adf + 12, a^2bcf \cdot 8, a^2c^2e \cdot 8; \\ & a^2bc + 4, a^2e^2 + 1, ac^2d \cdot 36, a^2bc^2 \cdot 24, a^2ef \cdot 6; \\ & a^2de + 2, a^2bcf + 4, a^2bce \cdot 6, ac^2d \cdot 12, a^2bcf \cdot 8, a^2c^3e + 8; \\ & a^2b^2 + 2, a^2c^2 + 6, a^2b^2d \cdot 2, a^2bcd \cdot 8, a^2bef \cdot 3, a^2ce^2 + 1, ac^3d \cdot 6, a^2bd^2 + 4, a^2c^2ef + 1 \\ & = a^4h \quad | x, y)^4 \end{aligned}$$

deg-orders 6.14, 7.17, 8.20, 9.23, 10.26.

**Covariant**  $I = a^2i$ : order 5:

$$\begin{aligned} I = a^2 & (bf \cdot 1, ce + 1; a^2b^2 \cdot 2, acd \cdot 6, a^2b^2c + 8, a^2ef \cdot 2; adf \cdot 15, a^2bef + 5, a^2c^2e \cdot 5; \\ & a^2bd \cdot 20, ac^2d + 8, a^2bce \cdot 5, acdf + 30, a^2bcf + 5, a^2c^3e \cdot 5; \\ & a^2c^2e \cdot 5; a^2b^2 + 2, a^2d^2 \cdot 12, a^2b^2cd \cdot 2, a^2b^2c^2 \cdot 6, a^2ce^2 \cdot 2, ac^3d \cdot 30, a^2bc^4 \cdot 8, a^2c^3ef \cdot 2; \\ & a^4b^4 + 1, a^2b^2cf + 3, a^2cde \cdot 5, a^2b^2cf + 1, a^2bc^2e \cdot 5, a^2c^2df \cdot 8, a^2bc^4 \cdot 1, a^2c^2e + 1 \quad | x, y)^5 \end{aligned}$$

$= a^2i$ , deg-orders 5.11, 6.14, 7.17, 8.20, 9.23, 10.26, 11.29.

**Covariant**  $J = a^4j$ : order 2:

$$\begin{aligned} J = a^4 & (a^2b^2 + 1, a^2d^2 + 9, a^2b^2c^2 \cdot 4, a^2b^2f \cdot 2, a^2ce^2 + 1; \\ & a^2b^2c \cdot 1, abdf \cdot 6, a^2cde + 6, a^2b^2cf \cdot 4, a^2bc^2e + 8, a^2ef \cdot 1 \quad | x, y)^2 = a^4j \end{aligned}$$

deg-orders 8.16, 9.19.

[Vol. X, printed page 365; PDF page 385. Table No. 97 (continued).]

**Covariant**  $K = a^4k$ : order 4:

$$K = a^4(ade + 3, a^2bcf + 2, a^2bce \cdot 2; a^2b^2 + 4, a^2d^2 \cdot 18, abcd \cdot 18, a^2b^2c^2 \cdot 16, a^2bef \cdot 5, a^2ce^2 + 1; \\ a^2bh + 1, abdf + 6, a^2b^2c^2 \cdot 2, a^2bc^2e \cdot 2, a^2ef \cdot 1; \\ a^2b^2d + 6, a^2b^2c + 2, a^2cd^2 \cdot 18, a^2b^2c^2 \cdot 30, a^2def \cdot 9, a^2b^2c^2 \cdot 8, a^2bcef \cdot 5, a^2c^4 + 3 \quad | x, y)^4 \\ = a^4k$$

deg-orders 7.13, 8.16, 9.19, 10.22.

**Covariant**  $L = a^2l$ : order 4 (deg-orders up to 14.38):

$$L = a^2(adf \cdot 3; a^2bd \cdot 3, a^2b^2c \cdot 7, ac^2d + 42, a^2bc^3 + 28, a^2cef + 7; \dots | x, y)^4$$

deg-orders 7.17, 8.20, 9.23, 10.26, 11.29, 12.32, 13.35, 14.38.

[The remaining covariants  $M, N, O, P, Q, R, S, T, U, V$  in this Table are reproduced in compact symbolic form in Table No. 98 below (divided and (mostly) segregate form), with their full numerical coefficient schedules; only the leading-coefficient identities are retained here.]

**Covariant**  $M = a^4m$ : order 4, deg-orders 10.22, 11.25, 12.28.

**Covariant**  $N = a^2n$ : order 4, deg-orders 9.19, 10.22, 11.25, 12.28, 13.31.

**Covariant**  $O = a^4o$ : order 4, deg-orders 11.21, 12.24.

**Covariant**  $P = a^2p$ : order 4, deg-orders 11.25, 12.28, 13.31, 14.34, 15.37, 16.40.

**Covariant**  $Q = a^6q$  (invariant), deg-order 14.30.

**Covariant**  $R = a^6r$ : order 2, deg-orders 13.27, 14.30, 15.33.

**Covariant**  $S = a^4s$ : order 2, deg-order 15.33.

**Covariant**  $T = a^8t$ : order 2, deg-orders 19.41, 20.44, 21.45.

**Covariant**  $U = a^{10}u$  (invariant), deg-order 12.0 (in this normalisation).

**Covariant**  $V = a^{12}v$ : deg-orders 22.46, 23.49.

[The intermediate detail (precise integer multipliers of each monomial term in every coefficient of every covariant) occupies six tightly-set printed pages of Table No. 97 in the original; the principal use of these schedules is verification of syzygies, and they are not used in the running argument that follows. The scan-image of the table is preserved alongside this transcription as a separate IA-linked artefact.]

[Vol. X, printed page 366; PDF page 386. Table No. 97 (continued).]

[Schedule of covariant  $N$  (order 4), deg-orders 9.19, 10.22, 11.25, 12.28, 13.31, with leading monomials  $a^2b^4$ ,  $a^2b^2e$ ,  $a^2bde$ ,  $a^2b^4$ ,  $a^2b^4e$  respectively; full coefficient lists in scan.]

[Schedule of covariant  $O$  (order 4,  $= a^4o$ ), deg-orders 11.21, 12.24, with leading monomials  $a^2b^2e + 3$  and  $a^2b^2d \cdot 6$ ; full coefficient lists in scan.]

[Schedule of covariant  $P$  (order 4): leading columns  $P = a^2(a^2bde \cdot 2, a^2d^2 \cdot 1; a^2b^4 \cdot \dots; a^2b^2e \cdot 5, a^2d^2e \cdot 9; a^2b^4d \cdot 8, a^2d^3 \cdot 72; a^2b^2de + 6, a^2b^4f + 5, \dots; a^2b^2 + 2, a^2b^2d^2 + 12, \dots \mid x, y)^4$ , deg-orders 11.25, 12.28, 13.31, 14.34, 15.37, 16.40.]

[Vol. X, printed page 367; PDF page 387. Table No. 97 (continued).]

**Covariant**  $Q = a^6q$  (invariant), deg-order 14.30:

$$Q = a^6(a^2b^3 \cdot 2, a^2b^2d^2 + 18, a^2b^2cd + 22, a^2bde^2 + 3, a^2cd^2 \cdot 54, a^2b^2c^2 + 12, \dots, a^2cef^2 + 1).$$

**Covariant**  $R = a^6r$  (order 2), deg-orders 13.27, 14.30, 15.33:

$$R = a^6(r_0; r_1; r_2 \mid x, y)^2$$

with  $r_0, r_1, r_2$  given as in Table 98 below (divided segregate form).

**Covariant**  $S = a^4s$  (order 2), deg-orders 15.33, 16.36, 17.39, 18.42:

$$S = a^4(s_0; s_1; s_2; s_3 \mid x, y)^2.$$



[Vol. X, printed page 368; PDF page 388. Table No. 97 (continued).]

**Covariant**  $T = a^8 t$  (order 2), deg-orders 19.41, 20.44, 21.45:

$$T = a^8 (t_0; t_1; t_2 \mid x, y)^2, \quad = a^8 t.$$

**Covariant**  $U = a^{10} u$  (invariant): the single coefficient is the polynomial in  $a, b, c, d, e, f$  of deg-order 21.45; for the simplified  $b = 0$  value see Art. 381 above.

[Vol. X, printed page 369; PDF page 389. Table No. 97 (concluded).]

**Covariant**  $V = a^{12}v$  (deg-orders 22.46, 23.49):

$$V = a^{12}(v_0; v_1 \mid x, y) = a^{12}v.$$

The coefficient  $v_1$  contains the monomial schedule listed on the right-hand of the original page; only the principal columns are reproduced in compact form in Table 98.

**Table No. 98.**

*Covariants of A, divided and (except as to a few coefficients) segregate.*

[Vol. X, printed page 370; PDF page 390.]

$A$  and  $B$  as given in Table 97 were divided and segregate.

$C$  was divided but not segregate: the divided and segregate form is

$$C = (c + 1; f + 3, a^2b + 3, c^2 \cdot 15; a^2b + 1, cf \cdot 10, c^3 + 15; a^2d + 6, a^2bc \cdot 3; \\ a^2i + 3, a^2cf + 3; a^4b^2 \cdot 1, a^2h + 1, a^2cd \cdot 4, a^2c^3 \cdot 1 \mid x, y)^6$$

deg-orders 2.6, 3.9, 4.12, 5.15, 6.18, 7.21, 8.24.

$D$  divided and segregate is

$$D = a(d + 1; i + 1; ab^2 \cdot 1, ah + 1, a^2cd \cdot 3; al \cdot 1, act \cdot 1 \mid x, y)^4$$

deg-orders 3.3, 4.6, 5.9, 6.12. An integer non-segregate form of the fractional coefficient is

$$\frac{ci \cdot 1}{df + 1}$$

$E$  was divided but not segregate: the divided and segregate form is

$$E = (e + 1; ad \cdot 6, a^2bc \cdot 10; ai + 12, a^2ce \cdot 10; a^2b^2 \cdot 8, ah \cdot 2, acd \cdot 24, a^2bc^2 + 20; \dots \mid x, y)^5$$

further columns:  $a^2be \cdot 5, a^2b^2 \cdot 1, ac^2 \cdot 12, a^2bc + 5; a^2c^2 \cdot 1, a^2bd + 6, a^2b^2c + 2, a^2cd^2 \cdot 2, a^2c^3d + 6, a^2bc^2 \cdot 2.$

deg-orders 3.6, 4.8, 5.11, 6.14, 7.17, 8.20.

[Vol. X, printed page 371; PDF page 391. Table No. 98 (continued).]

$F$  was divided but not segregate: the divided and segregate form is

$$F = (f + 1 ; a^2b + 2, a^2c \cdot 18 ; ah + 1, a^2c \cdot 36 ; a^2d + 34, a^2bc \cdot 42, a^2cf + 168 ; \dots | x, y)^4$$

remaining columns:  $a^2l + 40, a^2ce \cdot 35 ; a^4b^2 \cdot 16, a^2h \cdot 5, a^2cf \cdot 16 ; a^2be \cdot 21, a^2l \cdot 8, a^2cf \cdot 252, a^2ce + 189 ; \dots ; a^4b^2 \cdot 2, a^4bh \cdot 3, a^4cf \cdot 16, a^4cn \cdot 5, a^4cd + 16, a^4bc^2 + 4, a^4c^4 \cdot 2.$

Deg-orders 3.9, 4.12, 5.15, 6.18, 7.21, 8.24, 9.27, 10.30, 11.33, 12.36.

where for an integer non-segregate value of the fractional coefficient, see the original form of  $F$ .

$G$  as an invariant was divided and segregate,  $G = a^2g$ . (Deg-order 4.0.)

$H$  divided and segregate is

$$H = a^4(h + 1 ; be + 2, i \cdot 4 ; a^2g + 1, abd \cdot 12, a^2ch \cdot 6 ; a^2fk \cdot 2, abi \cdot 8, a^2bce \cdot 6, a^2cj + 12 ; a^2j + 2, a^2h + 4, a^2bh \cdot 5, a^2cp + 1, abcd + 12, a^2c^2h + 3 | x, y)^4$$

deg-orders 4.4, 5.7, 6.10, 7.13, 8.16. Where the fractional coefficients are

$$\begin{array}{l|l} abe + 2 & a^2f^2 + 2 \\ a^2bf + 4 & a^2d^2 + 6 \\ a^2bce \cdot 6 & abcd \cdot 2 \\ a^2cl + 4 & a^2b^2c \cdot 8 \\ & a^2bef \cdot 3 \\ & a^2c^2h + 1 \\ & a^2ce^2 + 1 \end{array}$$

[Vol. X, printed page 372; PDF page 392. Table No. 98 (continued).]

$I$  divided and segregate is

$$I = a^2(i + 1 ; ab^2 \cdot 2, ah + 2, a^2cd \cdot 18 ; al + 5, a^2cl \cdot 15 ; a^2bd \cdot 20, a^2c^2d + 60 ; \dots | x, y)^5$$

remaining columns:  $a^2b^2k \cdot 5, a^2bi \cdot 25, aci \cdot 30, a^2d^2 + 45 ; a^2ck \cdot 4, a^2b^2 + 10, a^2bh \cdot 8, a^2cg + 4, a^2bcd \cdot 6, a^2bc^2 \cdot 18, a^2c^2f \cdot 6, a^2c^2d \cdot 54 ; \dots ; a^2c^2t \cdot 3.$

deg-orders 4.6, 5.9, 6.12, 7.15, 8.18, 9.21, 10.24. Where the fractional coefficients are

$$\begin{array}{l|l|l} a^2de \cdot 5 & a^2b^2 + 2 & a^2b^2e + 1 \\ abf + 5 & a^2d^2 \cdot 12 & a^2bdf + 3 \\ a^2bce \cdot 5 & a^2bcd \cdot 2 & a^2cde \cdot 5 \\ a^2c^2i \cdot 5 & ab^2c \cdot 6 & ab^2c + 1 \\ a^2cdf + 30 & a^2c^2h \cdot 2 & a^2bc^2e \cdot 5 \\ & a^2ce^2 \cdot 2 & a^2cf \cdot 3 \\ & a^2c^2i \cdot 18 & a^2c^2i + 1 \\ & & a^2cdf \cdot 3 \end{array}$$

$J$  divided and segregate is

$$J = a^4(j, = a^4(x, y)^0).$$

(Deg-order 5.6 and 6.4.)

$K$  divided and segregate is

$$K = a^4(k + 1 ; aj \cdot 2, a^2b^2 + 6, a^2bh \cdot 9, a^2cg + 3 ; an + 1, a^2ck \cdot 3 ; a^2m \cdot 3, acj + 1, a^2b^2c \cdot 2, a^2bch + 3 | x, y)^4$$

deg-orders 5.3, 6.6, 7.9, 8.12.

[Vol. X, printed page 373; PDF page 393. Table No. 98 (continued).]

$L$  divided and (as to first six coefficients) segregate is

$$\frac{1}{3}L = a^2(l + 1 ; abd \cdot a^2k \cdot 4 ; a^2j \cdot 13, a^2c \cdot 3, a^4m \cdot 1, a^2bde \cdot 7, a^4b^2 \cdot 2 ; \dots | x, y)^4$$

further columns:  $abi + 1$ ,  $am$ ,  $a^4bf$ ,  $a^4b^2c \cdot 7$ ,  $a^4b^2d + 13$ ,  $a^4bd^3 + 3$ ;  $a^2ch + 7$ ,  $a^4cl \cdot 21$ ,  $a^4bh \cdot 5$ ,  $a^4ck \cdot 20$ ,  $a^4cj + 3$ ,  $a^4b^2c^2 + 14$ ,  $a^4b^2cd + 10$ ,  $a^4de^2 + 2$ ; . . .

deg-orders 5.7, 6.10, 7.13, 8.16, 9.19, 10.22, 11.25, 12.28. The last two coefficients have not been reduced to the segregate form.

$M$  divided and segregate is

$$M = a^4(m + 1 ; bk \cdot 1, l^2 \cdot 1 ; abj \cdot 11, adg + 1, a^2cm \cdot 1 | x, y)^4$$

deg-orders 6.2, 7.5, 8.8.

[Vol. X, printed page 374; PDF page 394. Table No. 98 (continued).]

$N$  divided and segregate is

$$N = a^4(n+1; cj+4; ap \cdot 6, cn \cdot 6; a^2bj \cdot 4, a^2dg+4, ab^4+8, a^2b^2h \cdot 12, a^2eg+4, a^2dm \cdot 8, a^2c^2j \cdot 4; a^4o+1, a^4bn \cdot 1, a^4cp+2, a^4c^2j+1 \mid x, y)^4$$

deg-orders 6.4, 7.7, 8.10, 9.13, 10.16.

$O$  divided and segregate is

$$O = a^4(o+1; b^2g+1, bm+6, c^2 \cdot 6, ch \cdot 1 \mid x, y)^4$$

deg-orders 7.1, 8.4.

$P$  divided and (as to first three coefficients) segregate is

$$P = a^2(p+1; abj+8, a^2dg \cdot 5, a^2bl \cdot 14, a^2b^2c+15, a^2bcg \cdot 5, a^2cm+10; \dots \mid x, y)^4$$

further columns:  $a^2o+7$ ,  $ab^4 \cdot 2$ ,  $o^2 \cdot 14$ ,  $a^2b^2ch \cdot 8$ ;  $a^2bm+8$ ,  $a^2b^2d+9$ ,  $a^4bcj+13$ ,  $a^4bdh \cdot 9$ ,  $a^4c^4 \cdot 81$ ,  $a^4b^2c \cdot 5$ ,  $a^4b^2ch \cdot 8$ , ...

and:  $a^2b^2k \cdot 3$ ,  $a^2bp \cdot 3$ ,  $a^2cj^2+2$ ,  $ab^2de+3$ ,  $a^2bl+3$ ,  $a^2bh+8$ ,  $a^4dh \cdot 27$ ,  $a^4cdh \cdot 3$ , ...;  $a^4b^2+; \dots$

deg-orders 7.8, 8.8, 9.11, 10.14, 11.17, 12.20. The last three coefficients have not been reduced to the segregate form.

[Vol. X, printed page 375; PDF page 395. Table No. 98 (continued).]

$Q$  as an invariant was divided and segregate,  $Q = a^6q$  (deg-order 8.0).

$R$  divided and segregate is

$$R = a^6(r + 1 ; aq + 1, a^2b^2j + 6, a^2bdg \cdot 5, a^2hj \cdot 1 ; abo \cdot 1, a^2gk \cdot 1, a^2s + 3, a^2cr \cdot 3 \mid x, y)^2$$

deg-orders 8.2, 9.5, 10.8. Where the fractional coefficients are

$$\begin{array}{l|l} b^2j + 2 & bdk + 3 \\ bdg \cdot 2 & bej + 1 \\ dm \cdot 6 & cr + 1 \\ & d^2j \cdot 1 \\ & dp + 3 \end{array}$$

$S$  divided and (as to the first three coefficients) segregate is

$$S = a^4(s + 1 ; agj \cdot 2, a^2b^2g + 2, a^2b^2m + 3, a^2bdj + 21, a^2b^2h \cdot 4, a^2cg^2 + 2, a^2cq \cdot 3 ; \dots \mid x, y)^2$$

further columns:  $abr \cdot 1, a^2do \cdot 1 ; a^2bq + 4, a^2b^2j + 4, a^2b^2k \cdot 4, a^2bdm \cdot 31, a^2d^2j \cdot 3, a^4b^2 + 4, a^4b^2cj + 16, a^4bd^2h \cdot 24, a^4dcn + 4, \dots$

deg-orders 9.3, 10.6, 11.9, 12.12. But the last coefficient is neither segregate nor integer.

[Vol. X, printed page 376; PDF page 396. Table No. 98 (concluded).]

$T$  divided and segregate is

$$T = a^8(t + 1 ; bgm + 1, k^2 + 4, dgj \cdot 3, kq \cdot 1 \mid x, y)^2$$

deg-orders 11.1, 12.4. Where the fractional coefficient is

$$\frac{b^2q \cdot 1}{hq + 1} \\ m^2 + 6$$

$U$  as an invariant was divided and segregate,  $U = a^{10}u$  (deg-order 12.0).

$V$  divided and segregate is

$$V = a^{12}(v + 1 ; bgr \cdot 5, bjo \cdot 10, gjk + 5, js \cdot 12, nq \cdot 9 \mid x, y)^6$$

deg-orders 13.1, 14.4. Where the fractional coefficient is

$$\frac{dt \cdot 6}{mr \cdot 6} \\ nq + 1$$

$W$  as an invariant was divided and segregate,  $W = a^{18}w$  (deg-order 18.0).



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*Derivatives. Art. Nos. 382 to 384, and Tables Nos. 99 and 100.*

**382.** I call to mind that any two covariants  $a$ ,  $b$ , the same or different, give rise to a set of derivatives  $(a, b)^1$ ,  $(a, b)^2$ ,  $(a, b)^3$ , &c., or, as I propose to write them,  $ab1$ ,  $ab2$ ,  $ab3$ , &c., viz.

$$\begin{aligned} ab1 &= d_x a \cdot d_y b - d_y a \cdot d_x b, \\ ab2 &= d_x^2 a \cdot d_y^2 b - 2 d_x d_y a \cdot d_x d_y b + d_y^2 a \cdot d_x^2 b, \\ ab3 &= d_x^3 a \cdot d_y^3 b - 3 d_x^2 d_y a \cdot d_x d_y^2 b + 3 d_x d_y^2 a \cdot d_x^2 d_y b - d_y^3 a \cdot d_x^3 b, \\ &\text{\&c.} \end{aligned}$$

or, as these are symbolically written,

$$ab1 = 12 a_1 b_2, \quad ab2 = 12^2 a_1 b_2, \quad ab3 = 12^3 a_1 b_2, \quad \text{\&c.};$$

where

$$12 = \frac{d}{d\xi_1} \cdot \frac{d}{d\eta_2} - \frac{d}{d\eta_1} \cdot \frac{d}{d\xi_2};$$

the differentiations  $\frac{d}{d\xi_1}$ ,  $\frac{d}{d\eta_1}$  applying to the  $a$ , and the  $\frac{d}{d\xi_2}$ ,  $\frac{d}{d\eta_2}$  applying to the  $b$ , but the suffixes being ultimately omitted: hence if  $\theta$  be the index of derivation, the derivative is thus a linear function of the differential coefficients of the order  $\theta$  of the two covariants  $a$  and  $b$  respectively: and we have the general property that any such derivative, if not identically vanishing, is a covariant. If the  $a$  and the  $b$  are one and the same covariant, then obviously every odd derivative is  $= 0$ ; so that in this case the only derivatives to be considered are the even derivatives  $aa2$ ,  $aa4$ , &c. Moreover, if the index of derivation  $\theta$  exceeds the order of either of the component covariants, then also the derivative is  $= 0$ : in particular, neither of the covariants must be an invariant. The degree of the derivative is evidently equal to the sum of the degrees of the component covariants; the order is equal to the sum of the orders less twice the index of derivation.

**383.** It was by means of the theory of derivatives that Gordan proved (for a binary quantic of any order) that the number of covariants was finite, and, in the particular case of the quintic, established the system of the 23 covariants. Starting from the quantic itself  $a$ , then the system of derivatives  $aa2$ ,  $aa4$ , &c., must include among itself all the covariants of the second degree, and if the entire system of these is, suppose,  $b$ ,  $c$ , &c., then the derivatives  $ab1$ ,  $ab2$ , &c.,  $ac1$ ,  $ac2$ , &c., must include among them all the covariants of the third degree, and so on for the higher degrees: and in this way, limiting by general reasoning the number of the independent covariants of each degree obtained by the successive steps, the foregoing conclusion is arrived at. But returning to the quintic, and supposing the system of the 23 covariants established, then knowing the deg-order of a derivative we know that it must be a linear function of the segregates of that deg-order; and we thus confirm, *à posteriori*, the results of the derivation theory. I annex the following Table No. 99, showing all the derivatives which present themselves, and for each of them the

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covariants as well congregate as segregate of the same deg-order: the congregates are distinguished each by two prefixed dots, *.bf*, &c. No further explanation of the arrangement is, I think, required. We see from the table in what manner the different covariants present themselves in connexion with the derivation-theory. Thus starting with the quintic itself *a*, we have the two derivatives *aa4*, *aa2*, which are in fact the covariants of the second degree (deg-orders 2.2 and 2.6 respectively) *b* and *c*. For the third degree we have the derivatives *ab2*, *ab1*, *ac5*, *ac4*, *ac3*, *ac2*, *ac1*: the deg-order of *ac5* is 3.1, and there being no covariants of this deg-order, *ac5* must, it is clear, vanish identically: *ab2* and *ac4* are each of them of the deg-order 3.3, but for this deg-order we have only the covariant *d*, and hence *ab2* and *ac4* must be each of them a numerical multiple of *d*; similarly, deg-order 3.5, *ab1* and *ac3* must be each of them a numerical multiple of *e*; deg-order 3.7, *ac2* must be a numerical multiple of *ab*; and deg-order 3.9, *ac1* must be a numerical multiple of *f*: the 7 derivatives, which *primâ facie* might give, each of them, a covariant of the third degree, thus give in fact only the 3 covariants *d*, *e*, *f*; and in order to show according to the theory of derivations that this is so, it is necessary to prove 1°, that *ac5* = 0; 2°, that *ac4* and *ab2* differ only by a numerical factor; 3°, that *ab1* and *ac3* differ only by a numerical factor; 4°, that *ac1* is a numerical multiple of *ab*: which being so, we have the 3 new covariants. The table shows that

for degrees    2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24  
No. of derivatives = 2, 7, 19, 29, 41, 46, 52, 46, 44, 35, 26, 19, 17, 12, 13, 6, 6, 3, 3, 1, 1, 0, 1

so that the whole number of derivatives is 429, giving the 22 covariants *b, c, . . . , w*. While it is very remarkable that (by general reasoning, as already mentioned, and with a very small amount of calculation) Gordan should have been able in effect to show this, the great excess of the number of derivatives over that of the covariants seems a reason why the derivations ought not to be made a basis of the theory.

It is to be remarked that we may consider derivatives *pq1*, *pq2*, &c., where *p, q* instead of being simple covariants are powers or products of covariants, but that these may be made to depend upon the derivatives formed with the simple covariants. (As to this see my paper “On the Derivatives of Three Binary Quantics,” *Quart. Math. Journal*, t. xv. (1877), pp. 157–168, [681].)

Table No. 99 (Index Table of Derivatives).

Deg. 2    (2 derivs.)

| Ord.      | 2 | 4        | 6        |
|-----------|---|----------|----------|
| <i>aa</i> |   | <i>b</i> | <i>c</i> |

Deg. 3    (7 derivs.)

| Ord.      | 1 | 3              | 5              | 7 | 9 |
|-----------|---|----------------|----------------|---|---|
| <i>ab</i> |   | <i>d</i> (= 4) | <i>e</i> (= 2) |   |   |
| <i>ac</i> | 5 | 4              | 3              | 2 | 1 |

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**Table No. 99** (continued).

Deg. 4 (19 derivs.) — covariants present:  $b^2, i, ad, ae, a^2b, h, bc, cd, ce, \dots$

| Ord. | 2 | 4 | 6 | 8 | 10 | 12 |
|------|---|---|---|---|----|----|
| $ad$ | 3 | 2 | 1 |   |    |    |
| $ae$ | 5 | 4 | 3 | 2 | 1  |    |
| $af$ |   | 5 | 4 | 3 | 2  | 1  |
| $bb$ | 2 |   |   |   |    |    |
| $bc$ |   | 2 | 1 |   |    |    |
| $ce$ | 6 | 4 |   | 2 |    |    |

Deg. 5 (29 derivs.) — covariants:  $j, k, ag, be, ab^2, ai, a^2d, bd, l, ah, ..bf, abc, cd, ce$

| Ord.  | 1 | 3 | 5 | 7 | 9 | 11 | 13 |
|-------|---|---|---|---|---|----|----|
| $ah$  | 4 | 3 | 2 | 1 |   |    |    |
| $ai$  | 5 | 4 | 3 | 2 | 1 |    |    |
| $bd$  | 2 | 1 |   |   |   |    |    |
| $be$  |   | 2 | 1 |   |   |    |    |
| $c-e$ |   | 2 | 1 |   |   |    |    |
| $cd$  | 3 | 2 | 1 |   |   |    |    |
| $ce$  | 4 | 3 | 2 | 1 |   |    |    |
| $cf$  | 6 | 5 | 4 | 3 | 2 | 1  |    |

Deg. 6 (41 derivs.) — covariants include  $bg, n, aj, a^2f, ak, av, abe, a^2b^2, a^2h, ci, acd, bh, ab^2c, bc^2, ag, ..df, ..d^2, ch, ..e^2, ..ef$

| Ord. | 2 | 4 | 6 | 8 | 10 | 12 | 14 |
|------|---|---|---|---|----|----|----|
| $aj$ |   |   |   | 1 |    |    |    |
| $ak$ |   | 3 | 2 | 1 |    |    |    |
| $al$ | 3 | 4 | 3 | 2 | 1  |    |    |
| $bh$ | 2 | 1 |   |   |    |    |    |
| $bi$ |   | 2 | 1 |   |    |    |    |
| $ch$ | 4 | 3 | 2 | 1 |    |    |    |
| $ci$ | 6 | 5 | 4 | 3 | 2  | 1  |    |
| $dd$ | 2 |   |   |   |    |    |    |
| $de$ | 3 | 2 | 1 |   |    |    |    |
| $df$ |   |   | 3 | 2 | 1  |    |    |
| $ee$ | 4 |   | 2 |   |    |    |    |
| $ef$ |   | 5 | 4 | 3 | 2  | 1  |    |
| $ff$ | 8 |   | 6 |   | 4  |    | 2  |

[Vol. X, printed page 380; PDF page 400. Table No. 99 (continued).]

Deg. 7 (46 derivs.) — covariants present at heads of columns:  $hj, hk, abg, an, a^2j, a^2k, dg, eg, am, ..b^2e, ab^3, abi, p, bM, bl, ..cde, ..ab^3, abh, ck, acg, ..ad^2, bcd, bce, ..dh, ..di, ..eh, cl, ..ei, ..fh, fg, ..by$  etc.

| Ord. | 1 | 3 | 5 | 7 | 9 | 11 | 13 |
|------|---|---|---|---|---|----|----|
| $am$ |   | 2 | 1 |   |   |    |    |
| $an$ | 4 | 3 | 2 | 1 |   |    |    |
| $bj$ | 1 |   |   |   |   |    |    |
| $bk$ | 2 | 1 |   |   |   |    |    |
| $bl$ |   |   | 2 | 1 |   |    |    |
| $cj$ |   |   | 1 |   |   |    |    |
| $ck$ |   | 3 | 2 | 1 |   |    |    |
| $cl$ | 6 | 5 | 4 | 3 | 2 | 1  |    |
| $dh$ | 3 | 2 | 1 |   |   |    |    |
| $di$ |   | 3 | 2 | 1 |   |    |    |
| $eh$ | 4 | 3 | 2 | 1 |   |    |    |
| $ei$ | 5 | 4 | 3 | 2 | 1 |    |    |
| $fh$ |   |   | 4 | 3 | 2 | 1  |    |
| $fi$ |   | 6 | 5 | 4 | 3 | 2  | 1  |

[The remainder of Table No. 99 (covering degrees 8 through 24 of derivatives, with the same column-by-deg-order layout) continues across the following printed pages and forms a single large index; it is not reproduced here. Total: 429 derivatives, yielding the system of 22 covariants  $b, c, \dots, w$  of the binary quintic.]

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End of pages 376–400 (PDF) of Volume X.  
Paper No. 693 (Tenth Memoir on Quantics) continues on PDF page 401.

# A Tenth Memoir on Quantics

(Continuation and conclusion, Paper No. 693)

*Philosophical Transactions of the Royal Society of London*, **169** (1878), pp. 603–662

[Vol. X, printed page 381; PDF page 401. Table No. 99 (continued) — Index Table of Derivatives of the binary quintic; degrees 8 through 24.]

[For each degree, the upper part of the printed table lists the covariant-symbols of the indicated deg-order, congregate prefixes marked by two dots; the lower part shows the deg-order distribution of the derivatives  $pqn$  where  $p, q$  run over the simple covariants of the quintic and  $n$  is the order-index. Owing to the very large number of entries (between 13 and 52 derivatives per degree) and the dense two-column arrangement of the original printed page, we reproduce here, for each remaining degree of Table No. 99, the structural skeleton (column heads in deg-order, row labels in derivative-symbol) faithfully transcribed; the symbolic congregate-content lists at column heads, which the printed page sets in small condensed type, are summarised in prose. The numerical entries in the body of each skeleton are reproduced verbatim from the scan.]

**Deg. 8** (52 derivs.) — column heads in deg-order: 2, 4, 6, 8, 10, 12, 14.

Covariant-symbols present at column 8:  $g^2, r, bk, bn, dj, ..dk, gh..ej, g^2$ ; at column 6:  $b^2g, bm, bn, adg, ..dk, g^2, beg, b^2i, bce, ..bde, cm, ..bd^2, ..ek$ ; at column 10:  $a^2j, abj, aeg, ..ap, ..b^2, ..bde, ..cl, en, fj, ..hi$ ; at columns 12 and 14: higher congregates  $a^2bh, ahn, ..ab^2e, ..adi, b^3c, acj, b^2ci, ..cde, a^2n, a^2ij, ..adh, bci, ..bd^2j, ..aeh, afg, beh, ..be^2, ..ed^2, ..d^3, ..fk$ , etc.

| Ord.      | 0 | 2 | 4 | 6 | 8 | 10 | 12 |
|-----------|---|---|---|---|---|----|----|
| <i>ao</i> |   | 1 |   |   |   |    |    |
| <i>ap</i> | 5 | 4 | 3 | 2 | 1 |    |    |
| <i>bm</i> | 2 | 1 |   |   |   |    |    |
| <i>bn</i> |   | 2 | 1 |   |   |    |    |
| <i>cm</i> |   |   | 2 | 1 |   |    |    |
| <i>cn</i> |   | 4 | 3 | 2 | 1 |    |    |
| <i>dj</i> |   | 1 |   |   |   |    |    |
| <i>dk</i> | 3 | 2 | 1 |   |   |    |    |
| <i>dl</i> |   |   | 3 | 2 | 1 |    |    |
| <i>ej</i> |   |   | 1 |   |   |    |    |
| <i>ek</i> |   | 3 | 2 | 1 |   |    |    |
| <i>el</i> |   | 6 | 4 | 3 | 2 | 1  |    |
| <i>fj</i> |   |   |   |   | 1 |    |    |
| <i>fk</i> |   |   |   | 3 | 2 | 1  |    |
| <i>fl</i> |   | 7 | 6 | 6 | 4 | 3  | 2  |
| <i>hh</i> | 4 |   | 2 |   |   |    |    |
| <i>hi</i> |   | 4 | 3 | 2 | 1 |    |    |
| <i>ii</i> | 6 |   | 4 |   | 2 |    |    |

**Deg. 9** (46 derivs.) — column heads 1, 3, 5, 7, 9, 11.

Covariants present:  $bo, c^2, ar, ab^2g, a^2o, bp, gk, aq, ..b^2h, ahn, \cdot, by, bce, adj, ..adk, bdg, bp, agh, ..aej, agi, dm.., co, b^2h, ..b^2e, b^2i, hj, dn.., bcj, em.., ..bdh, gi, cdg, bek, ..hk, ..cf, ..bdi, ..ij, ..en, ..beh, ..ik$ ,

$hfg, ceg, cp, ..d^2e, ..fin, ..hi.$

| Ord.      | 1 | 3 | 5 | 7 | 9 | 11 |
|-----------|---|---|---|---|---|----|
| <i>ar</i> |   | 2 | 1 |   |   |    |
| <i>bo</i> | 1 |   |   |   |   |    |
| <i>bp</i> |   | 2 | 1 |   |   |    |
| <i>co</i> |   |   | 1 |   |   |    |
| <i>cp</i> | 5 | 4 | 3 | 2 | 1 |    |
| <i>dm</i> | 2 | 1 |   |   |   |    |
| <i>dn</i> | 3 | 2 | 1 |   |   |    |
| <i>em</i> |   | 2 | 1 |   |   |    |
| <i>en</i> | 4 | 3 | 2 | 1 |   |    |
| <i>fm</i> |   |   |   | 2 | 1 |    |
| <i>fn</i> |   |   | 4 | 3 | 2 | 1  |
| <i>hj</i> |   | 1 |   |   |   |    |
| <i>hk</i> | 3 | 2 | 1 |   |   |    |
| <i>hl</i> |   | 4 | 3 | 2 | 1 |    |
| <i>ij</i> |   |   | 1 |   |   |    |
| <i>ik</i> |   | 3 | 2 | 1 |   |    |
| <i>il</i> | 6 | 5 | 4 | 3 | 2 | 1  |

[Vol. X, printed page 383; PDF page 403. Table No. 99 (continued).]

**Deg. 10** (44 derivs.) — column heads 0, 2, 4, 6, 8, 10, 12.

Covariants present:  $bq, br, b^2, abo, av, a^2t, ..do, c^2, as, a^2q, ..ab^2k, abeg, gm, gn, b^2g, bhn, b^2hi, ab^2j, abp, bdj, aco, f^2, jk, bgh, ..bdk, abd, ..adn, cf, ..bej, ..adm, ..aem, bgi, eq, cr, ahj, agl, ..d^2g, ..b^3, ..eo, ..deg, ..ahk, dp.., eh, hm.., ..aij, ..hn, b^2g, im.., ..b^2p, b^3c, bcm, ..b^2de, ..jl, ..bek, ben, ..bh^2, ..bd, cdj, .bfj, cgh, ..bhi, ..d^2h, ..cdk, ..ep, .cej, ..fo, cgi, in.., ..d^2l, ..kl, ..deh, ..dfg.$

| Ord.      | 0 | 2 | 4 | 6 | 8 | 10 |
|-----------|---|---|---|---|---|----|
| <i>as</i> | 3 | 2 | 1 |   |   |    |
| <i>br</i> | 2 | 1 |   |   |   |    |
| <i>cr</i> |   | 2 | 1 |   |   |    |
| <i>do</i> |   | 1 |   |   |   |    |
| <i>dp</i> | 3 | 2 | 1 |   |   |    |
| <i>eo</i> |   | 1 |   |   |   |    |
| <i>ep</i> | 5 | 4 | 3 | 2 | 1 |    |
| <i>fo</i> |   |   |   | 1 |   |    |
| <i>fp</i> |   | 6 | 4 | 3 | 2 | 1  |
| <i>hm</i> | 2 | 1 |   |   |   |    |
| <i>hn</i> | 4 | 3 | 2 | 1 |   |    |
| <i>im</i> |   | 2 | 1 |   |   |    |
| <i>in</i> |   | 4 | 3 | 2 | 1 |    |
| <i>jk</i> | 1 |   |   |   |   |    |
| <i>jl</i> |   |   | 1 |   |   |    |
| <i>kk</i> | 3 |   |   |   |   |    |
| <i>kl</i> |   | 3 | 2 | 1 |   |    |
| <i>ll</i> | 6 |   | 4 |   | 2 |    |

**Deg. 11** (35 derivs.) — column heads 1, 3, 5, 7, 9.

Covariants present:  $go, bgj, bs, b^2o, ahg^2, c^2, bgk, abq, ..ado, t, dg^2, dq, agm, agn, jm, ..dr, bs, ajk, ..b^2k, ..b^2eg, ef, b^2j, b^2dg, h^2p, gp, ..bdm, beo, ..ho, bhj, ..bdn, ..jn, cgj, ..bem, km.., d^2j, bgl, ..dgh, ..bhk, er.., ..bij, io.., cjk, c^2s, ..kn, ..d^2k, .dej, ..dgi, ..egh, ff, fq, hp.., lm..$

| Ord.      | 1 | 3 | 5 | 7 | 9 |
|-----------|---|---|---|---|---|
| <i>bs</i> | 2 | 1 |   |   |   |
| <i>cr</i> |   | 3 | 2 | 1 |   |
| <i>dr</i> | 2 | 1 |   |   |   |
| <i>er</i> |   | 2 | 1 |   |   |
| <i>fr</i> |   |   |   | 2 | 1 |
| <i>ho</i> |   | 1 |   |   |   |
| <i>hp</i> | 4 | 3 | 2 | 1 |   |
| <i>io</i> |   |   | 1 |   |   |
| <i>ip</i> | 5 | 4 | 3 | 2 | 1 |
| <i>jm</i> | 1 |   |   |   |   |
| <i>jn</i> |   | 1 |   |   |   |
| <i>km</i> | 2 | 1 |   |   |   |
| <i>kn</i> | 3 | 2 | 1 |   |   |
| <i>lm</i> | 2 | 1 |   |   |   |
| <i>ln</i> |   | 4 | 3 | 2 | 1 |

[Vol. X, printed page 385; PDF page 405. Table No. 99 (continued).]

**Deg. 12** (26 derivs.) — column heads 2, 4, 6, 8, 10, 12.

Covariants present:  $g^2r$ ,  $gr$ ,  $by$ ,  $ago$ ,  $at$ ,  $b^2r$ ,  $abgj$ ,  $ab^2o$ ,  $..bdo$ ,  $..adg^2$ ,  $jo$ ,  $cq$ ,  $bgn$ ,  $adq$ ,  $abgk$ ,  $bgm$ ,  $bjk$ ,  $abs$ ,  $ajm$ ,  $..adr$ ,  $hf$ ,  $..dgk$ ,  $aeq^2$ ,  $..ds$ ,  $b^2g$ ,  $aeq$ ,  $dgj$ ,  $egj$ ,  $cm$ ,  $agp$ ,  $fh$ ,  $b^2dj$ ,  $..hr$ ,  $b^2gh$ ,  $..aho$ ,  $hu$ ,  $..jp$ ,  $beg^2$ ,  $..ajn$ ,  $..ko$ ,  $mn..$ ,  $bcq$ ,  $..akm$ ,  $..bo^2$ ,  $..bdj^2g$ ,  $..beo$ ,  $b^2n$ ,  $..bhm$ ,  $..b^2r$ ,  $cgm$ ,  $..b^2ik$ ,  $..d^3m$ ,  $..b^2ej$ ,  $..dhj$ ,  $..b^2gi$ ,  $..egk$ ,  $bcr$ ,  $..es$ ,  $..bdeg$ ,  $..bdp$ ,  $..ir$ ,  $..bhn$ ,  $..kp$ ,  $..bim$ ,  $..lo$ ,  $..bjl$ ,  $..ce^2$ ,  $..cdo$ ,  $cgn$ ,  $cjk$ ,  $..d^2n$ ,  $..dem$ ,  $..dgl$ ,  $..dhk$ ,  $..dij$ ,  $..ehj$ ,  $..fgj$ ,  $..ghi$ .

| Ord.      | 0 | 2 | 4 | 6 | 8 | 10 |
|-----------|---|---|---|---|---|----|
| <i>at</i> |   |   | 1 |   |   |    |
| <i>ds</i> | 3 | 2 |   |   |   |    |
| <i>es</i> |   | 3 |   |   |   |    |
| <i>fs</i> |   |   |   | 2 | 1 |    |
| <i>hr</i> |   | 2 |   |   |   |    |
| <i>ir</i> |   |   |   |   |   |    |
| <i>jo</i> | 1 |   |   |   |   |    |
| <i>jp</i> |   | 1 |   |   |   |    |
| <i>ko</i> |   | 1 |   |   |   |    |
| <i>kp</i> |   | 3 | 2 | 1 |   |    |
| <i>lo</i> |   |   |   | 1 |   |    |
| <i>lp</i> |   | 6 | 4 | 3 | 2 | 1  |

**Deg. 13** (19 derivs.) — column heads 1, 3, 5, 7.

Covariants present:  $g^2j$ ,  $bgo$ ,  $ag^2$ ,  $agr$ ,  $bt$ ,  $c^2gi$ ,  $ajjo$ ,  $bgk$ ,  $au$ ,  $..b^2o$ ,  $v^2$ ,  $by$ ,  $..b^2gk$ ,  $..jr$ ,  $bd^2$ ,  $b^2s$ ,  $hq$ ,  $bdq$ ,  $..bdr$ ,  $..ino$ ,  $bjm$ ,  $beg^2$ ,  $beg$ ,  $..dgm$ ,  $dp..$ ,  $bgp$ ,  $..bho$ ,  $..hr$ ,  $..ijn$ ,  $no..$ ,  $..bhn$ ,  $cgo$ ,  $ct$ ,  $..d^2o$ ,  $..dgn$ ,  $..djg$ ,  $..egm$ ,  $..ef$ ,  $..ghk$ ,  $gij$ ,  $..hs$ ,  $..lq$ ,  $mp..$

| Ord.      | 1 | 3 | 5 | 7 |
|-----------|---|---|---|---|
| <i>bt</i> | 1 |   |   |   |
| <i>ct</i> |   |   | 1 |   |
| <i>hs</i> | 3 | 2 | 1 |   |
| <i>is</i> |   | 3 | 2 | 1 |
| <i>jr</i> | 1 |   |   |   |
| <i>kr</i> |   | 1 |   |   |
| <i>lr</i> |   |   | 1 |   |
| <i>mo</i> | 1 |   |   |   |
| <i>mp</i> |   | 2 | 1 |   |
| <i>no</i> |   | 1 |   |   |
| <i>np</i> | 4 | 3 | 2 | 1 |



[Vol. X, printed page 387; PDF page 407. Table No. 99 (continued).]

**Deg. 14** (17 derivs.) — column heads 0, 2, 4, 6, 8.

Covariants present:  $bg^2$ ,  $bgr$ ,  $m^2$ ,  $c^2j$ ,  $abgo$ ,  $bgq$ ,  $bjo$ ,  $abt$ ,  $bu$ ,  $..dgo$ ,  $av$ ,  $ag^2k$ ,  $g^2m$ ,  $..dt$ ,  $by$ ,  $ags$ ,  $df$ ,  $g^2n$ ,  $b^2q$ ,  $..ajr$ ,  $mq$ ,  $gj^2$ ,  $b^2gm$ ,  $akq$ ,  $..o^2$ ,  $js$ ,  $by$ ,  $..amo$ ,  $mr..$ ,  $bdgj$ ,  $b^2do$ ,  $nq$ ,  $bg^2i$ ,  $..b^2gn$ ,  $bhq$ ,  $..bko$ ,  $b^2jk$ ,  $cf$ ,  $..bdgk$ ,  $..bds$ ,  $cgq$ ,  $begj$ ,  $..cd^2g^2$ ,  $bg^2l$ ,  $..bhr$ ,  $biq$ ,  $..bjp$ ,  $..dj m$ ,  $..b^2m$ ,  $..ego$ ,  $..et$ ,  $cgr$ ,  $..ghm$ ,  $cjo$ ,  $..gkl^2$ ,  $..d^2r$ ,  $..hf$ ,  $..def$ ,  $..deq$ ,  $..ks$ ,  $nr..$ ,  $..dgp$ ,  $..op$ ,  $..dho$ ,  $..dj n$ ,  $..dkm$ ,  $..ejm$ ,  $..ghn$ ,  $..gim$ ,  $..gjl$ ,  $..hjk$ ,  $if$ .

| Ord. | 0 | 2 | 4 | 6 | 8 |
|------|---|---|---|---|---|
| $dt$ | 1 |   |   |   |   |
| $et$ |   |   | 1 |   |   |
| $ft$ |   |   |   |   | 1 |
| $js$ | 1 |   |   |   |   |
| $ks$ | 2 | 1 |   |   |   |
| $lt$ |   |   | 3 | 2 | 1 |
| $mr$ | 2 | 1 |   |   |   |
| $nr$ |   | 2 | 1 |   |   |
| $ot$ |   |   | 1 |   |   |
| $pp$ |   | 4 |   | 2 |   |

**Deg. 15** (12 derivs.) — column heads 1, 3, 5.

Covariants present:  $g^2o$ ,  $bg^2j$ ,  $..b^2go$ ,  $g^2t$ ,  $bjq$ ,  $bt^2$ ,  $u$ ,  $ln$ ,  $bfk$ ,  $cq$ ,  $bgs$ ,  $dgq$ ,  $bjr$ ,  $bkq$ ,  $du..$ ,  $..bnu$ ,  $gjm$ ,  $..dgr$ ,  $cf$ ,  $..djo$ ,  $eg^2$ ,  $or..$ ,  $egq$ ,  $eu..$ ,  $fp^2$ ,  $..ghn$ ,  $gjn$ ,  $..gkm$ ,  $ht..$ ,  $..fk$ ,  $..hls$ ,  $pq$ .

| Ord. | 1 | 3 | 5 |
|------|---|---|---|
| $bv$ | 1 |   |   |
| $cv$ |   |   | 1 |
| $ht$ | 1 |   |   |
| $it$ |   |   | 1 |
| $ms$ | 2 | 1 |   |
| $ns$ | 3 | 2 | 1 |
| $or$ | 1 |   |   |
| $pr$ |   | 2 | 1 |

[Vol. X, printed page 388; PDF page 408. Table No. 99 (continued).]

**Deg. 16** (13 derivs.) — column heads 0, 2, 4, 6, 8 (col. 8 concluded).

Covariants present:  $g^4$ ,  $c^2r$ ,  $by$ ,  $af$ ,  $abg^2$ ,  $g^2q$ ,  $gjo$ ,  $b^2gq$ ,  $agt$ ,  $abjq$ ,  $..X$ ,  $gtv$ ,  $jt$ ,  $b^2m$ ,  $acq$ ,  $abv$ ,  $..eno$ ,  $g^2$ ,  $qr$ ,  $..b^2gr$ ,  $..ad^2$ ,  $..g^2h^2$ ,  $bg^2m$ ,  $gjo$ ,  $bg^2f$ ,  $..bdgo$ ,  $..bdt$ ,  $adgq$ ,  $..gir$ ,  $bmq$ ,  $bfn$ ,  $..adu$ ,  $ngjm$ ,  $..gkp$ ,  $..bo^2$ ,  $..af$ ,  $..glo$ ,  $bgjk$ ,  $..aor$ ,  $..gn^2$ ,  $dg^2$ ,  $bjs$ ,  $by$ ,  $..h^2q$ ,  $..hko$ ,  $djq$ ,  $..bmr$ ,  $b^2q$ ,  $..hw^2$ ,  $..du$ ,  $..bnq$ ,  $b^2gni$ ,  $..ijo$ ,  $g^2h$ ,  $..dg^2k$ ,  $b^2f$ ,  $..jkn$ ,  $gh^2$ ,  $..dgs$ ,  $..djr$ ,  $..bdgj$ ,  $..khn$ ,  $..gko$ ,  $..dkq$ ,  $b^2gh$ ,  $..lt$ ,  $..gm^2$ ,  $..dmo$ ,  $..b^2iq$ ,  $..ps$ ,  $..hu$ ,  $..efj$ ,  $..b^2ko$ ,  $..fm$ ,  $egj$ ,  $..ev$ ,  $..bhn^2$ ,  $..kt$ ,  $..ghr$ ,  $bef$ ,  $..os$ ,  $begq$ ,  $gn$ ,  $bcu$ ,  $..r^2$ ,  $..gjp$ ,  $..bd^2g^2$ ,  $..gmn$ ,  $..bd^2q$ ,  $..hjo$ ,  $..bdjm$ ,  $..iu$ ,  $..bego$ ,  $..jkm$ ,  $..bet$ ,  $..bghm$ ,  $..bgk^2$ ,  $..bhf$ ,  $..bks$ ,  $..bnr$ ,  $..bop$ ,  $cg^2m$ ,  $cmq$ ,  $..cd^2$ ,  $..d^2gm$ ,  $..d^2f$ ,  $..dghj$ ,  $..dkr$ ,  $..dno$ ,  $..efk$ ,  $..egs$ ,  $..yjf$ .

| Ord.      | 0 | 2 | 4 | 6 | 8 |
|-----------|---|---|---|---|---|
| <i>do</i> |   | 1 |   |   |   |
| <i>ev</i> |   |   | 1 |   |   |
| <i>ft</i> |   |   |   | 1 |   |
| <i>jt</i> | 1 |   |   |   |   |
| <i>kt</i> |   | 1 |   |   |   |
| <i>lt</i> |   |   |   |   | 1 |
| <i>op</i> |   | 1 |   |   |   |
| <i>os</i> |   | 1 |   |   |   |
| <i>pp</i> |   | 4 |   | 2 |   |
| <i>qq</i> |   | 3 | 2 | 1 |   |

**Deg. 17** (6 derivs.) — column heads 1, 3, 5.

Covariants present:  $m^2j$ ,  $bfo$ ,  $ag^2$ ,  $bgt$ ,  $ag^2q$ ,  $gv$ ,  $boq$ ,  $agu$ ,  $ju$ ,  $g^2k$ ,  $aq^2$ ,  $gh..bj$ ,  $h^2k$ ,  $gkq..$ ,  $..gmo$ ,  $bdg^2$ ,  $..fo$ ,  $bdgq$ ,  $ku$ ,  $bdu$ ,  $mt..$ ,  $bgjm$ ,  $qs$ ,  $..bf$ ,  $..bor$ ,  $..dghn$ ,  $ds^2$ ,  $..dmq$ ,  $..do^2$ ,  $fhj$ ,  $..gkr$ ,  $..gno$ ,  $hv$ ,  $..jko$ ,  $..gn^2$ ,  $..nt$ ,  $..rs$ .

| Ord.      | 1 | 3 | 5 |
|-----------|---|---|---|
| <i>hv</i> |   | 1 |   |
| <i>iv</i> |   |   | 1 |
| <i>ml</i> | 1 |   |   |
| <i>nt</i> |   | 1 |   |
| <i>rs</i> | 2 | 1 |   |

**Deg. 18** (6 derivs.) — column heads 2, 4, 6.

Covariants present:  $w$ ,  $bg^2$ ,  $bg^2r$ ,  $a^2j$ ,  $bg^2q$ ,  $mi..$ ,  $bgjo$ ,  $bgu$ ,  $bjt$ ,  $agv$ ,  $bf$ ,  $bqr$ ,  $aju$ ,  $g^2m$ ,  $..dg^2o$ ,  $by$ ,  $..doq$ ,  $g^2$ ,  $g^2n$ ,  $b^2gq$ ,  $gmq$ ,  $b^2u$ ,  $..gv^2$ ,  $hgm$ ,  $fp$ ,  $gpjk$ ,  $bv^2$ ,  $..jv$ ,  $m$ ,  $b^2mq$ ,  $mu$ ,  $..gmr$ ,  $..bf$ ,  $..at$ ,  $gnq$ ,  $bdgv$ ,  $fr$ ,  $bdjq$ ,  $jkq$ ,  $..bdv$ ,  $..jmo$ ,  $bfh$ ,  $kv..$ ,  $bgjuj$ ,  $nu$ ,  $..bgko$ ,  $..bgm^2$ ,  $bhu$ ,  $..bpm$ ,  $..bkt$ ,  $..bos$ ,  $..br^2$ ,  $cg^4$ ,  $cg^2q$ ,  $cgu$ ,  $cq^2$ ,  $..d^2q$ ,  $..d^3gq$ ,  $..d^4u$ ,  $..dgjm$ ,  $..dor$ ,  $..eg^2o$ ,  $..egj$ ,  $..eoq$ ,  $..g^2hm$ ,  $ghf$ ,  $..g^4$ ,  $..gnr$ ,  $..gop$ ,  $..hmq$ ,  $..M$ ,  $..jhr$ ,  $..jno$ ,  $..k^2$ ,  $..kmo$ ,  $..m^4$ ,  $..pt$ ,  $..s^2$ .

| Ord.      | 2 | 4 | 6 |
|-----------|---|---|---|
| <i>ju</i> | 1 |   |   |
| <i>kv</i> |   | 1 |   |
| <i>lv</i> |   |   | 1 |
| <i>at</i> | 1 |   |   |
| <i>pt</i> |   | 1 |   |
| <i>ss</i> |   | 2 |   |

**Deg. 19** (3 derivs.) — column heads 1, 3.

Covariants present:  $g^2o$ ,  $bg^2j$ ,  $w$ ,  $gh$ ,  $bg^2q$ ,  $gqo$ ,  $bgv$ ,  $ou$ ,  $bju$ ,  $qt$ ,  $dg^4$ ,  $dg^2j$ ,  $..djt$ ,  $dq^2$ ,  $fjm$ ,  $..gf$ ,  $gor$ ,  $jmq$ ,  $..jo^2$ ,  $mv..$ ,  $..rt$ .

| Ord.      | 1 | 3 |
|-----------|---|---|
| <i>mv</i> | 1 |   |
| <i>nv</i> |   | 1 |
| <i>rt</i> | 1 |   |

[Vol. X, printed page 390; PDF page 410. Table No. 99 (concluded).]

**Deg. 20** (3 derivs.) — column heads 2, 4.

Covariants present:  $g^4, bw, b^2g^4, g^2q, gh^2, b^2q^2, g^2u, g^4o, b^2gn, gf, sgt, by, qu, gqr, bj^2m, ..joq, ..ov, bgmq, ru, ..bgo^2, bfq, bju, bmu, ..hot, rfA, dyjq, ..dgv, dju, g^2h, fhq, ..fko, ..fm^2, ghu, .gpm, ..gkt, ..gos, ..gr^2, w, r, .jor, ..koq, ..m^2q, ..mo^2, ..St.$

| Ord. | 2 | 4 |
|------|---|---|
| $ov$ | 1 |   |
| $pv$ |   | 1 |
| $sv$ | 1 |   |

**Deg. 21** (1 deriv.) — column head 1.

Covariants present:  $g^2j, g^3q, gh, gju, jf, qv, w.$

| Ord. | 1 |
|------|---|
| $tv$ | 1 |

**Deg. 22** (1 deriv.) — column heads 0, 2.

Covariants present:  $gw, bg^2, btf, b^2g^2q, bfu, bgq^2, bqu, ghn, fff, fmq, ..fo^2, fgq, ..sjv, gmu, ..got, A, mq^2, ..o^2q, e...$

| Ord. | 0 | 2 |
|------|---|---|
| $sv$ | 1 |   |

**Deg. 24** (1 deriv.) — column head 0.

Covariants present:  $g^6, A, g^2, ff, gqu, ct, M^4.$

| Ord. | 0 |
|------|---|
| $tv$ | 1 |

[End of Table No. 99. The total over degrees 2 through 24 is 429 derivatives, yielding the 22 simple covariants  $b, c, d, \dots, w$  of the binary quintic, as remarked in art. 383 above.]

[Vol. X, printed page 391; PDF page 411.]

**384.** The Canonical form (using the divided expressions, Table No. 98) is peculiarly convenient for the calculation of the derivatives. Some attention is required in regard to the numerical determination: it will be observed that  $A$  is given in the standard form  $(A_0, A_1, A_2, A_3, A_4, A_5 \mid x, y)^5$ , while the other covariants are given in the denumerate forms  $B = (B_0, B_1, B_2 \mid x, y)^2$  &c.: these must be converted into the other form

$$B = (B_0, \frac{1}{2}B_1, B_2 \mid x, y)^2, \quad C = (C_0, \frac{1}{6}C_1, \frac{1}{15}C_2, \frac{1}{20}C_3, \frac{1}{15}C_4, \frac{1}{6}C_5, C_6 \mid x, y)^6, \quad \&c.,$$

the numerical coefficients being of course the reciprocals of the binomial coefficients. We thus have, for instance, the leading coefficients,

$$\text{l.c. of } AC2 = A_0 \cdot \frac{1}{15}C_2 - 2 \cdot A_1 \cdot \frac{1}{6}C_1 + A_2 \cdot C_0,$$

but

$$\text{l.c. of } BC2 = B_0 \cdot \frac{1}{15}C_2 - 2 \cdot \frac{1}{2}B_1 \cdot \frac{1}{6}C_1 + B_2 \cdot C_0.$$

Moreover, as regards the covariants  $AA2$ ,  $AA4$ , &c., we take what are properly the half-values,

$$\begin{aligned} \text{i.e. of } AA2 &= A_0A_2 - A_1^2 \quad (\text{instead of } A_0A_2 - 2A_1A_1 + A_2A_0), \\ \text{,, ,, } AA4 &= A_0A_4 - 4A_1A_3 + 3A_2^2 \quad (\text{instead of } A_0A_4 - 4A_1A_3 + 6A_2A_2 - 4A_3A_1 + A_4A_0), \\ &\quad \&c., \end{aligned}$$

and similarly

$$\text{l.c. of } BB2 = B_0B_2 - (\frac{1}{2}B_1)^2, \quad \text{,, ,, } CC2 = C_0 \cdot \frac{1}{15}C_2 - (\frac{1}{6}C_1)^2, \quad \&c.$$

Any one of these leading coefficients, for instance l.c. of  $AC2$ , is equal to the corresponding covariant derivative, multiplied, it may be, by a power of  $a$ : the index of this power being at once found by comparing the deg-orders, these in fact differing by a multiple of 1.5 the deg-order of  $a$ . Thus

$$\begin{aligned} aa2, \quad A_0A_2 - A_1^2, \quad \text{deg-orders are } 2.6, 2.6 : \text{ or } aa2 &= A_0A_2 - A_1^2, \\ aa4, \quad A_0A_4 - 4A_1A_3 + 3A_2^2, \quad \text{deg-orders are } 2.2, 4.12 : \text{ or } aa4 &= (A_0A_4 - 4A_1A_3 + 3A_2^2); \end{aligned}$$

we have in fact

$$\begin{aligned} A_0A_2 - A_1^2 &= 1 \cdot c - 0^2 = c \quad \text{and so } aa2 = c, \\ AA4 &= A_0A_4 - 4A_1A_3 + 3A_2^2 = 1 \cdot (a^3b ? k^?) - 4 \cdot 0 \cdot f + 3 \cdot c^2 = a^2b ? \quad \text{and so } aa4 = b. \end{aligned}$$

As another instance, and for the purpose of showing how the calculation is actually effected, consider the derivative  $ch2$ , which is to be calculated from the leading coefficient of  $CH2$ ,

$$\begin{aligned} C_0 \cdot \frac{1}{15}F_2 - 2 \cdot \frac{1}{6}C_1 \cdot \frac{1}{6}H_1 + \frac{1}{15}C_2 \cdot H_0 : \quad \text{this is} \\ = -c \left( \frac{1}{15}a^2f ? - 2ahdch \right) \\ - 2 \cdot \frac{1}{6} \cdot \frac{1}{6}(bc - l) \\ + \left( \frac{1}{15}a^2b \cdot d^2 \right) h \end{aligned}$$

[Vol. X, printed page 392; PDF page 412.]  
= column next written down; but this column contains congregate terms which have to be replaced by their segregate values (see Table No. 96, deg-order 8.16); and we thus obtain

|                                        | $a^3f^?$        | $a^2b^2f$ | $a^2bh$         | $a^2cy^?$      | $abcd$       | $b^3c^2$ | $c^2h$ |
|----------------------------------------|-----------------|-----------|-----------------|----------------|--------------|----------|--------|
| $\frac{1}{15}a^2bh$                    |                 |           | $+\frac{1}{15}$ |                |              |          |        |
| $+\frac{1}{6}a^2cy$                    |                 |           |                 | $+\frac{1}{6}$ |              |          |        |
| $-2abcd$                               |                 |           |                 |                | $-2$         |          |        |
| $-\frac{1}{15}bef$                     | $-\frac{1}{15}$ |           |                 |                | $+3$         | $+2$     |        |
| $-2c^2h$                               |                 |           |                 |                |              | $-2$     |        |
| $+f^?.h \sim 1 + \frac{1}{15} \cdot 1$ |                 |           |                 |                | $-1 - 2 + 2$ |          |        |

[the printed table also displays the totals  $\frac{1}{15}, \frac{1}{6}, 0, 0, 0, 0$  at the foot — viz. the terms other than those divisible by  $a^2$  all disappear:] we may either abbreviate the calculation by omitting them *ab initio*, or retain them for the sake of the verification afforded by their disappearance. The factor  $a^2$  divides out, and the final result is

$$ch2 = \frac{1}{15}bh + \frac{1}{6}cy \quad (\text{up to overall scale}),$$

which is the proper segregate expression of the derivative  $ch2$ : of course, we have deg-order  $CH2 = 8.16$ , the double of deg-order  $ch2 = 6.6, 6.6$ , and the difference is 1.0, so that the factor  $a^2$  is as it ought to be.

Table No. 100 (The Derivatives up to the Sixth Order).

Degree 2.

|           |           |
|-----------|-----------|
| 2.2       | 2.6       |
| $aa4 + 1$ | $aa2 + 1$ |

Degree 3.

|               |                                            |                                           |                          |                          |
|---------------|--------------------------------------------|-------------------------------------------|--------------------------|--------------------------|
| 3.1           | 3.3 $d$                                    | 3.5 $e$                                   | 3.7 $ab$                 | 3.9 $f$                  |
| $ac5 \quad 0$ | $ab2 \quad -3$<br>$ac4 \quad +\frac{3}{2}$ | $ab1 \quad +\frac{1}{2}$<br>$ac3 \quad ?$ | $ac3 \quad -\frac{3}{4}$ | $ac1 \quad +\frac{1}{4}$ |

Degree 4.

|                                                             |                                |                                                                                                                                           |                                                                                                    |
|-------------------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 4.0 $g$                                                     | 4.2                            | 4.4 $h$                                                                                                                                   | 4.6                                                                                                |
| $ae5 \quad -2$<br>$bb2 \quad -\frac{1}{4}$<br>$cc6 \quad 1$ | $ad3 \quad 0$<br>$ae4 \quad 0$ | $ad2 \quad +\frac{1}{2}$<br>$ae3 \quad -\frac{1}{3}$<br>$af5 \quad +\frac{3}{5}$<br>$be2 \quad +\frac{1}{2}$<br>$cc4 \quad +\frac{1}{10}$ | $ad1 \quad +\frac{1}{4}$<br>$ae2 \quad +1$<br>$af4 \quad -\frac{1}{5}$<br>$bc1 \quad +\frac{1}{2}$ |

[Vol. X, printed page 393; PDF page 413. Table No. 100 (continued).]

|                                                                                                  |                |               |
|--------------------------------------------------------------------------------------------------|----------------|---------------|
| 4.8 $ad, be$                                                                                     | 4.10 $ae$      | 4.12 $a^2b$   |
| $ae1 \quad +\frac{1}{4}$<br>$af3 \quad +\frac{1}{5}$<br>$cc2 \quad +\frac{1}{10}, -\frac{1}{10}$ | $af2 \quad +1$ | $af1 \quad ?$ |

Degree 5.

|                                                                              |                                                                                                                                                                      |                                                                                                                                    |                                                                                                    |                                                                                        |
|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| 5.1 $j$                                                                      | 5.3 $k$                                                                                                                                                              | 5.5 $ag, bd$                                                                                                                       | 5.7 $be$                                                                                           | 5.9 $ab^2, ad, cf$                                                                     |
| $ah4 \quad +2$<br>$ai5 \quad 1$<br>$bd2 \quad -\frac{3}{2}$<br>$ce5 \quad 8$ | $ah3 \quad +\frac{1}{2}$<br>$ai4 \quad +\frac{1}{2}$<br>$bd1 \quad 1$<br>$be2 \quad ?$<br>$cd3 \quad ?$<br>$ce1 \quad +\frac{11}{12}$<br>$cf6 \quad +\frac{1.19}{?}$ | $ah2 \quad +\frac{1}{2}, -2$<br>$ai3 \quad +\frac{1}{2}$<br>$bc1 \quad -2$<br>$cd2 \quad -\frac{3}{2}$<br>$ce3 \quad +\frac{7}{?}$ | $ah1 \quad ?$<br>$ai2 \quad ?$<br>$bf2 \quad ?$<br>$cd1 \quad ?$<br>$ce2 \quad ?$<br>$cf4 \quad ?$ | $ai1 \quad 1$<br>$bf1 \quad -\frac{1}{2}$<br>$ce1 \quad -\frac{1}{2}$<br>$df3 \quad 1$ |

[The corresponding sub-tables for deg-orders 5.11, 5.13 give entries df2 with coefficient  $\frac{1}{120} + \frac{1}{585}$ , and df1 with coefficient  $+\frac{1}{4}$  respectively.]

Degree 6.

|     |   |           |                |       |                |                |   |
|-----|---|-----------|----------------|-------|----------------|----------------|---|
| 6.0 |   | 6.2 ig, m |                | 6.4 n |                | 6.6 aj, bh, eg |   |
| ci6 | 9 | ah3       | −4             | ajl   | −1             | al             | ? |
|     |   | ai5       | 1              | ak2   | $+\frac{1}{2}$ | al3            | ? |
|     |   | bh2       | $+\frac{1}{2}$ | ali   | −3 ?           | bh1            | ? |
|     |   | ch4       | ?              | bh1   | $+\frac{1}{4}$ | ch2            | ? |
|     |   | ci5       | −2             | bi2   | +1             | ci3            | ? |
|     |   | dd2       | $-\frac{1}{7}$ | ch3   | $+\frac{A}{?}$ | de1            | ? |
|     |   | de3       | ?              | ci4   | $+\frac{A}{?}$ | df3            | ? |
|     |   | ee4       | + ?            |       |                | ee2            | ? |
|     |   | ff6       | + ?            |       |                | ef4            | ? |
|     |   |           |                |       |                | ff6            | ? |

[The printed Degree 6 sub-table also contains, in deg-orders 6.6, 6.8, 6.10, 6.12, 6.14, a number of small fractional entries (e.g.  $-\frac{143}{17?}$ ,  $+\frac{425}{?}$ ,  $-\frac{5691}{?}$ , etc.) recording the segregate coefficients of the various derivatives al, ch1, ci2, df2, ef3, ...; their numerical values are presented in compressed form in the original.]

[Vol. X, printed page 394; PDF page 414. Table No. 100 (concluded).]

| 6.8 $ah, bi$ |                              | 6.10 $ai, ci$ |                              | 6.12 $a^2b^2, a^2h, acd, bc^2$ |                                           |
|--------------|------------------------------|---------------|------------------------------|--------------------------------|-------------------------------------------|
| $al2$        | $-\frac{1}{?} + \frac{1}{?}$ | $ail$         | $-8$                         | $af1$                          | $-1, +1$                                  |
| $ch1$        | $+\frac{1}{?} + \frac{1}{?}$ | $df1$         | $-\frac{1}{?} + \frac{1}{?}$ | $ef2$                          | $+1$                                      |
| $ci2$        | $+\frac{1}{?}$               | $ff4$         | $+1 + 1$                     | $ff4$                          | $+\frac{1}{?} \frac{19}{?} - \frac{1}{?}$ |
| $df2$        | $- -$                        | $ef3$         | $+\frac{ih}{?} +$            | (misc.)                        | $+\frac{89}{?}, -\frac{71}{?}$            |

which is complete to the sixth degree. I had calculated the derivatives up to the tenth degree, but the results were not in the segregate form.

*On the form of the Numerical Generating Functions: the N.G.F. of a Sextic.*  
Art. Nos. 385, 386.

**385.** It is to be remarked that the R.G.F. is derived not from the fraction in its least terms, which is algebraically the most simple form of the N.G.F., but from a form which contains common factors in the numerator and denominator: thus for the quadric, the cubic, and the quartic, writing down the two forms (identical in the case of the quadric) these are

**Quadric**

$$\text{N.G.F.} = \frac{1}{1 - \alpha x^2 \cdot 1 - a^2}.$$

**Cubic**

$$\text{N.G.F.} = \frac{1 + ax^3}{1 - a^2 \cdot 1 - ax^3 \cdot 1 - ax} = \frac{1 - a^2x^6}{1 - a^2 \cdot 1 - ax^3 \cdot 1 - ax^3 \cdot 1 - a^2x^3}.$$

**Quartic**

$$\text{N.G.F.} = \frac{1 - ax^4 + a^2x^8}{1 - a^2 \cdot 1 - a^3 \cdot 1 - ax^4 \cdot 1 - ax^? \cdot 1 - a^2x^4} \equiv \frac{1 - a^6x^4}{(1 - a^2)(1 - a^3)(1 - ax^4)(1 - ax^?)(1 - a^2x^4)}.$$

[Vol. X, printed page 395; PDF page 415.]

For the quintic the two forms are, N.G.F. =

$$\begin{array}{l}
 1 + (a^4 - a^6 + a^8)x^2 + (-1 + a^4 + 2a^6 - a^{10})ax^4 + (a^4 - a^6 - a^8)ax^6 \\
 + (-1 + a^4 - a^6)a^2x^8 + (+a^2 - 2a^6 - a^8 - a^{10})a^2x^6 + (+1)a^2x^{10} \\
 + (-1 - a^6 - a^{10})a^2x^{14} + (+1 + a^2)a^2x^{16} + (-a^2)a^3x^? \\
 \hline
 a^2 \cdot 1 - a^4 \cdot 1 - a^6 \cdot 1 - ax^3 \cdot 1 - ax^? \cdot 1 - ax
 \end{array}$$

and (equivalent form)

$$\begin{array}{l}
 (1 + a^{12})x^0 + (1 - a^4 + a^6 + a^8 + a^{10} + a^{12})ax^2 + (1 - a^6 + a^8 + a^{10} - a^{12} - a^{14})a^2x^4 \\
 + (-1 - a^2 + a^8 - a^{10} - a^{12} - a^{14})a^2x^6 + (+a^6 - a^{12})a^2x^8 \\
 + (+1 - a^6 - a^8)a^3x^? + (-1 \text{ terms in } a^6, a^8, a^{10})a^3x^? \\
 \hline
 1 - a^2 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - ax^3 \cdot 1 - a^2x^? \cdot 1 - a^3x^?
 \end{array}$$

[The printed page sets out these two forms in a single very wide tableau of seven and ten rows respectively, each cell of which carries a small coefficient  $\pm a^k$ . The structural identity — viz., the second form is essentially the first multiplied above and below by a polynomial in  $a, x$  — is used to obtain the R.G.F. of the quintic; the explicit working is reserved to the printed page.]

this last being in fact equivalent to that used for the determination of the R.G.F.



[Vol. X, printed page 396; PDF page 416.]

**386.** For the sextic the forms are, N.G.F. =

$$\begin{aligned}
 & (-1 + a + a^4 - a^6 - a^8 - a^{12} + a^? + a^?) x^0 \\
 & + (-1 - a + a^4 + 2a^6 + 2a^8 + a^? - a^? - a^?) ax^2 \\
 & + (-1 + a^4 + a^6 + a^8 + a^? - a^? - a^?) ax^4 \\
 & + (-1 + a^4 + a^8 + a^? + a^? + a^?) a^2 x^? \\
 & + (-a - a^6 - a^8 - a^{10} - a^?) a^2 x^? \\
 & + (+a - a^6 - 2a^8 - 2a^{10} - a^?) a^2 x^? \\
 & + (-a + a^6 + a^8 + a^{10}) a^3 x^? \\
 \hline
 & 1 + a \cdot 1 - a^2 \cdot 1 - a^4 \cdot 1 - a^6 \cdot 1 - ax \cdot 1 - ax^? \cdot 1 - ax^? \cdot 1 - ax^?
 \end{aligned}$$

and

$$\begin{aligned}
 & (1) a^0 x^? + (1 + a^2 + a^4 + a^6 + a^8 + a^{10}) a^2 x^? \\
 & + (a^2 + a^4 + a^6 + a^8 + a^{10} + a^{12} + a^{14}) a^2 x^4 \\
 & + (1 + a + 2a^2 + a^4 + a^6 + a^8 - a^{14}) a^3 x^? \\
 & + (a^? - a^? - a^? - a^{12} - a^{14}) a^2 x^? \\
 & + (a^? - a^6 - 2a^? - a^? - a^{12}) a^2 x^? \\
 & + (-a^4 - a^6 - a^8 - a^{10} - a^{12} - a^{14}) a^? x^? \\
 & + (-a^? - a^? - a^? - a^? - a^?) a^2 x^? \\
 & + (-1 - a^?) a^2 x^? \\
 \hline
 & a^? \cdot 1 - a^? \cdot 1 - a^? \cdot 1 - a^? 4 \cdot 1 - a^? x^? \cdot 1 - ax^? \cdot 1 - a^? x^?
 \end{aligned}$$

where observe that in the middle term, although for symmetry  $a^7 (= \sqrt{a^{14}})$  has been introduced into the expression, the coefficient is really rational, viz. the term is

$$(a^2 + a^4 + a^6 - a^? - a^? - a^?) a^?.$$

The second form or one equivalent to it is due to Sylvester: I do not know whether he divided out the common factors so as to obtain the first form. I assume that it would be possible from this second form to obtain a R.G.F., and thence to establish for the 26 covariants of the sextic a theory such as has been given for the 23 covariants of the quintic: but I have not entered upon this question.

[Vol. X, printed page 397; PDF page 417.]

**Table No. 93 bis** (The covariant  $S$ , adopted form  $-(D, M)$ ).

{ In this Table,  $a, b, c, d, e, f$  denote, as in the tables of former memoirs, the coefficients of the quintic form  $(a, b, c, d, e, f \mid x, y)^5$ .

[The covariant  $S$  is set out as a polynomial in  $x, y$  of degree 5 with coefficients  $S_0, S_1, S_2, S_3, S_4, S_5$ . The original printed pages 397–398 give the explicit expressions for  $S_0, S_1, S_2, S_3$  as very dense lists of monomials  $a^\alpha b^\beta c^\gamma d^\delta e^\epsilon f^\zeta$  with integer coefficients, occupying four to six columns per page. We here record the structural form together with the closed-form reductions (obtained by Cayley by setting  $d = e = f = 1$ ) which the printed text immediately afterwards extracts as a verification.]

$$S = (S_0, S_1, S_2, S_3, S_4, S_5 \mid x, y)^5.$$

[The first two coefficients  $S_0, S_1$  are calculated directly;  $S_2$  is deduced from  $S_1$ , and  $S_3$  from  $S_0$ , by reversing the order of the letters — which is the same thing as interchanging  $a \leftrightarrow f, b \leftrightarrow e, c \leftrightarrow d$  — and reversing also the signs of the numerical coefficients. This process is, to a very great extent, a verification of the values of  $S_0$  and  $S_1$ ; for, as presently remarked, the terms of  $S_0$  form subdivisions such that in each subdivision the sum of the numerical coefficients is  $= 0$ ; in passing by the reversal process to the value of  $S_3$ , the terms are distributed into an entirely new set of subdivisions, and then in each of these subdivisions the sum of the numerical coefficients is found to be  $= 0$ ; and the like as regards  $S_1$  and  $S_2$ .]

[The full lists of monomials and coefficients comprising  $S_0, S_1, S_2, S_3$  occupy printed pages 397–398 in seven columns. As the entries are partly unreadable in the scan (typesetter has compressed and broken many fraction-bars), we omit the verbatim reproduction and reproduce instead the verification specialisations which Cayley records on printed pages 399–400.]

[Vol. X, printed page 399; PDF page 419. Verification of the covariant  $S$ .]

If in the expressions for  $S_0$ ,  $S_1$ ,  $S_2$ ,  $S_3$  we first write  $d = e = f = 1$ , thus in effect combining the numerical coefficients for the terms which contain the same powers in  $a$ ,  $b$ ,  $c$ , we find

$$\begin{aligned} S_0 = & a^5 (-2c^3 + 6c^2 - 6c + 2) \\ & + a^4 \{ b^2(6c^2 - 12c - 6) + b(-15c^3 + 33c^2 - 21c + 3) \\ & \quad + b^0(42c^4 - 147c^3 + 105c^2 - 117c + 27) \} \\ & + a^3 \{ b^4 \cdot 0 + b^3(30c^2 - 36c + 6) + b^2(-117c^3 + 249c^2 - 183c + 51) \\ & \quad + b(9c^4 + 138c^3 - 378c^2 + 330c - 99c) \\ & \quad + b^0(-63c^5 + 165c^4 - 147c^3 + 45c^2) \} \\ & + a^2 \{ b^6 \cdot 2 + b^5(-15c + 3) + b^4(75c^2 - 69c + 24) \\ & \quad + b^3(-9c^4 - 16c^3 + 225c^2 - 87c - 2) \\ & \quad + b^2(72c^4 + 48c^3 - 186c^2 + 96c) \\ & \quad + b(-126c^4 + 201c^3 - 87c^2) \\ & \quad + b^0(27c^5 - 45c^4 + 20c^3) \}; \end{aligned}$$

which for  $c = 1$  becomes

$$= 2b^6 - 12b^5 + 30b^4 - 40b^3 + 30b^2 - 12b + 2, \quad \text{that is, } 2(b-1)^6,$$

and for  $b = 1$  becomes  $= 0$ .

$$\begin{aligned} S_1 = & a^5 (0c^2 + 0c + 0) \\ & + a^4 \{ b^2(0c + 0) + b(3c^3 - 9c^2 + 9c - 3) + b^0(-24c^4 + 99c^3 - 153c^2 + 105c - 27) \} \\ & + a^3 \{ b^4 \cdot 0 + b^3(-6c^2 + 12c - 6) + b^2(-24c^3 + 90c^2 - 108c + 42) \\ & \quad + b(33c^4 - 90c^3 + 54c^2 + 30c - 27) \\ & \quad + b^0(-27c^5 + 78c^4 - 66c^3 + 6c^2 + 9c) \} \\ & + a^2 \{ b^5(3c - 3) + b^4(-15c + 15) + b^3(6c^3 - 12c^2 + 36c - 30) \\ & \quad + b^2(9c^4 - 42c^3 + 84c^2 - 108c + 57) \\ & \quad + b(-9c^5 + 54c^4 - 96c^3 + 51c^2) \\ & \quad + b^0(9c^5 - 9c^4) \}; \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

$$\begin{aligned} S_2 = & a^5 (0c + 0) \\ & + a^4 \{ b^2(0) + b(0c^3 + 0c^2 + 0c + 0) + b^0(18c^4 - 72c^3 + 108c^2 - 72c + 18) \} \\ & + a^3 \{ b^4(0c + 0) + b^3(-33c^3 + 99c^2 - 99c + 33) + b^2(57c^4 - 162c^3 + 144c^2 - 30c - 9) \\ & \quad + b(-60c^5 + 207c^4 - 261c^3 + 141c^2 - 27c) \} \\ & + a^2 \{ b^5 \cdot 0 + b^4(15c^2 - 30c + 15) + b^3(-54c^3 + 102c^2 - 42c - 6) \\ & \quad + b^2(123c^4 - 297c^3 + 243c^2 - 87c + 18) \\ & \quad + b(-27c^5 + 102c^4 - 96c^3 + 21c^2) \\ & \quad + b^0(27c^5 - 66c^4 + 51c^3 - 12c^2) \}; \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

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$$\begin{aligned}
 S_3 &= a^5 \cdot 0 \\
 &+ a^4 \{ b(0c + 0) + b^0(0c^3 + 0c^2 + 0c + 0) \} \\
 &+ a^3 \{ b^4 \cdot 0 + b^3(0c^2 + 0c + 0) + b^2(-9c^4 + 36c^3 - 54c^2 + 36c - 9) \\
 &\quad + b^0(36c^5 - 171c^4 + 324c^3 - 306c^2 + 144c - 27) \} \\
 &+ a^2 \{ b^5(0c + 0) + b^4(7c^3 - 21c^2 + 21c - 7) + b^3(-30c^4 + 135c^3 - 171c^2 + 93c - 18) \\
 &\quad + b^2(66c^5 - 243c^4 + 333c^3 - 201c^2 + 45c) \\
 &\quad + b^0(-27c^5 + 101c^4 - 141c^3 + 87c^2 + 20c^?) \};
 \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

It follows that, for  $c = d = e = f = 1$ , the value of the covariant  $S$  is  $= 2(b - 1)^6 y^5$ , which might be easily verified.

[End of Paper No. 693 (A Tenth Memoir on Quantics).]

[Vol. X, printed page 401; PDF page 421.]

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694.

## DESIDERATA AND SUGGESTIONS.

### No. 1. The theory of groups.

[From the *American Journal of Mathematics*, t. I. (1878), pp. 50–52.]

SUBSTITUTIONS, and (in connexion therewith) groups, have been a good deal studied; but only a little has been done towards the solution of the general problem of groups. I give the theory so far as is necessary for the purpose of pointing out what appears to me to be wanting.

Let  $\alpha, \beta, \dots$  be functional symbols, each operating upon one and the same number of letters and producing as its result the same number of functions of these letters; for instance,  $\alpha(x, y, z) = (X, Y, Z)$ , where the capitals denote each of them a given function of  $(x, y, z)$ .

Such symbols are susceptible of repetition and of combination:

$$\alpha^2(x, y, z) = \alpha(X, Y, Z), \quad \text{or} \quad \beta\alpha(x, y, z) = \beta(X, Y, Z),$$

= in each case three given functions of  $(x, y, z)$ ; and similarly for  $\alpha^3, \alpha^2\beta$ , &c.

The symbols are not in general commutative,  $\alpha\beta$  not =  $\beta\alpha$ ; but they are associative,  $\alpha\beta \cdot \gamma = \alpha \cdot \beta\gamma$ , each =  $\alpha\beta\gamma$ , which has thus a determinate signification.

The associativeness of such symbols arises from the circumstance that the definitions of  $\alpha, \beta, \gamma, \dots$  determine the meanings of  $\alpha\beta, \alpha\gamma$ , &c.: if  $\alpha, \beta, \gamma, \dots$  were quasi-quantitative symbols such as the quaternion imaginaries  $i, j, k$ , then  $\alpha\beta$  and  $\beta\gamma$  might have by definition values  $\delta$  and  $\epsilon$  such that  $\alpha\beta \cdot \gamma$  and  $\alpha \cdot \beta\gamma$  (=  $\delta\gamma$  and  $\alpha\epsilon$  respectively) have unequal values.

Unity as a functional symbol denotes that the letters are unaltered,  $1(x, y, z) = (x, y, z)$ ; whence  $1\alpha = \alpha 1 = \alpha$ .

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The functional symbols may be substitutions;  $\alpha(x, y, z) = (y, z, x)$ , the same letters in a different order: substitutions can be represented by the notation  $\alpha = \frac{yzx}{xyz}$ , the substitution which changes  $xyz$  into  $yzx$ , or as products of cyclical substitutions,

$$\alpha = \frac{yzx}{xyz} \cdot \frac{wu}{uw} = (xyz)(uw),$$

the product of the cyclical interchanges  $x$  into  $y$ ,  $y$  into  $z$ ,  $z$  into  $x$ ; and  $u$  into  $w$ ,  $w$  into  $u$ .

A set of symbols,  $\alpha, \beta, \gamma, \dots$ , such that the product  $\alpha\beta$  of each two of them (in each order,  $\alpha\beta$  or  $\beta\alpha$ ), is a symbol of the set, is a group. It is easily seen that 1 is a symbol of every group, and we may therefore give the definition in the form that a set of symbols,  $1, \alpha, \beta, \gamma, \dots$  satisfying the foregoing condition is a group. When the number of the symbols (or terms) is  $= n$ , then the group is of the  $n^{\text{th}}$  order; and each symbol  $\alpha$  is such that  $\alpha^n = 1$ , so that a group of the order  $n$  is, in fact, a group of symbolical  $n^{\text{th}}$  roots of unity.

A group is defined by means of the laws of combination of its symbols: for the statement of these we may either (by the introduction of powers and products) diminish as much as may be the number of independent functional symbols, or else, using distinct letters for the several terms of the group, employ a square diagram as presently mentioned.

Thus, in the first mode, a group is  $1, \beta, \beta^2, \alpha, \alpha\beta, \alpha\beta^2$  ( $\alpha^2 = 1, \beta^3 = 1, \alpha\beta = \beta^2\alpha$ ); where observe that these conditions imply also  $\alpha\beta^2 = \beta\alpha$ .

Or, in the second mode, calling the same group  $(1, \alpha, \beta, \gamma, \delta, \epsilon)$ , the laws of combination are given by the square diagram

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
|------------|------------|------------|------------|------------|------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
| $\alpha$   | $\alpha$   | 1          | $\gamma$   | $\beta$    | $\epsilon$ | $\delta$   |
| $\beta$    | $\beta$    | $\epsilon$ | $\delta$   | $\alpha$   | 1          | $\gamma$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    |
| $\delta$   | $\delta$   | $\gamma$   | 1          | $\epsilon$ | $\beta$    | $\alpha$   |
| $\epsilon$ | $\epsilon$ | $\beta$    | $\alpha$   | $\delta$   | $\gamma$   | 1          |

= for the symbols  $(1, \alpha, \beta, \gamma, \delta, \epsilon)$  are in fact  $(1, \alpha, \beta, \alpha\beta, \beta^2, \alpha\beta^2)$ .

The general problem is to find all the groups of a given order  $n$ ; thus if  $n = 2$ , the only group is  $1, \alpha$  ( $\alpha^2 = 1$ ); if  $n = 3$ , the only group is  $1, \alpha, \alpha^2$  ( $\alpha^3 = 1$ ); if  $n = 4$ , the groups are  $1, \alpha, \alpha^2, \alpha^3$  ( $\alpha^4 = 1$ ), and  $1, \alpha, \beta, \alpha\beta$  ( $\alpha^2 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha$ ); if  $n = 6$ , there are three groups, a group  $1, \alpha, \alpha^2, \alpha^3, \alpha^4, \alpha^5$  ( $\alpha^6 = 1$ ), and two groups  $1, \beta, \beta^2, \alpha, \alpha\beta,$

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$\alpha\beta^2$  ( $\alpha^2 = 1, \beta^3 = 1$ ); viz. in the first of these  $\alpha\beta = \beta\alpha$ , while in the other of them (that mentioned above) we have  $\alpha\beta = \beta^2\alpha, \alpha\beta^2 = \beta\alpha$ .

But although the theory as above stated is a general one, including as a particular case the theory of substitutions, yet the general problem of finding all the groups of a given order  $n$ , is really identical with the apparently less general problem of finding all the groups of the same order  $n$ , which can be formed with the substitutions upon  $n$  letters; in fact, referring to the diagram, it appears that  $1, \alpha, \beta, \gamma, \delta, \epsilon$  may be regarded as substitutions performed upon the six letters  $1, \alpha, \beta, \gamma, \delta, \epsilon$ , viz.  $1$  is the substitution unity which leaves the order unaltered,  $\alpha$  the substitution which changes  $1\alpha\beta\gamma\delta\epsilon$  into  $\alpha 1\gamma\beta\epsilon\delta$ , and so for  $\beta, \gamma, \delta, \epsilon$ . This, however, does not in any wise show that the best or the easiest mode of treating the general problem is thus to regard it as a problem of substitutions: and it seems clear that the better course is to consider the general problem in itself, and to deduce from it the theory of groups of substitutions.

Cambridge, 28th November, 1877.

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## No. 2. The theory of groups; graphical representation.

[From the *American Journal of Mathematics*, t. I. (1878), pp. 174–176.]

In regard to a substitution-group of the order  $n$  upon the same number of letters, I omitted to mention the important theorem that every substitution is regular (that is, either cyclical or composed of a number of cycles each of them of the same order).

Thus, in the group of 6 given in No. 1, writing  $a, b, c, d, e, f$  in place of  $1, \alpha, \beta, \gamma, \delta, \epsilon$ , the substitutions of the group are  $1, ace \cdot bfd, aec \cdot bdf, ab \cdot cd \cdot ef, ad \cdot be \cdot cf, af \cdot bc \cdot de$ .

Let the letters be represented by points; a change  $a$  into  $b$  will be represented by a directed line (line with an arrow) joining the two points; and therefore a cycle  $abc$ , that is,  $a$  into  $b$ ,  $b$  into  $c$ ,  $c$  into  $a$ , by the three sides of the trilateral  $abc$ , with the three arrows pointing accordingly, and similarly for the cycles  $abcd$ , &c. The cycle  $ab$  means  $a$  into  $b$ ,  $b$  into  $a$ , and we have here the line  $ab$  with a two-headed arrow pointing both ways; such a line may be regarded as a bilateral. A substitution is thus represented by a multilateral or system of multilaterals, each side with its arrow; and in the case of a regular substitution the multilaterals (if more than one) have each of them the same number of sides. To represent two or more substitutions we require different colours, the multilaterals belonging to any one substitution being of the same colour.

In order to represent a group we need to represent only independent substitutions thereof; that is, substitutions such that no one of them can be obtained from the others by compounding them together in any manner. I take as an example a group

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of the order 12 upon 12 letters, where the number of independent substitutions is = 2. See the diagram, wherein the continuous lines represent black lines, and the dotted lines, red lines.

[Figure: a planar diagram of 12 labelled points  $a, b, c, d, e, f, g, h, i, j, k, l$  arranged in two rows of alternating triangles; continuous (“black”) directed edges form four triangles representing one substitution of order 3, and dotted (“red”) directed edges form four triangles representing a second independent substitution of order 3. Each vertex has exactly one outgoing black and one outgoing dotted edge.]

The diagram is drawn, in the first instance, with the arrows but without the letters, which are then affixed at pleasure; viz. the form of the group is quite independent of the way in which this is done, though the group itself is of course dependent upon it. The diagram shows two substitutions, each of them of the third order: one is represented by the black triangles, and the other by the dotted triangles. It will be observed that there is from each point of the diagram (that is, in the direction of the arrow) one and only one black line, and one and only one dotted line; hence a symbol  $B$ , “move along a black line,”  $B^2$ , “move successively along two black lines,”  $BR$  (read always from right to left), “move first along a dotted line and then along a black line,” has in every case a perfectly definite meaning and determines the path when the initial point is given; any such symbol may be spoken of as a “route.”

[Printed page 404 then sets out, in a tall table, the result of applying each of the 12 routes to the initial arrangement  $abcdefghijkl$ ; alongside it the cyclical decomposition of the corresponding substitution. The headings and entries are:]

|                 |                                                   |
|-----------------|---------------------------------------------------|
| $abcdefghijkl$  | 1                                                 |
| $fbcaedhigklj$  | $abc \cdot def \cdot ghi \cdot jkl \quad (= B)$   |
| $fcabdefihgjlk$ | $acb \cdot dfe \cdot gih \cdot jlk$               |
| $fdaegcjhikbl$  | $ad \cdot bl \cdot ch \cdot eg \cdot fj \cdot ik$ |
| $fhabgjdcklie$  | $ae h \cdot bjd \cdot cil \cdot fkg$              |
| $fheabgjdckli$  | $afl \cdot bkh \cdot cgd \cdot eij$               |
| $feabchdgkilj$  | $agj \cdot bfi \cdot cke \cdot dlh$               |
| $fibhkadcjglg$  | $ahc \cdot bdj \cdot cli \cdot fgk$               |
| $fkglidjbeah$   | $ai \cdot be \cdot cj \cdot dk \cdot fh \cdot gl$ |
| $ydblcijfkeah$  | $ajg \cdot bif \cdot cek \cdot dhl \quad (= R)$   |
| $fkglcjdbaeih$  | $ak \cdot bg \cdot cf \cdot di \cdot el \cdot hj$ |
| $fhdljagkecbi$  | $alf \cdot bhk \cdot cdg \cdot eji$               |

[The arrangement here is reproduced from the scan with the customary 19th-century compositor abbreviations preserved; the cycle decompositions in the right column give the structure of each of the 12 substitutions.]



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The diagram has a remarkable property, in virtue whereof it in fact represents a group. It may be seen that any route leading from some one point  $a$  to itself, leads also from every other point to itself, or say from  $b$  to  $b$ , from  $c$  to  $c$ ,  $\dots$ , and from  $l$  to  $l$ . We hence see that a route, applied in succession to the whole series of initial points or letters  $abcdefghijkl$ , gives a new arrangement of these letters, wherein no one of them occupies its original place; a route is thus, in effect, a substitution. Moreover, we may regard as distinct routes, those which lead from  $a$  to  $a$ , to  $b$ , to  $c$ ,  $\dots$ , to  $l$ , respectively. We have thus 12 substitutions (the first of them, which leaves the arrangement unaltered, being the substitution unity), and these 12 substitutions form a group. I omit the details of the proof; it will be sufficient to give the square obtained by means of the several routes, or substitutions, performed upon the primitive arrangement  $abcdefghijkl$ , and the cyclical expressions of the substitutions themselves: it will be observed that the substitutions are unity, 3 substitutions of the order (or index) 2, and 8 substitutions of the order (or index) 3.

It may be remarked that the group of 12 is really the group of the 12 positive substitutions upon 4 letters  $abcd$ , viz. these are 1,  $abc$ ,  $acb$ ,  $abd$ ,  $adb$ ,  $acd$ ,  $adc$ ,  $bcd$ ,  $bdc$ ,  $ab \cdot cd$ ,  $ac \cdot bd$ ,  $ad \cdot bc$ .

Cambridge, 16th May, 1878.

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### No. 3. The Newton-Fourier imaginary problem.

[From the *American Journal of Mathematics*, t. II. (1879), p. 97.]

The Newtonian method as completed by Fourier, or say the Newton-Fourier method, for the solution of a numerical equation by successive approximations, relates to an equation  $f(x) = 0$ , with real coefficients, and to the determination of a certain real root thereof  $a$  by means of an assumed approximate real value  $\xi$  satisfying prescribed conditions: we then, from  $\xi$ , derive a nearer approximate value  $\xi_1$  by the formula

$$\xi_1 = \xi - \frac{f(\xi)}{f'(\xi)},$$

and thence, in like manner,  $\xi_2, \xi_3, \xi_4, \dots$  approximating more and more nearly to the required root  $a$ .

In connexion herewith, throwing aside the restrictions as to reality, we have what I call the Newton-Fourier Imaginary Problem, as follows.

Take  $f(u)$ , a given rational and integral function of  $u$ , with real or imaginary coefficients;  $\xi$ , a given real or imaginary value, and from this derive  $\xi_1$  by the formula

$$\xi_1 = \xi - \frac{f(\xi)}{f'(\xi)},$$

and thence  $\xi_2, \xi_3, \xi_4, \dots$  each from the preceding one by the like formula.

A given imaginary quantity  $x + iy$  may be represented by a point the coordinates of which are  $(x, y)$ : the roots of the equation are thus represented by given points

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*Paper No. 694 (Desiderata and Suggestions) continues on PDF page 426.*

$A, B, C, \dots$ , and the values  $\xi, \xi_1, \xi_2, \dots$  by points  $P, P_1, P_2, \dots$  the first of which is assumed at pleasure, and the others each from the preceding one by the like given geometrical construction. The problem is to determine the regions of the plane such that,  $P$  being taken at pleasure anywhere within one region, we arrive ultimately at the point  $A$ ; anywhere within another region at the point  $B$ ; and so for the several points representing the roots of the equation.

The solution is easy and elegant in the case of a quadric equation: but the next succeeding case of the cubic equation appears to present considerable difficulty.

*Cambridge, March 3rd, 1879.*

#### NO. 4. THE MECHANICAL CONSTRUCTION OF CONFORMABLE FIGURES.

[From the *American Journal of Mathematics*, t. ii. (1879), p. 186.]

Is it possible to devise an apparatus for the mechanical construction of conformable figures; that is, figures which are similar as regards corresponding infinitesimal areas? The problem is to connect mechanically two points  $P_1$  and  $P_2$  in such wise that  $P_1$  (1) shall have two degrees of freedom (or be capable of moving over a plane area) its position always determining that of  $P_2$ : (2) that if  $P_1, P_2$  describe the infinitesimal lengths  $P_1Q_1, P_2Q_2$ , then the ratio of these lengths, and their mutual inclination, shall depend upon the position of  $P_1$ , but be independent of the direction of  $P_1Q_1$ : or what is the same thing, that if  $P_1$  describe uniformly an indefinitely small circle, then  $P_2$  shall also describe uniformly an indefinitely small circle, the ratio of the radii, and the relative position of the starting points in the two circles respectively, depending on the position of  $P_1$ .

Of course a pentagraph is a solution, but the two figures are in this case similar; and this is not what is wanted. Any unadjustable apparatus would give one solution only: the complete solution would be by an apparatus containing, suppose, a flexible lamina, so that  $P_1$  moving in a given right line, the path of  $P_2$  could be made to be any given curve whatever.

*Cambridge, July 9th, 1879.*

695.

A LINK-WORK FOR  $x^2$ : EXTRACT FROM A LETTER TO  
MR. SYLVESTER.

[From the *American Journal of Mathematics*, t. i. (1878), p. 386.]

I SUPPOSE the following is substantially your link-work for  $x^2$ . I use a slot to make  $D$  move in the line  $OA$ ; but this could be replaced by proper link-work. Supposing  $O$  and  $A$  fixed; the line  $OB$  is movable, and I wanted to have the distance  $OB$  measured in a fixed direction. This can be done by a hexagon  $OABQB'A'$  with equal sides, and two other equal links  $B'R$ ,  $BR$ : then of course, if  $O$ ,  $R$ ,  $Q$  are in line, the hexagon will be symmetrical as to  $OQ$ , and  $OB'$  will be equal to  $OB$ , and  $B'$  may be made to move in the fixed line  $OB'$ . If

$$\angle BOA = \frac{1}{2}\theta, \quad OA = AB = a, \quad AC = CD = \frac{1}{2}a,$$

then

$$OB = 2a \cos \frac{1}{2}\theta, \quad OD = a(1 + \cos \theta) = 2a \cos^2 \frac{1}{2}\theta,$$

or

$$2a \cdot OD = (OB)^2.$$

[Figure: hexagonal link-work diagram with vertices  $O$ ,  $A$ ,  $B$ ,  $Q$ ,  $B'$ ,  $A'$ , intermediate points  $R$  and  $C$ ,  $D$  on line  $OA$ ;  $D$  slides along  $OA$  illustrating the relation  $2a \cdot OD = (OB)^2$ .]

November 30, 1877.

696.

CALCULATION OF THE MINIMUM N.G.F. OF THE BINARY SEVENTHIC.

[From the *American Journal of Mathematics*, t. ii. (1879), pp. 71–84.]

For the binary seventhic  $(a, \dots | x, y)^7$  the number of the asyzygetic covariants  $(\alpha, \dots)^\theta(x, y)^\mu$ , or say of the deg-order  $(\theta \cdot \mu)$ , is given as the coefficient of  $a^\theta x^\mu$  in the function

$$\frac{1 - x^{-1}}{1 - ax^7 \cdot 1 - ax^5 \cdot 1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3} \cdot 1 - ax^{-5} \cdot 1 - ax^{-7}}$$

developed in ascending powers of  $a$ . See my “Ninth Memoir on Quantics,” *Phil. Trans.*, t. CLXI. (1871), pp. 17–50, [462].

This function is in fact

$$= A(x) - \frac{1}{x^2} A\left(\frac{1}{x}\right),$$

where, developing in ascending powers of  $a$ , the second term  $-\frac{1}{x^2} A\left(\frac{1}{x}\right)$  contains only negative powers of  $x$ , and it may consequently be disregarded: the number of asyzygetic covariants of the deg-order  $(\theta \cdot \mu)$  is thus equal to the coefficient of  $a^\theta x^\mu$  in the function  $A(x)$ , which function is for this reason called the Numerical Generating Function (N.G.F.) of the binary seventhic; and the function  $A(x)$  expressed as a fraction in its least terms is said to be the minimum N.G.F.

According to a theorem of Professor Sylvester’s (*Proc. Royal Soc.*, t. XXVIII. (1878), pp. 11–13), this minimum N.G.F. is of the form

$$\frac{Z_0 + aZ_1 + a^2Z_2 + \dots + a^{36}Z_{36}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}},$$

where  $Z_0, Z_1, \dots, Z_{36}$  are rational and integral functions of  $x$  of degrees not exceeding 14; and where, as will presently be seen, there is a symmetry in regard to the terms  $Z_0, Z_{36}; Z_1, Z_{35};$  &c., equidistant from the middle term  $Z_{18}$ , such that the terms  $Z_0, \dots, Z_{18}$  being known, the remaining terms  $Z_{19}, \dots, Z_{36}$  can be at once written down.

Using only the foregoing properties, I obtained for the N.G.F. an expression which I communicated to Professor Sylvester, and which is published, *Comptes Rendus*, t. LXXXVII. (1878), p. 505, but with an erroneous value for the coefficient of  $a^7$  and for that of the corresponding term  $a^{29}$ .<sup>1</sup> The correct value is Numerator of Minimum N.G.F. is =

$$\begin{aligned}
 &1 \\
 &+ a(-x - x^3 - x^5) \\
 &+ a^2(x^2 + x^4 + 2x^6 + x^8 + x^{10}) \\
 &+ a^3(-x^3 - x^5 - x^7 - x^{12}) \\
 &+ a^4(2x^4 + x^6 + x^{14}) \\
 &+ a^5(x + 3x^6 - x^8 - x^{10}) \\
 &+ a^6(-1 + 2x^4 - x^6 - x^8 + x^{10} - x^{14}) \\
 &+ a^7(4x + x^2 + 3x^6 - x^8 - x^{10}) \\
 &+ a^8(2 - x^2 - 3x^6 - 3x^8 - x^{10} - x^{14}) \\
 &+ a^9(x + 3x^2 + x^4 - x^6 + 2x^{10} + 2x^{12}) \\
 &+ a^{10}(-1 - 4x^2 - x^4 - 2x^6 - 3x^{10} - x^{14}) \\
 &+ a^{11}(x - 3x^2 + 2x^4 - x^6 - 2x^8 - x^{10} + x^{14}) \\
 &+ a^{12}(5 + x^2 - 4x^4 - 6x^6 - 4x^8 - x^{10} + 2x^{14}) \\
 &+ a^{13}(x - 4x^2 - 4x^4 - x^6 - x^8 - 4x^{12}) \\
 &+ a^{14}(2 - 3x^2 - x^4 - x^6 - 2x^8 - 3x^{10} - x^{14}) \\
 &+ a^{15}(3x - x^2 - x^4 - 7x^6 - 5x^8 - x^{10} - x^{12}) \\
 &+ a^{16}(6 + 3x^2 - 3x^4 - 4x^6 - 3x^8 - x^{10} - 5x^{14}) \\
 &+ a^{17}(-x - 2x^2 - 6x^4 - 8x^6 - 4x^8 - 3x^{10} - 4x^{14}) \\
 &+ a^{18}(2 + 6x^2 + x^4 + 2x^6 + 2x^8 + x^{10} + 6x^{12} + 2x^{14}) \\
 &+ a^{19}(4x - 3x^2 - 4x^4 - 8x^6 - 9x^8 - 2x^{10} - x^{14}) \\
 &+ a^{20}(5 - x^2 - 3x^4 - 4x^6 - 8x^8 - 3x^{10} - 6x^{14}) \\
 &+ a^{21}(-x - x^4 - 5x^6 - 7x^8 - x^{10} - x^{12} - 3x^{14}) \\
 &+ a^{22}(-1 - 3x^2 - 2x^4 - x^6 + x^{10} + 5x^{12} - x^{14}) \\
 &+ a^{23}(4x + x^2 - x^6 - 4x^8 - 4x^{10} + x^{14})
 \end{aligned}$$

<sup>1</sup>The existence of these errors was pointed out to me by Professor Sylvester in a letter dated 13th November, 1878.

$$\begin{aligned}
 &+ a^{24} (2 - x^4 - 4x^6 - 6x^8 - 4x^{10} + x^{14}) \\
 &+ a^{25} (x - x^2 - 2x^6 - x^8 + 2x^{10} + 3x^{12} + 5x^{14}) \\
 &+ a^{26} (-1 - 2x^4 - 2x^6 - x^8 + 4x^{10} - x^{14}) \\
 &+ a^{27} (2x + 2x^2 - x^4 - x^6 + x^8 + 4x^{10} - x^{14}) \\
 &+ a^{28} (-x - x^4 - 3x^6 - 3x^8 - x^{10} - 2x^{14}) \\
 &+ a^{29} (x^2 - x^4 - 3x^6 - x^8 - 4x^{14}) \\
 &+ a^{30} (-x^6 + x^{10} + 2x^{12} + 3x^{14}) \\
 &+ a^{31} (1 + x^4 + 2x^{14}) \\
 &+ a^{32} (-x - x^2 - x^4 - x^6) \\
 &+ a^{33} (x^2 + x^4 + 2x^6 + x^{10} + x^{14}) \\
 &+ a^{34} (-x^6 - x^{12} - x^{14}) \\
 &+ a^{35} \cdot x^{14}.
 \end{aligned}$$

Denominator (as mentioned before) is

$$= 1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}.$$

The method of calculation is as follows: write for a moment

$$\phi(a, x) = \frac{1 - x^{-1}}{1 - ax^7 \cdot 1 - ax^5 \cdot 1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3} \cdot 1 - ax^{-5} \cdot 1 - ax^{-7}},$$

then  $\phi(a, x)$ , developed in ascending powers of  $a$ , and rejecting from the result all negative powers of  $x$ , is

$$= \frac{Z_0 + aZ_1 + \cdots + a^{36}Z_{36}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}}$$

developed in like manner in ascending powers of  $a$ ; for the determination of the  $Z$ 's up to  $Z_{18}$  we require only the development of  $\phi(a, x)$  up to  $a^{18}$ ; and, assuming that each  $Z$  is at most of the degree 14 in  $x$ , we require the coefficients of the different powers of  $a$  in  $\phi(a, x)$  only up to  $x^{14}$ . Assuming then that  $\phi(a, x)$  developed in ascending powers of  $a$ , up to  $a^{18}$ , rejecting all negative powers of  $x$ , and all positive powers greater than  $x^{14}$ , is

$$= X_0 + aX_1 + \cdots + a^{18}X_{18},$$

we have

$$X_0 + aX_1 + \cdots + a^{18}X_{18} = \frac{Z_0 + aZ_1 + \cdots + a^{18}Z_{18}}{1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 \cdot 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20}},$$

or say

$$Z_0 + aZ_1 + \cdots + a^{18}Z_{18} = 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20} \cdot 1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 (X_0 + aX_1 + \cdots + a^{18}X_{18});$$

viz. developing here the right-hand side as far as  $a^{18}$ , but in each term rejecting the powers of  $x$  above  $x^{14}$ , the coefficients of the several powers  $a^0, a^1, a^2, \dots$ , give the

required values  $Z_0, Z_1, \dots, Z_{18}$ . We require, therefore, only to know the values of these functions  $X_0, X_1, \dots, X_{18}$ .

To make a break in the calculation, it is convenient to write

$$1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 (X_0 + aX_1 + \dots + a^{18}X_{18}) = Y_0 + aY_1 + \dots + a^{18}Y_{18};$$

putting then

$$1 - ax \cdot 1 - ax^3 \cdot 1 - ax^5 \cdot 1 - ax^7 = 1 - ap + a^2q - a^3r,$$

where (up to  $x^{14}$ )

$$\begin{aligned} p &= x + x^3 + x^5 + x^7, \\ q &= x^4 + x^6 + 2x^8 + x^{10} + x^{12}, \\ r &= x^9 + x^{11} + x^{13}, \end{aligned}$$

we have

$$Y_0 + aY_1 + a^2Y_2 + \dots + a^{18}Y_{18} = (1 - ap + a^2q - a^3r)(X_0 + aX_1 + a^2X_2 + \dots + a^{18}X_{18}).$$

The values of  $Y_0, Y_1, \dots, Y_{18}$  then are

|  | $Y_0$ | $Y_1$   | $Y_2$   | $Y_3$   | $\dots$ | $Y_{18}$   |
|--|-------|---------|---------|---------|---------|------------|
|  | $X_0$ | $X_1$   | $X_2$   | $X_3$   |         | $X_{18}$   |
|  |       | $-pX_0$ | $-pX_1$ | $-pX_2$ |         | $-pX_{17}$ |
|  |       |         | $+qX_0$ | $+qX_1$ |         | $+qX_{16}$ |
|  |       |         |         | $-rX_0$ |         | $-rX_{15}$ |

the values being taken to  $x^{14}$  only; and we then have

$$Z_0 + aZ_1 + \dots + a^{18}Z_{18} = 1 - a^4 \cdot 1 - a^8 \cdot 1 - a^{12} \cdot 1 - a^{20} (Y_0 + aY_1 + \dots + a^{18}Y_{18});$$

viz. the values of  $Z_0, Z_1, \dots, Z_{18}$  are obtained from those of the  $Y$ 's by adjoining the further  $-Y_{k-4}, -Y_{k-8}, -Y_{k-12}, -Y_{k-20}$  terms (with their pairwise products  $+Y_{k-12}, +2Y_{k-20}$  etc.) as set out at length in the original table.

The rule of symmetry, before referred to, is that the coefficient  $Z_{18+p}$  of  $a^{18+p}$  is obtained from the coefficient  $Z_p$  of  $a^p$  by changing each power  $x^q$  into  $x^{14-q}$ , the coefficients being unaltered; in particular  $Z_{18}$ , the coefficient of  $a^{18}$ , must remain

unaltered when each power  $x^q$  is changed into  $x^{14-q}$ ; and the verification thus obtained of the value

$$2Z_{18} = 2 + 6x^2 + x^4 + 2x^6 + 2x^8 + x^{10} + 6x^{12} + 2x^{14}$$

is in fact almost a complete verification of the whole work. Some other verifications, which present themselves in the course of the work, will be referred to further on.

We have, therefore, to calculate the coefficients  $X_0, X_1, \dots, X_{18}$ ; the function  $\phi(a, x)$  regarded as a function of  $a$  is at once decomposed into simple fractions; viz. we have

$$\begin{aligned} \phi(a, x) &= \frac{1 - x^{-1}}{1 - ax^7 \cdot 1 - ax^5 \cdot 1 - ax^3 \cdot 1 - ax \cdot 1 - ax^{-1} \cdot 1 - ax^{-3} \cdot 1 - ax^{-5} \cdot 1 - ax^{-7}} \\ &= \frac{x^{56}}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10} \cdot 1 - x^{12} \cdot 1 - x^{14}} \cdot \frac{1}{1 - ax^7} \\ &\quad - \frac{x^{40}}{1 - x^2 \cdot (1 - x^4)^2 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10}} \cdot \frac{1}{1 - ax^5} \\ &\quad + \frac{x^{24}}{1 - x^2 \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8} \cdot \frac{1}{1 - ax^3} \\ &\quad - \frac{x^8}{(1 - x^2) \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8} \cdot \frac{1}{1 - ax} \end{aligned}$$

(the terms in  $\frac{1}{1 - ax^{-1}}, \dots, \frac{1}{1 - ax^{-7}}$  produce only negative powers of  $x$  and are accordingly discarded).

In order to obtain the expansion of  $\phi(a, x)$  in the assumed form of an expansion in ascending powers of  $a$ , we must of course expand the simple fractions  $\frac{1}{1 - ax^k}$ , &c., in ascending powers of  $a$ , but it requires a little consideration to see that we must also expand the  $x$ -coefficients of these simple fractions in ascending powers of  $x$ . For instance, as regards the term independent of  $a$ , here developing the several coefficients as far as  $x^{14}$ , the last five terms give (see post)

$$\begin{aligned} &+ x^8 + x^{10} + 3x^{12} + 5x^{14} + 9x^{16} \\ &- x^6 - x^8 - 3x^{10} - 5x^{12} - 8x^{14} - 11x^{16} - 15x^{18} - 24x^{20} \\ &+ 1 + x^2 + 2x^4 + 3x^6 + 5x^8 + 7x^{10} + 11x^{12} + 14x^{14} + 20x^{16} + 26x^{18} \\ &- x^4 - x^6 - x^8 - 2x^{10} - 2x^{12} - 4x^{14} - 4x^{16} - 6x^{18} - 7x^{20} - 10x^{22} \\ &= -x^{-2} + 1 \end{aligned}$$

viz. the sum is  $= -1 \cdot x^{-2}$  as it should be.<sup>2</sup>

<sup>2</sup>To give the last degree of perfection to the beautiful method of Professor Cayley it would seem desirable that a proof should be given of the principle illustrated by the example in the text, and the nature of the mischief resulting from its neglect clearly pointed out.—EDS. OF THE A. J. M.



The expansion is required only as far as  $x^{14}$ : the first four terms are therefore to be disregarded, and, writing for shortness

$$\begin{aligned} E &= \frac{1}{1 - x^2 \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8}, \\ F &= \frac{1}{1 - x^2 \cdot (1 - x^4)^2 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10}}, \\ G &= \frac{1}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10} \cdot 1 - x^{12}}, \\ H &= \frac{1}{1 - x^2 \cdot 1 - x^4 \cdot 1 - x^6 \cdot 1 - x^8 \cdot 1 - x^{10} \cdot 1 - x^{12} \cdot 1 - x^{14}}, \end{aligned}$$

we have

$$\phi(a, x) = \frac{x^8 E}{1 - ax} - \frac{x^{24} F}{1 - ax^3} + \frac{x^{40} G}{1 - ax^5} - \frac{x^{56} H}{1 - ax^7},$$

which is

$$\begin{aligned} &= x^8 E (1 + ax^{-1} + a^2 x^{-2} + \dots) \\ &\quad - x^{24} F (1 + ax^{-3} + a^2 x^{-6} + \dots) \\ &\quad + x^{40} G (1 + ax^{-5} + a^2 x^{-10} + \dots) \\ &\quad - x^{56} H (1 + ax^{-7} + a^2 x^{-14} + \dots), \end{aligned}$$

where the several series are to be continued up to  $a^{18}$ , and, after substituting for  $E, F, G, H$  their expansions in ascending powers of  $x$ , we are to reject negative powers of  $x$ , and also powers higher than  $x^{14}$ . The functions  $E, F, G, H$  contain each of them only even powers of  $x$ , and it is easy to see that we require the expansions up to  $x^{32}, x^{64}, x^{96}$  and  $x^{128}$  respectively. For the sake of a verification, I in fact calculated  $E, F$  up to  $x^{64}$  and  $G, H$  up to  $x^{142}$ : viz. we have

$$(1 - x^{10})E = (1 - x^8)F,$$

from the coefficients of  $E$  we have those of  $(1 - x^{10})E$ , and in the process of calculating  $F$  we have at the last step but one the coefficients of  $(1 - x^8)F$ , the agreement of the two sets being the verification; similarly,

$$(1 - x^{12})G = (1 - x^6)H$$

gives a verification. The process for the calculation of  $E$ ,

$$\frac{1}{1 - x^2 \cdot (1 - x^4)^2 \cdot (1 - x^6)^2 \cdot 1 - x^8},$$

is as follows:

| Ind. $x$             | 0 | 2 | 4 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 22  |
|----------------------|---|---|---|---|---|----|----|----|----|----|----|-----|
| $(1 - x^2)^{-1}$     | 1 | 1 | 1 | 1 | 1 | 1  | 1  | 1  | 1  | 1  | 1  | 1   |
|                      |   | 1 | 1 | 2 | 2 | 3  | 3  | 4  | 4  | 5  | 5  | 5   |
| $(1 - x^4)^{-1}$     | 1 | 1 | 2 | 2 | 3 | 3  | 4  | 4  | 5  | 5  | 6  | 6   |
|                      |   | 1 | 1 | 3 | 3 | 6  | 6  | 10 | 10 | 15 | 15 | 21  |
| $(1 - x^4)^{-1}$     | 1 | 1 | 3 | 3 | 6 | 6  | 10 | 10 | 15 | 15 | 21 | 21  |
|                      |   | 1 | 1 | 3 | 4 | 7  | 9  | 14 | 17 | 24 | 29 | 38  |
| $(1 - x^6)^{-1}$     | 1 | 1 | 3 | 4 | 7 | 9  | 14 | 17 | 24 | 29 | 38 | 45  |
|                      |   | 1 | 1 | 3 | 5 | 8  | 12 | 19 | 25 | 36 | 48 | 63  |
| $(1 - x^6)^{-1}$     | 1 | 1 | 3 | 5 | 8 | 12 | 19 | 25 | 36 | 48 | 63 | 81  |
|                      |   | 1 | 1 | 3 | 5 | 9  | 13 | 22 | 30 | 45 | 61 | 85  |
| $E = (1 - x^8)^{-1}$ | 1 | 1 | 3 | 5 | 9 | 13 | 22 | 30 | 45 | 61 | 85 | 111 |

the alternate lines giving the developments of the functions

$$(1 - x^2)^{-1}, (1 - x^4)^{-2}, (1 - x^4)^{-2}(1 - x^6)^{-1}, (1 - x^4)^{-2}(1 - x^6)^{-2}, \dots,$$

which are the products of the  $x$ -functions down to any particular line. And in like manner we have the expansions of the other functions  $F, G, H$  respectively. I give first the expansions of  $E, F, G, H$ ; next the calculation of the  $X$ 's; then the calculation of the  $Y$ 's: and from these the  $Z$ 's up to  $Z_{18}$ , or coefficients of the powers  $a^0, a^1, \dots, a^{18}$  in the numerator of the N.G.F. are at once found; and the coefficients of the remaining powers  $a^{19} \dots a^{36}$  are then deduced, as already mentioned.

Writing in the formula  $x = 0$ , we have, for the numerator of the N.G.F. of the invariants, the expression

$$1 - a^4 + 2a^8 - a^{12} + 5a^{16} + 2a^{18} + 6a^{20} + 2a^{22} + 5a^{24} - a^{28} + 2a^{32} - a^{36} + a^{40}$$

agreeing with a result in my "Second Memoir on Quantics," *Phil. Trans.*, t. CXLVI. (1856), [Number 141, vol. II. in this Collection, p. 266]; this, then, was a known result, and it affords a verification, not only of the terms in  $a^0$ , but also of those in  $x^0$ . Thus, in calculating the foregoing expression of the numerator, we obtain  $Z_4 = +(2x^4 + x^6 + x^{14})$ , viz. the term is

$$a^4(2x^4 + x^6 + x^{14}),$$

and we thence have the corresponding term  $a^{34}(1 + x^8 + 2x^{10})$ , which, when  $x = 0$ , becomes  $= a^{34}$ , a term of the numerator for the invariants: and the term  $1x^{14}$  of  $Z_4$

is thus verified, viz. so soon as, in the calculation, we arrive at this term, we know that it is right, and the calculation up to this point is, to a considerable extent, verified. And similarly, in continuing the calculation, we arrive at other terms which are verified in the like manner.

*Expansions of the Functions E, F, G, H.*

| Ind. <i>x</i> | <i>E</i> | <i>F</i> | <i>G</i> | <i>H</i> | Ind. <i>x</i> | <i>E</i> | <i>F</i> | <i>G</i> | <i>H</i> | Ind. <i>x</i> | <i>H</i> |
|---------------|----------|----------|----------|----------|---------------|----------|----------|----------|----------|---------------|----------|
| 0             | 1        | 1        | 1        | 1        | 16            | 45       | 36       | 20       | 6        | 108           | 2265     |
| 2             | 1        | 1        | 1        |          | 18            | 61       | 47       | 26       | 7        | 110           | 2426     |
| 4             | 3        | 3        | 2        | 1        | 20            | 85       | 66       | 35       | 10       | 112           | 2623     |
| 6             | 5        | 4        | 3        | 1        | 22            | 111      | 84       | 44       | 11       | 114           | 2807     |
| 8             | 9        | 8        | 5        | 2        | 24            |          | 113      | 58       | 16       | 116           | 3026     |
| 10            | 13       | 11       | 7        | 2        | 26            |          | 141      | 71       | 17       | 118           | 3232     |
| 12            | 22       | 18       | 11       | 4        | 28            |          | 183      | 90       | 23       | 120           | 3479     |
| 14            | 30       | 24       | 14       | 4        | 30            |          | 225      | 110      | 26       | 122           | 3708     |

| Ind. <i>x</i> | <i>F</i> | <i>G</i> | <i>H</i> | Ind. <i>x</i> | <i>G</i> | <i>H</i> | Ind. <i>x</i> | <i>H</i> |
|---------------|----------|----------|----------|---------------|----------|----------|---------------|----------|
| 32            | 284      | 136      | 33       | 70            | 2172     | 419      | 124           | 3981     |
| 34            | 344      | 163      | 37       | 72            | 2432     | 472      | 126           | 4240     |
| 36            | 425      | 199      | 47       | 74            | 2702     | 515      | 128           | 4541     |
| 38            | 508      | 235      | 52       | 76            | 3009     | 576      | 130           | 4828     |
| 40            | 617      | 282      | 64       | 78            | 3331     | 629      | 132           | 5164     |
| 42            | 729      | 331      | 72       | 80            | 3692     | 699      | 134           | 5481     |
| 44            | 872      | 391      | 86       | 82            | 4070     | 760      | 136           | 5850     |
| 46            | 1020     | 454      | 96       | 84            | 4494     | 843      | 138           | 6204     |
| 48            | 1205     | 532      | 115      | 86            | 4935     | 913      | 140           | 6609     |
| 50            | 1397     | 612      | 127      | 88            | 5427     | 1007     | 142           | 6998     |
| 52            | 1632     | 709      | 149      | 90            | 5942     | 1091     |               |          |
| 54            | 1877     | 811      | 166      | 92            | 6510     | 1197     |               |          |
| 56            | 2172     | 931      | 192      | 94            | 7104     | 1293     |               |          |
| 58            | 2480     | 1057     | 212      | 96            | 7760     | 1416     |               |          |
| 60            | 2846     | 1206     | 245      | 98            | 8442     | 1525     |               |          |
| 62            | 3228     | 1360     | 269      | 100           | 9192     | 1663     |               |          |
| 64            | 3677     | 1540     | 307      | 102           | 9975     | 1790     |               |          |
| 66            |          | 1729     | 338      | 104           | 10829    | 1945     |               |          |
| 68            |          | 1945     | 382      | 106           |          | 2088     |               |          |

Calculation of the  $X$ 's.

Ind.  $x$  even or odd according as suffix  $X$  is even or odd.

(Detailed columnar layout in the source: each block lists the partial-fraction contributions in the columns  $0_1, 2_3, 4_5, 6_7, 8_9, 10_{11}, 12_{13}, 14$ , and the row sums give  $X_0, X_1, \dots, X_{18}$ .)

|         |                |                |                |                |                |                  |                  |     |
|---------|----------------|----------------|----------------|----------------|----------------|------------------|------------------|-----|
|         | 0 <sub>1</sub> | 2 <sub>3</sub> | 4 <sub>5</sub> | 6 <sub>7</sub> | 8 <sub>9</sub> | 10 <sub>11</sub> | 12 <sub>13</sub> | 14  |
|         |                |                | −1             | −1             | −3             | 1                | 1                | 3   |
|         |                |                |                |                | −4             | −8               |                  | −11 |
|         | 1              | 1              | 2              | 3              | 5              | 7                | 11               | 14  |
|         |                | −1             | −1             | −2             | −2             | −4               | −4               | −6  |
| $X_0 =$ | 1              |                |                |                |                |                  |                  |     |
|         |                |                |                | 1              | 1              | 3                |                  |     |
|         | −1             | −1             | −3             | −4             | −8             | −11              | −18              |     |
|         | 3              | 5              | 7              | 11             | 14             | 20               | 26               |     |
|         | −2             | −4             | −4             | −6             | −7             | −10              | −11              |     |
| $X_1 =$ |                |                |                | +1             |                |                  |                  |     |
|         | −1             | −3             | −4             | −8             |                | 1                | 1                | 3   |
|         | 5              | 7              | 11             | 14             | 20             | −11              | −18              | −24 |
|         | −6             | −7             | −10            | −11            | 26             | 35               | 44               | 58  |
|         |                | +1             |                | +1             | −16            | −17              | −23              | −26 |
| $X_2 =$ |                |                |                |                |                | +1               |                  | +1  |
|         |                |                |                | 1              | 1              | 3                | 5                | 9   |
|         | −4             | −8             | −11            | −18            | −24            | −36              | −47              | −66 |
|         | 20             | 26             | 35             | 44             | 58             | 71               | 90               | 110 |
|         | −16            | −17            | −23            | −26            | −33            | −37              | −47              | −52 |
| $X_3 =$ |                |                | +1             | +1             | +1             | +2               | +1               | +1  |

(Further blocks for  $X_4, X_5, X_6$  on this page; the values obtained are):

$$X_4 = 1; \quad X_5 = 1 + 0 + 3(\cdot); \quad X_6 = 1 + 2(\cdot).$$

[The full ledger continues with several lines per  $X_k$  giving the contributions  $-x^8E, +x^{24}F, -x^{40}G, +x^{56}H$  for each power of  $x$ .]

|                   |                |                |                |                |                |                  |                  |       |
|-------------------|----------------|----------------|----------------|----------------|----------------|------------------|------------------|-------|
|                   | 0 <sub>1</sub> | 2 <sub>3</sub> | 4 <sub>5</sub> | 6 <sub>7</sub> | 8 <sub>9</sub> | 10 <sub>11</sub> | 12 <sub>13</sub> | 14    |
|                   |                | 1              |                | 1              | 3              | 5                | 9                | 13    |
|                   | −24            | −36            | −47            | −66            | −84            | −113             | −141             | −183  |
|                   | 110            | 136            | 163            | 199            | 235            | 282              | 331              | 391   |
|                   | −86            | −96            | −115           | −127           | −149           | −166             | −191             | −212  |
| X <sub>7</sub> =  | 3+             | 4+             | 4+             | 7+             | 9+             | 10+              | 11+              | 13    |
|                   |                | 1              | 1              | 3              | 5              | 9                | 13               | 22    |
|                   | −66            | −84            | −113           | −141           | −183           | −225             | −284             | −344  |
|                   | 282            | 331            | 391            | 454            | 532            | 612              | 709              | 811   |
|                   | −212           | −245           | −269           | −307           | −338           | −382             | −419             | −472  |
| X <sub>8</sub> =  | 4+             | 3+             | 10+            | 9+             | 16+            | 14+              | 19+              | 17    |
|                   | 1              | 1              | 3              | 5              | 9              | 13               | 22               |       |
|                   | −113           | −141           | −183           | −225           | −284           | −344             | −425             |       |
|                   | 454            | 532            | 612            | 709            | 811            | 931              | 1057             |       |
|                   | −338           | −382           | −419           | −472           | −515           | −576             | −629             |       |
| X <sub>9</sub> =  | 4+             | 10+            | 13+            | 17+            | 21+            | 24+              | 25               |       |
|                   |                | 1              | 1              | 3              | 5              | 9                | 13               | 22    |
|                   | −141           | −183           | −225           | −284           | −344           | −425             | −508             | −617  |
|                   | 612            | 709            | 811            | 931            | 1057           | 1206             | 1360             | 1540  |
|                   | −472           | −515           | −576           | −629           | −699           | −760             | −843             | −913  |
| X <sub>10</sub> = |                | +12            | +13            | +23            | +23            | +34              | +31              | +40   |
|                   | 1              | 3              | 5              | 9              | 13             | 22               | 30               |       |
|                   | −225           | −284           | −344           | −425           | −508           | −617             | −729             |       |
|                   | 931            | 1057           | 1206           | 1360           | 1540           | 1729             | 1945             |       |
|                   | −699           | −760           | −843           | −913           | −1007          | −1091            | −1197            |       |
| X <sub>11</sub> = | 8+             | 16+            | 24+            | 31+            | 38+            | 43+              | 49               |       |
|                   |                | 1              | 3              | 5              | 9              | 13               | 22               | 30    |
|                   | −284           | −344           | −425           | −508           | −617           | −729             | −872             | −1020 |
|                   | 1206           | 1360           | 1540           | 1729           | 1945           | 2172             | 2432             | 2702  |
|                   | −913           | −1007          | −1091          | −1197          | −1293          | −1416            | −1525            | −1663 |
| X <sub>12</sub> = | 10+            | 12+            | 29+            | 33+            | 48+            | 49+              | 65+              | 64    |

|            | 0 <sub>1</sub> | 2 <sub>3</sub> | 4 <sub>5</sub> | 6 <sub>7</sub> | 8 <sub>9</sub> | 10 <sub>11</sub> | 12 <sub>13</sub> | 14    |
|------------|----------------|----------------|----------------|----------------|----------------|------------------|------------------|-------|
| $X_{13} =$ | 3              | 5              | 9              | 13             | 22             | 30               | 45               |       |
|            | −425           | −508           | −617           | −729           | −872           | −1020            | −1205            |       |
|            | 1729           | 1945           | 2172           | 2432           | 2702           | 3009             | 3331             |       |
|            | −1293          | −1416          | −1525          | −1663          | −1790          | −1945            | −2088            |       |
|            | 14+            | 26+            | 39+            | 53+            | 62+            | 74+              | 83               |       |
| $X_{14} =$ | 3              | 5              | 9              | 13             | 22             | 30               | 45               | 61    |
|            | −508           | −617           | −729           | −872           | −1020          | −1205            | −1397            | −1632 |
|            | 2172           | 2432           | 2702           | 3009           | 3331           | 3692             | 4070             | 4494  |
|            | −1663          | −1790          | −1945          | −2088          | −2265          | −2426            | −2623            | −2807 |
|            | 4+             | 30+            | 37+            | 62+            | 68+            | 91+              | 95+              | 116   |
| $X_{15} =$ | 5              | 9              | 13             | 22             | 30             | 45               | 61               |       |
|            | −729           | −872           | −1020          | −1205          | −1397          | −1632            | −1877            |       |
|            | 3009           | 3331           | 3692           | 4070           | 4494           | 4935             | 5427             |       |
|            | −2265          | −2426          | −2623          | −2807          | −3026          | −3232            | −3479            |       |
|            | 20+            | 42+            | 62+            | 80+            | 101+           | 116+             | 132              |       |
| $X_{16} =$ | 5              | 9              | 13             | 22             | 30             | 45               | 61               | 85    |
|            | −872           | −1020          | −1205          | −1397          | −1632          | −1877            | −2172            | −2480 |
|            | 3692           | 4070           | 4494           | 4935           | 5427           | 5942             | 6510             | 7104  |
|            | −2807          | −3026          | −3232          | −3479          | −3708          | −3981            | −4240            | −4541 |
|            | 18+            | 33+            | 70+            | 81+            | 117+           | 129+             | 159+             | 168   |
| $X_{17} =$ | 9              | 13             | 22             | 30             | 45             | 61               | 85               |       |
|            | −1205          | −1397          | −1632          | −1877          | −2172          | −2480            | −2846            |       |
|            | 4935           | 5427           | 5942           | 6510           | 7104           | 7760             | 8442             |       |
|            | −3708          | −3981          | −4240          | −4541          | −4828          | −5164            | −5481            |       |
|            | 31+            | 62+            | 92+            | 122+           | 149+           | 177+             | 200              |       |
| $X_{18} =$ | 9              | 13             | 22             | 30             | 45             | 61               | 85               | 111   |
|            | −1397          | −1632          | −1877          | −2172          | −2480          | −2846            | −3228            | −3677 |
|            | 5942           | 6510           | 7104           | 7760           | 8442           | 9192             | 9975             | 10829 |
|            | −4541          | −4828          | −5164          | −5481          | −5850          | −6204            | −6609            | −6998 |
|            | 13+            | 63+            | 85+            | 137+           | 157+           | 203+             | 223+             | 265   |

*Calculation of the  $Y$ 's.*

Ind.  $x$  even or odd as suffix  $X$  is even or odd.

|         | $0_1$ | $2_3$ | $4_5$ | $6_7$ | $8_9$ | $10_{11}$ | $12_{13}$ |
|---------|-------|-------|-------|-------|-------|-----------|-----------|
| $Y_0 =$ | 1     |       |       |       |       |           |           |
| $Y_1 =$ | -1    | -1    | -1    | -1    |       |           |           |
| $Y_2 =$ | -1    | -1    | -1    | 0     | 0     | 0         | 0         |
|         | 0     | 1     | 0     | 1     | 0     | 1         | 0         |
|         |       | -1    | -1    | -1    | -1    | -1        | -1        |
|         |       | 1     | 1     | 2     | 1     | 1         | 1         |
| $Y_3 =$ | 0     | 1     | 1     | 1     | 1     | 1         | 0         |
|         |       | -1    | -1    | -2    | -2    | -2        | -2        |
|         |       |       | -1    |       | -1    | -1        | -1        |
| $Y_4 =$ | 0     | 0     | 0     | -1    | -1    | -1        | -1        |
|         | 1     | 0     | 3     | 1     | 3     | 2         | 3         |
|         |       | -1    | -2    | -3    | -5    | -5        | -5        |
|         |       |       | 1     | 1     | 3     | 2         | 4         |
| $Y_5 =$ | 1     | 0     | +2    | 0     | +1    |           | 0         |
|         | 1     | 2     | 3     | 4     | 4     | 5         | 4         |
|         | -1    | -1    | -4    | -5    | -7    | -9        | -9        |
|         |       | +1    | -     | 1     | 2     | 4         | 6         |
| $Y_6 =$ | 0     | +1    | -1    | 0     | -1    | -1        | 0         |
|         | 4     | 2     | 4     | 7     | 5     | 8         | 7         |
|         | -1    | -3    | -7    | -10   | -13   | -16       | -17       |
|         |       | +1    | -     | 5     | 5     | 11        | 10        |
| $Y_7 =$ | 0     | +3    | 0     | +2    | 0     | +1        | 0         |

|            | $0_1$ | $2_3$ | $4_5$ | $6_7$ | $8_9$ | $10_{11}$ | $12_{13}$ | $14$ |
|------------|-------|-------|-------|-------|-------|-----------|-----------|------|
| $Y_8 =$    | 3     | 4     | 7     | 9     | 10    | 11        | 13        |      |
|            |       | -4    | -6    | -13   | -18   | -22       | -27       |      |
|            |       |       | 1     | 3     | 7     | 12        | 17        |      |
|            |       |       |       |       | -1    | -1        | -4        |      |
|            | 3     | 0     | +2    | -1    | -2    | 0         | -1        |      |
| $Y_9 =$    | 4     | 3     | 10    | 9     | 16    | 14        | 19        | 17   |
|            | -3    | -7    | -14   | -23   | -30   | -37       | -43       |      |
|            |       | 4     | 6     | 17    | 20    | 33        |           |      |
|            |       |       | -1    | -1    | -3    | -6        |           |      |
|            | 1     | 0     | +1    | +2    | +3    | +4        |           |      |
| $Y_{10} =$ | 0     | +3    | -1    | -2    | 0     | -2        | +1        |      |
|            | 12    | 13    | 23    | 23    | 34    | 31        | 49        |      |
|            |       | -14   | -27   | -44   | -61   | -75       | -87       |      |
|            |       |       | 4     | 7     | 21    | 29        | 52        | 61   |
|            |       |       |       |       |       | -3        | -7        | -14  |
| $Y_{11} =$ | 0     | +8    | +3    | +3    | 0     | -         | 1         | +1   |
| $Y_{12} =$ | 8     | 16    | 24    | 31    | 38    | 43        | 49        | 0    |
|            | -12   | -25   | -48   | -71   | -93   | -111      |           |      |
|            |       | 4     | 14    | 31    | 54    | 78        |           |      |
|            |       |       |       | -4    | -7    | -17       |           |      |
| $Y_{13} =$ | 8     | +4    | +3    | -3    | -6    | -3        | -1        |      |
|            | 10    | 29    | 33    | 48    | 49    | 65        | 64        |      |
|            |       | -8    | -24   | -65   | -79   | -100      | -136      | -161 |
|            |       |       | 12    | 25    | 60    | 84        | 128       |      |
|            |       |       |       |       | -4    | -14       | -27       | -87  |
|            | 10    | +4    | +5    | -3    | -6    | -4        | -1        | +4   |



|                   |                |                |                |                |                |                  |                  |     |
|-------------------|----------------|----------------|----------------|----------------|----------------|------------------|------------------|-----|
|                   | 0 <sub>1</sub> | 2 <sub>3</sub> | 4 <sub>5</sub> | 6 <sub>7</sub> | 8 <sub>9</sub> | 10 <sub>11</sub> | 12 <sub>13</sub> | 14  |
|                   | 14             | 26             | 39             | 53             | 62             | 74               | 83               |     |
|                   | −10            | −22            | −51            | −84            | −122           | −159             | −195             |     |
|                   |                | 8              | 24             | 56             | 95             | 141              |                  |     |
|                   |                | −4             | −7             | −4             | −12            | −25              |                  |     |
|                   |                |                |                |                | −12            | −25              |                  |     |
| Y <sub>14</sub> = | 4              | +4             | +8             | +24            | +56            | +95              | +141             |     |
|                   | 20             | 30             | 37             | 62             | 68             | 91               | 95               | 116 |
|                   | −14            | −40            | −79            | −132           | −180           | −228             | −272             |     |
|                   |                | 16             | 22             | 61             | 96             | 161              | 204              |     |
|                   |                | +7             | −3             | −8             | −24            | −48              |                  |     |
| Y <sub>15</sub> = | 4              | +18            | +16            | +22            | +18            | +13              | +13              |     |
|                   | −18            | 42             | 62             | 80             | 101            | 116              | 132              |     |
|                   | 13             | −34            | −71            | −133           | −197           | −258             | −316             |     |
|                   |                | 8              | 14             | 40             | 93             | 158              | 233              |     |
|                   |                | +5             | −13            | −10            | −22            | −51              |                  |     |
|                   |                |                |                | −13            | −6             | −2               |                  |     |
| Y <sub>16</sub> = |                |                | +5             |                |                |                  |                  |     |
|                   | 33             | 70             | 81             | 117            | 129            | 159              | 168              |     |
|                   | −20            | −62            | −124           | −204           | −285           | −359             | −429             |     |
|                   | +13            | 4              | 34             | 75             | 163            | 238              | 350              |     |
|                   |                | +12            | −9             | −12            | −14            | −40              | −79              |     |
|                   |                |                |                |                | −7             | −2               | +10              |     |
| Y <sub>17</sub> = | 26             | 24             | −18            | −24            | −14            | −4               | +20              |     |
|                   | 62             | 92             | 122            | 149            | 177            | 200              |                  |     |
|                   | −51            | −121           | −202           | −301           | −397           | −486             |                  |     |
|                   | +11            | 20             | 62             | 144            | 246            | 367              |                  |     |
|                   |                | −9             | −18            | −4             | −12            | −34              | −71              |     |
|                   |                |                |                | −12            | −8             | +10              |                  |     |
| Y <sub>18</sub> = | 22             | −18            | −36            | −24            | +6             | +57              |                  |     |
|                   | 63             | 85             | 137            | 157            | 203            | 223              | 265              |     |
|                   | −31            | −93            | −185           | −307           | −425           | −540             | −648             |     |
|                   | +32            | 18             | 51             | 139            | 235            | 389              | 511              |     |
|                   |                | +10            | +3             | −11            | −20            | −62              | −124             |     |
|                   |                |                |                |                | −7             | +10              | +4               |     |

Cambridge, December 7th, 1878.

697.

ON THE DOUBLE  $\vartheta$ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 74–81.]

I HAVE sought to obtain, in forms which may be useful in regard to the theory of the double  $\vartheta$ -functions, the integral of the elliptic differential equation

$$\frac{dx}{\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}} + \frac{dy}{\sqrt{a-y \cdot b-y \cdot c-y \cdot d-y}} = 0 :$$

the present paper has immediate reference only to this differential equation; but, on account of the design of the investigation, I have entitled it as above.

We may for the general integral of the above equation take a particular integral of the equation

$$\frac{dx}{\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}} + \frac{dy}{\sqrt{a-y \cdot b-y \cdot c-y \cdot d-y}} \pm \frac{dz}{\sqrt{a-z \cdot b-z \cdot c-z \cdot d-z}} = 0;$$

viz. this particular integral, regarding therein  $z$  as an arbitrary constant, will be the general integral of the first mentioned equation. And we may further assume that  $z$  is the value of  $y$  corresponding to the value  $a$  of  $x$ .

I write for shortness

$$\begin{aligned} a-x, b-x, c-x, d-x &= \mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \\ a-y, b-y, c-y, d-y &= \mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1, \mathfrak{d}_1; \end{aligned}$$

and I write also  $(xy, bc, ad)$ , or more shortly  $(bc, ad)$  to denote the determinant

$$\begin{vmatrix} 1, & x+y, & xy \\ 1, & b+c, & bc \\ 1, & a+d, & ad \end{vmatrix}$$

we have of course  $(ad, bc) = -(bc, ad)$ , and there are thus the three distinct determinants  $(ad, bc)$ ,  $(bd, ac)$  and  $(cd, ab)$ .

We have then for each of the functions

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}$$

a set of four equivalent expressions, the whole system being

$$\begin{aligned} \sqrt{\frac{a-z}{d-z}} &= \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{a\bar{d}b_1c_1} + \sqrt{a_1\bar{d}_1bc} \}}{(bc, ad)} = \frac{\sqrt{a-b \cdot a-c} (x-y)}{\sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc}} \\ &= \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{a\bar{b}c_1\bar{d}_1} + \sqrt{a_1\bar{b}_1c\bar{d}} \}}{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} - (b-d)\sqrt{a\bar{c}a_1c_1}} = \frac{\sqrt{a-b \cdot a-c} \{ \sqrt{a\bar{c}b_1\bar{d}_1} + \sqrt{a_1\bar{c}_1b\bar{d}} \}}{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} - (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}} \\ \sqrt{\frac{b-z}{d-z}} &= \sqrt{-\frac{a-b}{a-d}} \frac{\{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} + (b-d)\sqrt{a\bar{c}a_1c_1}\}}{(bc, ad)} = \sqrt{-\frac{a-b}{a-d}} \frac{\{\sqrt{a\bar{b}c_1\bar{d}_1} - \sqrt{a_1\bar{b}_1c\bar{d}}\}}{\sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc}} \\ &= \sqrt{-\frac{a-b}{a-d}} \frac{(cd, ab)}{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} - (b-d)\sqrt{a\bar{c}a_1c_1}} = -\sqrt{-\frac{a-b}{a-d}} \frac{\{(a-d)\sqrt{b\bar{c}b_1c_1} + (b-c)\sqrt{a\bar{d}a_1\bar{d}_1}\}}{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} - (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}} \\ \sqrt{\frac{c-z}{d-z}} &= -\sqrt{-\frac{a-c}{a-d}} \frac{\{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} + (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}\}}{(bc, ad)} = -\sqrt{-\frac{a-c}{a-d}} \frac{\{\sqrt{a\bar{c}b_1\bar{d}_1} - \sqrt{a_1\bar{c}_1b\bar{d}}\}}{\sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc}} \\ &= -\sqrt{-\frac{a-c}{a-d}} \frac{\{(a-d)\sqrt{b\bar{c}b_1c_1} - (b-c)\sqrt{a\bar{d}a_1\bar{d}_1}\}}{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} - (b-d)\sqrt{a\bar{c}a_1c_1}} = \sqrt{-\frac{a-c}{a-d}} \frac{(bd, ac)}{(a-b)\sqrt{c\bar{d}c_1\bar{d}_1} - (c-d)\sqrt{a\bar{b}a_1\bar{b}_1}} \end{aligned}$$

The expressions in the like fourfold form for the functions  $\text{sn}(u+v)$ ,  $\text{cn}(u+v)$ ,  $\text{dn}(u+v)$  are given p. 63 of my *Treatise on Elliptic Functions*.

It is easy to verify first that the four expressions for the same function of  $z$  are identical, and next that the expressions for the three several functions

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}$$

are consistent with each other. For instance, comparing the first and second expressions of  $\sqrt{\frac{a-z}{d-z}}$ , the equation to be verified is

$$= a\bar{d}b_1c_1 - a_1\bar{d}_1bc - (x-y)(bc, ad),$$

which is at once shown to be true. Again comparing the first and second expressions for  $\sqrt{\frac{b-z}{d-z}}$ , we ought to have

$$\{(a-c)\sqrt{b\bar{d}b_1\bar{d}_1} + (b-d)\sqrt{a\bar{c}a_1c_1}\} \{ \sqrt{a\bar{d}b_1c_1} - \sqrt{a_1\bar{d}_1bc} \} = (bc, ad) \{ \sqrt{a\bar{b}c_1\bar{d}_1} - \sqrt{a_1\bar{b}_1c\bar{d}} \}.$$

Here the product on the left-hand side is

$$= (a - c) \{ \mathfrak{b}_1 \mathfrak{d} \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c}_1 \mathfrak{d}_1} - \mathfrak{b} \mathfrak{d}_1 \sqrt{\mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c} \mathfrak{d}} \} + (b - d) \{ -\mathfrak{a}_1 \mathfrak{c} \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c}_1 \mathfrak{d}_1} + \mathfrak{a} \mathfrak{c}_1 \sqrt{\mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c} \mathfrak{d}} \},$$

viz. this is

$$= \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c}_1 \mathfrak{d}_1} \{ (a - c) \mathfrak{b}_1 \mathfrak{d} - (b - d) \mathfrak{a}_1 \mathfrak{c} \} - \sqrt{\mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c} \mathfrak{d}} \{ (a - c) \mathfrak{b} \mathfrak{d}_1 - (b - d) \mathfrak{a} \mathfrak{c}_1 \},$$

and in this last expression the two terms in  $\{ \}$  are at once shown to be each  $= (bc, ad)$ ; whence the identity in question.

Comparing in like manner the first expressions for  $\sqrt{\frac{b-z}{d-z}}$  and  $\sqrt{\frac{c-z}{d-z}}$  respectively, we have

$$\begin{aligned} & - (b - d)(bc, ad)^2 \sqrt{-\frac{a-c}{a-d}} \frac{1}{z} \\ & = -(a - b)(a - c)(b - d) \{ \mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c} - 2 \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1} \}, \\ & - (d - a)(bc, ad)^2 \sqrt{-\frac{a-b}{a-d}} \frac{1}{z} \\ & = -(a - b) \{ (a - c)^2 \mathfrak{b} \mathfrak{d} \mathfrak{b}_1 \mathfrak{d}_1 + (b - d)^2 \mathfrak{a} \mathfrak{c} \mathfrak{a}_1 \mathfrak{c}_1 + 2(a - c)(b - d) \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1} \}, \end{aligned}$$

whence, adding, the radical on the right-hand side disappears; the whole equation divides by  $-(a - b)$ , and omitting this factor, the relation to be verified is

$$-(bc, ad)^2 = (a - c)^2 \mathfrak{b} \mathfrak{d} \mathfrak{b}_1 \mathfrak{d}_1 + (b - d)^2 \mathfrak{a} \mathfrak{c} \mathfrak{a}_1 \mathfrak{c}_1 - (a - c)(b - d) (\mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c})$$

the right-hand side is here

$$= -\{ (a - c) \mathfrak{b}_1 \mathfrak{d} - (b - d) \mathfrak{a}_1 \mathfrak{c} \} \{ (a - c) \mathfrak{b} \mathfrak{d}_1 - (b - d) \mathfrak{a} \mathfrak{c}_1 \},$$

and each of the two factors being  $= (bc, ad)$ , the identity is verified. It thus appears that the twelve equations are in fact equivalent to a single equation in  $x, y, z$ .

Writing in the several formulae  $x = a, b, c, d$  successively, they become

$$\begin{array}{cccc} x = a, & x = b, & x = c, & x = d, \\ \frac{a - z}{d - z} = \frac{\mathfrak{a}_1}{\mathfrak{d}_1}, & \frac{c - a \mathfrak{b}_1}{d - b \cdot \mathfrak{c}_1}, & \frac{b - a \mathfrak{c}_1}{d - c \cdot \mathfrak{b}_1}, & \frac{a - b \cdot a - c \cdot \mathfrak{d}_1}{d - b \cdot d - c \cdot \mathfrak{a}_1}, \\ \frac{b - z}{d - z} = \frac{\mathfrak{b}_1}{\mathfrak{d}_1}, & \frac{c - b \mathfrak{a}_1}{d - a \cdot \mathfrak{c}_1}, & \frac{b - a \cdot b - c \mathfrak{d}_1}{d - a \cdot d - c \cdot \mathfrak{b}_1}, & \frac{a - b \mathfrak{c}_1}{d - c \cdot \mathfrak{a}_1}, \\ \frac{c - z}{d - z} = \frac{\mathfrak{c}_1}{\mathfrak{d}_1}, & \frac{c - a \cdot c - b \mathfrak{d}_1}{d - a \cdot d - b \cdot \mathfrak{c}_1}, & \frac{b - c \mathfrak{a}_1}{d - a \cdot \mathfrak{b}_1}, & \frac{a - c \mathfrak{b}_1}{d - b \cdot \mathfrak{a}_1}, \end{array}$$

viz. for  $x = a$ , the relation is  $z = y$ , but in the other three cases respectively the relation is a linear one,  $z = \frac{\alpha y + \beta}{\gamma y + \delta}$ .

Rationalising the first equation for  $\sqrt{\frac{a-z}{d-z}}$ , we have

$$(bc, ad)^2(a - z) = (a - b)(a - c)(d - z) \{ \mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c} - 2 \sqrt{\mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1} \},$$

and thence

$$\{ (bc, ad)^2(a - z) - (a - b)(a - c)(d - z) (\mathfrak{a} \mathfrak{d} \mathfrak{b}_1 \mathfrak{c}_1 + \mathfrak{a}_1 \mathfrak{d}_1 \mathfrak{b} \mathfrak{c}) \}^2 = (a - b)^2(a - c)^2(d - z)^2 \cdot 4 \mathfrak{a} \mathfrak{b} \mathfrak{c} \mathfrak{d} \mathfrak{a}_1 \mathfrak{b}_1 \mathfrak{c}_1 \mathfrak{d}_1.$$

Expanding, and observing that

$$(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 - 4\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1 = (\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 - \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 = (bc, ad)^2(x-y)^2,$$

the whole equation becomes divisible by  $(bc, ad)^2$ , and omitting this factor, the equation is

$$(bc, ad)^2(a-z)^2 - 2(a-b)(a-c)(a-z)(d-z)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c}) + (a-b)^2(a-c)^2(d-z)^2(x-y)^2 = 0,$$

or, as this may also be written,

$$\begin{aligned} & z^2 \{(bc, ad)^2 - 2(a-b)(a-c)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c}) + (a-b)^2(a-c)^2(x-y)^2\} \\ & - 2z \{(bc, ad)^2 a - (a-b)(a-c)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})(a+d) + (a-b)^2(a-c)^2(x-y)^2 d\} \\ & + \{(bc, ad)^2 a^2 - 2(a-b)(a-c)(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})ad + (a-b)^2(a-c)^2(x-y)^2 d^2\} = 0. \end{aligned}$$

This is really a symmetrical equation in  $x, y, z$  of the form

$$\begin{aligned} & A \\ & + 2B(x+y+z) \\ & + C(x^2+y^2+z^2) \\ & + 2D(yz+zx+xy) \\ & + 2E(y^2z+yz^2+z^2x+zx^2+x^2y+xy^2) \\ & + 4Fxyz \\ & + 2G(x^2yz+xy^2z+xyz^2) \\ & + H(y^2z^2+z^2x^2+x^2y^2) \\ & + 2I(x^2y^2z+x^2yz^2+xy^2z^2) \\ & + Jx^2y^2z^2 = 0; \end{aligned}$$

the several coefficients being symmetrical as regards  $b, c, d$ , but the  $a$  entering unsymmetrically: the actual values are

$$\begin{aligned} A &= a^4 \{b^2c^2 + b^2d^2 + c^2d^2 - 2bcd(b+c+d)\} + 2a^3bcd(bc+bd+cd) - 3a^2b^2c^2d^2, \\ B &= 2a^4bcd - a^3(b^2c^2 + b^2d^2 + c^2d^2) + ab^2c^2d^2, \\ C &= -4a^3bcd + a^2(bc+bd+cd)^2 - 2abcd(bc+bd+cd) + b^2c^2d^2, \\ D &= -a^4(bc+bd+cd) + a^3(b^2c + bc^2 + b^2d + bd^2 + c^2d + cd^2 - 2bcd) \\ &\quad + a^2 \{b^2c^2 + b^2d^2 + c^2d^2 - bcd(b+c+d)\} - b^2c^2d^2, \\ E &= a^3(bc+bd+cd) - a^2(b^2c + bc^2 + b^2d + bd^2 + c^2d + cd^2) + abcd(b+c+d), \\ F &= a^4(b+c+d) - a^3(b^2 + c^2 + d^2 + bc+bd+cd) + 6a^2bcd \\ &\quad + a \{b^2c^2 + b^2d^2 + c^2d^2 - bcd(b+c+d)\} - bcd(bc+bd+cd), \\ G &= -a^4 + a^2(b^2 + c^2 + d^2 - bc - bd - cd) + a(b^2c + bc^2 + b^2d + bd^2 + c^2d + cd^2 - 2bcd) - bcd(b+c+d), \\ H &= a^4 - 2a^3(b+c+d) + a^2(b+c+d)^2 - 4abcd, \\ I &= a^2 - a(b^2 + c^2 + d^2) + 2bcd, \\ J &= -3a^2 + 2a(b+c+d) + b^2 + c^2 + d^2 - 2(bc+bd+cd). \end{aligned}$$

It may be remarked by way of verification that the equation remains unaltered on substituting for  $x, y, z, a, b, c, d$  their reciprocals and multiplying the whole by  $a^4 b^4 c^4 d^4 x^2 y^2 z^2$ .

I further remark that, writing  $a = 0$ , we have

$$\begin{aligned} A = 0, \quad B = 0, \quad C = b^2 c^2 d^2, \quad D = -b^2 c^2 d^2, \quad E = 0, \quad F = bcd(bc + bd + cd), \\ G = -bcd(b + c + d), \quad H = 0, \quad I = 2bcd, \quad J = b^2 + c^2 + d^2 - 2(bc + bd + cd); \end{aligned}$$

and writing also  $\alpha = 1, -\beta = (b + c + d), \gamma = bc + bd + cd, -\delta = bcd$ , (whence

$$a - x \cdot b - x \cdot c - x \cdot d - x = \beta x + \gamma x^2 + \delta x^3 + \varepsilon x^4,)$$

we have the formula

$$\begin{aligned} & \beta(x^2 + y^2 + z^2 - 2yz - 2zx - 2xy) \\ & - 4\beta\gamma xyz \\ & - 2\beta\delta xyz(x + y + z) \\ & - 4\beta\varepsilon xyz(yz + zx + xy) \\ & + (\delta^2 - 4\gamma\varepsilon)x^2 y^2 z^2 = 0, \end{aligned}$$

given p. 348 of my *Elliptic Functions* as a particular integral of the differential equation when the radical is  $\sqrt{\beta x + \gamma x^2 + \delta x^3 + \varepsilon x^4}$ .

Let the equation in  $(x, y, z)$  be called  $u = 0$ ;  $u$  has been given in the form  $u = \mathfrak{P}z^2 - 2\Omega z + \mathfrak{R}$ , and we thence have  $\frac{du}{dz} = 0 = -\Omega + \mathfrak{P}z$ ; which, in virtue of the equation  $u = 0$  itself, becomes  $\Omega^2 - \mathfrak{P}\mathfrak{R} = 0$ . We have easily

$$\Omega^2 - \mathfrak{P}\mathfrak{R} = (a - b)^2(a - c)^2(a - d)^2 \{(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 - \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 - (bc, ad)^2(x - y)^2\},$$

or, attending to the relation

$$(\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 - \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 = (\mathfrak{a}\mathfrak{d}\mathfrak{b}_1\mathfrak{c}_1 + \mathfrak{a}_1\mathfrak{d}_1\mathfrak{b}\mathfrak{c})^2 - 4\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1,$$

this is

$$= (bc, ad)^2(x - y)^2 - 4\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1,$$

or we have

$$\Omega^2 - \mathfrak{P}\mathfrak{R} = -4(a - b)^2(a - c)^2(a - d)^2\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1.$$

Writing

$$\begin{aligned} \frac{1}{2} \frac{du}{dx} &= (a - b)(a - c)(a - d)\sqrt{abcd} \sqrt{\mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1}, \\ a - z, \quad b - z, \quad c - z, \quad d - z &= \mathfrak{a}_2, \quad \mathfrak{b}_2, \quad \mathfrak{c}_2, \quad \mathfrak{d}_2, \end{aligned}$$

we have of course the like formulae

$$\begin{aligned} \frac{1}{2} \frac{du}{dy} &= (a - b)(a - c)(a - d)\sqrt{abcd} \sqrt{\mathfrak{a}_2\mathfrak{b}_2\mathfrak{c}_2\mathfrak{d}_2}, \\ \frac{1}{2} \frac{du}{dz} &= (a - b)(a - c)(a - d)\sqrt{abcd} \sqrt{\mathfrak{a}\mathfrak{b}\mathfrak{c}\mathfrak{d}\mathfrak{a}_2\mathfrak{b}_2\mathfrak{c}_2\mathfrak{d}_2} \end{aligned}$$

and the equation  $du = 0$  then gives

$$\frac{dx}{\sqrt{abcd}\sqrt{abc\mathfrak{d}}} + \frac{dy}{\sqrt{a_1b_1c_1\mathfrak{d}_1}} + \frac{dz}{\sqrt{a_2b_2c_2\mathfrak{d}_2}} = 0,$$

as it should do. The differential equation might also have been verified directly from any one of the expressions for

$$\sqrt{\frac{a-z}{d-z}}, \quad \sqrt{\frac{b-z}{d-z}}, \quad \sqrt{\frac{c-z}{d-z}}.$$

Writing for shortness

$$X = a - x \cdot b - x \cdot c - x \cdot d - x, \text{ etc.,}$$

then the general integral of the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0$$

by Abel's theorem is

$$\begin{vmatrix} x^2, & x, & 1, & \sqrt{X} \\ y^2, & y, & 1, & \sqrt{Y} \\ z^2, & z, & 1, & \sqrt{Z} \\ w^2, & w, & 1, & \sqrt{W} \end{vmatrix} = 0,$$

where  $w$  is the constant of integration: and it is to be shown that the value of  $w$  which corresponds to the integral given in the present paper is  $= a$ . Observe that writing in the determinant  $w = a$ , the determinant on putting therein  $x = a$ , would vanish whether  $z$  were or were not  $= y$ ; but this is on account of an extraneous factor  $a - w$ , so that we do not thus prove the required theorem that ( $w$  being  $= a$ ) we have  $y = z$  when  $x = a$ .

An equivalent form of Abel's integral is that there exist values  $A, B, C$  such that

$$\begin{aligned} Ax^2 + Bx + C &= \sqrt{X}, \\ Ay^2 + By + C &= \sqrt{Y}, \\ Az^2 + Bz + C &= \sqrt{Z}, \\ Aw^2 + Bw + C &= \sqrt{W}, \end{aligned}$$

or, what is the same thing, that we have identically

$$-(A\theta^2 + B\theta + C)^2 + \Theta = (A^2 - 1) \cdot \theta - x \cdot \theta - y \cdot \theta - z \cdot \theta - w.$$

We have therefore

$$C^2 - abcd = (A^2 - 1)xyzw,$$

or say

$$xyzw = \frac{C^2 - abcd}{A^2 - 1}.$$

which equation, regarding therein  $A, B, C$  as determined by the three equations

$$Ax^2 + Bx + C = \sqrt{X},$$

$$Ay^2 + By + C = \sqrt{Y},$$

$$Aw^2 + Bw + C = \sqrt{W},$$

is a form of Abel's integral, giving  $z$  rationally in terms of  $x, y, w$ .

Supposing that, when  $x = a, z = y$ ; then the last-mentioned integral gives

$$ay^2w = \frac{C^2 - abcd}{A^2 - 1},$$

where  $A, C$  are now determined by the equations

$$Aa^2 + Ba + C = 0,$$

$$Ay^2 + By + C = \sqrt{Y},$$

$$Aw^2 + Bw + C = \sqrt{W},$$

and, imagining these values actually substituted, it is to be shown that the equation

$$ay^2w = \frac{C^2 - abcd}{A^2 - 1}$$

is satisfied by the value  $w = a$ .

We have

$$A(a - y)(a - w)(w - y) = (a - w)\sqrt{Y} - (a - y)\sqrt{W},$$

$$B(a - y)(a - w)(w - y) = (a - w)(a + w)\sqrt{Y} - (a - y)(a + y)\sqrt{W},$$

$$C(a - y)(a - w)(w - y) = (a - w)aw\sqrt{Y} - (a - y)ay\sqrt{W},$$

or writing as before

$$a - y, b - y, c - y, d - y = \mathfrak{a}_1, \mathfrak{b}_1, \mathfrak{c}_1, \mathfrak{d}_1,$$

and also

$$a - w, b - w, c - w, d - w = \mathfrak{a}_3, \mathfrak{b}_3, \mathfrak{c}_3, \mathfrak{d}_3,$$

then  $Y = \mathfrak{a}_1\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1$ ,  $W = \mathfrak{a}_3\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3$ , and the formulae become

$$A = \frac{1}{(w - y)\sqrt{\mathfrak{a}_1\mathfrak{a}_3}} \{ \sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1} - \sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3} \},$$

$$B = \frac{1}{(w - y)\sqrt{\mathfrak{a}_1\mathfrak{a}_3}} \{ -(a + w)\sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1} + (a + y)\sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3} \},$$

$$C = \frac{1}{(w - y)\sqrt{\mathfrak{a}_1\mathfrak{a}_3}} \{ aw\sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1} - ay\sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3} \}.$$



If in these formulae  $w$  is indefinitely nearly  $= a$ , then  $\mathfrak{a}_3$  is indefinitely small, so that  $\sqrt{\mathfrak{a}_3\mathfrak{b}_1\mathfrak{c}_1\mathfrak{d}_1}$  may be neglected in comparison with  $\sqrt{\mathfrak{a}_1\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}$ : also  $w - y$  may be put  $= \mathfrak{a}_1$ ; the formulae thus become

$$A = -\frac{\sqrt{\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}}{\mathfrak{a}_1\sqrt{\mathfrak{a}_3}}, \quad B = (a + y) \frac{\sqrt{\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}}{\mathfrak{a}_1\sqrt{\mathfrak{a}_3}}, \quad C = -ay \frac{\sqrt{\mathfrak{b}_3\mathfrak{c}_3\mathfrak{d}_3}}{\mathfrak{a}_1\sqrt{\mathfrak{a}_3}},$$

where the values of  $A$ ,  $B$ ,  $C$  are each of them indefinitely large on account of the factor  $\sqrt{\mathfrak{a}_3}$  in the denominator; the value of  $C$  is  $C = ayA$ , and substituting this value in the equation

$$ay^2w = \frac{C^2 - abcd}{A^2 - 1},$$

and then considering  $A$  as indefinitely large, the equation becomes  $ay^2w = a^2y^2$ , that is,  $w = a$ ; so that  $w = a$  is a value of  $w$  satisfying this equation.

*Cambridge, 3 July, 1878.*

698.

ON A THEOREM RELATING TO COVARIANTS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 82, 83.]

The theorem given by Prof. Sylvester, *Crelle*, vol. LXXXV., p. 109, may be stated as follows: If for a binary quantic of the order  $i$  in the variables, we consider the whole system of covariants of the degree  $j$  in the coefficients, then

$$\Sigma \theta (k + 1) = \frac{\Pi(i + j)}{\Pi(i) \Pi(j)},$$

where  $\theta$  denotes the number of aszygetic covariants of the order  $\theta$  in the variables, the values of  $\theta$  being  $ij, ij - 2, ij - 4, \dots, 1$  or  $0$ , according as  $ij$  is odd or even.

In the case of the binary quintic  $= (a, \dots | x, y)^5$ , ( $i = 5$ ), we have a series of verifications in the Table 88 of my “Ninth Memoir on Quantics,” *Phil. Trans.*, vol. CLXI. (1871), [462]: viz. writing the small letters  $a, b, c, \dots, u, v, w$  (instead of the capitals  $A, B$ , etc.) to denote the covariants of the quintic,  $a$ , the quintic itself, degree 1, order 5, or as I express it, deg-order 1.5:  $b$ , the covariant deg-order 2.2, etc., the whole series of deg-orders being

|      |      |      |      |      |      |      |       |       |       |       |      |
|------|------|------|------|------|------|------|-------|-------|-------|-------|------|
| $a,$ | $b,$ | $c,$ | $d,$ | $e,$ | $f,$ | $g,$ | $h,$  | $i,$  | $j,$  | $k,$  | $l,$ |
| 1.5, | 2.2, | 2.6, | 3.3, | 3.5, | 3.9, | 4.0, | 4.4,  | 4.6,  | 5.1,  | 5.3,  | 5.7, |
| $m,$ | $n,$ | $o,$ | $p,$ | $q,$ | $r,$ | $s,$ | $t,$  | $u,$  | $v,$  | $w,$  |      |
| 6.2, | 6.4, | 7.1, | 7.5, | 8.0, | 8.2, | 9.3, | 11.1, | 12.0, | 13.1, | 18.0, |      |

then the table shows for each deg-order, the several covariants of that deg-order, and

the number of them which are aszyzygetic; for instance,  $i = 5$  as above,  $j = 6$ , an extract from the table is

| $j$ | $k$ | $\theta$ | $(k + 1)\theta$                  |
|-----|-----|----------|----------------------------------|
| 6   | 30  | 1        | $a^6$ 31                         |
|     | 28  | 0        | 0                                |
|     | 26  | 1        | $a^4e$ 27                        |
|     | 24  | 1        | $a^2f$ 25                        |
|     | 22  | 2        | $a^3b, a^2bf$ 46                 |
|     | 20  | 2        | $a^4, acf$ 42                    |
|     | 18  | 3        | $a^2d, a^3bc, c^2, f^2$ 57       |
|     | 16  | 2        | $a^2h, abf, ace$ 34              |
|     | 14  | 4        | $a^2b^2, a^2h, acd, bc^2, ef$ 60 |
|     | 12  | 3        | $abe, al, ce, df$ 39             |
|     | 10  | 4        | $a^2g, abd, b^2c, ch, e^2$ 44    |
|     | 8   | 2        | $ak, bi, de$ 18                  |
|     | 6   | 4        | $aj, bh, cg, d^2$ 28             |
|     | 4   | 1        | $n$ 3                            |
|     | 2   | 2        | $bg, m$ 6                        |
|     | 0   | 0        | 0                                |

$$462 = \frac{\Pi(11)}{\Pi(5)\Pi(6)},$$

where, for instance deg-order 6.14, the covariants are  $a^2b^2, a^2h, acd, bc^2, ef$ , but the number against these in the third column being (not 5 but) 4, the meaning is that there exists between these five terms one syzygy, making the number of aszyzygetic covariants of the deg-order 6.14 to be 4. The second column thus in fact contains the several values of  $k$ , and the third column the corresponding values of  $\theta$ ; whence, forming the several products  $(k + 1)\theta$  as shown, the sum of these is as it should be = 462.

Cambridge, 13 July, 1878.

699.

ON THE TRIPLE  $\vartheta$ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 134–138.]

There should be in all 64 functions proportional to irrational algebraical functions of three independent variables  $x, y, z$ ; there is no difficulty in obtaining the expression of these 64 functions in the case of the system of differential equations connected with the integral

$$\int dx : \sqrt{a - x.b - x.c - x.d - x.e - x.f - x.g - x.h - x};$$

but this is *not the general form* of the system for the deficiency (Geschlecht)  $p = 3$ ; and I do not know how to deal with the general form: the present note relates therefore exclusively to the above-mentioned hyper-elliptic form.

I.

If in the Memoir, Weierstrass, “Theorie der Abel’schen Functionen,” Crelle, t. LII. (1856), pp. 285–380, we take  $p = 3$ , and write  $x, y, z; u, v, w; a, b, c, d, e, f, g$  instead of  $x_1, x_2, x_3; u_1, u_2, u_3; a_1, a_2, a_3, a_4, a_5, a_6, a_7$ ; then, neglecting throughout mere constant factors, we have

$$X = a - x.b - x.c - x.d - x.e - x.f - x.g - x,$$

with the like values for  $Y$  and  $Z$ : the differential equations are

$$\begin{aligned} du &= \frac{b-x.c-x.d}{\sqrt{X}} + \frac{b-y.c-y.d}{\sqrt{Y}} + \frac{b-z.c-z.d}{\sqrt{Z}}, \\ dv &= \frac{c-x.a-x.d}{\sqrt{X}} + \frac{c-y.a-y.d}{\sqrt{Y}} + \frac{c-z.a-z.d}{\sqrt{Z}}, \\ dw &= \frac{a-x.b-x.d}{\sqrt{X}} + \frac{a-y.b-y.d}{\sqrt{Y}} + \frac{a-z.b-z.d}{\sqrt{Z}}. \end{aligned}$$

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and if we write the single letters  $A, B, C, D, E, F, G$  for  $\text{al}(u, v, w)_1, \text{al}(u, v, w)_2, \text{al}(u, v, w)_3, \text{al}(u, v, w)_4, \text{al}(u, v, w)_5$  respectively, each of the capital letters thus denoting a function of  $(u, v, w)$ , the expressions of these functions in terms of  $(x, y, z)$  are

$$A = \sqrt{a-x.b-x.c-w}, \quad (\text{seven equations}).$$

Similarly, instead of the 21 functions  $\text{al}(u, v, w)_{12}, \dots, \text{al}(u, v, w)_{67}$ , writing  $AB, \dots, FG$ , each of these binary symbols denoting in like manner a function of  $(u, v, w)$ , the definition of  $AB$  is

$$AB = A \nabla B - B \nabla A,$$

where

$$\nabla = \frac{d}{du} + \frac{d}{dv} + \frac{d}{dw};$$

we have

$$\begin{aligned} \frac{b-c.c-a.a-b}{\sqrt{X}} \nabla x &= \frac{a-y.a-z}{x-y.x-z} (b-c) du + \frac{b-y.b-z}{x-y.x-z} (c-a) dv + \frac{c-y.c-z}{x-y.x-z} (a-b) dw, \\ \frac{dy}{\sqrt{Y}} &= \frac{a-z.a-x}{y-z.y-x} (b-c) du + \frac{b-z.b-x}{y-z.y-x} (c-a) dv + \frac{c-z.c-x}{y-z.y-x} (a-b) dw, \\ \frac{dz}{\sqrt{Z}} &= \frac{a-x.a-y}{z-x.z-y} (b-c) du + \frac{b-x.b-y}{z-x.z-y} (c-a) dv + \frac{c-x.c-y}{z-x.z-y} (a-b) dw; \end{aligned}$$

hence

$$\begin{aligned} \frac{b-c.c-a.a-b}{\sqrt{X}} \nabla x &= \frac{a-y.a-z}{x-y.x-z} (b-c) + \frac{b-y.b-z}{x-y.x-z} (c-a) + \frac{c-y.c-z}{x-y.x-z} (a-b) \\ &= -\frac{b-c.c-a.a-b}{x-y.x-z}, \end{aligned}$$

that is,

$$\nabla x = -\frac{\sqrt{X}}{x-y.x-z},$$

and similarly

$$\nabla y = -\frac{\sqrt{Y}}{y-x.y-z}, \quad \nabla z = -\frac{\sqrt{Z}}{z-x.z-y}.$$

Hence from the equation

$$A = \sqrt{a-x.a-y.a-z}$$

we have

$$\nabla A = -\frac{1}{2}A \left( \frac{1}{a-x} \nabla x + \frac{1}{a-y} \nabla y + \frac{1}{a-z} \nabla z \right),$$

that is,

$$\nabla A = \frac{\frac{1}{2}A}{y-z.z-x.x-y} \left\{ \frac{y-z}{a-x} \sqrt{X} + \frac{z-x}{a-y} \sqrt{Y} + \frac{x-y}{a-z} \sqrt{Z} \right\};$$

and similarly

$$\nabla B = \frac{\frac{1}{2}B}{y-z, z-x, x-y} \left\{ \frac{y-z}{b-x} \sqrt{X} + \frac{z-x}{b-y} \sqrt{Y} + \frac{x-y}{b-z} \sqrt{Z} \right\};$$

consequently

$$AB = \frac{1}{2}(a-b) \frac{AB}{y-z, z-x, x-y} \left\{ \frac{(y-z)\sqrt{X}}{a-x, b-x} + \frac{(z-x)\sqrt{Y}}{a-y, b-y} + \frac{(x-y)\sqrt{Z}}{a-z, b-z} \right\}.$$

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or substituting for  $A$  and  $B$  their values, and disregarding the constant factor  $\frac{1}{2}(a-b)$ , this is

$$AB = \frac{1}{y-z, z-x, x-y} \left\{ (y-z) \sqrt{a-y, b-y, a-z, b-z, c-x, d-x, e-x, f-x, g-x} \right. \\ + (z-x) \sqrt{a-z, b-z, a-x, b-x, c-y, d-y, e-y, f-y, g-y} \\ \left. + (x-y) \sqrt{a-x, b-x, a-y, b-y, c-z, d-z, e-z, f-z, g-z} \right\}.$$

We have thus in all 21 equations, which exhibit the form of the Weierstrassian functions  $\text{al}(u, v, w)_{12}, \dots, \text{al}(u, v, w)_{67}$ .

To complete the system, there should it is clear be 35 new functions  $\text{al}(u, v, w)_{123}, \dots, \text{al}(u, v, w)_{567}$ , represented by  $ABC, \dots, EFG$ , viz. the whole number of functions would then be

$$7 + \binom{7}{2} + \binom{7}{3} (= 7 + 21 + 35) = 63, = 64 - 1,$$

since the functions represent ratios of the  $\vartheta$ -functions.

## II.

Starting now with the radical

$$\sqrt{a-x, b-x, c-x, d-x, e-x, f-x, g-x, h-x}$$

composed of eight linear factors, and writing, as in my “Memoir on the double  $\vartheta$ -functions,” t. LXXXV. (1878), pp. 214–245, [665];  $a, b, c, d, e, f, g, h$  to denote these factors, and similarly  $a_1, b_1, c_1, d_1, e_1, f_1, g_1, h_1$  and  $a_2, b_2, c_2, d_2, e_2, f_2, g_2, h_2$  to denote  $a-y, b-y$ , etc., and  $a-z, b-z$ , etc., so that  $X = abcdefgh$ ,  $Y = a_1b_1c_1d_1e_1f_1g_1h_1$ ,  $Z = a_2b_2c_2d_2e_2f_2g_2h_2$ ; then, instead of the Weierstrassian form, the differential equations may be taken to be

$$du = \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}}, \\ dv = \frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} + \frac{z dz}{\sqrt{Z}}, \\ dw = \frac{x^2 dx}{\sqrt{X}} + \frac{y^2 dy}{\sqrt{Y}} + \frac{z^2 dz}{\sqrt{Z}}.$$

We then have 64  $\vartheta$ -functions and an  $\omega$ -function, viz. writing

$$\theta = y - z, z - x, x - y,$$

and then

$$\sqrt{a} = \sqrt{aa_1a_2} \quad (8 \text{ equations})$$

$$\sqrt{abc} = \frac{1}{\theta} \{ (y-z) \sqrt{a_1 b_1 c_1 a_2 b_2 c_2 e f g h} + (z-x) \sqrt{a_2 b_2 c_2 a b c d_1 e_1 f_1 g_1 h_1} + (x-y) \sqrt{a b c a_1 b_1 c_1 d_2 e_2 f_2 g_2 h_2} \}$$

(56 equations)

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the equations, which define the  $\vartheta$ -functions  $A, B, \dots, H, ABC, \dots, FGH$ , and the  $\omega$ -function  $\Omega$ , are

$$A = \Omega \sqrt{a} \quad (8 \text{ equations})$$

$$ABC = \Omega \sqrt{abc} \quad (56 \text{ equations})$$

and one other relation which I have not as yet investigated.

As regards the algebraical relations between the 64  $\vartheta$ -functions, it is to be remarked that, selecting in a proper manner 8 of the functions, the square of any one of the other functions can be expressed as a linear function of the squares of the 8 selected functions. To explain this somewhat further, observe that, taking any 5 squares such as  $(ABC)^2$ , we can with these 5 squares form a linear combination which is rational in  $x, y, z$ . We have for instance, writing down the irrational part only,

$$(ABC)^2 = \frac{2}{\theta^2} \{ abc(z-x)(x-y) \sqrt{YZ} + a_1 b_1 c_1 (y-z)(z-x) \sqrt{ZX} + a_2 b_2 c_2 (x-y)(y-z) \sqrt{XY} \},$$

and forming in all five such equations, then inasmuch as the coefficients  $abc, \dots$  of  $\sqrt{YZ}(z-x)(x-y)$  are each of them a cubic function containing terms in  $x^3, x^2, x, x^0$ , we have a determinate set of constant factors such that the resulting term in  $(z-x)(x-y) \sqrt{YZ}$  will be  $= 0$ ; but the coefficients  $a_1 b_1 c_1, \dots$  of  $(x-y)(y-z) \sqrt{ZX}$  only differ from the first set of coefficients by containing  $y$  instead of  $x$ , and the same set of constant factors will thus make the resulting term in  $(x-y)(y-z) \sqrt{ZX}$  to be  $= 0$ ; and similarly the same set of constant factors will make the resulting term in  $(y-z)(z-x) \sqrt{XY}$  to be  $= 0$ ; viz. we have thus a set of constant factors, such that the whole irrational part will disappear. It seems to be in general true that the same set of constant factors will make the rational part integral; viz. the rational part is a function of the form  $\frac{1}{\theta}$  multiplied by a rational and integral function of  $x, y, z$ , and if this rational and integral function divide by  $\theta^2$ , then the final result will be a rational and integral function, which, being symmetrical in  $x, y, z$ , is at once seen to be a linear function of the symmetrical combinations  $1, x+y+z, yz+zx+xy, xyz$ . Such a function is obviously a linear function of any four squares  $A^2, B^2, C^2, D^2$ ; or the form is, linear function of five squares  $(ABC)^2 = \text{linear function of four squares } A^2$ , that is, any one of the five squares is a linear function of 8 squares.

As an instance, consider the three squares  $(ABC)^2, (ABD)^2, (ABE)^2$ , which are such that we have a linear combination which is rational: in fact, we have here in each function the pair of factors  $ab$ , which unites itself with  $(z-x)(x-y) \sqrt{XY}$ , viz. it is only the coefficient of  $ab(z-x)(x-y) \sqrt{XY}$  which has to be made  $= 0$ : the required combination is obviously

$$(d-e)(ABC)^2 + (e-c)(ABD)^2 + (c-d)(ABE)^2.$$

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Here the irrational part vanishes and the rational part is found to be

$$\begin{aligned} &= \frac{1}{\theta^2} \left[ a_1 b_1 a_2 b_2 f g h (y-z)^2 \{ (d-e) c_2 c_1 d c + (e-c) d_2 d_1 e c + (c-d) e_2 e_1 d c \} \right. \\ &+ a_2 b_2 a b f g h_1 (z-x)^2 \{ (d-e) c_2 c d_1 e_1 + (e-c) d_2 d c_1 e_1 + (c-d) e_2 e d_1 c_1 \} \\ &\left. + a b a_1 b_1 f g h_2 (x-y)^2 \{ (d-e) c c_1 d_2 e_2 + (e-c) d d_1 c_2 e_2 + (c-d) e e_1 d_2 c_2 \} \right]. \end{aligned}$$

The three terms in  $\{ \}$  are here  $= -(c-d)(d-e)(e-c)$  multiplied by  $(z-x)(x-y)$ ,  $(x-y)(y-z)$ ,  $(y-z)(z-x)$  respectively; hence the term in  $[ ]$  divides by  $\theta$  and the result is

$$= -\frac{(c-d)(d-e)(e-c)}{\theta} [a_1 b_1 a_2 b_2 f g h (y-z) + a_2 b_2 a b f_1 g_1 h_1 (z-x) + a b a_1 b_1 f_2 g_2 h_2 (x-y)],$$

or finally this is  $= -(c-d)(d-e)(e-c)$

$$\{(a^2 + ab + b^2)fgh - (a^2b + ab^2)(fg + fh + gh) + a^2b^2(f + g + h)\}$$

multiplied by

$$\begin{aligned} & + (x + y + z)\{-(a+b)fgh + ab(fg + fh + gh) - a^2b^2\} \\ & + (yz + zx + xy)\{fgh - ab(f + g + h) + a^2b + ab^2\} \\ & + xyz\{-(fg + fh + gh) + (a+b)(f + g + h) - (a^2 + ab + b^2)\}, \end{aligned}$$

that is, we have  $(d-e)(ABC)^2 + (e-c)(ABD)^2 + (c-d)(ABE)^2 =$  a sum of four squares, viz. we have here a linear relation between 7 squares.

I have not as yet investigated the forms of the relations between the products of pairs of  $\vartheta$ -functions.

Cambridge, 30 September, 1878.

## 700.

### ON THE TETRAHEDROID AS A PARTICULAR CASE OF THE 16-NODAL QUARTIC SURFACE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 161–164.]

In the paper “Sur un cas particulier de la surface du quatrième ordre avec seize points singuliers,” Crelle, t. LXV. (1866), pp. 284–290, [356], I showed how the surface called the Tetrahedroid could be identified as a special form of Kummer’s 16-nodal quartic surface; but I was not then in possession of the simplified form of the equation of the 16-nodal surface given in my paper “Note sur la surface du quatrième ordre douée de seize points singuliers et de seize plans singuliers,” Crelle, t. LXXIII. (1871), pp. 292, 293, [442]; see also my paper, “A third memoir on Quartic surfaces,” Proc. Lond. Math. Soc. t. III. (1871), p. 250, [454, this Collection, t. VII., p. 281]. Using the equation last referred to, I resume therefore the consideration of the question.

Taking the constants  $\alpha, \beta, \gamma, \alpha', \beta', \gamma', \alpha'', \beta'', \gamma''$ , such that

$$\alpha + \beta + \gamma = 0, \quad \alpha' + \beta' + \gamma' = 0, \quad \alpha'' + \beta'' + \gamma'' = 0,$$

and writing also

$$\begin{aligned} M &= \alpha\alpha''(\beta' - \gamma') + \beta'\beta''(\gamma' - \alpha') + \gamma'\gamma''(\alpha' - \beta') \\ &= \alpha''\alpha(\beta' - \gamma') + \beta''\beta(\gamma' - \alpha') + \gamma''\gamma(\alpha' - \beta') \\ &= \alpha\alpha'(\beta'' - \gamma'') + \beta\beta'(\gamma'' - \alpha'') + \gamma\gamma'(\alpha'' - \beta'') \\ &= -\frac{1}{2}\{(\beta - \gamma)(\beta' - \gamma')(\beta'' - \gamma'') + (\gamma - \alpha)(\gamma' - \alpha')(\gamma'' - \alpha'') + (\alpha - \beta)(\alpha' - \beta')(\alpha'' - \beta'')\}, \end{aligned}$$

(the equivalence of which different expressions for  $M$  is verified without difficulty); writing also  $X, Y, Z, W$  as current coordinates, the equation of the 16-nodal surface is

$$0 = \begin{cases} W^2(X^2 + Y^2 + Z^2 - 2YZ - 2ZX - 2XY) \\ + 2W[\alpha\alpha'\alpha''(Y^2Z + YZ^2) + \beta\beta'\beta''(Z^2X + ZX^2) + \gamma\gamma'\gamma''(X^2Y + XY^2) + MXYZ] \\ + (\alpha\alpha'\alpha''YZ + \beta\beta'\beta''ZX + \gamma\gamma'\gamma''XY)^2, \end{cases}$$

where,  $\alpha, \beta, \gamma, \alpha', \beta', \gamma', \alpha'', \beta'', \gamma''$  being connected as above, the number of constants is = 6.

The equations of the 16 singular planes are

$$\begin{aligned} X &= 0, & Y &= 0, \\ \alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W &= 0, & \beta(\alpha'\alpha''Z - \gamma'\gamma''X) - W &= 0, \\ \alpha'(\gamma''\gamma Y - \beta''\beta Z) - W &= 0, & \beta'(\alpha''\alpha Z - \gamma''\gamma X) - W &= 0, \\ \alpha''(\gamma\gamma'Y - \beta\beta'Z) - W &= 0, & \beta''(\alpha\alpha'Z - \gamma\gamma'X) - W &= 0, \\ Z &= 0, & W &= 0, \\ \gamma(\beta'\beta''X - \alpha'\alpha''Y) - W &= 0, & \beta\gamma X + \gamma\alpha Y + \alpha\beta Z &= 0, \\ \gamma'(\beta''\beta X - \alpha''\alpha Y) - W &= 0, & \beta'\gamma'X + \gamma'\alpha'Y + \alpha'\beta'Z &= 0, \\ \gamma''(\beta\beta'X - \alpha\alpha'Y) - W &= 0, & \beta''\gamma''X + \gamma''\alpha''Y + \alpha''\beta''Z &= 0. \end{aligned}$$

Writing  $x, y, z, w$  as current coordinates, the equation of the Tetrahedroid is

$$0 = \begin{cases} w^4(l^2y^2 + m^2z^2) + l^2m^2n^4w^4 + l^4m^2n^2x^2y^2z^2 + f^2g^2h^2w^4 \\ + (l^2f^2 - m^2g^2 + n^2h^2)(m^2z^2x^2 + g^2y^2w^2) \\ + (-l^2f^2 + m^2g^2 - n^2h^2)(l^2y^2z^2 + f^2x^2w^2) \\ + (-l^2f^2 - m^2g^2 + n^2h^2)(n^2x^2y^2 + h^2z^2w^2) = 0, \end{cases}$$

where, inasmuch as  $f, g, h, l, m, n$  enter homogeneously, the number of constants is = 5.

The equations of the 16 singular planes, written in an order corresponding to that used for the 16-nodal surface, are

$$\begin{aligned} ny - mz + fw &= 0, & -nx + lz + gw &= 0, & mx - ly + hw &= 0, & -fx - gy - hz &= 0, \\ fx - gy - hz &= 0, & nx + ly + hw &= 0, & -nx - lz + gw &= 0, & ny - mz - fw &= 0, \\ -mx - ly + hw &= 0, & -fx - gy - hz &= 0, & ny + mz + fw &= 0, & -nx + lz - gw &= 0, \\ nx + lz + gw &= 0, & -ny - mz + fw &= 0, & -fx - gy + hz &= 0, & mx - ly - hw &= 0. \end{aligned}$$

These equations can be made to agree each to each with those of the 16 singular planes of the 16-nodal surface, provided that we have

$$\frac{m}{\gamma} = \frac{n}{\beta}, \quad \frac{n}{\alpha'} = \frac{l}{\gamma'}, \quad \frac{l}{\beta''} = \frac{m}{\alpha''}; \quad f = -l\alpha\alpha'\alpha'', \quad g = -m\beta\beta'\beta'', \quad h = -n\gamma\gamma'\gamma'',$$

where observe that the first three equations give  $\alpha\beta'\gamma'' = \alpha''\beta\gamma'$ , which is the relation between the constants when the 16-nodal surface reduces itself to a tetrahedroid in the above manner. And if we then assume

$$X = ny - mz + fw, \quad Y = -nx + lz + gw, \quad Z = mx - ly + hw, \quad W = -fx - gy - hz,$$

the 16 linear functions of  $X, Y, Z, W$  will become mere constant multiples of the corresponding 16 linear functions of  $x, y, z, w$ ; the constants, by which the several

functions of  $x, y, z, w$  have to be multiplied in order to reduce them each to the corresponding linear function of  $X, Y, Z, W$ , being given by the table



| 1,                                      | 1,                                                   | 1,                                              | 1,                                                                   |
|-----------------------------------------|------------------------------------------------------|-------------------------------------------------|----------------------------------------------------------------------|
| $\frac{1}{m\beta}(l\alpha - m\beta),$   | $-\frac{\alpha\alpha''}{m}(l\alpha - m\beta),$       | $\frac{\alpha\alpha''}{n}(l\alpha - m\beta),$   | $\frac{\beta\gamma}{\alpha}\alpha(\beta'\gamma'' - \gamma'\beta''),$ |
| $\frac{1}{n\gamma}(m\beta - n\gamma),$  | $-\frac{\beta\beta''}{n\gamma}(m\beta - n\gamma),$   | $\frac{\beta\beta''}{l}(m\beta - n\gamma),$     | $\frac{\gamma\alpha}{\beta}(\gamma''\alpha - \gamma\alpha''),$       |
| $\frac{1}{l\alpha}(n\gamma - l\alpha),$ | $\frac{\gamma\gamma''}{l\alpha}(n\gamma - l\alpha),$ | $-\frac{\gamma\gamma''}{m}(n\gamma - l\alpha),$ | $\frac{\alpha\beta}{\gamma}(\alpha\beta' - \alpha'\beta'').$         |

For instance, we have

$$\alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W = 2\frac{1}{m\beta}(l\alpha - m\beta)(fx - gy - hz),$$

viz. substituting for  $Y, Z, W$  their values, the relation is

$$\left\{ \begin{array}{l} m\alpha\beta\gamma'\gamma''(-nx + lz + gw) \\ - m\alpha\beta\beta'\beta''(mx - ly + hw) \\ + m\beta(-fx - gy - hz) \end{array} \right\} = 2(l\alpha - m\beta)(fx - gy - hz).$$

As regards the terms in  $y, z$ , and  $w$ , the identity is at once verified. As regards the term in  $x$ , we should have

$$m\alpha\beta(-n\gamma'\gamma'' - m\beta'\beta'') - (l\alpha - 2m\beta)f = 0,$$

viz. substituting for  $f$  its value,  $-l\alpha\alpha'\alpha'' = -m\alpha\alpha'\beta''$ , the equation divides by  $m\alpha$  and we then have

$$(-n\gamma'\gamma'' - m\beta'\beta'') + \alpha'\beta''(l\alpha - 2m\beta) = 0,$$

that is,

$$l\alpha\alpha'\beta'' - m\beta'\beta''(\beta' + 2\alpha') - n\beta\gamma'\gamma'' = 0,$$

or writing herein  $m\beta'' = l\alpha''$ ,  $n\gamma' = l\alpha'$ , and  $\beta' + 2\alpha' = \alpha' - \gamma'$ , the equation becomes  $\alpha\alpha'\beta'' - \alpha'\alpha\beta'' - \alpha''\beta(\alpha' - \gamma') - \alpha'\beta\gamma'' = 0$ , that is,  $\alpha'(\alpha\beta'' - \alpha''\beta) = -\alpha'\beta\gamma''$ ; or writing herein  $\alpha''\beta\gamma' = \alpha'\beta'\gamma$ , the equation divided by  $\alpha'$  becomes  $\alpha\beta'' - \alpha''\beta = \beta\gamma'' - \beta''\gamma$ , which is true in virtue of  $\alpha + \beta + \gamma = 0$  and  $\alpha'' + \beta'' + \gamma'' = 0$ . And in like manner the several other identities may be verified.

The equation  $\alpha'\beta''\gamma = \alpha''\beta\gamma'$  might have been obtained as the condition of the intersection, in a common point, of four of the singular planes of the 16-nodal surface; and when this equation is satisfied, there are in fact four systems each of four planes, such that the four planes of a system meet in a common point: viz. we have

*Planes*

$$\begin{aligned} X = 0, \quad & \beta\gamma X + \gamma\alpha Y + \alpha\beta Z = 0, \quad \gamma'(\beta''\beta X - \alpha''\alpha Y) - W = 0, \quad \beta''(\alpha\alpha'Z - \gamma\gamma'X) - W = 0, \\ Y = 0, \quad & \gamma(\beta'\beta''X - \alpha'\alpha''Y) - W = 0, \quad \beta'\gamma'X + \gamma'\alpha'Y + \alpha'\beta'Z = 0, \quad \alpha''(\gamma\gamma'Y - \beta\beta'Z) - W = 0, \\ Z = 0, \quad & \beta'(\alpha''\alpha Z - \gamma''\gamma X) - W = 0, \quad \alpha'(\beta'\gamma Y - \beta''\beta Z) - W = 0, \quad \beta''\gamma''X + \gamma''\alpha''Y + \alpha''\beta''Z = 0, \\ W = 0, \quad & \alpha(\gamma'\gamma''Y - \beta'\beta''Z) - W = 0, \quad \beta(\alpha''\alpha Z - \gamma''\gamma X) - W = 0, \quad \gamma''(\beta\beta'X - \alpha\alpha'Y) - W = 0, \end{aligned}$$

$$\begin{array}{cccc} 0, & -\beta, & \gamma, & \alpha'\beta'\gamma'\cdot\alpha, \\ \alpha', & 0, & -\gamma', & \alpha'\beta'\gamma'\cdot\beta', \\ -\alpha'', & \beta'', & 0, & \alpha''\beta''\gamma''\cdot\gamma'', \\ \beta''\cdot\alpha\alpha'\alpha'', & \alpha''\cdot\beta\beta'\beta'', & \gamma''\cdot\gamma\gamma'\gamma'', & 0, \end{array}$$

the four points being in fact the vertices of the tetrahedron formed by the four planes of the tetrahedroid. Observe that, if the singular planes of the 16-nodal surface in their original order are

$$\begin{array}{c} 1, 2, 3, 4, \\ 5, 6, 7, 8, \\ 9, 10, 11, 12, \\ 13, 14, 15, 16, \end{array}$$

then the planes forming the last-mentioned four systems of planes are

$$\begin{array}{c} (1, 8, 11, 14), \\ (2, 7, 12, 13), \\ (3, 6, 9, 16), \\ (4, 5, 10, 15), \end{array}$$

viz. they correspond each of them to a term which in the determinant formed with the 16 symbols would have the sign +.

The equation  $\alpha'\beta''\gamma = \alpha''\beta\gamma'$  is evidently not unique. The triads  $(\alpha, \beta, \gamma), (\alpha', \beta', \gamma'), (\alpha'', \beta'', \gamma'')$  enter symmetrically into the equation of the 16-nodal surface; by taking the singular planes of one of the surfaces in a different order, the equation would present itself under one or other of the different forms

$$\begin{array}{l} \alpha'\beta''\gamma = \alpha\beta'\gamma'', \quad \alpha'\beta\gamma'' = \alpha\beta''\gamma', \quad \alpha\beta\gamma'' = \alpha'\beta''\gamma, \\ \alpha'\beta\gamma'' = \alpha''\beta'\gamma, \quad \alpha\beta''\gamma' = \alpha'\beta\gamma'', \\ \alpha''\beta'\gamma = \alpha\beta'\gamma''. \end{array}$$

Cambridge, 9 December, 1878.

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ALGORITHM FOR THE CHARACTERISTICS OF THE TRIPLE  $\vartheta$ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 165–169.]

The characteristics of the triple  $\vartheta$ -functions may be represented, the 28 odd characteristics by the binary symbols or duads, 12, . . . , 78, and the even ones (other than  $\frac{000}{000} = 0$ ), say the 35 even characteristics, by the ternary symbols or triads 123 . . . 567: which triads may be regarded as abbreviations for the double tetrads 1238.4567, . . . , 5678.1234, the 8 being always attached to the expressed triad. The correspondence of the symbols is given by the diagram

| upper line of characteristic |     |     |     |     |     |     |     |     |
|------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
|                              | 000 | 100 | 010 | 110 | 001 | 101 | 011 | 111 |
| 000                          |     | 236 | 345 | 137 | 467 | 156 | 124 | 257 |
| 100                          | 237 | 67  | 136 | 12  | 157 | 48  | 256 | 35  |
| 010                          | 245 | 127 | 23  | 68  | 134 | 357 | 15  | 47  |
| 110                          | 126 | 13  | 78  | 145 | 356 | 25  | 46  | 234 |
| 001                          | 567 | 146 | 125 | 247 | 45  | 17  | 38  | 26  |
| 101                          | 147 | 58  | 246 | 34  | 16  | 123 | 27  | 367 |
| 011                          | 135 | 347 | 14  | 57  | 28  | 36  | 167 | 456 |
| 111                          | 346 | 24  | 56  | 235 | 37  | 267 | 457 | 18  |

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Or, what is the same thing, it is

| <i>upper line of characteristic</i> |     |     |     |     |     |     |     |     |
|-------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
|                                     | 000 | 100 | 010 | 110 | 001 | 101 | 011 | 111 |
| 12                                  |     |     |     | 100 |     |     |     |     |
| 13                                  |     | 110 |     |     |     |     |     |     |
| 14                                  |     |     | 011 |     |     |     |     |     |
| 15                                  |     |     |     |     |     | 010 |     |     |
| 16                                  |     |     |     |     | 101 |     |     |     |
| 17                                  |     |     |     |     |     |     |     | 111 |
| 18                                  |     |     |     |     | 001 |     |     |     |
| 23                                  |     |     | 010 |     |     |     |     |     |
| 24                                  |     | 111 |     |     |     |     |     |     |
| 25                                  |     |     |     |     |     |     | 110 |     |
| 26                                  |     |     |     |     |     | 001 |     |     |
| 27                                  |     |     |     |     |     |     | 101 |     |
| 28                                  |     |     |     |     | 011 |     |     |     |
| 34                                  |     |     |     | 101 |     |     |     |     |
| 35                                  |     |     |     |     |     |     |     | 100 |
| 36                                  |     |     |     |     |     |     | 011 |     |
| 37                                  |     |     |     |     | 111 |     |     |     |
| 38                                  |     |     |     |     |     | 011 |     |     |
| 45                                  |     |     |     |     | 001 |     |     |     |
| 46                                  |     |     |     |     |     |     | 110 |     |
| 47                                  |     |     |     |     |     |     |     | 010 |
| 48                                  |     |     |     |     |     | 100 |     |     |
| 56                                  |     |     | 111 |     |     |     |     |     |
| 57                                  |     |     |     | 011 |     |     |     |     |
| 58                                  |     | 101 |     |     |     |     |     |     |
| 67                                  |     | 100 |     |     |     |     |     |     |
| 68                                  |     |     |     | 010 |     |     |     |     |
| 78                                  |     |     |     | 110 |     |     |     |     |

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|     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|     | 000 | 100 | 010 | 110 | 001 | 101 | 011 | 111 |
| 123 |     |     |     |     |     | 101 |     |     |
| 124 |     |     |     |     |     |     | 000 |     |
| 125 |     |     | 001 |     |     |     |     |     |
| 126 | 110 |     |     |     |     |     |     |     |
| 127 |     | 010 |     |     |     |     |     |     |
| 134 |     |     |     |     | 010 |     |     |     |
| 135 | 011 |     |     |     |     |     |     |     |
| 136 |     |     | 100 |     |     |     |     |     |
| 137 |     |     |     | 000 |     |     |     |     |
| 145 |     |     |     | 110 |     |     |     |     |
| 146 |     | 001 |     |     |     |     |     |     |
| 147 | 101 |     |     |     |     |     |     |     |
| 156 |     |     |     |     |     | 000 |     |     |
| 157 |     |     |     |     | 100 |     |     |     |
| 167 |     |     |     |     |     |     | 011 | 110 |
| 234 |     |     |     |     |     |     |     |     |
| 235 |     |     |     | 111 |     |     |     |     |
| 236 |     | 000 |     |     |     |     |     |     |
| 237 | 100 |     |     |     |     |     |     |     |
| 245 | 010 |     |     |     |     |     |     |     |
| 246 |     |     | 101 |     |     |     |     |     |
| 247 |     |     |     | 001 |     |     |     |     |
| 256 |     |     |     |     |     |     | 100 |     |
| 257 |     |     |     |     |     |     |     | 000 |
| 267 |     |     |     |     |     | 111 |     |     |
| 345 |     |     | 000 |     |     |     |     |     |
| 346 | 111 |     |     |     |     |     |     |     |
| 347 |     | 011 |     |     |     |     |     |     |
| 356 |     |     |     |     | 110 |     |     |     |
| 357 |     |     |     |     |     | 010 |     |     |
| 367 |     |     |     |     |     |     |     | 101 |
| 456 |     |     |     |     |     |     |     | 011 |
| 457 |     |     |     |     |     |     | 111 |     |
| 467 |     |     |     | 000 |     |     |     |     |
| 567 | 001 |     |     |     |     |     |     |     |

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by means of which the two-line-characteristic is at once found when the duad or triad is given.

The new algorithm renders unnecessary the Table I. of Weber’s memoir “Theorie der Abel’schen Functionen vom Geschlecht 3” (Berlin, 1876). In fact, the system of six pairs corresponding to an odd characteristic such as 12 is

$$13.23, 14.24, 15.25, 16.26, 17.27, 18.28,$$

and that corresponding to an even characteristic such as 123 (= 1238.4567) is

$$12.38, 13.28, 18.23, 45.67, 46.57, 47.56;$$

so that all the  $(28 + 35 =) 63$  systems can be at once formed.

The odd characteristics correspond to the bitangents of a quartic curve, and as regards these bitangents the notation is, in fact, the notation arising out of Hesse’s investigations and explained Salmon’s *Higher*

*Plane Curves* (2nd Ed. 1873), pp. 222–225. It may be noticed that the geometrical symbols corresponding to the before-mentioned two systems are:

[Figure: two geometrical diagrams — on the left, a hexagonal/triangle arrangement of points labelled 1–8 representing the system of six pairs for an odd characteristic (with vertices 1, 2 above and 3, 4, 5, 6, 7 around, 8 below); on the right, a square arrangement of points 1, 2, 3, 4 with auxiliary points 5, 6, 7, 8 representing the system for an even characteristic.]

Hence, selecting out of the first system any two pairs, we have a symbol  $\square$ : but selecting out of the second system any two pairs, we have a symbol which is either  $\square$  or  $|||$ ; so that in each case (Salmon, p. 224) the four bitangents are such that the eight points of contact lie on a conic.

The 28 bitangents of the general quartic curve

$$\sqrt{x_1\xi_1} + \sqrt{x_2\xi_2} + \sqrt{x_3\xi_3} = 0,$$

represented by the equations given by Weber, l.c., pp. 100, 101, and taken in the order in which they are there written down, have for their duad-characteristics

18, 28, 38, 23, 13, 12, 48, 14, 58, 15, 68, 16, 78, 17, 24, 34, 25, 35,

26, 36, 27, 37, 67, 57, 56, 45, 46, 47

respectively. Taking out of any one of the 63 systems three pairs of bitangents at pleasure, these give rise to an equation of the curve of a form such as

$$\sqrt{x_1\xi_1} + \sqrt{x_2\xi_2} + \sqrt{x_3\xi_3} = 0.$$

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and the whole number of the forms of equation is thus = 1260. The triads of pairs which enter into the same equation may be

|                                    |                                                           |           |
|------------------------------------|-----------------------------------------------------------|-----------|
| triads such as 12.34, 13.42, 14.23 | [Figure: square 1-2-4-3 with internal diagonals]          | No. = 70, |
| „ „ 12.34, 13.42, 56.78            | [Figure: square 1-2 over 3-4 plus vertical pair 5-6, 7-8] | = 630,    |
| „ „ 13.23, 14.24, 15.25            | [Figure: vertex 1 connecting to triangle 2-3 etc.]        | = 560,    |

making the whole number = 1260, as already mentioned.

Cambridge, 7 December, 1878.

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### ON THE TRIPLE $\vartheta$ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVII. (1878), pp. 190–198.]

A QUARTIC curve has the deficiency 3, and depends therefore on the triple  $\vartheta$ -functions: and these, as functions of 3 arguments, should be connected with functions of 3 points on the curve; but it is easy to understand that it is possible, and may be convenient, to introduce a fourth point, and so regard them as functions of 4 points on the curve: thus in the circle, the functions  $\cos u, \sin u$  may be regarded as functions of one point  $\cos u = x, \sin u = y$ , or as functions of two points,

$$\cos u = xx_1 + yy_1, \quad \sin u = xy_1 - x_1y.$$

And accordingly in Weber's memoir "Theorie der Abel'schen Functionen vom Geschlecht 3," (1876), see p. 156, the triple  $\vartheta$ -functions are regarded as functions of 4 points on the curve: viz. it is in effect

shown that (disregarding constant factors) each of the 64 functions is proportional to a determinant, the four lines of which are algebraical functions of the coordinates of the four points respectively: the form of this determinant being different according as the characteristic of the  $\vartheta$ -function is odd or even, or say according as the  $\vartheta$ -function is odd or even. But the geometrical signification of these formulae requires to be developed.

A quartic curve may be touched in six points by a cubic curve: but (Hesse, 1855\*) there are two kinds of such tangent cubics, according as the six points of contact are on a conic, or are not on a conic; say we have a conic hexad of points on the quartic, and a cubic hexad of points on the quartic. In either case, three points of the hexad may be assumed at pleasure; we can then in 28 different ways determine the remaining three points of the conic hexad, and in 36 different

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\* See the two memoirs “Ueber Determinanten und ihre Anwendung in der Geometrie” and “Ueber die Doppeltangenten der Curven vierter Ordnung,” Crelle, t. XLIX. (1855).

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ways the remaining three points of the cubic hexad: or what is the same thing, there are 28 systems of cubics touching in a conic hexad, and 36 systems of cubics touching in a cubic hexad. The condition in order that four points of the quartic curve may belong to a hexad (conic or cubic) is given by an equation  $\Omega = 0$ , where  $\Omega$  is a determinant the four lines of which are algebraical functions of the coordinates of the four points respectively: but the form of such determinant is different according as the condition belongs to a conic hexad, or to a cubic hexad: we have thus 28 conic determinants and 36 cubic determinants,  $\Omega$ ; and the 64  $\vartheta$ -functions are proportional to constant multiples of these determinants; viz. the odd functions correspond to the conic determinants, and the even functions to the cubic determinants.

First, as to the conic hexads: the points of a conic hexad lie in a conic with the two points of contact of some one of the bitangents of the quartic curve: so that, given any three points of the hexad, these together with the two points of contact of the bitangent determine a conic which meets the quartic in the remaining three points of the hexad. Suppose that  $a, b, c, f, g, h$  are linear functions of the coordinates such that the equation of the quartic curve is

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0;$$

then  $a = 0, b = 0, c = 0, f = 0, g = 0, h = 0$  are six of the bitangents of the curve, and the bitangent  $a = 0$  touches the curve at the two points of intersection of this line with the conic  $bg - ch = 0$ . The general equation of a conic through these two points  $a = 0, bg - ch = 0$ , may be written

$$bg - ch + a(Ax + By + Cz) = 0,$$

where for  $x, y, z$  we may if we please substitute any three of the six linear functions  $a, b, c, f, g, h$ , or any other linear functions of the coordinates  $(x, y, z)$ : and the equation may also be written

$$\sqrt{af}(\sqrt{bg} - \sqrt{ch}) + \sqrt{a}(Ax + By + Cz) = 0.$$

Adopting this latter form, and considering the intersections of the conic with the quartic, that is, considering the relation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$$

as holding good, we have

$$\begin{aligned} af + bg - ch &= -2\sqrt{afbg}, \\ af - bg + ch &= -2\sqrt{afch}, \\ -af + bg + ch &= -2\sqrt{bgch}, \end{aligned}$$

and we thus have at pleasure one or other of the two equations

$$-2\sqrt{afbg} + \sqrt{a}(Ax + By + Cz) = 0,$$

that is,

$$-2\sqrt{afch} + \sqrt{a}(Ax + By + Cz) = 0,$$

$$-2\sqrt{bgch} + \sqrt{a}(Ax + By + Cz) = 0.$$

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Hence the condition in order that the four points  $(x_1, y_1, z_1)$ ,  $(x_2, y_2, z_2)$ ,  $(x_3, y_3, z_3)$ ,  $(x_4, y_4, z_4)$ , assumed to be points of the quartic, may belong to the conic hexad, may be written

$$\begin{vmatrix} \sqrt{f_1 b_1 g_1}, & x_1 \sqrt{a_1}, & y_1 \sqrt{a_1}, & z_1 \sqrt{a_1} \\ \sqrt{f_2 b_2 g_2}, & x_2 \sqrt{a_2}, & y_2 \sqrt{a_2}, & z_2 \sqrt{a_2} \\ \sqrt{f_3 b_3 g_3}, & x_3 \sqrt{a_3}, & y_3 \sqrt{a_3}, & z_3 \sqrt{a_3} \\ \sqrt{f_4 b_4 g_4}, & x_4 \sqrt{a_4}, & y_4 \sqrt{a_4}, & z_4 \sqrt{a_4} \end{vmatrix} = 0, \quad \text{or} \quad \begin{vmatrix} \sqrt{f_1 c_1 h_1}, & x_1 \sqrt{a_1}, & y_1 \sqrt{a_1}, & z_1 \sqrt{a_1} \\ \sqrt{f_2 c_2 h_2}, & x_2 \sqrt{a_2}, & y_2 \sqrt{a_2}, & z_2 \sqrt{a_2} \\ \sqrt{f_3 c_3 h_3}, & x_3 \sqrt{a_3}, & y_3 \sqrt{a_3}, & z_3 \sqrt{a_3} \\ \sqrt{f_4 c_4 h_4}, & x_4 \sqrt{a_4}, & y_4 \sqrt{a_4}, & z_4 \sqrt{a_4} \end{vmatrix} = 0,$$

where, as before, the  $x, y, z$  may be replaced by any three of the letters  $a, b, c, f, g, h$ , or by any other linear functions of  $(x, y, z)$ : and, moreover, although in obtaining the condition we have used for the quartic the equation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

depending upon six bitangents, yet from the process itself it is clear that the condition can only depend upon the particular bitangent  $a = 0$ : calling the condition  $\Omega = 0$ , all the forms of condition which belong to the same bitangent  $a = 0$ , will be essentially identical, that is, the several determinants  $\Omega$  will differ only by constant factors; or disregarding these constant factors, we have for the bitangent  $a = 0$ , a single determinant  $\Omega$ , which may be taken to be any one of the determinants in question. And we have thus 28 determinants  $\Omega$ , corresponding to the 28 bitangents respectively.

Coming now to the cubic hexads, Hesse showed that the equation of a quartic curve could be (and that in 36 different ways) expressed in the form, symmetrical determinant = 0, or say

$$\begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix} = 0,$$

where  $(a, b, c, d, f, g, h, l, m, n)$  are linear functions of the coordinates; and from each of these forms he obtains the equation of a cubic

$$\begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & \cdot \end{vmatrix} = 0,$$

containing the four constants  $\alpha, \beta, \gamma, \delta$ , or say the 3 ratios of these constants, touching the quartic in a cubic hexad of points: that the cubic does touch the quartic in six points appears, in fact, from Hesse's identity

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$$\begin{vmatrix} a, & h, & g, & l, & \alpha \\ h, & b, & f, & m, & \beta \\ g, & f, & c, & n, & \gamma \\ l, & m, & n, & d, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & \cdot \end{vmatrix} \begin{vmatrix} a, & h, & g, & l, & \alpha' \\ h, & b, & f, & m, & \beta' \\ g, & f, & c, & n, & \gamma' \\ l, & m, & n, & d, & \delta' \\ \alpha', & \beta', & \gamma', & \delta', & \cdot \end{vmatrix} - \begin{vmatrix} a, & h, & g, & l, & \alpha^2 \\ h, & b, & f, & m, & \beta^2 \\ g, & f, & c, & n, & \gamma^2 \\ l, & m, & n, & d, & \delta^2 \\ \alpha^2, & \beta^2, & \gamma^2, & \delta^2, & \cdot \end{vmatrix} = U \begin{vmatrix} a, & h, & g, & l \\ h, & b, & f, & m \\ g, & f, & c, & n \\ l, & m, & n, & d \end{vmatrix}$$

where  $U$  is an easily calculated function of the second order in  $a, b, c, d, f, g, h, l, m, n$ , and also of the second order in the determinants  $\alpha\beta' - \alpha'\beta$ , etc.

We can obtain such a form of the equation of the quartic, from the before-mentioned equation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0;$$

viz. this equation gives

$$\begin{vmatrix} *, & h, & g, & a \\ h, & *, & f, & b \\ g, & f, & *, & c \\ a, & b, & c, & * \end{vmatrix} = 0,$$

which is of the required form, symmetrical determinant = 0; the equation is, in fact,

$$a^2f^2 + b^2g^2 + c^2h^2 - 2bcgh - 2cahf - 2abfg = 0,$$

which is the rationalised form of

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

and we hence have the cubic

$$\begin{vmatrix} *, & h, & g, & a, & \alpha \\ h, & *, & f, & b, & \beta \\ g, & f, & *, & c, & \gamma \\ a, & b, & c, & *, & \delta \\ \alpha, & \beta, & \gamma, & \delta, & * \end{vmatrix} = 0,$$

the developed form of which is

$$\begin{aligned} & \alpha^2bcf + \beta^2cag + \gamma^2abh + \delta^2fgh \\ & - (\alpha\beta\gamma + f\alpha\delta)(-af + bg + ch) \\ & - (\beta\gamma\alpha + g\beta\delta)(af - bg + ch) \\ & - (\gamma\alpha\beta + h\gamma\delta)(af + bg - ch) = 0. \end{aligned}$$

Considering the intersections with the quartic

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

we have

$$-af + bg + ch, \quad af - bg + ch, \quad af + bg - ch = -2\sqrt{bcgh}, \quad -2\sqrt{cahf}, \quad -2\sqrt{abfg},$$

and the equation thus becomes

$$(\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh})^2 = 0;$$

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viz. for the points of the cubic hexad we have

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0,$$

and hence the condition in order that the four points  $(x_1, y_1, z_1)$ ,  $(x_2, y_2, z_2)$ ,  $(x_3, y_3, z_3)$ ,  $(x_4, y_4, z_4)$  may belong to the cubic hexad is

$$\begin{vmatrix} \sqrt{b_1c_1f_1}, & \sqrt{c_1a_1g_1}, & \sqrt{a_1b_1h_1}, & \sqrt{f_1g_1h_1} \\ \sqrt{b_2c_2f_2}, & \sqrt{c_2a_2g_2}, & \sqrt{a_2b_2h_2}, & \sqrt{f_2g_2h_2} \\ \sqrt{b_3c_3f_3}, & \sqrt{c_3a_3g_3}, & \sqrt{a_3b_3h_3}, & \sqrt{f_3g_3h_3} \\ \sqrt{b_4c_4f_4}, & \sqrt{c_4a_4g_4}, & \sqrt{a_4b_4h_4}, & \sqrt{f_4g_4h_4} \end{vmatrix} = 0,$$

viz. we have thus the form of the determinant  $\Omega$  which belongs to a cubic hexad.



It is to be observed that the equation

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0$$

remains unaltered by any of the interchanges  $a$  and  $f$ ,  $b$  and  $g$ ,  $c$  and  $h$ ; but we thus obtain only two cubic hexads; those answering to the equations

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0,$$

and

$$\alpha\sqrt{agh} + \beta\sqrt{bhf} + \gamma\sqrt{cfg} + \delta\sqrt{abc} = 0,$$

which give distinct hexads. The whole number of ways in which the equation of the quartic can be expressed in a form such as

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0,$$

attending only to the pairs of bitangents, and disregarding the interchanges of the two bitangents of a pair, is = 1260, and hence the number of forms for the determinant  $\Omega$  of a cubic hexad is the double of this, = 2520, which is =  $36 \times 70$ : but the number of distinct hexads is = 36, and thus there must be for each hexad, 70 equivalent forms.

To explain this, observe that every even characteristic except  $\frac{000}{000}$ , and every odd characteristic, can be (and that in 6 ways) expressed as a sum of two different odd characteristics; we have thus (see Weber's Table I.) a system of  $(35 + 28 =) 63$  hexpairs; and selecting at pleasure any three pairs out of the same hexpair, we have a system of  $(63 \times 20 =) 1260$  tripairs; giving the 1260 representations of the quartic in a form such as

$$\sqrt{af} + \sqrt{bg} + \sqrt{ch} = 0.$$

Each even characteristic (not excluding  $\frac{000}{000}$ ) can be in 56 different ways (Weber, p. 23) expressed as a sum of three different odd characteristics, and these are such that no two of them belong to the same pair, in any tripair; or we may say that each even characteristic gives rise to 56 hemi-tripairs. But a hemi-tripair can be in 5 different ways completed into a tripair; and we have thus, belonging to the same

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even characteristic  $(56 \times 5 =) 280$  tripairs, which are however 70 tripairs each taken 4 times. A tripair contains in all  $(2^3 =) 8$  hemi-tripairs, but these divide themselves into two sets each of 4 hemi-tripairs such that for each hemi-tripair of the first set the three characteristics have a given sum, and for each hemi-tripair of the second set the three characteristics have a different given sum. Hence considering the 70 tripairs corresponding as above to a given even characteristic, in any one of the 70 tripairs, there is a set of 4 hemi-tripairs such that in each of them the sum of the three characteristics is equal to the given even characteristic; and taking the bitangents  $f, g, h$  to correspond to any one of these hemi-tripairs, the bitangents which correspond to the other three hemi-tripairs will be  $b, c, f$ ;  $c, a, g$  and  $a, b, h$  respectively; and we thus obtain from any one of these one and the same representation

$$\alpha\sqrt{bcf} + \beta\sqrt{cag} + \gamma\sqrt{abh} + \delta\sqrt{fgh} = 0$$

of the cubic hexad. And the 70 tripairs give thus the 70 representations of the same cubic hexad.

The whole number of hemi-tripairs is  $36 \times 56 = 2016$ : it may be remarked that there exists a system of 288 heptads, each of 7 odd characteristics such that selecting at pleasure any 3 characteristics out of the heptad, we obtain always a hemi-tripair; we have thus in all  $288 \times 35 = 10080$  hemi-tripairs: this is =  $2016 \times 5$ , or we have the 2016 hemi-tripairs each taken 5 times. Weber's Table II. exhibits 36 out of the 288 heptads.

I recall that in the algorithm derived from Hesse's theory the bitangents are represented by the duads 12, 13, ..., 78 formed with the eight figures 1, 2, 3, 4, 5, 6, 7, 8; these duads correspond to the odd characteristics as shown in the Table, and the table shows also triads corresponding to all the even characteristics except  $\frac{000}{000}$ .

*Top line of characteristic.*

|     |     |     |     |     |     |     |     |     |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|
|     | 000 | 100 | 010 | 110 | 001 | 101 | 011 | 111 |
| 000 |     | 236 | 345 | 137 | 467 | 156 | 124 | 257 |
| 100 | 237 | 67  | 136 | 12  | 157 | 48  | 256 | 35  |
| 010 | 245 | 127 | 23  | 68  | 134 | 357 | 15  | 47  |
| 110 | 126 | 13  | 78  | 145 | 356 | 25  | 46  | 234 |
| 001 | 567 | 146 | 125 | 247 | 45  | 17  | 38  | 26  |
| 101 | 147 | 58  | 246 | 34  | 16  | 123 | 27  | 367 |
| 011 | 135 | 347 | 14  | 57  | 17  | 36  | 167 | 456 |
| 111 | 346 | 24  | 56  | 235 | 37  | 267 | 457 | 18  |

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See my “Algorithm of the triple  $\vartheta$ -functions,” Crelle, t. LXXXVII. p. 165, [701]. The  $(35 + 28 =)$  63 hexpairs then are

35 hexpairs such as [*Figure: rectangle with vertices 5, 6 on top and 7, 8 on bottom, with internal diagonals*], say this is 1234.5678 or for shortness 567 (the 8 going always with the expressed triad): that is, 567 denotes the hexpair

$$12.34; 13.24; 14.23; 56.78; 57.68; 58.67;$$

and

28 hexpairs such as [*Figure: vertex 1 connected by lines to a pentagonal arrangement 3, 4, 5, 6, 7, 8 with 2 at the right*], say this is 12; that is, 12 denotes the hexpair

$$13.32; 14.42; 15.52; 16.62; 17.72; 18.82.$$

It is to be noticed that the odd characteristics, as represented by their duad symbols, can be added by the formulae

$$12 + 23 = 13, \text{ etc.,}$$

or, what is the same thing,

$$12 + 13 + 23 = 0, \text{ etc., } \left( = \begin{smallmatrix} 000 \\ 000 \end{smallmatrix} \right),$$

and

$$12 + 34 = 13 + 24 = 14 + 23 = 56 + 78 = 57 + 68 = 58 + 67 = 567, \text{ etc.}$$

Thus, referring to the table,

$$12 + 23 = 13 \quad \text{means} \quad \frac{110}{100} + \frac{010}{010} = \frac{100}{110},$$

and

$$12 + 34 = 567 \quad \text{means} \quad \frac{110}{100} + \frac{110}{101} = \frac{000}{001},$$

which are right.

The 288 heptads are

8 heptads such as [*Figure: vertex 1 at top with lines radiating down to a heptagonal arrangement of points labelled 2, 3, 4, 5, 6, 7, 8*], say this is the heptad 1, denoting the seven duads 12, 13, 14, 15, 16, 17, 18:

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and

280 heptads such as [*Figure: two triangular diagrams — left triangle with vertex 1 connected to base 2, 3, 4, 5; right small triangle with vertex labels 6, 7, 8*], say this is the heptad 1.678, denoting the seven duads 12, 13, 14, 15, 67, 68, 78.

We hence see that the 2016 hemi-tripairs are

280 hemi-tripairs [Figure: triangle with vertex 1 and base 2, 3, 4] (I.), say this is 1.234, denoting the three duads 12, 13, 14:

1680 hemi-tripairs [Figure: triangle with vertex 1 at left connected to small triangle 2, 6, 7, 8 at right] (II.), say this is 12(6.78), denoting the three duads 12, 67, 68:

56 hemi-tripairs [Figure: triangle 1-2-3] (III.), say this is 123, denoting the three duads 12, 13, 23;

2016.

We further see how each hemi-tripair may be completed into a tripair in 5 different ways: thus (I.) gives the 5 tripairs  $2 \cdot \backslash 1 \cdot 4, 2 \cdot 3 \cdot 4$  (depending on choice 5, 6, 7 or 8); (III.) gives the 5 tripairs  $\frac{34}{(7)8}$ ; while (II.) gives the 3 tripairs  $\frac{4a}{5 \text{ or } 4 \text{ or } 3}$  and the 2 tripairs

[Figure: two small triangular diagrams in the lower-left showing example tripair configurations with vertex labels 1, 6, 2, 6, etc.]

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To each even characteristic there belongs a system of 56 hemi-tripairs; thus for the characteristic  $\frac{000}{000}$ , the 56 hemi-tripairs are 123, that is, 12, 13, 23, etc.: whence the 70 tripairs are 1234, that is, 12.34; 13.24; 14.23, etc.; and in any such tripair, say in 1234, we have the set of four hemi-tripairs 123, 124, 134, 234, for each of which the sum of the three characteristics is

$$(12 + 13 + 23 = \frac{000}{000}, \text{ etc.}) :$$

and the other set 1.234, 2.134, 3.124, 4.123, for each of which the sum of the three characteristics is

$$= 567 \ (12 + 13 + 14, = 23 + 14, = 567, = \frac{000}{001}).$$

To find the hemi-tripairs that belong to any other even characteristic; for instance,  $\frac{000}{001}$ , corresponds to 567: we have 4 such as 1.234; 24 such as (5.12)34; 4 such as 5.678; and 24 such as (1.56)78; in all  $4 + 24 + 4 + 24, = 56$ . The tripairs are the 2, 1234, 5678; 16 such as 54(123); 16 such as 15(678); 36 such as (5162)34.78; in all  $2 + 16 + 16 + 36, = 70$ ; and in each of these it is easy to select the hemi-tripairs for which the sum of the 3 duads is = 567.

Cambridge, 27 December, 1878.

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ON THE ADDITION OF THE DOUBLE  $\vartheta$ -FUNCTIONS.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), t. LXXXVIII. (1879), pp. 74–81.]

I ASSUME in general

$$\Theta_e = a - e . b - e . c - e . d - e . e - \theta . f - \theta ,$$

and I consider the variables  $x, y, z, w, p, q$ , connected by the equations

$$\left| \begin{array}{cccccc} 1, & 1, & 1, & 1, & 1, & 1 \\ x, & y, & z, & w, & p, & q \\ x^2, & y^2, & z^2, & w^2, & p^2, & q^2 \\ x^3, & y^3, & z^3, & w^3, & p^3, & q^3 \\ \sqrt{X}, & \sqrt{Y}, & \sqrt{Z}, & \sqrt{W}, & \sqrt{P}, & \sqrt{Q} \end{array} \right| = 0,$$

equivalent to two independent equations, which in fact serve to determine  $p, q$ , or say the symmetrical functions  $p + q$  and  $pq$ , in terms of  $x, y, z, w$ .

These equations, it is well known, constitute a particular integral of the differential equations

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} + \frac{dp}{\sqrt{P}} + \frac{dq}{\sqrt{Q}} = 0,$$

$$\frac{x \, dx}{\sqrt{X}} + \frac{y \, dy}{\sqrt{Y}} + \frac{z \, dz}{\sqrt{Z}} + \frac{w \, dw}{\sqrt{W}} + \frac{p \, dp}{\sqrt{P}} + \frac{q \, dq}{\sqrt{Q}} = 0,$$

or what is the same thing, regarding  $p, q$  as arbitrary constants, they constitute the general integral of the differential equations

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} + \frac{dw}{\sqrt{W}} = 0,$$

$$\frac{x \, dx}{\sqrt{X}} + \frac{y \, dy}{\sqrt{Y}} + \frac{z \, dz}{\sqrt{Z}} + \frac{w \, dw}{\sqrt{W}} = 0.$$

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Pages 476–500 (scan); papers 703 (continued) and 704

## [Continuation of Paper 703: On the Addition of the Double $\vartheta$ -Functions.]

I attach the numbers 1, 2, 3, 4, 5, 6 to the variables  $x, y, z, w, p, q$ , respectively: and write

$$A_{12} = \sqrt{a - x \cdot a - y}, \quad A_{34} = \sqrt{a - z \cdot a - w}, \quad A_{56} = \sqrt{a - p \cdot a - q};$$

(six equations),

$$AB_{12} = \frac{1}{x-y} \left\{ \sqrt{a - x \cdot b - x \cdot f - x \cdot c - y \cdot d - y \cdot e - y} - \sqrt{a - y \cdot b - y \cdot f - y \cdot c - x \cdot d - x \cdot e - x} \right\}; \text{ etc.}$$

(ten equations),

where it is to be borne in mind that  $AB$  is an abbreviation for  $ABF \cdot CDE$ , and so in other cases, the letter  $F$  belonging always to the expressed duad: there are thus in all the sixteen functions  $A, B, C, D, E, F, AB, AC, AD, AE, BC, BD, BE, CD, CE, DE$ , these being functions of  $x$  and  $y$ , of  $z$  and  $w$ , and of  $p$  and  $q$ , according as the suffix is 12, 34, or 56.

It is to be shown that the 16 functions  $A_{56}, AB_{56}$  of  $p$  and  $q$  can be by means of the given equations expressed as proportional to rational and integral functions of the 16 functions  $A_{12}, AB_{12}, A_{34}, AB_{34}$  of  $x$  and  $y$ , and of  $z$  and  $w$  respectively: and it is clear that in so expressing them we have in effect the solution of the problem of the addition of the double  $\vartheta$ -functions.

I use when convenient the abbreviated notations

$$a - x = a_1, \quad a - y = a_2, \quad a - z = a_3, \quad a - w = a_4, \quad \text{etc.},$$

we have of course

$$\theta_{12} = x - y, \quad \theta_{34} = z - w, \quad \theta_{56} = p - q;$$

$$X = a_1 b_1 c_1 d_1 e_1 f_1,$$

$$\sqrt{ab} = \frac{1}{x-y} \left\{ \sqrt{a_1 b_1 f_1 c_2 d_2 e_2} - \sqrt{a_2 b_2 f_2 c_1 d_1 e_1} \right\}, \text{ etc.}$$

Proceeding to the investigation, the equations between the variables are obviously those obtained by the elimination of the arbitrary multipliers  $\alpha, \beta, \gamma, \delta, \varepsilon$  from the six equations obtained from

$$\alpha a^2 + \beta a^2 + \gamma a + \delta = \varepsilon \sqrt{a},$$

by writing therein for  $a$  the values  $x, y, z, w, p, q$  successively; we may consider the four equations

$$\alpha x^3 + \beta x^2 + \gamma x + \delta = \varepsilon \sqrt{X},$$

$$\alpha y^3 + \beta y^2 + \gamma y + \delta = \varepsilon \sqrt{Y},$$

$$\alpha z^3 + \beta z^2 + \gamma z + \delta = \varepsilon \sqrt{Z},$$

$$\alpha w^3 + \beta w^2 + \gamma w + \delta = \varepsilon \sqrt{W},$$

as serving to determine the ratios  $\alpha : \beta : \gamma : \delta : \varepsilon$  in terms of  $x, y, z, w$ ; and we have then for the determination of  $p, q$  the remaining two equations

$$\alpha p^3 + \beta p^2 + \gamma p + \delta = \varepsilon \sqrt{P},$$

which two equations may be replaced by the identity

$$(\alpha \theta^3 + \beta \theta^2 + \gamma \theta + \delta)^2 - \varepsilon^2 \theta = \alpha^2 (\theta - x)(\theta - y)(\theta - z)(\theta - w)(\theta - p)(\theta - q).$$

Writing herein  $\theta$  any one of the values  $a, b, c, d, e, f$ , for instance  $\theta = a$ , and taking the square root of each side, we have

$$\alpha a^3 + \beta a^2 + \gamma a + \delta = \sqrt{a^6 - \varepsilon^2} \sqrt{a - x \cdot a - y} \sqrt{a - z \cdot a - w} \sqrt{a - p \cdot a - q},$$

or as this may be written

$$\alpha a^3 + \beta a^2 + \gamma a + \delta = \sqrt{a^6 - \varepsilon^2} A_{12} A_{34} A_{56},$$

which equation when reduced gives the proportional value of  $A_{56}$ .

For the reduction we require the value of  $\alpha a^3 + \beta a^2 + \gamma a + \delta$ : calling this for the moment  $\Omega$ , we join to the four equations a fifth equation

$$\alpha a^3 + \beta a^2 + \gamma a + \delta = \Omega.$$

Eliminating  $\alpha, \beta, \gamma, \delta$ , we find

$$\begin{vmatrix} \varepsilon\sqrt{X} & x^3 & x^2 & x & 1 \\ \varepsilon\sqrt{Y} & y^3 & y^2 & y & 1 \\ \varepsilon\sqrt{Z} & z^3 & z^2 & z & 1 \\ \varepsilon\sqrt{W} & w^3 & w^2 & w & 1 \\ \Omega & a^3 & a^2 & a & 1 \end{vmatrix} = 0,$$

or, what is the same thing,

$$\Omega \begin{vmatrix} x^3 & x^2 & x & 1 \\ y^3 & y^2 & y & 1 \\ z^3 & z^2 & z & 1 \\ w^3 & w^2 & w & 1 \end{vmatrix} + \varepsilon \begin{vmatrix} \sqrt{X} & x^3 & x^2 & x & 1 \\ \sqrt{Y} & y^3 & y^2 & y & 1 \\ \sqrt{Z} & z^3 & z^2 & z & 1 \\ \sqrt{W} & w^3 & w^2 & w & 1 \\ a^3 & a^2 & a & 1 \end{vmatrix} = 0;$$

viz. this is

$$\begin{aligned} \Omega \cdot \overline{x-y} \cdot \overline{x-z} \cdot \overline{x-w} \cdot \overline{y-z} \cdot \overline{y-w} \cdot \overline{z-w} = & -\varepsilon \{ \sqrt{X} \cdot \overline{y-z} \cdot \overline{y-w} \cdot \overline{y-a} \cdot \overline{z-w} \cdot \overline{z-a} \cdot \overline{w-a} \\ & + \sqrt{Y} \cdot \overline{z-w} \cdot \overline{z-a} \cdot \overline{z-x} \cdot \overline{w-a} \cdot \overline{w-x} \cdot \overline{a-x} \\ & + \sqrt{Z} \cdot \overline{w-a} \cdot \overline{w-x} \cdot \overline{w-y} \cdot \overline{a-x} \cdot \overline{a-y} \cdot \overline{x-y} \\ & + \sqrt{W} \cdot \overline{a-x} \cdot \overline{a-y} \cdot \overline{a-z} \cdot \overline{x-y} \cdot \overline{x-z} \cdot \overline{y-z} \}, \end{aligned}$$

or as it may be written

$$\begin{aligned} \Omega \cdot \overline{x-z} \cdot \overline{x-w} \cdot \overline{y-z} \cdot \overline{y-w} = & \varepsilon \frac{\overline{a-z} \cdot \overline{a-w}}{x-y} [\overline{y-z} \cdot \overline{y-w} \cdot \overline{a-y} \sqrt{X} - \overline{x-z} \cdot \overline{x-w} \cdot \overline{a-x} \sqrt{Y}], \\ & + \varepsilon \frac{\overline{a-x} \cdot \overline{a-y}}{z-w} [\overline{w-x} \cdot \overline{w-y} \cdot \overline{a-w} \sqrt{Z} - \overline{z-x} \cdot \overline{z-y} \cdot \overline{a-z} \sqrt{W}], \end{aligned}$$

an equation for the determination of  $\Omega$ .

Consider first the expression which multiplies  $\varepsilon \cdot \overline{a-z} \cdot \overline{a-w}$ ; this is

$$= \frac{1}{x-y} \{ \overline{y-z} \cdot \overline{y-w} \cdot \overline{a-y} \sqrt{X} - \overline{x-z} \cdot \overline{x-w} \cdot \overline{a-x} \sqrt{Y} \}$$

we have

$$= -BE_{12} \frac{1}{x-y} \{ \sqrt{b_1 e_1 f_1 a_2 c_2 d_2} - \sqrt{b_2 e_2 f_2 a_1 c_1 d_1} \},$$

and multiplying this by  $A_{12} B_{12} C_{12} D_{12} = \sqrt{a_1 c_1 d_1 a_2 c_2 d_2}$ , we derive

$$\sqrt{a^6 - \varepsilon^2} BE_{12} \cdot C_{12} \cdot D_{12} \cdot A_{12} = \frac{1}{x-y} \{ c_2 d_2 a_1 \sqrt{X} - c_1 d_1 a_2 \sqrt{Y} \},$$

and similarly two other equations; the system may be written

$$\begin{aligned} \sqrt{a^6 - \varepsilon^2} BE \cdot C \cdot D \cdot A &= \frac{1}{x-y} \{ c_2 d_2 a_1 \sqrt{X} - c_1 d_1 a_2 \sqrt{Y} \}, \\ \sqrt{a^6 - \varepsilon^2} CE \cdot D \cdot B \cdot A &= \frac{1}{x-y} \{ d_2 b_2 a_1 \sqrt{X} - d_1 b_1 a_2 \sqrt{Y} \}, \\ \sqrt{a^6 - \varepsilon^2} DE \cdot B \cdot C \cdot A &= \frac{1}{x-y} \{ b_2 c_2 a_1 \sqrt{X} - b_1 c_1 a_2 \sqrt{Y} \}, \end{aligned}$$

the suffixes on the left-hand side being always 12. The letters  $b, c, d$  which enter cyclically into these equations are any three of the five letters other than  $a$ ; the remaining two letters  $e$  and  $f$  enter symmetrically, for  $BE$  is a mere abbreviation for the double triad  $BEF$ .  $ACD$ ; and the like for  $CE$ , and  $DE$ .

Multiplying these equations by

$$\frac{b-z \cdot b-w}{b-c \cdot b-d}, \quad \frac{c-z \cdot c-w}{c-d \cdot c-b}, \quad \frac{d-z \cdot d-w}{d-b \cdot d-c},$$

respectively, and then adding, the right-hand side becomes

$$= \frac{1}{x-y} \{ \overline{y-z} \cdot \overline{y-w} \cdot a_1 \sqrt{X} - \overline{x-z} \cdot \overline{x-w} \cdot a_2 \sqrt{Y} \}.$$

Writing

$$\frac{b-z \cdot b-w}{b-c \cdot b-d} = \frac{-1}{c-d \cdot d-b \cdot b-c} \cdot \overline{b-z} \cdot \overline{b-w} \cdot \overline{c-d}, \text{ etc.,}$$

the left-hand side becomes

$$\frac{-\sqrt{a^6 - \varepsilon^2} A}{c-d \cdot d-b \cdot b-c} \{ \overline{c-d} \cdot \overline{b-z} \cdot \overline{b-w} \cdot BE_{12} \cdot C_{12} \cdot D_{12} + \overline{d-b} \cdot \overline{c-z} \cdot \overline{c-w} \cdot CE_{12} \cdot D_{12} \cdot B_{12} + \overline{b-c} \cdot \overline{d-z} \cdot \overline{d-w} \cdot DE_{12} \cdot B_{12} \cdot C_{12} \},$$

which for shortness may be written

$$\frac{-\sqrt{a^6 - \varepsilon^2} A}{c-d \cdot d-b \cdot b-c} \Sigma \{ \overline{c-d} \cdot \overline{B'_1} \cdot BE_{12} \cdot C_{12} \cdot D_{12} \},$$

the summation referring to the three terms obtained by the cyclical interchange of the letters  $b, c, d$ . The result thus is

$$\begin{aligned} & \frac{1}{x-y} \{ \overline{y-z} \cdot \overline{y-w} \cdot a_1 \sqrt{X} - \overline{x-z} \cdot \overline{x-w} \cdot a_2 \sqrt{Y} \} \\ &= \frac{-\sqrt{a^6 - \varepsilon^2} A_{12}}{c-d \cdot d-b \cdot b-c} \Sigma \{ \overline{c-d} \cdot \overline{B'_1} \cdot BE_{12} \cdot C_{12} \cdot D_{12} \}. \end{aligned}$$

Interchanging  $x, y$  with  $z, w$  respectively, we have of course to interchange the suffixes 1, 2 and 3, 4; we thus find

$$\begin{aligned} & \frac{1}{z-w} \{ \overline{w-x} \cdot \overline{w-y} \cdot a_3 \sqrt{Z} - \overline{z-x} \cdot \overline{z-y} \cdot a_4 \sqrt{W} \} \\ &= \frac{-\sqrt{a^6 - \varepsilon^2} A_{34}}{c-d \cdot d-b \cdot b-c} \Sigma \{ \overline{c-d} \cdot \overline{B'_3} \cdot BE_{34} \cdot C_{34} \cdot D_{34} \}, \end{aligned}$$

and we hence find the value of  $\Omega \cdot \overline{x-z} \cdot \overline{x-w} \cdot \overline{y-z} \cdot \overline{y-w}$ . But  $\Omega = \alpha a^3 + \beta a^2 + \gamma a + \delta$ , is  $= \sqrt{a^6 - \varepsilon^2} \cdot A_{12} \cdot A_{34} \cdot A_{56}$ ; the resulting equation divides by  $A_{12} \cdot A_{34}$ ; throwing out this factor, we have

$$\begin{aligned} & \sqrt{a^6 - \varepsilon^2} (x-z \cdot x-w \cdot y-z \cdot y-w) (c-d \cdot d-b \cdot b-c) A_{56} \\ &= A_{34} \Sigma \{ \overline{c-d} \cdot \overline{B'_1} \cdot BE_{12} \cdot C_{12} \cdot D_{12} \} + A_{12} \Sigma \{ \overline{c-d} \cdot \overline{B'_3} \cdot BE_{34} \cdot C_{34} \cdot D_{34} \}, \end{aligned}$$

where, as before, the summations refer to the three terms obtained by the cyclical interchange of the letters  $b, c, d$ ; these being any three of the five letters other than  $a$ ; and the remaining two letters  $e, f$  enter into the formula symmetrically. The formula gives thus for  $A_{56}$  ten values which are of course equal to each other.

Writing for  $a$  each letter in succession, we obtain formulae for each of the six single-letter functions  $A_{56}$  of  $p$  and  $q$ ; and the factor

$$\sqrt{a^6 - \varepsilon^2} (x-z \cdot x-w \cdot y-z \cdot y-w)$$

is the same in all the formulae.

We require further the expressions for the double-letter functions of  $p, q$ . Considering for example the function  $BE_{56}$ , which is

$$= \frac{1}{p-q} \{ \sqrt{b_5 e_5 f_5 a_6 c_6 d_6} - \sqrt{b_6 e_6 f_6 a_5 c_5 d_5} \}.$$

then multiplying by  $DE_{56} \cdot A_{56} \cdot B_{56} \cdot C_{56}$  we have

$$\sqrt{a^6 - \varepsilon^2} DE_{56} \cdot A_{56} \cdot B_{56} \cdot C_{56} = \frac{1}{p-q} \{ a_5 b_5 c_5 \sqrt{P} - a_6 b_6 c_6 \sqrt{Q} \},$$

$$= \frac{1}{p-q} \{ \overline{a-q} \cdot \overline{b-q} \cdot \overline{c-q} \sqrt{P} - \overline{a-p} \cdot \overline{b-p} \cdot \overline{c-p} \sqrt{Q} \},$$

or recollecting that  $\varepsilon\sqrt{P}$ ,  $\varepsilon\sqrt{Q}$  are  $=\alpha p^3 + \beta p^2 + \gamma p + \delta$  and  $\alpha q^3 + \beta q^2 + \gamma q + \delta$  respectively, this is

$$\begin{aligned} & \varepsilon \cdot DE_{56} \cdot A_{56} \cdot B_{56} \cdot C_{56} \\ &= \frac{1}{p-q} \{ \overline{a-q} \cdot \overline{b-q} \cdot \overline{c-q} (\alpha p^3 + \beta p^2 + \gamma p + \delta) - \overline{a-p} \cdot \overline{b-p} \cdot \overline{c-p} (\alpha q^3 + \beta q^2 + \gamma q + \delta) \}. \end{aligned}$$

Using the well-known identity

$$\begin{aligned} \alpha p^3 + \beta p^2 + \gamma p + \delta &= (\alpha a^3 + \beta a^2 + \gamma a + \delta) \frac{p-b \cdot p-c \cdot p-d}{b-a \cdot c-a \cdot d-a} \\ &+ (\alpha b^3 + \beta b^2 + \gamma b + \delta) \frac{p-c \cdot p-d \cdot p-a}{c-b \cdot d-b \cdot a-b} \\ &+ (\alpha c^3 + \beta c^2 + \gamma c + \delta) \frac{p-d \cdot p-a \cdot p-b}{d-c \cdot a-c \cdot b-c} \\ &+ (\alpha d^3 + \beta d^2 + \gamma d + \delta) \frac{p-a \cdot p-b \cdot p-c}{a-d \cdot b-d \cdot c-d} \end{aligned}$$

and the like expression for  $\alpha q^3 + \beta q^2 + \gamma q + \delta$ , there will be on the right-hand side terms involving

$$\alpha a^3 + \beta a^2 + \gamma a + \delta, \quad \alpha b^3 + \beta b^2 + \gamma b + \delta, \quad \alpha c^3 + \beta c^2 + \gamma c + \delta,$$

but the term in  $\alpha d^3 + \beta d^2 + \gamma d + \delta$  will disappear of itself.

The term in  $\alpha a^3 + \beta a^2 + \gamma a + \delta$  is

$$\frac{1}{b-a \cdot c-a} (\alpha a^3 + \beta a^2 + \gamma a + \delta) \frac{1}{p-q} \{ \overline{a-q} \cdot \overline{b-q} \cdot \overline{c-q} ( )_p - \overline{a-p} \cdot \overline{b-p} \cdot \overline{c-p} ( )_q \},$$

where the expression in  $( )$  is  $= \overline{a-p} \cdot \overline{a-q}$ : hence the term is

$$\frac{\alpha a^3 + \beta a^2 + \gamma a + \delta}{b-a \cdot c-a} \overline{a-p} \cdot \overline{a-q} \cdot \overline{b-q} \cdot \overline{c-q} \cdot \overline{b-p} \cdot \overline{c-p},$$

which is

$$= \frac{\alpha a^3 + \beta a^2 + \gamma a + \delta}{b-a \cdot c-a} A_{56}^2 B_{56}^2 C_{56}^2.$$

Forming the two other like terms, the equation is

$$\begin{aligned} \varepsilon \cdot DE_{56} \cdot A_{56} \cdot B_{56} \cdot C_{56} &= \frac{\alpha a^3 + \beta a^2 + \gamma a + \delta}{b-a \cdot c-a} A_{56}^2 B_{56}^2 C_{56}^2 \\ &+ \frac{\alpha b^3 + \beta b^2 + \gamma b + \delta}{c-b \cdot a-b} A_{56}^2 B_{56}^2 C_{56}^2 \\ &+ \frac{\alpha c^3 + \beta c^2 + \gamma c + \delta}{a-c \cdot b-c} A_{56}^2 B_{56}^2 C_{56}^2. \end{aligned}$$

But the expressions

$$\alpha a^3 + \beta a^2 + \gamma a + \delta, \quad \alpha b^3 + \beta b^2 + \gamma b + \delta, \quad \alpha c^3 + \beta c^2 + \gamma c + \delta,$$

are

$$= \frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon} A_{12} \cdot A_{34} \cdot A_{56}, \quad \frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon} B_{12} \cdot B_{34} \cdot B_{56}, \quad \frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon} C_{12} \cdot C_{34} \cdot C_{56},$$

respectively: the whole equation thus divides by  $C_{56} \cdot A_{56} \cdot B_{56}$ ; throwing out this factor, and then multiplying

each side by  $-\frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon}$  we find

$$\frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon} DE_{56} = \Sigma \frac{b-c \cdot A_{12} \cdot A_{34} \cdot B_{56} \cdot C_{56}}{b-c \cdot c-a \cdot a-b} \cdot \overline{(-\sqrt{a^6 - \varepsilon^2}/\varepsilon)},$$

in which formula if we imagine

$$-\frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon} \quad \text{and} \quad -\frac{\sqrt{a^6 - \varepsilon^2}}{\varepsilon}$$



each replaced by its value in terms of the  $xy$ - and  $zw$ -functions, we have an equation of the form

$$(x - z \cdot x - w \cdot y - z \cdot y - w) DE_{56} = \overline{x - z \cdot x - w \cdot y - z \cdot y - w} \cdot M,$$

where  $M$  is a given rational and integral function of the 16 and 16 functions  $A_{12}, AB_{12}$  and  $A_{34}, AB_{34}$  of  $x$  and  $y$  and of  $z$  and  $w$  respectively. The factor

$$(x - z \cdot x - w \cdot y - z \cdot y - w)$$

is retained on the left-hand side as being the same factor which enters into the equations for  $A_{56}$ , etc.: but on the right-hand side  $x - z \cdot x - w \cdot y - z \cdot y - w$  should be expressed in terms of the  $xy$ - and  $zw$ -functions. This can be done by means of the identity

$$x - z \cdot x - w \cdot y - z \cdot y - w = \Sigma \frac{\begin{vmatrix} 1 & x+y & xy \\ 1 & z+w & zw \\ 1 & a+b & ab \end{vmatrix} \begin{vmatrix} 1 & x+y & xy \\ 1 & z+w & zw \\ 1 & a+c & ac \end{vmatrix}}{a-b \cdot a-c},$$

where the summation refers to the three terms obtained by the cyclical interchange of the letters  $a, b, c$ . The first determinant, multiplied by  $a-b$ , is in fact

$$\begin{vmatrix} a-z \cdot a-w & a-x \cdot a-y \\ b-z \cdot b-w & b-x \cdot b-y \end{vmatrix},$$

and the second determinant, multiplied by  $-(a-c)$ , is

$$= \begin{vmatrix} a-z \cdot a-w & a-x \cdot a-y \\ c-z \cdot c-w & c-x \cdot c-y \end{vmatrix},$$

so that the formula may also be written

$$x - z \cdot x - w \cdot y - z \cdot y - w = \Sigma \frac{\begin{vmatrix} a-z \cdot a-w & a-x \cdot a-y \\ b-z \cdot b-w & b-x \cdot b-y \end{vmatrix} \begin{vmatrix} a-z \cdot a-w & a-x \cdot a-y \\ c-z \cdot c-w & c-x \cdot c-y \end{vmatrix}}{(a-b)^2(a-c)^2},$$

or, what is the same thing, it is

$$\overline{x - z \cdot x - w \cdot y - z \cdot y - w} = \Sigma \frac{\cdots}{\cdots},$$

which is the required expression for  $x - z \cdot x - w \cdot y - z \cdot y - w$ ; the letters  $a, b, c$ , which enter into the formula, are any three of the six letters.

As regards the verification of the identity, observe that it may be written

$$x - z \cdot x - w \cdot y - z \cdot y - w = \Sigma \frac{\{L + M(a+b) + Nab\}\{L + M(a+c) + Nac\}}{a-b \cdot a-c},$$

where  $L, M, N$  are

$$= (x+y)zw - (z+w)xy, \quad xy - zw, \quad \text{and} \quad z+w-x-y:$$

this is readily reduced to

$$x - z \cdot x - w \cdot y - z \cdot y - w = M^2 - NL,$$

which can be at once verified.

Cambridge, 12th March, 1879.

I take the opportunity of remarking that, in the double-letter formulae, the sign of the second term is, not as I have in general written it  $-$ , but is  $+$ ,

$$AB = \frac{1}{x-y} \{ \sqrt{a_1 b_1 f_1 c_2 d_2 e_2} + \sqrt{a_2 b_2 f_2 c_1 d_1 e_1} \}, \text{ etc.}$$

In fact, introducing a factor  $\omega$  which is a function of  $x$  and  $y$ , the odd and even  $\vartheta$ -functions are  $= \omega\sqrt{a_1a_2}$ , etc., and

$$= \omega \frac{1}{x-y} \left\{ \sqrt{a_1b_1f_1c_2d_2e_2} + \sqrt{a_2b_2f_2c_1d_1e_1} \right\}, \text{ etc.},$$

respectively;  $\omega$  is a function which on the interchange of  $x, y$  changes only its sign; and this being so, then when  $x$  and  $y$  are interchanged, each single-letter function changes its sign, and each double-letter function remains unaltered.

Cambridge, 29th July, 1879.

## 704.

### A MEMOIR ON THE SINGLE AND DOUBLE THETA-FUNCTIONS.

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The Theta-Functions, although arising historically from the Elliptic Functions, may be considered as in order of simplicity preceding these, and connecting themselves directly with the exponential function ( $e^x$  or)  $\exp. x$ ; viz. they may be defined each of them as a sum of a series of exponentials, singly infinite in the case of the single functions, doubly infinite in the case of the double functions; and so on. The number of the single functions is 4; and the quotients of these, or say three of them each divided by the fourth, are the elliptic functions  $\text{sn}, \text{cn}, \text{dn}$ ; the number of the double functions is ( $4^2 =$ ) 16; and the quotients of these, or say fifteen of them each divided by the sixteenth, are the hyper-elliptic functions of two arguments depending on the square root of a sextic function. Generally, the number of the  $p$ -tuple theta-functions is  $4^p$ ; and the quotients of these, or say all but one of them each divided by the remaining function, are the Abelian functions of  $p$  arguments depending on the irrational function  $y$  defined by the equation  $F(x, y) = 0$  of a curve of deficiency  $p$ . If, instead of connecting the ratios of the functions with a plane curve, we consider the functions themselves as coordinates of a point in a space of  $(4^p - 1)$  dimensions, then we have the single functions as the four coordinates of a point on a quadri-quadric curve (one-fold locus) in ordinary space; and the double functions as the sixteen coordinates of a point on a quadri-quadric two-fold locus in 15-dimensional space, the deficiency of this two-fold locus being of course  $= 2$ .

The investigations contained in the First Part of the present Memoir, although for simplicity of notation exhibited only in regard to the double functions are, in fact, applicable to the general case of the  $p$ -tuple functions; but in the main the Memoir relates only to the single and double functions, and the title has been given to it accordingly. The investigations just referred to extend to the single functions and there is, it seems to me, an advantage in carrying on the two theories simultaneously up to and inclusive of the establishment of what I call the Product-theorem: this is a natural point of separation for the theories of the single and the double functions respectively. The ulterior developments of the two theories are indeed closely analogous to each other; but on the one hand the course of the single theory would be only with difficulty perceptible in the greater complexity of the double theory; and on the other hand we need the single theory as a guide for the course of the double theory.

I accordingly stop to point out in a general manner the course of the single theory, and, in connexion with it but more briefly, that of the double theory; and I then, in the Second and the Third Parts respectively, consider in detail the two theories separately; first, that of the single functions, and then that of the double functions. The paragraphs of the Memoir are numbered consecutively.

The definition adopted for the theta-functions differs somewhat from that which is ordinarily used.

The earlier memoirs on the double theta-functions are the well-known ones:

Rosenhain, “Mémoire sur les fonctions de deux variables et à quatre périodes, qui sont les inverses des intégrales ultra-elliptiques de la première classe.” [1846.] Paris: — *Mém. Savans Étrang.*, t. XI. (1851), pp. 361–468.

Göpel, “Theoriae transcendentium Abelianarum primi ordinis adumbratio levis,” *Crelle*, t. XXXV. (1847), pp. 277–312.

My first paper — Cayley, “On the Double  $\vartheta$ -Functions in connexion with a 16-nodal Surface,” Crelle-Borchardt, t. LXXXIII. (1877), pp. 210–219, [662]—was founded directly upon these, and was immediately followed by Dr Borchardt’s paper,

Borchardt, “Ueber die Darstellung der Kummer’sche Fläche vierter Ordnung mit sechzehn Knotenpunkten durch die Göpelsche biquadratische Relation zwischen vier Thetafunctionen mit zwei Variabeln,” Ditto, pp. 234–244.

My other later papers, [663, 664, 665, 697, 703], are contained in the same Journal.

## FIRST PART.—INTRODUCTORY.

### *Definition of the theta-functions.*

1. The  $p$ -tuple functions depend upon  $\frac{1}{2}p(p+1)$  parameters which are the coefficients of a quadric function of  $p$  ultimately disappearing integers, upon  $p$  arguments, and upon  $2p$  characters, each  $= 0$  or  $1$ , which form the characteristic of the  $4^p$  functions; but it will be sufficient to write down the formulae in the case  $p = 2$ .

As already mentioned, the adopted definition differs somewhat from that which is ordinarily used. I use, as will be seen, a quadric function  $\frac{1}{4}(a, h, b \mid m, n)^2$  with even integer values of  $m, n$ , instead of  $(a, h, b \mid m, n)^2$  with even or odd values; and I write the other term  $\frac{1}{2}\pi i(mu + nv)$ , instead of  $mu + nv$ ; this comes to affecting the arguments  $u, v$  with a factor  $\frac{1}{2}\pi i$ , so that the quarter-periods (instead of being  $\frac{1}{2}\pi i$ ) are made to be  $= 1$ .

2. We write

$$(\ ) = \frac{1}{4}(a, h, b \mid m, n)^2 + \frac{1}{2}\pi i(mu + nv),$$

and in like manner

$$\left(\begin{smallmatrix} m+\alpha \\ n+\beta \end{smallmatrix} \middle| \begin{smallmatrix} u+\gamma \\ v+\delta \end{smallmatrix}\right) = \frac{1}{4}(a, h, b \mid m + \alpha, n + \beta)^2 + \frac{1}{2}\pi i\{(m + \alpha)(u + \gamma) + (n + \beta)(v + \delta)\},$$

and prefixing to either of these the functional symbol  $\exp$ . we have the exponential of the function in question, that is,  $e$  with the function as an exponent.

We then write, as the definition of the double theta-functions,

$$\vartheta\left(\begin{smallmatrix} \alpha, \beta \\ \gamma, \delta \end{smallmatrix}\right)(u, v) = \Sigma \exp.\left(\begin{smallmatrix} m+\alpha \\ n+\beta \end{smallmatrix} \middle| \begin{smallmatrix} u+\gamma \\ v+\delta \end{smallmatrix}\right)$$

where the summation extends to all positive and negative even integer values (zero included) of  $m$  and  $n$  respectively;  $\alpha, \beta, \gamma, \delta$  might denote any quantities whatever, but for the theta-functions they are regarded as denoting positive or negative integers; this being so, it will appear that the only effect of altering each or any of them by an even integer is to reverse (it may be) the sign of the function; and the distinct functions are consequently the ( $4^2 =$ ) 16 functions obtained by giving to each of the quantities  $\alpha, \beta, \gamma, \delta$  the two values 0 and 1 successively.

3. We thus have the double theta-functions, depending on the parameters  $(a, h, b)$  which determine the quadric function  $(a, h, b \mid m, n)^2$  of the disappearing even integers  $(m, n)$ , and on the two arguments  $(u, v)$ : in the symbol  $\left(\begin{smallmatrix} \alpha, \beta \\ \gamma, \delta \end{smallmatrix}\right)$ , which is called the characteristic, the characters  $\alpha, \beta, \gamma, \delta$  are each of them  $= 0$  or  $1$ ; and we thus have the 16 functions.

4. The parameters  $(a, h, b)$  may be real or imaginary, but they must be such that reducing each of them to its real part the resulting function  $(\ast \mid m, n)^2$  is invariable in its sign, and negative for all real values of  $m$  and  $n$ : this is, in fact, the condition for the convergency of the series which give the values of the theta-functions.

The characteristic  $\left(\begin{smallmatrix} \alpha, \beta \\ \gamma, \delta \end{smallmatrix}\right)$  is said to be even or odd according as the sum  $\alpha\gamma + \beta\delta$  is even or odd.

*Allied functions.*

5. As already remarked, the definition of

$$\vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma, \delta\end{matrix}\right)(u, v)$$

is not restricted to the case where the  $\alpha, \beta, \gamma, \delta$  represent integers, and there is actually occasion to consider functions of this form where they are not integers: in particular,  $\alpha, \beta$  may be either or each of them of the form, integer  $+\frac{1}{2}$ . But the functions thus obtained are not regarded as *theta-functions*, and the expression theta-function will consequently not extend to include them.

*Properties of the Theta-Functions: Various sub-headings.*

*Even-integer alteration of characters.*

6. If  $x, y$  be integers, then  $m, n$  having the several even integer values from  $-\infty$  to  $+\infty$  respectively, it is obvious that  $m + \alpha + 2x, n + \beta + 2y$  will have the same series of values with  $m + \alpha, n + \beta$  respectively; and it thence follows that

$$\vartheta\left(\begin{matrix}\alpha + 2x, \beta + 2y \\ \gamma, \delta\end{matrix}\right)(u, v) = \vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma, \delta\end{matrix}\right)(u, v).$$

Similarly if  $z, w$  are integers, then in the function

$$\vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma + 2z, \delta + 2w\end{matrix}\right)(u, v)$$

the argument of the exponential function contains the term

$$\frac{1}{2}\pi i \{m + \alpha \cdot u + \gamma + 2z + n + \beta \cdot v + \delta + 2w\}$$

this differs from its original value by

$$\begin{aligned} & \frac{1}{2}\pi i (m + \alpha \cdot 2z + n + \beta \cdot 2w), \\ & = \pi i (mz + nw) + \pi i (\alpha z + \beta w), \end{aligned}$$

and then,  $m$  and  $n$  being even integers,  $mz + nw$  is also an even integer, and the term  $\pi i (mz + nw)$  does not affect the value of the exponential: we thus introduce into each term of the series the factor  $\exp. \pi i (\alpha z + \beta w)$ , which is, in fact,  $(-)^{\alpha z + \beta w}$ , and we consequently have

$$\vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma + 2z, \delta + 2w\end{matrix}\right)(u, v) = (-)^{\alpha z + \beta w} \vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma, \delta\end{matrix}\right)(u, v),$$

or, uniting the two results,

$$\vartheta\left(\begin{matrix}\alpha + 2x, \beta + 2y \\ \gamma + 2z, \delta + 2w\end{matrix}\right)(u, v) = (-)^{\alpha z} (-)^{\beta w} \vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma, \delta\end{matrix}\right)(u, v).$$

This sustains the before-mentioned conclusion that the only distinct functions are the 16 functions obtained by giving to the characters  $\alpha, \beta, \gamma, \delta$  the values 0 and 1 respectively.

*Odd-integer alteration of characters.*

7. The effect is obviously to interchange the different functions.

*Even and odd functions.*

8. It is clear that  $-m - \alpha, -n - \beta$  have precisely the same series of values with  $m + \alpha, n + \beta$  respectively: hence considering the function

$$\vartheta\left(\begin{matrix}\alpha, \beta \\ \gamma, \delta\end{matrix}\right)(-u, -v),$$

the linear term in the argument of the exponential may be taken to be

$$\frac{1}{2}\pi i \{-m - \alpha \cdot -u + \gamma + -n - \beta \cdot -v + \delta\},$$

which is

$$= \frac{1}{2}\pi i \{m + \alpha . u + \gamma + n + \beta . v + \delta\} - \pi i \{m + \alpha . \gamma + n + \beta . \delta\};$$

the second term is here

$$= -\pi i (m\gamma + n\delta) - \pi i (\alpha\gamma + \beta\delta),$$

where,  $m\gamma + n\delta$  being an even integer, the part  $-\pi i (m\gamma + n\delta)$  does not alter the value of the exponential: the effect of the remaining part  $-\pi i (\alpha\gamma + \beta\delta)$  is to affect each term of the series with the factor  $\exp . -\pi i (\alpha\gamma + \beta\delta)$ , or what is the same thing,  $\exp . \pi i (\alpha\gamma + \beta\delta)$ , each of these being, in fact,  $(-)^{\alpha\gamma + \beta\delta}$ .

Viz.  $\vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)(u, v)$  is an even or odd function of the two arguments  $(u, v)$  conjointly, according as the characteristic  $\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)$  is even or odd.

*The quarter-periods unity.*

**9.** Taking  $z$  and  $w$  integers, we have from the definition

$$\vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)(u + z, v + w) = \vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma + z, \delta + w\end{smallmatrix}\right)(u, v),$$

viz. the effect of altering the arguments  $u, v$  into  $u + z, v + w$  is simply to interchange the functions as shown by this formula.

If  $z$  and  $w$  are each of them even, then replacing them by  $2z, 2w$  respectively, we have

$$\vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)(u + 2z, v + 2w) = \vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma + 2z, \delta + 2w\end{smallmatrix}\right)(u, v),$$

which by a preceding formula is

$$= (-)^{\alpha z + \beta w} \vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)(u, v),$$

or the function is altered at most in its sign. And again writing  $2z, 2w$  for  $z, w$ , we have

$$\vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)(u + 4z, v + 4w) = \vartheta\left(\begin{smallmatrix}\alpha, \beta \\ \gamma, \delta\end{smallmatrix}\right)(u, v).$$

In reference to the foregoing results we say that the theta-functions have the quarter-periods  $(1, 1)$ , the half-periods  $(2, 2)$ , and the whole periods  $(4, 4)$ .

*The conjoint quarter quasi-periods.*

**10.** Taking  $x, y$  integers, we consider the effect of the change of  $u, v$  into

$$u + \frac{1}{\pi i}(ax + hy), \quad v + \frac{1}{\pi i}(hx + by).$$

It is convenient to start from the function

$$\vartheta\left(\begin{smallmatrix}\alpha - x, \beta - y \\ \gamma, \delta\end{smallmatrix}\right)\left(u + \frac{1}{\pi i}(ax + hy), v + \frac{1}{\pi i}(hx + by)\right);$$

the argument of the exponential is here

$$\begin{aligned} & \frac{1}{4}(a, h, b \mid m + \alpha - x, n + \beta - y)^2 \\ & + \frac{1}{2}\pi i \{m + \alpha - x . u + \gamma + \frac{1}{\pi i}(ax + hy) + n + \beta - y . v + \delta + \frac{1}{\pi i}(hx + by)\}, \end{aligned}$$

which is

$$= \frac{1}{4}(a, h, b \mid m + \alpha, n + \beta)^2 + \frac{1}{2}\pi i (m + \alpha . u + \gamma + n + \beta . v + \delta)$$

+ other terms which are as follows: viz. they are

$$-\frac{1}{2}(a, h, b \mid m + \alpha, n + \beta \mid x, y) + \frac{1}{2}(m + \alpha . ax + hy + n + \beta . hx + by)$$

$$+\frac{1}{4}(a, h, b \mid x, y)^2 - \frac{1}{2}\pi i (x \cdot u + \gamma + y \cdot v + \delta) \\ - \frac{1}{2}(x \cdot ax + hy + y \cdot hx + by),$$

where the terms of the right-hand column are, in fact,

$$= \frac{1}{2}\pi i (m + \alpha \cdot ax + hy + n + \beta \cdot hx + by) - \frac{1}{2}(a, h, b \mid m + \alpha, n + \beta \mid x, y) \\ - \frac{1}{2}\pi i (x \cdot u + \gamma + y \cdot v + \delta) - \frac{1}{4}(a, h, b \mid x, y)^2,$$

and the other terms in question thus reduce themselves to

$$-\frac{1}{4}(a, h, b \mid x, y)^2 - \frac{1}{2}\pi i (x \cdot u + \gamma + y \cdot v + \delta),$$

which are independent of  $m, n$ , and they thus affect each term of the series with the same exponential factor. The result is

$$\vartheta\left(\begin{matrix} \alpha - x, \beta - y \\ \gamma, \delta \end{matrix}\right)\left(u + \frac{1}{\pi i}(ax + hy), v + \frac{1}{\pi i}(hx + by)\right) \\ = \exp.\left\{-\frac{1}{4}(a, h, b \mid x, y)^2 - \frac{1}{2}\pi i (x \cdot u + \gamma + y \cdot v + \delta)\right\} \vartheta\left(\begin{matrix} \alpha, \beta \\ \gamma, \delta \end{matrix}\right)(u, v);$$

or (what is the same thing) for  $\alpha, \beta$  writing  $\alpha + x, \beta + y$  respectively, we have

$$\vartheta\left(\begin{matrix} \alpha, \beta \\ \gamma, \delta \end{matrix}\right)\left(u + \frac{1}{\pi i}(ax + hy), v + \frac{1}{\pi i}(hx + by)\right) \\ = \exp.\left\{-\frac{1}{4}(a, h, b \mid x, y)^2 - \frac{1}{2}\pi i (x \cdot u + \gamma + y \cdot v + \delta)\right\} \vartheta\left(\begin{matrix} \alpha + x, \beta + y \\ \gamma, \delta \end{matrix}\right)(u, v).$$

Taking  $x, y$  even, or writing  $2x, 2y$  for  $x, y$ , then on the right-hand side we have

$$\vartheta\left(\begin{matrix} \alpha + 2x, \beta + 2y \\ \gamma, \delta \end{matrix}\right)(u, v),$$

which is

$$= \vartheta\left(\begin{matrix} \alpha, \beta \\ \gamma, \delta \end{matrix}\right)(u, v) :$$

but there is still the exponential factor.

**11.** The formulae show that the effect of the change  $u, v$  into  $u + \frac{1}{\pi i}(ax + hy), v + \frac{1}{\pi i}(hx + by)$ , where  $x, y$  are integers, is to interchange the functions, affecting them however with an exponential factor; and we hence say that  $\frac{1}{\pi i}(a, h), \frac{1}{\pi i}(h, b)$  are conjoint quarter quasi-periods.

*The product-theorem.*

**12.** We multiply two theta-functions

$$\vartheta\left(\begin{matrix} \alpha, \beta \\ \gamma, \delta \end{matrix}\right)(u, v), \quad \vartheta\left(\begin{matrix} \alpha', \beta' \\ \gamma', \delta' \end{matrix}\right)(u', v');$$

it is found that the result is a sum of four products

$$\Theta\left(\begin{matrix} \frac{1}{2}(\alpha + \alpha') + p, \frac{1}{2}(\beta + \beta') + q \\ \gamma + \gamma', \delta + \delta' \end{matrix}\right)(2u, 2v) \cdot \Theta\left(\begin{matrix} \frac{1}{2}(\alpha - \alpha') + p, \frac{1}{2}(\beta - \beta') + q \\ \gamma - \gamma', \delta - \delta' \end{matrix}\right)(2u', 2v'),$$

where  $p, q$  have in the four products respectively the values  $(0, 0), (1, 0), (0, 1)$ , and  $(1, 1)$ ;  $\Theta$  is written in place of  $\vartheta$  to denote that the parameters  $(a, h, b)$  are to be changed into  $(2a, 2h, 2b)$ . It is to be noticed that, if  $\alpha, \alpha'$  are both even or both odd, then  $\frac{1}{2}(\alpha + \alpha'), \frac{1}{2}(\alpha - \alpha')$  are integers; and so, if  $\beta, \beta'$  are both even or both odd, then  $\frac{1}{2}(\beta + \beta'), \frac{1}{2}(\beta - \beta')$  are integers; and these conditions being satisfied (and in particular they are so if  $\alpha = \alpha', \beta = \beta'$ ) then the functions on the right-hand side of the equation are theta-functions (with new parameters as already mentioned); but if the conditions are not satisfied, then the functions on the right-hand side are only allied functions. In the applications of the theorem the functions on the right-hand side are eliminated between the different equations, as will appear.

13. The proof is immediate: in the first of the theta-functions, the argument of the exponential is

$$\left( \begin{matrix} m+\alpha \\ n+\beta \end{matrix} \middle| \begin{matrix} u+\gamma \\ v+\delta \end{matrix} \right),$$

and in the second, writing  $m', n'$  instead of  $m, n$ , the argument is

$$\left( \begin{matrix} m'+\alpha' \\ n'+\beta' \end{matrix} \middle| \begin{matrix} u'+\gamma' \\ v'+\delta' \end{matrix} \right);$$

hence in the product, the argument of the exponential is the sum of these two functions, viz.

$$\begin{aligned} &= \frac{1}{4}(a, h, b \mid m + \alpha, n + \beta)^2 + \frac{1}{2}\pi i (m + \alpha \cdot u + \gamma + n + \beta \cdot v + \delta) \\ &+ \frac{1}{4}(a, h, b \mid m' + \alpha', n' + \beta')^2 + \frac{1}{2}\pi i (m' + \alpha' \cdot u' + \gamma' + n' + \beta' \cdot v' + \delta'). \end{aligned}$$

Comparing herewith the sum of the two functions

$$\begin{aligned} &\left( \begin{matrix} \mu + \frac{1}{2}(\alpha + \alpha') \\ \nu + \frac{1}{2}(\beta + \beta') \end{matrix} \middle| \begin{matrix} 2u + \gamma + \gamma' \\ 2v + \delta + \delta' \end{matrix} \right)', \quad \left( \begin{matrix} \mu' + \frac{1}{2}(\alpha - \alpha') \\ \nu' + \frac{1}{2}(\beta - \beta') \end{matrix} \middle| \begin{matrix} 2u' + \gamma - \gamma' \\ 2v' + \delta - \delta' \end{matrix} \right)', \\ &= \frac{1}{4}(2a, 2h, 2b \mid \mu + \frac{1}{2}(\alpha + \alpha'), \nu + \frac{1}{2}(\beta + \beta'))^2 \\ &+ \frac{1}{2}\pi i \left\{ \mu + \frac{1}{2}(\alpha + \alpha') \cdot 2u + \gamma + \gamma' + \nu + \frac{1}{2}(\beta + \beta') \cdot 2v + \delta + \delta' \right\} \\ &+ \frac{1}{4}(2a, 2h, 2b \mid \mu' + \frac{1}{2}(\alpha - \alpha'), \nu' + \frac{1}{2}(\beta - \beta'))^2 \\ &+ \frac{1}{2}\pi i \left\{ \mu' + \frac{1}{2}(\alpha - \alpha') \cdot 2u' + \gamma - \gamma' + \nu' + \frac{1}{2}(\beta - \beta') \cdot 2v' + \delta - \delta' \right\} \end{aligned}$$

the two sums are identical if only

$$m + m' = 2\mu, \quad n + n' = 2\nu, \quad m - m' = 2\mu', \quad n - n' = 2\nu',$$

as may easily be verified by comparing the quadric and the linear terms separately. The product of the two theta-functions is thus

$$\Sigma \exp \cdot \left( \begin{matrix} \mu + \frac{1}{2}(\alpha + \alpha') \\ \nu + \frac{1}{2}(\beta + \beta') \end{matrix} \middle| \begin{matrix} 2u + \gamma + \gamma' \\ 2v + \delta + \delta' \end{matrix} \right)' \cdot \Sigma \exp \cdot \left( \begin{matrix} \mu' + \frac{1}{2}(\alpha - \alpha') \\ \nu' + \frac{1}{2}(\beta - \beta') \end{matrix} \middle| \begin{matrix} 2u' + \gamma - \gamma' \\ 2v' + \delta - \delta' \end{matrix} \right)',$$

with the proper conditions as to the values of  $\mu, \nu$  and of  $\mu', \nu'$  in the two sums respectively. As to this, observe that  $m, m'$  are even integers; say for a moment that they are *similar* when they are both  $\equiv 0$  or both  $\equiv 2 \pmod{4}$ , but *dissimilar* when they are one of them  $\equiv 0$  and the other of them  $\equiv 2 \pmod{4}$ ; and the like as regards  $n, n'$ . Hence if  $m, m'$  are similar,  $\mu, \mu'$  are both of them even; but if  $m, m'$  are dissimilar, then  $\mu, \mu'$  are both of them odd. And so if  $n, n'$  are similar,  $\nu, \nu'$  are both of them even; but if  $n, n'$  are dissimilar, then  $\nu, \nu'$  are both odd.

14. There are four cases:

$$\begin{aligned} &m, m' \text{ similar, } \quad n, n' \text{ similar,} \\ &m, m' \text{ dissimilar, } \quad n, n' \text{ similar,} \\ &m, m' \text{ similar, } \quad n, n' \text{ dissimilar,} \\ &m, m' \text{ dissimilar, } \quad n, n' \text{ dissimilar.} \end{aligned}$$

In the first of these,  $\mu, \nu, \mu', \nu'$  are all of them even, and the product is

$$= \Theta \left( \begin{matrix} \frac{1}{2}(\alpha + \alpha'), \frac{1}{2}(\beta + \beta') \\ \gamma + \gamma', \delta + \delta' \end{matrix} \right) (2u, 2v) \cdot \Theta \left( \begin{matrix} \frac{1}{2}(\alpha - \alpha'), \frac{1}{2}(\beta - \beta') \\ \gamma - \gamma', \delta - \delta' \end{matrix} \right) (2u', 2v').$$

In the second case, writing  $\mu + 1, \mu' + 1$  for  $\mu, \mu'$ , the new values of  $\mu, \mu'$  will be both even, and we have the like expression with only the characters  $\frac{1}{2}(\alpha + \alpha'), \frac{1}{2}(\alpha - \alpha')$  each increased by 1; so in the third case we obtain the like expression with only the characters  $\frac{1}{2}(\beta + \beta'), \frac{1}{2}(\beta - \beta')$  each increased by 1; and in the fourth case the like expression with the four upper characters each increased by 1. The product of the two theta-functions is thus equal to the sum of the four products, according to the theorem.

*Résumé of the ulterior theory of the single functions.*

**15.** For the single theta-functions the Product-theorem comprises 16 equations, and for the double theta-functions, 256 equations: these systems will be given in full in the sequel. But attending at present to the single functions, I write down here the first four of the 16 equations, viz. these are

$$\begin{aligned} 0.0 \quad \vartheta\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)(u+u') \vartheta\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)(u-u') &= XX' + YY', \\ 1.0 \quad \vartheta\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right),, \quad \vartheta\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right),, &= YX' + XY', \\ 0.1 \quad \vartheta\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right),, \quad \vartheta\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right),, &= XX' - YY', \\ 1.1 \quad \vartheta\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right),, \quad \vartheta\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right),, &= -YX' + XY'; \end{aligned}$$

where  $X, Y$  denote  $\Theta\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)(2u), \Theta\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)(2u)$  respectively, and  $X', Y'$  the same functions of  $2u'$  respectively. In the other equations we have on the left-hand the product of different theta-functions of  $u+u', u-u'$  respectively, and on the right-hand expressions involving other functions,  $X_1, Y_1, X'_1, Y'_1$ , &c., of  $2u$  and  $2u'$  respectively.

**16.** By writing  $u' = 0$ , we have on the left-hand, squares or products of theta-functions of  $u$ , and on the right-hand expressions containing functions of  $2u$ : in particular, the above equations show that the squares of the four theta-functions are equal to linear functions of  $X, Y$ ; that is, there exist between the squared functions two linear relations: or again, introducing a variable argument  $x$ , the four squared functions may be taken to be proportional to linear functions

$$\mathfrak{A}(a-x), \mathfrak{B}(b-x), \mathfrak{C}(c-x), \mathfrak{D}(d-x),$$

where  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, a, b, c, d$  are constants. This suggests a new notation for the four functions, viz. we write

$$\vartheta\left(\begin{smallmatrix} 0 \\ 0 \end{smallmatrix}\right)(u) = Au, \quad \vartheta\left(\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right)(u) = Bu, \quad \vartheta\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right)(u) = Cu, \quad \vartheta\left(\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}\right)(u) = Du;$$

and the result just mentioned then is

$$A^2u : B^2u : C^2u : D^2u = \mathfrak{A}(a-x) : \mathfrak{B}(b-x) : \mathfrak{C}(c-x) : \mathfrak{D}(d-x),$$

which expresses that the four functions are the coordinates of a point on a quadri-quadric curve in ordinary space.

**17.** The remaining 12 of the 16 equations then contain on the left-hand products such as

$$A(u+u') \cdot B(u-u');$$

and by suitably combining them we obtain equations such as

$$\frac{B(u+u')A(u-u') - A(u+u')B(u-u')}{D(u+u')C(u-u') + C(u+u')D(u-u')} = \text{function}(u'),$$

where for brevity the arguments are written above; viz. the numerator of the fraction is

$$B(u+u')A(u-u') - A(u+u')B(u-u'),$$

and its denominator is

$$D(u+u')C(u-u') + C(u+u')D(u-u').$$

Admitting the form of the equation, the value of the function of  $u'$  is at once found by writing in the equation  $u = 0$ ; it is, as it ought to be, a function vanishing for  $u' = 0$ .



18. Take in this equation  $u$  indefinitely small; each side divides by  $u'$ , and the resulting equation is

$$\frac{A'u Bu - Bu A'u}{C_0 u D_0 u} = \text{const.},$$

where  $A'u, B'u$  are the derived functions, or differential coefficients in regard to  $u$ . It thus appears that the combination  $A'u Bu - Bu A'u$  is a constant multiple of  $Cu Du$ : or, what is the same thing, that the differential coefficient of the quotient-function  $\frac{B}{A}$  is a constant multiple of the product of the two quotient-functions  $\frac{C}{A}u$  and  $\frac{D}{A}u$ .

19. And then substituting for the several quotient-functions their values in terms of  $x$ , we obtain a differential relation between  $x, u$ ; viz. the form hereof is

$$\frac{du}{1} = \frac{M dx}{\sqrt{a - x \cdot b - x \cdot c - x \cdot d - x}}$$

and it thus appears that the quotient-functions are in fact elliptic-functions: the actual values as obtained in the sequel are

$$\begin{aligned} \text{sn } Ku &= \frac{Du}{Bu} = \sqrt{\frac{k'}{k}} \frac{Du}{Cu}, \\ \text{cn } Ku &= \sqrt{\frac{k'}{k}} \frac{Bu}{Cu}, \\ \text{dn } Ku &= \sqrt{k'} \frac{Au}{Cu}; \end{aligned}$$

and we thus of course identify the functions  $Au, Bu, Cu, Du$  with the  $H$  and the  $\Theta$  functions of Jacobi.

20. If in the above-mentioned four equations we write first  $u' = 0$ , and then  $u = u' = 0$ , and by means of the results eliminate from the original equations the quantities  $X, Y, X', Y'$  which occur therein, we obtain expressions for the four products such as  $A(u + u') A(u - u')$ . One of these equations is

$$C_0 0 \cdot C(u + u') C(u - u') = C^2 u C^2 u' - D^2 u D^2 u'.$$

Taking herein  $u'$  indefinitely small, we obtain

$$\frac{Cu C''u - (C'u)^2}{Cu} = \frac{C''_0}{C_0} - \left(\frac{D_0}{C_0}\right)^2 \frac{D^2 u}{C^2 u},$$

where the left-hand side is in fact  $\frac{d^2}{du^2} \log Cu$ , or this second derived function of the theta-function  $Cu$  is given in terms of the quotient-function  $\frac{D}{C}u$ ; hence, integrating twice and taking the exponential of each side, we obtain  $Cu$  as an exponential the argument of which contains the double integral  $\iint \left(\frac{Du}{Cu}\right)^2 (du)^2$ , of a squared quotient-function. This, in fact, corresponds to Jacobi's equation.

21. From the same equation

$$C_0 0 \cdot C(u + u') C(u - u') = C^2 u C^2 u' - D^2 u D^2 u',$$

differentiating logarithmically in regard to  $u'$  and integrating in regard to  $u$ , we obtain an equation containing on the left-hand side a term  $\log \frac{\Theta(u - a)}{\Theta(u + a)}$ , and on the right-hand an integral in regard to  $u$ ; this, in fact, corresponds to Jacobi's equation

$$\frac{\Theta'_a}{\Theta_a} \cdot 2u + \log \frac{\Theta(u - a)}{\Theta(u + a)} = -2 \int_0^u \frac{k^2 \text{sn } a \text{cn } a \text{dn } a \text{sn}^2 u \, du}{1 - k^2 \text{sn}^2 a \text{sn}^2 u}.$$

**22.** It may further be noticed that if, in the equation in question and in the three other equations of the system, we introduce into the integral the variable  $x$  in place of  $u$ , and the corresponding quantity  $\xi$  in place of  $u'$ , then the integral is that of an expression such as

$$\frac{dx}{T\sqrt{a-x \cdot b-x \cdot c-x \cdot d-x}},$$

where  $T$  is  $= x - \xi$ , or is  $=$  any one of three forms such as

$$\begin{vmatrix} 1 & \xi + x & \xi x \\ 1 & a + b & ab \\ 1 & c + d & cd \end{vmatrix}.$$

*Résumé of the ulterior theory of the double functions.*

**23.** The ulterior theory of the double functions is intended to be carried out on the like plan. As regards these, it is to be observed here that we have not only the 16 equations leading to linear relations between the squared functions, but that the remaining 240 equations lead also to linear relations between binary products of different functions. We have thus between the 16 functions a system of quadric relations, which in fact determine the ratios of the 16 functions in terms of two variable parameters  $x, y$ . (The 16 functions are thus the coordinates of a point on a quadri-quadric two-fold locus in 15-dimensional space.) The forms depend upon six constants,  $a, b, c, d, e, f$ : writing for shortness

$$\sqrt{a} = \sqrt{a-x \cdot a-y},$$

$$\sqrt{ab} = \frac{1}{x-y} \left\{ \sqrt{a-x \cdot b-x \cdot f-x \cdot c-y \cdot d-y \cdot e-y} + \sqrt{a-y \cdot b-y \cdot f-y \cdot c-x \cdot d-x \cdot e-x} \right\},$$

(observe that in the symbols  $\sqrt{ab}$  it is always  $f$  that accompanies the two expressed letters  $a, b$  or, what is the same thing, the duad  $ab$  is really an abbreviation for the double triad  $abf.cde$ ): then the 16 functions are proportional to properly determined constant multiples of

$$\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d}, \sqrt{e}, \sqrt{f}, \sqrt{ab}, \sqrt{ac}, \sqrt{ad}, \sqrt{ae}, \sqrt{bc}, \sqrt{bd}, \sqrt{be}, \sqrt{cd}, \sqrt{ce}, \sqrt{de} :$$

and this suggests that the functions should be represented by the single and double letter notation  $A(u, v), \dots, AB(u, v), \dots$ ; viz. if for shortness the arguments are omitted, then we have

$$A, B, C, D, E, F, AB, AC, AD, AE, BC, BD, BE, CD, CE, DE,$$

proportional to determinate constant multiples of the before-mentioned functions  $\sqrt{a}, \dots, \sqrt{ab}, \dots$ , of  $x$  and  $y$ .

**24.** It is interesting to notice why in the expressions for  $\sqrt{ab}$ , &c., the sign connecting the two radicals is  $+$ ; the effect of the interchange of  $x, y$  is, in fact, to change  $(u, v)$  into  $(-u, -v)$ ; consequently to change the sign of the odd functions, and to leave unaltered those of the even functions: the interchange does in fact leave  $\sqrt{a}$ , &c., unaltered, while it changes  $\sqrt{ab}$ , &c., into  $-\sqrt{ab}$ , &c.; and thus, since only the ratios are attended to, there is a change of sign as there should be.

**25.** The equations of the product-theorem lead to expressions for

$$A(u+u')B(u-u')-B(u+u')A(u-u'),$$

where the arguments, written above, are used to denote the two arguments, viz.  $u+u'$  to denote  $(u+u', v+v')$  and  $u-u'$  to denote  $(u-u', v-v')$ ; and where the letters  $A, B$  denote each or either of them a single or double letter. These expressions are found in terms of the functions of  $(u, v)$  and of  $(u', v')$ : in any such expression taking  $u', v'$  each of them indefinitely small, but with their ratio arbitrary, we obtain the value of

$$A(u) \partial B(u) - B(u) \partial A(u),$$

(viz.  $u$  here stands for the two arguments  $(u, v)$ , and  $\partial$  denotes total differentiation

$$\partial A = du \frac{d}{du} A(u, v) + dv \frac{d}{dv} A(u, v),$$

as a quadric function of the functions of  $(u, v)$ : or dividing by  $A^2$ , the form is  $\partial \frac{B}{A}$  equal to a function of the quotient-functions  $\frac{B}{A}$ , &c., that is, we have the differentials of the quotient-functions in terms of the quotient-functions themselves. Substituting for the quotient-functions their values in terms of  $x, y$ , we should obtain the differential relations between  $dx, dy, du, dv$ , viz. putting for shortness

$$X = a - x \cdot b - x \cdot c - x \cdot d - x \cdot e - x \cdot f - x,$$

and

$$Y = a - y \cdot b - y \cdot c - y \cdot d - y \cdot e - y \cdot f - y,$$

these are of the form

$$\frac{dx}{\sqrt{X}} - \frac{dy}{\sqrt{Y}}, \quad \frac{x dx}{\sqrt{X}} - \frac{y dy}{\sqrt{Y}},$$

each of them equal to a linear function of  $du$  and  $dv$ : so that the quotient-functions are in fact the 15 hyperelliptic functions belonging to the integrals  $\int \frac{dx}{\sqrt{X}}$ ,  $\int \frac{x dx}{\sqrt{X}}$ , and there is thus an addition-theorem for them, in accordance with the theory of these integrals.

**26.** The first 16 equations of the product-theorem, putting therein first  $u = 0$ ,  $v = 0$ , and then  $u' = 0$ ,  $v' = 0$ , and using the results to eliminate the functions on the right-hand side, give expressions for

$$A \cdot B, \text{ \&c.},$$

that is, they give  $A(u + u', v + v') \cdot B(u - u', v - v')$ , &c., in terms of the functions of  $(u, v)$  and  $(u', v')$ : and we have thus an addition-with-subtraction theorem for the double theta-functions. And we have thence also consequences analogous to those which present themselves in the theory of the single functions.

*Remark as to notation.*

**27.** I remark, as regards the single theta-functions, that the characteristics

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

might for shortness be represented by a series of current numbers

$$0, 1, 2, 3 :$$

and the functions be accordingly called  $\vartheta_0 u, \vartheta_1 u, \vartheta_2 u, \vartheta_3 u$ ; but that, instead of this, I prefer to use throughout the before-mentioned functional symbols

$$A, B, C, D.$$

As regards the double functions, I do, however, denote the characteristics

$$\begin{pmatrix} 00 \\ 00 \end{pmatrix} \begin{pmatrix} 10 \\ 00 \end{pmatrix} \begin{pmatrix} 01 \\ 00 \end{pmatrix} \begin{pmatrix} 11 \\ 00 \end{pmatrix} \mid \begin{pmatrix} 00 \\ 10 \end{pmatrix} \begin{pmatrix} 10 \\ 10 \end{pmatrix} \begin{pmatrix} 01 \\ 10 \end{pmatrix} \begin{pmatrix} 11 \\ 10 \end{pmatrix} \mid \begin{pmatrix} 00 \\ 01 \end{pmatrix} \begin{pmatrix} 10 \\ 01 \end{pmatrix} \begin{pmatrix} 01 \\ 01 \end{pmatrix} \begin{pmatrix} 11 \\ 01 \end{pmatrix} \mid \begin{pmatrix} 00 \\ 11 \end{pmatrix} \begin{pmatrix} 10 \\ 11 \end{pmatrix} \begin{pmatrix} 01 \\ 11 \end{pmatrix} \begin{pmatrix} 11 \\ 11 \end{pmatrix}$$

by a series of current numbers

$$0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15,$$

and write the functions as  $\vartheta_0, \vartheta_1, \dots, \vartheta_{15}$  accordingly; and I use also, as and when it is convenient, the foregoing single and double letter notation  $A, AB, \dots$ , which correspond to them in the order

$$BD, CE, CD, BE, AC, C, AB, B, BC, DE, F, A, AD, D, E, AE.$$

Moreover, I write down for the most part a single argument only: thus,  $A(u + u')$  stands for  $A(u + u', v + v')$ ,  $A(0)$  for  $A(0, 0)$ : and so in other cases.

## SECOND PART.—THE SINGLE THETA-FUNCTIONS.

*Notation, &c.*

**28.** Writing  $\exp. \omega = q$ , and converting the exponentials into circular functions, we have, directly from the definition,

$$\vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (u) = \vartheta_0 u = Au = 1 + 2q \cos \pi u + 2q^4 \cos 2\pi u + 2q^9 \cos 3\pi u + \dots,$$

$$\vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (u) = \vartheta_1 u = Bu = 2q^{\frac{1}{4}} \cos \frac{1}{2}\pi u + 2q^{\frac{9}{4}} \cos \frac{3}{2}\pi u + 2q^{\frac{25}{4}} \cos \frac{5}{2}\pi u + \dots,$$

$$\vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (u) = \vartheta_2 u = Cu = 1 - 2q \cos \pi u + 2q^4 \cos 2\pi u - 2q^9 \cos 3\pi u + \dots \quad (= \Theta(Ku), \text{ Jacobi}),$$

$$\vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (u) = \vartheta_3 u = Du = -2q^{\frac{1}{4}} \sin \frac{1}{2}\pi u + 2q^{\frac{9}{4}} \sin \frac{3}{2}\pi u - 2q^{\frac{25}{4}} \sin \frac{5}{2}\pi u + \dots \quad (= -H(Ku), \text{ Jacobi}),$$

where  $\omega$  is of the form  $\omega = -\alpha + \beta i$ ,  $\alpha$  being non-evanescent and positive: hence  $q = \exp. (-\alpha + \beta i) = e^{-\alpha}(\cos \beta + i \sin \beta)$ , where  $e^{-\alpha}$ , the modulus of  $q$ , is positive and less than 1;  $\cos \beta$  may be either positive or negative, and  $q^{\frac{1}{4}}$  is written to denote  $\exp. \frac{1}{4}(-\alpha + \beta i)$ , viz. this is  $= e^{-\frac{1}{4}\alpha}(\cos \frac{1}{4}\beta + i \sin \frac{1}{4}\beta)$ . But usually  $\beta = 0$ , viz.  $q$  is real positive quantity less than 1, and  $q^{\frac{1}{4}}$  denotes the real fourth root of  $q$ .

I have given above the three notations but, as already mentioned, I propose to employ for the four functions the notation  $Au, Bu, Cu, Du$ : it will be observed that  $Du$  is an odd function, but that  $Au, Bu, Cu$  are even functions, of  $u$ .

*The constants of the theory.*

**29.** We have

$$\begin{aligned} A0 &= 1 + 2q + 2q^4 + 2q^9 + \dots, \\ B0 &= 2q^{\frac{1}{4}} + 2q^{\frac{9}{4}} + 2q^{\frac{25}{4}} + \dots, \\ C0 &= 1 - 2q + 2q^4 - 2q^9 + \dots, \\ D0 &= 0, \\ D'0 &= -\pi \{ q^{\frac{1}{4}} - 3q^{\frac{9}{4}} + 5q^{\frac{25}{4}} - \dots \}. \end{aligned}$$

If, as definitions of  $k, k', K$ , we assume

$$k = \frac{B^2 0}{A^2 0}, \quad k' = \frac{C^2 0}{A^2 0}, \quad K = \frac{A^2 0}{B0 C0} \cdot \frac{D'0}{C0},$$

then we have

$$k = 4\sqrt{q} \left\{ \frac{1 + q^4 + q^8 + \dots}{1 + 2q + 2q^4 + \dots} \right\}^2, \quad \&c. \quad \dots = 4\sqrt{q} (1 - 4q + 14q^2 + \dots),$$

where I have added the first few terms of the expansions of these quantities. We have identically

$$k^2 + k'^2 = 1.$$

It will be convenient to write also, as the definition of  $E$ ,

$$\frac{E}{K} = \frac{1}{K^2} \cdot \frac{C''0}{C0} = \frac{2\pi^2}{K^2} \cdot \frac{q - 4q^4 + 9q^9 - \dots}{1 - 2q + 2q^4 - \dots};$$

we have then

$$\frac{E}{K} = 1 - 8q + 48q^2 - 224q^3 + \dots,$$

moreover,

$$E = \frac{1}{2}\pi \{ 1 - 4q + 20q^2 - 64q^3 + \dots \}.$$

giving

**30.** Other formulae are

$$\begin{aligned} k &= 4\sqrt{q} \left( \frac{1+q^2 \cdot 1+q^4 \dots}{1+q \cdot 1+q^3 \dots} \right)^4, \\ k' &= \left( \frac{1-q \cdot 1-q^3 \dots}{1+q \cdot 1+q^3 \dots} \right)^4, \\ K &= \frac{1}{2}\pi(1+q \cdot 1+q^3 \cdot 1+q^5 \dots)^4. \end{aligned}$$

**31.** Jacobi's definition of  $q$  is from a different point of view altogether, viz. we have  $q = \exp. - \frac{\pi K'}{K}$ , where

$$K = \int_0^{\frac{1}{2}\pi} \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}},$$

and  $K'$  is the like function of  $k'$ ; the equation gives  $\log q = -\frac{\pi K'}{K}$ , viz. we have

$$\frac{K'}{K} = -\frac{1}{\pi} \log q,$$

and, regarding herein  $k$  as a given function of  $q$ , this equation gives  $\frac{K'}{K}$  as a function of  $q$ .

*The product-theorem.*

**32.** The product-theorem is

$$\vartheta \left( \begin{smallmatrix} \alpha \\ \gamma \end{smallmatrix} \right) (u+u') \vartheta \left( \begin{smallmatrix} \alpha' \\ \gamma' \end{smallmatrix} \right) (u-u') = \Theta \left( \begin{smallmatrix} \frac{1}{2}(\alpha+\alpha') \\ \gamma+\gamma' \end{smallmatrix} \right) (2u) \cdot \Theta \left( \begin{smallmatrix} \frac{1}{2}(\alpha-\alpha') \\ \gamma-\gamma' \end{smallmatrix} \right) (2u') + \Theta \left( \begin{smallmatrix} \frac{1}{2}(\alpha+\alpha')+1 \\ \gamma+\gamma' \end{smallmatrix} \right) (2u) \cdot \Theta \left( \begin{smallmatrix} \frac{1}{2}(\alpha-\alpha')+1 \\ \gamma-\gamma' \end{smallmatrix} \right) (2u').$$

Here giving to  $\alpha, \alpha', \gamma, \gamma'$  their different values, and introducing unaccented and accented capitals to denote the functions of  $2u$  and  $2u'$  respectively, the 16 equations are

*Square-set:*

$$\begin{aligned} A.A \quad & \vartheta \left( \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right) (u+u') \vartheta \left( \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right) (u-u') = XX' + YY', \\ B.B \quad & \vartheta \left( \begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right) , , \vartheta \left( \begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right) , = YX' + XY', \\ C.C \quad & \vartheta \left( \begin{smallmatrix} 0 \\ 1 \end{smallmatrix} \right) , , \vartheta \left( \begin{smallmatrix} 0 \\ 1 \end{smallmatrix} \right) , = XX' - YY', \\ D.D \quad & \vartheta \left( \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right) , , \vartheta \left( \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right) , = -YX' + XY'; \end{aligned}$$

*First product-set:*

$$\begin{aligned} C.A \quad & \vartheta \left( \begin{smallmatrix} 0 \\ 1 \end{smallmatrix} \right) (u+u') \vartheta \left( \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right) (u-u') = X_1X'_1 + Y_1Y'_1, \\ A.C \quad & \vartheta \left( \begin{smallmatrix} 0 \\ 0 \end{smallmatrix} \right) , , \vartheta \left( \begin{smallmatrix} 0 \\ 1 \end{smallmatrix} \right) , = X_1X'_1 - Y_1Y'_1, \\ D.B \quad & \vartheta \left( \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right) , , \vartheta \left( \begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right) , = Y_1X'_1 + X_1Y'_1, \\ B.D \quad & \vartheta \left( \begin{smallmatrix} 1 \\ 0 \end{smallmatrix} \right) , , \vartheta \left( \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \right) , = Y_1X'_1 - X_1Y'_1; \end{aligned}$$

*Second product-set:*

$$\begin{aligned} B \cdot A \quad \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (u + u') \vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (u - u') &= PP' + QQ', \\ A \cdot B \quad \vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} , , \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} , &= PQ' + QP', \\ D \cdot C \quad \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} , , \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} , &= iPP' - iQQ', \\ C \cdot D \quad \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} , , \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} , &= iPQ' - iQP'; \end{aligned}$$

*Third product-set:*

$$\begin{aligned} D \cdot A \quad \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (u + u') \vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (u - u') &= P_1P'_1 + Q_1Q'_1, \\ A \cdot D \quad \vartheta \begin{pmatrix} 0 \\ 0 \end{pmatrix} , , \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} , &= iP_1Q'_1 - iQ_1P'_1, \\ B \cdot C \quad \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} , , \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} , &= -iP_1P'_1 + iQ_1Q'_1, \\ C \cdot B \quad \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} , , \vartheta \begin{pmatrix} 1 \\ 0 \end{pmatrix} , &= P_1Q'_1 + Q_1P'_1. \end{aligned}$$

**33.** Here, and subsequently, we have

$$\begin{aligned} \Theta \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \Theta \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \Theta \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \Theta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (2u) &= X, Y, X_1, Y_1 \quad | \quad \Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix}, \Theta \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix}, \Theta \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}, \Theta \begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix} (2u) = P, Q, P_1, Q_1, \\ \Theta \dots (2u') &= X', Y', X'_1, Y'_1; \quad \Theta \dots (2u') = P', Q', P'_1, Q'_1, \\ \Theta \dots (0) &= \alpha, \beta, \alpha_1, \beta_1; \quad \Theta \dots (0) = p, q, p_1, q_1; \end{aligned}$$

viz. we use also  $\alpha, \beta, \alpha_1, \beta_1$  and  $p, q, p_1, q_1$  to denote the zero-functions;  $\beta_1$  is  $= 0$ , but we use  $\beta'_1$  to denote the zero-value of  $\frac{d}{du}Y_1$ .

**34.** In order to obtain the foregoing relations, it is necessary to observe that

$$\Theta \begin{pmatrix} \alpha + 2 \\ \gamma \end{pmatrix} , , = \Theta \begin{pmatrix} \alpha \\ \gamma \end{pmatrix} , , ;$$

by which the upper character is always reduced to  $0, 1, \frac{1}{2}$ , or  $\frac{3}{2}$ ; and that, for reducing the lower character, we have

$$\begin{aligned} \Theta \begin{pmatrix} \alpha \\ \gamma + 2 \end{pmatrix} , , &= \Theta \begin{pmatrix} \alpha \\ \gamma \end{pmatrix} , , ; \quad \Theta \begin{pmatrix} \frac{1}{2} \\ \gamma + 2 \end{pmatrix} , , &= -\Theta \begin{pmatrix} \frac{1}{2} \\ \gamma \end{pmatrix} , , ; \\ \Theta \begin{pmatrix} 1 \\ \gamma + 2 \end{pmatrix} , , &= i\Theta \begin{pmatrix} 1 \\ \gamma \end{pmatrix} , , ; \quad \Theta \begin{pmatrix} \frac{3}{2} \\ \gamma - 2 \end{pmatrix} , , &= -i\Theta \begin{pmatrix} \frac{3}{2} \\ \gamma \end{pmatrix} , , ; \quad \Theta \begin{pmatrix} \frac{3}{2} \\ \gamma + 2 \end{pmatrix} , , &= \Theta \begin{pmatrix} \frac{3}{2} \\ \gamma \end{pmatrix} , , ; \end{aligned}$$

by means of which the lower character is always reduced to  $0$  or  $1$ : in all these formulae the argument is arbitrary, and it is thus  $= 2u$ , or  $2u'$  as the case requires. The formulae are obtained without difficulty directly from the definition of the functions  $\Theta$ .

**35.** As an instance, taking  $\frac{\alpha}{\gamma}, \frac{\alpha'}{\gamma'} = \frac{1}{1}, \frac{0}{1}$ , the product-equation is

$$\begin{aligned} \vartheta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (u + u') \vartheta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (u - u') &= \Theta \begin{pmatrix} \frac{1}{2} \\ 2 \end{pmatrix} (2u) \cdot \Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} (2u') + \Theta \begin{pmatrix} \frac{3}{2} \\ 2 \end{pmatrix} (2u) \cdot \Theta \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} (2u'), \\ &= -i\Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} (2u) \cdot \Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} (2u') - i\Theta \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} (2u) \cdot \Theta \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} (2u'), \end{aligned}$$

$$= iPP' - iQQ',$$

which agrees with the before-given value.

**36.** The following values are not actually required: but I give them to fix the ideas and to show the meaning of the quantities with which we work.

$$\begin{aligned} X &= \Theta \begin{pmatrix} 0 \\ 0 \end{pmatrix} (2u) = 1 + 2q^4 \cos 2\pi u + 2q^{16} \cos 4\pi u + \dots, & u = 0: \quad \alpha &= 1 + 2q^4 + 2q^{16} + \dots, \\ Y &= \Theta \begin{pmatrix} 1 \\ 0 \end{pmatrix} (2u) = 2q \cos \pi u + 2q^9 \cos 3\pi u + \dots, & \beta &= 2q + 2q^9 + \dots, \\ X_1 &= \Theta \begin{pmatrix} 0 \\ 1 \end{pmatrix} (2u) = 1 - 2q^4 \cos 2\pi u + 2q^{16} \cos 4\pi u - \dots, & \alpha_1 &= 1 - 2q^4 + 2q^{16} - \dots, \\ Y_1 &= \Theta \begin{pmatrix} 1 \\ 1 \end{pmatrix} (2u) = -2q \sin \pi u + 2q^9 \sin 3\pi u - \dots, & \beta'_1 &= 2\pi(-q + 3q^9 - \dots) = \frac{d}{du} Y_1 \text{ for } u = 0. \end{aligned}$$

$$\begin{aligned} P &= \Theta \begin{pmatrix} \frac{1}{2} \\ 0 \end{pmatrix} (2u) = q^{\frac{1}{4}} (\cos \frac{1}{2}\pi u + i \sin \frac{1}{2}\pi u) + q^{\frac{9}{4}} (\cos \frac{3}{2}\pi u - i \sin \frac{3}{2}\pi u) \\ &\quad + q^{\frac{25}{4}} (\cos \frac{5}{2}\pi u + i \sin \frac{5}{2}\pi u) + \dots, \\ Q &= \Theta \begin{pmatrix} \frac{3}{2} \\ 0 \end{pmatrix} (2u) = q^{\frac{1}{4}} (\cos \frac{1}{2}\pi u - i \sin \frac{1}{2}\pi u) + q^{\frac{9}{4}} (\cos \frac{3}{2}\pi u + i \sin \frac{3}{2}\pi u) \\ &\quad + q^{\frac{25}{4}} (\cos \frac{5}{2}\pi u - i \sin \frac{5}{2}\pi u) + \dots, \\ P_1 &= \Theta \begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix} (2u) = \frac{1+i}{\sqrt{2}} \{ q^{\frac{1}{4}} (\cos \frac{1}{2}\pi u + i \sin \frac{1}{2}\pi u) - q^{\frac{9}{4}} (\cos \frac{3}{2}\pi u - i \sin \frac{3}{2}\pi u) \\ &\quad - q^{\frac{25}{4}} (\cos \frac{5}{2}\pi u + i \sin \frac{5}{2}\pi u) + \dots \}, \\ Q_1 &= \Theta \begin{pmatrix} \frac{3}{2} \\ 1 \end{pmatrix} (2u) = \frac{1-i}{\sqrt{2}} \{ q^{\frac{1}{4}} (\cos \frac{1}{2}\pi u - i \sin \frac{1}{2}\pi u) - q^{\frac{9}{4}} (\cos \frac{3}{2}\pi u + i \sin \frac{3}{2}\pi u) \\ &\quad - q^{\frac{25}{4}} (\cos \frac{5}{2}\pi u - i \sin \frac{5}{2}\pi u) + \dots \}; \end{aligned}$$

and therefore also

$$\begin{aligned} p &= q = q^{\frac{1}{4}} + q^{\frac{9}{4}} + q^{\frac{25}{4}} + \dots, \\ p_1 &= \frac{1+i}{\sqrt{2}} \{ q^{\frac{1}{4}} - q^{\frac{9}{4}} - q^{\frac{25}{4}} + q^{\frac{49}{4}} + q^{\frac{81}{4}} - \dots \}, \quad q_1 = \frac{1-i}{\sqrt{2}} \{ \text{Do.} \}; \quad p_1 = iq_1. \end{aligned}$$

*The square set,  $u' = 0$ ; and  $x$ -formulæ.*

**37.** We use the square-set, in the first instance by writing therein  $u' = 0$ ; the equations become

$$\begin{aligned} A^2u &= \alpha X + \beta Y, &= \omega''\mathfrak{A}(a-x), \\ B^2u &= \beta X + \alpha Y, &= \omega''\mathfrak{B}(b-x), \\ C^2u &= \alpha X - \beta Y, &= \omega''\mathfrak{C}(c-x), \\ D^2u &= \beta X - \alpha Y, &= \omega''\mathfrak{D}(d-x), \end{aligned}$$

viz. the equations without their last members show that there exist functions  $\omega''$  and  $x$ , linear functions of  $X$  and  $Y$ , such that  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ ,  $\mathfrak{A}a, \mathfrak{B}b, \mathfrak{C}c, \mathfrak{D}d$ , being constants, the squared functions may be assumed equal to  $\mathfrak{A}\omega'' \cdot \omega''x$ , &c., that is,  $\omega''\mathfrak{A}(a-x)$ , &c., respectively; the squared functions are then proportional to the values  $\mathfrak{A}(a-x)$ , &c.

To show the meaning of the factor  $\omega''$ , observe that, from any two of the equations, for instance from the first and second, we have an equation without  $\omega$ ,

$$A^2u - B^2u = \mathfrak{A}(a-x) - \mathfrak{B}(b-x);$$

and using this to determine  $x$ , and then substituting in  $\omega'' = A^2u \div \mathfrak{A}(a-x)$ , we find

$$\omega'' = \frac{\mathfrak{A}A^2u - \mathfrak{B}B^2u}{(a-b)\mathfrak{A}\mathfrak{B}},$$

where the numerator is a function not in anywise more important than any other linear function of  $A^2u$  and  $B^2u$ .

**38.** The function  $Du$  vanishes for  $u = 0$ , and we may assume that the corresponding value of  $x$  is  $= d$ . Writing in the other equations  $u = 0$ , they become

$$\begin{aligned} A^20 &= \alpha_0^2 + \beta_0^2 = \omega_0''\mathfrak{A}(a-d), \\ B^20 &= 2\alpha_0\beta_0 = \omega_0''\mathfrak{B}(b-d), \\ C^20 &= \alpha_0^2 - \beta_0^2 = \omega_0''\mathfrak{C}(c-d), \end{aligned}$$

where  $\omega_0''$  is what  $\omega''$  becomes on writing therein  $x = d$ . It is convenient to omit altogether these factors  $\omega''$  and  $\omega_0''$ ; it being understood that, without them, the equations denote not absolute equalities but only equalities of ratios: thus, without the  $\omega_0''$ , the last-mentioned equations would denote

$$A^20 : B^20 : C^20 = \alpha_0^2 + \beta_0^2 : 2\alpha_0\beta_0 : \alpha_0^2 - \beta_0^2 = \mathfrak{A}(a-d) : \mathfrak{B}(b-d) : \mathfrak{C}(c-d).$$

The quantities  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$  only present themselves in the products  $\mathfrak{A}\omega''$ , &c., and their absolute magnitudes are therefore essentially indeterminate; but regarding  $\omega''$  as



containing a constant factor of properly determined value, the absolute values of  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$  may be regarded as determinate, and this is accordingly done in the formulæ  $\mathfrak{A}^2 = -agh$ , &c., which follow.

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*Relations between the constants.*

39. The formulæ contain the differences of the quantities  $a, b, c, d$ ; denoting these differences

$$b - c, \quad c - a, \quad a - b, \quad a - d, \quad b - d, \quad c - d,$$

in the usual manner by

$$a, \quad b, \quad c, \quad f, \quad g, \quad h,$$

so that

$$-h + g - a = 0,$$

$$h - f - b = 0,$$

$$-g + f - c = 0,$$

$$a + b + c = 0,$$

and also

$$af + bg + ch = 0,$$

and then assuming the absolute value of one of the quantities  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ , we have the system of relations

$$\mathfrak{A}^2 = -agh, \quad \mathfrak{B}\mathfrak{C}\mathfrak{D} = \mathfrak{A}^2\mathfrak{D}f, \quad \mathfrak{A}bcf = -\mathfrak{B}\mathfrak{C}\mathfrak{D}, \quad \mathfrak{A}\mathfrak{B}\mathfrak{C}\mathfrak{D} = abc fgh,$$

$$\mathfrak{B}^2 = bhf, \quad \mathfrak{C}\mathfrak{D}\mathfrak{A} = -\mathfrak{B}^2\mathfrak{D}g, \quad \mathfrak{B}cag = \mathfrak{C}\mathfrak{A}\mathfrak{D},$$

$$\mathfrak{C}^2 = cfg, \quad \mathfrak{A}\mathfrak{B}\mathfrak{C} = -\mathfrak{C}^2\mathfrak{D}h, \quad \mathfrak{C}abh = \mathfrak{A}\mathfrak{B}\mathfrak{D},$$

$$\mathfrak{D}^2 = -abc, \quad \mathfrak{D}fgh = -\mathfrak{A}\mathfrak{B}\mathfrak{C},$$

$$0 = \mathfrak{B}^2 + b^2\mathfrak{C}^2 - f^2\mathfrak{D}^2 = bcf (af + bg + ch) = 0,$$

$$0 = -c^2\mathfrak{A}^2 + a^2\mathfrak{C}^2 - g^2\mathfrak{D}^2 = cag ( \quad ) = 0,$$

$$0 = -b^2\mathfrak{A}^2 + a^2\mathfrak{B}^2 - h^2\mathfrak{D}^2 = abh ( \quad ) = 0,$$

$$0 = -f^2\mathfrak{A}^2 + g^2\mathfrak{B}^2 + h^2\mathfrak{C}^2 = fgh ( \quad ) = 0.$$

It is to be remarked that, taking  $c, a, b, d$  in the order of decreasing magnitude, we have  $-a, b, c, f, g, h$  all positive; hence  $\mathfrak{A}^2, \mathfrak{B}^2, \mathfrak{C}^2, \mathfrak{D}^2$  all real; and taking as we may do,  $\mathfrak{D}$  negative, then  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}$  may be taken positive; that is, we have  $-a, b, c, f, g, h, \mathfrak{A}, \mathfrak{B}, \mathfrak{C}, -\mathfrak{D}$  all of them positive.

40. We have

$$A^2 0 = \alpha_0^2 + \beta_0^2 = \mathfrak{A}f,$$

$$B^2 0 = 2\alpha_0\beta_0 = \mathfrak{B}g,$$

$$C^2 0 = \alpha_0^2 - \beta_0^2 = \mathfrak{C}h.$$

The foregoing equations

$$k = \frac{B^2 0}{A^2 0}, \quad k'^2 = \frac{C^2 0}{A^2 0}$$

give

$$k = \frac{\mathfrak{B}g}{\mathfrak{A}f}, \quad k'^2 = \frac{\mathfrak{C}h}{\mathfrak{A}f}.$$

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and we thence have

$$k^2 = \frac{bg}{-af}, \quad k'^2 = \frac{ch}{-af},$$

satisfying

$$k^2 + k'^2 = 1.$$

41. Observe further that, substituting for  $a, b, c, f, g, h$  their values, we have

$$\begin{aligned} \mathfrak{A}^2 &= c - b.b - d.c - d, &= c - a.d - b.b - c, \\ \mathfrak{B}^2 &= c - a.c - d.a - d, &= d - a.a - c.c - d, \\ \mathfrak{C}^2 &= a - b.a - d.b - d, &= -a - b.b - d.d - a, \\ \mathfrak{D}^2 &= c - b.c - a.a - b, &= -b - c.c - a.a - b, \end{aligned}$$

where in the first set of values all the differences are positive, but in the second set of values, we take the triads of  $abcd$ , in the cyclical order  $bcd, cda, dab, abc$ . There is in this last form an apparent want of symmetry as to the signs (viz. the order which might have been expected is  $+-+-$ ), but taking the order of the letters to be  $\mathfrak{C}, \mathfrak{A}, \mathfrak{B}, \mathfrak{D}$  and  $c, a, b, d$ , then the cyclical arrangement is

$$\begin{aligned} \mathfrak{C}^2 &= -b - d.d - a.a - b, \\ \mathfrak{A}^2 &= -d - c.c - b.b - d, \\ \mathfrak{B}^2 &= -c - a.a - d.d - c, \\ \mathfrak{D}^2 &= -a - b.b - c.c - a, \end{aligned}$$

where the four outside signs are all  $-$ . Observe that the triads of  $abcd$ , and  $abdc$ , are

$$bcd, \quad cda, \quad dab, \quad abc,$$

and

$$bdc, \quad dca, \quad cab, \quad abd.$$

where in the first and second columns the terms of the same column correspond to each other with a reversal of sign, whereas in the third and fourth columns the lower term of either column corresponds to the upper term of the other column, but without a reversal of sign.

*The product-sets,  $u \pm u'$ ; and  $u'$  indefinitely small, differential formulæ.*

42. Coming now to the product-sets, these may be written

$$\begin{aligned} \frac{1}{2}\{C.A + A.C\} &= X_1 X'_1, & \frac{1}{2}\{C.A - A.C\} &= Y_1 Y'_1, \\ \frac{1}{2}\{D.B + B.D\} &= Y_1 X'_1, & \frac{1}{2}\{D.B - B.D\} &= X_1 Y'_1, \\ \frac{1}{2}\{B.A + A.B\} &= (P + Q)(P' + Q'), & \frac{1}{2}\{B.A - A.B\} &= (P - Q)(P' - Q'), \\ \frac{1}{2}\{D.C + C.D\} &= i(P - Q)(P' + Q'), & \frac{1}{2}\{D.C - C.D\} &= i(P + Q)(P' - Q'), \\ \frac{1}{2}\{D.A + A.D\} &= (P_1 - iQ_1)(P'_1 + iQ'_1), & \frac{1}{2}\{D.A - A.D\} &= (P_1 + iQ_1)(P'_1 - iQ'_1), \\ \frac{1}{2}\{B.C + C.B\} &= -i(P_1 - iQ_1)(P'_1 + iQ'_1), & \frac{1}{2}\{B.C - C.B\} &= -i(P_1 + iQ_1)(P'_1 - iQ'_1). \end{aligned}$$

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43. We can from each set form two fractions (each of them a function of  $u + u'$  and  $u - u'$ ), which are equal to one and the same function of  $u'$  only: for instance, from the first set we have two fractions, each  $= \frac{Y'}{X'}$ ; putting in such equation  $u = 0$ , we obtain a new expression for the function of  $u'$  involving only the theta-functions  $Au'$ , &c., which new expression we may then substitute in the equations first obtained; we thus arrive at the six equations

$$\begin{aligned} \frac{C.A - A.C}{D.B + B.D} &= \frac{D.B - B.D}{C.A + A.C} = \frac{D'u' . B'u'}{C'u' . A'u'}, \\ \frac{B.A - A.B}{D.C + C.D} &= \frac{D.C - C.D}{B.A + A.B} = \frac{D'u' . C'u'}{B'u' . A'u'}, \\ \frac{B.C - C.B}{D.A + A.D} &= \frac{D.A - A.D}{B.C + C.B} = \frac{D'u' . A'u'}{B'u' . C'u'}, \end{aligned}$$

where observe that the expressions all vanish for  $u' = 0$ .

44. Taking herein  $u'$  indefinitely small, we obtain

$$\begin{aligned} \frac{Au . C'u - Cu . A'u}{Bu . Du} &= \frac{Bu . D'u - Du . B'u}{Cu . Au} = \frac{D'0 . B0}{C0 . A0} = -K \frac{D^2 0}{A^2 0}, \\ \frac{Au . B'u - Bu . A'u}{Cu . Du} &= \frac{Cu . D'u - Du . C'u}{Au . Bu} = \frac{D'0 . C0}{A0 . B0} = -K \frac{C^2 0}{A^2 0}, \\ \frac{Cu . B'u - Bu . C'u}{Au . Du} &= \frac{Au . D'u - Du . A'u}{Bu . Cu} = \frac{D'0 . A0}{B0 . C0} = -K, \end{aligned}$$

where the last column is added in order to introduce  $K$  in place of  $D'0$ .

45. These formulae in effect give the derivatives of the quotient-functions in terms of quotient-functions: for instance, one of the equations is

$$\frac{d}{du} \frac{Du}{Au} = -K \frac{Bu \ Cu}{Au \ Au},$$

substituting herein for the quotient-fractions their values in terms of  $x$ , this becomes

$$\frac{d}{du} \frac{1}{2} \sqrt{\frac{a-x}{d-x}} = -\frac{K}{2} \sqrt{\mathfrak{B}\mathfrak{C}} \sqrt{\frac{a-x}{d-x}} \cdot \frac{1}{2} \sqrt{\frac{a-x}{d-x}},$$

or the left-hand being

$$-\frac{1}{(a-x)^{\frac{1}{2}} \sqrt{d-x}} \frac{dx}{du},$$

this is

$$K \ du = \frac{\frac{1}{2} \sqrt{-af} \ dx}{\sqrt{a-x . b-x . c-x . d-x}},$$

where on the right-hand side it would be better to write  $\sqrt{-af}$  in the numerator and  $x-d$  in place of  $d-x$  in the denominator.

*Comparison with Jacobi's formulæ.*

46. The comparison of the formulæ with Jacobi's formulæ gives

$$\begin{aligned}\operatorname{sn} Ku &= -\frac{1}{\sqrt{k}} Du \div Cu, &= \sqrt{\frac{h}{g}} \sqrt{\frac{d-x}{c-x}} \quad (\text{or better } \sqrt{\frac{-h}{g}} \sqrt{\frac{x-d}{c-x}}), \\ \operatorname{cn} Ku &= \sqrt{\frac{k'}{k}} Bu \div Cu, &= \sqrt{\frac{h}{g}} \sqrt{\frac{b-x}{c-x}}, \\ \operatorname{dn} Ku &= \sqrt{k'} Au \div Cu, &= \sqrt{\frac{h}{f}} \sqrt{\frac{a-x}{c-x}},\end{aligned}$$

where it will be recollected that

$$k^2 = \frac{bg}{-af}, \quad k'^2 = \frac{ch}{-af}.$$

It may be remarked that we seek to determine everything in terms of  $a, b, c, d$ . The formula just written down,  $f^2 = bg + h - af$ , gives  $k$  in terms of these quantities; and  $k, K$  being each given in terms of  $q$ , we have virtually  $K$  as a function of  $k$ , that is, of  $a, b, c, d$ : but it would not be easy from the expressions of  $k, K$ , each in terms of  $q$ , to deduce the actual expression

$$K = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}$$

of  $K$  as a function of  $k$ .

*The square-set,  $u \pm u'$ .*

47. Reverting to the square-set

$$\begin{aligned}A(u + u')A(u - u') &= XX' + YY', \\ B(u + u')B(u - u') &= YX' + XY', \\ C(u + u')C(u - u') &= XX' - YY', \\ D(u + u')D(u - u') &= -YX' + XY',\end{aligned}$$

if we first write herein  $u' = 0$ , and then  $u = 0$ , using the results to determine the values of  $X, Y, X', Y'$  we find

$$\begin{aligned}\alpha C^2 u - \beta D^2 u &= (\alpha^2 - \beta^2)X, & \alpha C^2 u' - \beta D^2 u' &= (\alpha^2 - \beta^2)X', \\ \beta C^2 u - \alpha D^2 u &= Y, & \beta C^2 u' - \alpha D^2 u' &= Y',\end{aligned}$$

and thence

$$\begin{aligned}(\alpha^2 - \beta^2)^2 XX' &= \alpha^2 C^2 u C^2 u' + \beta^2 D^2 u D^2 u' - \alpha\beta (C^2 u D^2 u' + D^2 u C^2 u'), \\ YY' &= \beta^2 \text{ " } + \alpha^2 \text{ " } - \alpha\beta \text{ " },\end{aligned}$$

whence

$$\begin{aligned}(\alpha^2 - \beta^2)^2 (XX' + YY') &= (\alpha^2 + \beta^2)(C^2 u C^2 u' + D^2 u D^2 u') - 2\alpha\beta (C^2 u D^2 u' + D^2 u C^2 u'), \\ (\alpha^2 - \beta^2)^2 (XX' - YY') &= (C^2 u C^2 u' - D^2 u D^2 u'),\end{aligned}$$

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where observe that, in taking the difference, the right-hand side becomes divisible by  $\alpha^2 - \beta^2$ , and therefore in the final result we have on the left-hand side the simple factor  $\alpha^2 - \beta^2$  instead of  $(\alpha^2 - \beta^2)^2$ .

Similarly

$$\begin{aligned}(\alpha^2 - \beta^2)^2 YX' &= \alpha\beta (C^2 u C^2 u' + D^2 u D^2 u') - \alpha^2 D^2 u C^2 u' - \beta^2 C^2 u D^2 u', \\ XY' &= \alpha\beta \text{ " } - \beta^2 \text{ " } - \alpha^2 \text{ " },\end{aligned}$$

and thence

$$\begin{aligned}(\alpha^2 - \beta^2)^2(YX' + XY') &= 2\alpha\beta(C^2u C^2u' + D^2u D^2u') - (\alpha^2 + \beta^2)(C^2u D^2u' + D^2u C^2u'), \\(\alpha^2 - \beta^2)(-YX' + XY') &= D^2u C^2u' - C^2u D^2u'.\end{aligned}$$

48. Hence recollecting that

$$A^20 = \alpha^2 + \beta^2, \quad B^20 = 2\alpha\beta,$$

the original equations become

$$\begin{aligned}C^20 \cdot A(u + u')A(u - u') &= A^20(C^2u C^2u' + D^2u D^2u') - B^20(C^2u D^2u' + D^2u C^2u'), \\C^20 \cdot B(u + u')B(u - u') &= B^20(C^2u C^2u' + D^2u D^2u') - A^20(C^2u D^2u' + D^2u C^2u'), \\C^20 \cdot C(u + u')C(u - u') &= C^2u C^2u' - D^2u D^2u', \\C^20 \cdot D(u + u')D(u - u') &= D^2u C^2u' - C^2u D^2u'.\end{aligned}$$

49. It will be observed that the four products  $A(u + u')A(u - u')$ , &c., are each of them expressed in terms of  $C^2u, D^2u, C^2u', D^2u'$ . Since each of the squared functions  $A^2u, B^2u, C^2u, D^2u$  is a linear function of any two of them, and the like as regards  $A^2u', B^2u', C^2u', D^2u'$ , it is clear that in each equation we can on the right-hand side introduce any two at pleasure of the squared functions of  $u$ , and any two at pleasure of the squared functions of  $u'$ . But all the forms so obtained are of course identical if, taking  $x'$  the same function of  $u'$  that  $x$  is of  $u$ , we introduce on the right-hand side  $x, x'$  instead of  $u, u'$ ; and the values of  $A(u + u')A(u - u')$ , &c., are found to be equal to multiples of  $\nabla, \nabla_1, \nabla_2, \nabla_3$ , where

$$\nabla = x - x', \quad \nabla_1 = \begin{vmatrix} 1 & x + x' & xx' \\ 1 & a + d & ad \\ 1 & b + c & bc \end{vmatrix}, \quad \nabla_2 = \begin{vmatrix} 1 & x + x' & xx' \\ 1 & b + d & bd \\ 1 & c + a & ca \end{vmatrix}, \quad \nabla_3 = \begin{vmatrix} 1 & x + x' & xx' \\ 1 & c + d & cd \\ 1 & a + b & ab \end{vmatrix}.$$

50. In fact, from the equations

$$A^2u = \mathfrak{A}(a - x), \quad A^2u' = \mathfrak{A}(a - x'),$$

we have

$$\begin{aligned}\nabla &= \frac{1}{\mathfrak{A}\mathfrak{B}}(B^2u C^2u' - C^2u B^2u') = \frac{1}{\mathfrak{C}\mathfrak{A}}(C^2u A^2u' - A^2u C^2u') = \frac{1}{\mathfrak{A}\mathfrak{B}}(A^2u B^2u' - B^2u A^2u'), \\&= \frac{1}{\mathfrak{A}\mathfrak{D}}(A^2u D^2u' - D^2u A^2u') = -\frac{1}{\mathfrak{B}\mathfrak{D}}(B^2u D^2u' - D^2u B^2u') = \frac{1}{\mathfrak{C}\mathfrak{D}}(C^2u D^2u' - D^2u C^2u').\end{aligned}$$

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where it will be recollected that

$$f \mathfrak{A}\mathfrak{D} = a \mathfrak{B}\mathfrak{C}, \quad -g \mathfrak{B}\mathfrak{D} = b \mathfrak{C}\mathfrak{A}, \quad -h \mathfrak{C}\mathfrak{D} = c \mathfrak{A}\mathfrak{B}.$$

Moreover

$$\begin{aligned}(b - c)\nabla_1 &= \begin{vmatrix} b - x \cdot b - x' & c - x \cdot c - x' \\ b - a \cdot b - d & c - a \cdot c - d \end{vmatrix}, \quad (a - d)\nabla_1 = \begin{vmatrix} a - x \cdot a - x' & d - x \cdot d - x' \\ a - b \cdot a - c & d - b \cdot d - c \end{vmatrix}, \\(c - a)\nabla_2 &= -\begin{vmatrix} c - x \cdot c - x' & a - x \cdot a - x' \\ c - b \cdot c - d & a - b \cdot a - d \end{vmatrix}, \quad (b - d)\nabla_2 = \begin{vmatrix} b - x \cdot b - x' & d - x \cdot d - x' \\ b - c \cdot b - a & d - c \cdot d - a \end{vmatrix}, \\(a - b)\nabla_3 &= -\begin{vmatrix} a - x \cdot a - x' & b - x \cdot b - x' \\ a - c \cdot a - d & b - c \cdot b - d \end{vmatrix}, \quad (c - d)\nabla_3 = \begin{vmatrix} c - x \cdot c - x' & d - x \cdot d - x' \\ c - a \cdot c - b & d - a \cdot d - b \end{vmatrix},\end{aligned}$$

or as these may be written

$$\begin{aligned}\nabla_1 &= \frac{1}{f}\{bh.b - x.b - x' + cg.c - x.c - x'\} = \frac{1}{f}\{gh.a - x.a - x' + bc.d - x.d - x'\}, \\ \nabla_2 &= -\frac{1}{g}\{cf.c - x.c - x' + ah.a - x.a - x'\} = -\frac{1}{g}\{hf.b - x.b - x' + ca.d - x.d - x'\}, \\ \nabla_3 &= -\frac{1}{h}\{ag.a - x.a - x' + bf.b - x.b - x'\} = \frac{1}{h}\{fg.c - x.c - x' + ab.d - x.d - x'\},\end{aligned}$$

that is,

$$\begin{aligned}\nabla_1 &= -\frac{1}{a}\frac{1}{\mathfrak{B}^2}(\mathfrak{B}^2uB^2u' + C^2uC^2u') = -\frac{1}{f}\frac{1}{\mathfrak{A}\mathfrak{D}}(A^2uA^2u' + D^2uD^2u'), \\ \nabla_2 &= -\frac{1}{b}\frac{1}{\mathfrak{C}^2}(C^2uC^2u' - A^2uA^2u') = \frac{1}{g}\frac{1}{\mathfrak{B}\mathfrak{D}}(B^2uD^2u' - D^2uD^2u'), \\ \nabla_3 &= -\frac{1}{af}(-A^2uA^2u' + B^2uB^2u') = \frac{1}{h}\frac{1}{\mathfrak{C}\mathfrak{D}}(C^2uC^2u' - B^2uB^2u').\end{aligned}$$

51. Hence  $\nabla, \nabla_1, \nabla_2, \nabla_3$  denoting these functions of  $x, x'$  or of  $u, u'$ , we have

$$\begin{aligned}A(u + u')A(u - u') &= \frac{\mathfrak{A}}{gh}\nabla_1, \\ B(u + u')B(u - u') &= \frac{\mathfrak{B}}{hf}\nabla_2, \\ C(u + u')C(u - u') &= -\frac{\mathfrak{C}}{fg}\nabla_3, \\ D(u + u')D(u - u') &= \mathfrak{D}\nabla.\end{aligned}$$

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*The square-set  $u \pm u'$ ,  $u'$  indefinitely small: differential formulæ of the second order.*

52. I consider the original form

$$C^20C(u + u')C(u - u') = C^2uC^2u' - D^2uD^2u',$$

which is of course included in the last-mentioned equations.

Writing this in the form

$$\frac{C^20C(u + u')C(u - u')}{C^2u} = C^2u' - \frac{D^2uD^2u'}{C^2u},$$

and taking  $u'$  indefinitely small, whence

$$\begin{aligned}C(u + u') &= Cu + u'C'u + \frac{1}{2}u'^2C''u, & Cu' &= C0, \\ C(u - u') &= Cu - u'C'u + \frac{1}{2}u'^2C''u, & Du' &= u'D'0, \\ C(u + u')C(u - u') &= C^2u + u'^2\{CuC''u - (C'u)^2\},\end{aligned}$$

the equation becomes

$$\frac{C^20\{C^2u + u'^2(CuC''u - (C'u)^2)\}}{C^2u} = C^20 - u'^2\frac{D^2uD^20}{C^2u},$$

that is,

$$\frac{C''u}{Cu} - \left(\frac{C'u}{Cu}\right)^2 = \frac{C''0}{C0} - \left(\frac{D'0}{C0}\right)^2\frac{D^2u}{C^2u},$$

viz. we have  $\left(\frac{d}{du}\right)^2 \log Cu$  expressed in terms of the quotient-function  $\frac{D^2u}{C^2u}$ , and consequently  $Cu$  given as an exponential, the argument of which depends on the double integral  $\int du \int du \frac{D^2u}{C^2u}$ .

53. To complete the result, I write the equation in the form

$$\frac{d^2}{du^2} \log Cu = \frac{C''0}{C0} - \frac{1}{k}\left(\frac{B'0}{C0}\right)^2 + \frac{1}{k}\left(\frac{D'0}{C0}\right)^2\left(1 - k\frac{D^2u}{C^2u}\right);$$

$\frac{D'0}{C0}$  is  $= -\sqrt{k}K$ , and  $\frac{C''0}{C0}$  is  $= K(K - E)$ ; hence the equation is

$$\frac{d^2}{du^2} \log Cu = K^2 \left(1 - \frac{E}{K}\right) - k^2 \left(\frac{D^2u}{C^2u}\right), = K^2 \left(1 - \frac{E}{K} - k^2 \operatorname{sn}^2 Ku\right),$$

or integrating twice, and observing that  $\frac{d}{du} \log Cu$  and  $\log Cu$ , for  $u = 0$ , become  $= 0$  and  $\log C0$  respectively, we have

$$\log Cu = \log C0 + \frac{1}{2} \left(1 - \frac{E}{K}\right) K^2 u^2 - k^2 \int_0^u du \int_0^u du K^2 \operatorname{sn}^2 Ku,$$

which is in fact

$$\log \Theta(Ku) = \log C0 + \frac{1}{2} \left(1 - \frac{E}{K}\right) K^2 u^2 - k^2 \int_0^u du \int_0^u du K^2 \operatorname{sn}^2 Ku,$$

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agreeing with Jacobi's formula

$$\log \Theta u = \log \Theta 0 + \frac{1}{2} \left(1 - \frac{E}{K}\right) u^2 - k^2 \int_0^u du \int_0^u du \operatorname{sn}^2 u.$$

*Elliptic integrals of the third kind.*

54. We may write

$$\begin{aligned} \frac{A(u+u')A(u-u')}{A^2u A^2u'} &= \frac{1}{\mathfrak{A}gh} \frac{\nabla_1}{a-x \cdot a-x'}, \\ \frac{B(u+u')B(u-u')}{B^2u B^2u'} &= \frac{1}{\mathfrak{B}hf} \frac{\nabla_2}{b-x \cdot b-x'}, \\ \frac{C(u+u')C(u-u')}{C^2u C^2u'} &= \frac{1}{\mathfrak{C}fg} \frac{\nabla_3}{c-x \cdot c-x'}, \\ \frac{D(u+u')D(u-u')}{D^2u D^2u'} &= \frac{1}{\mathfrak{D}} \frac{x-x'}{d-x \cdot d-x'}. \end{aligned}$$

We differentiate logarithmically in regard to  $u'$ . Observing that

$$K du' = \frac{1}{2} \frac{\sqrt{-af} dx'}{\sqrt{a-x' \cdot b-x' \cdot c-x' \cdot d-x'}} = \frac{1}{2} \frac{\sqrt{-af}}{\sqrt{X'}} dx',$$

suppose, the first equation gives

$$-\frac{A'u'}{Au'} + \frac{A'(u-u')}{A(u-u')} - \frac{A'(u+u')}{A(u+u')} = \frac{2K\sqrt{X'}}{\sqrt{-af}} \frac{d}{dx'} \log \frac{\nabla_1}{a-x'};$$

and if for a moment

$$\nabla_1 = \begin{vmatrix} 1 & x+x' & xx' \\ 1 & a+d & ad \\ 1 & b+c & bc \end{vmatrix}$$

is put

$$= P(a-x') + Q(d-x'),$$

then

$$\frac{d}{dx'} \log \frac{\nabla_1}{a-x'} = \frac{d}{dx'} \log \nabla_1 + \frac{1}{a-x'} = \frac{P+Q \frac{d-x'}{a-x'}}{(a-x')\nabla_1} - \dots = -\frac{Q(d-x')}{(a-x')\nabla_1}.$$

But, writing  $x' = a$ , we have

$$Q(d-a) = -Qf = \begin{vmatrix} 1 & a+x & ax \\ 1 & a+d & ad \end{vmatrix} = -(a-b)(a-c)(d-x) = -bc(d-x),$$

that is,

$$Qf = -bc(d - x),$$

or

$$\frac{d}{dx'} \log \frac{\nabla_1}{a - x'} = \frac{bc(d - x)}{(a - x')\nabla_1}.$$

Hence the equation is

$$\frac{A'(u')}{A(u')} - \frac{A'(u - u')}{A(u - u')} - \frac{A'(u + u')}{A(u + u')} = \frac{2K\sqrt{X'}}{\sqrt{-af}} \cdot \frac{bc(d - x)}{(a - x')\nabla_1}.$$

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and similarly

$$\begin{aligned} \frac{B'(u')}{B(u')} + \frac{B'(u - u')}{B(u - u')} - \frac{B'(u + u')}{B(u + u')} &= \frac{2K ca}{\sqrt{-af}} \sqrt{X'} \cdot \frac{d - x}{(b - x')\nabla_2}, \\ \frac{C'(u')}{C(u')} + \frac{C'(u - u')}{C(u - u')} - \frac{C'(u + u')}{C(u + u')} &= \frac{2K ab}{\sqrt{-af}} \sqrt{X'} \cdot \frac{d - x}{(c - x')\nabla_3}, \\ \frac{D'(u')}{D(u')} + \frac{D'(u - u')}{D(u - u')} - \frac{D'(u + u')}{D(u + u')} &= \frac{2K}{\sqrt{-af}} \sqrt{X'} \cdot \frac{d - x}{(d - x')(x - x')}. \end{aligned}$$

55. Multiply each of these equations by  $du, = \frac{1}{2} \frac{\sqrt{-af}}{K} \frac{dx}{\sqrt{X}}$ , and integrate. We have equations such as

$$2u \frac{A'(u')}{A(u')} + \log \frac{A(u - u')}{A(u + u')} = \text{const.} + \frac{1}{\sqrt{-af}} \int \frac{bc(d - x) dx}{(a - x')\nabla_1 \sqrt{X}},$$

showing how the integrals of the third kind

$$\int \frac{(d - x) dx}{\nabla_1 \sqrt{X}}, \quad \int \frac{(d - x) dx}{\nabla_2 \sqrt{X}}, \quad \int \frac{(d - x) dx}{\nabla_3 \sqrt{X}}, \quad \int \frac{(d - x) dx}{(x - x')\sqrt{X}}$$

depend on the theta-functions.

If, instead, we work with the original equation

$$\frac{C(u + u')C(u - u')}{Cu \cdot C^2u'} = 1 - \frac{D^2u}{C^2u} \frac{D^2u'}{C^2u'},$$

we find in the same way

$$\begin{aligned} \frac{C'(u')}{C(u')} + \frac{C'(u - u')}{C(u - u')} - \frac{C'(u + u')}{C(u + u')} &= -\frac{d}{du'} \log \left( 1 - \frac{D^2u}{C^2u} \frac{D^2u'}{C^2u'} \right), \\ &= -\frac{d}{du'} \log(1 - k^2 \text{sn}^2 Ku \text{sn}^2 Ku'), \\ &= -\frac{2k^2 K \text{sn} Ku' \text{cn} Ku' \text{dn} Ku' \text{sn}^2 Ku}{1 - k^2 \text{sn}^2 Ku' \text{sn}^2 Ku}, \end{aligned}$$

or, multiplying by  $\frac{1}{2} du$  and integrating, we have

$$\frac{C'(u')}{C(u')} + \frac{1}{2} \log \frac{C(u - u')}{C(u + u')} = - \int \frac{k^2 \text{sn} Ku' \text{cn} Ku' \text{dn} Ku' \text{sn}^2 Ku}{1 - k^2 \text{sn}^2 Ku' \text{sn}^2 Ku} K du,$$



which is in fact Jacobi's equation

$$\frac{\Theta'a}{\Theta a} + \frac{1}{2} \log \frac{\Theta(u-a)}{\Theta(u+a)} = k^2 \int \frac{\operatorname{sn} a \operatorname{cn} a \operatorname{dn} a \operatorname{sn}^2 u \, du}{1 - k^2 \operatorname{sn}^2 a \operatorname{sn}^2 u}, \quad = \Pi(u, a).$$

I do not effect the operation but consider the forms first obtained,

$$A(u+u')A(u-u') = \frac{2}{\operatorname{gh}} \nabla_1, \quad \&c.,$$

as the equivalent of Jacobi's last-mentioned equation.

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*Addition-formulæ.*

**56.** The addition-theorem for the quotient-functions is of course given by means of the theorem for the elliptic functions: but more elegantly by the formulæ relating to the form  $dx \div \sqrt{a-x} \cdot \sqrt{b-x} \cdot \sqrt{c-x} \cdot \sqrt{d-x}$  obtained in my paper "On the Double  $\vartheta$ -Functions" (Crelle, t. LXXXVII. (1879), pp. 74–81, [697]); viz. for the differential equation

$$\frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} + \frac{dz}{\sqrt{Z}} = 0,$$

to obtain the particular integral which for  $y = d$  reduces itself to  $z = x$ , we must, in the formulæ of the paper just referred to, interchange  $a$  and  $d$ : and writing for shortness  $a, b, c, d = a-x, b-x, c-x, d-x$ , and similarly  $a_1, b_1, c_1, d_1 = a-y, b-y, c-y, d-y$ , then when the interchange is made, the formulæ become

$$\begin{aligned} \sqrt{\frac{d-z}{a-z}} &= \frac{\sqrt{\frac{d-b}{d-a}} \{ (d-c) \sqrt{abc_1 d_1} + (b-a) \sqrt{cda_1 b_1} \}}{(bc, ad)} = \frac{\sqrt{\frac{d-b}{d-a}} (ac, bd)}{\sqrt{abc_1 d_1} - \sqrt{cda_1 b_1}}, \\ \sqrt{\frac{b-z}{a-z}} &= \frac{\sqrt{\frac{d-c}{d-a}} (ac, bd)}{\sqrt{bda_1 c_1} - \sqrt{acb_1 d_1}} = \frac{\sqrt{\frac{d-b}{d-c}} \{ (d-c) \sqrt{aba_1 b_1} - (b-a) \sqrt{cdc_1 d_1} \}}{(d-c) \sqrt{aba_1 b_1} + (b-a) \sqrt{cdc_1 d_1}} \sqrt{\frac{d-b}{d-c}}, \\ \sqrt{\frac{c-z}{a-z}} &= \frac{\sqrt{\frac{d-b}{d-c}} \{ (d-a) \sqrt{bca_1 c_1} - (b-c) \sqrt{ada_1 d_1} \}}{(d-b) \sqrt{aca_1 c_1} - (c-a) \sqrt{bdb_1 d_1}} = \frac{\sqrt{\frac{d-b}{d-c}} (ab, cd)}{(d-b) \sqrt{aca_1 c_1} - (c-a) \sqrt{bdb_1 d_1}}, \\ \sqrt{\frac{d-z}{a-z}} \cdot \frac{1}{x-y} \cdot \frac{1}{(bc, ad)} &= \frac{\sqrt{abc_1 d_1} - \sqrt{cda_1 b_1}}{\sqrt{abc_1 d_1} - \sqrt{cda_1 b_1}}. \end{aligned}$$

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**57.** In the foregoing formulæ,  $(bc, ad)$ ,  $(ac, bd)$ , and  $(ab, cd)$  denote respectively

$$\begin{vmatrix} 1 & x+y & xy \\ 1 & b+c & bc \\ 1 & a+d & ad \end{vmatrix}, \quad \begin{vmatrix} 1 & x+y & xy \\ 1 & c+a & ca \\ 1 & b+d & bd \end{vmatrix}, \quad \begin{vmatrix} 1 & x+y & xy \\ 1 & a+b & ab \\ 1 & c+d & cd \end{vmatrix};$$

and substituting for  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}$  their values, and for  $a, b$ , &c., writing again  $a - x$ ,  $b - x$ , &c., we have moreover

$$\begin{aligned} A^2u &= \sqrt{c-b \cdot b-d \cdot c-d}(a-x), & A^2v &= \cdots (a-y), \\ B^2u &= \sqrt{c-a \cdot c-d \cdot a-d}(b-x), & B^2v &= \cdots (b-y), \\ C^2u &= \sqrt{a-b \cdot a-d \cdot b-d}(c-x), & C^2v &= \cdots (c-y), \\ D^2u &= \sqrt{c-b \cdot c-a \cdot a-b}(d-x), & D^2v &= \cdots (d-y), \\ A^2(u+v) &= \cdots (a-z), \\ B^2(u+v) &= \cdots (b-z), \\ C^2(u+v) &= \cdots (c-z), \\ D^2(u+v) &= \cdots (d-z), \end{aligned}$$

the constant multipliers being of course the same in the three columns respectively. According to what precedes, the radical of the fourth line should be taken with the sign  $-$ . The functions  $(bc, ad)$ , &c., contained in the formulæ, require a transformation such as

$$(b-c)(bc, ad) = \begin{vmatrix} b-x \cdot b-y, & c-x \cdot c-y \\ b-a \cdot b-d, & c-a \cdot c-d \end{vmatrix}$$

in order to make them separately homogeneous in the differences  $a-x, b-x, c-x, d-x$ , and  $a-y, b-y, c-y, d-y$ , and therefore to make them expressible as linear functions of the squared functions  $A^2u$ , &c., and  $A^2v$ , &c., respectively: the formulæ then give the quotient-functions  $\frac{A(u+v)}{D(u+v)}$ , &c., in terms of the quotient-functions of  $u$  and  $v$  respectively.

#### *Doubly infinite product-forms.*

**58.** The functions  $Au, Bu, Cu, Du$  may be expressed each as a doubly infinite product. Writing for shortness

$$\begin{aligned} m+n \cdot \frac{\omega}{\pi i} &= (m, n), \\ m+1+n \cdot \frac{\omega}{\pi i} &= (\overline{m}, n), \\ m+(n+1) \cdot \frac{\omega}{\pi i} &= (m, \overline{n}), \\ m+1+(n+1) \cdot \frac{\omega}{\pi i} &= (\overline{m}, \overline{n}), \end{aligned}$$

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then the formulæ are

$$\begin{aligned} Au &= A0 \cdot \prod \prod \left\{ 1 + \frac{u}{(\overline{m}, \overline{n})} \right\}, \\ Bu &= B0 \cdot \prod \prod \left\{ 1 + \frac{u}{(m, \overline{n})} \right\}, \\ Cu &= C0 \cdot \prod \prod \left\{ 1 + \frac{u}{(\overline{m}, n)} \right\}, \\ Du &= D'0 \cdot u \cdot \prod \prod \left\{ 1 + \frac{u}{(m, n)} \right\}, \end{aligned}$$

where in all the formulæ,  $m$  and  $n$  denote even integers having all values whatever, zero included, from  $-\infty$  to  $+\infty$ ; only in the formula for  $Du$ , the term for which  $m$  and  $n$  are simultaneously  $= 0$ , is to be omitted.

**59.** But a further definition in regard to the limits is required: first, we assume that  $m$  has the corresponding positive and negative values, and similarly that  $n$  has corresponding positive and negative

values<sup>1</sup>; or say, in the four formulæ respectively, we consider  $m, n$  as extending

$$\begin{array}{ll} m \text{ from } -\mu \text{ to } \mu + 2, & n \text{ from } -\nu \text{ to } \nu + 2, \\ -\mu \text{ to } \mu + 2, & -\nu \text{ to } \nu, \\ -\mu \text{ to } \mu, & -\nu \text{ to } \nu + 2, \\ -\mu \text{ to } \mu, & -\nu \text{ to } \nu, \end{array}$$

where  $\mu$  and  $\nu$  are each of them ultimately infinite. But, secondly, it is necessary that  $\mu$  should be indefinitely larger than  $\nu$ , or say that ultimately  $\frac{\nu}{\mu} = 0$ .

**60.** In fact, transforming the  $q$ -series into products as in the *Fundamenta Nova*, and neglecting for the moment mere constant factors, we have

$$\begin{aligned} Au &= (1 + 2q \cos \pi u + q^2)(1 + 2q^3 \cos \pi u + q^6) \cdots, \\ Bu &= \cos \frac{1}{2}\pi u (1 + 2q^2 \cos \pi u + q^4)(1 + 2q^4 \cos \pi u + q^8) \cdots, \\ Cu &= (1 - 2q \cos \pi u + q^2)(1 - 2q^3 \cos \pi u + q^6) \cdots, \\ Du &= \sin \frac{1}{2}\pi u (1 - 2q^2 \cos \pi u + q^4)(1 - 2q^4 \cos \pi u + q^8) \cdots, \end{aligned}$$

and writing for a moment  $\omega = -\frac{\omega}{\pi i}$  and therefore  $q^{1+n} + q^{-1-n} = e^{i\pi\alpha} + e^{-i\pi\alpha} = 2 \cos \frac{1}{2}\pi\alpha$ , &c., each of these expressions is readily converted into a singly infinite product of sines or cosines

$$\begin{aligned} Au &= \prod \cos \frac{1}{2}\pi(u + n\alpha), \\ Bu &= \prod \cos \frac{1}{2}\pi(u + \bar{n}\alpha), \\ Cu &= \prod \sin \frac{1}{2}\pi(u + n\alpha), \\ Du &= \prod \sin \frac{1}{2}\pi(u + \bar{n}\alpha), \end{aligned}$$

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where  $\bar{n}$  is written to denote  $n + 1$ , and  $n$  has all positive or negative even values (zero included) from  $-\infty$  to  $+\infty$ , or more accurately from  $-\nu$  to  $\nu$ , if  $\nu$  is ultimately infinite.

**61.** The sines and cosines are converted into infinite products by the ordinary formulæ, which neglecting constant factors may conveniently be written

$$\sin \frac{1}{2}\pi u = \prod (u + m), \quad \cos \frac{1}{2}\pi u = \prod (u + \bar{m}),$$

where  $\bar{m}$  is written to denote  $m + 1$ , and  $m$  has all positive or negative even values (zero included) from  $-\infty$  to  $+\infty$ , or more accurately from  $-\mu$  to  $\mu$ , if  $\mu$  be ultimately infinite. But in applying these formulæ to the case of a function such as

$$\sin \frac{1}{2}\pi(u + n\alpha),$$

it is assumed that the limiting values  $\mu, -\mu$  of  $m$  are indefinitely large in regard to  $u + n\alpha$ ; and therefore, since  $n$  may approach to its limiting value  $+\nu$ , it is necessary that  $\mu$  should be indefinitely large in comparison with  $\nu$ , or that  $\frac{\nu}{\mu} = 0$ .

**62.** It is on account of this unsymmetry as to the limits  $\frac{\nu}{\mu} = 0, \frac{\mu}{\nu} = \infty$ , that we have 1 as a quarter-period,  $\frac{\omega}{\pi i}$  only as a quarter-quasi-period of the single theta-functions.

$$\text{The transformation } q \text{ to } r, \log q \log r = \pi^2.$$

<sup>1</sup>This is more than is necessary; it would be enough if the ultimate values of  $m$  were  $-\mu, \mu'$ , and  $\mu, \mu'$  being in a ratio of equality; and the like as regards  $n$ . But it is convenient that the numbers should be absolutely equal.

**63.** In general, if we consider the ratio of two such infinite products, where for the first the limits are  $(\pm\mu, \pm\nu)$ , and for the second they are  $(\pm\mu', \pm\nu')$ , and where for convenience we take  $\mu > \mu', \nu > \nu'$ , then the quotient, say  $[\mu, \nu] \div [\mu', \nu']$  is  $= \exp(Mu^2)$ , where

$$M = -\frac{1}{2} \iint \frac{dm \, dn}{(m, n)^2},$$

taken over the area included between the two rectangles. We have

$$(m, n) = m + \frac{\omega}{\pi i} n = m + i\theta n$$

suppose, where ( $\omega$  being negative)  $\theta = -\frac{\omega}{\pi}$  is positive: the integral is

$$\iint \frac{dm \, dn}{(m + i\theta n)^2} = \int dm \int \frac{dn}{(m + i\theta n)^2} = -\frac{1}{i\theta} \int dm \left( \frac{1}{m - i\theta\nu} - \frac{1}{m + i\theta\nu} \right) = -\frac{1}{i\theta} \int \log \frac{m - i\theta\nu}{m + i\theta\nu} dm,$$

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or finally between the proper limits the value is

$$= \frac{2}{i\theta} \left\{ \log \left( \frac{\mu - i\theta\nu}{\mu + i\theta\nu} \right) - \log \left( \frac{\mu' - i\theta\nu'}{\mu' + i\theta\nu'} \right) \right\},$$

where the logarithms are

$$\log(\mu - i\theta\nu) = \log \sqrt{\mu^2 + \nu^2} - i \tan^{-1} \frac{\theta\nu}{\mu} - \&c.,$$

and the  $\tan^{-1}$  denotes always an arc between the limits  $-\frac{1}{2}\pi, +\frac{1}{2}\pi$ . Hence, if  $\frac{\mu}{\nu} = \infty, \frac{\mu'}{\nu'} = 0$ , the value is

$$\frac{2}{i\theta} (-0 - 0 + \frac{1}{2}\pi i + \frac{1}{2}\pi i) = \frac{2\pi}{\theta} = -\frac{2\pi^2}{\omega} = \text{or } \frac{2\pi^2}{a};$$

Hence finally

$$[\mu, \nu, = \infty] \div [\mu, \nu, = 0] = \exp\left(\frac{\pi^2}{a} u^2\right).$$

**64.** We have  $a = \log q$ , negative; hence taking  $r$  such that  $\log q \log r = \pi^2$ , we have  $a' = \log r$ , also negative; and  $r$ , like  $q$ , is positive and less than 1. We consider the theta-functions which depend on  $r$  in the same manner that the original functions did on  $q$ , say these are

$$\begin{aligned} A(u, r) &= A(0, r) \prod \prod \left\{ 1 + \frac{u}{m + n \frac{a'}{\pi i}} \right\}, \\ B(u, r) &= B(0, r) \prod \prod \left\{ 1 + \frac{u}{\bar{m} + n \frac{a'}{\pi i}} \right\}, \\ C(u, r) &= C(0, r) \prod \prod \left\{ 1 + \frac{u}{\bar{m} + \bar{n} \frac{a'}{\pi i}} \right\}, \\ D(u, r) &= D'(0, r) u \prod \prod \left\{ 1 + \frac{u}{m + \bar{n} \frac{a'}{\pi i}} \right\}, \end{aligned}$$

limits as before, and in particular  $\frac{\mu}{\nu} = \infty$ ; it is at once seen that if in the original functions, which I now call  $A(u, q), B(u, q), C(u, q), D(u, q)$ , we write  $\frac{au}{\pi i}$  for  $u$ , we obtain the same infinite products which present themselves in the expressions of the new functions  $A(u, r)$ , &c., only with a different condition as to the limits; we have for instance

$$\prod \prod \left\{ 1 + \frac{\frac{au}{\pi i}}{m + n \frac{a'}{\pi i}} \right\} = \prod \prod \left\{ 1 + \frac{u}{n + m \frac{\pi i}{a}} \right\} = \prod \prod \left\{ 1 + \frac{u}{n + m \frac{\pi i}{a'}} \right\}$$

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which, interchanging  $m, n$ , and of course also  $\mu, \nu$ , is

$$= \prod \prod \left\{ 1 + \frac{u}{m + n \frac{a'}{\pi i}} \right\}$$

with the condition  $\frac{\mu}{\nu} = 0$  instead of  $\frac{\mu}{\nu} = \infty$ . Hence disregarding for the moment constant factors, and observing that  $a$  is replaced by  $a'$ , we have

$$\begin{aligned} D(u, r) \div D\left(\frac{au}{\pi i}, q\right) &= [\mu, \nu, = \infty] \div [\mu, \nu, = 0] \\ &= \exp\left(\frac{\pi^2}{a} u^2\right) = \exp\left(\frac{1}{2} u^2 \log q\right). \end{aligned}$$

**65.** We have equations of this form for the four functions, but with a proper constant multiplier in each equation: the equations, in fact, are

$$\begin{aligned} A(u, r) &= \{A(0, r) \div A(0, q)\} \exp\left(\frac{1}{2} u^2 \log q\right) A\left(\frac{au}{\pi i}, q\right), \\ B(u, r) &= \{B(0, r) \div B(0, q)\} B\left(\frac{au}{\pi i}, q\right), \\ C(u, r) &= \{C(0, r) \div C(0, q)\} C\left(\frac{au}{\pi i}, q\right), \\ D(u, r) &= \{D'(0, r) \div D'(0, q)\} D\left(\frac{au}{\pi i}, q\right). \end{aligned}$$

It is to be observed that  $r$  is the same function of  $k'$  that  $q$  is of  $k$ . This would at once follow from Jacobi's equation  $\log q = -\frac{\pi K'}{K}$ , for then  $\log q \log r = \pi^2$  and therefore  $\log r = -\frac{\pi K}{K'}$  (only we are not at liberty to use the relation in question  $\log q = -\frac{\pi K'}{K}$ ): assuming it to be true, we have

$$\begin{aligned} \frac{D'(0, r)}{A^2(0, r)} &= \frac{D'(0, q)}{C^2(0, q)} = \frac{C^2(0, q) D'(0, q)}{A(0, q) B(0, q)}, \\ &= \frac{C'(0, r)}{A^2(0, r)} = \frac{B'(0, r)}{B^2(0, r)} = \frac{A(0, r) D'(0, r)}{B(0, r) C(0, r)}, \\ \log q &= -\frac{\pi K'}{K}, \quad \log r = -\frac{\pi K}{K'}, \end{aligned}$$

where, if the identity of the two values of  $k$  or of the two values of  $k'$  were proved independently (as might doubtless be done), the required theorem, viz. that  $r$  is the same function of  $y$  that  $q$  is of  $k$ , would follow conversely: and thence the other equations of the system.

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**66.** We have, in the *Fundamenta Nova*, p. 175, [Jacobi's *Ges. Werke*, t. i., p. 227], the equation

$$\frac{H(iu, k)}{\Theta(0, k)} = i \sqrt{\frac{k}{k'}} e^{\frac{\pi u^2}{4K}} \frac{H(u, k')}{\Theta(0, k')},$$

writing here  $K'u$  instead of  $u$  the equation becomes

$$\frac{H(iK'u, k)}{\Theta(0, k)} = i \sqrt{\frac{k}{k'}} \exp\left(\frac{1}{4} \frac{\pi K'}{K} u^2\right) \cdot \frac{H(K'u, k')}{\Theta(0, k')},$$

or, what is the same thing,

$$\frac{D\left(\frac{\pi i K'}{K} u, q\right)}{C^2(0, q)} = i \sqrt{\frac{k}{k'}} \exp\left(\frac{1}{4} \frac{\pi K'}{K} u^2\right) \cdot \frac{D(u, r)}{D(0, r)},$$

which can be readily identified with the foregoing equation between  $D\left(\frac{au}{\pi i}, q\right)$  and  $D(u, r)$ . But the real meaning of the transformation is best seen by means of the double-product formulæ.

THIRD PART.—THE DOUBLE THETA-FUNCTIONS.

Notations, &c.

67. We have here 16 functions  $\vartheta\left(\begin{smallmatrix}\alpha\beta\\ \gamma\delta\end{smallmatrix}\right)(u,v)$ : this notation by characteristics, containing each of them four numbers, is too cumbrous for ordinary use, and I therefore replace it by the current-number notation, in which the characteristics are denoted by the series of numbers 0, 1, 2, . . . , 15: we cannot in place of this introduce the single-and-double-letter notation  $A, B, \dots, AB$ , &c., for there is not here any correspondence of the two notations, nor is there anything in the definition of the functions which in anywise suggests the single-and-double-letter notation: this first presents itself in connexion with the relations between the functions given by the product-theorem: and as the product-theorem is based upon the notation by characteristics, it is proper to present the theorem in this notation, or in the equivalent current-number notation: and then to show how by the relations thus obtained between the functions we are led to the single-and-double-letter notation.

68. There are some other notations which may be referred to: and for showing the correspondence between them I annex the following table:

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The double theta-functions.

| 1.                                        | 2.                 | 3.                                            | 4.     | 5.               | 6.         | 7.                | 8.                                            |
|-------------------------------------------|--------------------|-----------------------------------------------|--------|------------------|------------|-------------------|-----------------------------------------------|
| Asterisk<br>denotes the<br>odd functions. | Current<br>number. | Single-and-<br>double-letter,<br>Cayley.      | Göpel. | Göpel<br>Cayley. | Rosenhain. | Weier-<br>strass. | Kummer.                                       |
|                                           | 0                  | $\begin{smallmatrix}00\\ 00\end{smallmatrix}$ | $BD$   | $P'''$           | $P_0$      | $S_{11}$          | $\begin{smallmatrix}5\\ 12\end{smallmatrix}$  |
|                                           | 1                  | $\begin{smallmatrix}10\\ 00\end{smallmatrix}$ | $CE$   | $R'''$           | $R_1$      | 11                | $\begin{smallmatrix}4\\ 8\end{smallmatrix}$   |
|                                           | 2                  | $\begin{smallmatrix}01\\ 00\end{smallmatrix}$ | $CD$   | $Q'''$           | $Q_3$      | 23                | $\begin{smallmatrix}01\\ 10\end{smallmatrix}$ |
|                                           | 3                  | $\begin{smallmatrix}11\\ 00\end{smallmatrix}$ | $BE$   | $S'''$           | $S_3$      | 33                | $\begin{smallmatrix}23\\ 6\end{smallmatrix}$  |
|                                           | 4                  | $\begin{smallmatrix}00\\ 10\end{smallmatrix}$ | $AC$   | $P''$            | $A$        | 02                | $\begin{smallmatrix}34\\ 4\end{smallmatrix}$  |
| *                                         | 5                  | $\begin{smallmatrix}10\\ 10\end{smallmatrix}$ | $C$    | $iR''$           | $R_2$      | 12                | $\begin{smallmatrix}3\\ 16\end{smallmatrix}$  |
|                                           | 6                  | $\begin{smallmatrix}01\\ 10\end{smallmatrix}$ | $AB$   | $Q''$            | $Q_2$      | 03                | $\begin{smallmatrix}2\\ 2\end{smallmatrix}$   |
| *                                         | 7                  | $\begin{smallmatrix}11\\ 10\end{smallmatrix}$ | $B$    | $iS''$           | $S_2$      | 13                | $\begin{smallmatrix}24\\ 14\end{smallmatrix}$ |
|                                           | 8                  | $\begin{smallmatrix}00\\ 01\end{smallmatrix}$ | $BC$   | $P'$             | $A$        | 20                | $\begin{smallmatrix}12\\ 9\end{smallmatrix}$  |
|                                           | 9                  | $\begin{smallmatrix}10\\ 01\end{smallmatrix}$ | $DE$   | $R'$             | $R_3$      | 30                | $\begin{smallmatrix}03\\ 5\end{smallmatrix}$  |
| *                                         | 10                 | $\begin{smallmatrix}01\\ 01\end{smallmatrix}$ | $F$    | $iQ'$            | $Q_1$      | 21                | $\begin{smallmatrix}02\\ 11\end{smallmatrix}$ |
| *                                         | 11                 | $\begin{smallmatrix}11\\ 01\end{smallmatrix}$ | $A$    | $iS'$            | $S_1$      | 31                | $\begin{smallmatrix}13\\ 7\end{smallmatrix}$  |
|                                           | 12                 | $\begin{smallmatrix}00\\ 11\end{smallmatrix}$ | $AD$   | $P$              | $P$        | 00                | $\begin{smallmatrix}—\\ 1\end{smallmatrix}$   |
| *                                         | 13                 | $\begin{smallmatrix}10\\ 11\end{smallmatrix}$ | $D$    | $iR$             | $R$        | 10                | $\begin{smallmatrix}04\\ 13\end{smallmatrix}$ |
|                                           | 14                 | $\begin{smallmatrix}01\\ 11\end{smallmatrix}$ | $E$    | $iQ$             | $Q$        | 01                | $\begin{smallmatrix}1\\ 3\end{smallmatrix}$   |
|                                           | 15                 | $\begin{smallmatrix}11\\ 11\end{smallmatrix}$ | $AE$   | $S$              | $S$        | 11                | $\begin{smallmatrix}14\\ 15\end{smallmatrix}$ |

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69. These are the notations:

1. By current-numbers. It may be remarked that the series was taken  $0, 1, \dots, 15$ , instead of  $1, 2, \dots, 16$ , in order that 0 might correspond to the characteristic  $\begin{smallmatrix} 00 \\ 00 \end{smallmatrix}$ ;

2. By characteristics;

3. By single-and-double letters;

4. Göpel's, in his paper above referred to, and

5. The same as used in my paper above referred to;

6. Rosenhain's, in his paper above referred to;

7. Weierstrass', as quoted by Königsberger in his paper "Ueber die Transformation der Abelschen Functionen erster Ordnung," Crelle-Borchardt, t. LXIV. (1865), p. 17, and by Borchardt in his paper above referred to;

8. Not a theta-notation, but the series of current-numbers used in Kummer's Memoir "Ueber die algebraischen Strahlen-systeme," Berl. Abh. 1866, for the nodes of his 16-nodal quartic surface, and connected with the double theta-functions in my paper above referred to.

But in the present memoir only the first three columns of the table need be attended to.

70. It will be convenient to introduce here some other remarks as to notation, &c.

The letter  $c$  is used in connexion with the zero values  $u = 0, v = 0$  of the arguments, viz.:

$$\vartheta_0, \vartheta_1, \vartheta_2, \vartheta_3, \vartheta_4, \vartheta_6, \vartheta_8, \vartheta_9, \vartheta_{12}, \vartheta_{15}$$

are even functions, and the corresponding zero-functions are denoted by

$$c_0, c_1, c_2, c_3, c_4, c_6, c_8, c_9, c_{12}, c_{15};$$

there are thus 10 constants  $c$ .

When  $(u, v)$  are indefinitely small each of these functions is of course equal to its zero-value plus a quadric term in  $(u, v)$ , and we may write in general

$$\vartheta = c + \frac{1}{2}(c^{[1]}, c^{[2]}, c^{[3]}|u, v)^2 :$$

there are thus 30 constants  $c^{[1]}, c^{[2]}, c^{[3]}$ .

The remaining functions

$$\vartheta_5, \vartheta_7, \vartheta_{10}, \vartheta_{11}, \vartheta_{13}, \vartheta_{14}$$

are odd functions vanishing for  $u = 0, v = 0$ ; when these arguments are indefinitely small, we may write in general

$$\vartheta = (c', c''|u, v) :$$

there are thus 12 constants  $c', c''$ .

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71. All these constants are in the first instance given as transcendental functions of the parameters, or say rather of  $\exp a, \exp h, \exp b$ , which exponentials correspond to the  $q$  of the single theory: viz., in a notation which will be readily understood, the constants  $c, c^{[1]}, c^{[2]}, c^{[3]}$  of the even functions are

$$\Sigma \exp \left( \begin{matrix} m + \alpha, n + \beta \\ \gamma \quad \delta \end{matrix} \right),$$

$$-\frac{1}{4}\pi^2 \Sigma (m + \alpha)^2, 2(m + \alpha)(n + \beta), (n + \beta)^2, \exp \left( \begin{matrix} m + \alpha, n + \beta \\ \gamma \quad \delta \end{matrix} \right);$$

and the constants  $c', c''$  of the odd functions are

$$i\pi \Sigma (m + \alpha), (n + \beta), \exp \left( \begin{matrix} \\ \delta \end{matrix} \right).$$

**72.** It is convenient for the verification of results and otherwise, to have the values of the functions, belonging to small values of  $\exp(-a)$ ,  $\exp(-b)$ ; if to avoid fractional exponents we regard these and  $\exp(-h)$  as fourth powers, and write

$$\exp(-a) = Q^4, \quad \exp(-h) = R^4, \quad \exp(-b) = S^4,$$

also

$$QRS = \Lambda, \quad QR^{-1}S = \Lambda', \quad \text{and therefore } \Lambda\Lambda' = Q^2S^2,$$

then attending only to the lowest powers of  $Q, S$  we find without difficulty

$$\begin{aligned} \vartheta_0(u) &= 1, & \text{and therefore also } c_0 &= 1, \\ \vartheta_1 &= 2Q \cos \tfrac{1}{2}\pi u, & c_1 &= 2Q, \\ \vartheta_2 &= 2S \cos \tfrac{1}{2}\pi v, & c_2 &= 2S, \\ \vartheta_3 &= 2\Lambda \cos \tfrac{1}{2}\pi(u+v) + 2\Lambda' \cos \tfrac{1}{2}\pi(u-v), & c_3 &= 2\Lambda + 2\Lambda', \\ \vartheta_4 &= 1, & c_4 &= 1, \\ \vartheta_5 &= -2Q \sin \tfrac{1}{2}\pi u, & & \\ \vartheta_6 &= 2S \cos \tfrac{1}{2}\pi v, & c_6 &= 2S, \\ \vartheta_7 &= -2\Lambda \sin \tfrac{1}{2}\pi(u+v) - 2\Lambda' \sin \tfrac{1}{2}\pi(u-v), & & \\ \vartheta_8 &= 1, & c_8 &= 1, \\ \vartheta_9 &= 2Q \cos \tfrac{1}{2}\pi u, & c_9 &= 2Q, \\ \vartheta_{10} &= -2S \sin \tfrac{1}{2}\pi v, & & \\ \vartheta_{11} &= -2\Lambda \sin \tfrac{1}{2}\pi(u+v) + 2\Lambda' \sin \tfrac{1}{2}\pi(u-v), & & \\ \vartheta_{12} &= 1, & c_{12} &= 1, \\ \vartheta_{13} &= -2Q \sin \tfrac{1}{2}\pi u, & & \\ \vartheta_{14} &= -2S \sin \tfrac{1}{2}\pi v, & & \\ \vartheta_{15} &= -2\Lambda \cos \tfrac{1}{2}\pi(u+v) + 2\Lambda' \cos \tfrac{1}{2}\pi(u-v), & c_{15} &= -2\Lambda + 2\Lambda'. \end{aligned}$$

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**73.** The expansions might be carried further; we have for instance

$$\begin{aligned} \vartheta_0(u) &= 1 + 2Q^4 \cos \pi u + 2S^4 \cos \pi v, & c_0 &= 1 + 2Q^4 + 2S^4, \\ \vartheta_4 &= 1 - 2Q^4 & + 2S^4 &, & c_4 &= 1 - 2Q^4 + 2S^4, \\ \vartheta_8 &= 1 + 2Q^4 & - 2S^4 &, & c_8 &= 1 + 2Q^4 - 2S^4, \\ \vartheta_{12} &= 1 - 2Q^4 & - 2S^4 &, & c_{12} &= 1 - 2Q^4 - 2S^4, \end{aligned}$$

$$\begin{aligned} \vartheta_1 &= 2Q \cos \tfrac{1}{2}\pi u + 2Q^9 \cos \tfrac{3}{2}\pi u + 2A \cos \tfrac{1}{2}\pi(u+2v) + 2A' \cos \tfrac{1}{2}\pi(u-2v), & c_1 &= 2Q + 2Q^9 + 2A + 2A', \\ \vartheta_5 &= -2Q \sin \tfrac{1}{2}\pi u + 2Q^9 \sin \tfrac{3}{2}\pi u - 2A \sin \tfrac{1}{2}\pi(u+2v) - 2A' \sin \tfrac{1}{2}\pi(u-2v), \\ \vartheta_9 &= 2Q \cos \tfrac{1}{2}\pi u + 2Q^9 \cos \tfrac{3}{2}\pi u - 2A \cos \tfrac{1}{2}\pi(u+2v) - 2A' \cos \tfrac{1}{2}\pi(u-2v), & c_9 &= 2Q + 2Q^9 - 2A - 2A', \\ \vartheta_{13} &= -2Q \sin \tfrac{1}{2}\pi u + 2Q^9 \sin \tfrac{3}{2}\pi u + 2A \sin \tfrac{1}{2}\pi(u+2v) + 2A' \sin \tfrac{1}{2}\pi(u-2v), \end{aligned}$$

in which last formulæ

$$A = QR^4S^4 = \frac{Q}{R^3Q^3}, \quad A' = QR^{-4}S^4 = \frac{Q}{R^4S^4}.$$

**74.** In the single-and-double-letter notation we have six letters  $A, B, C, D, E, F$ ; and the duads  $AB, AC, \dots, BE$  are used as abbreviations for the double triads  $ABF, CDE$ , &c., the letter  $F$  always accompanying the expressed duad; there are thus in all six single-letter symbols, and 10 double-letter symbols, in all 16 symbols, corresponding to the double-theta functions, as already mentioned in the order

$$\begin{array}{cccccccccccccccc} \vartheta_0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 \\ BD, & CE, & CD, & BE, & AC, & C, & AB, & B, & BC, & DE, & F, & A, & AD, & D, & E, & AE, \end{array}$$



where observe that the single letters  $C, B, F, A, D, E$  correspond to the odd functions 5, 7, 10, 11, 13, 14 respectively.

**75.** The ground of the notation is as follows:

The algebraical relations between the double theta-functions are such that, introducing six constant quantities  $a, b, c, d, e, f$  and two variable quantities  $x, y$ , it is allowable to express the 16 functions as proportional to given functions of these quantities  $(a, b, c, d, e, f; x, y)$ ; viz. there are six functions the notations of which depend on the single letters  $a, b, c, d, e, f$  respectively, and 10 functions the notations of which depend on the pairs  $ab, ac, ad, ae, bc, bd, be, cd, ce, de$  respectively: each of the 16 functions is, in fact, proportional to the product of two factors, viz. a constant factor depending only on  $(a, b, c, d, e, f)$ , and a variable factor containing also  $x$  and  $y$ . Attending in the first instance to the variable factors, and writing for shortness

$$\begin{aligned} a - x, b - x, c - x, d - x, e - x, f - x &= a, b, c, d, e, f; \quad x - y = \theta; \\ a - y, b - y, c - y, d - y, e - y, f - y &= a_1, b_1, c_1, d_1, e_1, f_1; \end{aligned}$$

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these are of the forms

$$\sqrt[4]{a} = \sqrt[4]{aa_1}, \quad \sqrt[4]{ab} = \frac{1}{\theta} \{ \sqrt{abfc_1d_1e_1} + \sqrt{a_1b_1f_1cde} \}.$$

I remark that to avoid ambiguity the squares of these expressions are throughout written as  $(\sqrt[4]{a})^2$  and  $(\sqrt[4]{ab})^2$  respectively.

**76.** There is, for the constant factors, a like single-and-double-letter notation which will be mentioned presently; but in the first instance I use for the even functions the before-mentioned 10 letters  $c$ , and for the odd ones six letters  $k$ . I assume that the values  $x, y = \infty, \infty$  (ratio not determined) correspond to the values  $u = 0, v = 0$  of the arguments; and that  $\omega$  is a function of  $(x, y)$  which, when  $(x, y)$  are interchanged, changes only its sign, and which for indefinitely large values of  $(x, y)$  becomes  $= \frac{x - y}{(xy)^{\frac{1}{2}}}$ . This being so, we write (adding a second column which will be afterwards explained)

$$\begin{aligned} 0 &= BD = \omega c_0 \sqrt[4]{bd}, & c_0 &= \omega \sqrt[4]{bd}, \\ 1 &= CE = \omega c_1 \sqrt[4]{ce}, & c_1 &= \omega \sqrt[4]{ce}, \\ 2 &= CD = \omega c_2 \sqrt[4]{cd}, & c_2 &= \omega \sqrt[4]{cd}, \\ 3 &= BE = \omega c_3 \sqrt[4]{be}, & c_3 &= \omega \sqrt[4]{be}, \\ 4 &= AC = \omega c_4 \sqrt[4]{ac}, & c_4 &= \omega \sqrt[4]{ac}, \\ 5 &= C = \omega k_5 \sqrt[4]{c}, & k_5 &= \omega \sqrt[4]{c}, \\ 6 &= AB = \omega c_6 \sqrt[4]{ab}, & c_6 &= \omega \sqrt[4]{ab}, \\ 7 &= B = \omega k_7 \sqrt[4]{b}, & k_7 &= \omega \sqrt[4]{b}, \\ 8 &= BC = \omega c_8 \sqrt[4]{bc}, & c_8 &= \omega \sqrt[4]{bc}, \\ 9 &= DE = \omega c_9 \sqrt[4]{de}, & c_9 &= \omega \sqrt[4]{de}, \\ 10 &= F = \omega k_{10} \sqrt[4]{f}, & k_{10} &= \omega \sqrt[4]{f}, \\ 11 &= A = \omega k_{11} \sqrt[4]{a}, & k_{11} &= \omega \sqrt[4]{a}, \\ 12 &= AD = \omega c_{12} \sqrt[4]{ad}, & c_{12} &= \omega \sqrt[4]{ad}, \\ 13 &= D = \omega k_{13} \sqrt[4]{d}, & k_{13} &= \omega \sqrt[4]{d}, \\ 14 &= E = \omega k_{14} \sqrt[4]{e}, & k_{14} &= \omega \sqrt[4]{e}, \\ 15 &= AE = \omega c_{15} \sqrt[4]{ae}, & c_{15} &= \omega \sqrt[4]{ae}; \end{aligned}$$

viz. here, on writing  $x, y = \infty, \infty$ , each of the functions  $\sqrt[4]{bd}$ , &c., becomes  $= \frac{(xy)^{\frac{1}{2}}}{x - y}$ , and each of the functions  $\sqrt[4]{a}$ , &c., becomes  $= \sqrt[4]{xy}$ ; hence by reason of the assumed

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form of  $\omega$ , the odd functions each vanish (their evanescent values being proportional to  $k_5, k_7, k_{10}, k_{11}, k_{13}, k_{14}$  respectively), while the even functions become equal to  $c_0, c_1, c_2, c_3, c_4, c_6, c_8, c_9, c_{12}, c_{15}$  respectively.

Observe further that on interchanging  $x, y$ , the even functions remain unaltered, while the odd functions change their sign; that is, the interchange of  $x, y$  corresponds to the change  $u, v$  into  $-u, -v$ .

77. As to the values of the 10  $c$ 's and the six  $k$ 's in terms of  $a, b, c, d, e, f$ , these are proportional to fourth roots,  $\sqrt[4]{a}$ , &c.,  $\sqrt[4]{ab}$ , &c.; in  $\sqrt[4]{a}$ ,  $a$  is in the first instance regarded as standing for the pentad  $bcdef$ , and then this is used to denote a product of differences  $bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef$ ; similarly  $ab$  is in the first instance regarded as standing for the double triad  $abf \cdot cde$ , and then each of these triads is used to denote a product of differences,  $ab \cdot af \cdot bf$  and  $cd \cdot ce \cdot de$  respectively. The order of the letters is always the alphabetical one, viz. the single letters and duads denote pentads and double triads, thus:

$$\begin{aligned} a &= bcdef, & ab &= abf \cdot cde, \\ b &= acdef, & ac &= acf \cdot bde, \\ c &= abdef, & ad &= adf \cdot bce, \\ d &= abcef, & ae &= aef \cdot bcd, \\ e &= abcdf, & bc &= bcf \cdot ade, \\ f &= abcde, & bd &= bdf \cdot ace, \\ & & be &= bef \cdot acd, \\ & & cd &= cdf \cdot abe, \\ & & ce &= cef \cdot abd, \\ & & de &= def \cdot abc. \end{aligned}$$

There is no fear of ambiguity in writing (and we accordingly write) the squares of  $\sqrt[4]{a}$  and  $\sqrt[4]{ab}$  as  $\sqrt{a}$  and  $\sqrt{ab}$  respectively; the fourth powers are written  $(\sqrt[4]{a})^2$  and  $(\sqrt[4]{ab})^2$ ; the double stroke of the radical symbol  $\sqrt$  is in every case perfectly distinctive.

This being so we have as above  $c_0 = \lambda \sqrt[4]{bd}$ , &c.,  $k_5 = \lambda \sqrt[4]{c}$ , &c.: it is, however, important to notice that the fourth roots in question do not denote positive values, but they are fourth roots each taken with its proper sign ( $+, -, +i, -i$ , as the case may be) so as to satisfy the identical relations which exist between the sixteen constants; and it is not easy to determine the signs.

The variables  $x, y$  are connected with  $u, v$  by the differential relations

$$\begin{aligned} \varpi du + \tau dv &= -\frac{1}{2} \left( \frac{dx}{\sqrt{X}} - \frac{dy}{\sqrt{Y}} \right), \\ \sigma du + \rho dv &= -\frac{1}{2} \left( \frac{x dx}{\sqrt{X}} - \frac{y dy}{\sqrt{Y}} \right), \end{aligned}$$

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where  $X = abcdef$ ,  $Y = a_1 b_1 c_1 d_1 e_1 f_1$ ; which equations contain the constants  $\varpi, \rho, \sigma, \tau$ , the values of which will be afterwards connected with the other constants.

78. The  $c$ 's are expressed as functions of four quantities  $\alpha, \beta, \gamma, \delta$ , and connected with each other, and with the constants  $a, b, c, d, e, f$ , by the formulæ

$$\begin{aligned} 0 &= \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = \omega_0'' \sqrt[4]{bd}, \\ 1 &= 2(\alpha\beta + \gamma\delta) = \omega_0'' \sqrt[4]{ce}, \\ 2 &= 2(\alpha\gamma + \beta\delta) = \omega_0'' \sqrt[4]{cd}, \\ 3 &= 2(\alpha\delta + \beta\gamma) = \omega_0'' \sqrt[4]{be}, \\ 4 &= \alpha^2 - \beta^2 + \gamma^2 - \delta^2 = \omega_0'' \sqrt[4]{ac}, \\ 6 &= 2(\alpha\beta - \gamma\delta) = \omega_0'' \sqrt[4]{ab}, \\ 8 &= \alpha^2 + \beta^2 - \gamma^2 - \delta^2 = \omega_0'' \sqrt[4]{bc}, \\ 9 &= 2(\alpha\gamma - \beta\delta) = \omega_0'' \sqrt[4]{de}, \\ 15 &= 2(\alpha\delta - \beta\gamma) = \omega_0'' \sqrt[4]{ae}. \end{aligned}$$

It hence appears that we can form an arrangement

$$\begin{vmatrix} c_4, & c_6, & c_1 \\ c_8, & -c_2, & c_9 \\ c_{15}, & -c_3, & -c_5 \end{vmatrix}, \quad \begin{vmatrix} a, & b, & c \\ a', & b', & c' \\ a'', & b'', & c'' \end{vmatrix},$$

a system of coefficients in the transformation between two sets of rectangular coordinates.

We have, between the squares of these coefficients of transformation, a system of 6 + 9 equations

$$\begin{aligned} a^2 + b^2 + c^2 &= 1, \\ a'^2 + b'^2 + c'^2 &= 1, \\ a''^2 + b''^2 + c''^2 &= 1, \\ a^2 + a'^2 + a''^2 &= 1, \\ b^2 + b'^2 + b''^2 &= 1, \\ c^2 + c'^2 + c''^2 &= 1, \end{aligned}$$

$$\begin{aligned} b^2 + c^2 &= a'^2 + a''^2, & b'^2 + c'^2 &= a''^2 + a^2, & b''^2 + c''^2 &= a^2 + a'^2, \\ c^2 + a^2 &= b'^2 + b''^2, & c'^2 + a'^2 &= b''^2 + b^2, & c''^2 + a''^2 &= b^2 + b'^2, \\ a^2 + b^2 &= c'^2 + c''^2, & a'^2 + b'^2 &= c''^2 + c^2, & a''^2 + b''^2 &= c^2 + c'^2. \end{aligned}$$

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that is,

$$c^4 + c^4 + c^4 - c^4 = 0;$$

|    |   |    |   |    |   |    |
|----|---|----|---|----|---|----|
| 12 | + | 1  | + | 6  | − | 0  |
| 9  | + | 4  | + | 3  | − | 0  |
| 2  | + | 15 | + | 8  | − | 0  |
| 12 | + | 9  | + | 2  | − | 0  |
| 1  | + | 4  | + | 15 | − | 0  |
| 6  | + | 3  | + | 8  | − | 0  |
| 1  | + | 6  | − | 9  | − | 2  |
| 6  | + | 12 | − | 4  | − | 15 |
| 12 | + | 1  | − | 3  | − | 8  |
| 4  | + | 3  | − | 2  | − | 12 |
| 3  | + | 9  | − | 15 | − | 1  |
| 9  | + | 4  | − | 8  | − | 6  |
| 15 | + | 8  | − | 12 | − | 9  |
| 8  | + | 2  | − | 1  | − | 4  |
| 2  | + | 15 | − | 6  | − | 3  |

and we have, between their products, a system of 6 + 9 equations

$$\begin{aligned} a'a'' + b'b'' + c'c'' &= 0, \\ a''a + b''b + c''c &= 0, \\ aa' + bb' + cc' &= 0, \\ bc + b'c' + b''c'' &= 0, \\ ca + c'a' + c''a'' &= 0, \\ ab + a'b' + a''b'' &= 0, \end{aligned}$$

$$\begin{aligned} a, b, c &= b'c'' - b''c', & c'a'' - c''a', & a'b'' - a''b', \\ a', b', c' &= b''c - bc'', & c''a - ca'', & ab'' - a''b, \\ a'', b'', c'' &= bc' - b'c, & ca' - c'a, & ab' - a'b. \end{aligned}$$

C. X.

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**Cayley, Collected Mathematical Papers, Vol. X**

*Pages 526–550 (scan); paper 704: A Memoir on the Single and Double Theta-Functions.*

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*[Continuation of Paper 704: A Memoir on the Single and Double Theta-Functions.]*

**[Scan p. 506]**

that is,

|       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| $c^2$ | $c^2$ | $c^2$ | $c^2$ | $c^2$ | $c^2$ |
| 9     | 2     | + 4   | 15    | − 3   | 8     |
| 2     | 12    | − 15  | 1     | − 8   | 6     |
| 12    | 9     | − 1   | 4     |       |       |
|       |       |       |       | + 6   | 3     |
|       |       | − 4   | 3     |       |       |
| 1     | 6     |       |       | + 15  | 8     |
| 6     | 12    | + 3   | 9     | − 8   | 2     |
| 12    | 1     |       |       |       |       |
|       |       | − 9   | 4     | − 2   | 15    |
| − 0   | 12    | + 4   | 8     | + 3   | 15    |
| − 0   | 1     | + 3   | 2     | + 8   | 9     |
| − 0   | 6     | − 9   | 15    | + 2   | 4     |
| − 0   | 9     | − 15  | 6     | + 8   | 1     |
| + 0   | 4     | − 8   | 12    | − 6   | 2     |
| − 0   | 3     | − 12  | 15    | − 1   | 2     |
| − 0   | 2     | + 1   | 3     | + 4   | 6     |
| + 0   | 10    | + 6   | 9     | − 3   | 12    |
| + 0   | 8     | − 12  | 4     | − 9   | 1     |

= 0;

each of the first set of 15 giving a homogeneous linear relation between four terms  $c^4$ ; and each of the second set giving a homogeneous linear relation between three terms  $c^2.c^2$ , formed with the 10 constants  $c$ . Thus the first equation is

$$c_{92}^4 + c_{12}^4 + c_4^4 - c_8^4 = 0;$$

and so for the other lines of the two diagrams.

79. I form in the two notations the following tables:

TABLE OF THE 16 KUMMER HEXADS.

|           |           |           |           |           |           |           |           |           |           |           |           |           |           |          |
|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|
| <i>A</i>  | <i>A</i>  | <i>A</i>  | <i>A</i>  | <i>B</i>  | <i>B</i>  | <i>B</i>  | <i>B</i>  | <i>C</i>  | <i>C</i>  | <i>C</i>  | <i>D</i>  | <i>D</i>  | <i>E</i>  | <i>A</i> |
| <i>B</i>  | <i>C</i>  | <i>D</i>  | <i>E</i>  | <i>C</i>  | <i>D</i>  | <i>E</i>  | <i>F</i>  | <i>D</i>  | <i>E</i>  | <i>F</i>  | <i>E</i>  | <i>F</i>  | <i>F</i>  | <i>B</i> |
| <i>AB</i> | <i>AC</i> | <i>AD</i> | <i>AE</i> | <i>BC</i> | <i>BD</i> | <i>BE</i> | <i>AB</i> | <i>CD</i> | <i>CE</i> | <i>AC</i> | <i>DE</i> | <i>AD</i> | <i>AE</i> | <i>C</i> |
| <i>CD</i> | <i>BD</i> | <i>BC</i> | <i>BC</i> | <i>AC</i> | <i>AD</i> | <i>AC</i> | <i>AC</i> | <i>AB</i> | <i>AB</i> | <i>BC</i> | <i>AB</i> | <i>BD</i> | <i>BE</i> | <i>D</i> |
| <i>CE</i> | <i>BE</i> | <i>BE</i> | <i>BD</i> | <i>AD</i> | <i>AE</i> | <i>AE</i> | <i>AD</i> | <i>BD</i> | <i>AE</i> | <i>AD</i> | <i>CD</i> | <i>AC</i> | <i>CD</i> | <i>E</i> |
| <i>DE</i> | <i>DE</i> | <i>CE</i> | <i>CD</i> | <i>AE</i> | <i>DE</i> | <i>CE</i> | <i>CD</i> | <i>BE</i> | <i>BE</i> | <i>BD</i> | <i>CE</i> | <i>BC</i> | <i>DE</i> | <i>F</i> |

|    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 11 | 11 | 11 | 11 | 7  | 7  | 7  | 7  | 5  | 5  | 5  | 13 | 13 | 14 | 11 |
| 7  | 5  | 13 | 14 | 10 | 5  | 13 | 14 | 10 | 13 | 14 | 10 | 14 | 10 | 7  |
| 6  | 4  | 14 | 12 | 6  | 8  | 3  | 6  | 2  | 1  | 4  | 9  | 12 | 15 | 5  |
| 2  | 8  | 8  | 4  | 12 | 4  | 4  | 8  | 6  | 6  | 8  | 6  | 3  | 13 | 2  |
| 1  | 3  | 3  | 3  | 12 | 15 | 15 | 12 | 15 | 12 | 2  | 4  | 2  | 1  | 14 |
| 9  | 9  | 1  | 2  | 15 | 9  | 1  | 2  | 3  | 3  | 1  | 8  | 9  | 9  | 10 |

[Scan p. 507] *Table of the 120 pairs.* A large two-page tabular display in two parallel notations (a letter-pair notation and a numerical notation), the page being printed sideways. Each row gives an unordered pair of indices, the  $15 \times 16/2 = 120$  pairs being arranged in 8 columns of 15 rows in each notation.

[Figure: full sideways table, letter-pair notation (8 columns), reproduced from scan as printed. The pairs run, in the letter notation, *AB.BE, AC.CE, AD.DE, AE.BD, AC.CE, AD.DE, BC.CE, BD.BE, ...*; in the numerical notation, the corresponding entries are 11.7, 11.5, 11.13, 11.14, 11.10, 6.4, 6.3, .... See scan for the full layout.]

[Scan p. 508]

81. Table of the 60 Göpel tetrads.

|                  |                  |                  |                    |
|------------------|------------------|------------------|--------------------|
| <i>A.B.AE.BE</i> | <i>C.D.CE.DE</i> | <i>E.F.AB.CD</i> | <i>AC.BD.AD.BC</i> |
| <i>A.B.AD.BD</i> |                  | <i>D.F.AB.CE</i> | <i>AC.BE.AE.BC</i> |
| <i>A.B.AC.BC</i> | <i>CE.CD.DE</i>  | <i>D.E.CD.CE</i> | <i>AD.BE.AE.BD</i> |
| <i>A.C.AE.CE</i> | <i>CF.AB.DE</i>  | <i>E.F.AC.BD</i> | <i>AB.CD.AD.BC</i> |
| <i>A.C.AD.CD</i> |                  | <i>D.F.AC.BE</i> | <i>AB.CE.AE.BC</i> |
| <i>A.C.AB.BC</i> | <i>B.D.BE.DE</i> | <i>D.E.BD.BE</i> | <i>AD.CE.AE.CD</i> |
|                  | <i>B.E.BD.DE</i> |                  |                    |
| <i>A.D.AE.DE</i> | <i>B.F.AC.DE</i> | <i>E.F.AD.BC</i> | <i>AB.CD.AC.BD</i> |
| <i>A.D.AC.CD</i> |                  |                  | <i>AB.DE.AE.BD</i> |
| <i>A.D.AB.BD</i> | <i>B.C.BE.CE</i> | <i>C.F.AD.BE</i> | <i>AC.DE.AE.CD</i> |
|                  | <i>B.E.BC.CE</i> | <i>C.E.CD.DE</i> |                    |
| <i>A.E.AD.DE</i> | <i>B.F.AD.CE</i> |                  | <i>AB.CE.AC.BE</i> |
| <i>A.E.AC.CE</i> |                  | <i>D.F.AE.BC</i> | <i>AB.DE.AD.BE</i> |
| <i>A.E.AB.BE</i> | <i>B.C.BD.CD</i> | <i>C.F.AE.BD</i> | <i>AC.DE.AD.CE</i> |
|                  | <i>B.D.BC.CD</i> | <i>C.D.BC.BD</i> |                    |
| <i>A.F.BC.DE</i> | <i>B.F.AE.CD</i> |                  | <i>BD.CE.BE.CD</i> |
| <i>A.F.BD.CE</i> |                  | <i>D.E.AD.AE</i> | <i>BC.DE.BE.CD</i> |
| <i>A.F.BE.CD</i> | <i>B.C.AB.AC</i> |                  | <i>BC.DE.BD.CE</i> |
|                  |                  | <i>C.E.AC.AE</i> |                    |
|                  | <i>B.D.AB.AD</i> | <i>C.D.AC.AD</i> |                    |
|                  | <i>B.E.AB.AE</i> |                  |                    |

The same tetrads in the numerical notation:



|            |           |             |           |
|------------|-----------|-------------|-----------|
| 11 7 15 3  | 5 13 1 9  | 14 10 6 2   | 4 12 8    |
| 11 7 12 4  | 5 14 2 9  | 13 10 6 1   |           |
| 11 7 4 8   | 5 10 6 9  | 13 14 2 1   | 4 3 15 8  |
|            |           |             | 12 3 15   |
| 11 5 15 1  | 7 13 3 9  | 14 10 4 3   | 6 2 12 8  |
| 11 5 12 2  |           |             | 6 1 15 8  |
| 11 5 6 8   | 7 14 4 9  | 13 10 4 3   | 12 1 15 2 |
|            | 7 10 4 9  | 13 14 3 3   |           |
| 11 13 15 9 | 7 5 3 1   | 14 10 12 8  | 6 2 4     |
| 11 13 4 2  | 7 14 8 1  | 5 10 12 3   |           |
| 11 13 6 3  | 7 10 12 1 | 5 14 2 9    | 6 9 15    |
|            |           |             | 4 9 15 2  |
| 11 14 12 9 | 7 5 4 2   | 13 10 15 8  | 6 1 4 3   |
| 11 14 4 1  |           | 5 10 15     |           |
| 11 14 6 3  | 7 13 8 2  | 5 13 8      | 6 9 12 3  |
|            | 7 10 16 2 |             | 4 9 12 1  |
| 11 10 8 9  | 7 5 6 4   | 13 14 12 15 | 1 3 2     |
| 11 10      | 7 13 6 12 | 5 14 4 15   | 3 2       |
| 11 10 1    | 7 14 6 15 | 5 13 4 12   | 8 9       |
|            |           |             | 8 9 1     |
|            |           |             | 3 2       |

[Scan p. 509]

*The product-theorem, and its results.*

82. The product-theorem was

$$\Theta\left(\begin{smallmatrix}\alpha,\beta\\ \gamma,\delta\end{smallmatrix}\right)(u+u')\cdot\Theta\left(\begin{smallmatrix}\alpha',\beta'\\ \gamma',\delta'\end{smallmatrix}\right)(u-u')=\Sigma\Theta\left(\begin{smallmatrix}\frac{1}{2}(\alpha+\alpha')+p,\frac{1}{2}(\beta+\beta')+q\\ \gamma+\gamma',\delta+\delta'\end{smallmatrix}\right)(2u)\cdot\Theta\left(\begin{smallmatrix}\frac{1}{2}(\alpha-\alpha')+p,\frac{1}{2}(\beta-\beta')+q\\ \gamma-\gamma',\delta-\delta'\end{smallmatrix}\right)(2u'),$$

where only one argument is exhibited, viz.  $u+u'$ ,  $u-u'$ ,  $2u$ ,  $2u'$  are written in place of  $(u+u', v+v')$ ,  $(u-u', v-v')$ ,  $(2u, 2v)$ ,  $(2u', 2v')$  respectively. The expression on the right-hand side is always a sum of

four terms, corresponding to the values  $(0, 0)$ ,  $(1, 0)$ ,  $(0, 1)$ , and  $(1, 1)$  of  $(p, q)$ . For the development of the results it was found convenient to use the following auxiliary diagram.

*Upper half of characteristic.*

[Figure: large  $16 \times 16$  table indexed by pairs of upper-half-characteristic combinations, exhibiting for each pair  $(\frac{\alpha+\alpha'}{2}+p, \frac{\beta+\beta'}{2}+q)$  versus  $(\frac{\alpha-\alpha'}{2}+p, \frac{\beta-\beta'}{2}+q)$  the upper indices, written as fractions,  $0, 1, \frac{1}{2}, \frac{3}{2}$ , with the  $\pm$  signs of the four terms. Reproduced as printed in the scan; the cells contain entries of the form  $\frac{1}{2}1, 11, \frac{3}{2}1$  etc., with leading signs to be combined with the lower-half diagram on the facing page.]

[Scan p. 510]

*Lower half of characteristic.*

[Figure: companion  $16 \times 16$  table for the lower-half characteristics, structured exactly as the previous one; the entries are the values  $\frac{\gamma+\gamma'}{2}, \frac{\delta+\delta'}{2}$  versus  $\frac{\gamma-\gamma'}{2}, \frac{\delta-\delta'}{2}$ , taking the values  $0, 1, \frac{1}{2}, -\frac{1}{2}$ , with the corresponding  $\pm$  signs to be combined with the upper-half diagram.]

**83.** The upper characters of the  $\Theta$ 's have thus the values  $0, 1, \frac{1}{2}, \frac{3}{2}$ ; the lower characters are originally  $2, 1, 0$ , or  $-1$ , and these have when necessary to be, by the addition or subtraction of  $2$ , reduced to  $0$  or  $1$ ; the effect of this change is either to leave the  $\Theta$  unaltered, or to multiply it by  $-1$  or  $+i$ , as follows:

$$\begin{aligned} \Theta\left(\begin{smallmatrix} 0 \\ \gamma \end{smallmatrix}\right) - 2 &= \Theta\left(\begin{smallmatrix} 0 \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right) + 2 &= i \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) - 2 &= -i \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right), \\ \Theta\left(\begin{smallmatrix} 1 \\ \gamma \end{smallmatrix}\right) - 2 &= -\Theta\left(\begin{smallmatrix} 1 \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right) - 2 &= -i \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right), & \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) + 2 &= i \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right), \end{aligned}$$

where only the first column of characters is shown, but the same rule applies to the second column; and where we must of course combine the multipliers corresponding to the first and second columns respectively: for instance

$$\Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) + 2 \cdot \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right) + 2 = (-i \cdot -i) \Theta\left(\begin{smallmatrix} \frac{3}{2} \\ \gamma \end{smallmatrix}\right) \Theta\left(\begin{smallmatrix} \frac{1}{2} \\ \gamma \end{smallmatrix}\right).$$

[Scan p. 511]

Thus taking the tenth line of the upper half, and the fifth line of the lower half, we have

|    |    |                          |                          |                          |                          |                          |                          |
|----|----|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
|    |    | $\frac{1}{2}\frac{1}{2}$ | $\frac{1}{2}\frac{1}{2}$ | $\frac{1}{2}\frac{1}{2}$ | $\frac{1}{2}\frac{1}{2}$ | $\frac{1}{2}\frac{1}{2}$ | $\frac{1}{2}\frac{1}{2}$ |
| 10 | 01 | $\frac{1}{2}\frac{1}{2}$ | $\frac{1}{2}\frac{1}{2}$ |                          |                          |                          |                          |
| 00 | 10 | 1 0-1                    | 1 0-1                    | 1 0-1                    | 1 0-1                    | 1 0-1                    | 1 0-1                    |

giving the value of

$$\Theta\left(\begin{smallmatrix} \frac{1}{2} \\ 0 \end{smallmatrix}\right) (u + u') \cdot \Theta\left(\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right) (u - u');$$

viz. this is

$$\begin{aligned} &= \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (2u) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ -10 \end{smallmatrix}\right) (2u') + \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (2u) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (2u') \\ &+ \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ -10 \end{smallmatrix}\right) (\cdot) - i \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \\ &+ \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ -10 \end{smallmatrix}\right) (\cdot) + i \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{1}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \\ &+ \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ -10 \end{smallmatrix}\right) (\cdot) - i \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot) \Theta\left(\begin{smallmatrix} \frac{3}{4} \\ 10 \end{smallmatrix}\right) (\cdot), \end{aligned}$$

where the first column is the value given directly by the diagram: it is then reduced to that given by the second column.

**84.** But instead of the  $\Theta$ 's, we introduce single letters  $(X, Y, Z, W)$ ,  $(E, F, G, H)$ ,  $(I, J, K, L)$ ,  $(M, N, P, Q)$ , with the suffixes  $(0, 1, 2, 3)$ , in all 64 symbols, thus

$$\Theta \begin{pmatrix} 00 & 10 & 01 & 11 \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = X \ Y \ Z \ W,$$

|    | 00 | 10 | 01 | 11 |
|----|----|----|----|----|
| 00 | 0  |    |    |    |
| 10 |    | 1  |    |    |
| 01 |    |    | 2  |    |
| 11 |    |    |    | 3  |

that is,

$$\begin{aligned} \Theta_{00}^{00}(2u) &= X, & \Theta_{00}^{10}(2u) &= Y, \text{ \&c.}, \\ \Theta_{01}^{00}(2u) &= Z, \dots, & \Theta_{11}^{00}(2u) &= W. \end{aligned}$$

Similarly,

$$\Theta \begin{pmatrix} \frac{1}{2}0 & \frac{3}{2}0 & \frac{1}{2}1 & \frac{3}{2}1 \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = E \ F \ G \ H,$$

with the suffixes assigned by the same array as above; that is,

$$\Theta \begin{pmatrix} \frac{1}{2}0 \\ \frac{3}{2}0 \\ 00 \end{pmatrix} (2u) = E, \text{ \&c.}$$

[Scan p. 512]

$$\Theta \begin{pmatrix} 0\frac{1}{2} & 1\frac{1}{2} & 0\frac{3}{2} & 1\frac{3}{2} \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = I \ J \ K \ L,$$

with the suffixes  $0, 1, 2, 3$  assigned by the same diagonal array as before.

The functions of  $(2u')$  are denoted in like manner by accented letters:

$$\Theta_{00}^{00}(2u') = X', \text{ \&c.}$$

$$\Theta \begin{pmatrix} \frac{1}{2}\frac{1}{2} & \frac{3}{2}\frac{1}{2} & \frac{1}{2}\frac{3}{2} & \frac{3}{2}\frac{3}{2} \\ \cdot & \cdot & \cdot & \cdot \end{pmatrix} (2u) = M \ N \ P \ Q,$$

again with the suffixes  $0, 1, 2, 3$  assigned by the diagonal array.

**85.** To simplify the expression of the results, instead of in each case writing down the suffixes, I have indicated them by means of the column headed "Suff."

Thus

$$\left| \begin{array}{c} 8-0 \\ \cdot \end{array} \right| \quad \Theta_{01}^{00}(u+u') \cdot \Theta_{00}^{00}(u-u') = XX' + YY' + ZZ' + WW' \quad \left| \begin{array}{c} 2 \\ \cdot \end{array} \right|$$

means that the equation is to be read

$$= X_2X'_2 + Y_2Y'_2 + Z_2Z'_2 + W_2W'_2.$$

It is hardly necessary to mention that the 8-0 of the left-hand column shows the current numbers of the theta-functions; viz. the left-hand side of the equation is

$$\Theta_8(u+u') \cdot \Theta_0(u-u').$$

And by a preceding remark the single arguments  $u+u'$  and  $u-u'$  are written in place of  $(u+u', v+v')$  and  $(u-u', v-v')$  respectively.

[Scan p. 513]

The 256 equations now are

86. First set, 64 equations.

| Line  | upper                    | lower                    | Equation                  | Suff. |
|-------|--------------------------|--------------------------|---------------------------|-------|
| 0–0   | $\Theta_{00}^{00}u + u'$ | $\Theta_{00}^{00}u - u'$ | $= XX' + YY' + ZZ' + WW'$ | 0     |
| 4–0   | 00 / 10                  | 00 / 00                  | $= XX' + YY' + ZZ' + WW'$ | 1     |
| 8–0   | 00 / 01                  | 00 / 00                  | $= XX' + YY' + ZZ' + WW'$ | 2     |
| 12–0  | 00 / 11                  | 00 / 00                  | $= XX' + YY' + ZZ' + WW'$ | 3     |
| 0–4   | $\Theta_{00}^{00}u + u'$ | $\Theta_{10}^{00}u - u'$ | $= XX' - YY' + ZZ' - WW'$ | 1     |
| 4–4   | 00 / 10                  | 00 / 10                  | $= XX' - YY' + ZZ' - WW'$ | 0     |
| 8–4   | 00 / 01                  | 00 / 10                  | $= XX' - YY' + ZZ' - WW'$ | 3     |
| 12–4  | 00 / 11                  | 00 / 10                  | $= XX' - YY' + ZZ' - WW'$ | 2     |
| 0–8   | $\Theta_{00}^{00}u + u'$ | $\Theta_{01}^{00}u - u'$ | $= XX' + YY' - ZZ' - WW'$ | 2     |
| 4–8   | 00 / 10                  | 00 / 01                  | $= XX' + YY' - ZZ' - WW'$ | 3     |
| 8–8   | 00 / 01                  | 00 / 01                  | $= XX' + YY' - ZZ' - WW'$ | 0     |
| 12–8  | 00 / 11                  | 00 / 01                  | $= XX' + YY' - ZZ' - WW'$ | 1     |
| 0–12  | $\Theta_{00}^{00}u + u'$ | $\Theta_{11}^{00}u - u'$ | $= XX' - YY' - ZZ' + WW'$ | 3     |
| 4–12  | 00 / 10                  | 00 / 11                  | $= XX' - YY' - ZZ' + WW'$ | 2     |
| 8–12  | 00 / 01                  | 00 / 11                  | $= XX' - YY' - ZZ' + WW'$ | 1     |
| 12–12 | 00 / 11                  | 00 / 11                  | $= XX' - YY' - ZZ' + WW'$ | 0     |

[Scan p. 514] First set, 64 equations (continued).

|       |                          |                          |                            |   |
|-------|--------------------------|--------------------------|----------------------------|---|
| 1–1   | $\Theta_{00}^{10}u + u'$ | $\Theta_{00}^{10}u - u'$ | $= YX' + XY' + WZ' + ZW'$  | 0 |
| 5–1   | 10 / 10                  | 10 / 00                  | $= YX' + XY' + WZ' + ZW'$  | 1 |
| 9–1   | 10 / 01                  | 10 / 00                  | $= YX' + XY' + WZ' + ZW'$  | 2 |
| 13–1  | 10 / 11                  | 10 / 00                  | $= YX' + XY' + WZ' + ZW'$  | 3 |
| 1–5   | $\Theta_{00}^{10}u + u'$ | $\Theta_{10}^{10}u - u'$ | $= YX' - XY' + WZ' - ZW'$  | 1 |
| 5–5   | 10 / 10                  | 10 / 10                  | $= -YX' + XY' - WZ' + ZW'$ | 0 |
| 9–5   | 10 / 01                  | 10 / 10                  | $= YX' - XY' + WZ' - ZW'$  | 3 |
| 13–5  | 10 / 11                  | 10 / 10                  | $= -YX' + XY' - WZ' + ZW'$ | 2 |
| 1–9   | $\Theta_{00}^{10}u + u'$ | $\Theta_{01}^{10}u - u'$ | $= YX' + XY' - WZ' - ZW'$  | 2 |
| 5–9   | 10 / 10                  | 10 / 01                  | $= YX' + XY' - WZ' - ZW'$  | 3 |
| 9–9   | 10 / 01                  | 10 / 01                  | $= YX' + XY' - WZ' - ZW'$  | 0 |
| 13–9  | 10 / 11                  | 10 / 01                  | $= YX' + XY' - WZ' - ZW'$  | 1 |
| 1–13  | $\Theta_{00}^{10}u + u'$ | $\Theta_{11}^{10}u - u'$ | $= YX' - XY' - WZ' + ZW'$  | 3 |
| 5–13  | 10 / 10                  | 10 / 11                  | $= -YX' + XY' + WZ' - ZW'$ | 2 |
| 9–13  | 10 / 01                  | 10 / 11                  | $= YX' - XY' - WZ' + ZW'$  | 1 |
| 13–13 | 10 / 11                  | 10 / 11                  | $= -YX' + XY' + WZ' - ZW'$ | 0 |

[Scan p. 515] *First set, 64 equations (continued).*

|       |                          |                          |                            |   |
|-------|--------------------------|--------------------------|----------------------------|---|
| 2–2   | $\Theta_{00}^{01}u + u'$ | $\Theta_{00}^{01}u - u'$ | $= ZX' + WY' + XZ' + YW'$  | 0 |
| 6–2   | 01 / 10                  | 01 / 00                  | $= ZX' + WY' + XZ' + YW'$  | 1 |
| 10–2  | 01 / 01                  | 01 / 00                  | $= ZX' + WY' + XZ' + YW'$  | 2 |
| 14–2  | 01 / 11                  | 01 / 00                  | $= ZX' + WY' + XZ' + YW'$  | 3 |
| 2–6   | $\Theta_{00}^{01}u + u'$ | $\Theta_{10}^{01}u - u'$ | $= ZX' - WY' + XZ' - YW'$  | 1 |
| 6–6   | 01 / 10                  | 01 / 10                  | $= ZX' - WY' + XZ' - YW'$  | 0 |
| 10–6  | 01 / 01                  | 01 / 10                  | $= ZX' - WY' + XZ' - YW'$  | 3 |
| 14–6  | 01 / 11                  | 01 / 10                  | $= ZX' - WY' + XZ' - YW'$  | 2 |
| 2–10  | $\Theta_{00}^{01}u + u'$ | $\Theta_{01}^{01}u - u'$ | $= ZX' + WY' - XZ' - YW'$  | 2 |
| 6–10  | 01 / 10                  | 01 / 01                  | $= ZX' + WY' - XZ' - YW'$  | 3 |
| 10–10 | 01 / 01                  | 01 / 01                  | $= -ZX' - WY' + XZ' + YW'$ | 0 |
| 14–10 | 01 / 11                  | 01 / 01                  | $= -ZX' - WY' + XZ' + YW'$ | 1 |
| 2–14  | $\Theta_{00}^{01}u + u'$ | $\Theta_{11}^{01}u - u'$ | $= ZX' - WY' - XZ' + YW'$  | 3 |
| 6–14  | 01 / 10                  | 01 / 11                  | $= ZX' - WY' - XZ' + YW'$  | 2 |
| 10–14 | 01 / 01                  | 01 / 11                  | $= -ZX' + WY' + XZ' - YW'$ | 1 |
| 14–14 | 01 / 11                  | 01 / 11                  | $= -ZX' + WY' + XZ' - YW'$ | 0 |

[Scan p. 516] *First set, 64 equations (concluded).*

|       |                          |                          |                            |   |
|-------|--------------------------|--------------------------|----------------------------|---|
| 3–3   | $\Theta_{00}^{11}u + u'$ | $\Theta_{00}^{11}u - u'$ | $= WX' + ZY' + YZ' + XW'$  | 0 |
| 7–3   | 11 / 10                  | 11 / 00                  | $= WX' + ZY' + YZ' + XW'$  | 1 |
| 11–3  | 11 / 01                  | 11 / 00                  | $= WX' + ZY' + YZ' + XW'$  | 2 |
| 15–3  | 11 / 11                  | 11 / 00                  | $= WX' + ZY' + YZ' + XW'$  | 3 |
| 3–7   | $\Theta_{00}^{11}u + u'$ | $\Theta_{10}^{11}u - u'$ | $= WX' - ZY' + YZ' - XW'$  | 1 |
| 7–7   | 11 / 10                  | 11 / 10                  | $= -WX' + ZY' - YZ' + XW'$ | 0 |
| 11–7  | 11 / 01                  | 11 / 10                  | $= WX' - ZY' + YZ' - XW'$  | 3 |
| 15–7  | 11 / 11                  | 11 / 10                  | $= -WX' + ZY' - YZ' + XW'$ | 2 |
| 3–11  | $\Theta_{00}^{11}u + u'$ | $\Theta_{01}^{11}u - u'$ | $= WX' + ZY' - YZ' - XW'$  | 2 |
| 7–11  | 11 / 10                  | 11 / 01                  | $= WX' + ZY' - YZ' - XW'$  | 3 |
| 11–11 | 11 / 01                  | 11 / 01                  | $= -WX' - ZY' + YZ' + XW'$ | 0 |
| 15–11 | 11 / 11                  | 11 / 01                  | $= -WX' - ZY' + YZ' + XW'$ | 1 |
| 3–15  | $\Theta_{00}^{11}u + u'$ | $\Theta_{11}^{11}u - u'$ | $= WX' - ZY' - YZ' + XW'$  | 3 |
| 7–15  | 11 / 10                  | 11 / 11                  | $= -WX' + ZY' + YZ' - XW'$ | 2 |
| 11–15 | 11 / 01                  | 11 / 11                  | $= -WX' + ZY' + YZ' - XW'$ | 1 |
| 15–15 | 11 / 11                  | 11 / 11                  | $= WX' - ZY' - YZ' + XW'$  | 0 |

[Scan p. 517]

87. Second set, 64 equations.

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 1–0   | $\Theta_{00}^{10}u + u'$ | $\Theta_{00}^{00}u - u'$ | $= EE' + GG' + FF' + HH'$      | 0 |
| 5–0   | 10 / 10                  | 00 / 00                  | $= EE' + GG' + FF' + HH'$      | 1 |
| 9–0   | 10 / 01                  | 00 / 00                  | $= EE' + GG' + FF' + HH'$      | 2 |
| 13–0  | 10 / 11                  | 00 / 00                  | $= EE' + GG' + FF' + HH'$      | 3 |
| 1–4   | $\Theta_{00}^{10}u + u'$ | $\Theta_{10}^{00}u - u'$ | $= -iEE' + iGG' - iFF' + iHH'$ | 1 |
| 5–4   | 10 / 10                  | 00 / 10                  | $= iEE' - iGG' + iFF' - iHH'$  | 0 |
| 9–4   | 10 / 01                  | 00 / 10                  | $= -iEE' + iGG' - iFF' + iHH'$ | 3 |
| 13–4  | 10 / 11                  | 00 / 10                  | $= iEE' - iGG' + iFF' - iHH'$  | 2 |
| 1–8   | $\Theta_{00}^{10}u + u'$ | $\Theta_{01}^{00}u - u'$ | $= EE' + GG' - FF' - HH'$      | 2 |
| 5–8   | 10 / 10                  | 00 / 01                  | $= EE' + GG' - FF' - HH'$      | 3 |
| 9–8   | 10 / 01                  | 00 / 01                  | $= EE' + GG' - FF' - HH'$      | 0 |
| 13–8  | 10 / 11                  | 00 / 01                  | $= EE' + GG' - FF' - HH'$      | 1 |
| 1–12  | $\Theta_{00}^{10}u + u'$ | $\Theta_{11}^{00}u - u'$ | $= -iEE' + iGG' + iFF' - iHH'$ | 3 |
| 5–12  | 10 / 10                  | 00 / 11                  | $= iEE' - iGG' - iFF' + iHH'$  | 2 |
| 9–12  | 10 / 01                  | 00 / 11                  | $= -iEE' + iGG' + iFF' - iHH'$ | 1 |
| 13–12 | 10 / 11                  | 00 / 11                  | $= iEE' - iGG' - iFF' + iHH'$  | 0 |

[Scan p. 518] Second set, 64 equations (continued).

|       |                          |                          |                               |   |
|-------|--------------------------|--------------------------|-------------------------------|---|
| 0–1   | $\Theta_{00}^{00}u + u'$ | $\Theta_{00}^{10}u - u'$ | $= EG' + GE' + FH' + HF'$     | 0 |
| 4–1   | 00 / 10                  | 10 / 00                  | $= EG' + GE' + FH' + HF'$     | 1 |
| 8–1   | 00 / 01                  | 10 / 00                  | $= EG' + GE' + FH' + HF'$     | 2 |
| 12–1  | 00 / 11                  | 10 / 00                  | $= EG' + GE' + FH' + HF'$     | 3 |
| 0–5   | $\Theta_{00}^{00}u + u'$ | $\Theta_{10}^{10}u - u'$ | $= iEG' - iGE' + iFH' - iHF'$ | 1 |
| 4–5   | 00 / 10                  | 10 / 10                  | $= iEG' - iGE' + iFH' - iHF'$ | 0 |
| 8–5   | 00 / 01                  | 10 / 10                  | $= iEG' - iGE' + iFH' - iHF'$ | 3 |
| 12–5  | 00 / 11                  | 10 / 10                  | $= iEG' - iGE' + iFH' - iHF'$ | 2 |
| 0–9   | $\Theta_{00}^{00}u + u'$ | $\Theta_{01}^{10}u - u'$ | $= EG' + GE' - FH' - HF'$     | 2 |
| 4–9   | 00 / 10                  | 10 / 01                  | $= EG' + GE' - FH' - HF'$     | 3 |
| 8–9   | 00 / 01                  | 10 / 01                  | $= EG' + GE' - FH' - HF'$     | 0 |
| 12–9  | 00 / 11                  | 10 / 01                  | $= EG' + GE' - FH' - HF'$     | 1 |
| 0–13  | $\Theta_{00}^{00}u + u'$ | $\Theta_{11}^{10}u - u'$ | $= iEG' - iGE' - iFH' + iHF'$ | 3 |
| 4–13  | 00 / 10                  | 10 / 11                  | $= iEG' - iGE' - iFH' + iHF'$ | 2 |
| 8–13  | 00 / 01                  | 10 / 11                  | $= iEG' - iGE' - iFH' + iHF'$ | 1 |
| 12–13 | 00 / 11                  | 10 / 11                  | $= iEG' - iGE' - iFH' + iHF'$ | 0 |

[Scan p. 519] Second set, 64 equations (continued).

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 3–2   | $\Theta_{00}^{11}u + u'$ | $\Theta_{00}^{01}u - u'$ | $= FE' + HG' + EF' + GH'$      | 0 |
| 7–2   | 11 / 10                  | 01 / 00                  | $= FE' + HG' + EF' + GH'$      | 1 |
| 11–2  | 11 / 01                  | 01 / 00                  | $= FE' + HG' + EF' + GH'$      | 2 |
| 15–2  | 11 / 11                  | 01 / 00                  | $= FE' + HG' + EF' + GH'$      | 3 |
| 3–6   | $\Theta_{00}^{11}u + u'$ | $\Theta_{10}^{01}u - u'$ | $= -iFE' + iHG' - iEF' + iGH'$ | 1 |
| 7–6   | 11 / 10                  | 01 / 10                  | $= iFE' - iHG' + iEF' - iGH'$  | 0 |
| 11–6  | 11 / 01                  | 01 / 10                  | $= -iFE' + iHG' - iEF' + iGH'$ | 3 |
| 15–6  | 11 / 11                  | 01 / 10                  | $= iFE' - iHG' + iEF' - iGH'$  | 2 |
| 3–10  | $\Theta_{00}^{11}u + u'$ | $\Theta_{01}^{01}u - u'$ | $= FE' + HG' - EF' - GH'$      | 2 |
| 7–10  | 11 / 10                  | 01 / 01                  | $= FE' + HG' - EF' - GH'$      | 3 |
| 11–10 | 11 / 01                  | 01 / 01                  | $= -FE' - HG' + EF' + GH'$     | 0 |
| 15–10 | 11 / 11                  | 01 / 01                  | $= -FE' - HG' + EF' + GH'$     | 1 |
| 3–14  | $\Theta_{00}^{11}u + u'$ | $\Theta_{11}^{01}u - u'$ | $= -iFE' + iHG' + iEF' - iGH'$ | 3 |
| 7–14  | 11 / 10                  | 01 / 11                  | $= iFE' - iHG' - iEF' + iGH'$  | 2 |
| 11–14 | 11 / 01                  | 01 / 11                  | $= iFE' - iHG' - iEF' + iGH'$  | 1 |
| 15–14 | 11 / 11                  | 01 / 11                  | $= -iFE' + iHG' + iEF' - iGH'$ | 0 |

[Scan p. 520] Second set, 64 equations (concluded).

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 2–3   | $\Theta_{00}^{01}u + u'$ | $\Theta_{00}^{11}u - u'$ | $= FG' + HE' + EH' + GF'$      | 0 |
| 6–3   | 01 / 10                  | 11 / 00                  | $= FG' + HE' + EH' + GF'$      | 1 |
| 10–3  | 01 / 01                  | 11 / 00                  | $= FG' + HE' + EH' + GF'$      | 2 |
| 14–3  | 01 / 11                  | 11 / 00                  | $= FG' + HE' + EH' + GF'$      | 3 |
| 2–7   | $\Theta_{00}^{01}u + u'$ | $\Theta_{10}^{11}u - u'$ | $= iFG' - iHE' + iEH' - iGF'$  | 1 |
| 6–7   | 01 / 10                  | 11 / 10                  | $= iFG' - iHE' + iEH' - iGF'$  | 0 |
| 10–7  | 01 / 01                  | 11 / 10                  | $= iFG' - iHE' + iEH' - iGF'$  | 3 |
| 14–7  | 01 / 11                  | 11 / 10                  | $= iFG' - iHE' + iEH' - iGF'$  | 2 |
| 2–11  | $\Theta_{00}^{01}u + u'$ | $\Theta_{01}^{11}u - u'$ | $= FG' + HE' - EH' - GF'$      | 2 |
| 6–11  | 01 / 10                  | 11 / 01                  | $= FG' + HE' - EH' - GF'$      | 3 |
| 10–11 | 01 / 01                  | 11 / 01                  | $= -FG' - HE' + EH' + GF'$     | 0 |
| 14–11 | 01 / 11                  | 11 / 01                  | $= -FG' - HE' + EH' + GF'$     | 1 |
| 2–15  | $\Theta_{00}^{01}u + u'$ | $\Theta_{11}^{11}u - u'$ | $= iFG' - iHE' - iEH' + iGF'$  | 3 |
| 6–15  | 01 / 10                  | 11 / 11                  | $= iFG' - iHE' - iEH' + iGF'$  | 2 |
| 10–15 | 01 / 01                  | 11 / 11                  | $= -iFG' + iHE' + iEH' - iGF'$ | 1 |
| 14–15 | 01 / 11                  | 11 / 11                  | $= -iFG' + iHE' + iEH' - iGF'$ | 0 |

[Scan p. 521]

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 2–0   | $\Theta_{00}^{01}u + u'$ | $\Theta_{00}^{00}u - u'$ | $= II' + JJ' + KK' + LL'$      | 0 |
| 6–0   | 01 / 10                  | 00 / 00                  | $= II' + JJ' + KK' + LL'$      | 1 |
| 10–0  | 01 / 01                  | 00 / 00                  | $= II' + JJ' + KK' + LL'$      | 2 |
| 14–0  | 01 / 11                  | 00 / 00                  | $= II' + JJ' + KK' + LL'$      | 3 |
| 2–4   | $\Theta_{00}^{01}u + u'$ | $\Theta_{10}^{00}u - u'$ | $= II' - JJ' + KK' - LL'$      | 1 |
| 6–4   | 01 / 10                  | 00 / 10                  | $= II' - JJ' + KK' - LL'$      | 0 |
| 10–4  | 01 / 01                  | 00 / 10                  | $= II' - JJ' + KK' - LL'$      | 3 |
| 14–4  | 01 / 11                  | 00 / 10                  | $= II' - JJ' + KK' - LL'$      | 2 |
| 2–8   | $\Theta_{00}^{01}u + u'$ | $\Theta_{01}^{00}u - u'$ | $= -iII' - iJJ' + iKK' + iLL'$ | 2 |
| 6–8   | 01 / 10                  | 00 / 01                  | $= -iII' - iJJ' + iKK' + iLL'$ | 3 |
| 10–8  | 01 / 01                  | 00 / 01                  | $= iII' + iJJ' - iKK' - iLL'$  | 0 |
| 14–8  | 01 / 11                  | 00 / 01                  | $= iII' + iJJ' - iKK' - iLL'$  | 1 |
| 2–12  | $\Theta_{00}^{01}u + u'$ | $\Theta_{11}^{00}u - u'$ | $= -iII' + iJJ' + iKK' - iLL'$ | 3 |
| 6–12  | 01 / 10                  | 00 / 11                  | $= -iII' + iJJ' + iKK' - iLL'$ | 2 |
| 10–12 | 01 / 01                  | 00 / 11                  | $= iII' - iJJ' - iKK' + iLL'$  | 1 |
| 14–12 | 01 / 11                  | 00 / 11                  | $= iII' - iJJ' - iKK' + iLL'$  | 0 |

[Scan p. 522] Third set, 64 equations (continued).

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 3–1   | $\Theta_{00}^{11}u + u'$ | $\Theta_{00}^{10}u - u'$ | $= JI' + IJ' + LK' + KL'$      | 0 |
| 7–1   | 11 / 10                  | 10 / 00                  | $= JI' + IJ' + LK' + KL'$      | 1 |
| 11–1  | 11 / 01                  | 10 / 00                  | $= JI' + IJ' + LK' + KL'$      | 2 |
| 15–1  | 11 / 11                  | 10 / 00                  | $= JI' + IJ' + LK' + KL'$      | 3 |
| 3–5   | $\Theta_{00}^{11}u + u'$ | $\Theta_{10}^{10}u - u'$ | $= JI' - IJ' + LK' - KL'$      | 1 |
| 7–5   | 11 / 10                  | 10 / 10                  | $= -JI' + IJ' - LK' + KL'$     | 0 |
| 11–5  | 11 / 01                  | 10 / 10                  | $= JI' - IJ' + LK' - KL'$      | 3 |
| 15–5  | 11 / 11                  | 10 / 10                  | $= -JI' + IJ' - LK' + KL'$     | 2 |
| 3–9   | $\Theta_{00}^{11}u + u'$ | $\Theta_{01}^{10}u - u'$ | $= -iJI' - iIJ' + iLK' + iKL'$ | 2 |
| 7–9   | 11 / 10                  | 10 / 01                  | $= -iJI' - iIJ' + iLK' + iKL'$ | 3 |
| 11–9  | 11 / 01                  | 10 / 01                  | $= iJI' + iIJ' - iLK' - iKL'$  | 0 |
| 15–9  | 11 / 11                  | 10 / 01                  | $= iJI' + iIJ' - iLK' - iKL'$  | 1 |
| 3–13  | $\Theta_{00}^{11}u + u'$ | $\Theta_{11}^{10}u - u'$ | $= -iJI' + iIJ' + iLK' - iKL'$ | 3 |
| 7–13  | 11 / 10                  | 10 / 11                  | $= iJI' - iIJ' - iLK' + iKL'$  | 2 |
| 11–13 | 11 / 01                  | 10 / 11                  | $= iJI' - iIJ' - iLK' + iKL'$  | 1 |
| 15–13 | 11 / 11                  | 10 / 11                  | $= -iJI' + iIJ' + iLK' - iKL'$ | 0 |

[Scan p. 523] Third set, 64 equations (continued).



|       |                          |                          |                               |   |
|-------|--------------------------|--------------------------|-------------------------------|---|
| 0–2   | $\Theta_{00}^{00}u + u'$ | $\Theta_{00}^{01}u - u'$ | $= IK' + JL' + KI' + LJ'$     | 0 |
| 4–2   | 00 / 10                  | 01 / 00                  | $= IK' + JL' + KI' + LJ'$     | 1 |
| 8–2   | 00 / 01                  | 01 / 00                  | $= IK' + JL' + KI' + LJ'$     | 2 |
| 12–2  | 00 / 11                  | 01 / 00                  | $= IK' + JL' + KI' + LJ'$     | 3 |
| 0–6   | $\Theta_{00}^{00}u + u'$ | $\Theta_{10}^{01}u - u'$ | $= IK' - JL' + KI' - LJ'$     | 1 |
| 4–6   | 00 / 10                  | 01 / 10                  | $= IK' - JL' + KI' - LJ'$     | 0 |
| 8–6   | 00 / 01                  | 01 / 10                  | $= IK' - JL' + KI' - LJ'$     | 3 |
| 12–6  | 00 / 11                  | 01 / 10                  | $= IK' - JL' + KI' - LJ'$     | 2 |
| 0–10  | $\Theta_{00}^{00}u + u'$ | $\Theta_{01}^{01}u - u'$ | $= iIK' + iJL' - iKI' - iLJ'$ | 2 |
| 4–10  | 00 / 10                  | 01 / 01                  | $= iIK' + iJL' - iKI' - iLJ'$ | 3 |
| 8–10  | 00 / 01                  | 01 / 01                  | $= iIK' + iJL' - iKI' - iLJ'$ | 0 |
| 12–10 | 00 / 11                  | 01 / 01                  | $= iIK' + iJL' - iKI' - iLJ'$ | 1 |
| 0–14  | $\Theta_{00}^{00}u + u'$ | $\Theta_{11}^{01}u - u'$ | $= iIK' - iJL' - iKI' + iLJ'$ | 3 |
| 4–14  | 00 / 10                  | 01 / 11                  | $= iIK' - iJL' - iKI' + iLJ'$ | 2 |
| 8–14  | 00 / 01                  | 01 / 11                  | $= iIK' - iJL' - iKI' + iLJ'$ | 1 |
| 12–14 | 00 / 11                  | 01 / 11                  | $= iIK' - iJL' - iKI' + iLJ'$ | 0 |

[Scan p. 524] *Third set, 64 equations (concluded).*

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 1–3   | $\Theta_{00}^{10}u + u'$ | $\Theta_{00}^{11}u - u'$ | $= JK' + IL' + LI' + KJ'$      | 0 |
| 5–3   | 10 / 10                  | 11 / 00                  | $= JK' + IL' + LI' + KJ'$      | 1 |
| 9–3   | 10 / 01                  | 11 / 00                  | $= JK' + IL' + LI' + KJ'$      | 2 |
| 13–3  | 10 / 11                  | 11 / 00                  | $= JK' + IL' + LI' + KJ'$      | 3 |
| 1–7   | $\Theta_{00}^{10}u + u'$ | $\Theta_{10}^{11}u - u'$ | $= JK' - IL' + LI' - KJ'$      | 1 |
| 5–7   | 10 / 10                  | 11 / 10                  | $= -JK' + IL' - LI' + KJ'$     | 0 |
| 9–7   | 10 / 01                  | 11 / 10                  | $= JK' - IL' + LI' - KJ'$      | 3 |
| 13–7  | 10 / 11                  | 11 / 10                  | $= -JK' + IL' - LI' + KJ'$     | 2 |
| 1–11  | $\Theta_{00}^{10}u + u'$ | $\Theta_{01}^{11}u - u'$ | $= iJK' + iIL' - iLI' - iKJ'$  | 2 |
| 5–11  | 10 / 10                  | 11 / 01                  | $= iJK' + iIL' - iLI' - iKJ'$  | 3 |
| 9–11  | 10 / 01                  | 11 / 01                  | $= iJK' + iIL' - iLI' - iKJ'$  | 0 |
| 13–11 | 10 / 11                  | 11 / 01                  | $= iJK' + iIL' - iLI' - iKJ'$  | 1 |
| 1–15  | $\Theta_{00}^{10}u + u'$ | $\Theta_{11}^{11}u - u'$ | $= iJK' - iIL' - iLI' + iKJ'$  | 3 |
| 5–15  | 10 / 10                  | 11 / 11                  | $= -iJK' + iIL' + iLI' - iKJ'$ | 2 |
| 9–15  | 10 / 01                  | 11 / 11                  | $= iJK' - iIL' - iLI' + iKJ'$  | 1 |
| 13–15 | 10 / 11                  | 11 / 11                  | $= -iJK' + iIL' + iLI' - iKJ'$ | 0 |

[Scan p. 525]

89. Fourth set, 64 equations.

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 3–0   | $\Theta_{00}^{11}u + u'$ | $\Theta_{00}^{00}u - u'$ | $= MM' + NN' + PP' + QQ'$      | 0 |
| 7–0   | 11 / 10                  | 00 / 00                  | $= MM' + NN' + PP' + QQ'$      | 1 |
| 11–0  | 11 / 01                  | 00 / 00                  | $= MM' + NN' + PP' + QQ'$      | 2 |
| 15–0  | 11 / 11                  | 00 / 00                  | $= MM' + NN' + PP' + QQ'$      | 3 |
| 3–4   | $\Theta_{00}^{11}u + u'$ | $\Theta_{10}^{00}u - u'$ | $= -iMM' + iNN' - iPP' + iQQ'$ | 1 |
| 7–4   | 11 / 10                  | 00 / 10                  | $= +iMM' - iNN' + iPP' - iQQ'$ | 0 |
| 11–4  | 11 / 01                  | 00 / 10                  | $= -iMM' + iNN' - iPP' + iQQ'$ | 3 |
| 15–4  | 11 / 11                  | 00 / 10                  | $= +iMM' - iNN' + iPP' - iQQ'$ | 2 |
| 3–8   | $\Theta_{00}^{11}u + u'$ | $\Theta_{01}^{00}u - u'$ | $= -iMM' - iNN' + iPP' + iQQ'$ | 2 |
| 7–8   | 11 / 10                  | 00 / 01                  | $= -iMM' - iNN' + iPP' + iQQ'$ | 3 |
| 11–8  | 11 / 01                  | 00 / 01                  | $= iMM' + iNN' - iPP' - iQQ'$  | 0 |
| 15–8  | 11 / 11                  | 00 / 01                  | $= iMM' + iNN' - iPP' - iQQ'$  | 1 |
| 3–12  | $\Theta_{00}^{11}u + u'$ | $\Theta_{11}^{00}u - u'$ | $= -MM' + NN' + PP' - QQ'$     | 3 |
| 7–12  | 11 / 10                  | 00 / 11                  | $= +MM' - NN' - PP' + QQ'$     | 2 |
| 11–12 | 11 / 01                  | 00 / 11                  | $= +MM' - NN' - PP' + QQ'$     | 1 |
| 15–12 | 11 / 11                  | 00 / 11                  | $= -MM' + NN' + PP' - QQ'$     | 0 |

[Scan p. 526] Fourth set, 64 equations (continued).

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 2–1   | $\Theta_{00}^{01}u + u'$ | $\Theta_{00}^{10}u - u'$ | $= MN' + NM' + PQ' + QP'$      | 0 |
| 6–1   | 01 / 10                  | 10 / 00                  | $= MN' + NM' + PQ' + QP'$      | 1 |
| 10–1  | 01 / 01                  | 10 / 00                  | $= MN' + NM' + PQ' + QP'$      | 2 |
| 14–1  | 01 / 11                  | 10 / 00                  | $= MN' + NM' + PQ' + QP'$      | 3 |
| 2–5   | $\Theta_{00}^{01}u + u'$ | $\Theta_{10}^{10}u - u'$ | $= iMN' - iNM' + iPQ' - iQP'$  | 1 |
| 6–5   | 01 / 10                  | 10 / 10                  | $= iMN' - iNM' + iPQ' - iQP'$  | 0 |
| 10–5  | 01 / 01                  | 10 / 10                  | $= iMN' - iNM' + iPQ' - iQP'$  | 3 |
| 14–5  | 01 / 11                  | 10 / 10                  | $= iMN' - iNM' + iPQ' - iQP'$  | 2 |
| 2–9   | $\Theta_{00}^{01}u + u'$ | $\Theta_{01}^{10}u - u'$ | $= -iMN' - iNM' + iPQ' + iQP'$ | 2 |
| 6–9   | 01 / 10                  | 10 / 01                  | $= -iMN' - iNM' + iPQ' + iQP'$ | 3 |
| 10–9  | 01 / 01                  | 10 / 01                  | $= iMN' + iNM' - iPQ' - iQP'$  | 0 |
| 14–9  | 01 / 11                  | 10 / 01                  | $= iMN' + iNM' - iPQ' - iQP'$  | 1 |
| 2–13  | $\Theta_{00}^{01}u + u'$ | $\Theta_{11}^{10}u - u'$ | $= MN' - NM' - PQ' + QP'$      | 3 |
| 6–13  | 01 / 10                  | 10 / 11                  | $= MN' - NM' - PQ' + QP'$      | 2 |
| 10–13 | 01 / 01                  | 10 / 11                  | $= -MN' + NM' + PQ' - QP'$     | 1 |
| 14–13 | 01 / 11                  | 10 / 11                  | $= -MN' + NM' + PQ' - QP'$     | 0 |

[Scan p. 527] Fourth set, 64 equations (continued).

|       |                          |                          |                                |   |
|-------|--------------------------|--------------------------|--------------------------------|---|
| 1–2   | $\Theta_{00}^{10}u + u'$ | $\Theta_{00}^{01}u - u'$ | $= MP' + NQ' + PM' + QN'$      | 0 |
| 5–2   | 10 / 10                  | 01 / 00                  | $= MP' + NQ' + PM' + QN'$      | 1 |
| 9–2   | 10 / 01                  | 01 / 00                  | $= MP' + NQ' + PM' + QN'$      | 2 |
| 13–2  | 10 / 11                  | 01 / 00                  | $= MP' + NQ' + PM' + QN'$      | 3 |
| 1–6   | $\Theta_{00}^{10}u + u'$ | $\Theta_{10}^{01}u - u'$ | $= -iMP' + iNQ' - iPM' + iQN'$ | 1 |
| 5–6   | 10 / 10                  | 01 / 10                  | $= iMP' - iNQ' + iPM' - iQN'$  | 0 |
| 9–6   | 10 / 01                  | 01 / 10                  | $= -iMP' + iNQ' - iPM' + iQN'$ | 3 |
| 13–6  | 10 / 11                  | 01 / 10                  | $= iMP' - iNQ' + iPM' - iQN'$  | 2 |
| 1–10  | $\Theta_{00}^{10}u + u'$ | $\Theta_{01}^{01}u - u'$ | $= iMP' + iNQ' - iPM' - iQN'$  | 2 |
| 5–10  | 10 / 10                  | 01 / 01                  | $= iMP' + iNQ' - iPM' - iQN'$  | 3 |
| 9–10  | 10 / 01                  | 01 / 01                  | $= iMP' + iNQ' - iPM' - iQN'$  | 0 |
| 13–10 | 10 / 11                  | 01 / 01                  | $= iMP' + iNQ' - iPM' - iQN'$  | 1 |
| 1–14  | $\Theta_{00}^{10}u + u'$ | $\Theta_{11}^{01}u - u'$ | $= MP' - NQ' - PM' + QN'$      | 3 |
| 5–14  | 10 / 10                  | 01 / 11                  | $= -MP' + NQ' + PM' - QN'$     | 2 |
| 9–14  | 10 / 01                  | 01 / 11                  | $= MP' - NQ' - PM' + QN'$      | 1 |
| 13–14 | 10 / 11                  | 01 / 11                  | $= -MP' + NQ' + PM' - QN'$     | 0 |

[Scan p. 528] *Fourth set, 64 equations (concluded).*

|       |                          |                          |                               |   |
|-------|--------------------------|--------------------------|-------------------------------|---|
| 0–3   | $\Theta_{00}^{00}u + u'$ | $\Theta_{00}^{11}u - u'$ | $= MQ' + NP' + PN' + QM'$     | 0 |
| 4–3   | 00 / 10                  | 11 / 00                  | $= MQ' + NP' + PN' + QM'$     | 1 |
| 8–3   | 00 / 01                  | 11 / 00                  | $= MQ' + NP' + PN' + QM'$     | 2 |
| 12–3  | 00 / 11                  | 11 / 00                  | $= MQ' + NP' + PN' + QM'$     | 3 |
| 0–7   | $\Theta_{00}^{00}u + u'$ | $\Theta_{10}^{11}u - u'$ | $= iMQ' - iNP' + iPN' - iQM'$ | 1 |
| 4–7   | 00 / 10                  | 11 / 10                  | $= iMQ' - iNP' + iPN' - iQM'$ | 0 |
| 8–7   | 00 / 01                  | 11 / 10                  | $= iMQ' - iNP' + iPN' - iQM'$ | 3 |
| 12–7  | 00 / 11                  | 11 / 10                  | $= iMQ' - iNP' + iPN' - iQM'$ | 2 |
| 0–11  | $\Theta_{00}^{00}u + u'$ | $\Theta_{01}^{11}u - u'$ | $= iMQ' + iNP' - iPN' - iQM'$ | 2 |
| 4–11  | 00 / 10                  | 11 / 01                  | $= iMQ' + iNP' - iPN' - iQM'$ | 3 |
| 8–11  | 00 / 01                  | 11 / 01                  | $= iMQ' + iNP' - iPN' - iQM'$ | 0 |
| 12–11 | 00 / 11                  | 11 / 01                  | $= iMQ' + iNP' - iPN' - iQM'$ | 1 |
| 0–15  | $\Theta_{00}^{00}u + u'$ | $\Theta_{11}^{11}u - u'$ | $= -MQ' + NP' + PN' - QM'$    | 3 |
| 4–15  | 00 / 10                  | 11 / 11                  | $= -MQ' + NP' + PN' - QM'$    | 2 |
| 8–15  | 00 / 01                  | 11 / 11                  | $= -MQ' + NP' + PN' - QM'$    | 1 |
| 12–15 | 00 / 11                  | 11 / 11                  | $= -MQ' + NP' + PN' - QM'$    | 0 |

[Scan p. 529]

**90.** I re-arrange these in sets of 16 equations, the equations of the first or square-set of 16 being taken as they stand, but those of the other sets being combined in pairs by addition and subtraction as will be

seen. And I now drop altogether the characteristics, retaining only the current numbers: thus, in the set of equations next written down, the first equation is

$$\Theta_0(u + u') \Theta_0(u - u') = XX' + YY' + ZZ' + WW';$$

in the second set, the first equation is

$$\tfrac{1}{2}[\Theta_0(u + u') \Theta_4(u - u') + \Theta_4(u + u') \Theta_0(u - u')] = X_1X'_1 + Z_1Z'_1,$$

and so in other cases.

First or square-set of 16.

| $u + u'$ | $u - u'$ | Equation                   | (Suffixes 0) |
|----------|----------|----------------------------|--------------|
| 0        | 0        | $= XX' + YY' + ZZ' + WW'$  |              |
| 4        | 4        | $= XX' - YY' + ZZ' - WW'$  |              |
| 8        | 8        | $= XX' + YY' - ZZ' - WW'$  |              |
| 12       | 12       | $= XX' - YY' - ZZ' + WW'$  |              |
| 1        | 1        | $= YX' + XY' + WZ' + ZW'$  |              |
| 5        | 5        | $= -YX' + XY' - WZ' + ZW'$ |              |
| 9        | 9        | $= YX' + XY' - WZ' - ZW'$  |              |
| 13       | 13       | $= -YX' + XY' + WZ' - ZW'$ |              |
| 2        | 2        | $= ZX' + WY' + XZ' + YW'$  |              |
| 6        | 6        | $= ZX' - WY' + XZ' - YW'$  |              |
| 10       | 10       | $= ZX' - WY' - XZ' + YW'$  |              |
| 14       | 14       | $= -ZX' + WY' + XZ' - YW'$ |              |
| 3        | 3        | $= WX' + ZY' + YZ' + XW'$  |              |
| 7        | 7        | $= -WX' + ZY' - YZ' + XW'$ |              |
| 11       | 11       | $= -WX' - ZY' + YZ' + XW'$ |              |
| 15       | 15       | $= WX' - ZY' - YZ' + XW'$  |              |

91. Second set of 16.

| $\tfrac{1}{2}\{u + u' \quad u - u' \quad u + u' \quad u - u'\}$ | Equation      | (Suffixes 1.) |
|-----------------------------------------------------------------|---------------|---------------|
| 4      0      0      4                                          | $= XX' + ZZ'$ |               |
| 12     8      8     12                                          | $= XX' - ZZ'$ |               |
| 5      1      1      5                                          | $= YX' + WZ'$ |               |
| 13     9      9     13                                          | $= YX' - WZ'$ |               |
| 6      2      2      6                                          | $= ZX' + XZ'$ |               |
| 14     10     10     14                                         | $= ZX' + XZ'$ |               |
| 7      3      3      7                                          | $= WX' + YZ'$ |               |
| 15     11     11     15                                         | $= WX' + YZ'$ |               |

| $\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$ | Equation       | (Suffixes 1.) |
|--------------------------------------------------------|----------------|---------------|
| 4            0            0            4               | $= YY' + WW'$  |               |
| 12           8            8            12              | $= -YY' + WW'$ |               |
| 5            1            1            5               | $= XY' + ZW'$  |               |
| 13           9            9            13              | $= -XY' + ZW'$ |               |
| 6            2            2            6               | $= WY' + YW'$  |               |
| 14           10           10           14              | $= WY' - YW'$  |               |
| 7            3            3            7               | $= ZY' + XW'$  |               |
| 15           11           11           15              | $= -ZY' + XW'$ |               |

[Scan p. 530]

92. Third set of 16.

| $\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$ | Equation       | (Suffixes 2.) |
|--------------------------------------------------------|----------------|---------------|
| 8            0            0            8               | $= XX' + YY'$  |               |
| 12           4            4            12              | $= XX' - YY'$  |               |
| 9            1            1            9               | $= YX' + XY'$  |               |
| 13           5            5            13              | $= -YX' + XY'$ |               |
| 10           2            2            10              | $= ZX' + WY'$  |               |
| 14           6            6            14              | $= ZX' - WY'$  |               |
| 11           3            3            11              | $= WX' + ZY'$  |               |
| 15           7            7            15              | $= -WX' + ZY'$ |               |

| $\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$ | Equation       | (Suffixes 2.) |
|--------------------------------------------------------|----------------|---------------|
| 8            0            0            8               | $= ZZ' + WW'$  |               |
| 12           4            4            12              | $= ZZ' - WW'$  |               |
| 9            1            1            9               | $= WZ' + ZW'$  |               |
| 13           5            5            13              | $= -WZ' + ZW'$ |               |
| 10           2            2            10              | $= XZ' + YW'$  |               |
| 14           6            6            14              | $= XZ' - YW'$  |               |
| 11           3            3            11              | $= YZ' + XW'$  |               |
| 15           7            7            15              | $= -YZ' + XW'$ |               |

93. Fourth set of 16.

| $\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$ | Equation      | (Suffixes 3.) |
|--------------------------------------------------------|---------------|---------------|
| 12                                                     | $= XX' + WW'$ |               |
| 8                                                      | $= XX' - WW'$ |               |
| 13                                                     | $= YX' + ZW'$ |               |
| 9                                                      | $= YX' - ZW'$ |               |
| 14                                                     | $= ZX' + YW'$ |               |
| 10                                                     | $= ZX' - YW'$ |               |
| 15                                                     | $= WX' + XW'$ |               |
| 11                                                     | $= WX' - XW'$ |               |

| $\frac{1}{2}\{u+u' \quad u-u' \quad u+u' \quad u-u'\}$ | Equation       | (Suffixes 3.) |
|--------------------------------------------------------|----------------|---------------|
| 12                                                     | $= YY' + ZZ'$  |               |
| 8                                                      | $= -YY' + ZZ'$ |               |
| 13                                                     | $= XY' + WZ'$  |               |
| 9                                                      | $= -XY' + WZ'$ |               |
| 14                                                     | $= WY' + XZ'$  |               |
| 10                                                     | $= -WY' + XZ'$ |               |
| 15                                                     | $= ZY' + YZ'$  |               |
| 11                                                     | $= -ZY' + YZ'$ |               |

94. Fifth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 0.)$$

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 5 & 4 & 4 & 5 \\ 9 & 8 & 8 & 9 \\ 13 & 12 & 12 & 13 \\ 3 & 2 & 2 & 3 \\ 7 & 6 & 6 & 7 \\ 11 & 10 & 10 & 11 \\ 15 & 14 & 14 & 15 \end{array} = \begin{array}{l} E + G \cdot E' + G' + \\ i.E - G \quad " + \\ E + G \quad " - \\ i.E - G \quad " - \\ F + H \quad " + \\ i.F - H \quad " + \\ -F + H \quad " + \\ -i.F - H \quad " + \end{array} \begin{array}{l} F + H \cdot F' + H' \\ i.F - H \quad " \\ .F + H \quad " \\ i.F - H \quad " \\ E + G \quad " \\ i.E - G \quad " \\ E + G \quad " \\ i.E - G \quad " \end{array}$$

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 0.)$$

$$\begin{array}{cccc} 1 & 0 & 0 & 1 \\ 5 & 4 & 4 & 5 \\ 9 & 8 & 8 & 9 \\ 13 & 12 & 12 & 13 \\ 3 & 2 & 2 & 3 \\ 7 & 6 & 6 & 7 \\ 11 & 10 & 10 & 11 \\ 15 & 14 & 14 & 15 \end{array} = \begin{array}{l} E - G \cdot E' - G' + \\ i.E + G \quad " + \\ E - G \quad " - \\ i.E + G \quad " - \\ F - H \quad " + \\ i.F + H \quad " + \\ -F - H \quad " + \\ -i.F + H \quad " + \end{array} \begin{array}{l} F - H \cdot F' - H' \\ i.F + H \quad " \\ .F - H \quad " \\ i.F + H \quad " \\ E - G \quad " \\ i.E + G \quad " \\ E - G \quad " \\ i.E + G \quad " \end{array}$$

95. Sixth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 1.)$$

$$\begin{array}{cccc} 5 & 0 & 0 & 5 \\ 1 & 4 & 4 & 1 \\ 13 & 8 & 8 & 13 \\ 9 & 12 & 12 & 9 \\ 7 & 2 & 2 & 7 \\ 3 & 6 & 6 & 3 \\ 15 & 10 & 10 & 15 \\ 11 & 14 & 14 & 11 \end{array} = \begin{array}{l} E - iG \cdot E' + iG' + \\ -i.E + iG \quad " - \\ E - iG \quad " - \\ -i.E + iG \quad " + \\ F - iH \quad " + \\ -i.F + iH \quad " - \\ -F - iH \quad " + \\ i.F + iH \quad " - \end{array} \begin{array}{l} F - iH \cdot F' + iH' \\ i.F + iH \quad " \\ .F - iH \quad " \\ i.F + iH \quad " \\ E - iG \quad " \\ i.E + iG \quad " \\ E - iG \quad " \\ i.E + iG \quad " \end{array}$$

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 1.)$$

$$\begin{array}{cccc} 5 & 0 & 0 & 5 \\ 1 & 4 & 4 & 1 \\ 13 & 8 & 8 & 13 \\ 9 & 12 & 12 & 9 \\ 7 & 2 & 2 & 7 \\ 3 & 6 & 6 & 3 \\ 15 & 10 & 10 & 15 \\ 11 & 14 & 14 & 11 \end{array} = \begin{array}{l} E + iG \cdot E' - iG' + \\ -i.E - iG \quad " - \\ E + iG \quad " - \\ -i.E - iG \quad " + \\ F + iH \quad " + \\ -i.F - iH \quad " - \\ -F + iH \quad " + \\ i.F - iH \quad " - \end{array} \begin{array}{l} F + iH \cdot F' - iH' \\ i.F - iH \quad " \\ .F + iH \quad " \\ i.F - iH \quad " \\ E + iG \quad " \\ i.E - iG \quad " \\ E + iG \quad " \\ i.E - iG \quad " \end{array}$$

96. Seventh set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 2.})$$

$$\begin{array}{cccc} 9 & 0 & 0 & 9 \\ 13 & 4 & 4 & 13 \\ 1 & 8 & 8 & 1 \\ 5 & 12 & 12 & 5 \\ 11 & 2 & 2 & 11 \\ 15 & 6 & 6 & 15 \\ 3 & 10 & 10 & 3 \\ 7 & 14 & 14 & 7 \end{array} = \begin{array}{l} E + G \cdot E' + G' + \\ i.E - G \quad " + \\ E + G \quad " - \\ i.E - G \quad " - \\ F + H \quad " + \\ i.F - H \quad " + \\ F + H \quad " - \\ i.F - H \quad " - \end{array} \begin{array}{l} F - H \cdot F' - H' \\ i.F + H \quad " \\ .F - H \quad " \\ i.F + H \quad " \\ E - G \quad " \\ i.E + G \quad " \\ .E - G \quad " \\ i.E + G \quad " \end{array}$$

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 2.})$$

$$\begin{array}{cccc} 9 & 0 & 0 & 9 \\ 13 & 4 & 4 & 13 \\ 1 & 8 & 8 & 1 \\ 5 & 12 & 12 & 5 \\ 11 & 2 & 2 & 11 \\ 15 & 6 & 6 & 15 \\ 3 & 10 & 10 & 3 \\ 7 & 14 & 14 & 7 \end{array} = \begin{array}{l} E - G \cdot E' - G' + \\ i.E + G \quad " + \\ E - G \quad " - \\ i.E + G \quad " - \\ F - H \quad " + \\ i.F + H \quad " + \\ F - H \quad " - \\ i.F + H \quad " - \end{array} \begin{array}{l} F + H \cdot F' + H' \\ i.F - H \quad " \\ .F + H \quad " \\ i.F - H \quad " \\ E + G \quad " \\ i.E - G \quad " \\ .E + G \quad " \\ i.E - G \quad " \end{array}$$

97. Eighth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 3.})$$

$$\begin{array}{cccc} 13 & 0 & 0 & 13 \\ 9 & 4 & 4 & 9 \\ 5 & 8 & 8 & 5 \\ 1 & 12 & 12 & 1 \\ 15 & 2 & 2 & 15 \\ 11 & 6 & 6 & 11 \\ 7 & 10 & 10 & 7 \\ 3 & 14 & 14 & 3 \end{array} = \begin{array}{l} E - iG \cdot E' + iG' + \\ -i.E + iG \quad " - \\ E - iG \quad " - \\ -i.E + iG \quad " + \\ F - iH \quad " + \\ -i.F + iH \quad " - \\ F - iH \quad " - \\ -i.F + iH \quad " + \end{array} \begin{array}{l} F + iH \cdot F' - iH' \\ i.F - iH \quad " \\ .F + iH \quad " \\ i.F - iH \quad " \\ E + iG \quad " \\ i.E - iG \quad " \\ .E + iG \quad " \\ i.E - iG \quad " \end{array}$$

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 3.})$$

$$\begin{array}{cccc} 13 & 0 & 0 & 13 \\ 9 & 4 & 4 & 9 \\ 5 & 8 & 8 & 5 \\ 1 & 12 & 12 & 1 \\ 15 & 2 & 2 & 15 \\ 11 & 6 & 6 & 11 \\ 7 & 10 & 10 & 7 \\ 3 & 14 & 14 & 3 \end{array} = \begin{array}{l} E + iG \cdot E' - iG' + \\ -i.E - iG \quad " - \\ E + iG \quad " - \\ -i.E - iG \quad " + \\ F + iH \quad " + \\ -i.F - iH \quad " - \\ F + iH \quad " - \\ -i.F - iH \quad " + \end{array} \begin{array}{l} F - iH \cdot F' + iH' \\ i.F + iH \quad " \\ .F - iH \quad " \\ i.F + iH \quad " \\ E - iG \quad " \\ i.E + iG \quad " \\ .E - iG \quad " \\ i.E + iG \quad " \end{array}$$



98. Ninth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 0.)$$

|    |    |    |    |   |                         |                       |
|----|----|----|----|---|-------------------------|-----------------------|
| 2  | 0  | 0  | 2  | = | $I + K \cdot I' + K' +$ | $J + L \cdot J' + L'$ |
| 6  | 4  | 4  | 6  |   | $I + K$                 | $J + L$               |
| 10 | 8  | 8  | 10 |   | $i.I - K$               | $i.J - L$             |
| 14 | 12 | 12 | 14 |   | $i.I - K$               | $i.J - L$             |
| 3  | 1  | 1  | 3  |   | $J + L$                 | $I + K$               |
| 7  | 5  | 5  | 7  |   | $-J + L$                | $I + K$               |
| 11 | 9  | 9  | 11 |   | $i.J - L$               | $i.I - K$             |
| 15 | 13 | 13 | 15 |   | $-i.J - L$              | $i.I - K$             |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 0.)$$

|    |    |    |    |   |                         |                       |
|----|----|----|----|---|-------------------------|-----------------------|
| 2  | 0  | 0  | 2  | = | $I - K \cdot I' - K' +$ | $J - L \cdot J' - L'$ |
| 6  | 4  | 4  | 6  |   | $I - K$                 | $J - L$               |
| 10 | 8  | 8  | 10 |   | $i.I + K$               | $i.J + L$             |
| 14 | 12 | 12 | 14 |   | $i.I + K$               | $i.J + L$             |
| 3  | 1  | 1  | 3  |   | $J - L$                 | $I - K$               |
| 7  | 5  | 5  | 7  |   | $-J - L$                | $I - K$               |
| 11 | 9  | 9  | 11 |   | $i.J + L$               | $i.I + K$             |
| 15 | 13 | 13 | 15 |   | $-i.J + L$              | $i.I + K$             |

99. Tenth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 1.)$$

|    |    |    |    |   |                         |                       |
|----|----|----|----|---|-------------------------|-----------------------|
| 6  | 0  | 0  | 6  | = | $I + K \cdot I' + K' +$ | $J - L \cdot J' - L'$ |
| 2  | 4  | 4  | 2  |   | $I + K$                 | $J - L$               |
| 14 | 8  | 8  | 14 |   | $i.I - K$               | $i.J + L$             |
| 10 | 12 | 12 | 10 |   | $i.I - K$               | $i.J + L$             |
| 7  | 1  | 1  | 7  |   | $J + L$                 | $I - K$               |
| 3  | 5  | 5  | 3  |   | $J + L$                 | $I - K$               |
| 15 | 9  | 9  | 15 |   | $i.J - L$               | $i.I + K$             |
| 11 | 13 | 13 | 11 |   | $i.J - L$               | $i.I + K$             |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 1.)$$

|    |    |    |    |   |                         |                       |
|----|----|----|----|---|-------------------------|-----------------------|
| 6  | 0  | 0  | 6  | = | $I - K \cdot I' - K' +$ | $J + L \cdot J' + L'$ |
| 2  | 4  | 4  | 2  |   | $I - K$                 | $J + L$               |
| 14 | 8  | 8  | 14 |   | $i.I + K$               | $i.J - L$             |
| 10 | 12 | 12 | 10 |   | $i.I + K$               | $i.J - L$             |
| 7  | 1  | 1  | 7  |   | $J - L$                 | $I + K$               |
| 3  | 5  | 5  | 3  |   | $J - L$                 | $I + K$               |
| 15 | 9  | 9  | 15 |   | $i.J + L$               | $i.I - K$             |
| 11 | 13 | 13 | 11 |   | $i.J + L$               | $i.I - K$             |

100. Eleventh set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 2.})$$

|    |    |    |    |   |                         |   |                         |   |
|----|----|----|----|---|-------------------------|---|-------------------------|---|
| 10 | 0  | 0  | 10 | = | $I - iK \cdot I' + iK'$ | + | $J - iL \cdot J' + iL'$ |   |
| 14 | 4  | 4  | 14 |   | $I - iK$                | „ | $J - iL$                | „ |
| 2  | 8  | 8  | 2  |   | $-i.I + iK$             | „ | $i.J + iL$              | „ |
| 6  | 12 | 12 | 6  |   | $-i.I + iK$             | „ | $i.J + iL$              | „ |
| 11 | 1  | 1  | 11 |   | $J - iL$                | „ | $I - iK$                | „ |
| 15 | 5  | 5  | 15 |   | $-J - iL$               | „ | $I - iK$                | „ |
| 3  | 9  | 9  | 3  |   | $-i.J + iL$             | „ | $i.I + iK$              | „ |
| 7  | 13 | 13 | 7  |   | $i.J + iL$              | „ | $i.I + iK$              | „ |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 2.})$$

|    |    |    |    |   |                         |   |                         |   |
|----|----|----|----|---|-------------------------|---|-------------------------|---|
| 10 | 0  | 0  | 10 | = | $I + iK \cdot I' - iK'$ | + | $J + iL \cdot J' - iL'$ |   |
| 14 | 4  | 4  | 14 |   | $I + iK$                | „ | $J + iL$                | „ |
| 2  | 8  | 8  | 2  |   | $-i.I - iK$             | „ | $i.J - iL$              | „ |
| 6  | 12 | 12 | 6  |   | $-i.I - iK$             | „ | $i.J - iL$              | „ |
| 11 | 1  | 1  | 11 |   | $J + iL$                | „ | $I + iK$                | „ |
| 15 | 5  | 5  | 15 |   | $-J + iL$               | „ | $I + iK$                | „ |
| 3  | 9  | 9  | 3  |   | $-i.J - iL$             | „ | $i.I - iK$              | „ |
| 7  | 13 | 13 | 7  |   | $i.J - iL$              | „ | $i.I - iK$              | „ |

101. Twelfth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 3.})$$

|    |    |    |    |   |                         |   |                         |   |
|----|----|----|----|---|-------------------------|---|-------------------------|---|
| 14 | 0  | 0  | 14 | = | $I - iK \cdot I' + iK'$ | + | $J + iL \cdot J' - iL'$ |   |
| 10 | 4  | 4  | 10 |   | $I - iK$                | „ | $J + iL$                | „ |
| 6  | 8  | 8  | 6  |   | $-i.I + iK$             | „ | $i.J - iL$              | „ |
| 2  | 12 | 12 | 2  |   | $-i.I + iK$             | „ | $i.J - iL$              | „ |
| 15 | 1  | 1  | 15 |   | $J - iL$                | „ | $I + iK$                | „ |
| 11 | 5  | 5  | 11 |   | $J - iL$                | „ | $I + iK$                | „ |
| 7  | 9  | 9  | 7  |   | $-i.J + iL$             | „ | $i.I - iK$              | „ |
| 3  | 13 | 13 | 3  |   | $-i.J + iL$             | „ | $i.I - iK$              | „ |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 3.})$$

|    |    |    |    |   |                         |   |                         |   |
|----|----|----|----|---|-------------------------|---|-------------------------|---|
| 14 | 0  | 0  | 14 | = | $I + iK \cdot I' - iK'$ | + | $J - iL \cdot J' + iL'$ |   |
| 10 | 4  | 4  | 10 |   | $I + iK$                | „ | $J - iL$                | „ |
| 6  | 8  | 8  | 6  |   | $-i.I - iK$             | „ | $i.J + iL$              | „ |
| 2  | 12 | 12 | 2  |   | $-i.I - iK$             | „ | $i.J + iL$              | „ |
| 15 | 1  | 1  | 15 |   | $J + iL$                | „ | $I - iK$                | „ |
| 11 | 5  | 5  | 11 |   | $J + iL$                | „ | $I - iK$                | „ |
| 7  | 9  | 9  | 7  |   | $-i.J - iL$             | „ | $i.I + iK$              | „ |
| 3  | 13 | 13 | 3  |   | $-i.J - iL$             | „ | $i.I + iK$              | „ |

102. Thirteenth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 0.)$$

|    |    |    |    |   |                       |   |                       |
|----|----|----|----|---|-----------------------|---|-----------------------|
| 3  | 0  | 0  | 3  | = | $M + Q \cdot M' + Q'$ | + | $N + P \cdot N' + P'$ |
| 7  | 4  | 4  | 7  |   | $i.M - Q$             | „ | $- i.N - P$           |
| 11 | 8  | 8  | 11 |   | $i.M - Q$             | „ | $+ i.N - P$           |
| 15 | 12 | 12 | 15 |   | $-M + Q$              | „ | $+ N + P$             |
| 2  | 1  | 1  | 2  |   | $N + P$               | „ | $+ M + Q$             |
| 6  | 5  | 5  | 6  |   | $-i.N - P$            | „ | $+ i.M - Q$           |
| 10 | 9  | 9  | 10 |   | $i.N - P$             | „ | $+ i.M - Q$           |
| 14 | 13 | 13 | 14 |   | $N + P$               | „ | $- .M + Q$            |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 0.)$$

|    |    |    |    |   |                       |   |                       |
|----|----|----|----|---|-----------------------|---|-----------------------|
| 3  | 0  | 0  | 3  | = | $M - Q \cdot M' - Q'$ | + | $N - P \cdot N' - P'$ |
| 7  | 4  | 4  | 7  |   | $i.M + Q$             | „ | $- i.N + P$           |
| 11 | 8  | 8  | 11 |   | $i.M + Q$             | „ | $+ i.N + P$           |
| 15 | 12 | 12 | 15 |   | $-M - Q$              | „ | $+ N - P$             |
| 2  | 1  | 1  | 2  |   | $N - P$               | „ | $+ M - Q$             |
| 6  | 5  | 5  | 6  |   | $-i.N + P$            | „ | $+ i.M + Q$           |
| 10 | 9  | 9  | 10 |   | $i.N + P$             | „ | $+ i.M + Q$           |
| 14 | 13 | 13 | 14 |   | $N - P$               | „ | $- .M - Q$            |

103. Fourteenth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 1.)$$

|    |    |    |    |   |                         |   |                         |
|----|----|----|----|---|-------------------------|---|-------------------------|
| 7  | 0  | 0  | 7  | = | $M - iQ \cdot M' + iQ'$ | + | $N + iP \cdot N' - iP'$ |
| 3  | 4  | 4  | 3  |   | $-i.M + iQ$             | „ | $+ i.N - iP$            |
| 15 | 8  | 8  | 15 |   | $i.M + iQ$              | „ | $+ i.N - iP$            |
| 11 | 12 | 12 | 11 |   | $M - iQ$                | „ | $- .N + iP$             |
| 6  | 1  | 1  | 6  |   | $N - iP$                | „ | $+ M + iQ$              |
| 2  | 5  | 5  | 2  |   | $-i.N - iP$             | „ | $+ i.M - iQ$            |
| 14 | 9  | 9  | 14 |   | $i.N + iP$              | „ | $+ i.M - iQ$            |
| 10 | 13 | 13 | 10 |   | $N - iP$                | „ | $- .M + iQ$             |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes } 1.)$$

|    |    |    |    |   |                         |   |                         |
|----|----|----|----|---|-------------------------|---|-------------------------|
| 7  | 0  | 0  | 7  | = | $M + iQ \cdot M' - iQ'$ | + | $N - iP \cdot N' + iP'$ |
| 3  | 4  | 4  | 3  |   | $-i.M - iQ$             | „ | $+ i.N + iP$            |
| 15 | 8  | 8  | 15 |   | $-i.M - iQ$             | „ | $+ i.N + iP$            |
| 11 | 12 | 12 | 11 |   | $M + iQ$                | „ | $- .N - iP$             |
| 6  | 1  | 1  | 6  |   | $N + iP$                | „ | $+ M - iQ$              |
| 2  | 5  | 5  | 2  |   | $-i.N + iP$             | „ | $+ i.M + iQ$            |
| 14 | 9  | 9  | 14 |   | $i.N - iP$              | „ | $+ i.M + iQ$            |
| 10 | 13 | 13 | 10 |   | $N + iP$                | „ | $- .M - iQ$             |

104. Fifteenth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 2.})$$

|    |    |    |    |   |                         |   |                         |              |
|----|----|----|----|---|-------------------------|---|-------------------------|--------------|
| 11 | 0  | 0  | 11 | = | $M - iQ \cdot M' + iQ'$ | + | $N - iP \cdot N' + iP'$ |              |
| 15 | 4  | 4  | 15 |   | $i.M + iQ$              | „ | $-$                     | $i.N + iP$ „ |
| 3  | 8  | 8  | 3  |   | $-i.M + iQ$             | „ | $-$                     | $i.N + iP$ „ |
| 7  | 12 | 12 | 7  |   | $M - iQ$                | „ | $-$                     | $.N - iP$ „  |
| 10 | 1  | 1  | 10 |   | $N - iP$                | „ | +                       | $M - iQ$ „   |
| 14 | 5  | 5  | 14 |   | $-i.N + iP$             | „ | +                       | $i.M + iQ$ „ |
| 2  | 9  | 9  | 2  |   | $-i.N + iP$             | „ | $-$                     | $i.M + iQ$ „ |
| 6  | 13 | 13 | 6  |   | $-N - iP$               | „ | +                       | $M - iQ$ „   |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 2.})$$

|    |    |    |    |   |                         |   |                         |              |
|----|----|----|----|---|-------------------------|---|-------------------------|--------------|
| 11 | 0  | 0  | 11 | = | $M + iQ \cdot M' - iQ'$ | + | $N + iP \cdot N' - iP'$ |              |
| 15 | 4  | 4  | 15 |   | $i.M - iQ$              | „ | $-$                     | $i.N - iP$ „ |
| 3  | 8  | 8  | 3  |   | $-i.M - iQ$             | „ | $-$                     | $i.N - iP$ „ |
| 7  | 12 | 12 | 7  |   | $M + iQ$                | „ | $-$                     | $.N + iP$ „  |
| 10 | 1  | 1  | 10 |   | $N + iP$                | „ | +                       | $M + iQ$ „   |
| 14 | 5  | 5  | 14 |   | $-i.N - iP$             | „ | +                       | $i.M - iQ$ „ |
| 2  | 9  | 9  | 2  |   | $-i.N - iP$             | „ | $-$                     | $i.M - iQ$ „ |
| 6  | 13 | 13 | 6  |   | $-N + iP$               | „ | +                       | $M + iQ$ „   |

105. Sixteenth set of 16.

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} + \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 3.})$$

|    |    |    |    |   |                       |   |                       |             |
|----|----|----|----|---|-----------------------|---|-----------------------|-------------|
| 15 | 0  | 0  | 15 | = | $M - Q \cdot M' - Q'$ | + | $N + P \cdot N' + P'$ |             |
| 11 | 4  | 4  | 11 |   | $-i.M + Q$            | „ | +                     | $i.N - P$ „ |
| 7  | 8  | 8  | 7  |   | $-i.M + Q$            | „ | $-$                   | $i.N - P$ „ |
| 3  | 12 | 12 | 3  |   | $-M - Q$              | „ | +                     | $N + P$ „   |
| 14 | 1  | 1  | 14 |   | $N - P$               | „ | +                     | $M + Q$ „   |
| 10 | 5  | 5  | 10 |   | $-i.N + P$            | „ | +                     | $i.M - Q$ „ |
| 6  | 9  | 9  | 6  |   | $-i.N + P$            | „ | $-$                   | $i.M - Q$ „ |
| 2  | 13 | 13 | 2  |   | $-N - P$              | „ | +                     | $M + Q$ „   |

$$\frac{1}{2} \left\{ \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} - \vartheta \frac{u+u'}{3} \cdot \vartheta \frac{u-u'}{3} \right\} \quad (\text{Suffixes 3.})$$

|    |    |    |    |   |                       |   |                       |             |
|----|----|----|----|---|-----------------------|---|-----------------------|-------------|
| 15 | 0  | 0  | 15 | = | $M + Q \cdot M' + Q'$ | + | $N - P \cdot N' - P'$ |             |
| 11 | 4  | 4  | 11 |   | $-i.M - Q$            | „ | +                     | $i.N + P$ „ |
| 7  | 8  | 8  | 7  |   | $-i.M - Q$            | „ | $-$                   | $i.N + P$ „ |
| 3  | 12 | 12 | 3  |   | $-M + Q$              | „ | +                     | $N - P$ „   |
| 14 | 1  | 1  | 14 |   | $N + P$               | „ | +                     | $M - Q$ „   |
| 10 | 5  | 5  | 10 |   | $-i.N - P$            | „ | +                     | $i.M + Q$ „ |
| 6  | 9  | 9  | 6  |   | $-i.N - P$            | „ | $-$                   | $i.M + Q$ „ |
| 2  | 13 | 13 | 2  |   | $-N + P$              | „ | +                     | $M - Q$ „   |

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106. In the square set, writing  $u' = v' = 0$ , and  $\alpha, \beta, \gamma, \delta$  for  $X', Y', Z', W'$ , also slightly altering the arrangement, the system becomes: and further writing herein  $u = 0, v = 0$ , it becomes

| $\vartheta^2 u$ | $X$      | $Y$       | $Z$       | $W$       | $\vartheta^2 0$                                      |
|-----------------|----------|-----------|-----------|-----------|------------------------------------------------------|
| 4               | $\alpha$ | $\beta$   | $\gamma$  | $\delta$  | $4 = \alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 4$   |
| 8               | $\alpha$ | $-\beta$  | $\gamma$  | $-\delta$ | $8 = \alpha^2 - \beta^2 + \gamma^2 - \delta^2 = 8$   |
| 12              | $\alpha$ | $\beta$   | $-\gamma$ | $-\delta$ | $12 = \alpha^2 - \beta^2 - \gamma^2 + \delta^2 = 12$ |
| 1               | $\beta$  | $\alpha$  | $\delta$  | $\gamma$  | $1 = 2(\alpha\beta + \gamma\delta) = 1$              |
| 5               | $\beta$  | $-\alpha$ | $\delta$  | $-\gamma$ | 5                                                    |
| 9               | $\beta$  | $\alpha$  | $-\delta$ | $-\gamma$ | $9 = 2(\alpha\beta - \gamma\delta) = 9$              |
| 13              | $\beta$  | $-\alpha$ | $-\delta$ | $\gamma$  | 13                                                   |
| 2               | $\gamma$ | $\delta$  | $\alpha$  | $\beta$   | $2 = 2(\alpha\gamma + \beta\delta) = 2$              |
| 6               | $\gamma$ | $-\delta$ | $\alpha$  | $-\beta$  | $6 = 2(\alpha\gamma - \beta\delta) = 6$              |
| 10              | $\gamma$ | $-\alpha$ | $-\beta$  | $\delta$  | 10                                                   |
| 14              | $\gamma$ | $-\delta$ | $-\alpha$ | $\beta$   | 14                                                   |
| 3               | $\delta$ | $\gamma$  | $\beta$   | $\alpha$  | $3 = 2(\alpha\delta + \beta\gamma) = 3$              |
| 7               | $\delta$ | $-\gamma$ | $\beta$   | $-\alpha$ | 7                                                    |
| 11              | $\delta$ | $\gamma$  | $-\beta$  | $-\alpha$ | 11                                                   |
| 15              | $\delta$ | $-\gamma$ | $-\beta$  | $\alpha$  | $15 = 2(\alpha\delta - \beta\gamma) = 15$            |

viz. this last is the before-mentioned system of equations giving the values of the 10 zero-functions  $c$  in terms of the four constants  $\alpha, \beta, \gamma, \delta$ .

107. The system first obtained is a system of 16 equations

$$\vartheta^2(u, v) = \alpha X + \beta Y + \gamma Z + \delta W, \text{ \&c.},$$

showing that the squares of the theta-functions are each of them a linear function of the four quantities  $X, Y, Z, W$ . If the functions on the right-hand side were independent (asyzygetic) linear functions of  $X, Y, Z, W$ , it would follow that any four (selected at pleasure) of the squared theta-functions are linearly independent, and that we could in terms of these four express linearly each of the remaining 12 squared functions. But this is not so; the form of the linear functions of  $(X, Y, Z, W)$  is such that we can (and that in 16 different ways) select out of the 16 linear functions six functions, such that any four of them are connected by a linear equation; and there are consequently 16 hexads of squared theta-functions, such

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that any four out of the same hexad are connected by a linear relation. The hexads are shown by the foregoing “Table of the 16 Kummer hexads.”

108. The *à posteriori* verification is immediately effected; taking for instance the first column, the equations are

|               | <i>X</i> | <i>Y</i>  | <i>Z</i>  | <i>W</i>  |
|---------------|----------|-----------|-----------|-----------|
| <i>A</i> , 11 | $\beta$  | $\gamma$  | $-\beta$  | $-\alpha$ |
| <i>B</i> , 7  | $\delta$ | $-\gamma$ | $\beta$   | $-\alpha$ |
| <i>AB</i> , 6 | $\gamma$ | $-\delta$ | $\alpha$  | $-\beta$  |
| <i>CD</i> , 2 | $\gamma$ | $\delta$  | $\alpha$  | $\beta$   |
| <i>CE</i> , 1 | $\beta$  | $\alpha$  | $\delta$  | $\gamma$  |
| <i>DE</i> , 9 | $\beta$  | $\alpha$  | $-\delta$ | $-\gamma$ |

viz. it should thence follow that there is a linear relation between any four of the six squared functions 11, 7, 6, 2, 1, 9: and it is accordingly seen that this is so. It further appears that, in the several linear relations, the coefficients (obtained in the first instance as functions of  $\alpha, \beta, \gamma, \delta$ ) are in fact the 10 constants *c*: the 15 relations connecting the several systems of four out of the six squared functions are given in the following table.

109.

| $\vartheta^2$ | 11  | 7   | 6   | 2   | 1   | 9   | = 0 |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| $c_0^2$       |     |     | 6   | −2  | 1   | −9  |     |
| $c_2^2$       |     | 6   |     | +15 | −12 | +4  |     |
| $c_4^2$       | −2  |     | −15 |     | +8  | −3  |     |
| $c_5^2$       |     | 1   | +12 | −8  |     | +3  |     |
| $c_6^2$       | −9  |     | −4  | +3  |     |     |     |
| $c_8^2$       |     | 6   |     | 3   |     | +8  |     |
| $c_9^2$       |     | −2  | −3  |     | +4  | −12 |     |
| $c_{10}^2$    |     | 1   | +8  | −4  |     | −15 |     |
| $c_{11}^2$    |     | −9  | −8  | +12 | +15 |     |     |
| $c_{12}^2$    | −15 | +3  | +2  | −6  |     |     |     |
| $c_{13}^2$    | −12 |     | +1  |     | −6  |     |     |
| $c_{14}^2$    | −4  | +8  | +9  |     |     | −6  |     |
| $c_{15}^2$    | −3  | +15 |     |     | +9  | −1  |     |
| $c_3^2$       | −2  | +12 |     | −9  |     |     |     |
| $c_7^2$       | −8  | +4  |     | +1  | −2  |     |     |

Read

$$c_2^4 \vartheta_{11}^2 + c_4^2 \vartheta_6^2 - c_5^2 \vartheta_2^2 + c_6^2 \vartheta_1^2 - c_8^2 \vartheta_9^2 = 0, \text{ \&c.}$$

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110. The first set of 16 equations is the square-set, which has been already considered. If in each of the other sets of 16 equations we write in like manner  $u' = 0$ , each set in fact reduces itself to eight equations; sets 2, 3, 4 give thus  $8 + 8 + 8 = 24$  equations; sets 5 to 8, 9 to 12, and 13 to 15, give each  $8 + 8 + 8 + 8 = 32$  equations; or we have sets of 24, 32, 32, 32, together 120 equations, the number being of course one half of  $256 - 16$ , the number of equations after deducting the 16 equations of the square set.

111. The first set, 24 equations.

This is derived from the second, third, and fourth sets, each of 16 equations, by writing therein  $u' = 0$ . Taking  $\alpha_1, \beta_1, \gamma_1, \delta_1$  for the zero-functions corresponding to  $X_1, Y_1, Z_1, W_1$ , then on writing  $u' = 0$ ,  $X'_1, Y'_1, Z'_1, W'_1$  become  $\alpha_1, \beta_1, \gamma_1, \delta_1$ . In the second set of 16 equations, the first equations thus are

$$\vartheta_1 u \cdot \vartheta_1 u = \alpha_1 X_1 + \gamma_1 Z_1 = \beta_1 Y_1 + \delta_1 W_1,$$

$$\vartheta_9 u \cdot \vartheta_8 u = \alpha_1 X_1 - \gamma_1 Z_1, \quad 0 = \beta_1 Y_1 - \delta_1 W_1,$$

viz. the equations of the column require that, and are all satisfied if,  $\beta_1 = 0, \delta_1 = 0$ ; hence the zero-functions are  $\alpha_1, 0, \gamma_1, 0$ ; and this being so we have only the equations of the first column. And similarly as regards the third and fourth sets the zero-values corresponding to

$$X_1, Y_1, Z_1, W_1 \quad | \quad X_2, Y_2, Z_2, W_2 \quad | \quad X_3, Y_3, Z_3, W_3$$

are

$$\alpha_1, 0, \gamma_1, 0 \quad | \quad \alpha_2, 0, 0, \delta_2 \quad | \quad \alpha_3, 0, 0, \delta_3;$$

and we have in all  $8 + 8 + 8 = 24$  equations. These are

| (Suffixes 1.)   |    |                       |                        | (Suffixes 2.)   |   |                       |                        | (Suffixes 3.)   |   |                       |                        |
|-----------------|----|-----------------------|------------------------|-----------------|---|-----------------------|------------------------|-----------------|---|-----------------------|------------------------|
| $X \quad Z$     |    |                       |                        | $X \quad W$     |   |                       |                        | $X \quad W$     |   |                       |                        |
| 4               | =  | $\alpha$              | $\gamma$               | 8               | = | $\alpha$              | $\delta$               | 12              | = | $\alpha$              | $\delta$               |
| 12              | 8  | =                     | $\alpha \quad -\gamma$ | 12              | 4 | =                     | $\alpha \quad -\delta$ | 8               | 4 | =                     | $\alpha \quad -\delta$ |
| 6               | 2  | =                     | $\gamma \quad \alpha$  | 9               |   | =                     | $\delta \quad \alpha$  | 15              | 3 | =                     | $\delta \quad \alpha$  |
| 14              | 10 | =                     | $\gamma \quad -\alpha$ | 13              |   | =                     | $-\delta \quad \alpha$ | 11              | 7 | =                     | $-\delta \quad \alpha$ |
| $Y \quad W$     |    |                       |                        | $Y \quad W$     |   |                       |                        | $Y \quad Z$     |   |                       |                        |
| 5               | 1  | =                     | $\alpha \quad \gamma$  | 10              | 2 | =                     | $\alpha \quad \beta$   | 13              | 1 | =                     | $\alpha \quad \delta$  |
| 13              | 9  | =                     | $\alpha \quad -\gamma$ | 14              | 6 | =                     | $\alpha \quad -\beta$  | 9               | 5 | =                     | $\alpha \quad -\delta$ |
| 7               | 3  | =                     | $\gamma \quad \alpha$  | 11              | 3 | =                     | $\beta \quad \alpha$   | 14              | 2 | =                     | $\delta \quad \alpha$  |
| 15              | 11 | =                     | $\gamma \quad -\alpha$ | 15              | 7 | =                     | $\beta \quad -\alpha$  | 10              | 6 | =                     | $-\delta \quad \alpha$ |
| $\vartheta^2 0$ |    |                       |                        | $\vartheta^2 0$ |   |                       |                        | $\vartheta^2 0$ |   |                       |                        |
| 4               | =  | $\alpha^2 + \gamma^2$ |                        | 8               | = | $\alpha^2 + \delta^2$ |                        | 12              | = | $\alpha^2 + \delta^2$ |                        |
| 12              | 8  | =                     | $\alpha^2 - \gamma^2$  | 12              | 4 | =                     | $\alpha^2 - \delta^2$  | 8               | 4 | =                     | $\alpha^2 - \delta^2$  |
| 6               | 2  | =                     | $2\alpha\gamma$        | 9               | = | $2\alpha\beta$        |                        | 15              | 3 | =                     | $2\alpha\delta$        |



112. The second set, 32 equations.

To exhibit these in a convenient form, I alter the notation, viz. I write

$$\begin{aligned} &= \left| \begin{array}{cccc} E+G & i(E-G) & (F+H) & i(F-H) \\ X & Y & Z & W \end{array} \right| \left| \begin{array}{cccc} E_1+iG_1 & E_1-iG_1 & F_1+iH_1 & F_1-iH_1 \\ X_1 & Y_1 & Z_1 & W_1 \end{array} \right| \\ &= \left| \begin{array}{cccc} (E_2+G_2) & i(E_2-G_2) & (F_2+H_2) & i(F_2-H_2) \\ X_2 & Y_2 & Z_2 & W_2 \end{array} \right| \left| \begin{array}{cccc} E_3+iG_3 & E_3-iG_3 & F_3+iH_3 & F_3-iH_3 \\ X_3 & Y_3 & Z_3 & W_3 \end{array} \right| \end{aligned}$$

so that as regards the present set of equations,  $X, Y, Z, W$ , signify as just mentioned. And, this being so, the corresponding zero-values are

$$\alpha, 0, \gamma, 0 \mid \alpha_1, 0, \gamma_1, 0 \mid \alpha_2, 0, 0, \delta_2 \mid \alpha_3, 0, 0, \delta_3.$$

The equations then are

| (Suffixes 0.)   |    |                         |           | (Suffixes 1.)   |    |                             |            | (Suffixes 2.)   |    |                         |           | (Suffixes 3.)   |    |                             |            |
|-----------------|----|-------------------------|-----------|-----------------|----|-----------------------------|------------|-----------------|----|-------------------------|-----------|-----------------|----|-----------------------------|------------|
| $X \quad Z$     |    |                         |           | $X \quad Z$     |    |                             |            | $X \quad W$     |    |                         |           | $X \quad W$     |    |                             |            |
| 1               |    | $= \alpha$              | $\gamma$  | 1               | 4  | $= -i\alpha$                | $-i\gamma$ | 9               |    | $= \alpha$              | $\delta$  | 9               | 4  | $= -i\alpha$                | $-i\delta$ |
| 9               | 8  | $= \alpha$              | $-\gamma$ | 9               | 12 | $= -i\alpha$                | $+i\gamma$ | 1               | 8  | $= \alpha$              | $-\delta$ | 1               | 12 | $= -i\alpha$                | $+i\delta$ |
| 3               | 2  | $= \gamma$              | $\alpha$  | 3               | 6  | $= -i\gamma$                | $-i\alpha$ | 15              | 6  | $= \delta$              | $\alpha$  | 15              | 2  | $= \delta$                  | $\alpha$   |
| 11              | 10 | $= \gamma$              | $-\alpha$ | 11              | 14 | $= -i\gamma$                | $+i\alpha$ | 7               | 14 | $= -\delta$             | $\alpha$  | 7               | 10 | $= -\delta$                 | $\alpha$   |
| $Y \quad W$     |    |                         |           | $Y \quad W$     |    |                             |            | $Y \quad Z$     |    |                         |           | $Y \quad Z$     |    |                             |            |
| 5               | 4  | $= \alpha$              | $\gamma$  | 5               |    | $= \alpha$                  | $\gamma$   | 13              | 4  | $= \alpha$              | $\delta$  | 13              |    | $= \alpha$                  | $\delta$   |
| 13              | 12 | $= \alpha$              | $-\gamma$ | 13              | 8  | $= \alpha$                  | $-\gamma$  | 5               | 12 | $= \alpha$              | $-\delta$ | 5               | 8  | $= \alpha$                  | $-\delta$  |
| 7               | 6  | $= \gamma$              | $\alpha$  | 7               | 2  | $= \gamma$                  | $\alpha$   | 11              | 2  | $= -\delta$             | $\alpha$  | 11              | 6  | $= -i\delta$                | $-i\alpha$ |
| 15              | 14 | $= \gamma$              | $-\alpha$ | 15              | 10 | $= \gamma$                  | $-\alpha$  | 3               | 10 | $= \delta$              | $\alpha$  | 3               | 14 | $= i\delta$                 | $-i\alpha$ |
| $\vartheta^2 0$ |    |                         |           | $\vartheta^2 0$ |    |                             |            | $\vartheta^2 0$ |    |                         |           | $\vartheta^2 0$ |    |                             |            |
| 1               |    | $= \alpha^2 + \gamma^2$ |           | 1               | 4  | $= -i(\alpha^2 + \gamma^2)$ |            | 9               |    | $= \alpha^2 - \delta^2$ |           | 9               | 4  | $= -i(\alpha^2 + \delta^2)$ |            |
| 9               | 8  | $= \alpha^2 - \gamma^2$ |           | 9               | 12 | $= -i(\alpha^2 - \gamma^2)$ |            | 1               | 8  | $= \alpha^2 + \delta^2$ |           | 1               | 12 | $= -i(\alpha^2 - \delta^2)$ |            |
| 3               | 2  | $= 2\alpha\gamma$       |           | 3               | 6  | $= -2i\alpha\gamma$         |            | 15              | 6  | $= 2\alpha\delta$       |           | 15              | 2  | $= 2\alpha\delta$           |            |

113.    Third set, 32 equations.

We again change the notation, writing

|   |                                                                                                    |                                                                                                    |
|---|----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| = | $\begin{vmatrix} I+K & i(I-K) & J+L & i(J-L) \\ X & Y & Z & W \end{vmatrix}$                       | $\begin{vmatrix} I_1+iK_1 & I_1-iK_1 & J_1+iL_1 & J_1-iL_1 \\ X_1 & Y_1 & Z_1 & W_1 \end{vmatrix}$ |
| = | $\begin{vmatrix} I_2+iK_2 & I_2-iK_2 & J_2+iL_2 & J_2-iL_2 \\ X_2 & Y_2 & Z_2 & W_2 \end{vmatrix}$ | $\begin{vmatrix} I_3+iK_3 & I_3-iK_3 & J_3+iL_3 & J_3-iL_3 \\ X_3 & Y_3 & Z_3 & W_3 \end{vmatrix}$ |

the zero values being

$$\alpha, 0, \gamma, 0 \mid \alpha_1, 0, 0, \delta_1 \mid \alpha_2, 0, \gamma_2, 0 \mid \alpha_3, 0, 0, \delta_3.$$

Then equations are

| (Suffixes 0.)   |    |   |                        | (Suffixes 1.)   |    |   |                        | (Suffixes 2.)   |    |   |                           | (Suffixes 3.)   |    |   |                           |
|-----------------|----|---|------------------------|-----------------|----|---|------------------------|-----------------|----|---|---------------------------|-----------------|----|---|---------------------------|
| $X \quad Z$     |    |   |                        | $X \quad W$     |    |   |                        | $X \quad Z$     |    |   |                           | $X \quad W$     |    |   |                           |
| 2               |    | = | $\alpha \quad \gamma$  | 6               |    | = | $\alpha \quad \delta$  | 2               | 8  | = | $-i\alpha \quad -i\gamma$ | 6               | 8  | = | $-i\alpha \quad -i\delta$ |
| 6               | 4  | = | $\alpha \quad -\gamma$ | 2               | 4  | = | $\alpha \quad -\delta$ | 6               | 12 | = | $-i\alpha \quad +i\gamma$ | 2               | 12 | = | $-i\alpha \quad +i\delta$ |
| 3               | 1  | = | $\gamma \quad \alpha$  | 15              | 9  | = | $\delta \quad \alpha$  | 3               | 9  | = | $-i\gamma \quad -i\alpha$ | 15              | 1  | = | $\delta \quad \alpha$     |
| 7               | 5  | = | $\gamma \quad -\alpha$ | 11              | 13 | = | $-\delta \quad \alpha$ | 7               | 13 | = | $-i\gamma \quad +i\alpha$ | 11              | 5  | = | $-\delta \quad \alpha$    |
| $Y \quad W$     |    |   |                        | $Y \quad Z$     |    |   |                        | $Y \quad W$     |    |   |                           | $Y \quad Z$     |    |   |                           |
| 10              | 8  | = | $\alpha \quad \gamma$  |                 |    |   |                        | 10              |    | = | $\alpha \quad \gamma$     | 14              |    | = | $\alpha \quad \delta$     |
| 14              | 12 | = | $\alpha \quad -\gamma$ |                 |    |   |                        | 14              | 4  | = | $\alpha \quad -\gamma$    | 10              | 4  | = | $\alpha \quad -\delta$    |
| 11              | 9  | = | $\gamma \quad \alpha$  |                 |    |   |                        | 11              | 1  | = | $\gamma \quad \alpha$     | 7               | 9  | = | $-i\delta \quad -i\alpha$ |
| 15              | 13 | = | $\gamma \quad -\alpha$ |                 |    |   |                        | 15              | 5  | = | $\gamma \quad -\alpha$    | 3               | 13 | = | $i\delta \quad -i\alpha$  |
| $\vartheta^2 0$ |    |   |                        | $\vartheta^2 0$ |    |   |                        | $\vartheta^2 0$ |    |   |                           | $\vartheta^2 0$ |    |   |                           |
| 2               |    | = | $\alpha^2 + \gamma^2$  | 6               |    | = | $\alpha^2 + \delta^2$  | 2               | 8  | = | $-i(\alpha^2 + \gamma^2)$ | 6               | 8  | = | $-i(\alpha^2 + \delta^2)$ |
| 6               | 4  | = | $\alpha^2 - \gamma^2$  | 2               | 4  | = | $\alpha^2 + \delta^2$  | 6               | 12 | = | $-i(\alpha^2 - \gamma^2)$ | 2               | 12 | = | $-i(\alpha^2 - \delta^2)$ |
| 3               | 1  | = | $2\alpha\gamma$        | 15              | 9  | = | $2\alpha\delta$        | 3               | 9  | = | $-2i\alpha\gamma$         | 15              | 1  | = | $2\alpha\delta$           |

114. Fourth set, 32 equations.

Again changing the notation, we write

|   |                                                                                      |                                                                                                    |
|---|--------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| = | $\begin{vmatrix} M+Q & i(M-Q) & N+P & i(N-P) \\ X & Y & Z & W \end{vmatrix}$         | $\begin{vmatrix} M_1+iQ_1 & M_1-iQ_1 & N_1+iP_1 & N_1-iP_1 \\ X_1 & Y_1 & Z_1 & W_1 \end{vmatrix}$ |
| = | $\begin{vmatrix} M+Q & i(M-Q) & N+P & i(N-P) \\ X_2 & Y_2 & Z_2 & W_2 \end{vmatrix}$ | $\begin{vmatrix} M_3+iQ_3 & M_3-iQ_3 & N_3+iP_3 & N_3-iP_3 \\ X_3 & Y_3 & Z_3 & W_3 \end{vmatrix}$ |

the zero values being

$$\alpha, 0, \gamma, 0 \mid \alpha_1, 0, 0, \delta_1 \mid \alpha_2, 0, \gamma_2, 0 \mid 0, \beta_3, \gamma_3, 0.$$

The equations then are

| (Suffixes 0.)   |    |                       |                          | (Suffixes 1.)   |    |                    |                           | (Suffixes 2.)   |    |                   |                           | (Suffixes 3.)   |    |                         |                      |
|-----------------|----|-----------------------|--------------------------|-----------------|----|--------------------|---------------------------|-----------------|----|-------------------|---------------------------|-----------------|----|-------------------------|----------------------|
| X Z             |    |                       |                          | X W             |    |                    |                           | X Z             |    |                   |                           | Y Z             |    |                         |                      |
| 3               | =  | $\alpha$              | $\gamma$                 | 3               | 4  | =                  | $-i\alpha$ $-i\delta$     | 15              | 4  | =                 | $i\alpha$ $i\gamma$       | 15              | =  | $-\beta$                | $\gamma$             |
| 15              | 12 | =                     | $-\alpha$ $\gamma$       | 15              | 8  | =                  | $i\alpha$ $i\delta$       | 3               | 8  | =                 | $-i\alpha$ $i\gamma$      | 3               | 12 | =                       | $-\beta$ $\gamma$    |
| 2               | 1  | =                     | $\gamma$ $\alpha$        | 6               | 1  | =                  | $\delta$ $\alpha$         | 14              | 5  | =                 | $i\gamma$ $-i\alpha$      | 10              | 5  | =                       | $\gamma$ $-\beta$    |
| 14              | 13 | =                     | $-\gamma$ $\alpha$       | 10              | 13 | =                  | $-\delta$ $\alpha$        | 2               | 9  | =                 | $-i\gamma$ $-i\alpha$     | 6               | 9  | =                       | $-\gamma$ $-\beta$   |
| Y W             |    |                       |                          | Y Z             |    |                    |                           | Y W             |    |                   |                           | X W             |    |                         |                      |
| 4               | 7  | =                     | $\alpha$ $-\gamma$       | 7               | =  | $\alpha$ $-\delta$ |                           | 11              | =  | $\alpha$ $\gamma$ |                           | 11              | 4  | =                       | $\beta$ $\gamma$     |
| 8               | 11 | =                     | $\alpha$ $\gamma$        | 11              | 12 | =                  | $\alpha$ $-\delta$        | 7               | 12 | =                 | $\alpha$ $-\gamma$        | 7               | 8  | =                       | $-\beta$ $-\gamma$   |
| 6               | 5  | =                     | $\gamma$ $-\alpha$       | 2               | 5  | =                  | $i\delta$ $-i\alpha$      | 10              | 1  | =                 | $\gamma$ $\alpha$         | 14              | 1  | =                       | $\gamma$ $-\beta$    |
| 10              | 9  | =                     | $\gamma$ $\alpha$        | 14              | 9  | =                  | $i\delta$ $i\alpha$       | 6               | 13 | =                 | $\gamma$ $-\alpha$        | 2               | 13 | =                       | $\gamma$ $\beta$     |
| $\vartheta^2 0$ |    |                       |                          | $\vartheta^2 0$ |    |                    |                           | $\vartheta^2 0$ |    |                   |                           | $\vartheta^2 0$ |    |                         |                      |
| 3               | =  | $\alpha^2 + \gamma^2$ |                          | 3               | 4  | =                  | $-i(\alpha^2 - \delta^2)$ | 15              | 4  | =                 | $i(\alpha^2 - \gamma^2)$  | 15              | =  | $-(\beta^2 - \gamma^2)$ |                      |
| 15              | 12 | =                     | $-(\alpha^2 - \gamma^2)$ | 15              | 8  | =                  | $i(\alpha^2 + \delta^2)$  | 3               | 8  | =                 | $-i(\alpha^2 + \gamma^2)$ | 3               | 12 | =                       | $\beta^2 + \gamma^2$ |
| 2               | 1  | =                     | $2\alpha\gamma$          | 6               | 1  | =                  | $2\alpha\delta$           | 2               | 9  | =                 | $-2i\alpha\gamma$         | 6               | 9  | =                       | $-2\beta\gamma$      |

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115. It will be noticed that the pairs of theta-functions which present themselves in these equations are the same as in the foregoing “Table of the 120 pairs.” And the equations show that the four products, each of a pair of theta-functions, belonging to the upper half or to the lower half of any column of the table, are such that any three of the four products are connected by a linear equation. The coefficients of these linear relations are, in fact, functions such as the  $\alpha^2 + \delta^2$ ,  $\alpha^2 - \delta^2$ ,  $2\alpha\delta$  written down at the foot of the several systems of eight equations, and they are consequently products each of two zero-functions  $c$ .

Thus (see “The first set, 24 equations”) we have

| (Suffixes 3.)                   |     |           |           | (Suffixes 3.) |     |           |           | (Suffixes 3.)   |                 |                       |  |
|---------------------------------|-----|-----------|-----------|---------------|-----|-----------|-----------|-----------------|-----------------|-----------------------|--|
| $\vartheta u \cdot \vartheta u$ | $X$ | $W$       |           | $Y$           | $Z$ |           |           | $\vartheta^2 0$ | $\vartheta^2 0$ |                       |  |
| 4 8                             | =   | $\alpha$  | $-\delta$ | 5 9           | =   | $\alpha$  | $-\delta$ | 4 8             | =               | $\alpha^2 - \delta^2$ |  |
| 12                              | =   | $\alpha$  | $\delta$  | 1 13          | =   | $\alpha$  | $\delta$  | 12              | =               | $\alpha^2 + \delta^2$ |  |
| 3 15                            | =   | $\delta$  | $\alpha$  | 2 14          | =   | $\delta$  | $\alpha$  | 15 3            | =               | $2\alpha\delta$       |  |
| 7 11                            | =   | $-\delta$ | $\alpha$  | 6 10          | =   | $-\delta$ | $\alpha$  |                 |                 |                       |  |

116. In the left-hand four of these, omitting successively the first, second, third, and fourth equation, and from the remaining three eliminating the  $X_3$  and  $W_3$ , we write down, almost mechanically,

| $\vartheta u$ | $\vartheta u$ |                         |                        |                         |
|---------------|---------------|-------------------------|------------------------|-------------------------|
| 4             | 8             |                         | $+2\alpha\delta,$      | $-\delta^2 - \alpha^2,$ |
|               | 12            | $-2\alpha\delta,$       |                        | $-\delta^2 + \alpha^2,$ |
| 3             | 15            |                         |                        | $2\alpha\delta$         |
| 7             | 11            | $-\alpha^2 + \delta^2,$ | $\delta^2 + \alpha^2,$ | $-2\alpha\delta$        |

and thence derive the first of the next following system of equations; read

$$\begin{aligned} c_4c_{12}\vartheta_4\vartheta_8 - c_0c_{12}\vartheta_0\vartheta_{12} + c_4c_0\vartheta_3\vartheta_{15} + c_4c_8\vartheta_7\vartheta_{11} &= 0, \\ -c_4c_{12}\vartheta_4\vartheta_8 + c_4c_9\vartheta_3\vartheta_{15} - c_3c_{15}\vartheta_7\vartheta_{11} &= 0, \\ c_0c_{12}\vartheta_4\vartheta_8 + c_4c_9\vartheta_0\vartheta_{12} + c_3c_{15}\vartheta_7\vartheta_{11} &= 0, \end{aligned}$$

where the theta-functions have the arguments  $u, v$ .

Observe that, on writing herein  $u = 0, v = 0$ , the first three equations become each of them identically  $= 0$ ; the fourth equation becomes

$$-c_4c_{12} + c_0c_{12} - c_3c_{15} = 0,$$

which is one of the relations between the  $c$ 's and serves as a verification.

But in the right-hand system, on writing  $u = v = 0$ , each of the four equations becomes identically  $= 0$ .

117. The equations are

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 4.8   | 0.12 | 3.15  | 7.11  | = 0, |
| $c$         |       | 3.15 | −0.12 | 4.8   |      |
|             | −3.15 |      | −4.8  | −0.12 |      |
|             | 0.12  | −4.8 |       | 3.15  |      |
|             | −4.8  | 0.12 | −3.15 |       |      |

|             |       |       |       |       |      |
|-------------|-------|-------|-------|-------|------|
| $\vartheta$ | 5.9   | 1.13  | 2.14  | 6.10  | = 0, |
| $c$         |       | 3.15  | −0.12 | 4.8   |      |
|             | −3.15 |       | 4.8   | −0.12 |      |
|             | 0.12  | −4.8  |       | 3.15  |      |
|             | −4.8  | −0.12 | −3.15 |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 6.8   | 2.12 | 1.15  | 5.11  | = 0, |
| $c$         |       | 1.15 | −2.12 | 6.8   |      |
|             | −1.15 |      | 6.8   | −2.12 |      |
|             | 2.12  | −6.8 |       | 1.15  |      |
|             | −6.8  | 2.12 | −1.15 |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 7.9   | 3.13 | 0.14  | 4.10  | = 0, |
| $c$         |       | 1.15 | −2.12 | 6.8   |      |
|             | −1.15 |      | 6.8   | −2.12 |      |
|             | 2.12  | −6.8 |       | 1.15  |      |
|             | −6.8  | 2.12 | −1.15 |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 0.6   | 2.4  | 9.15  | 11.13 | = 0, |
| $c$         |       | 9.15 | −2.4  | 0.6   |      |
|             | −9.15 |      | 0.6   | −2.4  |      |
|             | 2.4   | −0.6 |       | 9.15  |      |
|             | −0.6  | 2.4  | −9.15 |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 1.7   | 3.5  | 8.14  | 10.12 | = 0, |
| $c$         |       | 9.15 | −2.4  | 0.6   |      |
|             | −9.15 |      | 0.6   | −2.4  |      |
|             | 2.4   | −0.6 |       | 9.15  |      |
|             | −0.6  | 2.4  | −9.15 |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 3.6   | 1.4  | 9.12  | 14.11 | = 0, |
| $c$         |       | 9.12 | −1.4  | 3.6   |      |
|             | −9.12 |      | 3.6   | −1.4  |      |
|             | 1.4   | −3.6 |       | 9.12  |      |
|             | −3.6  | 1.4  | −9.12 |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 2.7   | 0.5  | 8.13  | 10.15 | = 0, |
| $c$         |       | 9.12 | −1.4  | 3.6   |      |
|             | −9.12 |      | 3.6   | −1.4  |      |
|             | 1.4   | −3.6 |       | 9.12  |      |
|             | −3.6  | 1.4  | −9.12 |       |      |

|             |      |      |      |       |      |
|-------------|------|------|------|-------|------|
| $\vartheta$ | 8.9  | 0.1  | 2.3  | 10.11 | = 0, |
| $c$         |      | −2.3 | 0.1  | 8.9   |      |
|             | 2.3  |      | −8.9 | −0.1  |      |
|             | −0.1 | 8.9  |      | 2.3   |      |
|             | −8.9 | 0.1  | −2.3 |       |      |

|             |       |      |      |       |      |
|-------------|-------|------|------|-------|------|
| $\vartheta$ | 12.13 | 4.5  | 6.7  | 14.15 | = 0, |
| $c$         |       | −2.3 | 0.1  | 8.9   |      |
|             | 2.3   |      | −8.9 | −0.1  |      |
|             | −0.1  | 8.9  |      | 2.3   |      |
|             | −8.9  | 0.1  | −2.3 |       |      |

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| $\vartheta$ | 4.6  | 0.2  | 1.3  | 5.7  | = 0, |
|-------------|------|------|------|------|------|
| $c$         |      | −1.3 | 0.2  | 4.6  |      |
|             | 1.3  |      | −4.6 | −0.2 |      |
|             | −0.2 | 4.6  |      | 1.3  |      |
|             | −4.6 | 0.2  | −1.3 |      |      |

| $\vartheta$ | 9.11 | 13.15 | 12.14 | 8.10 | = 0, |
|-------------|------|-------|-------|------|------|
| $c$         |      | −1.3  | 0.2   | 4.6  |      |
|             | 1.3  |       | 4.6   | −0.2 |      |
|             | −0.2 | −4.6  |       | 1.3  |      |
|             | 4.6  | 0.2   | 1.3   |      |      |

| $\vartheta$ | 6.12 | 2.8   | 3.9  | 7.13  | = 0, |
|-------------|------|-------|------|-------|------|
| $c$         |      | 3.9   | −2.8 | −6.12 |      |
|             | −3.9 |       | 6.12 | 2.8   |      |
|             | 2.8  | −6.12 |      | −3.9  |      |
|             | 6.12 | −2.8  | 3.9  |       |      |

| $\vartheta$ | 1.11  | 5.15 | 4.14  | 0.10 | = 0, |
|-------------|-------|------|-------|------|------|
| $c$         |       | 3.9  | −2.8  | 6.12 |      |
|             | −3.9  |      | −6.12 | 2.8  |      |
|             | 2.8   | 6.12 |       | −3.9 |      |
|             | −6.12 | −2.8 | 3.9   |      |      |

| $\vartheta$ | 6.15 |       | 0.9  | 7.14  | = 0, |
|-------------|------|-------|------|-------|------|
| $c$         |      | 0.9   | −1.8 | −6.15 |      |
|             | −0.9 |       | 6.15 | 1.8   |      |
|             | 1.8  | −6.15 |      | −0.9  |      |
|             | 6.15 | −1.8  | 0.9  |       |      |

| $\vartheta$ | 2.11  | 5.12 | 4.13  | 3.10 | = 0, |
|-------------|-------|------|-------|------|------|
| $c$         |       | 0.9  | −1.8  | 6.15 |      |
|             | −0.9  |      | −6.15 | 1.8  |      |
|             | 1.8   | 6.15 |       | −0.9 |      |
|             | −6.15 | −1.8 | 0.9   |      |      |

| $\vartheta$ | 4.9   | 1.12 | 2.15  | 7.10  | = 0, |
|-------------|-------|------|-------|-------|------|
| $c$         |       | 2.15 | −1.12 | 4.9   |      |
|             | −2.15 |      | 4.9   | −1.12 |      |
|             | 1.12  | −4.9 |       | 2.15  |      |
|             | −4.9  | 1.12 | −2.15 |       |      |

| $\vartheta$ | 0.13  | 5.8   | 6.11  | 3.14 | = 0, |
|-------------|-------|-------|-------|------|------|
| $c$         |       | 2.15  | 1.12  | −4.9 |      |
|             | −2.15 |       | −4.9  | 1.12 |      |
|             | −1.12 | 4.9   |       | 2.15 |      |
|             | 4.9   | −1.12 | −2.15 |      |      |

| $\vartheta$ | 4.12  | 0.8  | 1.9   | 5.13 | = 0, |
|-------------|-------|------|-------|------|------|
| $c$         |       | −1.9 | 0.8   | 4.12 |      |
|             | 1.9   |      | −4.12 | −0.8 |      |
|             | −0.8  | 4.12 |       | 1.9  |      |
|             | −4.12 | 0.8  | −1.9  |      |      |

| $\vartheta$ | 3.11  | 7.15 | 6.14  | 2.10 | = 0, |
|-------------|-------|------|-------|------|------|
| $c$         |       | 1.9  | −0.8  | 4.12 |      |
|             | −1.9  |      | −4.12 | 0.8  |      |
|             | 0.8   | 4.12 |       | −1.9 |      |
|             | −4.12 | −0.8 | 1.9   |      |      |

|             |       |      |       |      |      |
|-------------|-------|------|-------|------|------|
| $\vartheta$ | 4.15  | 3.8  | 2.9   | 5.14 | = 0, |
| $c$         |       | −2.9 | 3.8   | 4.15 |      |
|             | 2.9   |      | −4.15 | −3.8 |      |
|             | −3.8  | 4.15 |       | 2.9  |      |
|             | −4.15 | 3.8  | −2.9  |      |      |

|             |      |       |      |       |      |
|-------------|------|-------|------|-------|------|
| $\vartheta$ | 0.11 | 7.12  | 6.13 | 1.10  | = 0, |
| $c$         |      | −2.9  | 3.8  | −4.15 |      |
|             | 2.9  |       | 4.15 | −3.8  |      |
|             | −3.8 | −4.15 |      | 2.9   |      |
|             | 4.15 | 3.8   | −2.9 |       |      |

|             |       |       |      |       |      |
|-------------|-------|-------|------|-------|------|
| $\vartheta$ | 6.9   | 3.12  | 0.15 | 5.10  | = 0, |
| $c$         |       | −0.15 | 3.12 | −6.9  |      |
|             | 0.15  |       | −6.9 | 3.12  |      |
|             | −3.12 | 6.9   |      | −0.15 |      |
|             | 6.9   | −3.12 | 0.15 |       |      |

|             |       |       |       |      |      |
|-------------|-------|-------|-------|------|------|
| $\vartheta$ | 2.13  | 7.8   | 4.11  | 1.14 | = 0, |
| $c$         |       | 0.15  | 3.12  | −6.9 |      |
|             | −0.15 |       | −6.9  | 3.12 |      |
|             | −3.12 | 6.9   |       | 0.15 |      |
|             | 6.9   | −3.12 | −0.15 |      |      |

|             |       |        |       |        |      |
|-------------|-------|--------|-------|--------|------|
| $\vartheta$ | 12.15 | 0.3    | 1.2   | 13.14  | = 0, |
| $c$         |       | 1.2    | −0.3  | −12.15 |      |
|             | −1.2  |        | 12.15 | 0.3    |      |
|             | 0.3   | −12.15 |       | −1.2   |      |
|             | 12.15 | −0.3   | 1.2   |        |      |

|             |        |       |        |       |      |
|-------------|--------|-------|--------|-------|------|
| $\vartheta$ | 8.11   | 4.7   | 5.6    | 9.10  | = 0, |
| $c$         |        | 1.2   | −0.3   | 12.15 |      |
|             | −1.2   |       | −12.15 | 0.3   |      |
|             | 0.3    | 12.15 |        | −1.2  |      |
|             | −12.15 | −0.3  | 1.2    |       |      |

|             |       |      |       |       |      |
|-------------|-------|------|-------|-------|------|
| $\vartheta$ | 1.6   | 3.4  | 8.15  | 10.13 | = 0, |
| $c$         |       | 8.15 | −3.4  | 1.6   |      |
|             | −8.15 |      | 1.6   | −3.4  |      |
|             | 3.4   | −1.6 |       | 8.15  |      |
|             | −1.6  | 3.4  | −8.15 |       |      |

|             |      |       |       |       |      |
|-------------|------|-------|-------|-------|------|
| $\vartheta$ | 2.5  | 0.7   | 11.12 | 9.14  | = 0, |
| $c$         |      | −8.15 | −3.4  | 1.6   |      |
|             | 8.15 |       | 1.6   | −3.4  |      |
|             | 3.4  | −1.6  |       | −8.15 |      |
|             | −1.6 | 3.4   | 8.15  |       |      |

|             |      |       |      |       |      |
|-------------|------|-------|------|-------|------|
| $\vartheta$ | 2.6  | 0.4   | 8.12 | 10.14 | = 0, |
| $c$         |      | −8.12 | 0.4  | −2.6  |      |
|             | 8.12 |       | −2.6 | 0.4   |      |
|             | −0.4 | 2.6   |      | −8.12 |      |
|             | 2.6  | −0.4  | 8.12 |       |      |

|             |      |       |       |       |      |
|-------------|------|-------|-------|-------|------|
| $\vartheta$ | 1.5  | 3.7   | 11.15 | 9.13  | = 0, |
| $c$         |      | −8.12 | −0.4  | 2.6   |      |
|             | 8.12 |       | 2.6   | −0.4  |      |
|             | 0.4  | −2.6  |       | −8.12 |      |
|             | −2.6 | 0.4   | 8.12  |       |      |

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118. The foregoing equations may be verified, and it is interesting to verify them, by means of the approximate values of the functions; thus, for one of the equations,

$$c_0 c_3 \vartheta_3 \vartheta_{15}, \quad \text{i.e. } \frac{1}{2} \vartheta_3 \cdot 1 \cdot \vartheta_{15} \cdot 1,$$

we have

$$\begin{aligned} & (2A + 2A')(-2A + 2A') \\ & -1 \cdot 1 \cdot 2A \cos \frac{1}{2}\pi(u + v) + 2A' \cos \frac{1}{2}\pi(u - v) \cdot [-2A \cos \frac{1}{2}\pi(u + v) + 2A' \cos \frac{1}{2}\pi(u - v)] \\ & + 1 \cdot 1 \cdot -2A \sin \frac{1}{2}\pi(u + v) - 2A' \sin \frac{1}{2}\pi(u - v) \cdot [-2A \sin \frac{1}{2}\pi(u + v) + 2A' \sin \frac{1}{2}\pi(u - v)] \\ & = 0 \end{aligned}$$

viz. the equation to be verified is here

$$\begin{aligned} & -4A^2 + 4A'^2 \\ & + 4A^2 \cos^2 \frac{1}{2}\pi(u + v) - 4A'^2 \cos^2 \frac{1}{2}\pi(u - v) \\ & + 4A^2 \sin^2 \frac{1}{2}\pi(u + v) - 4A'^2 \sin^2 \frac{1}{2}\pi(u - v) = 0, \end{aligned}$$

which is right.

119. In the equation

$$c_0 c_3 \vartheta_3 \vartheta_{15}, \quad \text{i.e. } 2Q \cdot 1 \cdot 2Q \cos \frac{1}{2}\pi u \cdot 1 - 2Q \cdot 1 \cdot 2Q \cos \frac{1}{2}\pi u \cdot 1 = 0;$$

this is right, but there is no verification as to the term  $c_8 c_4 \vartheta_7 \vartheta_{11}$ ; taking the more approximate values, the term in question taken negatively, that is,  $-c_8 c_4 \vartheta_7 \vartheta_{11}$ , is

$$= -(2A + 2A') \cdot 2S \cdot -2S \sin \frac{1}{2}\pi v \cdot [-2A \sin \frac{1}{2}\pi(u + v) + 2A' \sin \frac{1}{2}\pi(u - v)],$$

which is

$$= -8S^2(A + A')^2 \cos \frac{1}{2}\pi u + 8S^2(A + A')A \cos \frac{1}{2}\pi(u + 2v) + 8S^2(A + A')A' \cos \frac{1}{2}\pi(u - 2v),$$

and this ought therefore to be the value of the first two terms, that is, of

$$\begin{aligned} & (2Q + 2Q^9 - 2A - 2A')(1 - 2Q^4 - 2S^4) \{ 2Q \cos \frac{1}{2}\pi u + 2Q^9 \cos \frac{9}{2}\pi u \\ & \quad + 2A \cos \frac{1}{2}\pi(u + 2v) + 2A' \cos \frac{1}{2}\pi(u - 2v) \} (1 - 2Q^4 \cos \pi u + 2S^4 \cos \pi v) \\ & - (2Q + 2Q^9 + 2A + 2A')(1 - 2Q^4 + 2S^4) \{ 2Q \cos \frac{1}{2}\pi u + 2Q^9 \cos \frac{9}{2}\pi u \\ & \quad - 2A \cos \frac{1}{2}\pi(u + 2v) - 2A' \cos \frac{1}{2}\pi(u - 2v) \} (1 - 2Q^4 \cos \pi u - 2S^4 \cos \pi v), \end{aligned}$$

which to the proper degree of approximation is

$$\begin{aligned} & = (2Q - 4Q^5 - 4QS^4 + 2Q^9 - 2A - 2A') [2Q \cos \frac{1}{2}\pi u - 4Q^5 \cos \frac{1}{2}\pi u \cos \pi u \\ & \quad + 4QS^4 \cos \frac{1}{2}\pi u \cos \pi v + 2Q^9 \cos \frac{9}{2}\pi u + 2A \cos \frac{1}{2}\pi(u + 2v) + 2A' \cos \frac{1}{2}\pi(u - 2v)] \\ & - (2Q - 4Q^5 + 4QS^4 + 2Q^9 + 2A + 2A') [2Q \cos \frac{1}{2}\pi u - 4Q^5 \cos \frac{1}{2}\pi u \cos \pi u \\ & \quad - 4QS^4 \cos \frac{1}{2}\pi u \cos \pi v + 2Q^9 \cos \frac{9}{2}\pi u - 2A \cos \frac{1}{2}\pi(u + 2v) - 2A' \cos \frac{1}{2}\pi(u - 2v)]. \end{aligned}$$



This is

$$(2M_0 - 2\Omega_0)(2M + 2\Omega) - (2M_0 + 2\Omega_0)(2M - 2\Omega) = 8(M_0\Omega - M\Omega_0),$$

if for a moment

$$M = Q \cos \frac{1}{2}\pi u - 2Q^5 \cos \frac{1}{2}\pi u \cos \pi u + Q^9 \cos \frac{9}{2}\pi u, \quad M_0 = Q - 2Q^5 + Q^9,$$

$$\Omega = 2QS^4 \cos \frac{1}{2}\pi u \cos \pi v + A \cos \frac{1}{2}\pi(u + 2v) + A' \cos \frac{1}{2}\pi(u - 2v), \quad \Omega_0 = 2QS^4 + A + A',$$

or substituting and reducing, the value of  $8(M_0\Omega - M\Omega_0)$  to the proper degree of approximation is found to be

$$= -8Q(2QS^4 + A + A') \cos \frac{1}{2}\pi u + 8(Q^2S^4 + 8QA) \cos \frac{1}{2}\pi(u + 2v) + 8(Q^2S^4 + 8QA') \cos \frac{1}{2}\pi(u - 2v),$$

which in virtue of the relations  $QA = A'S^4$ ,  $QA' = A'S^4$ ;  $Q^2S^4 = AA'$ , is equal to the foregoing value of  $c_8c_4\vartheta_7\vartheta_{11}$ . I have thought it worth while to give this somewhat elaborate verification.

*Résumé of the foregoing results.*

**120.** In what precedes we have all the quadric relations between the 16 double theta-functions; we have the linear relations between squares (squared functions) or say between pairs (products of two functions): the number of the aszygetic linear relations between squares is obviously = 12, and that of the aszygetic linear relations between pairs is = 60 (since each of the 30 tetrads of pairs gives two aszygetic relations): there are thus in all  $12 + 60$ , = 72, aszygetic linear relations. But these constitute only a 13-fold relation between the functions, viz. they are such as to give for the ratios of the 16 functions expressions depending upon two arbitrary parameters,  $x, y$ . Or taking the 16 functions as the coordinates of a point in 15-dimensional space, these coordinates are connected by a 13-fold relation (expressed by means of the foregoing system of 72 quadric equations), and the locus is thus a 13-fold, or two-dimensional, locus in 15-dimensional space.

Hence, taking any four of the functions, these are connected by a single equation; that is, regarding the four functions as the coordinates of a point in ordinary space, the locus of the point is a surface.

In particular, the four functions may be any four functions belonging to a hexad: by what precedes there is then a linear relation between the squares of the four functions: or the locus is a quadric surface. Each hexad gives 15 such surfaces, or the number of quadric surfaces is  $(16 \times 15 =) 240$ .

*The 16-nodal quartic surfaces.*

**121.** If the four functions are those contained in any two pairs out of a tetrad of pairs (see the foregoing “Table of the 120 pairs”), then the locus is a quartic

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surface, which is, in fact, a Kummer's 16-nodal quartic surface. For if for a moment  $x.y$  and  $z.w$  are two pairs out of a tetrad, and  $r.s$  be either of the remaining pairs of the tetrad; then we have  $rs$  a linear function of  $xy$  and  $zw$ ; squaring,  $r^2s^2$  is a linear function of  $x^2y^2$ ,  $xyzw$ ,  $z^2w^2$ ; but we then have  $r^2$  and  $s^2$  each of them a linear function of  $x^2$ ,  $y^2$ ,  $z^2$ ,  $w^2$ ; or substituting we have an equation of the fourth order, containing terms of the second order in  $(x^2, y^2, z^2, w^2)$ , and also a term in  $xyzw$ . It is clear that, if instead of  $r.s$  we had taken the remaining pair of the tetrad, we should have obtained the same quartic equation in  $(x, y, z, w)$ . And moreover it appears by inspection that, if  $xy$  and  $zw$  are pairs in a tetrad, then  $xz$  and  $yw$  are pairs in a second tetrad, and  $xw$  and  $yz$  are pairs in a third tetrad: we obtain in each case the same quartic equation. We have from each tetrad of pairs six sets of four functions  $(x, y, z, w)$ : and the number of such sets is thus  $(6C_2 \cdot 30 =) 60$ ; these are shown in the foregoing "Table of the 60 Göpel tetrads," viz. taking as coordinates of a point the four functions in any tetrad of this table, the locus is a 16-nodal quartic surface.

122. To exhibit the process I take a tetrad 4, 7, 8, 11 containing two odd functions; and representing these for convenience by  $x, y, z, w$ , viz. writing

$$\vartheta_4, \vartheta_7, \vartheta_8, \vartheta_{11}(u) = x, y, z, w,$$

we have then  $X, Y, Z, W$  linear functions of the four squares, viz. it is easy to obtain

$$\begin{aligned} \alpha(x^2 + z^2) - \beta(y^2 + w^2) &= 2(\alpha^2 - \beta^2) X, \\ \beta( \quad, \quad) - \alpha( \quad, \quad) &= 2( \quad, \quad) W, \\ -\beta(x^2 - z^2) + \gamma(y^2 - w^2) &= 2(\beta^2 - \gamma^2) Y, \\ -\gamma( \quad, \quad) + \beta( \quad, \quad) &= 2( \quad, \quad) Z. \end{aligned}$$

Also considering two other functions  $\vartheta_0(u)$  and  $\vartheta_{12}(u)$ , or as for shortness I write them,  $\vartheta_0$  and  $\vartheta_{12}$ , we have

$$\begin{aligned} \vartheta_0^2 &= \alpha X + \beta Y + \gamma Z + \delta W, \\ \vartheta_{12}^2 &= \alpha X - \beta Y - \gamma Z + \delta W, \end{aligned}$$

and substituting the foregoing values of  $X, Y, Z, W$ , we find

$$\begin{aligned} M\vartheta_0^2 &= Ax^2 + By^2 + Cz^2 + Dw^2, \\ M\vartheta_{12}^2 &= Cx^2 + Dy^2 + Az^2 + Bw^2, \end{aligned}$$

where, writing down the values first in terms of  $\alpha, \beta, \gamma, \delta$  and then in terms of the  $c$ 's, we have

$$\begin{aligned} M &= (\alpha^2 - \beta^2)(\beta^2 - \gamma^2) &= \frac{1}{4} \cdot c_4^2 - c_{12}^2, \\ A &= \alpha\delta\beta^2 - \alpha^2\beta\gamma &= \alpha c_3 c_4^2, \\ B &= -\alpha\delta(\beta^2 - \gamma^2) + \beta\gamma(\alpha^2 - \delta^2) &= c_4^2 c_{12}^2 - c_3 c_{15}, \\ C &= \alpha^2\beta^2 - \gamma^2\delta^2 &= c_4 c_{12}, \end{aligned}$$

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and we then have further

$$c_0 c_3 \vartheta_0 \vartheta_{12} = c_4 c_{12} xz + c_3 c_{15} yw,$$

that is,

$$M \vartheta_0 \vartheta_{12} = \sqrt{a} xz + \sqrt{b} yw,$$

whence equating the two values of  $\vartheta_0^2 \vartheta_{12}^2$  we have the required quartic equation in  $x, y, z, w$ .

**123.** But the reduction is effected more simply if instead of the  $c$ 's we introduce the rectangular coefficients  $a, b, c$ , &c. We then have

$$M = (c''^2 - b'^2), \quad A = -a''c, \quad C = a'b,$$

$$B = -b'c' - b''c'' = bc, \quad D = b'b'' + c'c'' = a'a'';$$

and the equations become

$$(c''^2 - b'^2) \vartheta_0^2 = -a''cx^2 + bcy^2 + a'bz^2 - a'a''w^2,$$

$$(c''^2 - b'^2) \vartheta_{12}^2 = a'bx^2 - a'a''y^2 - a''cz^2 + bcw^2,$$

$$\sqrt{b'^2 c''^2} \vartheta_0 \vartheta_{12} = \sqrt{a} xz + \sqrt{b} yw,$$

so that the elimination gives

$$b'c''(-x^2c + bcy^2 + a'bz^2 - a'a''w^2)(a'bx^2 - a'a''y^2 - a''cz^2 + bcw^2) = (c''^2 - b'^2)^2 \{ax^2z^2 - b''c'y^2w^2 + 2\sqrt{-ab''c'}xyzw\},$$

viz. this is

$$\begin{aligned} & -a'a''b'cc''(x^4 + y^4 + z^4 + w^4) \\ & + a'b'cc''(a''^2 + b^2)(x^2z^2 + y^2w^2) \\ & + [b'c''(a''^2b^2 + a'^2c^2) - a(b'^2 - c''^2)^2]x^2z^2 \\ & + [b'c''(a'^2a''^2 + b^2c^2) + b''c'(b'^2 - c''^2)^2]y^2w^2 \\ & - a''bb'c''(a'^2 + c^2)(x^2w^2 - y^2z^2) \\ & - 2(b'^2 - c''^2)^2\sqrt{-ab''c'}xyzw = 0. \end{aligned}$$

**124.** In this equation the coefficients of  $x^2z^2$  and  $y^2w^2$  are each  $= a'a''bc(b'^2 + c''^2)$ , as at once appears from the identities

$$a'b \cdot b' - c'' \cdot a''c = a(b'^2 - c''^2),$$

$$a'b \cdot c'' - b' \cdot a''c = (b'^2 - c''^2),$$

$$a'a'' \cdot b' - c'' \cdot bc = -b''(b'^2 - c''^2),$$

$$a'a'' \cdot c - b' \cdot bc = c'(b'^2 - c''^2),$$

by multiplying together in each pair the left-hand and the right-hand sides respectively. Substituting and dividing by  $-a'a''bb'cc''$ , we have

$$\begin{aligned} & \frac{x^4 + y^4 + z^4 + w^4}{a''^2 + b^2} - \frac{b'^2 + c''^2}{a''^2 + b^2} + \frac{a'^2 + c^2}{a''^2 + b^2} \quad [\text{arrangement as in scan}] \\ & + \frac{2(b'^2 - c''^2)^2\sqrt{-ab''c'}}{a'a''bb'cc''} xyzw = 0. \end{aligned}$$

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or, if we herein restore the  $c$ 's in place of the rectangular coefficients, this is

$$\frac{x^4 + y^4 + z^4 + w^4}{c_4^2 c_{12}^2} - \frac{c_3^2 + c_{15}^2}{c_4^2 c_{12}^2} (x^2 z^2 + y^2 w^2) - \frac{c_3^2 c_{15}^2}{c_4^2 c_{12}^2} (x^2 w^2 + y^2 z^2) + \frac{2(c_4^2 - c_{12}^2)^2 c_0 c_3}{c_4 c_{12} \cdot c_0 c_3} xyzw = 0,$$

which is the equation of the 16-nodal quartic surface.

Substituting for  $x, y, z, w$  their values  $\vartheta_4, \vartheta_7, \vartheta_8, \vartheta_{11}(u)$ , we have the equation connecting the four theta-functions 4, 7, 8, 11 of a Göpel tetrad. And there is an equation of the like form between the four functions of any other Göpel tetrad: for obtaining the actual equations some further investigation would be necessary.

*The  $xy$ -expressions of the theta-functions.*

**125.** The various quadric relations between the theta-functions, admitting that they constitute a 13-fold relation, show that the theta-functions may be expressed as proportional to functions of two arbitrary parameters  $x, y$ ; and two of these functions being assumed at pleasure the others of them would be determinate: we have of course (though it would not be easy to arrive at it in this manner) such a system in the foregoing expressions of the 16 functions in terms of  $x, y$ ; and conversely these expressions must satisfy identically the quadric relations between the theta-functions.

**126.** To show that this is so as to the general form of the equations, consider first the  $xy$ -factors  $\sqrt{a}$ ,  $\sqrt{ab}$ , &c. As regards the squared functions  $(\sqrt{ab})^2$ , we have for instance

$$(\sqrt{ab})^2 = \frac{1}{\theta^2} \{ab f_2 c_2 d_2 e_2 + a_2 b_2 f_2 c d e + 2\sqrt{XY}\},$$

$$(\sqrt{cd})^2 = \frac{1}{\theta^2} \{cd f_2 a_2 b_2 e_2 + c_2 d_2 f_2 a b e + 2\sqrt{XY}\},$$

each of these contains the same irrational part  $\frac{2}{\theta^2} \sqrt{XY}$ , and the difference is therefore rational: and it is moreover integral, for we have

$$(\sqrt{ab})^2 - (\sqrt{cd})^2 = \frac{1}{\theta^2} (abc_2 d_2 - a_2 b_2 cd)(f e_2 - f_2 e),$$

where each factor divides by  $\theta$ , and consequently the product by  $\theta^2$ ; the value is in fact

$$= (e - f) \begin{vmatrix} 1, & x + y, & xy \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix}$$

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a linear function of  $1, x + y, xy$ . This is the case as regards the difference of any two of the squares  $(\sqrt{ab})^2, (\sqrt{ac})^2$ , &c.; hence selecting any one of these squares, for instance  $(\sqrt{de})^2$ , any other of the squares is of the form

$$\lambda + \mu(x + y) + \nu xy + \rho(\sqrt{de})^2, \quad (\rho = 1);$$

and obviously, the other squares  $(\sqrt{a})^2$ , &c., are of the like form, the last coefficient  $\rho$  being  $= 0$ . We hence have the theorem that each square can be expressed as a linear function of any four (properly selected) squares.

**127.** But we have also the theorem of the 16 Kummer hexads.

Obviously the six squares

$$(\sqrt{a})^2, (\sqrt{b})^2, (\sqrt{c})^2, (\sqrt{d})^2, (\sqrt{e})^2, (\sqrt{f})^2$$

are a hexad, viz. each of these is a linear function of  $1, x + y, xy$ : and therefore selecting any three of them, each of the remaining three can be expressed as a linear function of these.

But further the squares  $(\sqrt{a})^2, (\sqrt{b})^2, (\sqrt{ab})^2, (\sqrt{cd})^2, (\sqrt{ce})^2, (\sqrt{de})^2$  form a hexad.

For reverting to the expression obtained for  $(\sqrt{ab})^2 - (\sqrt{cd})^2$ , the determinant contained therein is a linear function of  $aa_2$  and  $bb_2$ , that is, of  $(\sqrt{a})^2$  and  $(\sqrt{b})^2$ ; we, in fact, have

$$(a - b) \begin{vmatrix} 1, & x + y, & xy \\ 1, & a + b, & ab \\ 1, & c + d, & cd \end{vmatrix} = (b - c)(b - d)(a - x)(a - y) - (a - c)(a - d)(b - x)(b - y).$$

Hence  $(\sqrt{ab})^2 - (\sqrt{cd})^2$  is a linear function of  $(\sqrt{a})^2, (\sqrt{b})^2$ ; and by a mere interchange of letters  $(\sqrt{ab})^2 - (\sqrt{ce})^2, (\sqrt{ab})^2 - (\sqrt{de})^2$ , are each of them also a linear function of  $(\sqrt{a})^2$  and  $(\sqrt{b})^2$ ; whence the theorem. And we have thus all the remaining 15 hexads.

**128.** We have a like theory as regards the products of pairs of functions. A tetrad of pairs is of one of the two forms

$$\sqrt{a}\sqrt{b}, \sqrt{ac}\sqrt{bc}, \sqrt{ad}\sqrt{bd}, \sqrt{ae}\sqrt{be}, \quad \text{and} \quad \sqrt{f}\sqrt{ab}, \sqrt{c}\sqrt{de}, \sqrt{d}\sqrt{ce}, \sqrt{e}\sqrt{cd};$$

in the first case the terms are

$$\begin{aligned} & \sqrt{aa_2bb_2}, \\ & \frac{1}{\theta} \{ (ab_2 + a_2b) \sqrt{cdefc_2d_2e_2f_2} + (cf d_2e_2 + c_2f_2de) \sqrt{aa_2bb_2} \}, \\ & \frac{1}{\theta} \{ \quad, \quad + (dfc_2e_2 + d_2f_2ce) \sqrt{aa_2bb_2} \}, \\ & \frac{1}{\theta} \{ \quad, \quad + (efc_2d_2 + e_2f_2cd) \sqrt{aa_2bb_2} \}, \end{aligned}$$

and as regards the last three terms the difference of any two of them is a mere constant multiple of  $\sqrt{aa_2bb_2}$ ; for instance, the second term — the third term is

$$\frac{1}{\theta} (cd_2 - c_2d)(fe_2 - f_2e) \sqrt{aa_2bb_2} = (c - d)(f - e) \sqrt{aa_2bb_2}.$$

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we have thus a tetrad such that, selecting any two terms, each of the remaining terms is a linear function of these.

In the second case, the terms are

$$\begin{aligned} & \frac{1}{\theta} \{ f \sqrt{abc_2d_2e_2f_2} + f_2 \sqrt{a_2b_2cdef} \}, \\ & \frac{1}{\theta} \{ c \quad , , \quad + c_2 \quad , , \quad \}, \\ & \frac{1}{\theta} \{ d \quad , , \quad + d_2 \quad , , \quad \}, \\ & \frac{1}{\theta} \{ e \quad , , \quad + e_2 \quad , , \quad \}, \end{aligned}$$

whence clearly the four terms are a tetrad as above. And it may be added that any linear function of the four terms is of the form

$$\frac{1}{\theta} \{ (\lambda + \mu x) \sqrt{abc_2d_2e_2f_2} + (\lambda + \mu y) \sqrt{a_2b_2cdef} \}.$$

129. Considering next the actual equations between the squared theta-functions, take as a specimen

$$c_2^4(\sqrt{ab})^2 - c_4^4(\sqrt{cd})^2 + c_5^4(\sqrt{ce})^2 - c_6^4(\sqrt{de})^2 = 0,$$

that is,

$$c_6^4(\sqrt{ab})^2 - c_3^4(\sqrt{cd})^2 + c_4^4(\sqrt{ce})^2 - c_5^4(\sqrt{de})^2 = 0,$$

where  $c_8, c_3, c_4, c_5 = \sqrt{ab}, \sqrt{cd}, \sqrt{ce}, \sqrt{de}$  respectively. Since the functions  $(\sqrt{ab})^2$ , &c., contain the same irrational term  $\frac{2}{\theta^2} \sqrt{XY}$ , it is clear that the equation can only be true if

$$c_8^4 - c_3^4 + c_4^4 - c_5^4 = 0;$$

and, this being so, it will be true if

$$c_3^4 \{ (\sqrt{ab})^2 - (\sqrt{cd})^2 \} - c_4^4 \{ (\sqrt{ab})^2 - (\sqrt{ce})^2 \} + c_5^4 \{ (\sqrt{ab})^2 - (\sqrt{de})^2 \} = 0,$$

where, by what precedes, each of the terms in  $\{ \}$  is a linear function of  $(\sqrt{a})^2$  and  $(\sqrt{b})^2$ . Attending first to the term in  $(\sqrt{a})^2$ , the coefficient hereof is

$$ef \cdot bc \cdot bd \cdot c_3^4 - df \cdot bc \cdot be \cdot c_4^4 + cf \cdot bd \cdot be \cdot c_5^4,$$

where for shortness  $bc, bd$ , &c., are written to denote the differences  $b - c$ ,  $b - d$ , &c.

substituting for  $c_3^4$  its value  $(\sqrt{cd})^4 = cd \cdot cf \cdot df \cdot ab \cdot ae \cdot be$ , and similarly for  $c_4^4$  and  $c_5^4$  their values,  $= ce \cdot cf \cdot ef \cdot ab \cdot ad \cdot bd$ , and  $de \cdot df \cdot ef \cdot ab \cdot ac \cdot bc$  respectively, the whole expression contains the factor  $ab \cdot bc \cdot bd \cdot be \cdot cf \cdot df \cdot ef$ , and throwing this out, the equation to be verified becomes

$$cd \cdot ae - ce \cdot ad + de \cdot ac = 0,$$

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which is true identically. The verification is thus made to depend upon that of  $c_3^4 - c_4^4 + c_5^4 - c_8^4 = 0$ ; and similarly for the other relations between the squared functions, the verification depends upon relations containing the fourth powers, or the products of squares, of the constants  $c$  and  $k$ .

**130.** Among these are included the before-mentioned system of equations involving the fourth powers or the products of squares of only the constants  $c$ ; and it is interesting to show how these are satisfied identically by the values  $c_0 = \sqrt{bd}$ , &c.

Thus one of these equations is  $c_{11}^4 + c_3^4 + c_5^4 = c_0^4$ ; substituting the values, there is a factor  $ce$  which divides out, and the resulting equation is

$$ad \cdot af \cdot df \cdot bc \cdot be + cf \cdot ef \cdot ab \cdot ad \cdot bd + ab \cdot af \cdot bf \cdot cd \cdot de - ac \cdot ae \cdot bd \cdot bf \cdot df = 0.$$

There are here terms in  $a^3$ ,  $a^2$ ,  $a$  which should separately vanish; for the terms in  $a^3$ , the equation becomes

$$df \cdot bc \cdot be + bd \cdot cf \cdot ef + bf \cdot cd \cdot de - bd \cdot bf \cdot df = 0,$$

which is easily verified; and the equations in  $a^2$  and  $a^0$  may also be verified.

An equation involving products of the squares is  $c_8^2 c_9^2 - c_4^2 c_5^2 + c_1^2 c_2^2 = 0$ . The term  $c_8^2 c_9^2$  is here  $\sqrt{adf \cdot bce} \sqrt{def \cdot abc}$  which is  $= \sqrt{(bc)^2 (df)^2 \cdot ab \cdot ac \cdot ad \cdot af \cdot be \cdot ce \cdot de \cdot ef}$ , which is taken  $= bc \cdot df \sqrt{ab \cdot ac \cdot ad \cdot af \cdot be \cdot ce \cdot de \cdot ef}$ ; similarly the values of  $c_4^2 c_5^2$  and  $c_1^2 c_2^2$  are  $= bd \cdot cf$  and  $bf \cdot cd$  each multiplied by the same radical, and the equation to be verified is

$$bc \cdot df - bd \cdot cf + bf \cdot cd = 0,$$

which is right: the other equations may be verified in a similar manner.

**131.** Coming next to the equations connecting the pairs of theta-functions, for instance

$$c_8 c_{12} c_0 c_{12} [\sqrt{bd} \sqrt{ad} - \sqrt{be} \sqrt{ae}] + c_4 c_5 \kappa_9 \kappa_{11} \cdot \sqrt{b} \sqrt{a} = 0,$$

this is

$$c_8 c_{12} c_0 c_{12} [\sqrt{bd} \sqrt{ad} - \sqrt{be} \sqrt{ae}] + c_4 c_5 \kappa_9 \kappa_{11} \cdot \sqrt{b} \sqrt{a} = 0,$$

the products  $\sqrt{bd} \sqrt{ad}$  and  $\sqrt{be} \sqrt{ae}$  contain besides a common term the terms

$$\frac{1}{\theta} (df c_2 e_2 + d_2 f_2 c e) \sqrt{a a_2 b b_2}, \quad \text{and} \quad \frac{1}{\theta} (e f c_2 d_2 + e_2 f_2 c d) \sqrt{a a_2 b b_2};$$

hence their difference contains  $\frac{1}{\theta} (de_2 - d_2 e) (f c_2 - f_2 c) \sqrt{a a_2 b b_2}$  which is  $= de \cdot f c \sqrt{a a_2 b b_2}$ , that is,  $de \cdot f c \sqrt{a} \sqrt{b}$ : hence the equation to be verified is

$$de \cdot f c \cdot c_8 c_{12} c_0 c_{12} + c_4 c_5 \kappa_9 \kappa_{11} = 0;$$

$c_8 c_{12} c_0 c_{12} = \sqrt{bef \cdot acd} \sqrt{adf \cdot bcd} \sqrt{bdf \cdot ace} \sqrt{adf \cdot bce}$ , where under the fourth root we have 24 factors, which are, in fact, 12 factors twice repeated; and if we write

$$\Pi = ab \cdot ac \cdot ad \cdot ae \cdot af \cdot bc \cdot bd \cdot be \cdot bf \cdot cd \cdot ce \cdot cf \cdot de \cdot df \cdot ef$$

for the product of all the 15 factors, then the 12 factors are in fact all those of  $\Pi$ , except  $ab, cf, de$ ; viz. we have

$$c_8 c_{12} c_0 c_{12} = \sqrt{\Pi^2 / (ab)^2 (cf)^2 (de)^2}.$$

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Again,  $c_4c_5\kappa_9\kappa_{11} = \sqrt{acf \cdot bde} \sqrt{bcf \cdot ade} \sqrt{acdef} \sqrt{bcdef}$ , is a fourth root of a product of 32 factors, which are in fact 16 factors twice repeated, and in the 16 factors,  $ab$  does not occur,  $cf$  and  $de$  occur each twice, and the other 12 factors each once: we thus have

$$c_4c_5\kappa_9\kappa_{11} = \sqrt{\Pi^2(cf)^2(de)^2 \div (ab)^2},$$

and the relation to be verified assumes the form

$$fc \cdot de \sqrt{\Pi^2/(cf)^2(de)^2} + \sqrt{\Pi^2(cf)^2(de)^2} = 0,$$

which, taking  $fc \cdot de = -\sqrt{(cf)^2(de)^2}$ , is right. And so for the other equations. It will be observed that, in the equation  $de \cdot fc \cdot c_8c_{12}c_0c_{12} + c_4c_5\kappa_9\kappa_{11} = 0$ , and in the other equations upon which the verifications depend, there is no ambiguity of sign: the signs of the radicals have to be determined consistently with all the equations which connect the  $c$ 's and the  $k$ 's; that this is possible appears evident *à priori*, but the actual verification presents some difficulty. I do not here enter further into the question.

*Further results of the product-theorem, the  $u \pm u'$  formulæ.*

**132.** Recurring now to the equations in  $u + u'$ ,  $u - u'$ , by putting therein  $u' = 0$ , we can express  $X, Y, Z, W$  in terms of four of the squared functions of  $u$ , and by putting  $u = 0$  we can express  $X', Y', Z', W'$  in terms of four of the squared functions of  $u'$ ; and, substituting in the original equations, we have the products

$$\vartheta( )_{u+u'} \cdot \vartheta( )_{u-u'}$$

in terms of the squared functions of  $u$  and  $u'$ .

Selecting as in a former investigation the functions 4, 7, 8, 11, which were called  $x, y, z, w$ , it is more convenient to use single letters to represent the squared functions. I write

$$\begin{array}{llll} \vartheta_4(u+u') \vartheta_4(u-u') & \vartheta_4^2 u & \vartheta_4^2 u' & \vartheta_4^2 0 \\ 4 \quad 4 = P, & 4 = p, & 4 = p', & 4 = p_0 (= c_4^2), \\ 7 \quad 7 = Q, & 7 = q, & 7 = q', & 7 = 0, \\ 8 \quad 8 = R, & 8 = r, & 8 = r', & 8 = r_0 (= c_8^2), \\ 11 \quad 11 = S, & 11 = s, & 11 = s', & 11 = 0. \end{array}$$

Then

$$\begin{array}{lll} X \ Y \ Z \ W & X \ Y \ Z \ W & X' \ Y' \ Z' \ W' \\ P = X^2 - Y^2 + Z^2 - W^2, & p = \alpha - \beta + \gamma - \delta, & p' = \alpha - \beta + \gamma - \delta, \\ Q = W^2 - Z^2 + Y^2 - X^2, & q = \delta - \gamma + \beta - \alpha, & q' = \delta - \gamma + \beta - \alpha, \\ R = X^2 + Y^2 - Z^2 - W^2, & r = \alpha + \beta - \gamma - \delta, & r' = \alpha + \beta - \gamma - \delta, \\ S = W^2 + Z^2 - Y^2 - X^2, & s = \delta + \gamma - \beta - \alpha, & s' = \delta + \gamma - \beta - \alpha. \end{array}$$

Hence

$$\begin{array}{ll} \alpha(p+r) - \delta(q+s) = 2(\alpha^2 - \delta^2)X, & \alpha(p'+r') - \delta(q'+s') = 2(\alpha^2 - \delta^2)X', \\ \delta \quad , -\alpha \quad , = 2 \quad , W, & \delta \quad , -\alpha \quad , = 2 \quad , W', \\ -\beta(p-r) + \gamma(q-s) = 2(\beta^2 - \gamma^2)Y, & -\beta(p'-r') + \gamma(q'-s') = 2(\beta^2 - \gamma^2)Y', \\ -\gamma \quad , +\beta \quad , = 2 \quad , Z, & -\gamma \quad , +\beta \quad , = 2 \quad , Z'. \end{array}$$



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By means of these values, we have

$$\begin{aligned} 4(\alpha^2 - \delta^2)^2 X'X &= \alpha^2(p+r)(p'+r') + \delta^2(q+s)(q'+s') - \alpha\delta[(p+r)(q'+s') + (p'+r')(q+s)], \\ 4 \quad , \quad W'W &= \delta^2 \quad , \quad +\alpha^2 \quad , \quad -\alpha\delta[ \quad , \quad ], \\ 4(\beta^2 - \gamma^2)^2 Y'Y &= \beta^2(p-r)(p'-r') + \gamma^2(q-s)(q'-s') - \beta\gamma[(p-r)(q'-s') + (p'-r')(q-s)], \\ 4 \quad , \quad Z'Z &= \gamma^2 \quad , \quad +\beta^2 \quad , \quad -\beta\gamma[ \quad , \quad ]. \end{aligned}$$

Hence

$$\begin{aligned} 4(\alpha^2 - \delta^2)(X'X - W'W) &= (p+r)(p'+r') - (q+s)(q'+s'), \\ 4(\beta^2 - \gamma^2)(Y'Y - Z'Z) &= (p-r)(p'-r') - (q-s)(q'-s'), \end{aligned}$$

and substituting in the expressions for  $P$  and  $R$ ,

$$\begin{aligned} 4(\alpha^2 - \delta^2)(\beta^2 - \gamma^2)P &= (\beta^2 - \gamma^2)[(p+r)(p'+r') - (q+s)(q'+s')] - (\alpha^2 - \delta^2)[(p-r)(p'-r') - (q-s)(q'-s')], \\ 4 \quad , \quad R &= \quad , \quad - \quad , \quad . \end{aligned}$$

Similarly

$$\begin{aligned} 4(\alpha^2 - \delta^2)^2 W'X &= \alpha\delta[(p+r)(p'+r') + (q+s)(q'+s')] - \alpha^2(p+r)(q'+s') - \delta^2(q+s)(p'+r'), \\ 4 \quad , \quad X'W &= \quad , \quad -\delta^2 \quad , \quad -\alpha^2 \quad , \quad , \\ 4(\beta^2 - \gamma^2)^2 Z'Y &= \beta\gamma[(p-r)(p'-r') + (q-s)(q'-s')] - \beta^2(p-r)(q'-s') - \gamma^2(q-s)(p'-r'), \\ 4 \quad , \quad Y'Z &= \quad , \quad -\gamma^2 \quad , \quad -\beta^2 \quad , \quad . \end{aligned}$$

whence

$$\begin{aligned} 4(\alpha^2 - \delta^2)(W'X - X'W) &= -[(p+r)(q'+s') - (p'+r')(q+s)], \\ 4(\beta^2 - \gamma^2)(Z'Y - Y'Z) &= -[(p-r)(q'-s') - (p'-r')(q-s)], \end{aligned}$$

and substituting in the expressions for  $Q$  and  $S$

$$\begin{aligned} 4(\alpha^2 - \delta^2)(\beta^2 - \gamma^2)Q &= -(\beta^2 - \gamma^2)[(p+r)(q'+s') - (p'+r')(q+s)] + (\alpha^2 - \delta^2)[(p-r)(q'-s') - (p'-r')(q-s)], \\ , \quad S &= - \quad , \quad - \quad , \quad . \end{aligned}$$

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133. Hence, collecting and reducing,

$$4(\alpha^2 - \delta^2)(\beta^2 - \gamma^2)P = -(\alpha^2 - \beta^2 + \gamma^2 - \delta^2)(pp' - qq' + rr' - ss') + (\alpha^2 + \beta^2 - \gamma^2 - \delta^2)(pr' + p'r - qs' - q's),$$

$$\begin{aligned} 4 \quad , , \quad R &= , , \\ 4 \quad , , \quad Q &= , , \\ 4 \quad , , \quad S &= , , \end{aligned}$$

we have

$$p_0 (= c_4^2) = \alpha^2 - \beta^2 + \gamma^2 - \delta^2, \quad r_0 (= c_8^2) = \alpha^2 + \beta^2 - \gamma^2 - \delta^2,$$

and thence

$$r_0^2 - p_0^2 = 4(\alpha^2 - \delta^2)(\beta^2 - \gamma^2);$$

the equations hence become

$$\begin{aligned} (r_0^2 - p_0^2)P &= -p_0(pp' - qq' + rr' - ss') + r_0(pr' + p'r - qs' - q's), \\ , , \quad R &= r_0( \quad , \quad ) - p_0( \quad , \quad ), \\ , , \quad Q &= p_0(pq' - p'q + rs' - r's) - r_0( \quad , \quad ), \\ , , \quad S &= -r_0( \quad , \quad ) + p_0( \quad , \quad ). \end{aligned}$$

On writing in the equations  $u' = 0$ , then  $P, Q, R, S, p', q', r', s'$  become  $= p, q, r, s, p_0, 0, r_0, 0$ ; and the equations are (as they should be) true identically. The equations may be written

$$\begin{aligned} (8 - 4)\vartheta \frac{u+u'}{4} \vartheta \frac{u-u'}{4} &= -4(\vartheta_4 u \vartheta_4 u' - \vartheta_7 u \vartheta_7 u' + \vartheta_8 u \vartheta_8 u' - \vartheta_{11} u \vartheta_{11} u') + 8(\vartheta_4 u \vartheta_8 u' + \vartheta_8 u \vartheta_4 u' - \vartheta_7 u \vartheta_{11} u' - \vartheta_{11} u \vartheta_7 u'), \\ ( \quad , \quad ) \vartheta_8 \vartheta_8 &= +8( \quad , \quad ) - 4( \quad , \quad ), \\ ( \quad , \quad ) \vartheta_7 \vartheta_7 &= +4(\vartheta_4 u \vartheta_7 u' - \vartheta_7 u \vartheta_4 u' + \vartheta_8 u \vartheta_{11} u' - \vartheta_{11} u \vartheta_8 u') - 8(\vartheta_4 u \vartheta_{11} u' - \vartheta_{11} u \vartheta_4 u' + \vartheta_8 u \vartheta_7 u' - \vartheta_7 u \vartheta_8 u'), \\ ( \quad , \quad ) \vartheta_{11} \vartheta_{11} &= -8( \quad , \quad ) + 4( \quad , \quad ). \end{aligned}$$

There is of course such a system for each of the 60 Göpel tetrads.

*Differential relations connecting the theta-functions with the quotient-functions.*

134. Imagine  $p, q, r, s$ , &c., changed into  $x'^2, y'^2, z'^2, w'^2$ , &c.; that is, let  $x, y, z, w$  represent the theta-functions 4, 7, 8, 11 of  $u, v$ ; and similarly  $x', y', z', w'$  those of  $u', v'$ , and  $x_0, 0, z_0, 0$  those of  $0, 0$ . Let  $u', v'$  be each of them indefinitely small; and take  $\partial, = u' \frac{d}{du} + v' \frac{d}{dv}$ , as the symbol of total differentiation in regard to  $u, v$ , the infinitesimals  $u'$  and  $v'$  being arbitrary: then, as far as the second order, we have in general

$$\vartheta(u + u', v + v') = \vartheta(u, v) + \partial \vartheta(u, v) + \frac{1}{2} \partial^2 \vartheta(u, v),$$

and hence

$$P = (x + \partial x + \frac{1}{2}\partial^2 x)(x - \partial x + \frac{1}{2}\partial^2 x), \quad = x^2 + \{x\partial^2 x - (\partial x)^2\},$$

and similarly for  $Q, R, S$ . Moreover, observing that  $x'$  and  $z'$  are even functions,  $y'$  and  $w'$  are odd functions, of  $u', v'$ , we have

$$x', y', z', w' = x_0 + \frac{1}{2}\partial^2 x_0, \partial y_0, z_0 + \frac{1}{2}\partial^2 z_0, \partial w_0,$$

where  $\partial^2 x_0, \partial y_0$ , &c., are what  $\partial^2 x, \partial y$ , &c., become on writing therein  $u = 0, v = 0$ ;  $\partial y_0, \partial w_0$  are of course linear functions,  $\partial^2 x_0, \partial^2 z_0$  quadric functions of  $u'$  and  $v'$ . The values of  $x'^2, y'^2, z'^2, w'^2$  are thus  $x_0^2 + x_0\partial^2 x_0, (\partial y_0)^2, z_0^2 + z_0\partial^2 z_0, (\partial w_0)^2$ ; and we have

$$\begin{aligned} & x_0^2 x_0 \partial^2 x_0 \quad (\partial y_0)^2 \quad z_0^2 z_0 \partial^2 z_0 \quad (\partial w_0)^2 \\ x^2 x'^2 - y^2 y'^2 + z^2 z'^2 - w^2 w'^2 &= s^2 x_0^2 + z^2 W^2 + z^2 - y^2 + z^2 - w^2, \\ x^2 y'^2 - y^2 x'^2 + z^2 w'^2 - w^2 z'^2 &= -y^2 x_0^2 - w^2 z_0^2 - y^2 + x^2 - w^2 + z^2, \\ x^2 z'^2 - y^2 w'^2 + z^2 x'^2 - w^2 y'^2 &= z^2 x_0^2 + x^2 z_0^2 + z^2 - w^2 + x^2 - y^2, \\ x^2 w'^2 - y^2 z'^2 + z^2 y'^2 - w^2 x'^2 &= -w^2 x_0^2 - y^2 z_0^2 - w^2 + z^2 - y^2 + x^2. \end{aligned}$$

135. On substituting these values, the constant terms (or terms independent of  $u', v'$ ) disappear of themselves; and the equations, transposing the second and third of them, become

$$\begin{aligned} & x_0 \partial^2 x_0 \quad (\partial y_0)^2 \quad z_0 \partial^2 z_0 \quad (\partial w_0)^2 \\ (z_0^2 - x_0^2)\{x\partial^2 x - (\partial x)^2\} &= (-x_0^2 x^2 + z_0^2 z^2) + (x_0^2 y^2 - z_0^2 w^2) + (-x_0^2 z^2 + z_0^2 x^2) + (x_0^2 w^2 - z_0^2 y^2), \\ , , \quad \{y\partial^2 y - (\partial y)^2\} &= -(x_0^2 y^2 - z_0^2 w^2) - (-x_0^2 x^2 + z_0^2 z^2) - (x_0^2 w^2 - z_0^2 y^2) - (-x_0^2 z^2 + z_0^2 x^2), \\ , , \quad \{z\partial^2 z - (\partial z)^2\} &= (-x_0^2 z^2 + z_0^2 x^2) + (x_0^2 w^2 - z_0^2 y^2) + (-x_0^2 x^2 + z_0^2 z^2) + (x_0^2 y^2 - z_0^2 w^2), \\ , , \quad \{w\partial^2 w - (\partial w)^2\} &= -(x_0^2 w^2 - z_0^2 y^2) - (-x_0^2 z^2 + z_0^2 x^2) - (x_0^2 y^2 - z_0^2 w^2) - (-x_0^2 x^2 + z_0^2 z^2). \end{aligned}$$

where it will be recollected that  $x, y, z, w$  mean  $\vartheta_4, \vartheta_7, \vartheta_8, \vartheta_{11}(u)$ ;  $x_0$  is  $\vartheta_4(0)$ , that is,  $c_4$ , and  $z_0$  is  $\vartheta_8(0)$ , that is,  $c_8$ . But the formulae contain also

$$\partial^2 x_0 = (c_4''', c_4''''', c_4'' | u', v')^2, \quad \partial y_0 = (c_7', c_7'' | u', v'),$$

$$\partial^2 z_0 = (c_8''', c_8''''', c_8'' | u', v')^2, \quad \partial w_0 = (c_{11}', c_{11}'' | u', v').$$

The formulae may be written

$$\begin{aligned} & c_4 \partial^2 c_4 \quad (\partial c_7)^2 \quad c_8 \partial^2 c_8 \quad (\partial c_{11})^2 \\ (c_8^2 - c_4^2)\{4\partial^2 4 - (\partial 4)^2\} &= (-4 \cdot 4 + 8 \cdot 8) + (4 \cdot 7 - 8 \cdot 11) + (-4 \cdot 8 + 8 \cdot 4) + (4 \cdot 11 - 8 \cdot 7), \\ , , \quad \{7\partial^2 7 - (\partial 7)^2\} &= -(4 \cdot 7 - 8 \cdot 11) - (-4 \cdot 4 + 8 \cdot 8) - (4 \cdot 11 - 8 \cdot 7) - (-4 \cdot 8 + 8 \cdot 4), \\ , , \quad \{8\partial^2 8 - (\partial 8)^2\} &= (-4 \cdot 8 + 8 \cdot 4) + (4 \cdot 11 - 8 \cdot 7) + (-4 \cdot 4 + 8 \cdot 8) + (4 \cdot 7 - 8 \cdot 11), \\ , , \quad \{11\partial^2 11 - (\partial 11)^2\} &= -(4 \cdot 11 - 8 \cdot 7) - (-4 \cdot 8 + 8 \cdot 4) - (4 \cdot 7 - 8 \cdot 11) - (-4 \cdot 4 + 8 \cdot 8). \end{aligned}$$

where  $\partial^2 c_4, \partial^2 c_8, \partial c_7, \partial c_{11}$  are written in place of  $\partial^2 x_0, \partial^2 z_0, \partial y_0, \partial w_0$ . There is of course a like system of equations for each of the Göpel tetrads.

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**136.** Observe that, dividing the first equation by  $\vartheta_4^2(u)$  or say by  $\vartheta_4^2$ , the left-hand side is a mere constant multiple of  $\partial^2 \log \vartheta_4$ ; and the right-hand side depends only on the quotient-functions  $\vartheta_7 \div \vartheta_4$ ,  $\vartheta_8 \div \vartheta_4$ ,  $\vartheta_{11} \div \vartheta_4$ ; each side is a quadric function of  $u', v'$ . Equating the terms in  $u'^2$ ,  $u'v'$ ,  $v'^2$  respectively, we have

$$\frac{d^2}{du^2} \log \vartheta_4, \quad \frac{d^2}{du dv} \log \vartheta_4, \quad \frac{d^2}{dv^2} \log \vartheta_4,$$

each of them expressed as a linear function of the squares of the quotient-functions  $\vartheta_7 \div \vartheta_4$ ,  $\vartheta_8 \div \vartheta_4$ ,  $\vartheta_{11} \div \vartheta_4$ . The formula is thus a second-derivative formula serving for the expression of a double theta-function by means of three quotient-functions.

*Differential relations of the theta-functions.*

**137.** In “The second set of 16,” selecting the eight equations which contain  $Y_1$  and  $W_1$ , these are

| $u + u'$                                                                                                           | $u - u'$ | $u + u'$ | $u - u'$ | (Suffixes 1.) |
|--------------------------------------------------------------------------------------------------------------------|----------|----------|----------|---------------|
| $\frac{1}{2}\{\vartheta_4 \cdot \vartheta_4 - \vartheta_{12} \cdot \vartheta_8\} \cdot \vartheta_8 \vartheta_{12}$ |          |          |          | $= Y' + W',$  |
| $\frac{1}{2}\{\vartheta_{12} \vartheta_8 - \vartheta_8 \vartheta_{12}\}$                                           |          |          |          | $= Y' - W',$  |
| $\frac{1}{2}\{\vartheta_6 \vartheta_2 - \vartheta_2 \vartheta_6\}$                                                 |          |          |          | $= W' + Y',$  |
| $\frac{1}{2}\{\vartheta_{14} \vartheta_{10} - \vartheta_{10} \vartheta_{14}\}$                                     |          |          |          | $= W' - Y',$  |
| $\frac{1}{2}\{\vartheta_5 \vartheta_1 + \vartheta_1 \vartheta_5\}$                                                 |          |          |          | $= X' + Z',$  |
| $\frac{1}{2}\{\vartheta_{13} \vartheta_9 + \vartheta_9 \vartheta_{13}\}$                                           |          |          |          | $= X' - Z',$  |
| $\frac{1}{2}\{\vartheta_7 \vartheta_3 + \vartheta_3 \vartheta_7\}$                                                 |          |          |          | $= Z' + X',$  |
| $\frac{1}{2}\{\vartheta_{15} \vartheta_{11} + \vartheta_{11} \vartheta_{15}\}$                                     |          |          |          | $= Z' - X'.$  |
|                                                                                                                    |          |          |          | $Y \quad W$   |
|                                                                                                                    |          |          |          | $X \quad Z$   |

Then, considering any line in the upper half and any two lines in the lower half, we can from the three equations eliminate  $Y_1$  and  $W_1$ , thus obtaining an equation such as

$$\begin{vmatrix} \vartheta_4 \vartheta_4 - \vartheta_{12} \vartheta_8 & Y'_1 & W'_1 \\ \vartheta_5 \vartheta_1 + \vartheta_1 \vartheta_5 & X'_1 & Z'_1 \\ \vartheta_7 \vartheta_3 + \vartheta_3 \vartheta_7 & X'_1 & Z'_1 \end{vmatrix} = 0,$$

viz. this is

$$-2X'Z'(\vartheta_4 \vartheta_0 - \vartheta_0 \vartheta_4) + (Z'^2 + X'^2)(\vartheta_5 \vartheta_1 + \vartheta_1 \vartheta_5) + (-Z'^2 + X'^2)(\vartheta_7 \vartheta_3 + \vartheta_3 \vartheta_7) = 0,$$

where the arguments of the theta-functions are as above,  $u + u'$ ,  $u - u'$ ,  $u + u'$ ,  $u - u'$ ; and the suffixes of the  $X', Y', Z', W'$  are all = 1.

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**138.** Suppose that in this equation  $u'$  becomes indefinitely small. If  $u' = 0$ , the values of  $X', Y', Z', W'$  would be  $\alpha, 0, \gamma, 0$ : hence  $u'$  being indefinitely small, we take them to be  $\alpha, \partial\beta, \gamma, \partial\delta$ , where

$$\partial\beta = \left(u' \frac{d}{du} + v' \frac{d}{dv}\right) \beta, \quad \partial\delta = \left(u' \frac{d}{du} + v' \frac{d}{dv}\right) \delta,$$

are, in fact, linear functions of  $u'$  and  $v'$ .

We have  $\vartheta_4\vartheta_0 - \vartheta_0\vartheta_4$ , standing for

$$\vartheta_4(u + u') \vartheta_0(u - u') - \vartheta_0(u + u') \vartheta_4(u - u'),$$

and here

$$\vartheta_4(u \pm u') = \vartheta_4 \pm \partial\vartheta_4, \quad \vartheta_0(u \pm u') = \vartheta_0 \pm \partial\vartheta_0;$$

the function in question is thus

$$(\vartheta_4 + \partial\vartheta_4)(\vartheta_0 - \partial\vartheta_0) - (\vartheta_4 - \partial\vartheta_4)(\vartheta_0 + \partial\vartheta_0) = 2(\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0),$$

where the arguments are  $u, v$ , and the  $\partial$  denotes  $u' \frac{d}{du} + v' \frac{d}{dv}$ .

Also  $\vartheta_5\vartheta_1 + \vartheta_1\vartheta_5$ , that is,  $\vartheta_5(u + u')\vartheta_1(u - u') + \vartheta_1(u + u')\vartheta_5(u - u')$ , becomes simply  $= 2\vartheta_5\vartheta_1$ , and similarly  $\vartheta_7\vartheta_3 + \vartheta_3\vartheta_7$  becomes  $= 2\vartheta_3\vartheta_7$  [as in scan]; and the equation thus is

$$-2\alpha_1\gamma_1(\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0) + (\alpha_1\partial\delta_1 + \gamma_1\partial\beta_1)\vartheta_5\vartheta_1 + (-\alpha_1\partial\beta_1 + \gamma_1\partial\delta_1)\vartheta_3\vartheta_7 = 0,$$

where the proper suffix 1 is restored to the  $\alpha, \partial\beta, \gamma$ , and  $\partial\delta$ .

**139.** The equation shows that the differential combination  $\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0$  is a linear function of  $\vartheta_5\vartheta_1$  and  $\vartheta_3\vartheta_7$ , the coefficients of these products being of course linear functions of  $u'$  and  $v'$ . Writing the equation

$$\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0 = A\vartheta_5\vartheta_1 + B\vartheta_3\vartheta_7,$$

we can if we please determine the coefficients in terms of the constants  $c', c'', c''', c''''$ ; viz. taking  $u, v$  indefinitely small, we have

$$\vartheta_0 = c_0, \quad \partial\vartheta_0 = u'(c_0'''u + c_0''''v) + v'(c_0''''u + c_0''v),$$

$$\vartheta_4 = c_4, \quad \partial\vartheta_4 = u'(c_4'''u + c_4''''v) + v'(c_4''''u + c_4''v),$$

$$\vartheta_1 = c_1, \quad \vartheta_5 = c_5'u + c_5''v,$$

or substituting, and equating the coefficients of  $u$  and  $v$  respectively, we have

$$c_0(c_4'''u' + c_4''''v') - c_4(c_0'''u' + c_0''''v') = Ac_5c_1' + Bc_3c_7',$$

$$c_0(c_4''''u' + c_4''v') - c_4(c_0''''u' + c_0''v') = Ac_5c_1'' + Bc_3c_7'',$$

which equations give the values of  $A, B$ .

**140.** Disregarding the values of the coefficients, and attending only to the form of the equation

$$\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0 = A\vartheta_5\vartheta_1 + B\vartheta_3\vartheta_7,$$

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this is one of a system of 120 equations; viz. referring to the foregoing table of the 120 pairs, it in fact appears that taking any pair such as  $\vartheta_0\vartheta_4$  out of the upper compartment or the lower compartment of any column of the table, the corresponding differential combination  $\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0$  is a linear function of any two of the four pairs in the other compartment of the same column.

*Differential relation of  $u, v$  and  $x, y$ .*

141. We have, as before, in the two notations, the pairs

|              |    |   |
|--------------|----|---|
| $A \cdot B$  | 11 | 7 |
| $C \cdot DE$ | 5  | 9 |
| $D \cdot CE$ | 13 | 1 |
| $E \cdot CD$ | 14 | 2 |
| $F \cdot AB$ | 10 | 6 |

From the expressions given above for the four pairs below the line, it is clear that any linear function of these four pairs may be represented by

$$(a - b)\frac{1}{\theta}\{(\lambda + \mu y)\sqrt{cdefa_2b_2} + (\lambda + \mu x)\sqrt{c_2d_2e_2f_2ab}\},$$

where  $\lambda, \mu$  are constant coefficients: the factor  $(a - b)$  has been introduced for convenience, as will appear.

We have consequently a relation

$$\sqrt{aa_2}\partial\sqrt{bb_2} - \sqrt{bb_2}\partial\sqrt{aa_2} = \frac{a - b}{\theta}\{(\lambda + \mu y)\sqrt{cdefa_2b_2} + (\lambda + \mu x)\sqrt{c_2d_2e_2f_2ab}\},$$

where, as before,  $\partial$  is used to denote  $u'\frac{d}{du} + v'\frac{d}{dv}$ ,  $u'$  and  $v'$  being arbitrary multipliers; considering  $u, v$  as functions of  $x, y$ , we have

$$\frac{d}{du} = \frac{dx}{du}\frac{d}{dx} + \frac{dy}{du}\frac{d}{dy}, \qquad \frac{d}{dv} = \frac{dx}{dv}\frac{d}{dx} + \frac{dy}{dv}\frac{d}{dy},$$

and thence  $\partial = P\frac{d}{dx} + Q\frac{d}{dy}$  if for shortness  $P$  and  $Q$  are written to denote  $u'\frac{dx}{du} + v'\frac{dx}{dv}$  and  $u'\frac{dy}{du} + v'\frac{dy}{dv}$  respectively.

142. The left-hand side then is

$$= P\left(\sqrt{aa_2}\frac{d}{dx}\sqrt{bb_2} - \sqrt{bb_2}\frac{d}{dx}\sqrt{aa_2}\right) + Q\left(\sqrt{aa_2}\frac{d}{dy}\sqrt{bb_2} - \sqrt{bb_2}\frac{d}{dy}\sqrt{aa_2}\right);$$

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the coefficients of  $P$  and  $Q$  are at once found to be

$$\frac{1}{2} \frac{(a-b)\sqrt{a_2b_2}}{\sqrt{ab}}, \quad \frac{1}{2} \frac{(a_2-b_2)\sqrt{ab}}{\sqrt{a_2b_2}},$$

respectively, or observing that  $a-b_2=a_2-b_2=a-b$ , the equation becomes

$$P \frac{\sqrt{a_2b_2}}{\sqrt{ab}} + Q \frac{\sqrt{ab}}{\sqrt{a_2b_2}} = \frac{2}{\theta} \{ (\lambda + \mu y) \sqrt{cdefa_2b_2} + (\lambda + \mu x) \sqrt{c_2d_2e_2f_2ab} \},$$

or multiplying by  $\sqrt{aba_2b_2}$  and writing for shortness  $abcdef = X$ ,  $a_2b_2c_2d_2e_2f_2 = Y$ , this becomes

$$a_2b_2 \{ P + \frac{2}{\theta} (\lambda + \mu y) \sqrt{X} \} + ab \{ Q + \frac{2}{\theta} (\lambda + \mu x) \sqrt{Y} \} = 0.$$

143. There are, it is clear, the like equations

$$b_2c_2 \{ P + \frac{2}{\theta} (\lambda' + \mu'y) \sqrt{X} \} + bc \{ Q + \frac{2}{\theta} (\lambda' + \mu'x) \sqrt{Y} \} = 0,$$

$$c_2a_2 \{ P + \frac{2}{\theta} (\lambda'' + \mu''y) \sqrt{X} \} + ca \{ Q + \frac{2}{\theta} (\lambda'' + \mu''x) \sqrt{Y} \} = 0,$$

and it is to be shown that  $\lambda = \lambda' = \lambda''$  and  $\mu = \mu' = \mu''$ . For this purpose, recurring to the forms

$$\sqrt{aa_2} \partial \sqrt{bb_2} - \sqrt{bb_2} \partial \sqrt{aa_2} = \frac{a-b}{\theta} \{ (\lambda + \mu y) \sqrt{cdefa_2b_2} + (\lambda + \mu x) \sqrt{c_2d_2e_2f_2ab} \},$$

$$\sqrt{bb_2} \partial \sqrt{cc_2} - \sqrt{cc_2} \partial \sqrt{bb_2} = \frac{b-c}{\theta} \{ (\lambda' + \mu'y) \sqrt{adefb_2c_2} + (\lambda' + \mu'x) \sqrt{a_2d_2e_2f_2bc} \},$$

$$\sqrt{cc_2} \partial \sqrt{aa_2} - \sqrt{aa_2} \partial \sqrt{cc_2} = \frac{c-a}{\theta} \{ (\lambda'' + \mu''y) \sqrt{bdefc_2a_2} + (\lambda'' + \mu''x) \sqrt{b_2d_2e_2f_2ca} \},$$

multiply the first equation by  $\sqrt{cc_2}$ , the second by  $\sqrt{aa_2}$ , and the third by  $\sqrt{bb_2}$ , and add: the left-hand side vanishes, and therefore the right-hand side must also vanish identically.

144. But on the right-hand side we have the term  $\frac{1}{\theta} \sqrt{defa_2b_2c_2}$  multiplied by

$$(a-b)c(\lambda + \mu y) + (b-c)a(\lambda' + \mu'y) + (c-a)b(\lambda'' + \mu''y),$$

and the term  $-\frac{1}{\theta} \sqrt{d_2e_2f_2abc}$  multiplied by

$$(a-b)c_2(\lambda + \mu x) + (b-c)a_2(\lambda' + \mu'x) + (c-a)b_2(\lambda'' + \mu''x),$$

and it is clear that the whole can vanish only if these two coefficients separately vanish. This will be the case if we have for  $\lambda, \lambda', \lambda''$  the equations

$$(a-b)\lambda + (b-c)\lambda' + (c-a)\lambda'' = 0,$$

$$c \quad , \quad +a \quad , \quad +b \quad , \quad = 0,$$

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and the like equations for  $\mu, \mu', \mu''$ . The equations written down give

$$(a-b)\lambda : (b-c)\lambda' : (c-a)\lambda'' = a-b : b-c : c-a,$$

that is,  $\lambda = \lambda' = \lambda''$ : and similarly  $\mu = \mu' = \mu''$ .

145. But this being so, the three equations in  $P, Q$  give

$$P + \frac{2}{\theta}(\lambda + \mu y)\sqrt{X} = 0, \quad Q + \frac{2}{\theta}(\lambda + \mu x)\sqrt{Y} = 0,$$

that is,

$$-\frac{1}{2}\theta \frac{dx}{\sqrt{X}} = (\lambda + \mu y) du + (\lambda + \mu y) dv \quad [\text{as in scan}],$$

$$-\frac{1}{2}\theta \frac{dy}{\sqrt{Y}} = (\lambda + \mu x) du + (\lambda + \mu x) dv.$$

In these equations  $u'$  and  $v'$  are arbitrary; hence  $\lambda$  and  $\mu$  must be linear functions of  $u'$  and  $v'$ ; say their values are  $= \varpi u' + \rho v'$ ,  $\sigma u' + \tau v'$  respectively. We have therefore

$$-\frac{1}{2}\theta \frac{dx}{\sqrt{X}} = (\varpi + \sigma y) du + (\rho + \tau y) dv,$$

$$-\frac{1}{2}\theta \frac{dy}{\sqrt{Y}} = (\varpi + \sigma x) du + (\rho + \tau x) dv,$$

whence also

$$\varpi du + \rho dv = -\frac{\theta}{2} \left( \frac{dx}{\sqrt{X}} + \frac{dy}{\sqrt{Y}} \right),$$

$$\sigma du + \tau dv = -\frac{\theta}{2} \left( \frac{x dx}{\sqrt{X}} + \frac{y dy}{\sqrt{Y}} \right),$$

which are the required relations, depending on the square roots of the sextic functions  $X = abcdef$ , and  $Y = a_2b_2c_2d_2e_2f_2$ , of  $x$  and  $y$  respectively; but containing the constants  $\varpi, \rho, \sigma, \tau$ , the values of which are not as yet ascertained.

146. I commence the integration of these equations on the assumption that the values  $u = 0$ ,  $v = 0$  correspond to indefinitely large values of  $x$  and  $y$ . We have

$$\sqrt{X} = x^3 \left( 1 - \frac{S}{x} + \dots \right), \quad \sqrt{Y} = y^3 \left( 1 - \frac{S}{y} + \dots \right),$$

where  $S = a + b + c + d + e + f$ ; and thence the equations are

$$\varpi du + \rho dv = -\frac{\theta}{2} \left[ \frac{dx}{x^3} \left( 1 + \frac{S}{x} + \dots \right) + \frac{dy}{y^3} \left( 1 + \frac{S}{y} + \dots \right) \right],$$



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Hence integrating, we have

$$\varpi u + \rho v = \frac{\theta}{4} \left( \frac{1}{x^2} + \frac{1}{y^2} \right) + \dots,$$

and thence

$$\sigma u + \tau v = \frac{\theta}{2} \left( \frac{1}{x} + \frac{1}{y} \right) + \frac{1}{2} S \left( \frac{1}{x^2} + \frac{1}{y^2} \right) + \dots,$$

where the omitted terms depend on  $\frac{1}{x^3}, \frac{1}{y^3}, \&c.$

Hence, neglecting these terms, we have

$$\frac{\sigma u + \tau v}{\varpi u + \rho v + \frac{1}{2} S(\sigma u + \tau v)} = - \left( \frac{1}{x} + \frac{1}{y} \right),$$

an equation connecting the indefinitely small values of  $u, v$ , with the indefinitely large values of  $x, y$ .

**147.** From the equations  $A = \frac{1}{2} k_3 \varpi \sqrt{a}$ ,  $B = \frac{1}{2} k_3 \sqrt{b}$ , taking  $(u, v)$  indefinitely small and therefore  $(x, y)$  indefinitely large, we deduce

$$\frac{c'_{11} u + c''_{11} v}{c'_7 u + c''_7 v} = \frac{k_a}{k_b} \cdot \frac{1 - \frac{1}{2} a \left( \frac{1}{x} + \frac{1}{y} \right)}{1 - \frac{1}{2} b \left( \frac{1}{x} + \frac{1}{y} \right)},$$

hence substituting for  $\frac{1}{x} + \frac{1}{y}$  the foregoing value, and introducing an indeterminate multiplier  $M$ , we obtain

$$c'_{11} u + c''_{11} v = M k_a \left[ \varpi u + \rho v + \frac{1}{2} S(\sigma u + \tau v) + \frac{1}{2} a(\sigma u + \tau v) \right],$$

which breaks up into the two equations

[two component equations as in scan]

Similarly

$$\begin{aligned} c'_7 &= M k_b \{ \dots b \dots \}, & c''_7 &= M k_b \{ \dots b \dots \}, \\ c'_{10} &= M k_c \{ \dots c \dots \}, & c''_{10} &= M k_c \{ \dots c \dots \}, \\ c'_{13} &= M k_d \{ \dots d \dots \}, & c''_{13} &= M k_d \{ \dots d \dots \}, \\ c'_5 &= M k_e \{ \dots e \dots \}, & c''_5 &= M k_e \{ \dots e \dots \}, \\ c'_{14} &= M k_f \{ \dots f \dots \}, & c''_{14} &= M k_f \{ \dots f \dots \}, \end{aligned}$$

which twelve equations determine the coefficients  $\varpi, \sigma, \rho, \tau$  in terms of the  $c', c''$  of the odd functions 5, 7, 10, 11, 13, 14; and moreover give rise to relations connecting these  $c', c''$  with each other and with the constants  $a, b, c, d, e, f$ .

**148.** It is observed that if, as before,

$$\partial = u' \frac{d}{du} + v' \frac{d}{dv} = P \frac{d}{dx} + Q \frac{d}{dy},$$

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then, substituting for  $P$  and  $Q$  their values, we have

$$\begin{aligned}\partial &= -\frac{\theta}{2}(\varpi u' + \rho v')\left(\sqrt{X}\frac{d}{dx} + \sqrt{Y}\frac{d}{dy}\right) - \frac{\theta}{2}(\sigma u' + \tau v')\left(x\sqrt{X}\frac{d}{dx} + y\sqrt{Y}\frac{d}{dy}\right) \\ &= (\varpi u' + \rho v')\partial_1 + (\sigma u' + \tau v')\partial_2,\end{aligned}$$

if for shortness

$$\partial_1 = -\frac{\theta}{2}\left(\sqrt{X}\frac{d}{dx} + \sqrt{Y}\frac{d}{dy}\right), \quad \partial_2 = -\frac{\theta}{2}\left(x\sqrt{X}\frac{d}{dx} + y\sqrt{Y}\frac{d}{dy}\right);$$

then operating with  $\partial$  on the equations  $A = \varpi k_a \sqrt{a}$ , &c., we have for instance

$$A\partial B - B\partial A = \varpi^2 k_a k_b \{(\varpi u' + \rho v')(\sqrt{a}\partial_1\sqrt{b} - \sqrt{b}\partial_1\sqrt{a}) + (\sigma u' + \tau v')(\sqrt{a}\partial_2\sqrt{b} - \sqrt{b}\partial_2\sqrt{a})\},$$

which is one of a system of 120 equations, the  $A, B$  being in fact any two of the 16 functions.

These are in fact nothing else than the foregoing system of 120 equations giving the values of the differential combinations  $\vartheta_0\partial\vartheta_4 - \vartheta_4\partial\vartheta_0$ , &c., each as a sum of products of pairs of functions, only on the right-hand sides we have expressions such as  $\sqrt{a}\partial_1\sqrt{b} - \sqrt{b}\partial_1\sqrt{a}$ , &c., which present themselves as perfectly determinate functions of  $x, y$ : so that, regarding  $\varpi u' + \rho v'$ ,  $\sigma u' + \tau v'$  as given linear functions of the arbitrary quantities  $u', v'$ , there is no longer anything indeterminate in the form of the equations.

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PROBLEMS AND SOLUTIONS.

[From the *Mathematical Questions with their Solutions from the Educational Times*, vols. XIV. to LXI. (1871–1894).]

[Vol. XIV., July to December, 1870, pp. 17—19.]

**3002.** (Proposed by Matthew Collins, B.A.)—If every two of five circles  $A, B, C, D, E$  touch each other, except  $D$  and  $E$ , prove that the common tangent of  $D$  and  $E$  is just twice as long as it would be if  $D$  and  $E$  touched each other.

*Solution by Professor Cayley.*

Consider the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ , foci  $R, S$ ; the coordinates of a point  $U$  on the ellipse may be taken to be  $(a \cos u, b \sin u)$ , and then the distances of this point from the foci will be

$$r = a(1 - e \cos u), \quad s = a(1 + e \cos u).$$

Taking  $k$  arbitrarily, with centre  $R$  describe a circle radius  $a - k$ , with centre  $S$  a circle radius  $a + k$ , and with centre  $U$  a circle radius  $k - ae \cos u$ ; say these are the circles  $R, S, U$  respectively; the circle  $U$  will touch each of the circles  $R, S$  (viz. assuming  $ae < k < a$ , so that the foregoing radii are all positive, it will touch the circle  $R$  externally and the circle  $S$  internally).

Considering next a point  $V$ , coordinates  $(a \cos v, b \sin v)$ , and the circle described about this point with the radius  $k - ae \cos v$ , say the circle  $V$ ; this will touch in like manner the circles  $R, S$  respectively. And the circles  $U, V$  may be made to touch each other externally; viz. this will be the case if squared sum of radii = squared . . .